Finite group actions, loop structures, and profinite extensions

A mathematical development of *La Longue Marche*: LM4

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Working manuscript: INCOMPLETE. The numbered statements distinguish proved assertions, complete counterexamples, source theorems with their reductions, and the exact obligations still under investigation. This working record makes no assertion of completion of the task.

Notation and its manuscript concordance. Local generators, base points, coefficients and comparison maps are defined at their use. The table fixes the global symbols and the meaning of each variant.

Our notation	Meaning and Grothendieck's notation	Introducing passage / scans
$X, S, U = X - S$	Compact surface, marked set, punctured surface; X, S, U	§12: 91–92; §19: 204
$(g, \nu), I$	Genus, number and index set of punctures; $(g, \nu), I$	§13 bis: 115–116
π, π_0, P	Discrete surface group, chosen discretization, its completion; π, π_0, π	§12: 91; §28: 352
L_i, κ_i	Indexed peripheral subgroup and exterior source map; L_i, κ_i	§12: 91–99
$T = T_\pi$	Covariant orientation line; dual of the earlier cohomological line	91; 116; 130
A, A^0	Surface homeomorphism group and identity component; $A_{g,\nu}, A^0$	§19: 204–205
$M, \Gamma_{g,\nu}$	Geometric mapping class group; $\mathcal{T}_{g,\nu}, \Gamma_{g,\nu}$	193–203; 204–205
$!, +$	Pointwise marked-point preservation; orientation preservation	204–205
G, Γ, E	Finite action, operator group, lift extension; his Γ, G, E as locally specified	§15: 137–139; 307–312
Φ, μ	Maximal partial-section data and torsion multiplier; Φ, μ	150–160; 197–203
$SE, S\Gamma$	Special actual and exterior automorphism groups; $SE, S\Gamma$	§25: 296–300
$B, \overline{M}$	$\mathrm{Out}_{\mathrm{lac}}(P)$ and closure of a specified geometric group; $\widehat{\widehat{\mathcal{T}}}, \widehat{\mathcal{T}}$ where identified	318–320; 352–358
Discr, Prediscr	Actual, outer, or closed-stabilizer cosets as explicitly specified	§28: 352–358
GT_0, λ, f	The normalized centralizer of S_3, multiplier and commutator parameter	Two-group problem: 312, 316–317

Reading the proof record

We work in the order A–F of the supplied task. A group with loops retains its indexed exterior peripheral maps and its orientation line; this includes the exceptional cases where the image subgroups alone lose that data. All homomorphisms between profinite or pro-p groups are continuous. A section retains its prescribed quotient labels, and its equivalence is conjugacy by the named kernel. Every passage citation gives the manuscript section, folio, and scan; a folio followed by "verso" names its reverse side. A transcription alternative is retained when mathematical evidence bears on it. Our corrections are assertions about the displayed mathematics, not declarations that a doubtful reading has been settled.

The results register contains the full numbered assertions and dependencies. The readings register records the editorial consequences. The report gives the obligation-by-obligation state, source checks and computation provenance. An "Open" label records an unfinished obligation in this working manuscript; it does not replace a requested proof. A "Known" label includes the derivation from the precise source theorem. The code uses exact finite arithmetic, and the proofs identify what the computations certify and what requires a separate infinite-group argument.

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1 Groups with loops and their rigidity

We use $[x,y]=xyx^{-1}y^{-1}$. A peripheral subgroup is indexed even when two indices give the same subgroup. The notation $\mathrm{Out}_{\mathrm{lac}}$ means preservation of the indexed peripheral conjugacy classes, with their permitted permutations; in hyperbolic type these permutations are determined by the automorphism. We specify the additional data needed in the exceptional types below. All profinite homomorphisms and automorphisms in this section are continuous.

A.1. [Proved] The presentation, ranks, and exceptional types. Let X be a connected closed oriented topological surface of genus g, let S consist of ν distinct points, and put $U=X\setminus S$. A discrete group with loops of type (g,ν) consists of a group π, a finite index set I of cardinality ν, a free rank-one integral orientation module T, and exterior homomorphisms $\kappa_i : T\to\pi$. Here an exterior homomorphism is a conjugacy class, under conjugation in the target, of actual homomorphisms. After choosing a generator of T, representatives $c_i=\kappa_i(1)$, an ordering of I, and the handle generators, these data admit the presentation

$$\pi_{g,\nu}=\left\langle a_1,b_1,\dots,a_g,b_g,c_1,\dots,c_\nu \;\middle|\; \prod_{j=1}^{g}[a_j,b_j]c_1\cdots c_\nu=1\right\rangle,\qquad L_i=\langle c_i\rangle.$$

The profinite version uses a free rank-one $\widehat{\mathbb{Z}}$-module T, continuous exterior homomorphisms, and the corresponding presentation in profinite groups, namely the quotient of the free profinite group by the closed normal subgroup generated by this relator. When I is empty one additionally identifies T with the dual of $H^2(\pi;\mathbb{Z})$, or its continuous profinite counterpart. This is the covariant orientation convention of scan 116. More precisely, an earlier cohomological marking $\kappa : H^2(\pi;\mathbb{Z})\xrightarrow{\sim} T_{\mathrm{coh}}$ on scan 91 gives, by dualization, $\kappa^\vee : T_{\mathrm{coh}}^\vee\xrightarrow{\sim} H_2(\pi;\mathbb{Z})$; the covariant module used here is $T=T_{\mathrm{coh}}^\vee$. The same typed dual map specifies the profinite convention. For an isomorphism of groups with loops the data are an isomorphism $f:\pi\to\pi_1$, a bijection $s:I\to I_1$, and an isomorphism $t:T\to T_1$ such that $f\kappa_i$ is conjugate to $\kappa_{s(i)}t$ for every i, with the corresponding dual-cohomology compatibility when I is empty. An exterior group with loops regards two such f as equivalent when they differ by target inner conjugation, with the same s,t. Thus its isomorphism sets are explicitly the quotient of the preceding isomorphism sets by $\mathrm{Inn}(\pi_1)$. In all nontrivial admitted types $L_i=\kappa_i(T)$ recovers its source by A.2; the one-puncture sphere requires the original source map. For $\nu>0$ the group is free of rank $2g+\nu-1$, and its abelianization has that rank. For $\nu=0$ the abelianization has rank $2g$. The abelian types are exactly

$$(0,0),\quad (0,1),\quad (0,2),\quad (1,0),$$

with groups $1,1,\mathbb{Z},\mathbb{Z}^2$, respectively. The same list holds for their profinite completions. The peripheral maps from the source orientation group $\mathbb{Z}$ to π are injective except at $(0,1)$, among the types admitted in scans 91–92. The closed sphere is explicitly excluded there.

Manuscript: 12; 44–48; 91–100; 13 bis; 56–57; 115–118.

Proof. Eliminate c_ν when $\nu>0$; the remaining generators have no relation. Abelianizing the displayed presentation leaves only $c_1+\cdots+c_\nu=0$, giving both ranks. A free group of rank at least two is nonabelian, as witnessed by its quotient S_3, generated by (12) and (123). A closed genus-g group for $g\geq 2$ maps onto the free group on $a_1,\dots,a_g$ by killing all the b_j. The same finite quotient witnesses noncommutativity after completion. For $\nu\geq 2$, each peripheral generator belongs to a free basis obtained by eliminating a different peripheral generator. For $\nu=1$, $g\geq 1$, the nonempty cyclically reduced word $\prod[a_j,b_j]$ is nontrivial and has infinite order. Its image has infinite order in every pro-p completion: map the first pair into the upper unitriangular 3×3 matrices over $\mathbb{Z}_p$, and all other pairs to 1; the commutator is $1+E_{13}$. This also proves that its procyclic closure in the full completion is $\widehat{\mathbb{Z}}$, since every prime component

injects. At $(0,1)$ the peripheral word is 1, so the source map $\mathbb{Z} \to 1$ is not injective. These calculations prove the correction $\nu \leq 2$ to the bracket $\nu \leq 3$ on scan 116. This is mathematical evidence of an error, not a changed reading of the visible numeral. □

Manuscript: §12, folios 44–48, scans 91–100; §13 bis, folios 56–57, scans 115–118.

A.2. [Proved] Canonical comparison of the orientation modules. Assume $\nu > 0$ and $\pi \neq 1$. There is a canonical exact sequence

$$0 \longrightarrow T_\pi \longrightarrow \bigoplus_{i \in S} L_i \longrightarrow \pi^{\mathrm{ab}} \longrightarrow H_1(X;\mathbb{Z}) \longrightarrow 0, \qquad T_\pi = \ker\left(\bigoplus_i L_i \longrightarrow \pi^{\mathrm{ab}}\right).$$

Each projection $T_\pi \to L_i$ is an isomorphism of rank-one integral modules. If $M \subset U$ is obtained by removing small open discs at the punctures, the boundary morphism and evaluation give canonical isomorphisms

$$H_2(M, \partial M; \mathbb{Z}) \xrightarrow{\sim} T_\pi, \qquad T_\pi \xrightarrow{\sim} \mathrm{Hom}\big(H_c^2(U;\mathbb{Z}), \mathbb{Z}\big).$$

For $\nu = 0$, $g \geq 1$, the corresponding formula is $T_\pi = \mathrm{Hom}(H^2(\pi;\mathbb{Z}),\mathbb{Z}) = H_2(X;\mathbb{Z})$. Filling the discs identifies both orientation lines with $H_2(X;\mathbb{Z})$, naturally for homeomorphisms of the marked surface. All these comparisons persist with $\widehat{\mathbb{Z}}$ and continuous relative group cohomology. At $(0,1)$ the kernel formed from the image subgroup $L_1 = 1$ is zero, although $H_c^2(\mathbb{R}^2;\mathbb{Z}) = \mathbb{Z}$; the source peripheral morphism is therefore essential. At $(0,0)$ group cohomology $H^2(1;\mathbb{Z}) = 0$ cannot supply the surface orientation line.

Manuscript: 13 bis; 56–57; 116–118; 14; 64 verso; 130.

Proof. The relevant portion of the integral homology sequence of the compact pair is

$$0 \to H_2(M, \partial M) \xrightarrow{\partial} H_1(\partial M) \to H_1(M) \to H_1(M, \partial M).$$

The initial zero follows from $H_2(M) = 0$. A relative fundamental class has boundary the sum of the consistently oriented boundary circles. Identifying each boundary circle with its peripheral subgroup, the boundary morphism identifies $H_2(M, \partial M)$ with the stated kernel and has degree 1 on each factor. The cokernel is $H_1(X)$ by attaching the discs and using the homology sequence. Universal coefficients for the pair, whose homology is free, identify $H^2(M, \partial M)$ with the dual of $H_2(M, \partial M)$; excision identifies the former with $H_c^2(U)$. Let $D \subset X$ be the union of the filled discs. The canonical isomorphisms $H_2(X) \xrightarrow{\sim} H_2(X, D) \xrightarrow{\sim} H_2(M, \partial M)$ are, respectively, the pair map and excision. Their inverse is the filling comparison, sending a relative fundamental class to the closed fundamental class and proving compatibility without a chosen sign. Conjugating a peripheral subgroup changes its identification by an inner normalizer action, which is trivial by A.3; hence the sequence is independent of paths to the boundary. For the annulus the same conclusion follows directly because its group is abelian.

For clarity, the profinite compact-support notation is the relative continuous cochain complex

$$C^*_{\mathrm{rel}}(\widehat{\pi}; (\widehat{L}_i); \mathbb{Z}/n) = \mathrm{Cone}\left(C^*_{\mathrm{cont}}(\widehat{\pi}; \mathbb{Z}/n) \longrightarrow \bigoplus_i C^*_{\mathrm{cont}}(\widehat{L}_i; \mathbb{Z}/n)\right)[-1].$$

For $\nu > 0$, $\widehat{\pi}$ has cohomological dimension one and each $\widehat{L}_i \simeq \widehat{\mathbb{Z}}$; thus its degree-two group is the cokernel of restriction on H^1. In the displayed presentation the restriction matrix is the transpose of the integral peripheral inclusion matrix. Its cokernel is canonically $\mathrm{Hom}(T_\pi, \mathbb{Z}/n)$. Taking the inverse limit over n gives the claimed duality. For a closed surface of genus at least two, the continuous duality pairing identifies $H^2(\widehat{\pi}; \widehat{\mathbb{Z}})$ with the completed surface orientation cohomology; this is the surface Poincare duality statement in [46, Proposition 1.3]. At genus

one, for each prime p the Koszul resolution of the trivial module over $\mathbb{Z}_p[[\mathbb{Z}_p^2]]$ has differentials given by the two generators minus 1. Applying trivial coefficients $\mathbb{Z}/p^r$ makes both differentials zero, so the cup product identifies degree two with the exterior square of degree one, exactly as for the discrete torus. The other prime factors have zero positive-degree cohomology with these coefficients. The Chinese remainder theorem gives the comparison for every finite coefficient $\mathbb{Z}/n$. The transition maps on degree one are surjective, so their first derived inverse limit vanishes; consequently passing to inverse limits introduces no extra degree-two term. The canonical map from the discrete presentation to its completion induces the identity on the surface relation and on every peripheral source; consequently the finite-coefficient comparisons commute with filling. The two trivial-group exceptions follow by the explicit cohomology calculations in the statement. □

Manuscript: §13 bis, folios 56–57, scans 116–118; §14, folio 64 verso, scan 130. We read $H^2_!$ on scan 130 as the relative complex just specified, rather than as ordinary cohomology of the underlying abstract profinite group.

A.3. [Proved] Discrete peripheral rigidity in every hyperbolic type. For $2g+\nu>2$, the group $\pi=\pi_{g,\nu}$ has trivial centre and $C_\pi(H)=1$ for every finite-index subgroup H. For every peripheral subgroup and every integer $n\geq 1$,

$$N_\pi(L_i)=C_\pi(L_i)=C_\pi(L_i^n)=L_i.$$

Distinct peripheral indices give distinct, noncommensurable conjugacy classes.

Manuscript: 12; 47 verso–48; 98–99; 13 bis; 57; 117–118.

Proof. For $\nu>0$, the free action of a free group on its Cayley tree shows that a nonidentity element has an axis and that its centralizer is the infinite cyclic group of translations generated by its primitive root. An element normalizing this cyclic group preserves the axis. A reversal of the axis fixes a vertex or an edge midpoint, which is incompatible with the free, inversion-free action; the normalizer is the same cyclic group. For $\nu\geq 2$ each peripheral element is primitive in a free basis. For $\nu=1$, the cyclic word $\prod[a_j,b_j]$ contains the positive letter a_1 exactly once. A proper cyclic power has the number of occurrences of every signed letter divisible by its exponent, so this word is not a proper power. This proves the peripheral formulas including the marginal case of scan 117.

For the closed case, the hyperbolic polygon realization of [20, Theorem 1.2] embeds π_g as a torsion-free cocompact group of orientation-preserving isometries of the hyperbolic plane. Every nonidentity element is hyperbolic. Its centralizer preserves its axis and acts discretely by translations, hence is cyclic. An axis reversal would be an order-two elliptic element and is absent. A finite-index subgroup is a surface group of genus $1+[\pi:H](g-1)\geq 2$. In the punctured case the covering has free rank $1+[\pi:H](2g+\nu-2)\geq 2$. Thus a finite-index subgroup cannot lie in a cyclic centralizer. This proves $C_\pi(H)=1$, including $H=\pi$.

For distinct punctures with $\nu\geq 3$, their primitive abelianization vectors span distinct lines; for $\nu=2$, $g\geq 1$, choose the free basis containing c_1. The cyclically reduced lengths of c_1 and $c_2=(\prod[a_j,b_j]c_1)^{-1}$ are 1 and $4g+1$, and their abelianizations force any commensurability to match equal absolute exponents. They cannot be conjugate. This proves the final assertion; at $\nu=1$ it is vacuous. □

Manuscript: §12, folios 47 verso–48, scans 98–99; §13 bis, folio 57, scans 117–118.

A.4. [Proved] Exact exceptional-case table. The discrete and profinite exceptional groups have the following properties; in the profinite column replace $\mathbb{Z}$ by $\widehat{\mathbb{Z}}$.

(g,ν)	π	$Z(\pi)$	$C_\pi(H)$ (H finite index/open)	peripherals
$(0,0)$	1	1	1	none
$(0,1)$	1	1	1	$L_1 = 1$
$(0,2)$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$L_1 = L_2 = \pi$
$(1,0)$	$\mathbb{Z}^2$	$\mathbb{Z}^2$	$\mathbb{Z}^2$	none

Consequently the peripheral normalizer and power-centralizer equalities hold in the plane and annulus, while distinctness of the two peripheral classes fails on the annulus. The nonzero vector 1 on the annulus and $(1,0)$ on the torus witness failure of both centre and finite-index-centralizer assertions. The plane's failure is the noninjective source map, not a nontrivial centre. For the standard base-element formula of scan 501, the annulus satisfies the equality since its basis element generates its whole group. On the torus $C(\langle(1,0)\rangle^n) = \mathbb{Z}^2$ for every $n \geq 1$, and $(0,1)$ witnesses its failure to equal $\langle(1,0)\rangle$. The same vector works after completion. The sphere and plane have no nontrivial base elements to test.

Manuscript: 20; 111–115 verso; 222–230; 12; 47 verso; 98; 13 bis; 56 verso; 116.

Proof. A.1 identifies the groups and peripheral images. Every subgroup of an abelian group is centralized by the entire group. The assertions now follow by direct substitution. For the profinite version the listed elements remain nonzero, as their reductions modulo 2 show. □

Manuscript: §20, folios 111–115 verso, scans 222–230; omissions on scans 98 and 116.

A.5. [Proved] Peripheral decomposition groups and transporters. Let $1 \to \pi \to E \xrightarrow{p} \Gamma \to 1$ arise from an action on a hyperbolic marked surface. Choose one peripheral subgroup L_s and the corresponding peripheral lift group $E_s \subset E$; thus $E_s \cap \pi = L_s$ and $p(E_s) = \Gamma_s$, the stabilizer of that puncture. Define

$$\mathrm{Trans}_E(L_s^d, L_s) = \{e \in E : eL_s^d e^{-1} \subseteq L_s\} \qquad (d \geq 1).$$

Then $N_E(L_s) = \mathrm{Trans}_E(L_s^d, L_s) = E_s$. The same assertions hold for the corresponding profinite extension and closed peripheral groups. On the annulus these assertions fail for $z \mapsto 1/z$: with $E = \mathbb{Z} \rtimes C_2$ and the involution acting by inversion, $E_s = \mathbb{Z}$ but $N_E(L_s) = E$ and every transporter is E. More generally for every annulus action both normalizer and transporter equal E, while E_s is the inverse image of the end stabilizer. Thus their equality with E_s holds exactly when the action fixes both ends. On the one-puncture sphere the peripheral mixed group maps onto $E = \Gamma$ with kernel $\mathbb{Z}$; using its image, both equalities hold, whereas identifying that mixed group itself with a subgroup is invalid. The closed sphere and torus have no peripheral statement.

Manuscript: 9; 33; 69; 12; 48; 99.

Proof. A normalizer element maps to Γ_s, since the peripheral conjugacy classes are distinct. Choose $e_s \in E_s$ with the same image. Then $e_s^{-1}e$ belongs to $N_\pi(L_s) = L_s$, proving the normalizer equality. For the transporter assertion, $eL_s^d e^{-1} \subset L_s$ implies $L_s^d \subset e^{-1}L_s e$. The latter is abelian, so it lies in $C_\pi(L_s^d) = L_s$, by A.3 or A.7. In the discrete case a peripheral generator cannot be a proper power, so the inclusion of peripheral groups $e^{-1}L_s e \subset L_s$ is an equality. In the profinite case use compactness of E: a closed subgroup L of a compact group cannot be properly contained in an inner conjugate. Indeed $aLa^{-1} \subset L$ gives $a^k La^{-k} \subset L$ for all $k \geq 1$, and a net of positive powers of a converges to a^{-1} inside its procyclic closure. Passing to the limit gives the reverse conjugation inclusion and hence equality. Thus the transporter normalizes L_s in both cases. This argument only uses the exact power-centralizer statement and avoids replacing

the one-sided edge statement in [46, Proposition 1.2(i)] by a stronger unstated assertion about chosen subgroup representatives. The peripheral lift groups for the action on a punctured disc map onto the puncture stabilizer by construction, so the preceding choice of e_s is legitimate. For the annulus the involution swaps the two punctures, whose inertia subgroups both equal π. Its lift is the explicit element in the normalizer outside E_s. Replacing $\mathbb{Z}$ with $\widehat{\mathbb{Z}}$ gives the completed counterexample. □

Manuscript: §9, folio 33, scan 69; §12, folio 48, scan 99. In the plane, the map from the peripheral mixed group to E has a kernel $\mathbb{Z}$; writing that mixed group as a subgroup would already discard essential data.

A.6. [Proved] Centre-free extension reconstruction and its abelian correction. For a centre-free discrete group π and a homomorphism $\rho : \Gamma \to \mathrm{Out}(\pi)$, the extension inducing ρ is, up to a unique isomorphism inducing the identity on its prescribed kernel and quotient,

$$E_\rho = \{(a, \gamma) \in \mathrm{Aut}(\pi) \times \Gamma : [a] = \rho(\gamma)\}.$$

The same formula holds for a topologically finitely generated centre-free profinite π and continuous ρ, with the profinite topology. For $A = \mathbb{Z}$ or $\mathbb{Z}^2$ and a prescribed action $\rho : \Gamma \to \mathrm{Aut}(A)$, extension classes with their kernel and quotient identified are instead classified by $H^2(\Gamma; A_\rho)$. Their automorphisms inducing the identity on both ends form $Z^1(\Gamma; A_\rho)$. For finite Γ this statement holds with $A = \widehat{\mathbb{Z}}$ or $\widehat{\mathbb{Z}}^2$ and continuous cochains. Thus both the annulus and the torus require the abelian correction.

Manuscript: 12; 47 verso; 98; 17; 77–80; 155–160.

Proof. An extension E maps to E_ρ by $e \mapsto (\mathrm{conj}_e, p(e))$. Its kernel is $Z(\pi) = 1$. Given (a, γ), choose e_0 above γ. The automorphism $a\,\mathrm{conj}_{e_0}^{-1}$ is inner, implemented by a unique element of π, so the map is onto. An isomorphism inducing the identity on both ends fixes both components of this formula and is unique. In the profinite case this continuous bijection between compact Hausdorff groups is an isomorphism.

For abelian A, choose a set-theoretic section t of the quotient with $t(1) = 1$ and write $t(g)t(h) = c(g, h)t(gh)$. Associativity is exactly $c(g, h) + c(gh, k) = \rho(g)c(h, k) + c(g, hk)$. Changing t by a function $b : \Gamma \to A$ adds its coboundary. Conversely this identity makes $(a, g)(b, h) = (a + \rho(g)b + c(g, h), gh)$ associative. An automorphism fixing both ends has the form $(a, g) \mapsto (a + b(g), g)$, where the homomorphism equation says b is a 1-cocycle. For finite Γ all choices of section are continuous. For example $\Gamma = C_2$, $A = \mathbb{Z}$ with trivial action has the two extension classes $\mathbb{Z} \times C_2$ and $\mathbb{Z}$, with kernel $\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ in the latter. They induce the same outer action and are not isomorphic as extensions; reduction modulo 2 detects the different value of $t(\sigma)^2$. □

Manuscript: §12, folio 47 verso, scan 98; §17, folios 77–80, scans 155–160.

A.7. [Known] The full profinite rigidity statements, with reductions. For the full profinite completion $\Pi = \widehat{\pi}_{g,\nu}$ of a hyperbolic type, Π is torsion-free, $C_\Pi(H) = 1$ for every open subgroup H, and

$$N_\Pi(\widehat{L}_i) = C_\Pi(\widehat{L}_i^n) = \widehat{L}_i \quad (n \in \mathbb{Z}_{>0}).$$

Distinct peripheral classes are noncommensurable. The same conclusions hold for the pro-p completions, for every prime p, including 2 and 3. The four exceptional types have exactly the completed table of A.4, and A.5 supplies the extension and transporter assertions.

Manuscript: 13 bis; 57 verso; 118; 35; 253–254; 499–501.

Proof. Take the full formation of finite groups, respectively the formation of finite p-groups, in [47, Remark 1.2.2 and Proposition 1.4]. A hyperbolic complex curve with the prescribed topological presentation identifies its geometric group with Π, respectively the pro-p completion. Proposition 1.4 asserts precisely the vanishing of centralizers of open subgroups, with no properness assumption, so taking $H = \Pi$ gives centre-freeness. Remark 1.2.2 gives finite ℓ-cohomological dimension for every relevant prime, hence excludes an element of order ℓ: such an element would give a closed C_ℓ subgroup, whose ℓ-cohomological dimension is infinite. Every nontrivial finite-order element has a prime-order power; this proves torsion-freeness.

For peripherals, [45, Lemma 1.3.7] gives $\mathrm{Comm}_\Pi(\widehat{L}_i) = \widehat{L}_i$. A normalizer is contained in that commensurator. An element centralizing $\widehat{L}_i^n$ conjugates the finite-index subgroup $n\widehat{L}_i$ to itself, so it too belongs to the commensurator. Conversely the procyclic group $\widehat{L}_i$ centralizes every power. This proves both equalities. In the semi-graph of the smooth pointed curve, peripherals are the cusp edge groups. Proposition 1.2(i) of [46] says that an intersection open in one edge group identifies the underlying edge; hence distinct punctures cannot have commensurable inertia groups. This gives the needed distinction of peripheral indices; A.5 proves its transporter assertion separately using power-centralizers and compactness. No claim about centralizers of arbitrary non-peripheral elements is needed here.

For $\nu \geq 2$ there is also a direct source for the power-centralizer assertion: each peripheral generator can be made a free basis element, and [49, Lemma (2.1.2)] gives the centralizer of every nonidentity element of its procyclic closure. For $\nu = 1$ the commensurator argument covers precisely the case that this basis argument does not cover. For every $\nu \geq 1$, each handle generator a_j, b_j belongs to the free basis obtained by eliminating c_ν. Applying that same lemma proves equation (4) of scan 501 for all these base elements, including the once-punctured case. The basis condition is material if the phrase "systeme de generateurs on scan 500 is read literally without a cardinality restriction: in a free profinite group on a, b, the redundant generating list a, b, a^2 has $C(a^2) = \overline{\langle a \rangle} \supsetneq \overline{\langle a^2 \rangle}$. The last index is 2, detected in abelianization modulo 2. For the pro-p version use a, b, a^p and index p. The stated source theorem applies to a member of a free basis; it does not supply the assertion for every redundant generating list. □

Manuscript: §13 bis, folio 57 verso, scan 118; §35, folios 253–254, scans 499–501.

A.8. [Proved] Lemme 5 from open-subgroup centralizers alone. Let P be a discrete or profinite group for which the centralizer of every finite-index, respectively open, subgroup is trivial. Let $H \subset P$ be such a subgroup and let $u \in \mathrm{Aut}(P)$ preserve H. An automorphism restricting to the identity on H is the identity on P. An automorphism whose restriction equals conj_h for some $h \in H$ equals conj_h on P.

Manuscript: 35; 252 verso–254; 498–501.

Proof. Take the finite-index normal core $N = \bigcap_{x \in P} xHx^{-1} \subset H$. If $u|_H = \mathrm{id}$, then for $x \in P$ and $n \in N$,

$$u(x)nu(x)^{-1} = u(xnx^{-1}) = xnx^{-1}.$$

Thus $x^{-1}u(x) \in C_P(N) = 1$ for every x. For the second assertion apply the first to $\mathrm{conj}_{h^{-1}} u$. In the profinite case the core is an intersection of finitely many open subgroups and is open. This proof uses neither peripheral generators nor their power-centralizers. □

Manuscript: §35, folio 252 verso, scan 498, Lemme 5; folios 253–254, scans 500–501, proposed proof.

A.9. [Proved] Faithfulness ascends along a finite etale cover. Let k be a characteristic-zero field, U/k a hyperbolic geometrically connected smooth curve, and $V \to U$ a finite etale

geometrically connected covering defined over k. Put $\Gamma = \mathrm{Gal}(\overline{k}/k)$ and choose compatible geometric base points. Faithfulness of $\Gamma \to \mathrm{Out}(\pi_1(U_{\overline{k}}))$ implies faithfulness of $\Gamma \to \mathrm{Out}(\pi_1(V_{\overline{k}}))$.

Manuscript: 35; 253; 499–500.

Proof. The covering gives an open embedding $\Pi_V \hookrightarrow \Pi_U$ of arithmetic fundamental groups, compatible with their surjections onto the same Γ. Their geometric kernels are an open subgroup $H \subset P$. Let $\gamma \in \Gamma$ act trivially in $\mathrm{Out}(H)$ and choose a lift $e \in \Pi_V$. Conjugation by e preserves H and restricts there to conjugation by some $h \in H$. By A.7 and A.8 its action on P is also conjugation by h. Thus γ acts trivially in $\mathrm{Out}(P)$ and equals 1. A change of compatible base points conjugates these homomorphisms and does not change their kernels. The proof does not use the structure of finite subgroups of Γ. □

Manuscript: §35, folio 253, scans 499–500, Corollaire 2.

A.10. [Known] Closed-surface base elements: two separate source reductions. Let $\Pi = \widehat{\pi}_g$ with $g \geq 2$, let x be any of the standard surface generators a_i, b_i, and let $\Lambda = \overline{\langle x \rangle}$. Then $\Lambda \simeq \widehat{\mathbb{Z}}$ and, for every positive integer n,

$$C_\Pi(\Lambda^n) = N_\Pi(\Lambda^n) = \Lambda.$$

Manuscript: 35; 254; 501.

Proof. The standard generator maps to a primitive vector of $\Pi^{\mathrm{ab}} = \widehat{\mathbb{Z}}^{2g}$; hence its procyclic closure injects there and is $\widehat{\mathbb{Z}}$. For the first source reduction, x is represented by a nonseparating simple closed curve, and is therefore algebraically simple in the terminology of [8, Definition 1.3]. Proposition 3.6 of that paper gives $N_\Pi(\Lambda^n) = \Lambda$. Centralizers lie in normalizers and contain Λ, proving the centralizer equality.

For the second reduction, pinch that curve on the genus-g surface. The resulting stable one-node curve has a normalization of genus $g-1$ with two marked inverse branches; at $g = 2$ this is a twice-marked genus-one component, still hyperbolic. Its associated semi-graph of anabelioids has one edge group whose image in the smoothing's profinite group is exactly Λ, up to the chosen base-path conjugation. Proposition 1.2(ii) of [46] makes this edge group commensurably terminal. An element normalizing Λ^n commensurates Λ, proving the same normalizer equality. These are independent applications of the two stated source results, and neither replaces a closed surface group by a free group. □

Manuscript: §35, folio 254, scan 501, equation (4).

A.11. [Proved] What the abstract profinite presentation specifies. The full profinite presentation required by scan 318, equation (4), already determines a profinite group with loops isomorphic to $\widehat{\pi}_{g,\nu}$ with its standard loops. Hence A.7 and A.10 apply to every abstract object satisfying that presentation axiom. Merely requiring generators to satisfy the displayed equation would not suffice: $P = \widehat{\mathbb{Z}}^2$, $c_1 = (1, 0)$, $c_2 = (0, 1)$, $c_3 = (-1, -1)$ satisfies the $(0, 3)$ relation, has three injective procyclic peripheral subgroups and the rank-one orientation kernel, but has centre P.

Manuscript: 26; 161; 318–319.

Proof. The universal property gives a continuous homomorphism from the quotient of the free profinite group by the closed normal closure of the relator to P. The assertion that this is a defining relation makes this homomorphism an isomorphism; its inverse sends the specified base to the standard base. This gives exactly the comparison map in scan 319, equation (5). For the

weaker condition the explicit abelian example has $c_1 + c_2 + c_3 = 0$ and generates P, but has the additional relation $[c_1, c_2] = 1$. Its peripheral inclusion matrix has kernel generated by $(1, 1, 1)$, and its centre contains $(1, 0) \neq 0$. The finite reductions and the rank calculation are checked by the accompanying code. □

Manuscript: §26, folio 161, scans 318–319, equations (4)–(5). The words "relation generatrice" are indispensable; weakening them changes the objects by the explicit quotient above.

A.12. [Proved] The discrete filling isomorphism and all compatibility maps. Let $U = X \setminus S$ be hyperbolic, choose $b \in U$, and put $U' = U \setminus \{b\}$, $\pi' = \pi_1(U')$, and $\pi = \pi'/\langle\langle L_b \rangle\rangle$. Choose the peripheral representative $L_b \subset \pi'$. The normalization of representatives defines an isomorphism of discrete groups

$$d_b : \mathrm{Out}_{\mathrm{lac}}(\pi'; b) \xrightarrow{\sim} \mathrm{Aut}_{\mathrm{lac}}(\pi).$$

On both sides old punctures may be permuted and orientation may be reversed. It identifies the pure orientation-preserving subgroups, whose common index is $2\nu!$, where $\nu = |S|$. It commutes with forgetting the distinguished puncture, with the maps to $\mathrm{Out}_{\mathrm{lac}}(\pi)$, with orientation characters, with the permutation maps to $\mathfrak{S}_\nu$, and with point-pushing and inner conjugation on their kernels.

Manuscript: 18; 93–94; 186–189.

Proof. An outer class fixing $[L_b]$ has a representative α preserving L_b itself: conjugate any representative by an element taking its peripheral image back to L_b. Two normalized representatives differ by conjugation by an element of $N_{\pi'}(L_b) = L_b$, by A.3. Such conjugation becomes the identity after killing L_b. Thus α induces a well-defined actual automorphism $d_b([\alpha])$ of the specified quotient π. Products of normalized representatives are normalized, so d_b is a homomorphism. Its definition also proves compatibility with the permutation of the remaining loops and with the orientation module, using A.2's filling comparison.

We give the group-theoretic identification of this homomorphism with the topological one. The Dehn–Nielsen–Baer theorem identifies the extended mapping class group of a hyperbolic punctured surface with the outer automorphisms preserving its peripheral conjugacy classes, and in the closed case with $\mathrm{Out}(\pi)$; the precise versions are [20, Theorems 8.1 and 8.8]. The evaluation map of the homeomorphism group of (X, S) at b is a locally trivial fibration over U: over a sufficiently small disc, move b to a nearby point by a homeomorphism supported in that disc, which constructs a local section. Its connecting map pushes b round a loop. The identity component of this homeomorphism group is contractible in hyperbolic type, by the surface homeomorphism-group theorem in [60, Lemma 2.1 and Theorem 1.1]; hence the homotopy sequence gives

$$1 \to \pi \xrightarrow{\mathrm{push}} \mathrm{Out}_{\mathrm{lac}}(\pi'; b) \xrightarrow{\mathrm{fill}} \mathrm{Out}_{\mathrm{lac}}(\pi) \to 1.$$

One may alternatively use the extended Birman exact sequence, [20, Theorem 4.6 and the immediately following extended-group remark]. Surjectivity follows directly by moving the image of b back to b along an arc avoiding S.

Choose the convention that the point-push associated to a based loop γ moves the base-point track in the direction for which the induced based map is $\eta \mapsto \gamma\eta\gamma^{-1}$. This can also be checked on the boundary of the homotopy square traversed by η under the isotopy. It proves $d_b \circ \mathrm{push} = \mathrm{conj}$. The filled mapping class gives the same outer automorphism as d_b by construction. Thus the last sequence maps to

$$1 \to \pi \xrightarrow{\mathrm{conj}} \mathrm{Aut}_{\mathrm{lac}}(\pi) \to \mathrm{Out}_{\mathrm{lac}}(\pi) \to 1$$

by the identity on the end groups. Injectivity and surjectivity of d_b follow by the elementary exact-sequence argument: a kernel element belongs to point-pushing and maps trivially by the

injective inner map, and any target element can first be matched in the outer quotient and then in the kernel.

Finally every permutation of old punctures is realized by half-twists supported in small discs along arcs, and an orientation-reversing homeomorphism can be chosen to fix the finite marked set pointwise. Hence the map to $\{\pm 1\} \times \mathfrak{S}_\nu$ is onto; its kernel is precisely the pure orientation-preserving group. Its index is $2\nu!$. This also identifies the subgroups denoted $\mathcal{T}^{00}$ on scan 188 and proves the compatibility checks left on scan 189. □

Manuscript: §18, folios 93–94, scans 186–189. On scan 187 the inequality must be $2g+\nu' \geq 4$ when ν' counts punctures before filling; the meaningful alternative ≤ 4 contradicts both examples and the stated hyperbolicity requirement. The mathematical argument supports the first alternative without settling the handwriting.

A.13. [Proved] Filling into the two abelian targets. Exactly one of the drilled and filled groups is abelian in the two cases $(0,3) \to (0,2)$ and $(1,1) \to (1,0)$. In the first, the homomorphism on automorphisms of the underlying groups with peripheral subgroup classes is

$$C_2 \times C_2 = \mathrm{Out}_{\mathrm{lac}}(F_2; [\langle xy \rangle]) \longrightarrow \mathrm{Aut}(\mathbb{Z}) = C_2,$$

with kernel of order two. Its nontrivial element has representative $x \mapsto y^{-1}$, $y \mapsto x^{-1}$. Its restriction to the positive-orientation subgroup is an isomorphism. Retaining the two indexed peripheral source maps and their orientation line on the annulus enlarges the target automorphism group to $C_2 \times C_2$ and restores the full isomorphism. In the second case the filling map is the isomorphism $\mathrm{GL}_2(\mathbb{Z}) \to \mathrm{GL}_2(\mathbb{Z})$, with positive subgroup $\mathrm{SL}_2(\mathbb{Z})$.

Manuscript: 18; 94–97; 189–194.

Proof. A.1 lists the two cases. For a three-punctured sphere the extended mapping class group is $\mathfrak{S}_3 \times C_2$: a homeomorphism fixing the three punctures is isotopic, in each orientation class, to the identity or a fixed reflection, by cutting along three joining arcs and applying the disc isotopy. The stabilizer of the distinguished puncture is $C_2 \times C_2$. On $F_2 = \langle x, y \rangle$, the old-puncture transposition $x \leftrightarrow y$ and the reflection $x \mapsto x^{-1}, y \mapsto y^{-1}$ both induce inversion after imposing $xy = 1$. Their product is the displayed nontrivial kernel element. It takes the distinguished loop to its inverse and reverses orientation. Direct free-word and finite-group calculations appear in the attached program.

For an annulus with ordered end sources $\kappa_1(1) = t$ and $\kappa_2(1) = t^{-1}$, an automorphism includes a sign $\epsilon \in \{\pm 1\}$ on T and a permutation $s \in \mathfrak{S}_2$. Compatibility forces the underlying map $t \mapsto t^{\epsilon \operatorname{sgn}(s)}$. Thus there are four automorphisms of this full data, and the map forgetting s, ϵ is exactly the preceding quotient. This calculation specifies the information lost by retaining only subgroup images.

For the torus, every matrix in $\mathrm{GL}_2(\mathbb{Z})$ is realized by a linear torus homeomorphism fixing the origin. A based torus homeomorphism inducing the identity matrix is isotopic relative to that point to the identity. Indeed the restriction of evaluation to the translation subgroup $\mathbb{R}^2/\mathbb{Z}^2$ is the identity of the torus, so evaluation is surjective on fundamental groups and the point-pushing kernel is zero. The unbased kernel is the identity component: isotope the images of two primitive curves intersecting once back to those curves, use the identity matrix to remove their remaining twists, and cut along them to a disc, where the Alexander isotopy finishes the argument. Removing the fixed point identifies this mapping class group with the peripheral-preserving outer group of the once-punctured torus by Dehn–Nielsen–Baer. Filling induces abelianization, hence the asserted identity map on matrices. Its orientation sign is the determinant. □

Manuscript: §18, folios 94–97, scans 189–194. The abelian cases in the margin of scan 194 refer to the filled types; the corresponding drilled groups are both F_2.

A.14. [Known] Primitive loops in finite-step solvable quotients. Let Σ be a nonempty set of primes, F a free pro-Σ group of finite rank at least two, and $F^{[j]}$ its closed derived series. Put $F_m = F/F^{[m]}$, $m \geq 2$. For any element x of a free basis and any positive integer n,

$$C_{F_m}(x^n) = N_{F_m}(\overline{\langle x \rangle}) = \overline{\langle x \rangle} \simeq \mathbb{Z}_\Sigma, \qquad \mathbb{Z}_\Sigma = \prod_{p \in \Sigma} \mathbb{Z}_p.$$

Every open subgroup of F_m has trivial centralizer. For $\nu \geq 2$, all peripheral loops of the pro-Σ group of type (g, ν) satisfy these formulas in every F_m. Their indexed conjugacy classes are pairwise distinct in all hyperbolic types.

Manuscript: 13 bis; 57–57 verso; 117–118; strengthened PartA(c).

Proof. The power-centralizer formula is [62, Theorem 1.5, finite-rank case]; the source explicitly treats every nonzero integer power, including powers divisible by 2 or 3. We checked its finite-rank proof: Proposition 1.3 uses the finite semidirect product $(\mathbb{Z}/N)[G/\langle \bar{x} \rangle] \rtimes (G \times \langle \bar{x} \rangle)$ to force the centralizer into $\langle x \rangle F_m^{[m-1]}$; the coefficient at the distinguished coset forces the first G coordinate into $\langle \bar{x} \rangle$. At $m = 2$, the Blanchfield–Lyndon injection of F'/F'' into $\mathbb{Z}_\Sigma[[F^{\mathrm{ab}}]]^r$ and injectivity of multiplication by $\bar{x}^n - 1$ remove the intersection with F_m'. That injectivity is proved by passing from C_M to C_{kM}: invariant coefficients downstairs are divisible by every Σ-integer k, so are zero. The finite-rank induction passes to finite abelian covers where a positive power of x is a free basis element. These are exactly the finite-rank hypotheses used here; no infinite-rank inverse-limit assertion is used.

If u normalizes $\langle x \rangle$, write $uxu^{-1} = x^\lambda$, $\lambda \in \mathbb{Z}_\Sigma^\times$. The primitive abelianization coordinate gives $\lambda = 1$, so the normalizer lies in the already computed centralizer. An open subgroup contains positive powers of two distinct free generators; their cyclic centralizers intersect trivially because their abelianization coordinates are independent. This proves the open-centralizer assertion directly.

Each peripheral generator for $\nu \geq 2$ belongs to a free basis after eliminating another boundary generator. For $\nu \geq 3$ distinct peripheral abelianization lines separate the conjugacy classes. When $\nu = 2$ and $g \geq 1$, kill c_1 and all but one handle pair, mapping onto a free metabelian group on a, b. The image of c_1 is 1 and that of c_2 is $[a, b]^{-1} \neq 1$; the latter has infinite order in the upper unitriangular 3×3 group over $\mathbb{Z}_\Sigma$. Hence they cannot be conjugate, or commensurable with finite indices. These maps factor through every F_m, $m \geq 2$. □

Manuscript: §13 bis, scans 117–118, and the strengthened finite-step task in Part A(c).

A.15. [Proved] The compact intersection calculation used in descent. Let (H_α) be a decreasing directed family of closed subgroups of a profinite group P, and put $K = \bigcap_\alpha H_\alpha$. Then

$$K^{[j]} = \bigcap_\alpha H_\alpha^{[j]} \quad (j \geq 0).$$

For a closed subgroup $L \subset P$ and decreasing closed normal subgroups N_α of P, one also has $\bigcap_\alpha LN_\alpha = L(\bigcap_\alpha N_\alpha)$.

Manuscript: Auxiliary compact-group lemma for finite-step PartA(c).

Proof. For any open normal $V \subset P$, the images of H_α in the finite group P/V eventually stabilize. Their eventual image equals that of K: each compatible coset fibre is a decreasing family of nonempty compact sets. Taking a fixed number of derived subgroups commutes with homomorphic images, and taking the inverse limit of all finite images recovers the closed derived subgroup. This proves the first equality. For the second equality, an element z of every LN_α gives nonempty closed sets $L \cap zN_\alpha$; compactness makes their intersection nonempty. A member of that intersection gives the required single factorization of z. □

A.16. [Proved] The one-puncture solvable threshold, with a finite certificate. Let F be the pro-Σ completion of $\pi_{g,1}$, $g \geq 1$, let $c = \prod_{j=1}^{g}[a_j, b_j]$ (the inverse peripheral generator), and write c for its image in $F_m = F/F^{[m]}$. For every positive integer n,

$$\begin{aligned} m = 2: &\quad C_{F_2}(c^n) = N_{F_2}(\overline{\langle c\rangle}) = F_2', \\ m \geq 3: &\quad C_{F_m}(c^n) = N_{F_m}(\overline{\langle c\rangle}) = \overline{\langle c\rangle}. \end{aligned}$$

In the first line the inclusion $\overline{\langle c\rangle} \subset F_2'$ is strict. Thus peripheral rigidity fails at the smallest solvable depth $m = 2$ and is restored at every $m \geq 3$, for the full profinite and every pro-p version.

Manuscript: 13 bis; 57 verso; 118; strengthened PartA(c).

Proof. Let $H \subset F$ run over open normal subgroups containing F', with $|F/H| \geq 3$. They form a family whose intersection is F'. The corresponding finite abelian covering of the once-punctured surface has exactly $d = |F/H|$ punctures, because $c \in F'$ has trivial image in the deck group. Choose the cusp above the original based peripheral loop. Its loop is c itself; the other loops are its conjugates by deck representatives. The submodule of H^{ab} generated by these cusps is

$$\mathbb{Z}_\Sigma^{F/H}/\mathbb{Z}_\Sigma\left(\sum_{q\in F/H} e_q\right),$$

with the deck group acting by left translation. This follows either from the relative surface homology sequence of A.2 or directly from the free-group peripheral presentation. For $d \geq 3$ the lines generated by distinct e_q in this module are distinct; moreover $ne_q = ae_1$, for $0 \neq n \in \mathbb{Z}$ and $a \in \mathbb{Z}_\Sigma$, forces $q = 1$ and $a = n$. To check this last assertion, lift the equality to the free module and compare a coordinate different from both 1 and q; its coefficient forces the possible multiple of $\sum e_q$ to be zero, after which the coordinates at q and 1 give the result.

For $m \geq 2$, $F^{[m]} \subset H'$, so the natural map $H \to H^{\mathrm{ab}}$ factors through the image of H in F_m. Let $u \in F$ represent an element of $C_{F_m}(c^n)$. The relation $uc^nu^{-1} = c^n$ modulo $F^{[m]}$ implies $ne_{\bar{u}} = ne_1$ in the cusp submodule of H^{ab}. Hence $u \in H$ for every such H, so its image belongs to F_m'. The same argument for a normalizer, using $ucu^{-1} = c^\lambda$ modulo $F^{[m]}$, $\lambda \in \mathbb{Z}_\Sigma^\times$, gives $u \in F'$ and $\lambda = 1$. The subgroup $\overline{\langle c\rangle}$ is indeed $\mathbb{Z}_\Sigma$: the unitriangular 3×3 quotient used in A.1 proves injectivity of all its prime components. Thus its normalizer equals its centralizer.

When $m = 2$, the derived subgroup is abelian, contains c, and centralizes it. This gives equality with F_2'. Now let $m \geq 3$ and let $uF^{[m]}$ centralize c^n. We have proved $u \in F' \subset H$. Since $F' \subset H$, $F^{[m]} = (F')^{[m-1]} \subset H^{[m-1]}$; therefore there is a well-defined continuous epimorphism

$$H/F^{[m]} \longrightarrow H/H^{[m-1]}.$$

The group H is free pro-Σ, and c is a member of a free basis because this cover has at least two punctures. By A.14, applied at depth $m - 1 \geq 2$, the image of u in this quotient belongs to $\overline{\langle c\rangle}$. Consequently

$$u \in \bigcap_H \overline{\langle c\rangle} H^{[m-1]} = \overline{\langle c\rangle}\left(\bigcap_H H^{[m-1]}\right) = \overline{\langle c\rangle}(F')^{[m-1]} = \overline{\langle c\rangle}F^{[m]},$$

where both interchanges are proved in A.15. This proves the second line.

For strictness in the first line, set $a = a_1$, $b = b_1$, and kill the remaining handle pairs. In $\mathrm{UT}_4(\mathbb{F}_p)$ take

$$x = 1 + E_{12} + E_{23}, \qquad y = 1 + E_{34}.$$

Their group is metabelian; direct multiplication gives $[x, y] = 1+E_{24}$ and $x[x, y]x^{-1} = 1+E_{24}+E_{14}$. The second matrix is not a power of the first. Thus $aca^{-1} \in F_2'$ centralizes c but does not belong

to its procyclic closure. For every $p \in \Sigma$ this quotient factors through F_2 and proves strictness; the attached program independently enumerates and verifies the cases $p = 2, 3, 5$. In particular there is no excluded residue-characteristic caveat. □

Manuscript: §13 bis, scan 118, in the strengthened finite-step question of Part A(c). This assertion does not assert a new literature priority; the searched sources and the proof audit are listed in the report.

A.17. [Proved] Centre and torsion in all finite-step surface quotients. Let P be the full profinite or pro-Σ completion of a hyperbolic surface group, open or closed, and let $P_m = P/P^{[m]}$. For every $m \geq 2$, P_m is torsion-free and centre-free. This verifies the form of [62, Theorem 2.9 and Corollary 2.10] needed here by the following argument from finite covers and transfer.

Manuscript: Strengthened PartA(c); Yamaguchi2026 Theorem2.9 and Corollary2.10.

Proof. Every open subgroup of P is the pro-Σ group of a finite covering surface, so its abelianization is a free $\mathbb{Z}_\Sigma$-module. A closed subgroup K is the intersection of its open neighbourhood subgroups; by A.15 its abelianization is the inverse limit of their abelianizations. More explicitly, a compatible system in those abelianizations lifts to a compatible coset system in the compact group, and the kernel is the intersection of the closed derived subgroups. Thus K^{ab} is torsion-free. Applying this to each $P^{[j]}$ and inducting in

$$1 \to (P^{[m-1]})^{\mathrm{ab}} \to P_m \to P_{m-1} \to 1$$

proves torsion-freeness, starting with the torsion-free abelianization.

We also need the faithful action of a finite deck group $Q = H/N$ on N^{ab}, for open $N \trianglelefteq H \subset P$. Choose $\ell \in \Sigma$. If a subgroup Q_0 acts trivially on H^1 of the covering surface with $\mathbb{Q}_\ell$ coefficients, transfer gives an isomorphism of that H^1 with the H^1 of the quotient by Q_0. The two surfaces are both proper or both nonproper, and their Euler characteristics are consequently equal. Multiplicativity of Euler characteristic gives $\chi(Y) = |Q_0|\chi(Y/Q_0)$. Both are negative, so $Q_0 = 1$. The comparison with $N^{\mathrm{ab}} \otimes_{\mathbb{Z}_\Sigma} \mathbb{Q}_\ell$ is the canonical duality between first homology and first cohomology. Thus the deck action on N^{ab} is faithful.

Put $K = P^{[m-1]}$, $A = K/K'$, and $Q = P/K'$. For an open normal subgroup $N \subset Q$ containing A, its preimage $\widetilde{N} \subset P$ contains K, hence $K' \subset \widetilde{N}'$. Therefore $N^{\mathrm{ab}} = \widetilde{N}^{\mathrm{ab}}$, and the preceding deck faithfulness implies that an element of $Z(Q)$ belongs to every such N. Their intersection is A.

We finish by proving $A^{P/K} = 0$. For any open normal $V \subset P$ containing K, write $j : V^{\mathrm{ab}} \to P^{\mathrm{ab}}$ for inclusion and $t : P^{\mathrm{ab}} \to V^{\mathrm{ab}}$ for transfer. The composite tj on the P/V-invariants is multiplication by $[P : V]$, which is injective because V^{ab} is torsion-free. Hence inclusion is injective on those invariants. Passing to the inverse limit over V shows that $K^{\mathrm{ab},P/K} \to P^{\mathrm{ab}}$ is injective. But $K \subset P'$, so this map is zero. Therefore $A^{P/K} = 0$. Since $Z(Q) \subset A$, the centre is zero, as claimed. This proof checks precisely the transfer and finite-cover steps used in the cited unrefereed result. □

A.18. [Proved] The remaining affine solvable rigidity list and a closed bound. For all hyperbolic types with $\nu > 0$, and all $m \geq 2$, the maximal m-step-solvable full profinite or pro-Σ quotient has trivial centre and trivial centralizers of open subgroups, and is torsion-free. Peripheral conjugacy classes remain distinct. For $\nu \geq 2$ all peripheral normalizer and integer-power-centralizer equalities hold at every $m \geq 2$; for $\nu = 1$ their exact values are those of A.16. In an extension obtained by quotienting a geometric action extension by the mth closed derived subgroup of its kernel, the equality $N_E(L_s) = E_s$ holds in the former cases and at $m \geq 3$ in the latter. At $\nu = 1, m = 2$ its correct value is $N_E(L_s) = F_2' E_s$, with the intersection $F_2' \cap E_s = L_s$.

In all these cases every transporter $\mathrm{Trans}_E(L_s^d, L_s)$, $d \geq 1$, equals the same displayed normalizer. Since the kernels are centre-free by A.17, the extension reconstruction of A.6 also applies to all these solvable quotients. Standard free-basis generators satisfy the power-centralizer formula at every $m \geq 2$ by A.14. For a closed hyperbolic group P and an open subgroup $H \subset P_m$, the proven universal bound is

$$C_{P_m}(H) \subseteq P_m^{[m-1]}.$$

Manuscript: 13 bis; 57 verso; 118; 35; 252 verso–254; 498–501; strengthened PartA(c).

Proof. The first assertions are A.14, A.16, and A.17. For the extension calculation, the quotient map is injective on the peripheral group, since the peripheral generator has infinite order in P_2 by A.1. The image of a geometric peripheral lift group still maps onto the puncture stabilizer. The kernel of the normalizer map to that stabilizer is the normalizer calculated in A.14 or A.16, so the same lifting argument as A.5 proves the equalities. In the metabelian one-puncture case there is just one puncture, so the stabilizer is the whole acting group and multiplication by E_s gives the asserted $F_2' E_s$.

It remains to justify the one-sided transporter equality in these quotients. Suppose $eL_i^d e^{-1} \subset L_i$. The action sends L_i to a conjugate of some L_j and its chosen generator to a conjugate of c_j^{λ}, with $\lambda \in \mathbb{Z}_\Sigma^\times$. For $\nu \geq 3$, the primitive abelianization lines imply $j = i$. For $\nu = 2$, $g \geq 1$, the quotient killing c_i and all but one handle pair sends c_j to a nonzero commutator of infinite order in $\mathrm{UT}_3(\mathbb{Z}_\Sigma)$; thus $j \neq i$ would contradict the given containment. Choose an element of E_i with the same image as e in the acting group and divide it out. The resulting element u lies in P_m and satisfies $uc_i^d u^{-1} = c_i^a$ for some $a \in \mathbb{Z}_\Sigma$. The primitive abelianization coordinate forces $a = d$, so A.14 puts u in L_i. This proves the equality for $\nu \geq 2$. For $\nu = 1$, the acting group already fixes the only puncture. After the same division by E_i, apply the finite abelian cover calculation in A.16 to $uc^d u^{-1} = c^a$. It forces $a = d$ and reduces the transporter to the integer-power centralizer calculated there. This gives E_i at $m \geq 3$ and $F_2' E_i$ at $m = 2$.

The closed bound recorded here is strengthened to triviality in A.19 below. For this bound, put $A = P_m^{[m-1]}$ and choose any open normal $N \subset P_m$ containing A. The group N^{ab} is the torsion-free abelianization of its preimage in P, as in A.17. If z centralizes H, its conjugation action fixes the image of $H \cap N$ in N^{ab}. This image has finite index, so some nonzero integer multiple of every element is fixed. Torsion-freeness then makes the action trivial on all of N^{ab}. Deck faithfulness from A.17 implies $z \in N$. Intersecting all these N yields $z \in A$. □

Manuscript: §13 bis, scan 118; §35, scans 498–501; additional finite-step task A(c).

A.19. [Proved] Closed finite-step power centralizers and slimness. Let P be the pro-Σ completion of a closed oriented surface group of genus $g \geq 2$, let $m \geq 2$, and put $P_m = P/P^{[m]}$. Let x be represented by a nonseparating simple closed curve, in particular by any standard handle generator. Then for every positive integer n,

$$C_{P_m}(x^n) = N_{P_m}(\overline{\langle x^n \rangle}) = \overline{\langle x \rangle} \simeq \mathbb{Z}_\Sigma.$$

Consequently $C_{P_m}(H) = 1$ for every open subgroup H. This includes the full profinite and every pro-p quotient, without a restriction at $p = 2$ or 3.

Manuscript: 35; 254; 501; strengthened PartA(c).

Proof. We first give the completed module construction required for the metabelian case. Let F be a free pro-Σ group on $z_1, \ldots, z_r$, let $A = F/F'$, and put $R = \mathbb{Z}_\Sigma[[A]]$. The continuous homomorphism

$$\mu : F \longrightarrow R^r \rtimes A, \qquad z_i \longmapsto (e_i, \bar{z}_i),$$

exists by freeness; A acts by multiplication on the coefficient module, so the product in the target is $(v,a)(w,b) = (v + aw, ab)$. Its kernel is exactly F''. The inclusion $F'' \subset \ker \mu$ follows because the target is metabelian. To prove the reverse inclusion without assuming any exactness result about completed chain complexes, take an arbitrary finite metabelian quotient G of F. Write $K = G'$, $Q = G/K$, and choose a section $t : Q \to G$ with $t(1) = 1$. There is an injective homomorphism

$$G \longrightarrow K^Q \rtimes Q, \qquad g \longmapsto (f_g, \bar{g}), \qquad f_g(q) = t(q)\, g\, t(q\bar{g})^{-1}.$$

Indeed $f_{gh}(q) = f_g(q) f_h(q\bar{g})$, and, when $\bar{g} = 1$, evaluation at $q = 1$ recovers g. Use additive notation for the finite abelian group K. The finite module K^Q is an R-module through $A \to Q$ and right translation on Q. Sending e_i to the function coordinate of the image of z_i gives a continuous R-linear map $R^r \to K^Q$. The resulting homomorphism of semidirect products makes the homomorphism from F factor through μ, since it agrees on every free generator. Hence $\ker \mu$ maps trivially to every finite metabelian quotient of F. These quotients separate F/F'', proving the assertion. In particular the function coordinate gives an injective R-linear morphism

$$F'/F'' \longhookrightarrow R^r.$$

Its image is a closed R-submodule: conjugation gives the A-action, integral powers give the scalar action, and continuity extends these to the completed group ring.

Apply this construction with $r = 2g$ and

$$P = F/\langle\langle r_0 \rangle\rangle, \qquad r_0 = \prod_{i=1}^{g} [a_i, b_i], \qquad A = P^{\mathrm{ab}} = \mathbb{Z}_\Sigma^{2g}.$$

The function coordinate of r_0 is

$$v = (1 - \bar{b}_1, \bar{a}_1 - 1, \ldots, 1 - \bar{b}_g, \bar{a}_g - 1) \in R^{2g}.$$

This is obtained by multiplying the four factors of each commutator in $R^{2g} \rtimes A$. Since $r_0 \in F'$, its closed normal closure maps in F'/F'' onto exactly Rv: its conjugates give the A-multiples of v, their products give their integral sums, and their closure is Rv. The last module is closed because it is the image of the compact ring R. Thus the natural quotient maps induce the injective comparison

$$B := P'/P'' \longhookrightarrow R^{2g}/Rv. \tag{A.19.1}$$

Indeed P'' is the image of F'' and the kernel in F'/F'' is precisely the just calculated module. This establishes the needed completed Fox-module comparison directly.

Choose a geometric basis with $x = a_1$, and set $h = \bar{a}_1^n - 1$. We use two elementary facts about these completed group rings. First, multiplication by h on R is injective. Second, multiplication by $\bar{a}_2 - 1$ on R/hR is injective. Here

$$R/hR \xrightarrow{\sim} \mathbb{Z}_\Sigma[[A/\overline{\langle \bar{a}_1^n \rangle}]] \tag{A.19.2}$$

is the map induced by the group quotient. Its kernel is the closed ideal generated by h; that ideal equals hR since hR is compact. To prove the injectivity assertions, one can use the following coefficient calculation. In a completed group ring of $\mathbb{Z}_\Sigma \times D$, consider an element invariant under the chosen infinite cyclic coordinate, or under its nth power. At finite coefficient level $\mathbb{Z}/k$ and finite coordinate level C_M, enlarge the coordinate level to C_{kM}, taking M divisible by the Σ-part of n in the second case. Invariance makes the coefficients constant on each orbit. The projection down to C_M sums k equal coefficients on each orbit fibre and therefore is zero modulo k. This works with every finite quotient of the remaining factor D. All finite images of the original element vanish, so the element is zero. Apply this argument first to the a_1 coordinate of R, and

then to the independent a_2 coordinate in the group of (A.19.2). It applies to arbitrary Σ, because k and the chosen coordinate orders have prime factors in Σ.

These ring facts show that multiplication by h is injective on R^{2g}/Rv. If $hz = vt$, inspect the b_2 coordinate, which is $hz_{b_2} = (\bar{a}_2 - 1)t$. Reducing modulo h and using the second injectivity gives $t = ht'$ for some $t' \in R$. The first injectivity then gives $z = vt'$ in every coordinate. Consequently (A.19.1) implies

$$B^{\langle x^n \rangle} = 0. \tag{A.19.3}$$

For completeness, we next constrain the abelianization of an arbitrary centralizer element, at every depth $m \geq 2$. Put $V = \mathbb{Z}_\Sigma^{2g}$ with its ordered handle basis and put $W = \bigwedge^2 V/\mathbb{Z}_\Sigma\omega$, where $\omega = \sum_i a_i \wedge b_i$. For every Σ-integer k, form the finite class-two group with underlying set $(V/kV) \times (W/kW)$ and product

$$(q, w)(q', w') = (q + q', w + w' + \beta(q, q')), \qquad \beta(q, q') = \sum_{i<j} q_i q'_j \, z_i \wedge z_j.$$

Its commutator is $(0, q \wedge q')$. The standard surface relation maps to $\omega = 0$, so sending the handle generators to their coordinate vectors defines a quotient homomorphism from P. The finite group is metabelian and a Σ-group; the homomorphism therefore factors through every P_m, $m \geq 2$. If $u \in P_m$ centralizes a_1^n and has abelianization q, these homomorphisms give $n(q \wedge a_1) = 0$ in W. The module W is free because ω is primitive, so multiplication by the nonzero integer n is injective. Write $q \wedge a_1 = t\omega$. The coefficient of $a_2 \wedge b_2$ on the left is zero; on the right it is t. Hence $t = 0$, and the remaining coordinates give $q = \lambda a_1$ for some $\lambda \in \mathbb{Z}_\Sigma$. The procyclic group generated by a_1 injects into V, so dividing u by a_1^λ puts it in P'_m without changing its centralization property. At $m = 2$, (A.19.3) makes this remaining element trivial, proving the centralizer formula.

We now induct on m, simultaneously for every genus at least two and every nonseparating simple closed curve. Suppose $m \geq 3$, and divide a centralizer element as just explained so that it is represented by $u \in P'$. Let H run over all open normal subgroups of P containing P', and let d be the order of x in P/H. The corresponding finite abelian covering of the closed surface has genus at least two. The based lift of the simple curve x is represented by x^d, traversed once. This lift is a simple nonseparating curve: its integral homology class pushes forward to $d[x] \neq 0$, whereas a separating curve has zero homology class. It is therefore a standard handle generator for some geometric basis of the covering surface group H.

Since $P' \subset H$, we have $P^{[m]} \subset H^{[m-1]}$. The quotient morphism

$$H/P^{[m]} \longrightarrow H/H^{[m-1]}$$

is consequently defined. The image of u centralizes $(x^d)^n$ in its target. By the induction hypothesis at depth $m - 1$, that image belongs to $\overline{\langle x^d \rangle}$. Hence

$$u \in \bigcap_H \overline{\langle x \rangle} H^{[m-1]} = \overline{\langle x \rangle} \left(\bigcap_H H \right)^{[m-1]} = \overline{\langle x \rangle} P^{[m]},$$

by A.15 and $\bigcap_H H = P'$. The intersection of $\overline{\langle x \rangle}$ with P' is trivial, again by its primitive abelianization coordinate. Since $u \in P'$, its image in P_m is trivial. This completes the induction.

If an element normalizes $\overline{\langle x^n \rangle}$, its conjugation sends x^n to $x^{n\lambda}$ for a unit $\lambda \in \mathbb{Z}_\Sigma^\times$. Abelianization gives $n\lambda = n$, whence $\lambda = 1$, so the normalizer equals the calculated centralizer. Finally an open subgroup contains positive integer powers of a_1 and b_1. Their calculated centralizers have trivial intersection because they inject into distinct independent abelianization coordinates. This proves slimness. □

Manuscript: §35, scan 501, equation (4), and the finite-step strengthening in Part A(c). The proof establishes the exact module comparison and all closed-group descent maps used; it does not rely on the unknown general slimness statement mentioned before Proposition 2.6 in [62], nor claim a literature priority.

A.20. [Proved] The typed completion comparison for filling. For the two hyperbolic groups in A.12, put $\Pi' = \widehat{\pi'}$ and $\Pi = \widehat{\pi}$. There is a commutative diagram of profinite groups

$$\begin{array}{ccc} \widehat{\mathrm{Out}_{\mathrm{lac}}(\pi'; b)} & \xrightarrow{\widehat{d_b}} & \widehat{\mathrm{Aut}_{\mathrm{lac}}(\pi)} \\ \downarrow c_{\mathrm{out}} & & \downarrow c_{\mathrm{aut}} \\ \mathrm{Out}_{\mathrm{lac}}(\Pi'; b) & \xrightarrow{d_b^{\mathrm{cont}}} & \mathrm{Aut}_{\mathrm{lac}}(\Pi). \end{array}$$

The upper horizontal map is an isomorphism. The lower horizontal map is the normalized filling homomorphism on all continuous loop automorphisms. The vertical maps are induced by extending discrete automorphisms to group completions. The same construction with pro-p completions of the surface groups still has the full profinite completions of the discrete automorphism groups in its upper row. These assertions specify the precise consequence of the discrete filling theorem; no isomorphism assertion for the lower horizontal or vertical arrows is used.

Manuscript: 18; 93–94; 186–189; comparison with profinite PartA(c).

Proof. The presentation gives $\Pi'/\overline{\langle\langle L_b \rangle\rangle} = \Pi$. By A.7, $N_{\Pi'}(L_b) = L_b$. Normalize an outer representative to preserve L_b, then pass to this quotient. Two normalized representatives differ by conjugation by an element of L_b, which dies in the quotient. This proves that d_b^{cont} is a well-defined homomorphism. It is continuous: the closed subgroup of $\mathrm{Aut}(\Pi')$ preserving L_b is compact, its quotient onto the indicated outer group has kernel the inner automorphisms by L_b, and the continuous quotient action factors through this quotient. The same proof works for the pro-p groups using A.7.

Every discrete automorphism extends uniquely to an automorphism of its completion; extensions of inner automorphisms remain inner. Since the resulting target automorphism groups are profinite, the extension maps themselves extend to the full profinite completions of the discrete automorphism groups. Normalizing a discrete representative and then completing gives the same quotient automorphism as completing and then normalizing. Thus the square commutes on the dense discrete subgroup and therefore everywhere. A.12 implies that its upper arrow is an isomorphism. The quotient maps $\widehat{\pi} \twoheadrightarrow \pi^{(p)}$ and $\widehat{\pi'} \twoheadrightarrow (\pi')^{(p)}$ have characteristic kernels and commute with filling, giving the further comparison from the full square to its pro-p version.

For odd p, replacing the upper groups by their pro-p completions would in general fail to define these same maps: an orientation-reversing involution is killed by every homomorphism to a pro-p group, whereas its action on the pro-p orientation line is multiplication by $-1 \neq 1$. This is an explicit obstruction to that proposed factorization, rather than an identification of two differently typed completion operations. □

2 Finite actions, sections, and rotation data

In this section $U = X \setminus S$ is an orientable connected surface of finite type, unless another hypothesis is explicitly stated. A universal covering $p : \widetilde{U} \to U$ is fixed, and $\pi = \mathrm{Deck}(p)$ acts on the left. For a faithful action of a finite group G on U, we write E for the group of all lifts of its elements. Thus $1 \to \pi \to E \to G \to 1$ is an extension of *discrete* groups. A partial section means a finite subgroup of E projecting isomorphically onto a subgroup of G; its equivalence relation throughout this section is conjugacy by π. We use G^+ for the orientation-preserving subgroup. The claims below also specify the corrections required when $G^+ = 1$.

2.1 The plane lemma and the section correspondence

B.1. [Proved] Finite plane actions and simultaneous local models. Every faithful action of a finite group H on $\mathbb{R}^2$ by homeomorphisms is conjugate to an orthogonal action. When $H^+ \neq 1$, H^+ is cyclic, H is cyclic or dihedral, and $(\mathbb{R}^2)^H$ is a single point. A nontrivial orientation-reversing involution has a fixed line. Every stabilizer in a finite action on a surface with universal cover $\mathbb{R}^2$ has an invariant disc on which the whole stabilizer acts by an orthogonal action. These conclusions supply the two local assertions used in §15, folios 70–74, scans 141–149, and §33, scan 440.

Manuscript: 15; 70-74; 141-149; 33; 222; 440.

Proof. We use precisely Kolev's Theorem 1.2: a compact subgroup of $\mathrm{Homeo}(S^2)$ is conjugate to a closed subgroup of $O(3)$ [39]. Each homeomorphism of $\mathbb{R}^2$ extends to its one-point compactification, and the finite group fixes the added point ∞. After the stated conjugacy the group fixes a point q of the round sphere. Its action on $q^\perp$ gives an injective homomorphism into $O(2)$. Stereographic projection from q is equivariant and identifies the action on $S^2 \setminus \{q\}$ with that linear action. Finite subgroups of $SO(2)$ are the cyclic groups of roots of unity: an element of least positive angle generates the group, by division with remainder of angles. An orientation-reversing orthogonal matrix in dimension two is a reflection and conjugates every rotation to its inverse. This proves the cyclic/dihedral assertion and all assertions about fixed sets.

For the local statement, let H stabilize $x \in U$ and choose $\widetilde{x} \in p^{-1}(x)$. For each $h \in H$ there is a unique lift fixing $\widetilde{x}$. Uniqueness makes the assignment a homomorphism, and it is injective because its projection to U is the original action. The preceding plane argument linearizes these lifts simultaneously. Pull back a sufficiently small round disc about the common fixed point; choose it inside an evenly covered neighbourhood of $\widetilde{x}$. Its projection is the required H-invariant disc. The same assertion for a sphere follows directly from Kolev's theorem. In particular the fixed locus of an anti-involution is locally a diameter, with the involution interchanging its two sides. No differentiability of the original action has entered the argument. The single-element versions are Constantin–Kolev, Lemma 2.5, Theorems 3.1 and 4.1, and Corollary 4.2 [16]. □

B.2. [Refuted] The omitted orientation hypothesis in scans 146–147. The displayed lemma on scan 147, taken without its preceding orientation hypothesis, is false. Its correct form is: an orientation-preserving finite-order homeomorphism u of $\mathbb{C}$ commuting with $T(z) = z + 1$ and satisfying $u(0) = 0$ is the identity. The quotient $\mathbb{C}/\mathbb{Z}$ provides a complete proof along the route proposed in the manuscript.

Manuscript: 15; 72-73; 146-147.

Proof. Complex conjugation is a counterexample to the unqualified sentence: it has order two, fixes zero, and commutes with T. For the corrected statement, $u(k) = k$ for every integer k. B.1 gives at most one fixed point for a nonidentity orientation-preserving periodic plane homeomorphism, so $u = 1$.

Here is the quotient argument. Commutation with T gives a finite-order homeomorphism $\bar{u}$ of the annulus $A = \mathbb{C}/\mathbb{Z}$, inducing $+1$ on $\pi_1(A) = \mathbb{Z}$. It preserves the orientation of A. The two end loops, oriented as boundary loops of a compact subannulus, represent opposite homology classes. Consequently an orientation-preserving map inducing $+1$ on $H_1(A;\mathbb{Z})$ fixes each end. Filling the ends extends $\bar{u}$ to an orientation-preserving periodic homeomorphism of S^2 fixing both added points and [0]. By Constantin–Kolev, Theorem 4.1, a nonidentity such map has exactly two fixed points. Hence $\bar{u} = 1$. Its lift u is a deck transformation T^k; $u(0) = 0$ forces $k = 0$. On scan 146 the equation $G = G^\circ$ is exactly what makes the lifted action orientation

preserving. The mathematical repair therefore restores an inherited hypothesis; it does not justify a spectral-sequence argument. □

B.3. [Proved] Sections and isolated fixed points. Let $\widetilde{U} \cong \mathbb{R}^2$. A faithful finite action on U with $G^+ \neq 1$ has a bijection

$$U^G \longrightarrow \mathrm{Sec}(E \to G)/\pi.$$

For every section subgroup K, $C_\pi(K) = 1$. Existence of a section forces G to be cyclic or dihedral. More generally, every nontrivial finite subgroup of E whose orientation-preserving part is nontrivial is contained in exactly one maximal finite subgroup of E. For an orientation-preserving action the map from points with nontrivial stabilizer to maximal partial-section classes is a G-equivariant bijection. These are the theorem and corollary of §15, folios 73–75, scans 148–151, with their stated orientation hypotheses.

Manuscript: 15; 73-75; 148-151.

Proof. For $x \in U^G$, the lifts of G fixing a chosen point over x form a section. Changing that point conjugates the section by the corresponding element of π. Conversely a section K is a finite faithful group of plane homeomorphisms. B.1 supplies a unique fixed point $\widetilde{x}$. Its image is fixed by G, and the section fixing $\widetilde{x}$ is K. These constructions are inverse: conjugate sections have deck-translated fixed points and hence the same image in U. An element of $C_\pi(K)$ preserves the unique fixed point. A deck transformation with a fixed point is the identity. This proves the centralizer assertion. B.1 gives the possible finite groups.

Let $K \leq E$ have nontrivial positive part and fix $\widetilde{x}$. The stabilizer $E_{\widetilde{x}}$ is finite, since it intersects π trivially and injects into the finite group G. Any finite subgroup containing K has a fixed point by B.1; that point must be $\widetilde{x}$. It is therefore contained in $E_{\widetilde{x}}$. This proves uniqueness as an actual subgroup, before any conjugacy quotient. In the orientation-preserving case every nontrivial partial section has this property. Applying the same construction to each stabilizer G_x gives the final bijection and its equivariance. The argument also applies to a nonorientable U whenever the finite lifted group has nontrivial positive part; in particular the cardinality condition $|G| \geq 3$ on scan 149 ensures it. □

B.4. [Proved] Reflection sections, ovals, and punctured ovals. Let σ be an orientation-reversing involution of U with $\widetilde{U} \cong \mathbb{R}^2$, and let $E \to C_2$ be its lift extension. There is a bijection

$$\pi_0(U^\sigma) \longrightarrow \mathrm{Sec}(E \to C_2)/\pi.$$

For the section s associated with a component Z, inclusion induces an isomorphism $\pi_1(Z) \cong C_\pi(s)$. Thus the centralizer is 1 for a line and $\mathbb{Z}$ for an oval. This includes punctures fixed or permuted by σ and proves §33, folios 222–225, scans 440–445, in the required noncompact cases.

Manuscript: 33; 222-225; 440-445.

Proof. Each order-two lift s is a reflection of the plane by B.1. Its fixed set L is a line. The restriction $p : L \to U^\sigma$ is locally a homeomorphism onto the real locus, by the local model in B.1. Its image is the entire component Z containing any of its points. To verify this without a properness assumption, lift a path in Z starting at $p(\widetilde{x})$: applying s gives another lift with the same initial point, so uniqueness of path lifting fixes the lifted path. The same argument for a homotopy shows that $L \to Z$ is its universal covering. Every section has such a fixed line, so every section class arises. Two fixed lines projecting to the same component contain lifts of the same point; a deck transformation taking one lift to the other conjugates the associated involutions. The two lifts fixing the resulting point agree, because lifts of a specified base homeomorphism are unique once the value at one point is fixed.

A deck transformation preserving L normalizes $\langle s\rangle$ and hence centralizes s. Conversely a deck transformation centralizing s preserves L. Therefore the deck group of $L \to Z$ is exactly $C_\pi(s)$. Since $L \cong \mathbb{R}$, this proves both the asserted isomorphism and injectivity of $\pi_1(Z) \to \pi$. A finite-type real locus is a finite union of circles and open intervals after its real punctures are removed. Each interval is homeomorphic to $\mathbb{R}$. This argument is also the explicit reduction of Mochizuki's Theorem 3.6 and Corollary 3.7 [44] to Grothendieck's formulation: his full centralizer is $\pi_1(Z) \times C_2$, and intersecting it with π gives the centralizer used here. □

For $G = 1$ the section set has one element and its centralizer is π; the comparison is the bijection $\pi_0(U) \to \{*\}$. The map from individual points is not injective. The hypothesis on the universal cover has content: on S^2 a nontrivial rotation has two fixed points while its lift extension, with kernel 1, has one section; the antipodal involution has no fixed point and still has one section. These examples explain exactly why the same formula does not extend to that excluded surface.

B.5. [Proved] Recovering the incidence and cyclic order of real punctures. For a hyperbolic $U = X \setminus S$ with anti-involution σ, its discrete extension together with its indexed peripheral subgroups determines the following finite graph. Its vertices are the section classes of B.4, its edges are the real punctures, and each edge is attached to its two adjacent real half-branches. Every punctured oval gives one cycle, allowing a loop or two parallel edges. The graph recovers which punctures lie on each oval and their cyclic order up to reversal. Unpunctured ovals are the vertices with infinite cyclic centralizer. The claim uses the extension and peripheral incidence maps; a naked set of section classes alone has no such incidence information.

Manuscript: 33; 221-225; 437,440-445.

Proof. A puncture is real exactly when its C_2-stabilizer is all of C_2. For such a puncture its peripheral extension is

$$E_i = \langle t, s \mid s^2 = 1, ; sts^{-1} = t^{-1}\rangle, \qquad L_i = \langle t\rangle.$$

Indeed the local anti-involution of the filled disc is reflection. On the logarithmic covering $z = \exp(2\pi i w)$ of the punctured disc its two types of order-two lifts are $w \mapsto -\overline{w}$ and $w \mapsto 1 - \overline{w}$. Algebraically the reflections $t^k s$ have two L_i-conjugacy classes, distinguished by the parity of k, since conjugation by t^m adds $2m$. The two inclusions of these classes into the section set of E are precisely the two real half-branches. They can have the same image. Attach the edge i using these two maps. Walking through the intervals of a punctured circle gives a cycle whose edge labels are its punctures. Conversely these cycles determine the partition of the punctures and their circular order, with the orientation of each circle unchosen. B.4 identifies every remaining vertex with an unpunctured oval. A nonreal pair of punctures has trivial stabilizer and supplies no edge. Finally any two configurations on circles with the same circular order are related by a homeomorphism: map successive closed arcs monotonically, and extend over collars. Thus this recovery specifies the topological position of the punctures, including the single-puncture and single-interval case, without specifying their analytic positions. □

2.2 Finite automorphisms and the withdrawn assertions

B.6. [Proved] The fundamental loop lemma of §18. Let $\pi = \pi_1(U)$ be hyperbolic and discrete, with its indexed peripheral structure and orientation module. A finite-order orientation-preserving automorphism of π fixing a peripheral index and normalizing its subgroup L_i is the identity. Consequently the peripheral extension attached to a nontrivial finite-order positive outer automorphism fixing i cannot split. This proves the lemma and corollary of scans 193–194 without adding realizability to the hypotheses.

Manuscript: 18; 96-98; 193-196.

Proof. The based Dehn–Nielsen–Baer correspondence identifies the given finite cyclic group with a finite subgroup of the mapping class group with one additional marked base point. Kerckhoff's realization theorem realizes this group by homeomorphisms fixing that point and preserving the marked punctures. The original paper explicitly treats cusps in §IV, pp. 255–256, in addition to the orientation-reversing convention on p. 235 [36]; hence the punctured case used here is included. The section determined by the fixed base point identifies the induced genuine action on π with the given one.

An orientation-preserving automorphism fixing i acts as $+1$ on $L_i \cong \mathbb{Z}$, so it fixes a generator ℓ of L_i. A nontrivial finite positive cyclic group of plane lifts has one fixed point by B.1. Commutation with the nontrivial deck transformation ℓ would give the second fixed point $\ell\widetilde{x}$, a contradiction. Thus the automorphism is the identity. A splitting of the peripheral extension gives a finite-order lift normalizing L_i; the result just proved makes its outer class trivial.

The margin on scan 194 writes the *filled* type as (g, ν) and the acting group as type $(g, \nu+1)$. For $(g, \nu) = (0, 2)$ or $(1, 0)$ the acting group is still a hyperbolic free group of rank two, so the same proof applies. For a literal annulus, a positive automorphism fixing one of its two indexed ends acts trivially on $\pi = \mathbb{Z}$. Forgetting the indices permits the positive end interchange, which induces -1 on $\mathbb{Z}$ while normalizing its common peripheral subgroup. For a literal unpunctured torus there is no peripheral index, so the displayed loop lemma has no instance. These are the precise comparisons required by the marginal remark, also reflected in A.13. □

B.7. [Refuted] The unrestricted fixed-subgroup question, with its replacement. The assertion $\pi^G = 1$ for every finite $G \leq \mathrm{Aut}_{\mathrm{lac}}(\pi)$ is false. For a hyperbolic discrete π, a nontrivial finite genuine loop action has fixed subgroup 1 when its positive part is nontrivial. Its remaining possibility is a reflection action of C_2; its fixed subgroup is 1 or $\mathbb{Z}$, as in B.4. For any finite-type faithful geometric action there are finitely many classes of full and partial sections. This answers questions (c) and (e), scans 197–198.

Manuscript: 18; 98-99; 197-198.

Proof. A finite subgroup of $\mathrm{Aut}_{\mathrm{lac}}(\pi)$ intersects $\mathrm{Inn}(\pi)$ trivially: π is torsion-free and centre-free. Its outer image is therefore finite and is realized by Kerckhoff. The canonical centre-free extension identifies the given genuine action with a section of its geometric lift extension. B.3 and B.4 now give the complete list of fixed subgroups. An explicit counterexample is the double of an orientable one-holed torus, a genus-two surface, with the reflection interchanging the two halves. Its fixed oval is essential and a base point on it gives a genuine order-two automorphism with fixed subgroup $\mathbb{Z}$.

For finiteness, compactify U by its finitely many ends. Every nonidentity positive periodic homeomorphism has isolated fixed points by B.1. The resulting closed discrete subset of the compact surface is finite. A reflection has a compact one-dimensional fixed submanifold, hence finitely many components; removing finitely many punctures preserves finiteness. Apply B.3 or B.4 to each of the finitely many subgroups of G. This proves the asserted finiteness. For the abelian groups the genuine actions can be read directly: a nonidentity positive finite-order element of $SL_2(\mathbb{Z})$ fixes no nonzero lattice vector, whereas a determinant -1 involution fixes a primitive rank-one sublattice. For $\mathbb{Z}$, the nonidentity automorphism is -1 and its fixed subgroup is zero. The trivial group of operators fixes all of π in every type. □

B.8. [Refuted] Unique maximal sections fail for reflections, already in type $(0, 3)$**.** The orientation-free extension of question (d), scan 198, and of the expectation on scan 155 is false.

Put

$$E = \langle s_0, s_1, s_2 \mid s_j^2 = 1, (s_0 s_1)^2 = 1, (s_1 s_2)^3 = 1 \rangle.$$

There is a geometric extension $1 \to \pi_{0,3} \to E \to S_3 \times C_2 \to 1$. The reflection subgroup $\langle s_1 \rangle$ is contained in two distinct maximal finite subgroups $\langle s_0, s_1 \rangle \cong C_2^2$ and $\langle s_1, s_2 \rangle \cong S_3$, which are not $\pi_{0,3}$-conjugate. The valid uniqueness statement is B.3, requiring a nontrivial positive part.

Manuscript: 15; 77; 155; 18; 99-101; 198,201-203.

Proof. Take a hyperbolic triangle with angles $\pi/2, \pi/3, 0$ and reflect in its three sides. Reflected triangles tile the plane; the local tilings at the two finite vertices give exactly the displayed relations. Equivalently its group is the amalgam $C_2^2 *_{C_2} S_3$, with common subgroup generated by s_1. Write $S_3 = \langle r, a \mid r^3 = a^2 = 1, ara = r^{-1} \rangle$, and let τ generate the central C_2. Define the surjection

$$s_0 \mapsto \tau, \qquad s_1 \mapsto a\tau, \qquad s_2 \mapsto ra\tau.$$

It is injective on both vertex groups. A finite subgroup of this amalgam fixes a vertex of its Bass–Serre tree: the finite orbit spans a finite tree, whose centre is fixed (there are no edge inversions). Thus the kernel is torsion-free. The character sending each s_j to -1 factors through the exponent of τ, so the kernel preserves orientation. The group Euler characteristic is $1/4 + 1/6 - 1/2 = -1/12$, and the index is 12; hence the kernel has Euler characteristic -1. The stabilizer of the cusp is $\langle s_0, s_2 \rangle$; its image has order four. The kernel therefore has $12/4 = 3$ cusps, and its quotient surface has genus zero. This constructs the asserted geometric extension.

Each vertex group is maximal finite, again by the tree action: a finite overgroup would fix a vertex, and a vertex group of order greater than the edge group fixes only its vertex. Their intersection is $\langle s_1 \rangle$ by the normal form for the amalgam. Their orders differ, and their generated group is all of the infinite amalgam. This proves every claimed failure. The finite quotient, its vertex and cusp images, and the Euler-characteristic calculation are checked in `code/check_part_b.py`. Grothendieck's proof on scans 201–203 applies when the finite overgroups are positive cyclic groups: they centralize their intersection, and B.3 makes the kernel of that centralizer map trivial. A reflection has a line of fixed points, so that argument supplies neither of these requirements in the generalization just refuted. □

B.9. [Proved] The valid normalizer assertion. For a hyperbolic discrete group with loops, a nontrivial finite subgroup $K \leq \mathrm{Aut}^+_{\mathrm{lac}}(\pi)$ is cyclic and

$$N_{\mathrm{Aut}^+_{\mathrm{lac}}(\pi)}(K) = C_{\mathrm{Aut}^+_{\mathrm{lac}}(\pi)}(K).$$

An orientation-reversing element of its full normalizer acts on K by inversion. Every finite positive overgroup of K is cyclic. These are the valid assertions within the passage labelled "Ultra fort!" on scans 198–199.

Manuscript: 18; 99-100; 198-200.

Proof. Realize the outer image of K and use its given section. B.3 makes K cyclic. The boundary rotation character of B.12 is injective on K and is natural for loop automorphisms with their action on the orientation module. A positive normalizer therefore preserves every value of that character, so fixes every element of K. A negative normalizer changes its sign, so inverts every element. The same realization argument proves cyclicity of each finite positive overgroup. This proof compares a normalizing automorphism on the canonical oriented boundary; it makes no assertion that an infinite normalizer acts as a finite isometry group. □

B.10. [Refuted] The withdrawn uniqueness of topological roots. The lemma marked "faux!" on scan 200 is false even for the punctured torus: two different finite-order homeomorphisms fixing the base point can have the same nontrivial square and induce the same automorphism of π_1.

Manuscript: 18; 100; 200.

Proof. On $U = \mathbb{R}^2/\mathbb{Z}^2 \setminus \{0\}$ choose $p = (1/2, 1/2)$ and let $r(x, y) = (-y, x)$, $h = r^2$. Choose a small disc D whose four r-translates are disjoint and avoid both 0 and p. Choose a nonidentity homeomorphism f_0 supported in D and isotopic to the identity there relative to its boundary; for example use a radial bump rotation in disc coordinates. Put $f = f_0(hf_0h^{-1})$. Its two disjoint supports show that $fh = hf$; the same supports show that $fr \neq rf$, since r moves their union to the disjoint pair rD, r^3D. Hence $v = frf^{-1} \neq r$, but $v^2 = h$ and $v(p) = p$. The isotopy of f fixes 0 and p, so $f_* = 1$ on $\pi_1(U, p)$ and therefore $v_* = r_*$. This verifies the exact data of the withdrawn statement. What survives is uniqueness up to conjugacy by the identity component in the realization comparison, not equality of the homeomorphisms. □

B.11. [Refuted] Torsion in the centralizer need not be a subgroup. The assertion following the normalizer assertion on scan 199, that the torsion elements of the centralizer form one largest finite cyclic overgroup, is false for the hyperbolic type $(1, 1)$.

Manuscript: 18; 99-100; 198-200.

Proof. Keep U, p, r, h as in B.10. The essential simple curve $c = \{x + y = 1/2 \pmod 1\}$ avoids $0, p$ and is invariant under h. The based mapping class T_c commutes with h, since conjugation of a positive Dehn twist by an orientation-preserving map is the positive twist about the image curve. Thus the two genuine automorphisms induced by r and $T_c r T_c^{-1}$ have order four and common square h_*; both centralize h_*. On $H_1(U; \mathbb{Z})$, with a consistent choice of twist sign, their matrices are

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad TST^{-1} = \begin{pmatrix} -2 & -5 \\ 1 & 2 \end{pmatrix}, \qquad T = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}.$$

Both square to $-I$, while their product has trace -6 and determinant one. Its eigenvalues are $-3 \pm \sqrt{8}$, so it has infinite order. The corresponding product of automorphisms has infinite order as well. These integral equalities are checked by the supplied program. Consequently the centralizer's torsion set is not closed under multiplication, and there is no finite subgroup containing all its finite overgroups of $\langle h_* \rangle$. This coexists with the exact equality of normalizer and centralizer in B.9. □

2.3 The multiplier and the abelian surfaces

B.12. [Proved] An intrinsic discrete formula for the multiplier. For a geometric finite extension of a hyperbolic discrete surface group, E with its peripheral structure and orientation module T determines μ_E on all torsion elements. On E^+ it is the rotation number of its canonical oriented boundary action. More explicitly, let $\widetilde{\mathrm{Homeo}}^+(S^1) \to \mathrm{Homeo}^+(S^1)$ be the central extension by the positively oriented deck translation z. For $u \in E^+$ of order n and any lift $\widetilde{u}$, write $\widetilde{u}^n = z^k$. Then

$$\mu_E(u) = \frac{k}{n} \pmod{\mathbb{Z}} \quad \text{in } T \otimes \mathbb{Q}/\mathbb{Z}.$$

For a nonidentity orientation-reversing torsion element we put $\mu_E(u)$ equal to the unique nonzero element of order two. With this convention the formula is natural under isomorphisms preserving

the loops and transporting T, satisfies $\mu(u^m) = m\mu(u)$, and has the same order as u. This answers the discrete intrinsicity question of §15, folios 77–79, scans 155–160.

Manuscript: 15; 77-79; 155-160.

Proof. Choose an oriented finite-area hyperbolic uniformization of U. Its boundary at infinity is a circle. We explain the ambiguity in this choice. A marking isomorphism of surface groups preserving their peripheral subgroups and the orientation is induced by an orientation-preserving homeomorphism, by the Dehn–Nielsen–Baer correspondence. We can choose that marking quasiconformal: on a compact core smooth it, and on each cusp use standard cylindrical coordinates and extend with bounded dilatation. Its lift extends to an oriented boundary homeomorphism. This extension is uniquely determined by its π-equivariance. Indeed it must carry the attracting and repelling endpoints of every hyperbolic element to those of the corresponding element. These endpoints are dense in the boundary circle of a finite-area surface group. For completeness, density follows from its full limit set: a boundary arc meets translates of a fundamental ideal polygon arbitrarily near infinity; conjugating either of two hyperbolic elements with disjoint endpoints by the corresponding translates gives endpoints in that arc. Thus two equivariant boundary maps agree on a dense set and hence everywhere.

For each automorphism of π preserving the peripheral structure, Dehn–Nielsen–Baer provides the boundary map conjugating its deck action. The preceding uniqueness makes composition exact and makes this circle, together with its E-action, independent of the uniformization up to a unique π-equivariant boundary isomorphism. The boundary orientation is recorded by T, so changing orientation acts by -1. Realize the finite action and choose an invariant conformal structure, using B.1's cyclic and dihedral local charts on the quotient. Its hyperbolic metric makes a finite-order positive lift an elliptic isometry. In the disc model centred at its fixed point it is the same rotation on the tangent circle and on the circle at infinity. The displayed formula therefore gives its local multiplier.

Replacing $\widetilde{u}$ by $z^j\widetilde{u}$ replaces k by $k + nj$, so the value is independent of the lift. An orientation-preserving finite-order circle homeomorphism of order n is conjugate to a rotation by k/n with $\gcd(k, n) = 1$; this also follows by ordering the finite orbit on the circle and extending a homeomorphism of one fundamental interval around that orbit. This proves the order and power statements. A negative finite plane isometry is a reflection, of order two, so its assigned value has exactly the required power property and is unchanged by sign reversal.

The normalizer formula suggested in the margin of scan 159 is recovered as follows. Drill the orbit of the fixed point of u. Above the new puncture the positive peripheral normalizer is infinite cyclic, generated by a positively oriented element v, with peripheral subgroup generated by v^n. A lift of u to this normalizer is v^a modulo $\langle v^n\rangle$, and $\mu(u) = a/n$. In a local coordinate this is the equality between the rotation $z \mapsto e^{2\pi ia/n}z$ and monodromy in $z \mapsto z^n$. Harvey's Theorem 7 [29] computes exactly this inverse-residue relation from the elliptic generators of the orbifold epimorphism. His convention uses an inverse deck transformation, accounting for the sign visible in his proof on pp. 393–394. Our local-coordinate calculation also applies to cusps and to the positive cyclic part of a dihedral stabilizer; B.1 supplies the reflection part. □

B.13. [Refuted] Forgetting loops can change the multiplier. An abstract extension, with its orientation line regarded as an additional rank-one module, need not determine the multiplier after the peripheral structure is forgotten. There are isomorphic extensions with kernel a free group of rank two and quotient C_3 whose geometric multipliers disagree, even after allowing either isomorphism between their orientation lines.

Manuscript: 15; 77-79; 155-160.

Proof. Let $E_0 = \langle u, v \mid u^3 = v^3 = 1 \rangle = C_3 * C_3$. Realize it as the positive orbifold group of a sphere with one cusp and two cone points of order three; choose u, v as its positive local generators. The epimorphisms

$$q_+(u) = q_+(v) = 1, \qquad q_-(u) = 1, \quad q_-(v) = -1 \quad \text{(values in } \mathbb{Z}/3)$$

are injective on every finite subgroup, because finite subgroups of a free product are conjugate into its factors (the tree argument in B.8 applies). Their kernels are torsion-free of index three and Euler characteristic -1. The cusp generator is $(uv)^{-1}$. Its image under q_+ is nonzero, so there is one cusp and the kernel is a once-punctured torus group. Its image under q_- is zero, so there are three cusps and the kernel is a pair-of-pants group. Both have free rank two.

The automorphism $\psi(u) = u,\ \psi(v) = v^{-1}$ satisfies $q_-\psi = q_+$ and is an isomorphism of the two abstract extensions. The positive local generator values in each realization are $\mu(u) = \mu(v) = 1/3$. Thus ψ preserves the value at u but changes the value at v to $-1/3$. Reversing the isomorphism of orientation lines would instead break the equality at u. No orientation-line identification repairs both. The isomorphism sends the cusp word uv to uv^{-1}, displaying precisely which peripheral comparison has been discarded. The finite monodromy calculation is included in the computational record. □

B.14. [Proved] Finite annulus and torus actions, and their multipliers. Every finite topological action on the torus is topologically conjugate to an affine action on V/Λ, where $\Lambda \cong \mathbb{Z}^2$ and $V = \Lambda \otimes \mathbb{R}$. For a fixed finite operator group G such an action is specified by a homomorphism $A : G \to GL(\Lambda)$ and a cocycle $b \in Z^1(G, V/\Lambda)$, subject to faithfulness of $g \mapsto (x \mapsto A_g x + b_g)$. Translation conjugacy corresponds exactly to a coboundary, and

$$H^1(G, V/\Lambda) \xrightarrow{\sim} H^2(G, \Lambda)$$

is the connecting homomorphism. Linear changes of the lattice give the further conjugacies. Every finite annulus action has a simultaneous normal form

$$(x, t) \longmapsto (\epsilon_g x + a_g, \eta_g t), \qquad x \in \mathbb{R}/\mathbb{Z}, \quad t \in \mathbb{R}, \quad \epsilon_g, \eta_g \in \{1, -1\}.$$

These descriptions determine the multiplier of every finite-order element in the respective lift extensions.

Manuscript: 15; 78-79; 157-159; 17; 89-90; 179-181.

Proof. We first justify simultaneous conformalization. B.1 gives genuine cyclic or dihedral charts for every finite stabilizer. The quotient is a compact surface orbifold after the ends are added; its interior special points have cyclic charts, and its mirror edges and corners have reflection and dihedral charts. Choose a smooth metric on the quotient away from these charts, choose round cone and mirror metrics in them, and patch in collars using an invariant partition of unity on their finite local covers. The local charts make the patched lifts smooth. Pullback is a smooth invariant conformal structure on the original surface. This construction proves the required passage from the original topological action; it does not average a merely topological metric and call the result Riemannian.

Uniformization of the torus gives $\mathbb{C}/\Lambda$. Every holomorphic automorphism lifts to an entire map f with $f(z + \lambda) - f(z) \in \Lambda$. Its derivative is periodic and hence bounded on $\mathbb{C}$, so it is constant; $f(z) = az + c$. An antiholomorphic automorphism is obtained by composing such a map with complex conjugation on its source. Hence every element is real affine on the same lattice. The action law is precisely $b_{gh} = b_g + A_g b_h$. Averaging cochains over the finite group gives $H^q(G, V) = 0$ for $q > 0$. The exact sequence $0 \to \Lambda \to V \to V/\Lambda \to 0$ therefore gives the asserted isomorphism. In concrete terms, lifts $v_g \in V$ of b_g give the extension cocycle $v_g + A_g v_h - v_{gh} \in \Lambda$.

Conjugating by translation adds a coboundary. Moreover any topological conjugacy between two affine actions can be replaced by an affine conjugacy. A lift of the conjugacy has form $Bx + f(x)$, with $B \in GL_2(\mathbb{Z})$ and f periodic. Average its equivariance equation over the torus with Haar measure. Affine lattice automorphisms preserve that measure, so $Bx + \int f$ satisfies the same equation modulo Λ.

For the annulus, extend the action to its two-point sphere compactification and apply Kolev. In the resulting conformal sphere choose a Möbius coordinate sending the two ends to zero and infinity. The maps are then of the forms $z \mapsto az$, a/z, $a\overline{z}$, or $a/\overline{z}$. On the logarithmic radial coordinate they act affinely with slopes $\eta = \pm 1$. A finite group of affine transformations of the real line fixes the average of any orbit; translation to this common fixed point removes every radial constant. The angular coordinate gives the claimed $\epsilon x + a$. This proves the displayed simultaneous normal form. The action on π_1 is ϵ_g and its orientation sign is $\epsilon_g \eta_g$.

For the promised computation on the torus, an element of the lift extension is $x \mapsto Ax + v$. It has finite order exactly when A has finite order, say n, and $(1 + A + \cdots + A^{n-1})v = 0$. Averaging its orbit gives a fixed point, and translation to that point conjugates it to A. A nonidentity positive finite-order matrix has trace $-2, -1, 0, 1$, giving orders $2, 3, 4, 6$, respectively. For order two its multiplier is $1/2$. For the other orders it is $+1/n$ or $-1/n$ according to the sign of $\det(w, Aw)$ for any nonzero w; this sign is independent of w, as is seen after averaging an inner product and writing A as a rotation. A negative finite-order matrix is a reflection and receives $1/2$. A lift with $A = I$ has finite order only when it is the identity. For the annulus a lift is $(x, t) \mapsto (\epsilon x + a + k, \eta t)$. With $\epsilon = -1$ its square is the identity; with $\epsilon = 1$ finite order requires $a + k = 0$, leaving the identity or $t \mapsto -t$. Thus every nonidentity torsion lift on an annulus has order two and multiplier $1/2$. □

B.15. [Refuted] The "only counterexample" of scan 179. The assertion that a faithful homotopically trivial finite annulus action failing to be a translation action must be the single anti-involution $z \mapsto 1/\overline{z}$ is false. For the two non-simply-connected abelian types, the exact replacement is: homotopically trivial torus actions are translation actions; homotopically trivial annulus actions are the finite subgroups of $SO(2) \times C_2$ in the normal form of B.14. Those contained in $SO(2)$ are translations, while those with nontrivial second projection include two different types of anti-involution and groups of arbitrarily large order.

Manuscript: 17; 89-90; 179-181.

Proof. On the torus, homotopic triviality is $A_g = I$ for every g, so B.14 proves the translation statement and the implication doubted on scans 180–181. On the annulus it is $\epsilon_g = 1$, which gives the asserted subgroup. For each $n \geq 1$ the maps

$$r(z) = e^{2\pi i/n} z, \qquad s(z) = 1/\overline{z}$$

generate a faithful $C_n \times C_2$ action, and every element induces the identity on $\pi_1(\mathbb{C}^*)$. The fixed circle of s excludes a translation realization: a nonidentity translation in any group is fixed-point free. In addition $s_-(z) = -1/\overline{z}$ is an involution inducing $+1$ on π_1, but it has no fixed point, since its fixed-point equation is $|z|^2 = -1$. It cannot be conjugate to s, and it cannot be a translation of a connected topological group: every such translation is isotopic to the identity along a path in that group and hence preserves orientation. These formulas give the required full counterexamples, not merely a gap in the argument.

More explicitly let the rotation subgroup have order n and generator $(x, t) \mapsto (x + 1/n, t)$. A second-coset generator can be chosen as $(x, t) \mapsto (x + m/(2n), -t)$, with $m = 0$ or 1 after multiplying by a rotation. Its square is the mth rotation. This exhausts the finite subgroups projecting onto C_2, and includes both the reflection and glide cases. For the omitted simply connected surfaces, finite plane actions are the orthogonal actions of B.1 and are all homotopically

trivial; positive sphere actions are likewise homotopic to the identity. Nontrivial rotations fixing a point already refute any extension of the translation formulation to those types. □

2.4 Anti-involutions and the later tube construction

B.16. [Proved] Complete topological classification of anti-involutions. Conjugacy classes of anti-involutions of a closed oriented genus-g surface are classified by the orientability of the quotient and its number b of boundary components. The possibilities are

$$\begin{aligned} &\text{nonorientable quotient:} && 0 \le b \le g, \quad h = g + 1 - b \ge 1 \text{ crosscaps;} \\ &\text{orientable quotient:} && b \ge 1, \quad g = 2h + b - 1, \quad h \ge 0. \end{aligned}$$

There are $(g+1) + \lfloor (g+2)/2 \rfloor = \lfloor (3g+4)/2 \rfloor$ classes. For ν unlabelled invariant punctures the additional data are an integer $r \ge 0$ and an unordered list $(q_1, \dots, q_b)$ of nonnegative integers with $2r + \sum q_i = \nu$: r is the number of interior punctures of the quotient, and q_i counts punctures on its ith boundary circle.

Manuscript: 33; 221-222; 437-440.

Proof. By B.1 the quotient $Y = X/\sigma$ is a compact surface with boundary X^σ. The complement of the real locus covers its interior with degree two; the Euler characteristic of the boundary is zero. Hence $\chi(X) = 2\chi(Y)$. If Y is nonorientable with h crosscaps, $\chi(Y) = 2 - h - b$, yielding $g = h + b - 1$; if orientable of genus h it yields $g = 2h + b - 1$. In the latter case b must be positive, since the oriented double of a closed orientable quotient would be disconnected. Conversely form the orientation double over the interior of each allowed Y and join the two local sides at its boundary. The result is a connected closed orientable surface, with the deck interchange extended to reflection along the boundary; its genus is given by the displayed formula. The quotient recovers Y, so the classification of compact surfaces proves both existence and uniqueness of these conjugacy classes. This also proves the complete list underlying Gross–Harris, Proposition 3.1 [26], including the two corrections in the margins of scan 439. Grothendieck's nonorientable parameter γ is $h - 1$; thus his formula $g = \gamma + b$ agrees exactly with ours.

An anti-involution of a punctured finite-type surface extends to its end compactification. Punctures in its quotient lie either in the interior, with two-point inverse image, or on the boundary, with one-point inverse image. Any two finite interior configurations of the same cardinality are related by an isotopy supported away from the boundary. Boundary components may be permuted, and any two unlabelled configurations of the same cardinality on one circle are related by a circle homeomorphism, extended over a collar. These facts prove completeness of the punctured invariants. With individually labelled punctures one must instead retain their assignments and their circular orders; B.5 provides the corresponding incidence reconstruction. □

B.17. [Proved] Real moduli components, with the group and marking convention fixed. Let $\mathcal{M}_{g,\nu}$ mean the real moduli stack of smooth proper geometrically connected genus-g curves with an *unlabelled* degree-ν reduced puncture divisor. We retain the stack and its real forms also in the four unstable types. Its complex mapping class group is $\mathcal{T}^+_{g,\nu}$, allowing permutations of the punctures. Write $p_b(N)$ for the number of partitions of N into at most b parts, with $p_0(0) = 1$ and $p_0(N) = 0$ for $N > 0$. Then

$$\#\pi_0(\mathcal{M}_{g,\nu}(\mathbb{R})) = \sum_{b=0}^{g} \sum_{r=0}^{\lfloor \nu/2 \rfloor} p_b(\nu - 2r) + \sum_{\substack{1 \le b \le g+1 \\ b \equiv g+1 \ (2)}} \sum_{r=0}^{\lfloor \nu/2 \rfloor} p_b(\nu - 2r).$$

For closed hyperbolic surfaces this is $\lfloor (3g+4)/2 \rfloor$. In the four unstable types $(0,0)$, $(0,1)$, $(0,2)$ and $(1,0)$ the counts are, respectively, $2, 1, 3, 3$. In the last case no origin is prescribed. Distinct

topological anti-involution classes remain distinct in the extended mapping class group in these hyperbolic types. The last assertion fails for $(0,0)$, $(0,2)$, and $(1,0)$.

Manuscript: 33; 222; 439-440.

Proof. C.5 and C.6, finite-action realization and uniqueness up to the identity component, identify conjugacy classes of anti-involutions on the fixed punctured surface with conjugacy classes of orientation-reversing involutions in $\mathcal{T}_{g,\nu}$. One can use positive conjugacy here: an orientation-reversing conjugator can be multiplied by the target involution to become positive. Mochizuki, Corollary 3.8, identifies these conjugacy classes with the connected components of the real moduli *stack* [44]. His Example 3.3 uses the full modular group of a punctured surface, which is the unlabelled convention just specified. B.16 supplies the list. For fixed b, r, unordered nonnegative lists of length b summing to $\nu - 2r$ are exactly the partitions counted by $p_b(\nu - 2r)$. Summing over the two quotient families proves the formula. The supplied code enumerates these numbers for $2g + \nu \leq 6$ and checks the closed formula for $0 \leq g \leq 20$; the proof holds for all g.

We verify the unstable component counts without using mapping-class injectivity. A real genus-zero curve is a split or nonsplit real conic; analytically these are the sphere with $z \mapsto \overline{z}$ or with $z \mapsto -1/\overline{z}$. To see these are the two possibilities, conformalize its anti-involution as in B.14 and apply the two quotient types in B.16; uniformization identifies the resulting conformal double with the sphere. A real point exists exactly in the first type. There is no complex moduli parameter in genus zero. With one puncture the point is real, and $PGL_2(\mathbb{R})$ acts transitively on the real projective line, giving one component. With two unlabelled punctures the possibilities are two real points on the split conic, one conjugate pair on the split conic, or one conjugate pair on the nonsplit conic. The first is transitive under $PGL_2(\mathbb{R})$, the second under its action on the upper half-plane, and the third under the rotations commuting with the antipodal map. These give three components. With no puncture there are the two conic components.

For an unpointed genus-one curve, B.14 and B.16 give the three anti-involution types with zero, one, or two real ovals. Each type has a connected parameter space of complex structures. Explicit representatives are

b	Λ_t	$\sigma(z)$
0	$\mathbb{Z} + it\mathbb{Z}$	$\overline{z} + \frac{1}{2}$
1	$\mathbb{Z} + (\frac{1}{2} + it)\mathbb{Z}$	$\overline{z}$
2	$\mathbb{Z} + it\mathbb{Z}$	$\overline{z}$

$(t > 0)$.

To obtain this list from an invariant lattice, rotate a primitive real period to 1. Conjugation of a second period forces its real part modulo $\mathbb{Z}$ to be 0 or $1/2$. The translation cocycle of B.14 is then removed by a change of origin, except for the nonzero half-period class in the rectangular case. That class gives the first row. Thus every real genus-one curve occurs in one of these connected families. The number of real ovals is locally constant in a smooth proper real family: local trivialization can be chosen invariant under the involution by an invariant horizontal metric and its path lifting. Distinct rows consequently belong to different components. This proves the number three, with no origin added to the moduli problem. Direct substitution of the four unstable types into the displayed partition formula gives exactly these counts.

The mapping-class exceptions can be checked directly. On S^2 reflection and the antipodal map have different fixed sets, but both represent the orientation-reversing mapping class. On $\mathbb{R}^2/\mathbb{Z}^2$ the maps $(x, y) \mapsto (x, -y)$ and $(x, y) \mapsto (x + 1/2, -y)$ have respectively two fixed circles and no fixed points, but both induce $\mathrm{diag}(1, -1)$ and hence the same mapping class. On the annulus, the reflection and glide anti-involutions of B.15 also have the same mapping class: the path $(x, t) \mapsto (x + s/2, -t)$ joins them through homeomorphisms preserving the two ends as a set. These examples establish the exact failure of the unrestricted injectivity hoped for on scan 440. For labelled moduli spaces or a coarse moduli space, the quotient and its real locus have to

be specified separately; the displayed formula is for the stack and puncture convention stated above. □

B.18. [Proved] The two centralizer corollaries in both quotient families. Let $g \geq 2$, and let an anti-involution of X_g have $b > 0$ ovals, with outer class σ. There are exactly b classes of order-two genuine loop automorphisms lifting σ, under conjugacy by π_g. For each lift u, $\pi_g^u \cong \mathbb{Z}$. The positive centralizer of σ acts transitively on those b classes. More precisely, writing $A = \mathrm{Aut}_{\mathrm{lac}}(\pi_g)$ and $O = \mathrm{Out}_{\mathrm{lac}}(\pi_g)$, there is an exact sequence

$$1 \longrightarrow \pi_g^u \longrightarrow C_{A^+}(u) \longrightarrow \mathrm{Stab}_{C_{O^+}(\sigma)}([u]) \longrightarrow 1,$$

and the last group has index b in $C_{O^+}(\sigma)$. The corresponding full centralizer image also has index b. This proves Corollaries 1 and 2, folios 225–226, scans 445–448, for both orientable and nonorientable quotients.

Manuscript: 33; 225-226; 445-448.

Proof. B.4 proves the count and the fixed subgroup. Any permutation of the boundary components of a compact connected quotient surface is realized by a homeomorphism: join two components by an embedded arc, interchange them in a neighbourhood of that arc and the two collars, and use transpositions. Each quotient homeomorphism lifts to the oriented double; among its two lifts choose the orientation-preserving one. It commutes with the reflection. Consequently the positive centralizer is transitive on the ovals, and hence on the section classes.

For exactness, the centre of π_g is trivial, so $\mathrm{Inn}(\pi_g)$ is identified with π_g. Its intersection with $C_A(u)$ is exactly π_g^u. An outer class α centralizing σ preserves $[u]$ precisely when one, hence every, lift $\widetilde{\alpha}$ satisfies $\widetilde{\alpha} u \widetilde{\alpha}^{-1} = vuv^{-1}$ for an inner automorphism v. Then $v^{-1}\widetilde{\alpha}$ centralizes u, proving surjectivity onto the displayed stabilizer. Orbit–stabilizer gives index b in both versions. For $b = 0$ there are no order-two genuine lifts and these assertions about a chosen lift have no instance. With punctures the same exact sequence holds, but the orbit size records the boundary decorations from B.16; it need not be the total number of ovals when different ovals carry different puncture data. □

B.19. [Proved] The tube twists, their maximality property, and their normalizer. For $g \geq 2$, the elements ρ_j of §33, scans 453–470, are full Dehn twists about a cut system $c_1, \ldots, c_g$: disjoint nonseparating curves with connected complement. They generate $H \cong \mathbb{Z}^g$. This is not a maximal free abelian subgroup of $\mathcal{T}_g$: it is contained in a multitwist subgroup of rank $3g - 3$. It has the precise maximality property of being a full multitwist group of largest rank whose supporting multicurve has connected complement. Furthermore

$$N_{\mathcal{T}_g}(H) = \mathrm{Stab}_{\mathcal{T}_g}(\{c_1, \ldots, c_g\}), \qquad N_{\mathcal{T}_g}(H)/C_{\mathcal{T}_g}(H) \cong S_g \times C_2.$$

The action on H permutes its basis and multiplies every basis vector by the same orientation sign. Cutting and capping gives an exact sequence

$$1 \longrightarrow H \longrightarrow N_{\mathcal{T}_g}(H) \longrightarrow \mathcal{T}_{0,2g}^{\mathrm{pairs}} \longrightarrow 1,$$

where the last extended sphere mapping class group preserves the partition of its $2g$ punctures into the g unordered gluing pairs.

Manuscript: 33; 229-237; 453-470.

Proof. In the coordinates of scan 461 the map is $\rho_j(z, t) = (-ze^{\pi i t}, t)$ on $S^1 \times [-1, 1]$ and is the identity outside the tube. Its angular displacement increases from 0 to 2π, so its relative

boundary mapping class is one full Dehn twist, with sign fixed by that coordinate orientation. The complement of the tubes in his construction is a sphere with $2g$ boundary circles, establishing the cut-system assertion.

Twists in disjoint collars commute. They are independent: cut the surface along the c_j, choose transverse curves dual to these cuts, and measure the integer winding of the resulting arcs when the collars are glued back. A product $\prod T_{c_j}^{n_j}$ changes the jth such winding by n_j; an isotopy cannot change these relative-end winding differences. This proves $H \cong \mathbb{Z}^g$. Complete the $2g$-holed sphere to a pants decomposition by adding $2g-3$ essential disjoint curves. The same winding argument gives a rank-$3g-3$ multitwist group containing H. In contrast, cutting k disjoint curves while keeping the complement connected reduces the genus to $g-k$, so $k \le g$. This proves the stated maximality among connected-complement multitwist systems.

For the normalizer, conjugation satisfies $fT_cf^{-1} = T_{f(c)}^{\epsilon(f)}$. A single twist can equal a product of twists about the c_j only by being one of these twists to the corresponding power. Here is the needed support argument. Such a twist commutes with every T_{c_j}. The elementary intersection formula $i(T_d(c), c) = i(c, d)^2$ for a single twist forces d to be disjoint from each c_j; otherwise its conjugation changes T_{c_j}. Add d to the disjoint system if it is not already there and apply the preceding independent-winding argument. The resulting nontrivial relation is impossible. Thus a normalizer permutes precisely the c_j, proving the first equality. Conversely every such permutation of curves normalizes H. Every permutation of the handles is realized positively, and a reflection of the standard cut-system model reverses the global orientation while preserving each curve. This gives the full image $S_g \times C_2$. Its kernel is exactly the centralizer.

A representative preserving the multicurve can be isotoped to preserve its collars. Its restriction to the cut surface, after capping each boundary to a puncture, preserves the gluing pairs. Conversely a homeomorphism of the capped sphere preserving those pairs can be made radial in small puncture discs and then extended over the paired collars. Two such extensions differ by integer collar twists. An isotopy on the capped sphere lifts after allowing exactly these twists, by the isotopy extension theorem on collars. This proves both surjectivity and the stated kernel, and types the comparison with the complement suggested on scan 470. □

B.20. [Refuted] Presentation (52) and its exact correction. For $g \ge 2$, presentation (52) on scan 469 does not present the stated $\mathbb{Z}^g$-by-finite group. Put $r = \varepsilon_0\varepsilon_1$ and $\rho_j = r^j\tau'\tau r^{-j}$, $0 \le j < g$. The printed presentation gives

$$F_g \rtimes (D_{4g} \times \langle\tau\rangle),$$

where $D_{4g} = \langle r, \varepsilon_1 \mid r^{2g} = \varepsilon_1^2 = 1, \varepsilon_1 r \varepsilon_1 = r^{-1}\rangle$ permutes the free generators through its action on $\mathbb{Z}/g$, and τ inverts each free generator. Adding the relations $[\rho_i, \rho_j] = 1$ for $i < j$ gives exactly $\mathbb{Z}^g \rtimes (D_{4g} \times C_2)$ with the manuscript's specified action.

Manuscript: 33; 237; 469.

Proof. The relations say that τ commutes with D_{4g}, that τ' commutes with ε_1 and r^g, and that both τ, τ' are involutions. Set $\rho_0 = \tau'\tau$. Then $\tau\rho_0\tau = \rho_0^{-1}$, $\varepsilon_1\rho_0\varepsilon_1 = \rho_0$, and $r^g\rho_0 r^{-g} = \rho_0$. Its conjugates are indexed by the g cosets of $\langle\varepsilon_1, r^g\rangle$ in D_{4g}, with no further relation among them. To check the last assertion directly, form the displayed semidirect product with a free basis $\rho_0, \dots, \rho_{g-1}$ and define $\tau' = \rho_0\tau$ in it. All printed relations hold. The reverse map sends these free generators to $r^j\tau'\tau r^{-j}$; the two maps agree inversely on all generators. This is a proof by explicit mutually inverse homomorphisms. Abelianizing only the free factor adds exactly the stated commutators and gives the corrected group.

There is also a finite certificate. For $g = 2$ set $\varepsilon_0 = (13)$, $\varepsilon_1 = \tau = 1$, $\tau' = (12)$ in S_3. All nine displayed relators evaluate to 1, but $\rho_0 = (12)$ and $\rho_1 = (23)$ do not commute. The program

checks every one of these evaluations. Relations (33), scan 454, already assert commutation for the geometric twists; they are exactly the missing input to (52). □

B.21. [Proved] The one-holed-torus subgroups and the proved part of the analogy. For $g \geq 2$, choose a one-holed torus W_j containing c_j and a transverse curve d_j with $i(c_j, d_j) = 1$. Its group fixing its boundary pointwise embeds by extension by the identity into $\mathcal{T}_g^+$. It is

$$B_3 = \langle A_j, B_j \mid A_j B_j A_j = B_j A_j B_j \rangle,$$

and capping the boundary gives

$$1 \longrightarrow \langle T_{\partial W_j} \rangle \cong \mathbb{Z} \longrightarrow B_3 \longrightarrow SL_2(\mathbb{Z}) \longrightarrow 1, \qquad (A_j B_j)^6 = T_{\partial W_j}^{\pm 1}.$$

This extension has no section. For disjoint chosen W_j, their supported subgroups commute and, together with H, preserve their boundary multicurve; they do not generate $\mathcal{T}_g^+$. On explicit generators the tube subgroup and these two twists act by

$$\begin{array}{ll} \rho_j : & a_j \mapsto a_j, \quad b_j \mapsto b_j a_j, \\ B_j : & a_j \mapsto a_j b_j^{-1}, \quad b_j \mapsto b_j, \end{array}$$

fixing the other handle generators of $\pi_g = \langle a_j, b_j \mid \prod_j [a_j, b_j] = 1 \rangle$.

Manuscript: 33; 236-237; 467-470.

Proof. Choose the based handle loops so that their commutator is the boundary loop. The displayed substitutions preserve $[a_j, b_j]$ exactly, and are the effect of crossing the appropriate twist collar. Twists in curves meeting once obey the braid relation: in these generators substitution gives $ABA = BAB$. The sixfold product $(AB)^6$ conjugates both generators by $[a, b]^{-1}$, which is the boundary twist with this collar convention. Both word equalities are checked by free reduction in `code/free_words.py`; the stated substitutions also give their direct verification. Filling the boundary gives the standard two elementary matrices generating $SL_2(\mathbb{Z})$. Their action on primitive slopes, reduced by the Euclidean algorithm, shows that the filled mapping class group is exactly $SL_2(\mathbb{Z})$. The kernel of filling is the boundary winding: cut along two based generating arcs, isotope a kernel representative to fix them, and read its remaining integer twist in the boundary collar. The kernel is therefore the infinite cyclic group generated by the boundary twist. The braid group on three strands has the same central extension: with $x = ABA, y = AB$ it is $\langle x, y \mid x^2 = y^3 \rangle$, and its quotient by y^6 is $SL_2(\mathbb{Z})$. The comparison sends the kernel generator to the boundary twist, so it is an isomorphism of these extensions.

Extension by the identity is injective because an isotopy to the identity on the closed surface would fix the essential boundary and, after collar adjustment, give an isotopy on the one-holed torus; a nonzero boundary twist remains nontrivial by the winding argument in B.19. The braid group is torsion-free: $\langle x, y \mid x^2 = y^3 \rangle$ is the amalgam of two infinite cyclic groups along their subgroups of indices two and three; a finite subgroup fixes a vertex in its tree and is conjugate into an infinite cyclic group. A section from $SL_2(\mathbb{Z})$ would inject its order-four element, contradicting this fact.

Disjoint supports commute. Each supported homeomorphism fixes the chosen boundary curves, and so does every generator of H. An essential curve crossing one of those boundaries has a twist which does not preserve it, by the intersection formula in B.19. Hence the generated subgroup is proper. The exact analogy proved here is the homomorphism to the usual rank-two integral representation and its central boundary-winding extension, with braid and disjoint-support relations. For overlapping choices of the strips one must specify their intersections before computing the remaining relations; no unspecified choice of strips supplies additional relations by itself. □

2.5 Computations and the limits of the requested table

B.22. [Refuted] A finite subgroup alone does not index the requested abelian table. For all types in Part B(f), a table indexed only by a finite subgroup of the mapping class group cannot assign unique splitting numbers or Φ. The type $(1,0)$ and the single subgroup $\langle \mathrm{diag}(1,-1)\rangle$ already have two different answers. The additional extension datum is precisely the class in $H^2(G,\pi)$ in the abelian cases of B.14.

Manuscript: 15; 78-79; 157-159; prompt B.f.

Proof. On the torus compare $\sigma_0(x,y)=(x,-y)$ and $\sigma_1(x,y)=(x+1/2,-y)$. Both are faithful involutions with the same homomorphism to $GL_2(\mathbb{Z})$, but the first fixes two circles and the second fixes nothing. The corresponding extensions, with commuting lattice generators a,b, have relations

$$\widetilde{\sigma}a\widetilde{\sigma}^{-1}=a, \qquad \widetilde{\sigma}b\widetilde{\sigma}^{-1}=b^{-1}, \qquad \widetilde{\sigma}^2=a^{\epsilon}, \quad \epsilon=0 \text{ or } 1.$$

The square of $a^m b^n\widetilde{\sigma}$ is $a^{2m+\epsilon}$. For $\epsilon=1$ there is no order-two lift. For $\epsilon=0$ such lifts require $m=0$, and conjugation by b changes n by two. Thus there are exactly two section classes in the first case and zero in the second. These are the two parity classes of the extension cocycle. B.14 proves that retaining the extension class gives the correct affine classification, with all translation conjugacies accounted for. Thus the counterexample identifies the missing index and its exact replacement. □

B.23. [Proved] Complete finite-subgroup table in type $(0,3)$. The ten conjugacy classes in the extended group $\mathcal{T}_{0,3}\cong S_3\times C_2$, and their discrete section invariants, are the following. In the table F counts full-section classes, P counts all nontrivial partial-section classes with their subgroup images distinguished, $M=|\Phi|$ counts maximal partial-section classes, and O counts the group orbits on Φ.

G	$\lvert G\rvert$	F	P	M	O
1	1	1	0	0	0
C_2 (positive)	2	1	1	1	1
C_2 (horizontal mirror)	2	3	3	3	3
C_2 (vertical mirror)	2	1	1	1	1
C_3	3	2	2	2	2
C_2^2	4	1	6	3	2
C_6	6	0	5	5	2
S_3 (positive)	6	0	5	5	2
S_3 (mixed)	6	2	7	2	2
$S_3\times C_2$	12	0	16	5	2

The attached CSV gives every maximal stabilizer; the JSON table gives every partial section and all of its multiplier values. In particular it checks the reflection failure of maximal-section uniqueness from B.8 inside this whole list.

Manuscript: 15; 75-77; 150-155; prompt B.f.

Proof. Place the three punctures at the vertices of an equatorial equilateral triangle. The positive rotation group of that triangle is S_3: it is generated by the 120-degree vertical rotation r and a half-turn a around an axis through a puncture. Reflection τ in the equator commutes with both. The mapping class group map to permutations and orientation is an isomorphism. For its kernel, fix three arcs forming a tree through the punctures by isotopy; its complement is a disc, and Alexander's disc isotopy makes the remaining map isotopic to the identity. This proves completeness of the ambient finite group and its stated realization.

The program represents this group exactly by pairs $(p, e) \in S_3 \times C_2$. Starting with the trivial subgroup it repeatedly adjoins every group element and takes closure, until no new subgroup occurs. Every subgroup is generated by its elements, so this terminating procedure is exhaustive. It finds sixteen subgroups and ten conjugacy classes, represented in the displayed order by

$$1,\ \langle a\rangle,\ \langle\tau\rangle,\ \langle a\tau\rangle,\ \langle r\rangle,\ \langle a,\tau\rangle,\ \langle r,\tau\rangle,\ \langle r,a\rangle,\ \langle r,a\tau\rangle,\ \langle r,a,\tau\rangle.$$

Here is the independent geometric input to its section calculation. Denote the poles by N, S, and the three surviving opposite equatorial axis points by Q_i; the other three axis points are the removed punctures. The fixed set of $\langle r\rangle$ is $\{N, S\}$, and that of a positive half-turn is its single surviving Q_i. The horizontal mirror has three open intervals, each containing one Q_i. A vertical mirror has one real line after its single puncture is removed, and this line contains N, S, Q_i. Intersect these fixed sets to find the fixed sets of every subgroup. B.3 and B.4 turn their components into exactly its section classes. A partial section on a mirror interval is contained in a point stabilizer precisely when that point lies on the interval; this determines all inclusions and hence all maximal classes. The code records and checks these inclusions and the conjugation action of every element.

Every reflection and positive half-turn has multiplier $1/2$. For r, the values at N, S are respectively $1/3, -1/3$, and its powers give all remaining values. This computes all entries in the attached JSON. The geometry also verifies that all the centralizers occurring for nontrivial sections here are trivial: every mirror component is a line, and every positive section fixes an isolated point. The absence of a common finite overgroup for the two vertex stabilizers along a vertical mirror is retained in the inclusion data. □

B.24. [Proved] Completed finite-subgroup catalogue and exact indexing. Every finite-subgroup conjugacy class in the requested range $2g + \nu \le 6$ has now been enumerated. For the hyperbolic types, the numbers of classes in the extended mapping class group are

(g, ν)	# classes	proofs and executed tables
$(0,3), (0,4), (0,5), (0,6)$	$10, 30, 20, 57$	$B.23, B.25, B.26$
$(1,1), (1,2), (1,3), (1,4)$	$13, 40, 38, 115$	$B.25, B.27$
$(2,0), (2,1), (2,2)$	$92, 32, 121$	$B.28, B.29$
$(3,0)$	292	$B.33, B.37$–$B.41$.

(B.24.1)

Each table includes the full and partial section classes, multipliers, centralizers, extension incidences, maximal section classes Φ, and their group orbits and stabilizers. In the four exceptional abelian types, the finite mapping-class subgroup counts are respectively $2, 2, 5, 13$ for sphere, plane, annulus and torus. Their injectively realized finite-action counts are $3, 2, 7, 17$; the exact covering extension is retained as part of the section index in B.36, as required by the explicit failures of extension-independent indexing in B.22. The families with a nontrivial kernel on passage to mapping classes are described in B.14. The exact tube words, stabilizer action, strip relations and arbitrary transversal obstruction formerly required here are proved and executed in B.30–B.32.

Manuscript: §18; folios 96–101; scans 193–203; §33; folios 222–237; scans 440–470; Part B(e) and B(f) of the task.

Proof. The first three rows of (B.24.1) are precisely the complete catalogues cited there. We spell out the closed genus-three reconciliation to record the coverage and the absence of double counting. The positive portion has 55 classes by B.33. For a mixed finite action let H be its positive subgroup. The positive quotient has genus zero, one, two or three, by the positive classification. The genus-zero case gives either a mirror disc (145 classes, B.37) or a projective plane (11 classes, B.39). The genus-one case gives an annulus or Möbius band (57 classes, B.40), or a Klein bottle (6 classes, B.41). The genus-two case is the free positive C_2 action and gives 12

mixed classes in B.38. The genus-three case has trivial positive part and gives its six negative involution classes in B.38. Thus the number of mixed classes is

$$145 + 11 + 57 + 6 + 12 + 6 = 237, \qquad 55 + 237 = 292. \tag{B.24.2}$$

Every conjugacy descends to homeomorphisms of both the positive quotient and the full quotient orbifold. The positive quotient genus, and the full quotient's boundary and orientability, separate the listed portions. Within each portion the cited proofs calculate exactly the geometric kernel-normalizer orbits, so these counts are irreducible counts of surface-action conjugacy classes.

B.34 supplies an independent hyperelliptic construction and is counted through its explicit overlaps, not added a second time. All 79 of its rows are identified by monodromy isomorphisms: 18 with positive B.33 rows, 57 with B.37 rows, and four with B.39 rows. The executed program `reconcile_genus3_catalogue.py` records these 79 maps and checks the corresponding section counts. Its unified `type_3_0_complete_catalogue.csv` has 292 different class identifiers, and `type_3_0_complete_sections.json` contains the full finite group and section model for every row. The exceptional replacement in B.36 is complete with its stated realization index; it retains, in particular, the sphere antipodal action's algebraic splitting without identifying it with a geometric fixed point. This proves the claimed coverage with exactly the comparisons and indices used by the computations. □

B.25. [Proved] Complete executed catalogues in types $(1,1)$ and $(0,4)$. The extended mapping class groups of types $(1,1)$ and $(0,4)$ have, respectively, 13 and 30 conjugacy classes of finite subgroups, including the trivial subgroup. The tables below give every class and the numbers of full splittings, nontrivial partial splittings, maximal partial section classes Φ, and group orbits on Φ. The accompanying exact-arithmetic files give every section, its image subgroup, its multiplier on every element, its centralizer, all inclusion relations, the group stabilizers, and the puncture stabilizers. These are exhaustive catalogues for the two specified surface types.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; Part B(f) of the task; no further manuscript assertion.

Proof. Put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For the punctured torus use their linear actions on $\mathbb{R}^2/\mathbb{Z}^2 \setminus \{0\}$. The homology map identifies its extended mapping class group with $\mathrm{GL}_2(\mathbb{Z})$: surjectivity is supplied by linear maps, and a map acting trivially on homology can be isotoped to fix two transverse basis curves, after which the complementary punctured disc is handled by the Alexander isotopy. This is the punctured version of [20, Theorem 2.5 and pp. 54–55].

For the four-punctured sphere use the pillowcase

$$(\mathbb{R}^2/\mathbb{Z}^2 \setminus (\tfrac{1}{2}\mathbb{Z}^2/\mathbb{Z}^2))/\{p \sim -p\}.$$

Write $U(p) = p + (1/2, 0)$ and $V(p) = p + (0, 1/2)$. The extended mapping class group is $V_4 \rtimes \mathrm{PGL}_2(\mathbb{Z})$ with these affine representatives. The map to $\mathrm{PGL}_2(\mathbb{Z})$ is obtained by lifting through the double branched torus cover and acting on torus homology. Its kernel consists of the four half translations: a kernel element fixes all slope classes; after a half translation it fixes the four punctures and two filling curves, so the Alexander isotopy on their complementary discs kills it. Linear maps split the homomorphism and act on half translations by reduction modulo two. This is the explicit morphism of [20, Proposition 2.7 and the extended statement on p. 57].

Here are finite amalgams that make enumeration complete. The matrices S, R generate $\mathrm{SL}_2(\mathbb{Z})$ by the Euclidean algorithm, since $S^{-1}R = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Modulo $\{\pm I\}$ their images have orders

two and three, with no further relation: on the real projective line, $\bar{S}$ sends the negative half-line to the positive half-line, and both nonidentity powers of $\bar{R}$ send the positive half-line into the negative half-line. The free-product ping-pong argument gives $\mathrm{PSL}_2(\mathbb{Z}) = C_2 * C_3$. Conjugation by F fixes $\bar{S}$ and inverts $\bar{R}$. Consequently

$$\mathrm{PGL}_2(\mathbb{Z}) = V_4 *_{C_2} S_3, \qquad \mathrm{GL}_2(\mathbb{Z}) = \langle S, F\rangle *_{\langle -I, F\rangle} \langle R, F\rangle = D_8 *_{V_4} D_{12},$$

where D_k has order k. The second identity follows also by lifting the first through their common central subgroup $\langle -I\rangle$; the normal form for the amalgam leaves no additional kernel. Adding the normal group of half translations gives the pillowcase amalgam

$$(V_4 \rtimes V_4) *_{V_4 \rtimes C_2} (V_4 \rtimes S_3),$$

whose vertex groups have orders 16 and 24, and whose edge group has order 8. The two vertex groups are $D_8 \times C_2$ and S_4.

A finite subgroup of an amalgam fixes a vertex in its Bass–Serre tree. Indeed the convex hull of a finite orbit is a finite tree; its centre is fixed, and the action preserves the two vertex types, so an edge centre also gives fixed endpoints. Two vertex subgroups are conjugate in the amalgam precisely through the equivalence generated by conjugacy in each vertex group and identification of edge subgroups. To prove the nontrivial direction, a conjugacy makes the subgroup fix both endpoints of the intervening path; it fixes every edge on that path, which decomposes the conjugacy into those successive vertex and edge steps. Thus finite enumeration in the two vertices, followed by these exact fusion steps, is a complete conjugacy calculation in the infinite group. The program finds 10 and 16 actual subgroups in the torus vertices, and 35 and 30 in the pillowcase vertices. Fusion leaves exactly 13 and 30 classes. All subgroup lists are saturated under adjoining an arbitrary vertex-group element, so the enumeration has no search bound that could omit a subgroup.

We next justify the computation of sections. Let $Q = (\frac{1}{12}\mathbb{Z}^2)/\mathbb{Z}^2$. Every isolated fixed point needed for either model lies in Q. For the torus it solves $(A - I)p = 0$; for the pillowcase it solves

$$(A - \epsilon I)p = -t/2 \pmod{\mathbb{Z}^2}, \qquad \epsilon \in \{1, -1\}, \quad t \in (\mathbb{Z}/2)^2.$$

For every positive matrix in the finite vertices, the inverse matrices and the particular solutions of the nonsingular equations have denominators dividing 12. The program certifies this by exact fractions for every matrix and translation, rather than inferring it from a grid count. The singular case for a nonidentity positive element is a nontrivial translation and contributes no solutions to that sign.

For a reflection, the same equations are affine circle equations. The greatest common divisor of the nonzero matrix entries is one or two, so every component meets Q. On each such circle the primitive integral kernel direction, divided by 12, joins consecutive sampled points. In the pillowcase the plus and minus circles meet only at two-torsion points, all of which are included in Q. Quotienting the sampled graph by $p \sim -p$ therefore gives a subdivision of the true fixed circles. Deleting the prescribed punctures gives exactly their connected components. A component adjacent to a deleted point is a line; the others are ovals. All rotations and all punctures are retained as vertices, so no incidence needed for restriction of sections is lost.

B.3 and B.4 identify these fixed components with section classes. For a partial section with a rotation use its fixed point; for a reflection use its fixed line or oval. Inclusion of image subgroups and containment of the fixed point in the fixed component determine exactly when one section extends another. Applying each affine element gives the exact action on this finite section poset. This computes Φ, its orbits and its stabilizers without replacing pointwise section data by conjugacy of image subgroups in the quotient. Centralizers are 1 for isolated points and lines and $\mathbb{Z}$ for ovals, by B.3–B.4.

At a torus fixed point the local positive derivative is A. At a nonbranch pillowcase fixed point it is ϵA, with the unique sign specified by $Ap + t/2 = \epsilon p$. For these matrices the traces $-2, -1, 0, 1$ give orders $2, 3, 4, 6$; the half-turn has multiplier $1/2$, and otherwise the sign of $\det(e_1, Ae_1)$ selects $+1/n$ or $-1/n$. Reflections have multiplier $1/2$. At a removed pillowcase branch point the quotient coordinate squares a torus coordinate, so the positive rotation angle doubles. The program records these puncture data separately. It checks the order and power identities for every element of every section and checks that every section containing a rotation has exactly one maximal overgroup. The reflection classes with several maximal extensions are retained explicitly.

Finally, C.5–C.6 show that the chosen finite actions compute the action attached to each mapping class subgroup up to conjugacy, so these fixed-component computations depend on the listed subgroup class, not on an unrecorded realization choice. □

In the tables f is the number of full splitting classes, p the number of nontrivial partial splitting classes with image subgroup retained, $\phi = |\Phi|$, and $o = |\Phi/\Gamma|$. Juxtaposition in the generator column means composition of the matrices or affine maps defined above. For $(0, 4)$ the letters S, R, F mean their pillowcase classes. The complete output files are `type_1_1_finite_subgroups.csv`, `type_0_4_finite_subgroups.csv`, and the corresponding `partial_sections.json` files, produced by `catalogue_low_complexity.py`.

Type $(1, 1)$

Class	Group	Generators	$\lvert\Gamma^+\rvert$	f	p	ϕ	o
T01	1	1	1	1	0	0	0
T02	C_2	SF	1	2	2	2	2
T03	C_2	SSF	1	1	1	1	1
T04	C_2	SS	2	3	3	3	3
T05	C_3	RR	3	2	2	2	2
T06	C_2^2	SS, SF	2	3	10	3	3
T07	C_2^2	SS, SSF	2	1	6	3	2
T08	C_4	S	4	1	4	3	2
T09	S_3	RR, RRF	3	2	7	2	2
T10	S_3	RR, RF	3	0	5	5	2
T11	C_6	R	6	0	5	5	2
T12	D_8	S, SF	4	1	15	3	2
T13	D_{12}	R, RRF	6	0	16	5	2

Type $(0, 4)$

Class	Group	Generators	$\lvert\Gamma^+\rvert$	f	p	ϕ	o
S01	1	1	1	1	0	0	0
S02	C_2	SF	1	4	4	4	4
S03	C_2	SFV	1	1	1	1	1
S04	C_2	$SFUV$	1	0	0	0	0
S05	C_2	F	1	2	2	2	2
S06	C_2	V	2	2	2	2	2
S07	C_2	S	2	0	0	0	0
S08	C_3	RR	3	1	1	1	1

Class	Group	Generators	$\lvert\Gamma^+\rvert$	f	p	ϕ	o
S09	C_2^2	V, SF	2	2	9	4	3
S10	C_2^2	V, SFU	2	0	3	3	2
S11	C_2^2	UV, SF	2	0	6	6	3
S12	C_2^2	UV, SFV	2	2	6	2	2
S13	C_2^2	UV, F	2	2	8	2	2
S14	C_2^2	SF, F	2	0	6	6	3
S15	C_2^2	$SFUV, F$	2	0	2	2	1
S16	C_4	FU	2	0	2	2	1
S17	C_2^2	V, U	4	0	6	6	3
S18	C_2^2	UV, S	4	0	2	2	1
S19	C_4	SU	4	2	4	2	2
S20	S_3	RR, FR	3	1	8	4	2
S21	C_2^3	V, U, SF	4	0	18	6	3
S22	C_2^3	UV, SF, F	4	0	12	6	2
S23	$C_4 \times C_2$	FU, SU	4	0	8	6	2
S24	D_8	FU, V	4	0	12	6	2
S25	D_8	SU, SFV	4	2	16	2	2
S26	D_8	FU, SFV	4	0	6	2	1
S27	D_8	SU, V	8	0	8	6	2
S28	A_4	RR, RUR	12	0	10	10	2
S29	$D_8 \times C_2$	FU, SU, V	8	0	28	6	2
S30	S_4	FUR, URF	12	0	32	10	2

B.26. [Proved] Complete executed spherical catalogues through six punctures. For $\nu = 3, 4, 5, 6$, the extended mapping class group of the sphere with ν unlabelled punctures has respectively

$$10, \quad 30, \quad 20, \quad 57$$

conjugacy classes of finite subgroups, including the trivial subgroup. The exact finite parameterization below gives every class once. The executed files `type_0_ν_orbifold_catalogue.csv` and `type_0_ν_orbifold_sections.json`, for each of these four values of ν, give all partial-section classes, all their inclusion relations, Φ, its group action and stabilizers, the centralizers, the multiplier on each section element, and the puncture stabilizers. Thus these files settle the whole computation requested in B(f) for the hyperbolic spherical types with $2g + \nu \leq 6$.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

Proof. Extend an action over the punctures. By B.1 and the sphere theorem of [39, Theorem 1.2], a finite topological action on S^2 is conjugate to an orthogonal action. Write $Q = S^2/G$ with its local orbifold groups. A point has cyclic positive stabilizer, or a dihedral stabilizer with mirrors; the mirror-only stabilizer has order two. The covering formula is

$$\chi_{\mathrm{orb}}(Q) = \chi(|Q|) - \sum_{j=1}^{r}(1 - 1/m_j) - \frac{1}{2}\sum_{j=1}^{k}(1 - 1/n_j) = \frac{2}{N}, \qquad N = |G|. \tag{B.26.1}$$

Here the m_j are interior cone orders and the n_j are corner orders. One proves this formula directly by a triangulation whose vertices include all cone and corner points: count each cell in the covering with weight equal to the reciprocal of its stabilizer order. Mirrors are boundary strata; they introduce no additional term after the boundary cells and their endpoints are counted together.

Since (B.26.1) is positive, the underlying compact surface is S^2, $\mathbb{RP}^2$, or a disc. The sphere has at most three cones. A single cone is impossible, since its local generator would be trivial in the orbifold presentation. Two cones have equal orders, since the two local generators are inverses and both local cyclic groups inject. For three cones the inequality $1/m_1 + 1/m_2 + 1/m_3 > 1$ gives precisely $(2,2,n), (2,3,3), (2,3,4), (2,3,5)$. The projective plane has at most one cone. For a disc, positivity gives $2r + k \leq 3$. A single corner without a cone is impossible; two corners without a cone have equal orders, by the same local-injectivity argument. Three corners give the same four triples. One cone and one corner give $2/m + 1/n > 1$, hence $(m,n) = (2,n)$ or $(3,2)$. Consequently the following table is exhaustive. In its notation $S(m_1, \ldots, m_r)$ is a sphere with cones, $P(m)$ is a projective plane with one cone, $D(m_1, \ldots, m_r; n_1, \ldots, n_k)$ is a mirror disc with the specified interior cones and cyclically ordered corners, and an empty list means no such singularities.

Q	N	Q	N
$S()$	1	$P()$	2
$S(n,n)$	n	$P(n)$	$2n$
$S(2,2,n)$	$2n$	$D(;)$	2
$S(2,3,3)$	12	$D(;n,n)$	$2n$
$S(2,3,4)$	24	$D(n;)$	$2n$
$S(2,3,5)$	60	$D(;2,2,n)$	$4n$
		$D(2;n)$	$4n$
		$D(;2,3,3)$	24
		$D(;2,3,4)$	48
		$D(;2,3,5)$	120
		$D(3;2)$	24

(B.26.2)

Every integer n in this table is at least two. For $3 \leq \nu \leq 6$ only $2 \leq n \leq 6$ can occur. Indeed, in every family depending on n, the union of all point orbits of size strictly smaller than n has at most two points; an invariant set of at least three punctures therefore contains an orbit of size at least n. This assertion also follows immediately by dividing N by each local group order in (B.26.2). In the two icosahedral rows every point orbit has at least twelve points. Those rows cannot occur in the present range. These are proved finite bounds, rather than bounds inferred from the output.

We now give the exact decoration rule used in the program. A cone orbit may be removed or retained; put $c_j \in \{0,1\}$ accordingly. A corner orbit similarly has $d_j \in \{0,1\}$. For each open mirror edge let $e_i \geq 0$ be the number of removed orbits on that edge; a cornerless mirror circle is one edge for this purpose. Finally let $a \geq 0$ be the number of regular interior orbits removed. The single equation is

$$\sum_j c_j \frac{N}{m_j} + \sum_j d_j \frac{N}{2n_j} + \sum_i e_i \frac{N}{2} + aN = \nu. \tag{B.26.3}$$

On a sphere, permutations of cones of equal order identify decorations. On a mirror disc, identify them under precisely the rotations and reflections of the cyclic corner list that preserve its order labels. The corresponding edge counts move with their edges. For two corners this rule includes the independent exchanges of the two corners and the two edges. For one or zero corners it introduces no further identification of the single edge count. There is at most one interior cone in every remaining disc or projective-plane row.

For clarity, this rule also proves completeness of the conjugacy classification. A homeomorphism of an interval moves any two unlabelled finite subsets of the same size into one another, and the same assertion holds in the connected regular interior of each quotient surface. Such moves extend to the quotient surface by disjoint small discs and strips. The only remaining data are therefore the selected singular strata and the cyclic mirror-edge counts in (B.26.3),

modulo the explicitly stated symmetries. Conversely every quotient homeomorphism preserves those strata, their orders, and their cyclic incidence. Since S^2 is the universal orbifold cover, a homeomorphism between the decorated quotient orbifolds lifts to a conjugacy between their covering actions; this follows by lifting paths and their local cone or mirror charts. A conjugacy of actions descends to such a quotient homeomorphism. Thus no extra equivalence of decorations is possible. Finally C.5 and C.6 identify these topological action classes with conjugacy classes of finite subgroups in the extended mapping class group of the punctured sphere. The induced action there is faithful: a nonidentity positive periodic map isotopic to the identity would fix every puncture individually, whereas a nonidentity sphere rotation has exactly two fixed points; a negative map cannot be isotopic to the identity. Here $\nu \geq 3$ is essential.

Here are certificates for the finite group calculations. The source `catalogue_spheres.py` constructs the cyclic and dihedral families as regular permutation groups, the tetrahedral and octahedral positive groups as A_4, S_4, their needed extensions by an explicitly given central involution, and the full octahedral group as signed permutations of three axes. All generators, products, orders and signs are recorded as finite permutations. The program verifies each orbifold relation, injection of every local cyclic or dihedral group, and $N\chi_{\mathrm{orb}}(Q) = 2$ using rational arithmetic. These checks prove that the permutation group is the entire orbifold group, as follows. The kernel of the displayed quotient of the orbifold presentation produces a covering with no cone, corner or mirror singularity because all local groups inject. Its orientation character is trivial, and its Euler characteristic is two. It is therefore a closed orientable sphere; its fundamental group is trivial. This proves injectivity of the quotient presentation as well as its already checked surjectivity. In particular these certificates do not merely exhibit a possibly proper finite quotient of an unexamined presentation.

We describe how the section data are recovered without numerical sampling. Cone and corner lifts are cosets of their cyclic and dihedral stabilizers. In a disc presentation choose reflections $r_0, \ldots, r_k$ with

$$(r_j r_{j+1})^{n_j} = 1, \qquad r_k = e r_0 e^{-1}, \qquad x_1 \cdots x_r e = 1. \tag{B.26.4}$$

Indices of the corner orders are shifted here to start at zero. The code stores $r_0, \ldots, r_{k-1}$ and derives the final reflection. An edge lift is a coset of $\langle r_i \rangle$ and meets the adjacent corner cosets; the closing incidence at edge zero is multiplied by e^{-1}. With no corners it instead joins the corresponding two cosets of $\langle r_0 \rangle$ by that same rule. Insert the prescribed removed vertices along each edge, retaining two auxiliary vertices in each complementary segment. These auxiliary vertices prevent a one-edge loop from collapsing when represented as an ordinary finite adjacency graph. This is exactly the lifted mirror locus with a finite subdivision.

For any reflection $r \in G$, keep the vertices whose stabilizers contain r. Before punctures are deleted the program verifies degree two at every such vertex. Its components are consequently the full fixed circles of r. Deleting the selected puncture vertices gives all the fixed lines and ovals, with their exact incidence. For a subgroup whose positive part is nontrivial, the fixed points are precisely the retained cone and corner lifts whose stabilizers contain that subgroup. A negative involution without a mirror has an empty fixed set. Thus B.3 and B.4 supply every section class, with centralizer 1 for a point or a line and $\mathbb{Z}$ for an oval. A section over H extends a section over $K < H$ exactly when its fixed component is contained in the latter fixed component. Hence the same finite incidence calculation gives the entire partial-section poset and all its maximal elements, including the reflection cases with several maximal extensions. Left multiplication on cosets gives the group action and every stabilizer without a choice of orbit representatives.

The multiplier is exact as well. Orient a local lift of each cone, and at a corner use the positive generator $r_j r_{j+1}$ determined by the oriented boundary order. Write this generator as u, of order m. At the point represented by a coset gD, an element h of its positive stabilizer satisfies $g^{-1}hg = u^b$, and

$$\mu(h) = \operatorname{sgn}(g)\frac{b}{m} \pmod{\mathbb{Z}}. \tag{B.26.5}$$

Replacing g by another coset representative leaves this value unchanged: a positive local element commutes with u, while a negative local element both reverses the sign and inverts u. For a reflection the value is $1/2$, and for the identity it is zero. Formula (B.26.5) is the local instance of B.12, and applies to all sections in the files. The code checks its order and every power identity, as well as uniqueness of the maximal extension for every section with nontrivial positive part. All nontrivial subgroups of each finite permutation group are enumerated by saturation under adjoining each group element; this stops only when the subgroup set is closed under every possible enlargement.

The following compact census is part of the executed output; each entry is the number of conjugacy classes of that group order. A dash denotes zero. Complete individual rows, rather than just this census, are in the CSV files specified in the statement.

$\nu\backslash N$	1	2	3	4	5	6	8	10	12	16	20	24	48	total
3	1	3	1	1	–	3	–	–	1	–	–	–	–	10
4	1	6	1	11	–	1	7	–	1	1	–	1	–	30
5	1	4	1	4	1	3	1	3	1	–	1	–	–	20
6	1	7	1	14	1	9	8	1	9	1	–	4	1	57

(B.26.6)

As an independent check, the program compares the multisets of all six columns consisting of group order, positive order, full-section count, nontrivial partial-section count, maximal-section count, and group-orbit count, for $\nu = 3, 4$, against the previous permutation and amalgam calculations in B.23 and B.25. Both multisets agree, with multiplicities. The check and its inputs are attached; the completeness proof above does not depend on that agreement. □

B.27. [Proved] Complete executed punctured-torus catalogues through four punctures. The extended mapping class groups of a torus with $\nu = 1, 2, 3, 4$ unlabelled punctures have, respectively,

$$13, \quad 40, \quad 38, \quad 115$$

conjugacy classes of finite subgroups. The files `type_1_`ν`_affine_catalogue.csv`, together with their `affine_sections.json` companions and `torus_affine_normal_forms.json`, list every class once. They specify each group by its complete finite set of rational affine transformations, its punctures, every full and nontrivial partial section class, all extension incidences, Φ and its group orbits and stabilizers, the centralizers, and μ on every section and puncture-stabilizer element. Thus the torus portion of B(f) is complete for every hyperbolic type in the requested range.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; Part B(f) of the task; no further manuscript assertion.

Proof. We give the finite classification underlying the program `catalogue_tori.py`; this also specifies the tables without relying on abstract finite-group names. By B.14 extend over the punctures and put the action in affine form on $V/\Lambda = \mathbb{R}^2/\mathbb{Z}^2$. Write

$$1 \longrightarrow T \longrightarrow G \longrightarrow H \longrightarrow 1, \qquad T \subset V/\Lambda, \quad H \subset \mathrm{GL}_2(\mathbb{Z}). \tag{B.27.1}$$

Here T consists exactly of the translations in G. It acts freely on V/Λ, so each T-orbit on the puncture set has $|T|$ elements. Consequently $|T| \mid \nu$ and in particular $|T| \leq 4$. The complete list of subgroups of V/Λ of order at most four consists of the trivial subgroup, the three cyclic subgroups of order two, the four cyclic subgroups of order three, the six cyclic subgroups of order four, and $(\frac{1}{2}\Lambda)/\Lambda$. Indeed a group of order at most four is cyclic or C_2^2; a cyclic subgroup is generated by a point of that exact order, and division by the number of generators gives precisely these counts. All fifteen groups lie in $(\frac{1}{12}\Lambda)/\Lambda$. The thirteen possibilities for H, up to integral conjugacy, are exactly those proved in B.25. For each we retain only the H-invariant T.

We now enumerate the affine extensions, including the actions with no common fixed point. If H contains a nonidentity positive element, choose a generator r of its cyclic positive subgroup. Every affine map with linear part r has a fixed point, since $I-r$ is invertible on V and is therefore onto on V/Λ. Translation of that point to the origin makes this generator $(r,0)$. A positive cyclic H then gives exactly $G=\langle T,(r,0)\rangle$. For a dihedral H choose a negative generator f and write its lift as (f,b). The two relations in H give exactly

$$(I-r)b\in T,\qquad (I+f)b\in T. \tag{B.27.2}$$

Conversely these equations make the subgroup generated by $T,(r,0),(f,b)$ a finite group with kernel T and quotient H: every word reduces to a translation followed by r^j or r^jf, and its linear part distinguishes these cosets. Values of b modulo T are enough. The finite set $(I-r)^{-1}T$ therefore supplies every candidate, with no bound on words or finite quotients involved. Translations of the origin preserving this normal form are exactly a with $(I-r)a\in T$; they change b by $(I-f)a$ modulo T.

The remaining nontrivial linear groups have $H=\langle f\rangle$ of order two. Put $W=(V/\Lambda)/T$. Their extensions are represented by $b\in W$ with $(I+f)b=0$, modulo $(I-f)W$. Every such class has a representative in $W[2]$. To verify this assertion directly, choose a lift $v\in V$ of b for the enlarged lattice $\Lambda_T=\{v\in V : v \bmod \Lambda\in T\}$. Then $(I+f)v\in\Lambda_T$. Write $v=v_+ + v_-$ in the real $+1$ and -1 eigenspaces of f. Subtracting $(I-f)(v_-/2)$ removes v_-, and the remaining v_+ satisfies $2v_+\in\Lambda_T$. Thus the program uses precisely

$$2b\in T,\qquad (I+f)b\in T. \tag{B.27.3}$$

For two candidates b,b' the translation-conjugacy test is also exact. Let w be a primitive integral row vector satisfying $wf=w$. The image of $I-f$ on V/Λ is the connected circle $\ker w$: its real image is that eigenspace, and multiplication by any nonzero integer is onto on a circle. Hence the two candidates are equivalent precisely when $w(b-b')\in w(T)$ in $\mathbb{R}/\mathbb{Z}$. This includes both reflections and fixed-point-free glide reflections.

Integral changes of basis are handled without truncating an infinite matrix group. For $H=1$ or $\{\pm I\}$, use the generators S,R,F of B.25 on the complete fifteen-element list of possible T's. Their orbit calculation is the full $\mathrm{GL}_2(\mathbb{Z})$-conjugacy calculation. For a fixed T, a spanning tree of this finite orbit and Schreier's formula give generators for its entire integral stabilizer. Explicitly, with a chosen representative B_U carrying T to U, the generator associated to a matrix $A\in\{S,R,F,S^{-1},R^{-1}\}$ and the edge $U\mapsto AU$ is

$$B_{AU}^{-1}AB_U. \tag{B.27.4}$$

For every other H its integral normalizer is finite. Here is an exhaustiveness check for the small matrix enumeration in the code. For a reflection conjugate to $\mathrm{diag}(-1,1)$, a commuting integral unimodular matrix is diagonal with entries ±1. For a reflection conjugate to $\left(\begin{smallmatrix}0&-1\\-1&0\end{smallmatrix}\right)$, the equation is $A=\left(\begin{smallmatrix}a&b\\b&a\end{smallmatrix}\right)$, with $(a-b)(a+b)=\pm1$, giving four solutions. A normalizer of the four-group generated by this reflection and $-I$ may also exchange the two eigenspaces; the same calculation gives eight solutions. A group with a rotation of order four has its normalizer in the eight signed permutation matrices, since a commuting matrix has determinant $a^2+b^2=1$. A group with rotation of order three or six has its normalizer in the twelve matrices for the hexagonal lattice: the commuting equation gives the norm $a^2+ab+b^2=1$, and one reflection supplies the other coset. Testing these finite sets against H gives the full normalizer. Every resulting matrix has entries of absolute value at most one, so the code's box of entries from -2 to 2 contains every solution and introduces none outside the tested normalizer equations. Applying these matrices and the origin changes above fuses all and only affine-conjugate candidates. The result is 93 affine action classes with $|T|\le 4$ before choosing punctures; some admit no invariant set of a given required size.

We explain why decorating these affine actions gives topological conjugacy, including all permutations of the singular strata. B.14 proves that a topological conjugacy between finite affine actions can be replaced by an affine conjugacy. The replacement also induces the same permutation on the strata used here. In detail write a lift of the conjugacy as $h(x) = Ax + \delta(x)$ with δ periodic, and put $\bar{\delta} = \int_{V/\Lambda} \delta$. The equivariance equations and invariance of Haar measure show that $x \mapsto Ax + \bar{\delta}$ has the same equivariance. The straight interpolation of these two lifts is an equivariant homotopy. An isolated fixed point of a nontrivial positive subgroup stays constant under this homotopy. Each reflection circle stays in the same fixed-set component. The degree on every such circle is fixed by the homology map A; these circles are essential, since they lift to parallel lines. Together with the images of the isolated vertices this determines the permutation of the intervening open mirror intervals. Thus the affine normalizer gives exactly the possible stratum permutations.

To compute them, let $\mathcal{V}$ be the union of the fixed points of all positive nontranslation elements. It is a finite set. The fixed circles of the reflections, cut at $\mathcal{V}$, are open mirror intervals or uncut mirror circles. Take the G-orbits of the vertices and of these mirror components; call their numbers v and e. There is one regular interior stratum in the quotient, since its interior after removal of finitely many cone points is connected. A vertex orbit may be removed once, a mirror-component orbit may contain any nonnegative number of puncture orbits, and the regular interior may contain any nonnegative number. Thus the decoration is

$$(d_1, \dots, d_v; c_1, \dots, c_e; a), \qquad d_i \in \{0, 1\},\ c_j, a \geq 0,$$

subject to the exact equation

$$\sum_i d_i |V_i| + \sum_j c_j \frac{|G|}{2} + a|G| = \nu. \tag{B.27.5}$$

Divide these finitely many decorations by the stratum permutations of the affine normalizer computed above. Counts are sufficient on an open mirror interval or circle: an increasing interval map, or a circle map of the prescribed degree, moves any two unlabelled finite sets of the same size into each other. The moves extend through small strips in the quotient. In the regular interior use disjoint moving discs; its configuration space of finitely many unlabelled points is connected. Such quotient isotopies lift through the finite orbifold covering. They preserve all cone and corner charts and therefore give equivariant homeomorphisms upstairs. Conversely a conjugacy preserves the displayed orbit sizes and the incidence of these strata. This proves completeness and injectivity of the decorated classification.

All computations use exact integers on $Q = (\frac{1}{144}\mathbb{Z}^2)/\mathbb{Z}^2$. We record why this finite mesh loses no component or conjugacy. Coordinates of T have denominator dividing its order $k \leq 4$. For the rotations of orders $2, 3, 4, 6$, the inverses of $I - r$ have denominators $2, 3, 2, 1$, respectively. Thus all solutions needed for (B.27.2), all normalized origin changes, and all isolated fixed points have denominators dividing $2k$ or $3k$. The representatives (B.27.3) have denominator dividing $2k$. For a reflection normalizer, the origin equation is a rank-one integral equation with nonzero Smith invariant 1 or 2; every connected solution component has a representative with denominator dividing $4k$. The integer 144 is divisible by each of these integers for $1 \leq k \leq 4$. The code separately checks the inverse matrix and particular-solution denominators for every positive affine element. On each reflection circle, moving by its primitive integral direction divided by 144 connects consecutive sampled points. Every isolated intersection and every selected puncture is retained as a vertex. This graph is a subdivision of the actual fixed-circle arrangement, so its components after removing punctures are exactly the ovals and lines of the action. The mesh has enough unused representatives on each eligible stratum to place the at most four punctures; the executed program explicitly verifies the prescribed orbit counts and total cardinality for every decoration.

For each unremoved vertex p, enumerate all subgroups of its finite cyclic or dihedral stabilizer that contain a positive nonidentity element. Each gives the section class with fixed point p. For each reflection enumerate the components of its fixed circles after deleting the punctures. B.3–B.4 identify exactly these data with all nontrivial partial-section classes, and identify their centralizers as 1 at points and lines and $\mathbb{Z}$ at ovals. Containment of image subgroups and reverse containment of their fixed components give all extension relations. Applying every affine matrix computes the group action on this poset and hence Φ, its orbit decomposition and every stabilizer. At a positive fixed point the local matrix is its actual linear part. Its trace $-2, -1, 0, 1$ gives order $2, 3, 4, 6$, and the sign of $\det(e_1, Ae_1)$ gives the sign of the angle except for a half-turn. Reflections have multiplier $1/2$. The program checks the order and power identities on every element of every section. It retains rather than deletes the reflection section classes with several maximal extensions.

Finally a nonidentity finite affine map acting trivially in the punctured-torus mapping class group would have trivial homology matrix and hence be a nonidentity translation. It would also fix each puncture, contradicting freeness of translations. Thus each action gives a faithful finite mapping class subgroup. The realization and uniqueness statements C.5–C.6 identify their conjugacy classes with the topological action classes just enumerated. The computation returns the four counts in the statement. For $\nu = 1$, the multiset of group orders, positive orders, full splitting counts and partial splitting counts agrees, class for class, with the independent amalgam calculation of B.25. A first complete run with denominator 72 and the final run with denominator 144 also give the same four counts; the proof of completeness uses the latter mesh. □

B.28. [Proved] Complete executed extended genus-two catalogue. The extended mapping class group of the closed orientable genus-two surface has exactly 92 conjugacy classes of finite subgroups. Of these, 57 contain the hyperelliptic involution and 35 do not; 21 classes preserve orientation. The files `type_2_0_extended_catalogue.csv` and `type_2_0_extended_sections.json` specify every group by an exact finite permutation representation and give every full and partial splitting, its multiplier and centralizer, all extension incidences, Φ, and its group orbits and stabilizers.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

Proof. Let ι be the hyperelliptic mapping class. The genus-two Birman–Hilden homomorphism, with orientation reversal allowed, is

$$1 \longrightarrow \langle \iota \rangle \longrightarrow \mathcal{T}_{2,0} \longrightarrow \mathcal{T}_{0,6} \longrightarrow 1. \tag{B.28.1}$$

We recall its geometric content, to specify every comparison used here. The involution is the deck map of the double cover of S^2 branched at six points. The covering is determined by the homomorphism from the fundamental group of the six-times-punctured sphere to C_2 that sends each peripheral generator to the nonidentity element; the product relation is respected because six is even. Every homeomorphism preserving the branch set preserves this homomorphism, and hence has exactly two lifts, differing by ι. The genus-two hyperelliptic mapping class is central: the usual chain of five nonseparating curves is invariant under the involution, their Dehn twists generate the oriented genus-two mapping class group, and a reflection of the standard branched-cover model commutes with ι and supplies the negative coset. The Birman–Hilden isotopy property identifies the kernel in (B.28.1) with these two deck classes; it also makes the lifting map onto. This is the hyperelliptic Birman–Hilden description used in B.25, in genus two instead of the four-point pillowcase; see [20, §9.4.2, p. 257, with the proof in §9.4.3].

For a finite $K \subset \mathcal{T}_{2,0}$ put $E = K\langle \iota \rangle$ and let H be its image in $\mathcal{T}_{0,6}$. The group E is exactly the full inverse image of H in (B.28.1), and has order $2|H|$. Thus B.26 supplies its complete list

of 57 conjugacy classes. A subgroup K with this fixed E either equals E, or is a complement to $\langle\iota\rangle$. Every complement is normal in E and is uniquely the kernel of a homomorphism

$$\chi : E \longrightarrow \mathbf{F}_2, \qquad \chi(\iota) = 1. \tag{B.28.2}$$

Indeed $E = K \times \langle\iota\rangle$ for a complement, giving this homomorphism; conversely the kernel of (B.28.2) has index two and maps isomorphically onto H. Because ι is central in the whole mapping class group, conjugate K's have conjugate associated E's. For a fixed E, their further fusion is therefore exactly the action of its normalizer on the characters (B.28.2).

Here are explicit presentations of all the finite groups E. They are also used to calculate the sections, so the statement is independent of the names assigned to abstract finite groups. Let $Q = S^2/H$ be one of the spherical orbifolds in B.26, with its six-branch-point decoration. Write $z = \iota$, $z^2 = 1$, and require z to commute with all generators. An interior cone of order m_j has a generator x_j with

$$x_j^{m_j} = z^{d_j}, \tag{B.28.3}$$

where $d_j = 1$ exactly when its orbit is a branch orbit. Let a be the number of regular interior branch orbits. For a spherical underlying quotient the remaining relation is

$$\prod_j x_j = z^a. \tag{B.28.4}$$

For a projective-plane quotient, with crosscap generator u, it is $u^2 \prod_j x_j = z^a$.

For a mirror disc, expand the cyclic corner list by inserting each branch orbit on an open mirror edge as an extra entry $(m, d) = (1, 1)$. An original corner contributes (m, d) with its original order m and its branch flag d. The distinction is retained as a color. Let the resulting cyclic list be $(m_j, d_j)_{j=0}^{L-1}$. There are reflections $R_0, \ldots, R_{L-1}$ and a boundary element e, with

$$\begin{aligned} R_j^2 &= 1, \\ \prod_i x_i\, e &= z^a, \\ (R_j R_{j+1})^{m_j} &= z^{d_j}, \\ R_L &= e R_0 e^{-1}. \end{aligned} \tag{B.28.5}$$

When $L = 0$ there is one reflection R_0 and the last relation is $eR_0e^{-1} = R_0$. A newly inserted mirror branch entry says exactly $R_{j+1} = R_j z$.

These are presentations, rather than only relations satisfied by some finite images. To verify this, use the branched cover described above. Its full group of lifts of H is a group of order $2|H|$. A small positive circle about a branch point has deck monodromy z; about an unbranched point it has trivial deck monodromy. This gives (B.28.3). The product of the boundary loops gives (B.28.4), or its crosscap or disc version. At a branch point on a mirror, the local map is $w \mapsto w^2$: the two reflections on the adjacent lifted half-branches differ by $w \mapsto -w$. This gives precisely the inserted entry in (B.28.5). At a corner the positive local cyclic generator has its order doubled exactly when it is branched, giving the other entries. These local monodromies and z generate the full lift group. Conversely quotienting the displayed presentation by $z = 1$ gives the original spherical orbifold presentation, which is H by the local-injectivity and Euler-characteristic certificate in B.26. Since z is central of order at most two, the presented group has at most $2|H|$ elements. Its surjection onto the geometric lift group proves equality and the presentation assertion.

We determine the entire normalizer action needed in (B.28.2), not just conjugation inside the finite group. A normalizing mapping class descends through (B.28.1) and through the H-action to an orbifold homeomorphism of Q preserving its branch decoration. Conversely any such

orbifold homeomorphism lifts to S^2, because S^2 is the universal orbifold cover of Q, and then to the double branched cover, because the six peripheral mod-two monodromies are preserved. Thus its lifts give the whole normalizer action, with inner automorphisms and the deck involution as the harmless lifting ambiguities. Both act trivially on characters to $\mathbf{F}_2$.

On a sphere, an orbifold homeomorphism sends each local cone generator to a conjugate of a local generator with the same original order and branch flag, or to its inverse when it reverses orientation. Characters ignore conjugation and inversion. All permutations among cones of the same pair (m_j, d_j) are realized by homeomorphisms of the sphere. Point-pushing and pure braids conjugate the local generators and therefore act trivially on these characters. The ordinary branch generators already equal the central element z. Consequently the exact action on (B.28.2) is the permutation action on these colored cone coordinates. The character equations are

$$m_j\chi(x_j) = d_j, \qquad \sum_j \chi(x_j) = a \pmod 2. \tag{B.28.6}$$

This includes, for example, the two elliptic complements in a four-group lying over a sphere involution with no fixed branch points: exchanging the two cone points exchanges those complements. They must be fused, and the program does so.

On a mirror disc there is at most one interior cone. The normalizer preserves the cyclic order on its boundary, up to reversal. It therefore induces exactly the rotations and reversals of the expanded colored corner list that preserve its colors. Each such symmetry is realized by a disc homeomorphism preserving the interior decoration. To see its action explicitly, extend the reflection sequence by $R_{j+L} = eR_je^{-1}$. A cyclic shift by s sends R_j to R_{j+s} and preserves e; a reversal sends R_j to R_{s-j} and e to e^{-1}. The interior cone generator is sent to a conjugate of itself or its inverse. Modulo two these formulas are literal permutations of the reflection coordinates and leave the interior coordinate fixed. They respect every relation in (B.28.5). Every remaining disc mapping class comes from moving the interior marked points or from a boundary twist; its effect on local generators is conjugation and hence is trivial on the characters. The character equations are therefore

$$\begin{aligned} m_i\chi(x_i) &= d_i, \\ \chi(e) &= a + \sum_i \chi(x_i), \\ m_j(\chi(R_j) + \chi(R_{j+1})) &= d_j \pmod 2. \end{aligned} \tag{B.28.7}$$

The program enumerates these bit vectors and divides by exactly the colored dihedral permutations just proved. For a projective-plane quotient, the six-branch-point equation allows only the three cases $P()$ with $a = 3$, $P(2)$ with $(d, a) = (1, 1)$, and $P(3)$ with $(d, a) = (0, 1)$. Their full lifts are respectively C_4, C_8, C_{12}, with z the involution. None has a character taking z to one. Thus no further complement fusion is needed in these cases.

Executing (B.28.6)–(B.28.7) gives 35 character orbits, in addition to the 57 full groups. The following order distribution is a compact view of the complete 92-row table:

$\lvert K\rvert$	1	2	3	4	5	6	8	10
#	1	7	1	13	1	11	14	3
$\lvert K\rvert$	12	16	20	24	32	48	96	
#	17	8	1	9	1	4	1	

(B.28.8)

In particular its seven involution classes consist of the hyperelliptic class, the elliptic positive class, and the five negative classes predicted independently by B.16.

We finish by specifying the verification of all section data. For each of the 57 presentations, the program performs an exact Todd–Coxeter enumeration and obtains the regular action on exactly

$2|H|$ cosets. It evaluates every relator in these permutations and checks that z is nonidentity of order two. In the quotient orbifold for the genus-two action, an originally branched cone of order m now has order $2m$, a regular branch orbit contributes a cone of order two, and a branched corner has its order doubled. Every local cyclic or dihedral group injects into the computed finite group. The orbifold Euler characteristic multiplied by $2|H|$ is -2, and the covering subgroup is the kernel of a monodromy homomorphism compatible with orientation. Its torsion-free covering surface is therefore orientable of genus two. The program also checks that z has exactly six section classes.

The cone and corner cosets, together with the lifted mirror coset graph of B.26, give every fixed point and reflection oval. Enumerating subgroups of these local stabilizers and conjugating them by the finite group is enough: every finite subgroup with a nonempty fixed set is contained in such a local stabilizer. The section multiplier is read from the power of the corresponding positive local generator; conjugating by a negative element reverses that angle. For each complement, retain exactly the section classes whose image subgroup lies in its character kernel. This restriction is exact because the lift extension for the complement is the inverse image of that subgroup in the full lift extension. Recompute maximality in the retained poset, intersect the section stabilizers with the complement, and compute its actual orbits. Thus no maximality or orbit count is inferred merely by restricting the full-group summary. All order and power identities for μ and all unique-maximal-extension claims for sections containing a rotation are checked. Finally the 21 positive rows agree with the independent primary-source oriented genus-two catalogue used in B.33 on the complete multiset of five invariants: group order, full sections, nontrivial partial sections, $|\Phi|$, and $|\Phi/K|$. □

B.29. [Proved] Complete executed punctured genus-two catalogues. The extended mapping class groups of types $(2,1)$ and $(2,2)$ have respectively 32 and 121 conjugacy classes of finite subgroups. Their positive subgroups form 10 and 23 of these classes. The files `type_2_`ν`_extended_catalogue.csv` and `type_2_`ν`_extended_sections.json`, for $\nu = 1, 2$, list every class with its exact group, puncture orbits and stabilizers, all section classes and their incidences, Φ, its group action and stabilizers, centralizers, and all section and puncture multipliers.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

Proof. Fill the punctures of a finite action on the genus-two surface. The resulting finite subgroup in the closed mapping class group is faithful: a finite homeomorphism isotopic to the identity on a closed hyperbolic surface is the identity, by B.6. Thus B.28 supplies the complete list of closed actions that must be decorated. For such an action K, keep the notation $E = K\langle\iota\rangle$, $H = E/\langle\iota\rangle$, and $Q = S^2/H$ from that proof.

The strata relevant to punctures are intrinsic to K. Let $\mathcal{V}_K$ be the set of points whose stabilizer in K contains a positive nonidentity element. Take its K-orbits. Cut the fixed circles of the reflections in K at $\mathcal{V}_K$ and take the K-orbits of the resulting open intervals or uncut circles. These are the mirror strata. There is one connected regular interior in the quotient S_2/K. A vertex orbit is selected zero or one time; a mirror stratum receives any nonnegative number of puncture orbits; so does the regular interior. Their weights are, respectively,

$$|Kp|, \qquad |K|/2, \qquad |K|. \tag{B.29.1}$$

The weighted sum must equal ν. Points fixed by ι but having no positive stabilizer in K are ordinary points of these K-strata. They introduce no extra puncture label. In particular punctures need not be preserved by ι when K is a complement in E.

We prove the normalizer permutations used to identify these decorations. Every mapping class normalizing K normalizes E, since ι is central in $\mathcal{T}_{2,0}$. At the homeomorphism level it may be represented by a normalizer of the chosen finite action E without changing its action on the intrinsic K-strata. Indeed compare the two commuting involutions ι and $h\iota h^{-1}$ after conjugating K by h. The finite actions of the abstract extension $K\langle\iota\rangle$ have the same mapping classes. Uniqueness of realization in C.6 gives a conjugating homeomorphism isotopic to the identity and commuting with K. C.7 gives an isotopy through such commuting homeomorphisms. It fixes the isolated K-fixed points individually and preserves every reflection component. Composing with it makes h normalize the chosen E and leaves the required stratum permutation unchanged. Thus it suffices, and is necessary, to compute the action of the normalizer of E that preserves K.

The full-group case $K = E$ is direct. A cone in Q has the color consisting of its original order and its branch flag. An ordinary branch orbit has its own color $(1, 1)$. The normalizer permits all permutations of identical interior cone colors and exactly the dihedral symmetries of the expanded colored boundary list of B.28. Each is realized by a quotient orbifold homeomorphism and hence lifts. Pure interior braids conjugate local generators, so they leave the corresponding E-orbits unchanged. These observations prove that this finite permutation group is the whole normalizer action on the strata.

Suppose now $K = \ker\chi$ with $\chi(z) = 1$. For a local stabilizer $L \subset E$, its point orbit is E/L. It is a single K-orbit when $\chi(L) = \mathbf{F}_2$. When $\chi(L) = 0$, it consists of two K-orbits, labelled

$$b = \chi(g), \qquad gL \in E/L. \tag{B.29.2}$$

This is well defined precisely in the latter case. It gives the exact orbit comparison, including the ambiguity in choosing a lift of a quotient homeomorphism: composing with z simultaneously changes every available sheet label b to $b + 1$. The code includes this deck permutation.

An interior cone with generator x contributes an intrinsic K-vertex only when $K \cap \langle x\rangle$ is nontrivial. For cone permutations, retain exactly the equal-color permutations that also preserve the character values on their local generators. On a sphere quotient, every split cone pair has an additional independent interchange. Here is the explicit lift proving that assertion. Choose a branch local generator y with $\chi(y) = 1$; one exists because the branch decoration is nonempty. An ordinary branch generator is z. A branched old cone has $y^m = z$ with m odd whenever a character with $\chi(z) = 1$ exists, and therefore also has $\chi(y) = 1$. For a split cone generator x, we have $\chi(x) = 0$. A pure braid about these two quotient points acts by

$$x \longmapsto (xy)x(xy)^{-1}, \qquad y \longmapsto xyx^{-1}. \tag{B.29.3}$$

The other local generators are unchanged. The character is preserved. The conjugator for x has character one, so (B.29.2) interchanges its two K-orbits; the conjugator for y has character zero. Thus it leaves the other split pairs unchanged. This proves the claimed independent interchanges. Together with cone permutations they generate the whole stratum action: pure braids generate the remaining mapping classes of a punctured sphere, and their action on the two-element fibers can only be these interchanges.

For a disc quotient there is at most one old interior cone. An ordinary interior branch point permits exactly the same interchange (B.29.3) of that split cone pair, supported away from the boundary. When there is no such interior branch point, the disc has at most one interior marked cone; its pure braid group introduces no additional permutation. Boundary permutations are computed as follows. Let L be the length of the expanded corner list, write $R_{j+L} = eR_je^{-1}$, and put $\epsilon = \chi(e)$. A colored cyclic shift sends a corner index j to $s + j$ and an edge index i to $s + i$. A reversal sends them to $s - j - 1$ and $s - i$, respectively. For either image write the resulting integer as $qL + t$, $0 \le t < L$. Its local generator is conjugated by e^q; hence the sheet label changes by

$$b \longmapsto b + q\epsilon. \tag{B.29.4}$$

Retain precisely the colored symmetries that preserve the reflection character values, and include the global deck interchange. These formulas give both vertex and mirror-component permutations. For a cornerless boundary the same rule is its circular covering with monodromy $\chi(e)$; the lifted mirror graph records whether the two local sheet labels belong to one or two components.

There are no omitted boundary permutations. An orbifold homeomorphism of the disc preserves its cyclic boundary order up to reversal. After its colored dihedral permutation has been removed, its remaining mapping class is an interior braid or a boundary twist. An interior braid has the local conjugators already described and fixes the boundary. A boundary twist has the conjugator e, already represented by (B.29.4) and the deck permutation. Inner automorphisms coming from K leave K-orbits unchanged. The projective-plane cases in B.28 have no complements; their full-group decorations require only permutations of identical interior cone colors. Thus every case has an explicit complete finite normalizer permutation group.

We spell out how this calculation handles smooth points of the K-mirror locus that were corners for E. The code starts with the exact finite mirror coset graph of the full action E. It finds the vertices with positive stabilizer in K, removes only these vertices to form the open mirror strata, and computes the fixed circles of each actual reflection in K. Thus an E-corner that has only a reflection stabilizer in K lies inside a single K-mirror stratum. The sheet and boundary formulas above are applied to an ordinary edge sample in each such component. They determine its whole component by connectedness. This is also a direct check that a puncture can move through a noncritical Weierstrass point whenever the K-stratum allows it.

Enumerate all vectors with the weights (B.29.1) summing to $\nu = 1$ or 2, and divide by the proved finite permutation group. These are precisely the topological action classes with punctures. Indeed unlabelled finite subsets of an interval or circle with the same cardinality are related by a homeomorphism of prescribed degree, and the regular interior configuration space is connected. Extend such moves by small discs and strips in the quotient and lift the isotopies through the finite orbifold cover. Conversely an equivariant homeomorphism preserves the intrinsic strata, their weights, and a stratum permutation in the normalizer just computed. C.5–C.6 then identify these action classes with the finite-subgroup conjugacy classes in the punctured mapping class group.

The executed program `catalogue_genus2_punctures.py` returns 32 and 121 classes. It places the punctures on the exact coset graph, with one midpoint inserted in every edge before any vertex is removed. This subdivision guarantees that even two punctures at formerly adjacent samples leave the intervening open interval represented. The program checks that an element stabilizing an edge fixes both its endpoints, so the inserted midpoint has exactly the edge stabilizer and creates no artificial critical point. It records regular puncture orbits by the free K-set K, with trivial stabilizer.

All positive section classes from B.28 are retained exactly when their fixed point has not been removed and their image subgroup lies in K. For every reflection, remove the punctures from its actual fixed-circle graph and recompute the resulting ovals and lines. These operations give all partial-section classes by B.3–B.4. Their inclusion poset, maximal classes, K-action, stabilizers and centralizers are then recomputed. Local multipliers at punctures are retained from the positive local generators or assigned $1/2$ to a reflection. The positive portions contain 10 and 23 classes and agree with the independent primary-source calculation in B.33 on the full multiset of group order, full sections, partial sections, $|\Phi|$, and $|\Phi/K|$. The negative involutions provide another independent check: there are 4 classes with one puncture and 11 with two, exactly the decorations of the five closed real types in B.16. □

2.6 The full tube action and the strip subgroups

B.30. [Proved] Explicit outer words for the antipodally paired tube model. For $g \geq 2$, put $n = 2g$ and label the equatorial holes by $j \in \mathbb{Z}/n$, with j glued by a tube to $j + g$. The following presentation and substitutions compute the entire exterior action requested on scan 467:

$$\pi_g = \langle x_0, \dots, x_{g-1}, p_0, \dots, p_{g-1} \mid R = 1 \rangle, \qquad R = x_0 \cdots x_{g-1} \prod_{j=0}^{g-1} (p_j^{-1} x_j^{-1} p_j).$$

Extend the notation, with all subscripts read modulo n, by

$$x_{j+g} = p_j^{-1} x_j^{-1} p_j, \qquad p_{j+g} = p_j^{-1},$$
$$u_j = (x_0 \cdots x_j)^{-1}, \quad u_{n-1} = 1, \qquad q_j = u_j^{-1} p_j u_{j+g-1}.$$

Let $r = \varepsilon_0 \varepsilon_1$ and $s = \varepsilon_1$ as in B.20. Representatives of their outer automorphisms, and of τ and ρ_k, are specified on the $2g$ displayed generators by

	x_j	p_j
r	x_{j+1}	p_{j+1}
s	$u_{-j}^{-1} x_{-j} u_{-j}$	q_{-j}
τ	$u_{j+g-1}^{-1} x_{j+g}^{-1} u_{j+g-1}$	q_j^{-1}
ρ_k	x_j	$x_j^{\delta_{jk}} p_j$.

The map to $\mathrm{Out}(\pi_g)$ defined by these substitutions is the geometric action of $\mathbb{Z}^g \rtimes (D_{4g} \times C_2)$ of B.20. The finite factor acts by

$$r \rho_j r^{-1} = \rho_{j+1}, \qquad s \rho_j s^{-1} = \rho_{-j}, \qquad \tau \rho_j \tau^{-1} = \rho_j^{-1},$$

with indices in these three formulas taken modulo g.

Manuscript: §33; folios 234–236; scans 463–467.

Proof. We specify the base paths so that the formulas can be used as words, rather than only as unbased pictures. Let N, S be the north and south poles in the sphere with the n discs removed. Draw disjoint meridian arcs $d_j : N \to S$ through the gap immediately following hole j. The loop about that hole, based at N, is $x_j = d_{j-1} d_j^{-1}$. Collapse the maximal-tree edge d_{n-1} and identify N, S by that edge. The remaining arcs become $u_j = (x_0 \cdots x_j)^{-1}$, because the product telescopes to $d_{n-1} d_j^{-1}$. In particular $\prod_{j=0}^{n-1} x_j = 1$.

Let p_j run from N to the northern side of hole j, through the tube to hole $j + g$, and back to N on its northern side. The opposite traversal is $p_{j+g} = p_j^{-1}$. Its boundary identification reads $x_{j+g} = p_j^{-1} x_j^{-1} p_j$. Van Kampen therefore gives exactly the presentation in the claim. The parallel path based at S, on the southern side of the tube, is $q_j = d_j^{-1} p_j d_{j+g-1}$ after inserting the northern paths, which becomes the displayed formula on collapsing the tree edge. These identities can also be taken as a two-vertex cellular description: the sphere faces give the telescoping boundary relation and each tube face gives the indicated boundary conjugation.

Rotation about the polar axis sends hole j to $j + 1$, preserves N, and sends p_j to p_{j+1}. This proves the r row. The half-turn s about the equatorial axis through hole 0 exchanges N, S and sends hole j to $-j$. The southern boundary loop at hole k is $d_k^{-1} x_k d_k$, and the southern tube path is q_k, giving the s row. The antipodal map exchanges N, S, sends j to $j + g$, and reverses orientation. It therefore inverts the southern boundary loop and reverses the corresponding tube path. The boundary-loop formula written with u_{j+g-1} equals that written with u_{j+g}, since $u_{k-1} = x_k u_k$. This proves the τ row. Finally a positive full twist of tube k inserts x_k into the

crossing p_k and leaves all other chosen paths unchanged. This proves the ρ_k row. Each row comes from an actual homeomorphism, so it is an automorphism of the stated group and has the prescribed orientation sign.

For s and τ the chosen path from N to the image of N is d_{n-1}. We consequently get actual automorphisms only after this base-path choice. For example their squares are

$$s^2 = \mathrm{Inn}(x_0), \qquad \tau^2 = \mathrm{Inn}(x_0 \cdots x_{g-1}),$$

whereas $r^{2g} = 1$ as an actual automorphism. These formulas follow by concatenating the base path with its image: $s(d_{n-1}) = d_0^{-1}$ and $\tau(d_{n-1}) = d_{g-1}^{-1}$. Thus the displayed group relation $s^2 = \tau^2 = 1$ holds in the outer group with precisely the stated inner ambiguity. The geometric symmetries satisfy $srs = r^{-1}$ and commute with the antipodal involution on the surface. They preserve and permute the tube collars; a positive symmetry preserves twist sign and the antipodal map reverses it. This proves every relation of the asserted semidirect product in the outer group. Changing the maximal-tree edge conjugates each based comparison by its corresponding base-path word and leaves this exterior action unchanged.

The accompanying programs check all relator images for $2 \le g \le 8$; r, s carry R to a conjugate of R, and τ carries it to a conjugate of R^{-1}. They additionally check all the finite-factor and twist-conjugacy relations in the surface group, with the actual inner conjugators above. For instance $srsr = 1$ and $\tau r \tau r^{-1} = \mathrm{Inn}(x_0 \cdots x_g)$. The word checker uses the symmetrized relator: its largest common piece has length one, so the length-$4g$ presentation satisfies $C'(1/6)$ for $g \ge 2$, and the Dehn reduction is exact. These finite certificates supplement the cellular proof for arbitrary g. □

B.31. [Proved] The eight symmetries of one tube. The stabilizer of tube T_0 in $D_{4g} \times \langle \tau \rangle$ is $\langle r^g, s, \tau \rangle \cong C_2^3$. There are coordinates $(\theta, t) \in (\mathbb{R}/2\pi\mathbb{Z}) \times [-1, 1]$ on that tube in which its three generators act as

$$r^g(\theta, t) = (-\theta, -t), \qquad s(\theta, t) = (\theta + \pi, t), \qquad \tau(\theta, t) = (\theta + \pi, -t).$$

These formulas agree with the outer action in B.30 and with the manuscript's free antipodal extension, after the explicit coordinate change $\theta_{\mathrm{new}} = \theta_{\mathrm{old}} + \pi t/2$.

Manuscript: §33; folios 229–230; scans 453–454; §33; folios 233–236; scans 461–467.

Proof. The stabilizer of the unordered pair $\{0, g\}$ under the polygon's dihedral group is $\{1, r^g, s, r^g s\}$: a stabilizing rotation is either 1 or r^g, and every stabilizing reflection is one of these followed by s. The independent central antipodal factor doubles it. In a symmetric tube, r^g interchanges its ends and reverses its angular direction, while s preserves both ends and rotates the cross section by π. The displayed representatives commute, are involutions, and have eight distinct combinations; the antipodal representative is free. They extend the same boundary maps as the corresponding sphere symmetries and hence give the desired tube extension.

The formula of scan 453, retained through scans 461–462, is $\tau_{\mathrm{old}}(\theta, t) = (\theta + \pi + \pi t, -t)$. Conjugating it by the stated change gives exactly $(\theta + \pi, -t)$; this also transports both boundary identifications. Thus it is an explicit isomorphism of the two actions. The full twist is $(\theta, t) \mapsto (\theta + \pi + \pi t, t)$ in either coordinate. Conjugation by the two positive generators preserves its class, while conjugation by the negative generator inverts its class, agreeing with B.30. All eight maps and their products are checked symbolically in `code/verify_tube_local.py`. □

B.32. [Proved] Relations for arbitrary strips and a uniform obstruction to generation. Choose for each c_j one simple transversal d_j, using a strip in the complement of the other tubes, so that $i(c_i, d_j) = \delta_{ij}$. Let W_j be a regular neighbourhood of $c_j \cup d_j$, a one-holed torus, and put

$A_j = T_{c_j} = \rho_j$, $B_j = T_{d_j}$. Its supported subgroup is the central extension $\langle A_j, B_j\rangle \cong B_3$ of B.21, with quotient $SL_2(\mathbb{Z})$. The exact pairwise relations are

$$[A_i, A_j] = 1, \quad [A_i, B_j] = 1 \ (i \neq j), \qquad A_j B_j A_j = B_j A_j B_j.$$

For distinct i, j, the two supported subgroups commute precisely when $i(d_i, d_j) = 0$. For intersection one, B_i, B_j obey the braid relation and do not commute; for intersection at least two they freely generate a free group of rank two. For every such choice of one strip per tube and every $g \geq 2$, these g supported subgroups, together with H, generate a proper subgroup of $\mathcal{T}_g^+$.

Manuscript: §33; folios 236–237; scans 467–470.

Proof. Each W_j has genus one and one boundary component because the two core curves meet transversely once. The extension-by-identity map and its faithful boundary-twist kernel are the explicit maps of B.21. The disjointness conditions give the first two commutation relations, and the calculation on its two free generators in B.21 proves the braid relation. For any two essential simple curves a, b, their twists commute exactly when they are disjoint: conjugation sends T_b to $T_{T_a(b)}$, and $i(T_a(b), b) = i(a, b)^2$. Hence intersecting curves cannot commute. If $i(a, b) = 1$, their regular neighbourhood is a one-holed torus, so the same braid calculation applies.

For $k = i(a, b) \geq 2$ the freedom assertion has an elementary ping-pong proof. On isotopy classes of essential curves set

$$X_a = \{x : i(x, a) < i(x, b)\}, \qquad X_b = \{x : i(x, b) < i(x, a)\}.$$

Both are nonempty and disjoint. Counting crossings in a twist collar and cancelling only crossings already present outside it gives

$$i(T_a^m(x), b) \geq |m|\, i(x, a)k - i(x, b).$$

For $m \neq 0$ and $x \in X_b$ this is strictly greater than $i(x, a) = i(T_a^m(x), a)$, so $T_a^m(X_b) \subset X_a$. The symmetric inclusion holds for powers of T_b. The ping-pong argument on a reduced alternating word proves that it is nontrivial: apply it to the appropriate one of X_a, X_b, inserting one further nonzero power from the opposite factor when the first and last factors agree. Thus $\langle T_a, T_b\rangle \cong \mathbb{Z} * \mathbb{Z}$. These statements compute the required pairwise relations for arbitrary intersections of the selected strips; no disjoint-strip convention is being used to conclude the general generation assertion.

For that assertion work on $V = H_1(X_g; \mathbb{F}_2)$ with its nonsingular intersection pairing. The $2g$ classes $[c_j], [d_j]$ form a basis: pairing a linear relation first with the $[c_j]$ kills all coefficients of the $[d_j]$, and pairing next with the $[d_j]$ kills the rest. There is a unique quadratic refinement $q : V \to \mathbb{F}_2$ with $q([c_j]) = q([d_j]) = 1$. Explicitly, for this ordered basis (v_i),

$$q\Big(\sum_i a_i v_i\Big) = \sum_i a_i + \sum_{i<j} a_i a_j \langle v_i, v_j\rangle.$$

It obeys $q(v + w) = q(v) + q(w) + \langle v, w\rangle$ by expansion. A twist about a curve of class v acts by $x \mapsto x + \langle x, v\rangle v$; substitution gives

$$q(T_v x) - q(x) = \langle x, v\rangle (q(v) + 1).$$

Every generator A_j, B_j therefore preserves q, without any restriction on the intersections among the d_j.

Join c_1 and c_2 by an embedded arc in their connected complement and form a band sum e. This is a simple curve of nonzero class $[e] = [c_1] + [c_2]$, so it is nonseparating. Since the two cores are disjoint, $q([e]) = 0$. Taking $x = [d_1]$ gives $\langle x, [e]\rangle = 1$, so T_e fails to preserve q. Consequently T_e belongs to $\mathcal{T}_g^+$ and lies outside the group generated by the strip subgroups. This proves

properness for all choices under discussion. The quadratic calculation has been exhaustively checked for every possible matrix of transversal intersections modulo two in genera two, three and four; the general identity is the displayed algebraic proof.

The rank-one comparison is thus the specified central quotient $B_3 \to SL_2(\mathbb{Z})$ and its action on the homology of W_j, together with the above braid and intersection relations. The subgroup they generate fixes the explicit quadratic refinement. This supplies the precise algebraic maps and the explicit obstruction governing the proposed generation in scan 470. □

B.33. [Proved] Executed positive catalogues in genera two and three. For $(g, \nu) = (2,0), (2,1), (2,2), (3,0)$, the numbers of conjugacy classes of finite orientation-preserving subgroups, allowing conjugacy in the full extended mapping class group, are respectively

$$21, \qquad 10, \qquad 23, \qquad 55.$$

The trivial subgroup is included. The files `oriented_higher_genus_catalogue.csv` and `oriented_higher_genus_sections.json` give each class, its actual finite permutation group, every subgroup's number of sections, all nontrivial partial-section classes and their inclusion relations, the G-set Φ, its stabilizers, and every value of μ on every section. Punctures are unlabelled. These assertions concern positive subgroups; they do not enumerate subgroups containing orientation-reversing elements.

Manuscript: §15; folios73–75; scans148–151; §17; folios77–79verso; scans155–160; §18; folios96–101; scans193–203; Part B(f) of the task; positive higher-genus cases.

Proof. We specify the classification input and the computation using it. Karabáš–Nedela–Skyvová [35, Theorem 2.2, Corollary 2.3, Theorem 4.1 and §8] identify the positive topological classes with the orbits of smooth epimorphisms from the Fuchsian group

$$\Delta = \langle x_1, \dots, x_r, a_1, b_1, \dots, a_h, b_h \mid x_i^{m_i} = 1,\ [a_1, b_1] \cdots [a_h, b_h] x_1 \cdots x_r = 1 \rangle$$

under target automorphisms and orientation-preserving orbifold automorphisms. Here smooth means that every image of x_i has order exactly m_i. Their §8 gives a complete genus-two list of 21 classes and a genus-three list of 55 classes in 49 group/signature pairs, including the trivial actions. We opened the accompanying data [34], downloaded the two original JSON files, and retained their attribution and SHA-256 digests. The attached program reads those exact files. It verifies that each generating vector generates the group of the stated order, each elliptic generator has its specified order, every presentation relation holds, and

$$|G| \left(2 - 2h - \sum_i (1 - 1/m_i) \right) = 2 - 2g.$$

The conversion from the source's right-acting permutations to the left-acting permutations in our code is the isomorphism sending each permutation to its inverse. For a handle word the two commutator conventions are compared explicitly; replacing both handle generators by their inverses converts $a^{-1}b^{-1}ab$ to the convention $aba^{-1}b^{-1}$. The files retain the actual generating vector, so this convention does not affect a recorded elliptic rotation.

The local-order checks imply that the kernel of each epimorphism is torsion-free: every finite-order element of a Fuchsian group is conjugate to a power of an elliptic generator. The covering formula then identifies its quotient surface as genus g. Conversely a positive finite action determines this epimorphism by lifting paths in the quotient. The two constructions are inverse modulo the stated source and target automorphisms. Finally C.5–C.6 identify these action classes with conjugacy classes of finite mapping-class subgroups. This supplies the exact reduction from the classification input to the groups asked for here.

We also recompute the equivalence relation whenever $h = 0$, rather than using only the published class count. For an ordered generating vector $(u_1, \dots, u_r)$ with product one, the braid generator acts by

$$\sigma_i : (u_i, u_{i+1}) \longmapsto (u_i u_{i+1} u_i^{-1}, u_i).$$

Use adjacent such generators when $m_i = m_{i+1}$ and all pure braid generators, obtained by moving the jth entry next to the ith, applying σ_i^2, and reversing the moves. Theorem 4.1 of the cited paper proves that these transformations induce the full orientation-preserving orbifold automorphism action after inner automorphisms have been divided out. A generating vector is compared modulo $\mathrm{Aut}(G)$ by its rooted, edge-coloured Cayley graph: breadth-first enumeration with the ordered generators gives a canonical transition table. Equality of tables gives the explicit bijection of vertices preserving every labelled multiplication edge; induction on words proves that it is the unique target-group isomorphism carrying the generating vectors to one another. Thus the canonicalization does not approximate group automorphisms.

To allow conjugacy by negative homeomorphisms, we add the reflection of the marked sphere. Its effect on the tuple is

$$(u_i)_{i=1}^r \longmapsto \left(p_{i-1} u_i^{-1} p_{i-1}^{-1}\right)_{i=1}^r, \qquad p_{i-1} = u_1 \cdots u_{i-1}.$$

The product of the new tuple is $(u_1 \cdots u_r)^{-1} = 1$, as successive cancellation shows. The program computes the orbits both before and after adding this transformation; it recovers the published planar counts and finds no fusion. All nonplanar group/signature pairs in these two genera have only one positive class in the input classification. Reversing orientation preserves the group and signature and hence fixes that class. Consequently the positive counts remain 21 and 55 under conjugacy in the extended group.

For genus two we add all puncture configurations with $\nu \le 2$. An invariant finite puncture set consists of selected elliptic orbits and a number a of ordinary orbits. Write $c_i \in \{0, 1\}$ for selection of the ith elliptic orbit. The exact equation is

$$\sum_i c_i \frac{|G|}{m_i} + a|G| = \nu. \tag{B.33.1}$$

The program enumerates every solution. It carries the bits c_i along under each braid move, exchanging the two bits when the corresponding two elliptic entries are exchanged. It then takes orbits of these decorated generating vectors under the same coloured braids, target automorphisms, and reflection. These are exactly the decorated topological classes: an equivariant homeomorphism descends to the quotient orbifold, and its action on the covering is the compatibility of its induced automorphism with the epimorphism just described. Conversely that compatibility lifts the quotient homeomorphism to an equivariant homeomorphism. Ordinary unlabelled punctures can be moved to any other such configuration through paths in the regular part of the quotient; point-pushing the paths is isotopic to the identity after these additional points are forgotten and so preserves the original covering kernel. Thus a records all their remaining data.

There are just two nonplanar genus-two entries. For the trivial group, there is one configuration for each ν. The other has group C_2 and signature $(1; 2, 2)$. Its two elliptic images are the unique involution, and exchanging their small discs preserves the monodromy after an adjustment of the two handle paths. The four possible pairs of handle images in C_2^2 are connected by point-pushing either branch point around a handle: crossing the branch cut adds its nonzero monodromy to the transverse handle image. These two pushes change either handle coordinate independently. Hence the exchange lifts and only the number of selected branch orbits matters. These elementary moves account for all four vectors in the supplied data and prove the nonplanar decoration step used by the code. Equation (B.33.1) and the decorated orbits give 10 and 23 classes at one and two punctures.

We now derive every section entry from the checked finite groups. A retained elliptic orbit is the left G-set $G/\langle u_i\rangle$. Its point at the coset $v\langle u_i\rangle$ has stabilizer $v\langle u_i\rangle v^{-1}$ and positive generator vu_iv^{-1}. B.3 gives exactly one kernel-conjugacy class of sections for every point fixed by a specified nontrivial subgroup $K \leq G$. Such a subgroup is cyclic because it is contained in the cyclic point stabilizer. Conversely every such containment gives a section. Thus the exact count is

$$\#\operatorname{Sec}(E_K \to K)/\pi = \sum_{i:c_i=0} \#\{v\langle u_i\rangle : K \leq v\langle u_i\rangle v^{-1}\}. \tag{B.33.2}$$

The formula is used for every subgroup, which the program enumerates by saturation under adjoining each remaining group element. It checks that no noncyclic subgroup has a section. The trivial subgroup has one section, and it is not included in the nontrivial partial-section totals.

The maximal partial sections are precisely these individual elliptic points, with G acting by left multiplication on the cosets. A partial section lies below another exactly when they have the same point and its subgroup is contained in the other subgroup. This determines the whole inclusion relation and proves uniqueness of a maximal extension in the positive case. Each nontrivial section has centralizer one by B.3. The total number of nontrivial partial-section classes is independently checked against

$$\sum_{i:c_i=0} \frac{|G|}{m_i}(\tau(m_i) - 1),$$

where $\tau(m)$ is the number of positive divisors of m. Finally, for $h = vu_i^b v^{-1}$ the value of μ at this section is $b/m_i \pmod{\mathbb{Z}}$. This lists every power of every local generator and is exactly the intrinsic multiplier identified in B.12. Puncture stabilizers are the removed cyclic groups or the trivial group for ordinary orbits. All these data are computed with integer permutations and rational fractions. The output has 109 individual action rows and 59 independent planar braid-orbit checks. □

B.34. [Proved] The complete genus-three subcatalogue containing a hyperelliptic involution. There are exactly 79 conjugacy classes of finite subgroups of the extended mapping class group of a closed orientable genus-three surface that contain a hyperelliptic involution. Exactly 18 preserve orientation and 61 contain orientation-reversing elements. The latter 61 classes are listed below and are specified, together with every section and its extension incidences, in `type_3_0_hyperelliptic_mixed_catalogue.csv` and `type_3_0_hyperelliptic_sections.json`.

Manuscript: §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; the genus-three hyperelliptic subcatalogue is an added result, not a further manuscript assertion.

Proof. Fix a hyperelliptic involution z and its branched quotient $p : S_3 \longrightarrow S^2$, with branch set P of size eight. On the unbranched complements this is the connected double cover classified by the homomorphism

$$\pi_1(S^2 \setminus P) \longrightarrow C_2 \quad \text{sending each of the eight meridians to 1.} \tag{B.34.1}$$

Their sum is zero modulo two. Every homeomorphism of the sphere preserving P preserves (B.34.1), and therefore has exactly two lifts. The Birman–Hilden isotopy property gives the exact sequence

$$1 \longrightarrow \langle z\rangle \longrightarrow C_{\mathcal{T}_{3,0}}(z) \longrightarrow \mathcal{T}_{0,8} \longrightarrow 1. \tag{B.34.2}$$

Here the left group is the centralizer of this specified involution; the map is induced by the quotient p. The lifting and isotopy argument is precisely the fully ramified hyperelliptic case of [20,

§9.4]. Allowing orientation reversal adds the negative lifts of the same sphere homeomorphisms, with the same kernel.

The involution z is the unique hyperelliptic involution inside any finite subgroup K containing it. To verify the assertion directly, realize K by isometries and use the resulting invariant complex structure. A hyperelliptic involution acts as -1 on $H^1(S_3;\mathbb{R})$: its invariant subspace is the pullback of $H^1(S^2;\mathbb{R}) = 0$ under its quotient, as follows by averaging cochains, or by the transfer outside the branch points and filling their meridians. If z' were another, the finite-order holomorphic map zz' would act trivially on H^1. A nonidentity such map has isolated fixed points, each of positive local Lefschetz index; a fixed-point-free map has Lefschetz number zero. But its Lefschetz number from homology is $2-6=-4$. Thus $zz'=1$, proving uniqueness. Consequently every element of K centralizes z, and any conjugacy between two such finite groups maps their hyperelliptic involutions to one another. Once z is fixed, (B.34.2) therefore gives a bijection

$$\left\{\begin{array}{c}\text{finite subgroup conjugacy classes}\\ \text{in } \mathcal{T}_{3,0} \text{ containing } z\end{array}\right\} \longleftrightarrow \left\{\begin{array}{c}\text{finite subgroup conjugacy classes}\\ \text{in } \mathcal{T}_{0,8}\end{array}\right\}. \quad \text{(B.34.3)}$$

The inverse takes the full preimage, of order twice the sphere group. The inclusion of every kernel element follows from the assumption $z \in K$. This also proves that no additional fusion occurs when the chosen hyperelliptic involution is forgotten.

For completeness, B.26's decorated spherical-orbifold enumeration applies verbatim to eight marked points: a selected cone orbit has size $|H|/m$, a selected corner orbit has size $|H|/(2m)$, an orbit on an open mirror edge has size $|H|/2$, and a regular orbit has size $|H|$. Their sum is eight. The cyclic and dihedral rotation orders are at most eight, since every non-polar point orbit has at least that many points. The tetrahedral and octahedral cases are included. An icosahedral orbit has at least twelve points, so cannot occur. The spherical classification and the homeomorphism classification of these decorations in B.26 prove exhaustiveness and irredundancy. Executing the integer weight equation gives 79 decorated quotients, 18 spherical and 61 with a negative element.

For each decorated quotient, the full lift is given by the exact central extension presentation (B.28.3)–(B.28.5), now with eight branch points. Its presentation proof uses (B.34.1) in exactly the same way: the displayed group surjects onto the full geometric lift group of order $2|H|$, while its quotient by the central element z is H, so it has at most $2|H|$ elements. The two groups are therefore isomorphic by that surjection. For a projective-plane quotient its generators include both z and the crosscap element; their relation is $u^2\prod x_i = z^a$. In particular no cyclicity assumption from the six-branch-point case is imposed.

The program `catalogue_genus3_hyperelliptic.py` was executed. For every presentation it computes an exact regular permutation representation of degree $2|H|$, verifies every defining relator, checks that z has order two, and checks injectivity on every local cyclic or dihedral stabilizer. The resulting orbifold Euler characteristic is $-4/|K|$, so its oriented surface kernel has genus three. Exactly eight point sections have image $\langle z\rangle$, providing an independent branch-number check. The finite coset incidence graph of B.26 then computes every point and oval fixed component and every partial section. Each point section has trivial centralizer in the surface group, each oval section has infinite-cyclic centralizer, and the rotation multipliers, extensions, maximal sections Φ, group orbits and stabilizers are recorded explicitly in the JSON file. The calculation retains all intersections of reflection ovals at dihedral stabilizers. □

In the following table, b is both the G3H class suffix and the decorated sphere-eight row number in `type_3_0_hyperelliptic_catalogue.csv`; N is the group order; p counts all nontrivial partial splitting classes; and m, o are respectively $|\Phi|$ and $|K\backslash\Phi|$. The CSV gives the cone, corner, mirror and ordinary branch decorations of every row, and the JSON gives its finite group generators and all relators.

b	N	p	m	o
2	4	8	8	4
3	4	10	10	5
4	4	12	8	5
5	4	16	8	6
6	4	20	8	7
7	4	24	8	8
11	8	12	12	3
12	8	14	14	4
13	8	20	12	4
14	8	28	12	5
15	8	14	8	3
16	8	22	8	4
17	8	20	12	4
18	8	26	12	5
19	8	34	12	6
20	8	24	12	5
21	8	32	12	6
22	8	32	12	6
23	8	24	8	4
24	8	32	8	5
25	8	32	8	5
29	12	12	8	2
30	12	14	10	3
31	12	24	8	3
32	12	30	8	3
33	12	42	8	4
34	12	42	8	4
39	16	16	12	2
40	16	28	20	3
41	16	40	20	4
42	16	28	12	3
43	16	18	14	3
44	16	32	12	3
45	16	38	20	4
46	16	52	20	5
47	16	50	20	5
48	16	48	12	4
49	16	48	12	4
50	16	36	12	3
51	16	48	12	4
52	16	44	12	4
56	24	54	20	3
57	24	36	8	2
58	24	28	8	2
59	24	68	20	4
60	24	66	20	4
61	24	66	8	3
62	24	72	8	3
65	28	45	10	3
66	32	60	28	3

b	N	p	m	o
67	32	44	12	2
68	32	36	12	2
69	32	82	28	4
70	32	84	28	4
71	32	76	12	3
72	32	72	12	3
74	48	54	20	2
75	48	120	20	3
76	48	120	20	3
78	64	136	28	3
79	96	210	44	3

The mixed-class order distribution is

$\lvert K\rvert$	4	8	12	16	24	28	32	48	64	96
#	6	15	6	14	7	1	7	3	1	1.

(B.34.4)

The scope of (B.34.3) is the condition that the subgroup itself contain a hyperelliptic involution. A finite subgroup that only centralizes an involution outside itself is represented by a complement in a lift group; its conjugacy after forgetting the external involution requires an additional calculation. No such complement is counted among these 79 classes.

B.36. [Proved] Complete exceptional catalogues and their realization fibers. For the four exceptional types, the numbers of finite-subgroup conjugacy classes in the extended mapping class group, and the numbers of finite action classes inducing these subgroups isomorphically, are as follows:

(g,ν)	$\mathcal{T}_{g,\nu}$	finite subgroup classes	realized action classes
$(0,0)$	C_2	2	3
$(0,1)$	C_2	2	2
$(0,2)$	C_2^2	5	7
$(1,0)$	$\mathrm{GL}_2(\mathbb{Z})$	13	17

(B.36.1)

The executed files `exceptional_finite_subgroups_and_lifts.csv` and `exceptional_sections.json` give all 29 action rows over these 22 mapping class subgroup rows. They retain the extension, all full and partial section classes, Φ and its stabilizers, centralizers, and every defined local multiplier. The request in B(f) for one set of section data attached only to a finite mapping class subgroup is false in these types: the displayed realization fibers are required. For the antipodal action on S^2, the algebraic section exists and the fixed-point locus is empty; there is no fixed-point rotation multiplier for that section.

Manuscript: §15; folios 68–75; scans 137–151; §17; folios 88–90; scans 179–181; §19; folios 111–115; scans 222–230; Part B(f) of the task; no further manuscript assertion.

Proof. We first identify the groups in (B.36.1) and the comparison being classified. Orientation is the component invariant for $\mathrm{Homeo}(S^2)$ and for $\mathrm{Homeo}(S^2,\{\infty\})$. An orientation-preserving homeomorphism is isotopic to the identity: fix a disc and its complementary disc by the Alexander isotopy, with the marked point kept fixed in the second case. For the annulus, the two component invariants are the sign ϵ on its fundamental group and the sign η of its permutation of the two ends. Their four representatives are

$$(x,t)\longmapsto(\epsilon x,\eta t),\qquad x\in\mathbb{R}/\mathbb{Z},\quad t\in\mathbb{R}. \tag{B.36.2}$$

The kernel of these invariants is connected: compactify the two ends, isotope an arc joining them to a fixed arc, and apply the Alexander isotopy on the complementary disc. Winding of the arc is absorbed by rotating a puncture neighborhood, since its end is a puncture and has no prescribed boundary parametrization. For the torus, homology gives $\pi_0 \operatorname{Homeo}(T^2) = \mathrm{GL}_2(\mathbb{Z})$. Surjectivity is given by linear maps. A homologically trivial homeomorphism is first isotoped to fix the origin, then the two basis curves, and then their complementary disc; translations lie in the identity component. B.25 gives its thirteen finite subgroup classes.

In this statement a realized action over a finite subgroup $H \subset \pi_0 \operatorname{Homeo}(U)$ means a finite subgroup $G \subset \operatorname{Homeo}(U)$ for which the projection $G \to \pi_0 \operatorname{Homeo}(U)$ is an isomorphism onto a conjugate of H. Actions are identified by homeomorphism conjugacy. This specifies the exact fiber of the comparison map. The additional finite actions having a nontrivial kernel in that projection are the full affine families already classified in B.14; their kernels are retained in that result and are not silently identified with an injective realization.

On S^2, a negative involution is conjugate either to a reflection or to the antipodal map. Indeed the sphere theorem in B.1 conjugates the finite action into $O(3)$. An orthogonal involution has eigenvalues ± 1; negative determinant leaves precisely the patterns $(1, 1, -1)$ and $(-1, -1, -1)$. Their fixed sets are a circle and the empty set. Together with the trivial action this gives three rows over the two mapping class subgroups. On the plane, a negative involution is conjugate to a reflection by B.1, giving two rows including the trivial action. For both surfaces $\pi = 1$, so the lift extension is simply $E = G$. For the negative C_2 there is exactly one full section, exactly one nontrivial partial section, and one maximal class, with centralizer 1 in π and stabilizer C_2. For the trivial group there is one full section and no nontrivial partial section. These are algebraic assertions about the extension and include the antipodal action. In that action $p = -p$ implies $p = 0$, which has no solution on the unit sphere. The program verifies $A^2 = I$, $\det A = -1$, and $\det(I - A) = 8$ for $A = -I_3$. It therefore records zero fixed components next to the one algebraic full section.

A reflection has the local multiplier $1/2$. For the antipodal section no fixed-point germ exists from which to read such a multiplier; the table marks that entry as undefined. It separately records the formal value $1/2$ obtained by extending the orientation-sign convention to every negative involution. This convention has a stated source and is not substituted for a nonexistent fixed-point comparison. In particular the circle fixed by a sphere reflection has trivial image in $\pi_1(S^2)$, so the hyperbolic oval-centralizer formula $\mathbb{Z}$ does not transfer to this case.

For the annulus use the simultaneous normal form of B.14. Let

$$a(x,t) = (-x,t), \qquad b(x,t) = (x,-t), \qquad b'(x,t) = (x+\tfrac{1}{2},-t), \qquad c(x,t) = (-x,-t). \tag{B.36.3}$$

The three nontrivial mapping class elements are represented by a, b, c. A lift of the class of a or c has an arbitrary angular shift, removed by translation conjugacy. A lift of the class of b has angular shift 0 or $1/2$, because its square is the identity. These give b and b'; their fixed sets distinguish them. For the whole four-group, normalize its angular reflection to a. Commutation forces the radial generator's angular shift to be 0 or $1/2$, giving exactly $\langle a, b\rangle$ and $\langle a, b'\rangle$. Conjugation cannot exchange the three nontrivial mapping class elements, since their component group is abelian. The fixed oval or its absence again distinguishes these two actions. This proves the seven-row classification.

Here are all its section counts; f, p, ϕ, o mean full sections, nontrivial partial sections, $|\Phi|$,

and $|\Phi/G|$:

action	f	p	ϕ	o
1	1	0	0	0
$\langle a\rangle$	2	2	2	2
$\langle b\rangle$	1	1	1	1
$\langle b'\rangle$	0	0	0	0
$\langle c\rangle$	2	2	2	2
$\langle a,b\rangle$	2	7	2	2
$\langle a,b'\rangle$	0	4	4	2

(B.36.4)

To verify them, the fixed sets of a, b, b', c are respectively two lines, one oval, the empty set, and two points. In the split four-group, each of its two global fixed points extends the corresponding point section and the incident reflection sections; the oval section has both maximal extensions. In the glide four-group, the two angular-reflection lines and the two fixed points of ab' are disjoint. They are all maximal, and the group interchanges each pair. Thus (B.36.4) also gives the required partial-section distinction. Every nonidentity torsion element in these sections has multiplier $1/2$. The centralizer in $\pi = \mathbb{Z}$ is $\mathbb{Z}$ for the radial oval section and 1 for every point or line section.

These calculations can also be read directly from the extension. For an affine element write its chosen angular shift as $a_g \in [0, 1)$ and let ϵ_g be its action on $\pi = \mathbb{Z}$. The program records the integer cocycle

$$c(g,h) = a_g + \epsilon_g a_h - a_{gh} \quad \text{and} \quad (n,g)(m,h) = (n + \epsilon_g m + c(g,h), gh). \tag{B.36.5}$$

It checks the cocycle identity on every triple. In the universal cover $\mathbb{R}^2$, a finite subgroup of the lift extension fixes a point; its fixed set is affine and connected. Passing to the deck quotient identifies its conjugacy class with the corresponding fixed component just listed. This proves the section calculation without using a center-free hypothesis.

For the torus apply the exact affine classification in B.27 and retain precisely the normal forms whose translation kernel has order one. The linear projection of such a finite group is faithful, which is exactly the comparison stipulated above. Conversely every injective realized mapping class subgroup has this property and is included by B.14. There are seventeen affine forms over the thirteen integral linear classes. In the order of the linear classes in B.27, the fiber sizes are

$$1, 2, 1, 1, 1, 3, 1, 1, 1, 1, 1, 2, 1. \tag{B.36.6}$$

The proof of B.27 enumerates all cocycles and all affine conjugacies without a word bound. Its averaging comparison in B.14 proves that no further topological conjugacy is missing. Run the same exact fixed-point and reflection-circle computation with no punctures removed. Every isolated point section has centralizer 1 in $\mathbb{Z}^2$; every reflection oval has centralizer the primitive invariant copy of $\mathbb{Z}$. The trivial section has centralizer $\mathbb{Z}^2$. Local rotation matrices give the multipliers, and the complete section poset gives the full and partial counts, Φ, and every stabilizer. The resulting seventeen rows and their exact rational matrices are included in the JSON.

The executed program `catalogue_exceptional.py` carries out all these finite calculations, records the two annulus-end stabilizers and the plane's puncture stabilizer, and verifies the four mapping class counts and the four realization-fiber totals. Consequently it supplies a complete corrected computational answer for the exceptional types, including the precise fibers where B.22 prevents attaching one extension or one section table to a bare mapping class subgroup. □

B.37. [Proved] Complete genus-three mirror-disc quotient catalogue. There are exactly 145 conjugacy classes of finite subgroups K of $\mathcal{T}_{3,0}$ for which $H = K \cap \mathcal{T}_{3,0}^{+}$ has spherical quotient and the induced involution on S_3/H has a fixed circle. Exactly 57 contain a hyperelliptic

involution, and 88 do not. The files `type_3_0_reflection_quotient_catalogue.csv`, `type_3_0_reflection_quotient_certificates.json`, and `type_3_0_reflection_quotient_sections.json` give every class, its exact finite group, and all its section data.

Manuscript: §18; folios 96–101; scans 193–203; Part B(f) of the task; genus-three mirror-disc quotient subcatalogue.

Proof. The quotient by the induced involution is a mirror disc Q. Let $(m_1, \dots, m_k)$ be its cyclic list of corner orders and $(n_1, \dots, n_l)$ its list of interior cone orders. The orientation double of Q is the spherical orbifold with cone multiset

$$\{m_1, \dots, m_k, n_1, n_1, \dots, n_l, n_l\}. \tag{B.37.1}$$

The involution on a sphere with a fixed circle is conjugate to a reflection: cut along that circle into two discs and extend a boundary parametrization over either disc, then use the involution to define the parametrization on the other disc. Cone points on its circle become the corners of Q, while its other cone points occur in exchanged pairs. Thus the corner and interior data in (B.37.1) exhaust its quotient types. The corner list is considered up to cyclic permutation and reversal; the interior list is unordered.

Use the following presentation of $\Lambda = \pi_1^{\mathrm{orb}}(Q)$:

$$\left\langle x_1, \dots, x_l, e, c_0, \dots, c_k \;\middle|\; \begin{array}{c} x_i^{n_i} = 1, \quad x_1 \cdots x_l e = 1, \\ c_j^2 = 1, \quad (c_{j-1}c_j)^{m_j} = 1, \quad c_k = e c_0 e^{-1} \end{array} \right\rangle. \tag{B.37.2}$$

For $k = 0$ there is one reflection c_0, commuting with e. The orientation kernel $\Delta = \Lambda^+$ is the orbifold group of (B.37.1). Put $t = c_0$, $Z_j = c_{j-1}c_j$, $P_j = Z_1 \cdots Z_j$, $U_i = x_{l+1-i}^{-1}$, and $V_i = tU_it$. Schreier rewriting, or the doubled fundamental polygon, gives the spherical generating vector

$$(Z_1, \dots, Z_k, U_1, \dots, U_l, V_l^{-1}, \dots, V_1^{-1}), \qquad P_k\, U_1 \cdots U_l\, (V_1 \cdots V_l)^{-1} = 1. \tag{B.37.3}$$

For example $e = U_1 \cdots U_l$ and $P_k = tc_k = tete^{-1} = (V_1 \cdots V_l)e^{-1}$, which proves the product relation with its stated order. Conjugation by t acts on these generators by

$$\psi(Z_j) = P_{j-1} Z_j^{-1} P_{j-1}^{-1}, \qquad \psi(U_i) = V_i, \qquad \psi(V_i) = U_i. \tag{B.37.4}$$

These formulas include every base-path conjugation.

The input [34], whose completeness is established in [35, Theorem 2.2, Corollary 2.3, and the planar calculation], contains every $\mathrm{Aut}(H)$-class of smooth epimorphisms $\theta : \Delta \to H$ in genus three. Its 197 vectors cover all 49 genus-three signature/group pairs, including those with positive quotient genus. Here we use precisely the spherical-quotient pairs. The original input and its attribution are retained in `research_data/actions-genus3.json`; B.33 verifies its orders and surface signatures. A prescribed ordering (B.37.3) is obtained from the published sorted ordering by explicit adjacent Hurwitz moves. This is a bijection of epimorphism sets, since each move has its displayed inverse in the program.

For each vector the program tests whether its kernel is invariant under (B.37.4). More explicitly, it attempts to define

$$\alpha(\theta(a)) = \theta(\psi(a)) \qquad (a \in \Delta) \tag{B.37.5}$$

by propagation on the finite Cayley graph. Agreement on every edge is exactly the condition that (B.37.5) be well defined. Surjectivity then makes it an automorphism and (B.37.4) gives $\alpha^2 = 1$. The resulting full action group is explicitly

$$K = H \rtimes_\alpha \langle t \rangle, \quad t^2 = 1. \tag{B.37.6}$$

Its epimorphism from (B.37.2) sends x_i to their recovered vector entries and c_j to $t\theta(P_j)$. Its kernel is exactly $\ker\theta \subset \Delta$. In particular no choice of a further extension class is omitted: the orbifold kernel determines $\Lambda/\ker\theta$ and hence (B.37.6). Conversely every action in the claim supplies this invariant kernel and therefore occurs.

We now prove the precise fusion calculation. Two epimorphisms from the same orbifold group have the same kernel exactly when a unique isomorphism between their finite images intertwines them: define that isomorphism by $\theta(a) \mapsto \theta'(a)$, which is well defined exactly when the kernels agree. Thus the published $\mathrm{Aut}(H)$-normalization records kernels without loss. A conjugacy of the surface actions descends to a decorated-orbifold homeomorphism of Q carrying one kernel to the other. Conversely such a quotient homeomorphism lifts through the two regular orbifold coverings because it carries their defining kernels to one another; its lift is the required surface conjugacy. This is an exact equivalence, so it remains to calculate the full action of these disc homeomorphisms on the finite kernel set.

A disc homeomorphism permutes its colored boundary corners cyclically, or reverses their cyclic order. After composing with that boundary map it fixes the boundary pointwise up to isotopy. The remaining mapping class is a braid of the interior cone points. To see this, forget the interior points and isotope the boundary-fixed disc map to the identity by the Alexander isotopy; the tracks of the points are its braid, and an isotopy of those tracks gives precisely an isotopy of the punctured disc map by isotopy extension. The colored braid subgroup is generated by pure braid generators and adjacent exchanges among equal colors after the colors are sorted. The program uses the standard words $\sigma_{j-1}\cdots\sigma_{i+1}\sigma_i^2\sigma_{i+1}^{-1}\cdots\sigma_{j-1}^{-1}$ and those equal-color exchanges. On interior generators an adjacent positive move is

$$(x_i, x_{i+1}) \longmapsto (x_i x_{i+1} x_i^{-1}, x_i), \tag{B.37.7}$$

fixing e and every boundary reflection. The inverse is also used when forming the pure braid words.

Here are the boundary moves, including their effect on the chosen reflection. Extend the sequence by $c_{j+k} = e c_j e^{-1}$. One cyclic shift has

$$\begin{aligned} (Z_1,\dots,Z_k) &\longmapsto (Z_2,\dots,Z_k, eZ_1e^{-1}),\\ (x_1,\dots,x_l) &\longmapsto (x_1,\dots,x_l),\\ t &\longmapsto tZ_1, \qquad \alpha \longmapsto \mathrm{Inn}(Z_1^{-1})\alpha. \end{aligned} \tag{B.37.8}$$

A reversal has

$$\begin{aligned} (Z_1,\dots,Z_k) &\longmapsto (Z_k^{-1},\dots,Z_1^{-1}),\\ (x_1,\dots,x_l) &\longmapsto (x_l^{-1},\dots,x_1^{-1}),\\ t &\longmapsto tP_k, \qquad \alpha \longmapsto \mathrm{Inn}(P_k^{-1})\alpha. \end{aligned} \tag{B.37.9}$$

They follow by sending c_j respectively to c_{j+1} and c_{k-j} in (B.37.2); in the reversal e goes to e^{-1}. Only the shifts and shifted reversals preserving the corner-order list are self-homeomorphisms of the chosen colored quotient. A fixed braid restores the sorted interior order after a reversal. Equations (B.37.7)–(B.37.9), together with the preceding braid argument, prove both that every programmed fusion is realized and that every actual conjugacy is included.

The program computes the finite orbits under these moves. It identifies two generating vectors under $\mathrm{Aut}(H)$ by their rooted, edge-colored Cayley graphs: starting at the identity, visit vertices in generator order and record each colored successor. Equality defines a unique group isomorphism mapping each generator to the corresponding generator, and such an isomorphism plainly gives equality. Thus this is an exact isomorphism test rather than a coarse group invariant. The computation gives 145 orbits.

For each orbit the second program constructs the regular action of (B.37.6) explicitly from $(h,\epsilon)(h',\epsilon') = (h\alpha^{\epsilon}(h'),\epsilon+\epsilon')$. It checks (B.37.2), every local stabilizer injection, and $|K|\chi^{\mathrm{orb}}(Q) = -4$. The resulting cover is therefore a closed orientable genus-three surface. The finite coset incidence graph then gives every fixed point and oval, multiplier, section, extension, maximal section, centralizer, orbit and stabilizer as in B.26. These checks passed for all 145 classes.

Finally the overlap with B.34 is identified by actual monodromy isomorphisms. In each of B.34's 57 mirror-disc groups the program extracts the positive group and the generators x_i, Z_j and conjugation by c_0. It applies (B.37.8)–(B.37.9) and the interior braids to put the signature in the same ordering, and locates its rooted colored Cayley graph in the computed kernel orbit. The file `genus3_hyperelliptic_reflection_crosscheck.json` records the resulting bijection with 57 different rows and equality of all eight reported section-count columns. Thus these are exactly the common classes, leaving 88 additional ones. □

The 145 classes have the following order distribution:

$\lvert K\rvert$	4	6	8	12	14	16	18	24
#	5	3	36	13	3	35	1	15
$\lvert K\rvert$	28	32	42	48	64	96	192	336
#	1	13	1	11	2	4	1	1.

(B.37.10)

Together B.34 and B.37 therefore give 149 distinct mixed classes. The additional four are the projective-plane quotients in B.34. The scope of B.37 requires both a spherical positive quotient and a fixed-circle involution on it; the remaining quotient types are listed in the accompanying computation report.

B.38. [Proved] The free positive involution and the six negative involutions in genus three. There are exactly twelve conjugacy classes of mixed finite subgroups of $\mathcal{T}_{3,0}$ whose positive subgroup is a free involution. Each is a four-group. Their quotient signatures and numbers are

underlying quotient	$S_{0,3}$	$S_{1,1}$	$N_{1,2}$	$N_{2,1}$	$N_{3,0}$
number of classes	4	1	4	2	1,

(B.38.1)

where $S_{h,b}$ is orientable of genus h with b mirror boundary components, and $N_{q,b}$ is nonorientable with q crosscaps and b mirror boundary components. There are no cones or corners. There are also exactly six mixed classes whose positive subgroup is trivial: the negative involutions with quotient signatures

$$S_{0,4},\quad S_{1,2},\quad N_{1,3},\quad N_{2,2},\quad N_{3,1},\quad N_{4,0}. \tag{B.38.2}$$

All eighteen classes, their exact monodromies, complete section posets, Φ, stabilizers, centralizers and multipliers appear in `type_3_0_free_positive_and_anti_catalogue.csv` and its `sections.json` companion.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; §33; folios 222–225; scans 440–445; Part B(f) of the task; no further manuscript assertion.

Proof. Let X be the closed orientable genus-three surface. First let $G^+ = \langle a\rangle$ have order two and act freely, and let G contain negative elements. Then $|G| = 4$. A group of order four is cyclic or a four-group: an element of order four gives the first case; otherwise $(xy)^2 = 1$ for all x, y, which gives commutativity and the second case. There is no nontrivial positive point stabilizer. A negative point stabilizer is generated by a reflection, so $Q = X/G$ has no cone or corner points. Its orbifold Euler characteristic is $-4/4 = -1$. The complete list of compact underlying surfaces with this Euler characteristic is exactly (B.38.1).

The cyclic possibility cannot occur. A negative element of order four has no fixed point, since a negative local finite stabilizer element is a reflection of order two. Its square is free by

assumption, so the whole action would be free. The quotient would then be $N_{3,0}$. Its fundamental group has the presentation

$$\langle u_1, u_2, u_3 \mid u_1^2 u_2^2 u_3^2 = 1 \rangle.$$

An orientation-compatible monodromy to $C_4 = \mathbb{Z}/4$ sends each crosscap generator to an odd residue. Each square has image 2, so the relator has image $6 = 2 \pmod 4$, a contradiction. The program checks all eight assignments of odd residues. Thus G is the four-group $V_4 = \mathbf{F}_2^2$, with orientation given by its second coordinate, and $a = (1, 0)$.

We classify all four-group monodromies, including their full topological equivalence relation. The mirror orbifold group of an orientable $S_{h,b}$ has generators a_i, b_i, e_j, c_j and relations

$$c_j^2 = 1, \qquad [e_j, c_j] = 1, \qquad \prod_i [a_i, b_i] \prod_j e_j = 1. \tag{B.38.3}$$

The negative c_j is the reflection on the jth boundary, and the positive e_j traverses that boundary. For $N_{q,b}$ replace the handles by crosscap generators u_i, and replace the last relation by

$$\prod_i u_i^2 \prod_j e_j = 1. \tag{B.38.4}$$

The orientation character ε is zero on handles and e_j, and one on crosscaps and c_j. An orientation-compatible epimorphism to V_4 is exactly

$$q_\alpha : \Pi_Q \longrightarrow V_4, \qquad q_\alpha(x) = (\alpha(x), \varepsilon(x)), \quad \alpha \in H^1(\Pi_Q; \mathbf{F}_2) \setminus \{0, \varepsilon\}. \tag{B.38.5}$$

In coordinates write the handle or crosscap values as η_i, the boundary values as $\beta_j = \alpha(e_j)$, and the reflection values as $\gamma_j = \alpha(c_j)$. The only linear relation is $\sum_j \beta_j = 0$. The two automorphisms of V_4 preserving orientation identify α with $\alpha + \varepsilon$.

Every map (B.38.5) is injective on local reflection groups. Its kernel is therefore torsion-free; one proves this either from the universal hyperbolic orbifold cover or by cutting along arcs to its reflection polygon, where every finite stabilizer is a displayed local reflection group. The kernel lies in the orientation kernel, and its Euler characteristic is $4(-1) = -4$. It is consequently the fundamental group of a closed orientable genus-three surface. Conversely every action under consideration yields exactly this construction. Two such actions are conjugate precisely when an orbifold homeomorphism of Q carries one monodromy kernel to the other, or equivalently when it carries the monodromy to the other after an automorphism of V_4. This follows by lifting paths in the two covers, with the local reflection charts giving the extension across mirrors. Thus the following finite cohomology-orbit calculation is an exact classification of actions.

We prove the entire mapping class action needed in this calculation. A boundary permutation permutes the pairs (β_j, γ_j). The interior homology action preserves the mod-two intersection pairing, its radical generated by boundary classes, and the orientation character. It introduces no reflection coordinate: an interior loop is mapped to an interior loop, whereas each reflection generator is mapped to a conjugate of the reflection at the corresponding boundary. Thus the γ_j are affected only by boundary permutations and by the global replacement $\alpha \mapsto \alpha + \varepsilon$.

For $S_{0,3}$ the interior homology is generated by its boundary classes, with their single sum relation, so boundary permutations are the whole action. When $\beta = (0, 0, 0)$, surjectivity leaves one nonconstant reflection-color pattern up to permutation and global color reversal. When β has two ones, distinguish the boundary with value zero and use global color reversal to make its reflection value zero. The two remaining reflection values have weights zero, one or two. This gives $1 + 3 = 4$ classes.

For $S_{1,1}$ the boundary value is zero. Global color reversal sets its reflection value to zero. The remaining nonzero pair of handle values is a single orbit under $\mathrm{Sp}_2(\mathbf{F}_2)$, generated by the two Dehn twists in a pair of basis curves. This gives one class.

For $N_{1,2}$ write u for its one-sided core and e for either mod-two boundary class. Its intersection form has $u \cdot u = 1$, with e in the radical. An automorphism preserving the boundary classes and the orientation character sends u either to u or to $u + e$. Both possibilities are realized. Push one boundary disc once around the one-sided core and return it as an unparametrized disc; its boundary orientation may reverse. A core arc crossing the track once acquires one boundary circuit. Thus, modulo two, this homeomorphism sends u to $u + e$ and leaves both reflection coordinates unchanged. The boundary exchange is realized away from the core. These maps generate every permitted action on the displayed homology and boundary data. After global color reversal sets $\eta = \alpha(u)$ to zero, the case $\beta = (0, 0)$ has two nonzero reflection patterns up to exchange. The case $\beta = (1, 1)$ has two patterns: the boundary push followed by global reversal interchanges $\gamma = (0, 0)$ and $(1, 1)$, while $(0, 1)$ and $(1, 0)$ are exchanged by the boundary permutation. This gives four classes. The boundary push is necessary for this count.

For $N_{2,1}$ the boundary value is zero. Set its reflection value to zero by global reversal. The intersection form on the two crosscap classes is the identity matrix over $\mathbf{F}_2$; its orthogonal group has exactly the identity and the exchange of those two classes. They are realized by exchanging the two crosscaps. The nonzero crosscap vectors therefore have two orbits, represented by $(1, 0)$ and $(1, 1)$.

For $N_{3,0}$ the mod-two intersection form is the identity matrix in three crosscap classes, and its characteristic vector is $(1, 1, 1)$. An orthonormal basis cannot contain that characteristic vector, since it pairs to one with every vector of odd self-intersection. Its members must therefore be the three standard basis vectors. The full orthogonal group is their permutation group, realized by crosscap permutations. Global addition of $\varepsilon = (1, 1, 1)$ exchanges weights one and two, leaving one orbit after $0, \varepsilon$ are excluded. This proves all five numbers in (B.38.1).

We now compute the sections explicitly. A mirror boundary with data (β_j, γ_j) has local reflection $c_j = (\gamma_j, 1)$ and boundary monodromy $e_j = (\beta_j, 0)$. Its preimage is described by the two cosets $V_4/\langle c_j \rangle$; one trip around the boundary permutes these cosets by e_j. Consequently it gives two fixed ovals when $\beta_j = 0$, and one when $\beta_j = 1$. The two-oval case has an oval stabilizer of order two, $\langle c_j \rangle$, and the one-oval case has stabilizer $V_4 = \langle c_j, e_j \rangle$. Both are one V_4-orbit of section classes. There are no point sections because the positive involution is free, and there are no full sections because V_4 cannot fix a point without giving a positive local stabilizer. Every reflection section is maximal, has centralizer $\mathbb{Z}$ in the surface group, and has multiplier $1/2$ on its nonidentity element. Thus

$$|\Phi| = \#\{\text{nontrivial partial sections}\} = 2b - \#\{j : \beta_j = 1\}, \qquad |\Phi/V_4| = b. \tag{B.38.6}$$

The program constructs the actual finite coset graph, with three subdivisions per boundary traversal, computes its components and the group action, and verifies every count and stabilizer in this formula.

Finally a mixed group with trivial positive subgroup is C_2. Its quotient has orbifold Euler characteristic -2 and no cones or corners. An orientable quotient must have mirror boundary; otherwise its orientation character would be trivial and could not give the required negative covering. Solving $2 - 2h - b = -2$ with $b > 0$, or $2 - q - b = -2$ with $q > 0$, gives precisely (B.38.2). For each quotient the monodromy is forced: it is the orientation character, taking every crosscap and local reflection to the nonidentity element and every handle and boundary traversal to zero. Its orientation double is connected, torsion-free, and has Euler characteristic -4. The classification of compact quotient surfaces and path lifting prove that each signature gives one conjugacy class and that distinct signatures give distinct classes. There are b fixed ovals, hence b full sections, b nontrivial partial sections, and b maximal classes, each fixed by the whole C_2 and with surface-group centralizer $\mathbb{Z}$. The case $N_{4,0}$ has none. These are the six genus-three real types of B.16 in an executed form.

The code `catalogue_genus3_free_positive.py` enumerates every cohomology vector in (B.38.5), applies exactly the proved generators of the equivalence relation, checks surjectivity, all

quotient relations, orientation compatibility, and the covering Euler characteristic, and records every orbit. Its second calculation constructs all six orientation doubles in (B.38.2). This supplies the complete eighteen-row catalogue and its section data. □

B.39. [Proved] Complete genus-three projective-plane quotient catalogue. There are exactly 11 conjugacy classes of mixed finite genus-three actions whose positive quotient is a sphere and whose induced involution on that sphere is fixed-point-free. Their full quotients are projective planes with cone points. Four classes are identified with B.34 and seven are additional. Consequently B.37 and the present claim give exactly 156 mixed classes with spherical positive quotient.

Manuscript: §18; folios 96–101; scans 193–203; Part B(f) of the task; genus-three projective-plane quotient subcatalogue.

Proof. A fixed-point-free sphere involution has projective-plane quotient: its quotient is a closed surface of Euler characteristic one, and its orientation cover is the given sphere. Cone points are exchanged in pairs. Thus each multiplicity in the positive cone multiset must be even. Inspection of the complete input in B.33 leaves exactly O3.4, O3.9, O3.11, O3.15, O3.21, O3.22, and O3.25. Their paired cone orders are respectively $(2,2,2,2)$, $(2,2,2)$, $(4,4)$, $(2,6)$, $(2,4)$, $(2,4)$, and $(2,3)$.

For a projective plane with cone orders $n_1,\dots,n_l$ use

$$\Lambda = \langle x_1,\dots,x_l,d \mid x_i^{n_i}=1,\quad x_1\cdots x_l d^2 = 1\rangle. \tag{B.39.1}$$

Set $X_i = x_i$, $Y_i = dx_i d^{-1}$ and $A = d^2 = (X_1\cdots X_l)^{-1}$. Its orientation double has the spherical vector

$$v = (X_1,\dots,X_l,Y_l^{-1},\dots,Y_1^{-1}),\qquad \prod_i X_i = \prod_i Y_i. \tag{B.39.2}$$

Conjugation by d is

$$X_i \longmapsto Y_i,\qquad Y_i \longmapsto AX_iA^{-1}. \tag{B.39.3}$$

For every published $\mathrm{Aut}(H)$-normalized vector in these seven cases, the program tests that (B.39.3) descends to an automorphism α of its finite positive image H. The same Cayley-graph propagation as in B.37 gives an exact kernel-invariance test. It also verifies $\alpha^2 = \mathrm{Inn}(A)$ and $\alpha(A) = A$. The full finite image is then the set $H\times\{0,1\}$ with product

$$(h,\epsilon)(k,\delta) = \big(h\alpha^{\epsilon}(k)A^{\epsilon\delta},\epsilon+\delta \bmod 2\big). \tag{B.39.4}$$

The two identities just verified prove associativity; the element $d=(1,1)$ has square A and induces α on H. Thus (B.39.4) is exactly $\Lambda/\ker\theta$, where θ is the positive epimorphism (B.39.2).

For clarity, the full source fusion is as follows. The colored mapping class group of the punctured projective plane is generated by the colored disc braids and by sliding each puncture once around a one-sided loop. One direct proof cuts the projective plane along a projective line disjoint from the marked disc: a moving puncture crossing the cut is separated off as a one-sided slide, after which all the remaining tracks form a braid in the disc. Equivalently, forgetting punctures successively expresses their point-pushing loops by peripheral loops and the one-sided generator. The former are the pure disc braids. The unmarked projective-plane mapping class group is trivial: isotope the image of a projective line to that line, adjust its parametrization by a projective rotation, and apply the disc Alexander isotopy after cutting. This proves that these point motions exhaust all mapping classes. Color-preserving permutations are supplied by the same equal-color braid generators used in B.37.

The slide of the last puncture has the exact substitutions

$$x_i \longmapsto x_i\quad (i<l),\qquad x_l \longmapsto (x_l d)x_l^{-1}(x_l d)^{-1},\qquad d\longmapsto x_l d. \tag{B.39.5}$$

They are obtained by tracking the meridian through a neighborhood of that puncture and the one-sided loop. The two-cone case, including its geometric puncture-slide interpretation, is verified independently in [25, Lemma 3.19]. Other punctures outside this neighborhood are fixed, so the same formula applies to all l used here. It respects the product relation directly: if $B = x_1 \cdots x_{l-1}$ and $x = x_l$, then

$$B(xdx^{-1}d^{-1}x^{-1})(xd)^2 = Bxd^2 = 1.$$

In the finite positive group its effect is

$$x_l \longmapsto x_l\alpha(x_l)^{-1}x_l^{-1}, \qquad \alpha \longmapsto \mathrm{Inn}(x_l)\alpha, \qquad A \longmapsto x_l\alpha(x_l)A. \tag{B.39.6}$$

To slide any other puncture, a specified sequence of adjacent braids moves it to the last position, applies (B.39.6), and moves it back. The full word preserves its color even when the intermediate braids exchange different colors. Equations (B.39.5)–(B.39.6) and the disc braids therefore give exactly the source mapping-class action. The kernel-to-cover conjugacy argument of B.37 applies without change.

Executing these finite orbits gives respectively $1, 3, 3, 1, 2, 0, 1$ classes in the seven signature/group pairs above. For every class, `catalogue_genus3_projective_quotients.py` constructs the regular representation of (B.39.4), checks every relation in (B.39.1), injectivity of all local cyclic groups, and Euler characteristic -4. The finite point-coset calculation gives every partial section and its multiplier, centralizer and extensions. No negative element has a fixed point, since its image on the positive quotient sphere is fixed-point-free. Thus all nontrivial sections in these rows have positive cyclic image and are point sections; their surface-group centralizers are trivial.

Finally the four projective-plane groups in B.34 are compared using their actual cone generators and conjugation by their crosscap element. After a fixed ordering braid, their doubled vectors are identified in the finite kernel orbits by the rooted colored Cayley graph isomorphism test. The resulting exact identifications are

$$\mathrm{G3H}_{002} \leftrightarrow \mathrm{G3P}_{001}, \quad \mathrm{G3H}_{011} \leftrightarrow \mathrm{G3P}_{002}, \quad \mathrm{G3H}_{029} \leftrightarrow \mathrm{G3P}_{008}, \quad \mathrm{G3H}_{039} \leftrightarrow \mathrm{G3P}_{010}. \tag{B.39.7}$$

All seven applicable section-count columns agree under this bijection. Hence precisely seven classes are additional to B.34. The quotient involution on the positive sphere has a fixed circle in B.37 and has empty fixed set here; a conjugacy induces a sphere conjugacy carrying these fixed sets to each other. Thus none of these eleven rows can be conjugate to a B.37 row, proving the total $145 + 11 = 156$. □

Here p counts nontrivial partial section classes, and m, o are $|\Phi|$ and $|K\backslash\Phi|$. The final column i counts negative involutions in the displayed finite group; all of them act freely.

class	$\lvert K\rvert$	cone orders	p	m	o	i
001	4	$(2,2,2,2)$	8	8	4	2
002	8	$(2,2,2)$	12	12	3	0
003	8	$(2,2,2)$	12	12	3	2
004	8	$(2,2,2)$	12	12	3	4
005	8	$(4,4)$	8	4	2	0
006	8	$(4,4)$	8	4	2	2
007	8	$(4,4)$	8	4	2	4
008	12	$(2,6)$	12	8	2	2
009	16	$(2,4)$	16	12	2	0
010	16	$(2,4)$	16	12	2	0
011	24	$(2,3)$	20	20	2	4

(B.39.8)

Every row has zero full splittings, since its full image contains a negative element and no such element fixes a point. The JSON `type_3_0_projective_quotient_sections.json` includes all finite permutations, local generators, multipliers and section incidences behind this table.

B.40. [Proved] Complete annulus and Möbius-band quotients in genus three. There are exactly 57 mixed finite genus-three action classes whose full quotient is an annulus or a Möbius band with its cone and mirror corner structure. There are 39 annulus classes and 18 Möbius-band classes. Their positive quotient is a torus. Every group, generating vector, quotient signature and complete section poset is given in `type_3_0_torus_mirrors_catalogue.csv`, `type_3_0_torus_mirrors_certificates.json`, and `type_3_0_torus_mirrors_sections.json`.

Manuscript: §18; folios 96–101; scans 193–203; Part B(f) of the task; genus-three annulus and Mobius quotient cases.

Proof. An involution reversing the orientation of a torus has annulus, Möbius-band or Klein-bottle quotient. Indeed its fixed set consists of disjoint circles, and its quotient has Euler characteristic zero; the orientation double is the torus. The classification of compact surfaces leaves the orientable annulus, the nonorientable Möbius band, and the closed nonorientable Klein bottle. Their orientation doubles give respectively two, one, and zero fixed circles. Cone points on a fixed circle become mirror corners, and other cone points are paired to interior cones. The Klein-bottle cases are B.41.

The complete positive input of B.33 leaves the following seven signature/group pairs. In each row r is the number of positive quotient cone points, all of order m; the last two columns are the executed numbers of full annulus and Möbius-band classes.

positive row	H	m	r	annulus	Möbius
O3.3	C_2	2	4	10	3
O3.5	C_3	3	2	5	4
O3.7	C_4	2	2	8	4
O3.8	C_2^2	2	2	9	4
O3.12	S_3	3	1	3	1
O3.17	D_8	2	1	3	1
O3.18	Q_8	2	1	1	1

(B.40.1)

Here D_8 has order eight. For l interior cones, the total number of corners is $r-2l$. On an annulus distribute them as $k_1 \le k_2$ with $k_1+k_2 = r-2l$; on a Möbius band place all $k = r-2l$ on its single boundary. All their orders are m, so these integer choices already classify the colored quotient signatures.

We first give an exhaustive construction of the possible oriented finite groups, rather than assuming a small-group name list. Choose a negative coset representative t in an index-two extension of H. Let $\alpha(h) = tht^{-1}$ and $A = t^2$. Its complete extension data satisfy

$$\alpha^2 = \mathrm{Inn}(A), \qquad \alpha(A) = A. \tag{B.40.2}$$

Conversely the product of (B.39.4), with these data, defines that extension:

$$(h,\epsilon)(k,\delta) = (h\alpha^{\epsilon}(k)A^{\epsilon\delta}, \epsilon+\delta \bmod 2). \tag{B.40.3}$$

Every oriented group isomorphism restricts to some $\beta \in \mathrm{Aut}(H)$ and sends t to some kt' in the negative coset. Therefore the entire isomorphism action on (B.40.2) is generated by transport by β and by

$$(\alpha, A) \longmapsto (\mathrm{Inn}(k)\alpha,\ k\alpha(k)A). \tag{B.40.4}$$

The converse follows by sending the displayed generators accordingly in (B.40.3), so this is an exact fusion test. The program enumerates $\mathrm{Aut}(H)$ by assigning images to a generating set and

propagating on its complete Cayley graph, enumerates every pair (B.40.2), and takes precisely the finite orbits (B.40.4). For the seven H's in (B.40.1), it obtains respectively $2, 2, 4, 3, 1, 4, 4$ oriented extension classes.

Here are the full source presentations and enumeration conditions. For an annulus, put $X = x_1 \cdots x_l$ and use positive generators x_i, e_1, e_2 with

$$x_i^m = 1, \qquad X e_1 e_2 = 1. \tag{B.40.5}$$

On its jth boundary there are reflections $c_{j,0}, \ldots, c_{j,k_j}$ satisfying

$$c_{j,s}^2 = 1, \quad (c_{j,s-1} c_{j,s})^m = 1, \qquad c_{j,k_j} = e_j c_{j,0} e_j^{-1}. \tag{B.40.6}$$

For $k_j = 0$, there is one reflection commuting with e_j. For a Möbius band there is one negative crosscap generator d, one boundary with (B.40.6), and the relation

$$X e d^2 = 1, \qquad e = X^{-1} d^{-2}. \tag{B.40.7}$$

In each finite extension the program assigns every possible positive order-m element to every x_i, every positive element to e_1 in the annulus, and every negative element to d in the Möbius band. It then assigns every sequence of negative involutions to the mirrors, requiring exact order m for each adjacent product and the stated closing relation. It retains precisely the vectors generating the whole finite group. Two involutions whose product has exact order m generate a dihedral group of order $2m$; thus all local groups inject. The orientation character takes the specified signs on every generator, so the kernel is oriented and contains no reflection. The signature equation gives $|K|\chi^{\mathrm{orb}}(Q) = -4$; its torsion-free surface kernel therefore has genus three. Conversely every action with this quotient yields one of these enumerated epimorphisms.

Target automorphisms are removed by the exact rooted colored Cayley graph test of B.37, now applied to the full NEC generating vector. Its prescribed generator signs force any resulting isomorphism to preserve the orientation character. For source fusion, the following are explicit complete generators. The interior braids are (B.37.7), with all other generators fixed. A cyclic boundary move sends $c_{j,s}$ to $c_{j,s+1}$, extending the reflection sequence by (B.40.6), while fixing x_i, e_j, and the crosscap generator. After k_j moves it conjugates that boundary's reflection sequence by e_j and fixes the other generators, which supplies the boundary Dehn twist where it is needed.

For the annulus, pushing the last interior point once around the first boundary, with $a = x_l$ and $e = e_1$, has

$$\begin{aligned} a &\longmapsto (ae)a(ae)^{-1}, \\ e_1 &\longmapsto a e_1 a^{-1}, \\ c_{1,s} &\longmapsto a c_{1,s} a^{-1}. \end{aligned} \tag{B.40.8}$$

It fixes the other x_i, e_2 and the second reflection sequence. This is the square of the Hurwitz exchange between the puncture meridian and the first boundary loop; in particular it preserves ae_1 exactly. The interior braids conjugate this to pushes of each other interior point. When $k_1 = k_2$, interchange of the two boundary components is

$$\begin{aligned} e_1 &\longmapsto e_1 e_2 e_1^{-1}, \qquad e_2 \longmapsto e_1, \\ c_{1,s} &\longmapsto e_1 c_{2,s} e_1^{-1}, \quad c_{2,s} \longmapsto c_{1,s}, \end{aligned} \tag{B.40.9}$$

fixing the interior generators. This is their orientation-preserving boundary exchange, with the chosen connecting paths.

The remaining annulus generator reverses the orientation of its fundamental region. For the peripheral list $(w_1, \ldots, w_{l+2}) = (x_1, \ldots, x_l, e_1, e_2)$, write $W_i = w_1 \cdots w_i$. It sends

$$w_i \longmapsto W_{i-1} w_i^{-1} W_{i-1}^{-1}. \tag{B.40.10}$$

For the first boundary its reflection sequence is reversed and conjugated by X; for the second it is reversed and conjugated by Xe_1. Reversing a cycle means $(c_0, \dots, c_{k-1}) \mapsto (c_k, c_{k-1}, \dots, c_1)$, with $c_k = ec_0e^{-1}$; an empty cycle retains its single reflection. These substitutions verify (B.40.5)–(B.40.6) directly.

For the Möbius band, a one-sided slide of the last interior point is particularly explicit. Set

$$a = x_l, \qquad a' = (ad)a^{-1}(ad)^{-1}.$$

The substitution is

$$\begin{aligned} x_l &\longmapsto a', & d &\longmapsto ad, \\ c_s &\longmapsto a'^{-1} a\, c_s\, a^{-1} a'. \end{aligned} \tag{B.40.11}$$

It fixes the other interior generators and determines e by (B.40.7). To obtain it geometrically, regard the boundary temporarily as the last peripheral loop in the projective-plane presentation: exchange (a, e) by its Hurwitz move, slide the now-last puncture by (B.39.5), and undo that exchange. This gives exactly (B.40.11), including the displayed transport of every boundary reflection. Finally a reversal of the core has

$$\begin{aligned} (x_1, \dots, x_l) &\longmapsto (x_l^{-1}, \dots, x_1^{-1}), \\ d &\longmapsto d^{-1}, \qquad c_s \longmapsto X c_{k-s} X^{-1}. \end{aligned} \tag{B.40.12}$$

Here e goes to $Xe^{-1}X^{-1}$, so the closing relation is preserved.

We justify that this is the whole source action. First prescribe the cyclic permutations, and possible reversal or interchange, of the marked boundary components. The boundary moves above realize exactly these permutations. Upon forgetting the interior points, a remaining annulus map is generated by its core twist, and a remaining Möbius-band map by reversal of its core. Cutting along a spanning arc gives this classification: the annulus permits the integer winding of the arc, while cutting the Möbius band reduces a boundary-fixed map to a disc isotopy. The boundary reversal is (B.40.12). The core twist is already among the full cyclic boundary moves when both boundaries are marked; when a boundary is unmarked its rotation isotopes it away in the unpunctured quotient. The kernel of forgetting interior points consists of their braids. An interior point in an annulus minus the other points has fundamental group generated by the puncture meridians and the core loop; in a Möbius band the corresponding extra generator is one-sided. Forgetting points one at a time, and using isotopy extension, therefore shows that their braid groups are generated by the ordinary interior braids and respectively (B.40.8) or (B.40.11). This also includes any twist that becomes trivial only after the interior points are forgotten. Thus every geometric source automorphism is included, and each included automorphism is realized by the maps just described. The kernel-to-cover lifting argument of B.37 proves that the computed orbits are exactly the surface conjugacy classes.

Execution gives the 57 classes in (B.40.1). There is an independent check on the source generators in every one-cone, no-corner case: the complete primary automorphism lists in [25, Lemmas 3.7 and 3.8] give exactly the same partitions of the entire target-normalized epimorphism sets. The file `check_torus_mirror_generators_output.json` records all six comparisons, comparing full orbit memberships and not just their numbers.

The program then computes all section invariants. Its mirror incidence graph has a separate closed cycle for each quotient boundary, with endpoint gluing by that boundary's actual element e_j. At every corner it uses the full local dihedral group, and checks that the fixed subgraph of each reflection has degree two at every vertex. Consequently its connected components are exactly the fixed ovals, including all their intersections with other reflection fixed sets. Cone and corner cosets give the isolated fixed points. The procedure of B.26 then supplies every section, multiplier, centralizer, extension incidence, maximal section, group orbit and stabilizer. All 57 presentations, local injections, Euler characteristics and complete section calculations passed. □

The 57 classes have orders $4, 6, 8, 12, 16$ with multiplicities $13, 9, 25, 4, 6$, respectively. The full quotient surfaces in this claim have boundary, whereas those in B.41 are closed. A conjugacy of actions descends to a homeomorphism of their full quotient orbifolds and therefore preserves boundary. Their positive quotients have genus one, whereas those in B.37 and B.39 have genus zero and those in B.38 have genus two or three. The induced homeomorphism of positive quotients preserves genus. These exact quotient maps prove that the four portions contribute disjoint conjugacy classes.

B.41. [Proved] The six genus-three actions with Klein-bottle quotient. There are exactly six mixed finite-subgroup classes in $\mathcal{T}_{3,0}$ whose quotient orbifold has underlying Klein bottle. Their positive subgroups, full groups, cone orders, and section counts are

G^+	G	cone orders	f	p	$\lvert\Phi\rvert$	$\lvert\Phi/G\rvert$
C_2	C_4	$2,2$	0	4	4	2
C_2	C_2^2	$2,2$	0	4	4	2
C_3	C_6	3	0	2	2	1
C_4	C_8	2	0	4	4	1
C_4	$C_4 \times C_2$	2	0	4	4	1
C_2^2	D_8	2	0	4	4	1

(B.41.1)

Here f and p denote full and nontrivial partial section classes, and D_8 has order eight. The files `type_3_0_klein_quotient_catalogue.csv` and `type_3_0_klein_quotient_sections.json` contain every group operation, monodromy, local multiplier, section stabilizer and centralizer. All centralizers in the surface group are trivial. In the primary positive catalogue, these rows lie over O3.3, O3.5, O3.7 and O3.8, respectively in numbers $2, 1, 2, 1$.

Manuscript: §15; folios 73–75; scans 148–151; §15; folios 77–79; scans 155–160; §18; folios 96–101; scans 193–203; Part B(f) of the task; no further manuscript assertion.

Proof. The orientation double of a Klein bottle is a torus. An interior cone of order m in the quotient has two lifts of order m in that double, and there are no mirrors or corners. The complete positive catalogue in B.33 therefore leaves precisely the positive actions

$$(C_2; 2, 2, 2, 2), \quad (C_3; 3, 3), \quad (C_4; 2, 2), \quad (C_2^2; 2, 2), \tag{B.41.2}$$

with torus underlying quotient. The other torus-positive rows have a single cone and cannot arise by this double-cover comparison. Thus the full groups have orders four, six or eight.

We enumerate all oriented group extensions, rather than presupposing a list of abstract groups. For a fixed positive group H choose a negative representative t, put $\alpha(h) = tht^{-1}$ and $A = t^2 \in H$. Associativity is equivalent to

$$\alpha^2 = \mathrm{Inn}(A), \qquad \alpha(A) = A. \tag{B.41.3}$$

Conversely the set $H \times \{0, 1\}$ with normal form ht^i and relations $th = \alpha(h)t$, $t^2 = A$ is a group for every pair in (B.41.3); checking the two possible negative factors verifies associativity. Changing t to kt changes the pair to

$$(\mathrm{Inn}(k)\alpha,\ k\alpha(k)A), \tag{B.41.4}$$

and changing the identification of H by $\beta \in \mathrm{Aut}(H)$ changes it to $(\beta\alpha\beta^{-1}, \beta(A))$. These are all orientation-preserving isomorphisms of the extensions, because any such isomorphism restricts to an automorphism of H and sends a negative representative to one negative coset representative.

For $H = C_2$ this gives C_4 and C_2^2. For $H = C_3$, the two possible conjugation actions give C_6 and S_3. Explicitly the identity action allows the square of t to be removed using (B.41.4), while inversion forces $A = 1$. For $H = C_4 = \langle a \rangle$, the identity action has two square classes, $A = 1$ or

a, giving $C_4 \times C_2$ and C_8. Inversion allows $A = 1$ or a^2, giving D_8 and Q_8. For $H = C_2^2$, the identity action gives C_2^3 or $C_4 \times C_2$ according as $A = 1$ or $A \neq 1$. Every nonidentity order-two automorphism of H is conjugate to a transvection. Its fixed subgroup equals the image of $1 + \alpha$, so (B.41.4) removes either allowable square. This gives one further oriented extension, D_8, with positive subgroup a four-group. These arguments prove all the extension counts $2, 2, 4, 3$ used in the computation.

First take a Klein bottle with one cone of order m. Its orbifold group is

$$\Lambda_m = \langle d_1, d_2 \mid (d_1^2 d_2^2)^m = 1 \rangle. \tag{B.41.5}$$

Put $y = d_1$, $z = d_1 d_2$. Then y is negative, z is positive, and a positive local cone generator is

$$x = (yzy^{-1}z)^{-1}. \tag{B.41.6}$$

A smooth orientation-compatible epimorphism to an oriented finite group G is therefore exactly a pair (Y, Z) such that Y is negative, Z is positive, $\langle Y, Z \rangle = G$, and the image of (B.41.6) has order m. Gromadzki–Szepietowski [25, Lemma 3.3] prove that the full outer automorphism group of Λ_m is generated by

$$(y, z) \longmapsto (yz, z), \qquad (y, z) \longmapsto (y^{-1}, z), \qquad (y, z) \longmapsto (y, z^{-1}). \tag{B.41.7}$$

Their proof identifies these maps with the Dehn twist and puncture and crosscap slides of the one-pointed Klein bottle. The automorphism assertion concerns Λ_m itself and does not impose their later nonorientable-kernel hypothesis. We apply that assertion to the orientation-compatible epimorphisms above; every generator in (B.41.7) preserves the orientation character. Quotienting the finite set of pairs also by $\mathrm{Aut}(G)$ preserving orientation is therefore exactly topological conjugacy of the surface actions.

For $H = C_3$ the cone order is three. In the S_3 extension a negative element inverts every positive element, so (B.41.6) is the identity. In C_6 it equals Z^{-2}; Z must generate the positive C_3. The first move in (B.41.7) runs through the entire negative coset, and the last move exchanges the two choices of Z. Thus there is one class, with full group C_6.

For $H = C_4$ the cone order is two. The negative elements of D_8 and Q_8 invert H, again making (B.41.6) trivial. In the two abelian extensions it is Z^{-2}, so Z must generate C_4. For $C_4 \times C_2$, multiplication of Y by powers of Z sets Y to the order-two generator of the other factor. For C_8, an automorphism sets the odd generator Y to a chosen generator, and inversion of Z identifies its two remaining possibilities. Each gives exactly one class.

For $H = C_2^2$, the elementary abelian extension C_2^3 cannot be generated by two elements. In the abelian extension $C_4 \times C_2$, every $Z \in H$ has square one, so (B.41.6) is trivial. Write the remaining extension as

$$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle, \qquad H = \langle r^2, s \rangle.$$

The admissible Z are $s, r^2 s$, and every negative Y gives the cone image r^2. An orientation-preserving automorphism exchanges the two choices of Z; the first move in (B.41.7) changes a negative reflection to a negative rotation, and inversion then identifies r and r^{-1}. All eight pairs are one orbit. This gives the final row in (B.41.1).

It remains to handle the two order-two cones in the first case of (B.41.2). For either extension $G = C_4$ or C_2^2, let h be the nonidentity positive element. Both cone generators must map to h. The two crosscap generators may independently map to either negative element. Every one of these four assignments is surjective and respects the relation

$$d_1^2 x_1 d_2^2 x_2 = 1. \tag{B.41.8}$$

They are all equivalent. Here is a geometric source move with its exact word comparison, so no unproved transitivity is needed. In a one-crosscap neighborhood of a cone, slide that cone once around the one-sided core. On the free generators (d, x), with exterior boundary word d^2x, the move is

$$d \longmapsto dx, \qquad x \longmapsto (dx)^{-1}x^{-1}(dx). \tag{B.41.9}$$

It preserves d^2x exactly and sends the cone generator to a conjugate of its inverse. It is an automorphism: first replace the free basis (d, x) by (dx, x), then invert and conjugate the second member by the first. Cutting the crosscap neighborhood along its core gives the stated cone slide; the unchanged exterior boundary permits extension by the identity to the other crosscap. In either abelian target, where x has order two, this replaces the image of d by its product with h and leaves the cone image unchanged. Apply it independently at the two cones to connect all four assignments. This gives one class for each of C_4 and C_2^2. Their different exponents distinguish the two full groups.

For each surviving monodromy the local cone groups inject. The orbifold covering subgroup is therefore torsion-free, and it lies in the orientation kernel by construction. The Klein-bottle orbifold Euler characteristic is $-\sum_j(1 - 1/m_j)$; multiplying it by $|G|$ gives -4 in all six cases. The covering surface is consequently closed, orientable, and of genus three. This proves existence and faithfulness of every listed action, as well as completeness.

There are no mirrors, so no reflection sections. For each cone generator x_j of prime order $m_j = 2$ or 3, the points of its critical orbit are exactly $G/\langle x_j\rangle$. At the coset represented by g, the stabilizer is $g\langle x_j\rangle g^{-1}$. Write $\varepsilon : G \to \{\pm 1\}$ for orientation. The multiplier map $\mu_{j,g}$ on the section represented by that point is

$$\mu_{j,g}(gx_j^k g^{-1}) = \varepsilon(g)\frac{k}{m_j} \pmod{\mathbb{Z}}. \tag{B.41.10}$$

Every such point supplies one nontrivial partial section, which is maximal; its centralizer in the surface group is trivial by B.3. No whole G is a local stabilizer, so there are no full sections. The resulting coset counts are exactly (B.41.1). The program records the permutations of these cosets, their stabilizers and the multiplier of every local element. In the C_6 row the two point generators have multipliers $1/3$ and $-1/3$ under the fixed positive generator; the negative coset exchanges the two.

The executed standard-library program `catalogue_genus3_klein.py` independently enumerates all pairs (B.41.3), checks associativity of their finite multiplication tables and the orientation homomorphism, enumerates every smooth generating monodromy, and computes its orbit under the complete source and target automorphism actions. It retains the unsuccessful oriented extensions with their empty monodromy lists, so the exclusions above are part of the reproducible computation. □

3 C. Equivariant comparison, drilling, and the topology of special groups

Throughout this part, $U = X \setminus S$ is connected, X is a closed orientable surface, and S is a finite set of cardinality ν. We put $A = \mathrm{Homeo}(X, S)$ with its compact-open topology and $K = \pi_0(A)$. Thus A permits both permutations of S and orientation reversal; A^0 fixes S pointwise and preserves orientation. We call the type hyperbolic when $2g - 2 + \nu > 0$. A superscript u means the centralizer of the image of the specified homomorphism $u : \Gamma \to A$. In particular, $(A^0)^u = A^0 \cap A^u$; it is not, by definition, the identity component of A^u. We distinguish these two groups until connectedness is proved.

3.1 The categories and their exact fidelity conditions

C.1. [Refuted] The unrestricted full target has too many arrows. The literal comparison from surfaces and homeomorphisms to the full 2-category of peripheral groupoid diagrams and arbitrary functors is not locally essentially surjective, already for $\Gamma = 1$ and type $(0, 3)$. The correct target for a realization biequivalence is the 2-groupoid of such diagrams with equivalences as 1-arrows.

Manuscript: §23; folios 134–136; scans 267–270.

Proof. For the three-punctured sphere put $\pi = \langle x, y \rangle$ and take the peripheral diagram

$$\coprod_{i=1}^{3} B\mathbb{Z} \longrightarrow B\pi, \qquad 1 \longmapsto x,\ y,\ (xy)^{-1}.$$

The identity on its three-object index set, the trivial homomorphism on each copy of $\mathbb{Z}$, and the trivial homomorphism $\pi \to \pi$ define a strictly commuting endomorphism of this diagram. Its map on $\pi_{\mathrm{ab}} = \mathbb{Z}^2$ is zero. A homeomorphism induces an automorphism on this abelianization. Natural isomorphisms between functors induce inner conjugations, which act trivially on abelianization. Consequently this endomorphism cannot become isomorphic to the image of a homeomorphism. This is a counterexample to the literal fullness assertion, not to its core version.

Here is the precise core version we use. An admissible surface is a finite disjoint union of finite-type orientable surfaces, each genus-zero component having at least one puncture, as on scan 264, folio 133. The source 2-groupoid has strict Γ-actions as objects, equivariant homeomorphisms as 1-arrows, and endpoint-preserving homotopy classes of equivariant isotopies as 2-arrows. An object of the target is a diagram of groupoids over $B\Gamma$

$$D_\Gamma \longrightarrow U_\Gamma \longrightarrow B\Gamma.$$

Its homotopy fibre over the base object is a peripheral diagram $\coprod_{i \in I} B\mathbb{Z} \to B\pi$ with the surface presentation $\prod_{j=1}^{g} [a_j, b_j] \prod_{i=1}^{\nu} c_i = 1$; the ith map sends 1 to c_i. The fibre condition includes the common orientation module and permits the noninjective map in type $(0, 1)$ and the repeated image in type $(0, 2)$. This is the condition M_{lac}. For $I = \varnothing$ we use the closed surface group, with $g \geq 1$.

A target 1-arrow is a pair of groupoid equivalences over $B\Gamma$ together with the invertible natural transformation expressing commutativity of the peripheral square. A 2-arrow is a pair of natural isomorphisms, with identity projection to $B\Gamma$, satisfying that square's compatibility. Composition is composition of functors with pasting of these specified natural transformations. Equivalently these are equivalences of the corresponding opfibrations over $[1] \times B\Gamma$ that preserve the displayed fibres. The topological comparison takes the fundamental groupoids of the homotopy quotients of a Γ-invariant punctured collar system and of U. Two collar systems have a common smaller system; the induced equivalences and natural transformations specify the independence of this choice. No arbitrary functor is silently replaced by an equivalence. □

C.2. [Proved] The coset category and the quotient correction. For a hyperbolic peripheral pair $(\pi; (L_i)_{i \in I})$, the category $\mathscr{C}$ with objects u, p_i, automorphism groups $\mathrm{Aut}(u) = \pi$, $\mathrm{Aut}(p_i) = 1$, and $\mathrm{Hom}(p_i, u) = \pi / L_i$ determines the pair up to an isomorphism and simultaneous inner conjugation. Its equivalence groupoid has components $\mathrm{Out}_{\mathrm{lac}}(\pi)$ and trivial automorphism groups. For an action with extension $1 \to \pi \to E \to \Gamma \to 1$, the corresponding category over $B\Gamma$ uses the bimodule E/L_i, with right action of $E_i/L_i = \Gamma_i$, not E/E_i.

Manuscript: §23; folios 133–136; scans 264–270.

Proof. The stabilizer in $\mathrm{Aut}(u)$ of a selected arrow $p_i \to u$ is L_i. Changing that arrow to aL_i conjugates its stabilizer by a. An equivalence fixes the unique object with nontrivial noncyclic endomorphism group and permutes the other objects. Its restriction at u is an automorphism α of π. Equivariance of its maps on Hom-sets says precisely that $\alpha(L_i)$ is conjugate to $L_{\sigma(i)}$. Such a map of transitive sets is determined by a coset whose stabilizer has this property. The equality $N_\pi(L_i) = L_i$ makes that coset unique. A natural transformation between two such equivalences has its component $a \in \pi$ at u, its identity components at the p_i, and imposes $\alpha' = \mathrm{Int}(a)\alpha$ and the corresponding multiplication of all cosets by a. An automorphism of an equivalence therefore lies in $Z(\pi) = 1$. This proves the assertion about the equivalence groupoid and gives both comparison constructions explicitly.

For the geometric exit interpretation, the space of exits from p_i to u is the homotopy fibre of the link map $BL_i \to B\pi$. Its components are π/L_i and its fundamental groups are the kernels of $L_i \to \pi$; its higher homotopy groups vanish. The peripheral injection makes those components contractible. Thus the ordinary category above has exactly these exit mapping spaces. This also identifies the exact failure of this ordinary model for the plane: the fibre of $B\mathbb{Z} \to *$ is $B\mathbb{Z}$, not a discrete space. More explicitly the Grothendieck category of $BL_i \to B\pi$ has automorphism groups L_i, π and forward Hom-set π. Its comparison with the exit category is identity at the generic group, trivial at the special endomorphism group, and $\pi \to \pi/L_i$ on forward arrows. Conversely the translation groupoid of the generic action on the exit Hom-set is $(\pi/L_i)//\pi \simeq BL_i$, with its map to $B\pi$. Thus the peripheral diagram is recovered by this explicit link construction; these two displayed categories are not silently identified. On classifying spaces the forward comparison collapses each link to its special point, realizing the filling map $B\pi \to B\pi \cup_{\coprod BL_i} \coprod *$.

In the action category take one special object for each Γ-orbit of punctures, choosing i in that orbit. The generic object has automorphism group E, the special object has automorphism group Γ_i, and composition on E/L_i is left multiplication by E and right multiplication by E_i/L_i. These actions commute. The homotopy fibre over the base object of $B\Gamma$ has special objects indexed by Γ/Γ_i for each chosen representative, and hence recovers every upstairs puncture. An equivalent unskeletonized model uses all punctures and the nonempty Γ-transporter sets between those in the same orbit. The projection of a morphism eL_i to $B\Gamma$ is the image $\bar{e} \in \Gamma$; it is well-defined because $L_i \subset \pi$. It is not well-defined on E/E_i when $\Gamma_i \neq 1$: representatives e and ea, $a \in E_i$, can have different images in Γ.

For a geometric example take $U = \mathbf{P}^1 \setminus \{0, \infty, 1, -1\}$, $\Gamma = C_2$ acting by $z \mapsto -z$, and quotient coordinate $w = z^2$. Then $E = \pi_1(\mathbf{P}^1 \setminus \{0, 1, \infty\}) = F_2$. With a a loop about $w = 0$, the upstairs inertia is $L_0 = \langle a^2 \rangle$ and $E_0 = \langle a \rangle$. The right C_2 action on $E/\langle a^2 \rangle$ is nontrivial, whereas replacing it by $E/\langle a \rangle$ erases that action. This is the exact morphism obstruction to the quotient-category inference in the prompt. □

C.3. [Proved] The four levels and the coefficient ladder. Let A be a locally path-connected topological group whose components are 1-types. For the comparison from strict actions to weak actions in the categorical group $\Pi_1 A$, the first three fidelity conditions are, on each comparison $M \to N$ of transport spaces,

0	$\pi_1 M \to \pi_1 N$ injective at every point,
1	$\pi_0 M \to \pi_0 N$ injective and $\pi_1 M \to \pi_1 N$ isomorphic,
2	$\pi_0 M \to \pi_0 N$ bijective and $\pi_1 M \to \pi_1 N$ isomorphic.

Level 3 adds essential surjectivity on action objects. When A^0 is simply connected, these give exactly scans 279–280: vanishing of $\pi_1((A^0)^u)$; connectedness of $(A^0)^u$; conjugacy of two lifts of the same $\Gamma \to K$ by an element of A^0; and existence of a lift.

Manuscript: §23; folios 136–137; scans 271–273; §23; folios 140–141; scans 279–281.

Proof. The source Hom-groupoid is $\Pi_1 M$. A functor of groupoids is faithful exactly when it is injective on the automorphism group at each object. It is fully faithful exactly when that condition is an isomorphism and two source objects whose images are connected are already connected. It is an equivalence exactly when it is fully faithful and every target component is hit. Applying these three elementary tests proves the table, including the cumulative meaning of each level.

When $\pi_1(A^0) = 0$, the categorical group $\Pi_1 A$ is equivalent to the discrete group K. For strict homomorphisms $u, v : \Gamma \to A$ the map of transport spaces is

$$\{a \in A : au(\gamma)a^{-1} = v(\gamma)\ (\gamma \in \Gamma)\} \longrightarrow \{k \in K : k\bar{u}(\gamma)k^{-1} = \bar{v}(\gamma)\}.$$

A nonempty source is a torsor under A^u. Its fibres over points of the target are torsors under $(A^0)^u$. The first two lines of the table are therefore the first two conditions stated. For the third, given a target transporter k, choose $a \in A$ representing k. The two actions aua^{-1} and v have the same projection to K; conjugating them by an element of A^0 produces the prescribed transporter. Conversely, apply transporter surjectivity to the identity element of K. This also verifies the stronger marginal correction on scan 280.

For comparison with the GALOIS ladder, first take the 1-truncations of the transport spaces; fixed spaces are not assumed to inherit the ambient group's higher truncation. On a map of 1-types $M \to N$, set-valued constant coefficients test bijectivity on π_0; full faithfulness of restriction tests that bijectivity together with surjectivity on π_1; equivalence tests bijectivity on π_0 and an isomorphism on π_1. For groupoid-valued coefficients all three tests coincide with a weak equivalence, because $\pi_i M = \pi_i N = 0$ for $i \geq 2$; the same is true for 2-type coefficients. These assertions follow by applying the long exact sequence to the homotopy fibres: an r-connected fibre gives isomorphisms through degree r and surjectivity in degree $r + 1$. Thus level 2 is the $r = 0$ equivalence test on the transport spaces. Levels 0 and 1 do not equal a rung in that ladder: their π_1-injectivity and π_0-injectivity tests have been written out above, whereas each rung of the coefficient ladder demands a component bijection. Level 3 concerns objects rather than this map of Hom-spaces. This is a comparison of the specified maps, not an identification based only on the words "faithful" and "equivalence". □

C.4. [Proved] Coherent actions and rectification on spaces. For a discrete group P, the space of self-homotopy equivalences of BP has components indexed by $\mathrm{Out}(P)$, each with homotopy type $BZ(P)$. In particular, for the centre-free surface groups the map from outer actions to coherent actions is an equivalence of action groupoids. Every coherent action of a discrete group on a space has a strict realization on a space of the same homotopy type. For $P = \mathbb{Z}$ or $\mathbb{Z}^2$, the obstruction class in degree 3 is zero and the coherent actions with a specified linear action are classified by $H^2(\Gamma, P)$.

Manuscript: §23; folios 131–132; scans 261–263; §23; folios 135–141; scans 268–281.

Proof. The groupoid of functors $BP \to BP$ has homomorphisms $P \to P$ as objects and conjugating elements of P as arrows. Its equivalence subgroupoid has automorphisms as objects. The stabilizer of an automorphism is $Z(P)$, and the orbits are $\mathrm{Out}(P)$. The nerve computes the mapping space because nerves of groupoids represent 1-types. Delooping the resulting categorical group has $\pi_1 = \mathrm{Out}(P)$, $\pi_2 = Z(P)$ and no higher homotopy groups. Its Postnikov class is the usual extension obstruction in $H^3(\mathrm{Out}(P); Z(P))$. For P abelian the map $\mathrm{Aut}(P) \to \mathrm{Out}(P)$ is the identity and gives a strict section; the categorical group is $BP \rtimes \mathrm{Aut}(P)$, so this Postnikov class vanishes. A choice of coherence is then a 2-cocycle with values in the specified Γ-module P; changing it by a coherent equivalence changes the cocycle by a coboundary. This proves the stated classification without asserting a nonzero obstruction for either abelian surface group.

For rectification, represent a coherent action by a fibration $p : Y \to B\Gamma$ with the prescribed homotopy fibre. Pull it back along $E\Gamma \to B\Gamma$. The pullback $Y' = E\Gamma \times_{B\Gamma} Y$ has a strict deck action of Γ. Its projection to the contractible space $E\Gamma$ has fibre the original homotopy type, so the inclusion of any chosen fibre is a homotopy equivalence. Its homotopy quotient recovers $Y \to B\Gamma$. This construction applies simultaneously to peripheral diagrams by the diagrammatic pullback. It produces a space model; the additional demand that the model be the original surface is exactly the realization problem addressed below. For continuous profinite actions the objects and the homotopy quotient must instead be specified in a continuous/profinite homotopy theory; the preceding construction makes no untyped transfer to that setting. □

3.2 Finite actions on hyperbolic surfaces

C.5. [Proved] Existence, with punctures and orientation reversal. For a hyperbolic type and every finite group Γ, each homomorphism $\rho : \Gamma \to K$ lifts to a homomorphism $u : \Gamma \to A$. Its image can be chosen to consist of isometries of a complete finite-area hyperbolic metric on U.

Manuscript: §23; folios 136, 140–141; scans 270–271, 279–280; §24; folio 147; scan 293.

Proof. Let $H = \rho(\Gamma)$, a finite subgroup of the extended mapping class group. Kerckhoff's Theorem 5, together with the explicit extension in §IV, pp. 255–256, of the original paper [36], gives a fixed hyperbolic structure for H. That extension explicitly permits cusps; the paper's mapping-class convention explicitly includes orientation-reversing maps. We use the original paper, rather than inferring the punctured case from the announcement.

Choose the isometry in each class of H. It is unique. Indeed an isometry isotopic to the identity has a lift to $\mathbb{H}^2$ commuting with the surface deck group. That deck group contains two hyperbolic elements with different endpoint pairs, so an isometry commuting with both fixes at least three points of the ideal boundary and is the identity. The same argument covers a finite-area punctured surface; its non-elementary deck group again contains those elements. Products of the chosen isometries represent products in H, and uniqueness therefore makes the choices a homomorphism $H \to A$. Composing with ρ proves the claim. The isometries extend to the end compactification and permute S, so the target really is the specified group A. □

C.6. [Proved] Uniqueness up to the identity component. Let $u, v : \Gamma \to A$ be homomorphisms with Γ finite, U hyperbolic, and $\bar{u} = \bar{v} : \Gamma \to K$. There is an $h \in A^0$ such that $hu(\gamma)h^{-1} = v(\gamma)$ for every γ. This holds for topological actions without a smoothness assumption.

Manuscript: §23; folios 136, 140–141; scans 270–271, 279–280.

Proof. We first justify the analytic model for either action. For a point $x \in U$ and its finite stabilizer H, lift each element of H to the universal cover with the chosen lift of x fixed. Uniqueness of a lift with a specified value makes these lifts a finite group. Compactifying $\widetilde{U} \simeq \mathbb{R}^2$ by one point gives a finite action on S^2 with two common fixed points. Kolev's compact-sphere theorem [39, Theorem 1.2] conjugates that action into $O(3)$. A subgroup of $O(3)$ fixing two distinct points is contained in the orthogonal group of the perpendicular plane; choosing a sufficiently small round disc at the first point and pulling it back yields a common invariant disc with an $O(2)$ action. Choosing it small enough makes its translates by distinct cosets of H disjoint. At a puncture we apply the same finite-stabilizer disc argument on the compactification, as in the finite-disc theorem proved from the same source.

The quotient consequently has ordinary charts, cone charts, mirror charts, and corner charts with their finite cyclic or dihedral local groups. A finite triangulation respecting these strata supplies a smooth orbifold structure: smooth each triangle, use collar charts along its edges, and

use the displayed linear quotient charts at its vertices. Pulling the charts to X makes the original finite action smooth for a compatible smooth structure. Average a metric and use its conformal class. Uniformization of the punctured conformal surface gives a finite-type structure for which each operator is conformal or anticonformal. Thus we have conformal structures c_u, c_v on the same marked topological surface.

The extremal quasiconformal map $h : (U, c_u) \to (U, c_v)$ in the identity isotopy class is unique, by the finite-type Teichmüller theorem [4, §3(A)–(D)]. For each γ, the map $v(\gamma)^{-1}hu(\gamma)$ is orientation preserving: $u(\gamma)$ and $v(\gamma)$ have the same orientation sign. Pre- and postcomposition by these conformal or anticonformal maps leaves the maximal dilatation unchanged. The equality of the two mapping-class homomorphisms says that this map is isotopic to the identity, with the punctures retained. Uniqueness therefore gives $v(\gamma)^{-1}hu(\gamma) = h$. The quasiconformal map extends across the finite punctures; hence $h \in A^0$ and has the asserted conjugacy property. □

C.7. [Proved] The finite centralizer component, including mirrors. For a finite action u on a hyperbolic U, the group $(A^0)^u$ is contractible. In particular it is the identity component of A^u.

Manuscript: §§23–24; folios 140–144; scans 279–286.

Proof. Replace Γ by its image H. The preceding local construction makes H a group of hyperbolic isometries. Let π be the surface deck group and let E be the full group of lifts of H to $\mathbb{H}^2$. It is a discrete group of isometries, possibly with reflections; its quotient Q is a finite-type hyperbolic orbifold. Write D for descent of a centralizing homeomorphism to the labelled underlying surface of Q, with cone points, corner points, mirror boundary, and punctures retained.

We give the injectivity argument at the place where an orientation-preserving-only Birman–Hilden theorem would not suffice. For $f \in (A^0)^u$, the identity isotopy supplies a lift $\widetilde{f}$ that commutes with π. Since f commutes with H, conjugation by $\widetilde{f}$ induces an automorphism α of E that is the identity on π. For $e \in E$ and $a \in \pi$,

$$\alpha(e)a\alpha(e)^{-1} = \alpha(eae^{-1}) = eae^{-1}.$$

Thus $e^{-1}\alpha(e)$ centralizes π, and hence is the identity by the distinct-endpoint argument in C.5. Consequently $\alpha = \mathrm{id}_E$. The orbifold Dehn–Nielsen–Baer injectivity theorem, with its mirror and corner structure, makes $D(f)$ isotopic to the identity through homeomorphisms preserving those structures; the relevant precise form is [38, Theorem 2 and Proposition 3]. At the algebraic step in their Lemma 3 we have used the calculation just written: centralizers of individual reflections need not be cyclic, but the centralizer of the non-elementary surface subgroup is trivial. Thus that inapplicable sentence of their proof is not being used. There is an orientation hypothesis in that source when the labelled underlying quotient is orientable, and we verify it here. The canonical π-equivariant lift $\widetilde{f}$ extends by the identity on $\partial_\infty\mathbb{H}^2$. For a compact surface this follows from bounded displacement on a compact fundamental domain. For a punctured surface use a compact truncation and disjoint invariant horodiscs: displacement is bounded on the truncated part, and preservation of each end sends sufficiently deep cusp horodiscs into the corresponding cusp horodiscs. A sequence escaping through distinct horodiscs has the same ideal limit before and after the map, since their visual diameters tend to zero; a sequence in one horodisc has its cusp endpoint as its only ideal limit. These observations prove the claimed continuous extension. Because $\alpha = \mathrm{id}$, $\widetilde{f}$ preserves each reflection axis. It fixes that axis's two ideal endpoints individually, so preserves its linear orientation, and, being orientation preserving on $\mathbb{H}^2$, preserves its two sides. It therefore preserves every mirror chamber. On such a chamber the quotient orientation is the one induced from $\mathbb{H}^2$; hence $D(f)$ preserves it. Without mirrors an orientable quotient is covered orientably and the same assertion is immediate. Thus the theorem is applied inside its stated homeomorphism group.

Let $\mathcal{H}(Q)^0$ denote the identity component of that labelled homeomorphism group. Every path starting at its identity lifts uniquely, starting at the identity upstairs, to a path centralizing H. Lift first on the regular covering locus. The local cyclic and dihedral models extend the lift over cone points, corners and mirrors; continuity of the extended lift follows in the charts from the finite quotient map. The same construction for a parameter family proves the path and homotopy lifting properties. The only ambiguity for an endpoint is a central deck transformation. Such a deck transformation cannot belong to A^0 unless it is the identity, by C.5's isometry argument. It follows that descent identifies $(A^0)^u$ with $\mathcal{H}(Q)^0$, as soon as the latter is simply connected; the construction is continuous in local isotopy charts, so this is an isomorphism of topological groups.

We check that this group is contractible, including the small underlying quotients. End-compactify the quotient and retain its finitely many interior and boundary marked points. Yagasaki's compact-pair Lemma 2.1 [60] gives contractibility outside its explicit exceptional pairs. A marked point on a boundary component is covered by Lemma 2.1(ii). With no such point, the exceptional disk has at most one interior marked point, the annulus and Möbius band have none, the sphere has at most two, the projective plane at most one, and the torus and Klein bottle have none. Each of these has nonnegative orbifold Euler characteristic: an interior cone subtracts $1 - 1/m$, an interior puncture subtracts 1, and a corner or boundary puncture would already be a boundary marked point. But $\chi_{\mathrm{orb}}(Q) = \chi(U)/|H| < 0$. None occurs. The homeomorphism group with labels has the same identity component as the group fixing all these finite marks pointwise. Yagasaki's theorem therefore applies. This proves contractibility, and also supplies the simple connectedness used in the lifting construction. □

C.8. [Proved] The finite core biequivalence and homotopy fixed points. On hyperbolic surfaces, for every finite Γ the corrected comparison of C.1 has all four fidelity levels and is a biequivalence. For a specified action u, the comparison

$$A^u \longrightarrow A^{h\Gamma} = \mathrm{Map}_\Gamma(E\Gamma, A), \qquad a \longmapsto (\text{constant map with value } a),$$

where Γ acts on A by conjugation through u, is a weak homotopy equivalence. In standard CW models it is a homotopy equivalence.

Manuscript: §23; folios 136–141; scans 270–281.

Proof. The nonequivariant peripheral comparison identifies K with $\mathrm{Out}_{\mathrm{lac}}(\pi)$ and has trivial 2-automorphism groups, by the peripheral Dehn–Nielsen–Baer theorem [20, Theorems 8.1 and 8.8] and $Z(\pi) = 1$. C.7 gives levels 0 and 1, C.6 gives level 2 by the transporter argument of C.3, and C.5 gives level 3. These arguments also treat a nonfaithful homomorphism: its finite image is the action group, and a finite-order homeomorphism in A^0 is the identity. Disconnected admissible surfaces reduce to representatives of the finitely many orbits of connected components and their stabilizers, with the stated transporters providing the arrows between choices. This is the desired biequivalence on its hyperbolic part.

The map $A \to K$ is Γ-equivariant. Its fibres are contractible by the same homeomorphism-group theorem with $H = 1$. Taking derived homotopy fixed points gives a weak equivalence $A^{h\Gamma} \to K^\Gamma = C_K(\bar{u}(\Gamma))$, the latter a discrete space. The constant-map comparison followed by this map is exactly $\pi_0(A^u) \to C_K(\bar{u}(\Gamma))$. Its surjectivity follows by applying C.6 to aua^{-1} and u for any representative a of a centralizing mapping class; its injectivity and its contractible fibres are C.7. Thus the displayed map is a weak equivalence. The source components have CW homotopy type by the ANR statement for quotient homeomorphism groups in [60]; taking the usual CW model of the derived target turns it into a homotopy equivalence. We state the model convention because scan 281 writes "homotopisme" without specifying an infinite-dimensional mapping-space model. □

3.3 The exceptional types and infinite operator groups

C.9. [Proved] Corrected abelian tests and a torus calculation. For $\pi_1(A^0) \neq 0$, replacement of the four tests by connectedness and uniqueness of lifts of $\Gamma \to K$ loses the coherent action data. The valid tests are C.3 with weak transport spaces. At a specified strict action the target automorphism space is the 1-type $A^{h\Gamma}$ and

$$\pi_1(A^{h\Gamma}, 1) = \pi_1(A, 1)^{\Gamma}.$$

For $A = \mathrm{Homeo}(T^2)$ and $u(C_2) = \{1, -1\}$, the group $(A^0)^u$ has four components, all contractible, whereas the right-hand invariant fundamental group is zero. Thus the naive connectedness test fails; the corrected comparison does not fail in this example.

Manuscript: §23; folio 141; scans 280–281.

Proof. The homotopy-fixed-point groupoid of a 1-type has loops equal to the fixed automorphisms of its base object, proving the displayed equality. Its components include the $H^1(\Gamma; \pi_1 A)$ coherence data, with possible obstruction for an individual component of A. Hence it is not legitimate to test only $\pi_0 A$ when $\pi_1 A \neq 0$.

For the torus, evaluation at 0 gives $A^0 \cong T^2 \times A^0_{\{0\}}$ by $f \mapsto (f(0), t_{-f(0)} f)$. The second factor is contractible by the homeomorphism-group theorem for a torus with a marked point. In the centralizer of -1, $f(0)$ is one of the four points of $T^2[2]$. Each is realized by translation. In the fibre $f(0) = 0$, regarding the torus as once punctured makes it hyperbolic; the fibre is contractible by C.7. This proves the four-component assertion. Conjugation by -1 acts as $-I$ on $\pi_1(A^0) = \mathbb{Z}^2$, so its invariants vanish. The inclusion of translations $T^2 \hookrightarrow A^0$ is equivariant and a weak equivalence; its homotopy fixed points have components $H^1(C_2; \mathbb{Z}^2_-) = (\mathbb{Z}/2)^2$ and zero higher homotopy groups. The four translations just identified realize exactly these four components.

For the sphere, the peripheral groupoid diagram is empty over $*$ and cannot encode its homotopy type; $\pi_2(S^2) = \mathbb{Z}$ already records the failure of asphericity. The plane, annulus and torus have $A^0 \simeq S^1, S^1, T^2$, respectively; in the plane the peripheral map $B\mathbb{Z} \to *$ retains the circle of automorphism 2-arrows, and in the annulus the two ends remain indexed even though their subgroup images coincide. These exact maps explain why collapsing to a bare outer group would erase data. The complete finite-action comparison for these models, including its parametric centralizer proof, is C.10. □

C.10. [Proved] Every finite action in the exceptional aspherical types. For $U = \mathbb{R}^2$, $S^1 \times \mathbb{R}$, or T^2, and every finite strict action, the fixed-to-homotopy-fixed comparison in C.8 is a weak equivalence and is a homotopy equivalence on CW models. The corrected peripheral 2-groupoid comparison has all four fidelity levels for these types. The tests are those of C.3; ordinary outer actions alone do not retain the required coherence classes.

Manuscript: §23; folios 136–141; scans 270–281.

Proof. We prove the parametric assertion, not merely classification of individual actions. Write respectively

$$G = O(2), \qquad G = (\mathbb{R}/\mathbb{Z}) \rtimes (C_2 \times C_2), \qquad G = (\mathbb{R}^2/\mathbb{Z}^2) \rtimes \mathrm{GL}_2(\mathbb{Z}).$$

The annulus element (a, ϵ, η) sends (θ, t) to $(\epsilon\theta + a, \eta t)$; the first sign acts on the circle factor and the second on the two ends. In every case $G \to A$ is a weak equivalence. Here is a direct check of the identity components. On the torus, evaluation at 0 splits A^0 as T^2 times its point-fixing subgroup. An identity isotopy can be made point fixing by postcomposing at time s with

translation by the negative of the trace of 0. Thus that subgroup is its own identity component, and is contractible by the punctured-torus case of the homeomorphism-group theorem [60, Theorem 1.1 and Lemma 2.1]. The same adjustment on the annulus uses angular rotation and translation of its real coordinate. Its point-fixing subgroup is the identity group of a hyperbolic three-ended sphere and is contractible, so $A^0 \simeq S^1$. On the plane first fix 0 by translations and then fix 1 using multiplication by a nonzero complex number. The remaining identity group fixes $0, 1, \infty$ and is contractible; hence $A^0 \simeq S^1$ with generator the rotation circle. The component groups are respectively C_2, $C_2 \times C_2$, and $\mathrm{GL}_2(\mathbb{Z})$: orientation and end signs give the first two, and the torus action on its lattice gives the third. Cutting the plane or annulus along a ray or spanning arc verifies the stated kernels; the torus case is the genus-one Dehn–Nielsen–Baer theorem.

By the simultaneous finite normal forms proved in B.1 and B.14, conjugate the given finite image H into G. We will prove

$$C_G(H) \longrightarrow C_A(H) \tag{C.10.1}$$

is a weak equivalence by examining its fibres. Conjugation by the chosen homeomorphism transports both the fixed and derived-fixed comparisons, so this reduction is an isomorphism of the actual diagrams.

Torus displacement. For $f \in C_A(H)$ let $M \in \mathrm{GL}_2(\mathbb{Z})$ be its lattice action and lift it as $F(x) = Mx + d(x)$, with d periodic. Put $b = \int_{T^2} d(x)\,dx$ and $g(x) = Mx + b$ modulo $\mathbb{Z}^2$. Changing the lift changes b by an integral vector and leaves g unchanged. For $\gamma(x) = A_\gamma x + t_\gamma$ the commutation equation is

$$d(A_\gamma x + t_\gamma) - A_\gamma d(x) = (I - M)t_\gamma + k_\gamma, \qquad k_\gamma \in \mathbb{Z}^2.$$

Integration preserves its left-hand change of variables, because the affine operator preserves Haar measure. It gives the same equation with d replaced by b. Thus g commutes with H and $g^{-1}f \in A^0$ has a unique lift of zero mean displacement. Consequently multiplication is a homeomorphism of spaces

$$C_G(H) \times \mathcal{K} \;\xrightarrow{\sim}\; C_A(H),$$

where $\mathcal{K}$ consists of these zero-mean maps. The averaging map need not be a group homomorphism; only this explicit space factorization is used. A zero-mean lift commutes exactly with every lift of H: the possible integral commutator above has zero average and is therefore zero.

First divide by the finite translation subgroup $N = H \cap T^2$. If a represents a member of N of order m, commutation implies $F(x + a) = F(x) + a + k$. Iterating m times and using $ma \in \mathbb{Z}^2$ forces $k = 0$ for a zero-mean identity lift. Hence its displacement is periodic for the enlarged covering lattice. Descent and lift identify $\mathcal{K}$ with the same zero-mean space on $T' = T^2/N$ for H/N. This quotient group has faithful linear part, a finite subgroup P of $\mathrm{GL}_2(\mathbb{Z})$.

When P contains a nonidentity positive operator, its positive subgroup K is cyclic, generated by an affine rotation with $I - A$ invertible. It has a nonempty finite fixed set R on T', stable under H/N. A zero-mean lift commutes with each lift of that rotation and fixes its unique Euclidean fixed point; thus it fixes every point of R. Choose $p \in R$. The point-fixing argument above places our map in $\mathrm{Homeo}(T', p)^0$. C.7, applied to the cyclic action on $T' \setminus \{p\}$, supplies an equivariant identity isotopy. This isotopy fixes all of R, since R is finite. Our map therefore belongs to $\mathrm{Homeo}(T', R)^0$. Apply C.7 again, now to the full H/N action on the hyperbolic $T' \setminus R$. Its identity centralizer is exactly $\mathcal{K}$: conversely an equivariant identity isotopy relative to R has commuting identity lifts, whose mean is fixed by A and hence is zero. This proves contractibility of $\mathcal{K}$.

The remaining linear groups are $P = 1$ and a group of order two generated by a reflection. For $P = 1$, the space factorization on the identity component is $\mathrm{Homeo}(T')^0 \cong T' \times \mathcal{K}$. The first factor is a section of the mean map. Since the left side has the homotopy type T', the mean map

is an isomorphism on π_1 and $\mathcal{K}$ is weakly contractible. It is a retract of the ANR on the left, so it has CW homotopy type and is contractible.

For a reflected or glide-reflected torus the underlying quotient is an annulus, a Möbius band, or a Klein bottle. This exhausts the cases: the action reverses orientation, has no isolated rotation point, and the quotient has Euler characteristic zero; its boundary consists of mirrors. A zero-mean lift F commutes exactly with the Euclidean orbifold group E. Each lifted reflection line is preserved, and bounded displacement prevents reversal of its infinite directions. In particular the compact annulus quotient's two boundary circles cannot be interchanged. The explicit straight homotopy

$$F_s(x) = (1-s)F(x) + sx$$

is E-equivariant, because every element of E is affine. On each reflection line it stays on that same line. It therefore descends to a homotopy of the quotient as a surface with boundary, preserving its boundary throughout. Its quotient domain is compact, so it is proper. Epstein's Theorem 6.3, with the topological Appendix [18], applies with exactly this proper, boundary-preserving homotopy and gives an identity isotopy. This also handles the boundary-free Klein bottle; Theorem 5.8 of that source separately verifies the Möbius-band case. Thus no unverified inference from an abstract outer automorphism to an exceptional-surface isotopy is needed here.

Descent consequently puts $\mathcal{K}$ in the identity group of the labelled quotient. Every identity isotopy there lifts; among the two lifts exactly one preserves the orientation upstairs. This identifies the identity component of $C_{A^0}(H/N)$ with the quotient identity homeomorphism group. By Yagasaki's actual exception list its homotopy type is S^1 in all three cases. The mean factorization on that component is $T_0'^P \times \mathcal{K}$, with $T_0'^P \cong S^1$. Here P acts linearly on the translation torus; this fixed translation circle is present even for an affine glide with no fixed point. The section by translations makes the mean map surjective on π_1; a surjection $\mathbb{Z} \to \mathbb{Z}$ is an isomorphism. Hence $\mathcal{K}$ is weakly contractible, and is contractible by the same ANR-retract argument. This completes (C.10.1) for the torus, including free glide reflections.

Annulus displacement. Lift $f \in C_A(H)$ to $F(\theta,t) = (\epsilon_f\theta + d(\theta,t), F_2(\theta,t))$ and put $b = \int_0^1 d(\theta,0)\,d\theta$. Define $g(\theta,t) = (\epsilon_f\theta + b, \eta_f t)$, where η_f is its end sign. The commutation equation, integrated on the invariant core $t = 0$, proves that g commutes with H. It gives the same continuous factorization $C_A(H) \cong C_G(H) \times \mathcal{K}$, with zero-mean identity lifts in $\mathcal{K}$. Divide first by the finite rotation subgroup. The resulting faithful sign group is a subgroup of $C_2 \times C_2$.

When it contains $(-1,-1)$, that positive involution has two fixed points. Its subgroup is normal. A zero-mean lift fixes each of the corresponding fixed points in $\mathbb{R} \times \mathbb{R}$. The same two applications of C.7 as in the torus argument, first marking one fixed point and then the whole invariant fixed set, identify $\mathcal{K}$ with a contractible centralizer on a hyperbolic punctured annulus. An equivariant identity lift in that punctured model has mean zero because its angular linear sign is -1, proving equality in both directions.

A subgroup without $(-1,-1)$ is trivial, generated by $(-1,1)$, or generated by $(1,-1)$. In the first case the factorization and $A^0 \simeq S^1$ give contractibility as above. An angular reflection has a strip as quotient, with its two mirror boundary lines and two ends. Exact commutation of a zero-mean lift preserves both lines and both ends. The same straight homotopy in $\mathbb{R} \times \mathbb{R}$ is equivariant for all the affine covering operators and preserves the boundary lines. It is proper on the strip quotient: for $t \to +\infty$ both t and $F_2(\theta,t)$ tend uniformly to $+\infty$ on a compact angular fundamental interval, and for $t \to -\infty$ both tend to $-\infty$. Every convex combination has the same uniform property. Epstein's proper boundary-homotopy theorem therefore places the descended map in the identity group. That group is contractible by Yagasaki, so $\mathcal{K}$ is contractible. Finally a radial reflection or radial glide reflection has quotient a half-open annulus or a punctured projective plane, respectively. The straight homotopy is again equivariant; it preserves the fixed mirror line in the reflection case and has the just-verified properness at the quotient's end. Epstein's Theorem 6.3, or Theorem 5.8 for the open Möbius band, gives the

required identity isotopy. Descent and isotopy lifting identify the upstairs centralizer identity component with that group. Yagasaki gives S^1 in both cases. The mean factorization has a section by the fixed angular circle, hence again forces $\mathcal{K}$ to be contractible. These cases include every affine finite action and retain the end permutation throughout.

Plane centralizers. For a cyclic rotation group of order $n > 1$, a commuting map fixes 0. Descent through $z \mapsto z^n$ is a covering of the orientation-preserving centralizer onto $\mathrm{Homeo}^+(\mathbb{R}^2, 0)$, with deck group C_n. The base is connected with homotopy type S^1. A loop of one rotation downstairs lifts from the identity to the deck rotation by $2\pi/n$. Its monodromy is therefore onto C_n, so the covering centralizer is connected and the upstairs rotation circle induces the isomorphism on π_1; all higher homotopy groups vanish. A negative map centralizes the rotation only for $n = 2$, as its action on the punctured-plane fundamental group inverts the deck generator. In that case conjugation supplies the second component. These are precisely the components and circle types of $C_{O(2)}(C_n)$.

For a reflection, descent has half-plane quotient. A commuting map has two independent signs: its orientation on the fixed line and its orientation on the plane. The four diagonal orthogonal maps realize them. Each fibre is the identity homeomorphism group of the half-plane, which is contractible by Yagasaki. For a dihedral group with rotation order $n > 2$, a centralizing map preserves plane orientation and each reflection line. Preservation of cyclic order of their rays permits only the identity permutation or simultaneous reversal of every ray; these are realized by I and $-I$. For $n = 2$ the two perpendicular reflection lines permit four independent ray-sign choices, realized by the four diagonal orthogonal maps. After dividing by the matching map, descent is to a sector homeomorphism fixing its vertex and both boundary rays. This group is contractible: the sector is a half-plane with a marked boundary point, and its identity group is covered by Yagasaki's boundary-point case. The boundary rays also make the lift unique. The trivial group case was verified at the start. This proves (C.10.1) for every finite subgroup of $O(2)$.

It remains to identify the derived comparison, including all components. The following elementary torus calculation does so for each of our three G. Let $T = V/\Lambda$, with V real and Λ a lattice, and let a finite group act on a T-torsor by affine maps. Its translation part is a cocycle $b \in Z^1(\Gamma, T)$, where T and Λ have the specified linear Γ-module structures. The homotopy-fixed space is the section space of its Borel fibration. Its obstruction is the class $\delta[b] \in H^2(\Gamma, \Lambda)$ of the extension obtained by lifting the affine maps to V. Averaging cochains gives $H^j(\Gamma, V) = 0$ for $j > 0$, so $\delta : H^1(\Gamma, T) \to H^2(\Gamma, \Lambda)$ is an isomorphism. Thus a homotopy fixed point exists exactly when b is a coboundary, which is exactly existence of an actual fixed point. After choosing one,

$$\pi_0(T^{h\Gamma}) = H^1(\Gamma, \Lambda), \qquad \pi_1(T^{h\Gamma}) = \Lambda^\Gamma,$$

and there are no higher groups. The exact sequence

$$0 \to \Lambda^\Gamma \to V^\Gamma \to T^\Gamma \to H^1(\Gamma, \Lambda) \to 0$$

identifies the actual fixed components and their fundamental groups with exactly these groups. The constant-map comparison induces these identifications. Apply this to every invariant component of G, each of which is just such an affine torus torsor. Noninvariant components contain neither fixed nor homotopy-fixed points. It follows that $G^\Gamma \to G^{h\Gamma}$ is a weak equivalence. The equivariant weak equivalence $G \to A$, together with (C.10.1), now proves the claimed comparison for A by the commutative square of fixed and derived-fixed spaces.

Finally BG is the split 2-type with fundamental group $\pi_0 G$, second group $\Lambda = \pi_1 G$, and zero Postnikov invariant. Coherent finite actions with a specified homomorphism to $\pi_0 G$ are therefore classified by $H^2(\Gamma, \Lambda)$. The same connecting isomorphism realizes each by a T-valued cocycle, hence by a strict homomorphism $\Gamma \to G$. It also identifies coherent equivalences in the identity component with coboundaries, namely conjugacies by translations. The fixed-space calculation supplies their full transport-groupoid comparison; a prescribed nonidentity transporter

component is treated by composing with its representative in G. Thus the corrected four fidelity levels all hold. We finish by checking that this is the actual peripheral comparison, not just an abstract replacement of A. For the plane the diagram is $B\mathbb{Z} \to *$. Its equivalence 2-group has component group $\{\pm 1\}$, 2-automorphism group $\mathbb{Z}$, and sign action. For the annulus use $B\mathbb{Z} \amalg B\mathbb{Z} \to B\mathbb{Z}$ with the two maps $+1, -1$. An equivalence is specified on components by a generic sign ϵ and an end permutation of sign η. Its peripheral sign is forced to be $\epsilon\eta$. The two square conjugators can be removed by peripheral natural transformations; the surviving automorphism of such a diagram equivalence is one diagonal $\mathbb{Z}$, on which the generic sign ϵ acts. Thus its 2-group has exactly the two-sign component group and lattice action of the annulus G displayed above. For the torus, the empty peripheral diagram over $B\mathbb{Z}^2$ has component group $\mathrm{GL}_2(\mathbb{Z})$ and 2-automorphism group $\mathbb{Z}^2$ with its standard action. In all three cases strict diagram functors with zero square-conjugators give a section, so the Postnikov class is zero. The actual comparison maps a full plane or annulus rotation to the generator traced by a peripheral or generic basepoint, and torus translation loops to the two lattice generators. It is therefore an isomorphism on these specified π_1 groups, as well as on components. This proves that the models and comparisons used throughout the proof are the ones induced by C.1's peripheral functor. In particular the annulus end permutation has never been forgotten. □

C.11. [Proved] Infinite cyclic failures at levels one and two. For every hyperbolic surface containing a disc, there is a strict $\mathbb{Z}$-action for which $(A^0)^u$ is disconnected. There also are two strict $\mathbb{Z}$-actions with the same mapping-class homomorphism that are not conjugate. Thus finite-group hypotheses cannot be removed from levels 1 and 2.

Manuscript: §23; folios 140–141; scans 279–281.

Proof. In a coordinate disc choose two disjoint small round discs centered at $q_\pm = \pm 1$. Define f to rotate each small disc by angle $\sqrt{2}(1 - r/r_0)$ at radius $r < r_0$, and to be the identity elsewhere. This is a homeomorphism isotopic to the identity through the same formula with the angle multiplied by a parameter. The two centres are isolated fixed points; all other fixed points lie outside the interiors of these discs. At radius $r_0/2$ the rotation angle is $\sqrt{2}/2$, whose ratio to 2π is irrational, so f has infinite order. Let h rotate the larger disc by angle π on a concentric disc containing the two small discs, tapering continuously to angle zero before the outer boundary. It is also isotopic to the identity. It exchanges the two small discs by rigid rotation and therefore commutes with f.

Set $u(1) = f$. Every homeomorphism commuting with f preserves its fixed set and sends isolated fixed points to isolated fixed points. Along a path in the centralizer the induced permutation of $\{q_-, q_+\}$ is constant. The identity fixes both and h exchanges them. They therefore belong to different components of $(A^0)^u$. This proves the level-1 failure by the exact test C.3. The action u and the trivial $\mathbb{Z}$-action both project to the trivial homomorphism into K, but a conjugate of $f \neq \mathrm{id}$ cannot be the identity. This proves the level-2 failure. The formulas are checked on an explicit grid in the supplied program; the isolation and irrationality arguments prove the assertions beyond that grid. □

C.12. [Open] The requested infinite-group zero-level example. For the admissible surfaces and the exact comparison of C.1–C.3, we have not produced an infinite operator group whose level-0 comparison has a nontrivial kernel, nor proved universal injectivity. For a hyperbolic surface the required kernel is $\pi_1(C_{A^0}(u(\Gamma)), 1)$, since the target is zero. The disconnected centralizer in C.11 proves level-1 failure. The enlarged sphere comparison in C.25 has its stated nontrivial kernel, but the unpunctured sphere is excluded from C.1's object class. The requested admissible infinite-group zero-level decision remains unsettled here.

Manuscript: §23; folios 140–141; scans 279–281.

The attempted construction using a supported radial rotation produces isolated fixed points and a component obstruction, not a nonzero loop. Adding finitely many marked fixed points still gives aspherical configuration spaces and cannot by that argument produce the required π_2 of an evaluation base. No implication from the examples for levels 1 and 2 to level 0 is asserted. The independent infinite-group non-realization theorem C.23 gives level-3 failure and does not close this missing zero-level test. In particular, the possibility that level 0 holds for all operator groups has not been ruled out by this return.

Scope of the remaining infinite-group obligation. The source fixed in C.1 is the admissible-surface category of scan 264, folio 133: every genus-zero compact component meets the finite puncture set. The unpunctured sphere is therefore absent. The last sentence of Part C(b) asks for an infinite-group example at each fidelity level within this specified comparison. An example on one of its admitted abelian types would meet that existential request; a hyperbolic example would settle the more specific question in C.12. The examples C.11 and C.23 supply the independent obstructions at levels 1, 2, and 3. The level-0 example inside the admitted source remains unprovided.

Here is the exact kernel that is required. For an arbitrary discrete operator group and a strict action $u : \Gamma \to A$, put $H = u(\Gamma)$ and let N_u be its weak automorphism groupoid in $\Pi_1 A$. At its identity object, a 2-automorphism is precisely a loop class in A invariant under conjugation by every $u(\gamma)$. The comparison is thus the inclusion-induced map

$$\pi_1(C_A(H), 1) \longrightarrow \pi_1(N_u, 1) = \pi_1(A, 1)^{\Gamma}. \tag{C.12.1}$$

Pointwise commutation proves that the image lies in these invariants. Every based loop and every based homotopy in the source lies in A^0, so its source can equally be written $\pi_1(C_{A^0}(H), 1)$. For hyperbolic types the target vanishes, giving exactly C.12's test. For the plane, annulus, and torus the target is the invariant subgroup of $\mathbb{Z}$, $\mathbb{Z}$, or $\mathbb{Z}^2$, respectively; the required obstruction is a nonzero kernel of (C.12.1).

C.39 proves that every class in this kernel represented by a loop whose values commute pairwise is zero. Its contraction is global on the plane, including arbitrary changes and accumulation of orbit types, and its explicitly lifted filling disk descends to every connected admissible surface. Thus a remaining example must involve a class without a commuting-valued representative. C.40 constructs such a representative for every loop whose generated closure, in the actual compact-open topology, is locally compact. No representative reduction for arbitrary loops is proved here. C.41 proves the exact inclusion injective for every operator group with a dense orbit, without a local compactness assumption on either group. C.36–C.37 separately provide the stated contraction on the closed union of dimension-two proper-image open orbits for general loops. C.42 identifies the actual boundary quotient as an open map and proves that every component of every real lifted fibre is a properly embedded line; it also supplies an exact example with disconnected whole circle fibres. C.43 proves that the real lifted fibre is connected and supplies actual product coordinates. C.44 gives the five downstairs covering forms, with source a plane or annulus. C.45 gives the exact relation between two quotients and its three planar coordinate domains. C.46 constructs a simultaneous intrinsic contraction when the full set of lifted boundary fibre partitions has exactly two members, retaining every operator and quotient relation. C.47 proves that three independent actual quotients force the full compatible group on the source plane to be locally compact and Lie, with its actual topology. C.48 gives the simultaneous intrinsic contraction for the full one-foliation family, including every component of the proper operator group and the dense lattice case. C.49 identifies the actual closed lifted generated group in the three-quotient case and contracts each whole orbit closure, with exact overlap agreement and finite pasting. C.50 supplies an exact uniform word bound and extends these contractions through arbitrary accumulation, giving a joint contraction on $X_* = \overline{U_2 \cup U_3}$. C.51 extends the intrinsic contractions through the entire frontier when O/Q_O is compact and

the full boundary-quotient family has at most two members, with the exact median boundary value and finite pasting. For an empty family every frontier point is then fixed. C.52 supplies exact agreement with the prescribed exponential values on X_* by a natural hybrid target rule and diagonal frontier continuity. C.53 enlarges that region to $X^\dagger$ using all global nonopen quotient maps, and controls every accumulation of all H-translates of finitely many remaining cocompact representatives. The remaining requirements concern noncocompact quotients, arbitrarily many H-inequivalent remaining components, and extension through nonopen regions outside these closed sets.

C.25 proves such a kernel for the separately enlarged $\Pi_1 \operatorname{Homeo}(S^2)$ comparison. Its relationship with the admitted punctured comparisons can be computed explicitly. Write $p_\pm$ for the poles of its irrational rotation. In the identity component its centralizer C fixes both poles, and it is the same subgroup in the unmarked, once-marked, and twice-marked sphere homeomorphism groups. Forgetting marks induces

$$\begin{aligned} \pi_1 C = \mathbb{Z} \xrightarrow{1} \pi_1 \operatorname{Homeo}(S^2, \{p_+, p_-\}) = \mathbb{Z} \xrightarrow{1} \pi_1 \operatorname{Homeo}(S^2, \{p_+\}) = \mathbb{Z}, \\ \pi_1 \operatorname{Homeo}(S^2, \{p_+\}) = \mathbb{Z} \xrightarrow{\text{mod } 2} \pi_1 \operatorname{Homeo}(S^2) = \mathbb{Z}/2. \end{aligned} \tag{C.12.2}$$

The same full axis-rotation loop generates the first three groups by the plane and annulus comparisons of C.10; its inclusion into $SO(3)$ gives the last map. Conjugation by the irrational rotation is based homotopic to the identity in each ambient group, so these are also the weak-transporter target groups in (C.12.1). Thus puncturing one or both poles removes C.25's kernel. Every nonpolar point has an infinite orbit, so this action admits no other finite invariant puncture set.

Keeping the peripheral-diagram recipe while adding the sphere gives yet another explicitly related target: its diagram is $\varnothing \to *$, whose equivalence 2-group is terminal. The comparison factors through the functor $\Pi_1 \operatorname{Homeo}(S^2) \to \operatorname{Eq}(\varnothing \to *)$, which sends the identity 2-automorphism group $\mathbb{Z}/2$ to zero. For the irrational action its fundamental-group map is the composite $\mathbb{Z} \to \mathbb{Z}/2 \to 0$; C.25 computes the first arrow. Neither enlargement changes the admitted object class fixed in C.1. Consequently C.25 is retained with its exact stated comparison, while the admissible infinite-group zero-level obligation remains recorded here.

Exact lift comparison for the remaining zero-level question. The compact-open topology agrees with the marked compactification topology used in C.1. More precisely, restriction $\operatorname{Homeo}(X, S) \to \operatorname{Homeo}(X \setminus S)$ is a topological-group isomorphism. A homeomorphism extends uniquely to the finite end compactification, giving the algebraic inverse. For its continuity at f_0, choose disjoint connected target end neighborhoods W_j of small diameter and complementary compact core C. Choose a source core K with $f_0^{-1}(C) \subset \operatorname{int} K$, and connected source tails V_i whose f_0-images lie in the prescribed W_j. The conditions $f^{-1}(C) \subset \operatorname{int} K$ and $f(z_i) \in W_j$ for one point $z_i \in V_i$ are open: inversion is continuous in the compact-open homeomorphism group of a locally compact, locally connected surface. Connectedness forces $f(V_i) \subset W_j$, since this image avoids C and meets W_j. This controls the extended maps uniformly on all tails. Compact-open control on the remaining compact core controls them on the rest of X, proving continuity of the inverse.

Let U be the hyperbolic surface of C.12, let $A^0 = \operatorname{Homeo}(U)^0$, let H be the image of the specified operator group, and let $p : \widetilde{U} \to U$ be the universal covering with deck group π. Write $\widetilde{H}$ for the group of all lifts of elements of H, regarded as actual homeomorphisms of $\widetilde{U}$. There is a continuous injective homomorphism

$$j : A^0 \longrightarrow \operatorname{Homeo}(\widetilde{U}), \qquad p\, j(f) = f\, p,$$

characterized by $j(f)\delta = \delta j(f)$ for every $\delta \in \pi$. It identifies the precise centralizers:

$$j\big(C_{A^0}(H)\big) = j(A^0) \cap C_{\operatorname{Homeo}(\widetilde{U})}(\widetilde{H}).$$

In particular every based loop in the left-hand centralizer lifts to a closed loop of homeomorphisms commuting with the entire lift group.

Proof. Lift an isotopy from id to f starting at $\mathrm{id}_{\widetilde{U}}$. Conjugation on the discrete deck group is constant during the lift, so its endpoint commutes with π. Two such endpoints differ by a deck transformation centralizing π; they are equal because $Z(\pi) = 1$ by Part A. Uniqueness also makes j a homomorphism and an injection. For continuity, fix $\widetilde{u} \in \widetilde{U}$. The evaluation map $A^0 \to U$ lifts continuously to $\widetilde{U}$, since A^0 is contractible and locally path connected by the surface homeomorphism-group theorem. This lift gives $j(f)(\widetilde{u})$, and the local lifting property of a covering gives continuous dependence of the lifted homeomorphism in the compact-open topology.

Suppose f centralizes H and $\widetilde{h}$ is any lift of $h \in H$. Then $[j(f), \widetilde{h}]$ is a deck transformation. Its conjugation on π is the commutator of the identity automorphism with the automorphism induced by $\widetilde{h}$, hence is trivial. Centre-freeness makes this deck transformation the identity. This proves one inclusion in the displayed equality; descent proves the other. Finally $j(\mathrm{id}) = \mathrm{id}$, so a based loop has the same endpoint after lifting. □

This comparison does not supply a contraction inside the right-hand centralizer. Choosing a hyperbolic metric and interpolating from $j(f)(x)$ to x along geodesics has two unproved properties: the arbitrary lift group $\widetilde{H}$ need not preserve these geodesics, and an interpolated map need not be a homeomorphism. The first prevents asserting equivariance; the second prevents asserting that the interpolation lies in $j(A^0)$. Neither property follows from the displayed centralizer equality. The universal vanishing of $\pi_1(C_{A^0}(H))$, and the requested counterexample to it, therefore remain unresolved by this argument.

Current orbitwise reduction and the remaining compatibility problem. C.26 settles the globally transitive centralizer case on a closed hyperbolic surface and also gives an explicit closed subgroup with an ambient-null but intrinsically nontrivial loop; its latter example is not asserted to be a centralizer. For the canonical plane lift above, C.28 identifies the orbits of the group generated by the loop: points, intrinsic lines, circles, and open surface orbits. C.27 gives an order-natural contraction on each intrinsic line. C.29 proves that every circle restriction has winding zero by the actual plane-loop evaluation map, and contracts that restriction in its circle centralizer. C.31 supplies inverse and track control, and C.34 proves joint continuity of the contractions and their inverses on the entire closed union Y of the nonopen orbits. This is a based contraction through homeomorphisms of Y, commuting with every homeomorphism that commutes with the original loop. In particular C.34 settles the hyperbolic centralizer loop whenever its lifted generated group has no open orbit.

The open orbits have also been treated intrinsically. C.30 gives the exact restricted operator group, including its restriction kernel, and proves that its closed image acts freely and properly. C.32 handles a discrete image, and C.33 treats every locally compact closed image supplied by C.30. In each case the restricted loop has a based null-homotopy inside the centralizer of the full restricted operator group. These results give contractions on all individual open orbits and on the whole nonopen locus. The remaining step is to choose these contractions compatibly across the frontiers of the open components, with joint continuity of the resulting homeomorphisms and their inverses.

C.35 proves that this step cannot require extending the existing median prescription unchanged. Its explicit plane action has the boundary restriction map

$$\mathbb{R}^2 \longrightarrow \mathbb{R}^3, \qquad (A, B) \longmapsto (A, B, A - B).$$

At the displayed parameter the prescribed three edge medians are $(1, 0, 0)$, which lies outside its image because $r_1 - r_2 - r_3 = 1$. Thus that prescribed homeomorphism of Y has no equivariant

extension over an adjacent open orbit. The original loop in the same example has the explicit global equivariant contraction given in (C.35.5). C.36 constructs compatible choices on closures of open orbits whose closed proper operator image has dimension two, using the actual open quotient onto their simply transitive Lie centralizers. Scaling the logarithm of each based universal-cover lift respects the exact common boundary action. C.37 supplies a uniform three-word estimate and proves joint continuity, including inverses, on

$$X_2 = \overline{\bigcup_{\dim Q_O = 2} O},$$

including every infinite accumulation of those open orbits. Its bounded residual equations determine the unique logarithmic limit on line orbits. The actual null residual word isotopies and the two-point configuration map determine the real lifted limit on circle orbits, including its winding.

C.49–C.50 subsequently enlarge this closed set to X_*, including all open orbits with at least three actual boundary quotient foliations. C.51 proves whole-closure extension for the remaining open orbits with compact operator quotient, using the median frontier prescription and finite pasting. C.52 makes these extensions exactly relative to X_*. C.53 enlarges the exponential region to $X^\dagger$ and proves continuity across all accumulations of the operator translates of finitely many remaining cocompact representatives. The outstanding problem includes noncocompact operator quotients, infinitely many H-inequivalent remaining representatives, and extension through exterior regions consisting of nonopen orbits. The hybrid target prescription has been proved continuous on the treated closed sets; its transverse continuity on all of the nonopen locus has not been proved. C.35 requires allowing changes to the original median prescription. These exact remaining requirements, and the general kernel in (C.12.1), remain unresolved here.

3.4 Configuration invariants and drilling

C.13. [Proved] The invariant braid group and nonabelian cohomology. Let a finite group act faithfully and orientation preservingly on a hyperbolic U, and let $R = U^!$ be its finite set of points with nontrivial stabilizer. Put $V = U \setminus R$, $B = \{f \in A^0 : f|_R = \mathrm{id}\}$, and $B^0 = \mathrm{Homeo}(V)^0$. With the action by conjugation, we have

$$(B/B^0)^\Gamma = 1, \qquad H^1(\Gamma, B^0) = 1, \qquad (A^0)^\Gamma = B^\Gamma \cong \mathrm{Homeo}(V/\Gamma)^0.$$

Here the quotient group identifies canonically with $\pi_1(\mathrm{Conf}_R(U), R)$ by evaluation, and its invariant subgroup is the H^0 asked for on scan 285.

Manuscript: §24; folios 142–144; scans 282–286.

Proof. Evaluation $A^0 \to \mathrm{Conf}_R(U)$ is a locally trivial principal B-bundle, by the local extension theorem for embeddings of a finite set. Its total space is contractible. Its homotopy exact sequence gives $\pi_0 B \cong \pi_1 \mathrm{Conf}_R(U)$, with the specified base configuration, and its identity component is the identity component after retaining the new punctures. The construction is equivariant for Γ, so this is the stated action on the braid group.

An invariant component $b \in B/B^0$ is a mapping class of V in the kernel of filling R that centralizes the action of Γ on V. By C.6 and its transporter version, choose a representative f on V commuting with Γ. It extends across the punctures and induces the identity mapping class on U. Thus $f \in (A^0)^\Gamma$, and C.7 joins it to the identity by an equivariant isotopy on U. Such an isotopy fixes each point of R: it preserves each finite critical-point set and starts with the identity permutation. Restriction to V proves $b = 1$.

For a nonabelian 1-cocycle $a_\gamma \in B^0$, the maps $v(\gamma) = a_\gamma u(\gamma)$ form another finite action on V. The cocycle equation is exactly the homomorphism equation for v. Both actions induce

the same homomorphism into the mapping class group of V. By C.6 they are conjugate by an element $b \in B^0$; equivalently $a_\gamma = bu(\gamma)b^{-1}u(\gamma)^{-1}$. Hence the cocycle is a coboundary and $H^1(\Gamma, B^0) = 1$.

Finally C.7 gives connectedness. Descent to the quotient with its branch points retained is precisely descent to V/Γ with its punctures retained; the identity lift construction in C.7 is the displayed isomorphism. In particular the unprimed V in the transcription of the identity-component line on scan 286 must be compared with the quotient V' of scans 282–285. The two homeomorphism groups need not be isomorphic; the descent map identifies the quotient one. □

C.14. [Refuted] Finite critical configurations do not include mirrors. The finite-set configuration formulation of C.13 does not extend by replacing an orientation-preserving action by an arbitrary action and keeping $R = U^!$ equal to all points with nontrivial stabilizer. A reflection has a one-dimensional fixed locus, and deleting finitely many points cannot make that action free.

Manuscript: §24; folios 142–148; scans 282–294.

Proof. Take complex conjugation on $U = \mathbf{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$. It acts faithfully on this hyperbolic surface. Its fixed set is $\mathbf{P}^1(\mathbb{R}) \setminus \{0, 1, \infty\}$, a union of three open arcs. Every point of these arcs has nontrivial stabilizer. A finite deletion leaves points in each arc. At any remaining fixed point, the lift of conjugation fixing a selected universal-cover point is an element of order two in the lift extension. Therefore the drilled extension still has torsion. This explicit counterexample refutes both the finiteness premise and a torsion-free finite-drilling assertion for such an action. The corrected geometry retains mirror boundary in the quotient or cuts along the mirror locus; it is not the operation of adding finitely many puncture loop classes. The equations C.13 were stated on scans 282–286 after the explicit restriction $\Gamma = \Gamma^+$. □

C.15. [Proved] The oriented drilling equivalence and the bijection. Fix a finite group Γ and faithful orientation-preserving outer actions on discrete hyperbolic groups with loops. Let $\Phi(\pi, \Gamma)$ be the set of π-conjugacy classes of nontrivial maximal finite subgroups of the canonical extension. Filling a specified Γ-stable set J of punctures with nontrivial stabilizers induces an equivalence between the category of such drilled actions and the category of filled actions with a specified subset $J \subset \Phi(\pi, \Gamma)$. In particular

$$\Phi(\pi', \Gamma) \amalg J \xrightarrow{\sim} \Phi(\pi, \Gamma)$$

is a canonical Γ-equivariant bijection. Drilling all of Φ produces a torsion-free extension, uniquely up to the unique exterior isomorphism over the specified filled object.

Manuscript: §24; folios 144–147; scans 286–293.

Proof. Realize the action by C.5. A finite nontrivial subgroup of the lift group acts on $\mathbb{H}^2$ by orientation-preserving isometries and has a unique fixed point. Its stabilizer at that point is the unique maximal finite subgroup containing it. Conversely every point of U with nontrivial stabilizer gives that subgroup after choosing a lift. Changing the lift conjugates by π. This identifies Φ with the finite critical-point set of the realized action, including its Γ action. It also proves the assertion about maximality, rather than assuming it for arbitrary finite subgroups containing reflections.

Deleting the points of J gives the drilled surface. Remaining critical points and deleted points partition the original critical-point set, which proves the displayed bijection. The group map is the epimorphism $\pi' \to \pi$ obtained by killing the normal closure of exactly the new peripheral subgroups. Local punctured discs show that its induced map on lift groups sends a cusp stabilizer

to the corresponding finite point stabilizer. Thus the geometric bijection is the algebraic map of scan 290, with the same conjugacy ambiguities.

To prove fullness, realize an equivariant outer isomorphism of the filled peripheral groups by an equivariant homeomorphism, using C.6's transporter conclusion. It preserves the selected subset of Φ by the preceding construction. Its restriction to the complements gives the desired drilled outer isomorphism. To prove faithfulness, represent a drilled equivariant outer automorphism in the kernel by an equivariant homeomorphism of the drilled surface and extend it across the new punctures. It is isotopic to the identity on the filled surface. By C.7 there is an equivariant identity isotopy there. Every point of J stays fixed throughout, since it is a member of the finite critical set, so the isotopy restricts to the drilled surface. The kernel is therefore trivial. These two arguments also prove uniqueness over a fixed filled isomorphism, not merely uniqueness of isomorphism classes.

After all critical points are removed, the finite action is free. Its lift extension acts freely on $\mathbb{H}^2$ and is torsion-free, since a nontrivial finite-order isometry fixes a point. The operations of deleting and filling are inverse by the just-proved full faithfulness. The only use of geometric realization in this proof is C.5 together with the equivariant homeomorphism and isotopy results C.6–C.7; the comparison map on groups and the kernel argument have been specified. □

C.16. [Refuted] Nonfaithful outer actions obstruct torsion-free drilling. The unrestricted existence claim on scans 286–288 fails for the trivial outer action of C_2 on the group of type $(0, 3)$. The faithful, orientation-preserving version is C.15.

Manuscript: §24; folios 144–145; scans 286–288.

Proof. Suppose a finite-puncture drilling of that action had torsion-free extension. Realize its finite outer action on the drilled hyperbolic surface by C.5. Filling the specified punctures extends these homeomorphisms to a finite action on the filled three-punctured sphere. The induced mapping class of every element is trivial. The finite-order rigidity argument of C.5 makes every filled homeomorphism the identity. Its restriction to the dense drilled surface is therefore the identity. The upstairs outer action is trivial as well. For a centre-free kernel its canonical extension is $\pi' \times C_2$, which contains $\{1\} \times C_2$. This contradicts torsion-freeness. The argument proves the same obstruction for every nonfaithful outer action after restricting to a nontrivial element of its finite kernel. □

C.17. [Proved] Torsion-free finite extensions with peripheral data. Let π be a discrete hyperbolic surface group, possibly punctured, and let $1 \to \pi \to E \to \Gamma \to 1$ be a torsion-free extension by a finite group whose outer action preserves the loop structure. Then E is the fundamental group of a finite-type surface, possibly nonorientable. Its peripheral subgroups are the centralizers in E of the peripheral subgroups of π. With their conjugacy classes it is a surface group with loops in this extended sense. The associated compact surface and boundary groups form a Poincaré duality pair of dimension two with its orientation local system.

Manuscript: §24; folios 147–148; scans 292–294; §28; folio 178; scan 353.

Proof. The centre-free extension is the pullback of $\mathrm{Aut}(\pi) \to \mathrm{Out}(\pi)$. The centralizer of π in E maps isomorphically to the kernel of the outer action: a representative inducing an inner automorphism can be multiplied by the inverse inner element to centralize π, and this representative is unique because $Z(\pi) = 1$. This centralizer is finite, so torsion-freeness forces it to be trivial. Hence the outer action is faithful.

Apply C.5 and identify E with the group of lifts of the realized action, by uniqueness of the centre-free extension. A nontrivial stabilizer of a point gives a finite subgroup of E, so the

action on U is free. Its quotient $W = U/\Gamma$ is an ordinary surface and $\mathbb{H}^2 \to W$ is its universal covering. This gives the required isomorphism $E \cong \pi_1(W)$, respecting the specified subgroup π and quotient Γ.

A stabilizer of an end of U cannot contain an orientation-reversing operator. The local finite-group model at its compactifying point would then contain a reflection and fixed points arbitrarily close to that point, contrary to freeness on U. Its local action is therefore cyclic rotation. Consequently an upstairs peripheral subgroup is a finite-index subgroup of the downstairs peripheral infinite cyclic group. In the hyperbolic universal cover its centralizer is exactly that full cusp group. These are all the cusps, since every end downstairs has a preimage upstairs. Thus the stated peripheral classes, including their full subgroups rather than merely inclusions, are obtained.

Choose disjoint cusp neighbourhoods in W and truncate them to obtain a compact surface M with boundary. The inclusions of boundary circles induce precisely these cyclic groups. Lift a triangulation to the contractible universal cover. The cellular cap-product map with the relative orientation class gives the chain equivalence between cochains and complementary relative chains; the local sign is the orientation character of W. This is the Poincaré–Lefschetz duality pair, and its group-pair chain model is the relative cellular complex because both the universal cover and the lifted boundary components are contractible. This proves the duality-pair assertion rather than inferring it from virtual freeness alone. The exact geometric input is the finite realization theorem; a separate proof from abstract duality axioms is not substituted for this argument. □

C.18. [Proved] The discrete oriented realization criterion. For a faithful orientation-preserving finite outer action on a discrete hyperbolic group with loops, the criterion of scan 294 is valid with the word admissible meaning the full-faithfulness and uniqueness conditions of scan 293. The drilling in question is at the canonical set $\Phi(\pi, \Gamma)$, and its extension carries the quotient loop structure described in C.17.

Manuscript: §24; folios 147–148; scans 293–294.

Proof. A realized action can be drilled at its critical points. C.15 proves that the resulting action is admissible and its extension torsion-free; C.17 identifies that extension with a group with loops. This verifies every requirement of the stated criterion.

Conversely let $\pi' \to \pi$ be the stipulated filling map and let E' be its extension, with the group-with-loops structure required in the criterion. Realize E' by a compact surface with its boundary circles, then take its covering corresponding to π'. The comparison of peripheral subgroups in the definition ensures that this cover has exactly the prescribed upstairs loop classes. Its deck group is Γ, acting freely before filling. The cyclic end stabilizers extend across each new puncture by the local map $z \mapsto z^m$. Filling the selected ends thus gives a Γ-action on the surface whose peripheral group is π. The canonical map of lift extensions identifies its outer action with the original one. Admissibility makes this construction independent, up to the uniquely specified exterior isomorphism, of the chosen drilled object. This proves the converse by the typed covering and filling maps. The mirror counterexample C.14 explains the necessary restriction of this formulation. □

3.5 Special homeomorphisms: the topology and the extension class

C.19. [Refuted] Escaping support invalidates the compact-open version. Let $SA \subset A$ consist of homeomorphisms equal to the identity on some neighbourhood of every puncture, and give it the subspace compact-open topology. The canonical special-group comparison of scan 304 does not identify $\pi_0(SA)$ with the boundary-framed mapping class group. A peripheral

boundary twist is connected to the identity inside this SA. In particular, the annulus conclusion $\pi_0(SA) = \mathbb{Z}$ on scan 304 is false in this topology.

Manuscript: §25; folios 151–153; scans 301–304.

Proof. Use a coordinate $z = re^{i\theta}$ on a disc about a puncture at 0. Choose a continuous function $\eta : [0, \infty) \to [0, 1]$ equal to 0 on $[0, 1/3]$ and to 1 on $[2/3, \infty)$. For $0 < s \leq 1$ define

$$h_s(re^{i\theta}) = r \exp(i\theta + 2\pi i\eta(r/s)), \qquad h_0 = \mathrm{id},$$

and extend by the identity outside the coordinate disc. Each h_s is identity near the puncture and outside the annulus $s/3 < r < 2s/3$. It is a boundary twist when viewed on a fixed compact surface obtained by truncating the puncture. Nevertheless $\sup_z |h_s(z) - z| \leq 4s/3$, and the same bound holds for its inverse. Thus $s \mapsto h_s$ is a continuous path in the stated subspace topology from the identity to that twist.

For the annulus every compactly supported orientation-preserving homeomorphism fixes both ends and is represented on a compact annulus by an integer twist; cutting along a spanning arc and applying the Alexander isotopy of a disc proves this integer classification. The isotopy to the corresponding integer twist can be chosen in that fixed compact annulus. The path above then kills that twist in SA. Hence SA is path connected in the compact-open topology, proving the explicit contradiction. The programme checks the formula and its uniform bound; the bound itself proves convergence for the full space, not just on a finite sample.

The fixed-boundary formulation below retains a single compact surface throughout the isotopy. That is the required correction. No change of topology is silently made in this refutation. □

C.20. [Proved] The fixed-boundary repair and its class. Let M be an oriented compact surface obtained by replacing the $\nu > 0$ punctures of a hyperbolic type by boundary circles. Put $A_\partial = \mathrm{Homeo}^+(M \text{ rel } \partial M)$. Then A_∂^0 is contractible, $\pi_0(A_\partial) \cong S\Gamma_{g,\nu}^{!+}$ for the framed group of E.1–E.2, and there is a central extension

$$1 \longrightarrow \mathbb{Z}^\nu \longrightarrow \pi_0(A_\partial) \longrightarrow \Gamma_{g,\nu}^{!+} \longrightarrow 1.$$

The ith kernel generator is the positive boundary twist. Its extension class is the vector of Euler classes of the oriented tangent circle lines at the marked points, with the sign fixed by that generator. For the annulus the kernel is the quotient of $\mathbb{Z}^2$ by the diagonal rotation relation; for the once-punctured sphere it is zero.

Manuscript: §25; folios 152–154; scans 302–306.

Proof. Allow boundary rotations first, retaining each boundary component setwise. Restriction to the boundary is a locally trivial fibration onto $\prod_i \mathrm{Homeo}^+(S^1)$: extend a small boundary isotopy across collars. Its fibre is A_∂. Each boundary homeomorphism group deformation retracts onto rotations by lifting to increasing maps of $\mathbb{R}$ and linearly interpolating the periodic displacement. Thus the base has homotopy type $(S^1)^\nu$. Capping each boundary by a disc with its centre marked identifies the free-boundary mapping classes with the pure oriented punctured mapping classes. The same collar-extension construction gives the homotopy comparison on identity components.

The identity component of the free-boundary group is contractible in a hyperbolic type by Yagasaki's compact-pair theorem. The homotopy sequence therefore injects $\pi_1((S^1)^\nu) = \mathbb{Z}^\nu$ into the components of the fibre and identifies the quotient as stated. Lifting one rotation in the ith collar gives exactly the boundary twist. Its conjugation action is trivial because the boundary circles are retained individually and their orientations are fixed. Choose a point on each boundary circle. A boundary-fixing homeomorphism acts on the framed peripheral diagram of E.2, and an isotopy relative to the boundary gives its prescribed natural isomorphism. This

defines $\pi_0(A_\partial) \to S\Gamma^{!+}_{g,\nu}$. On the quotient it is the peripheral Dehn–Nielsen–Baer isomorphism. A positive boundary twist changes the path from the generic basepoint to that boundary point by one peripheral turn; hence the induced map on kernels sends the ith boundary generator to the ith generator of T^I. It is an isomorphism on both kernel and quotient, so the two short exact sequences give the asserted isomorphism of extensions. Contractibility of A^0_∂ follows directly from Lemma 2.1(ii), with the nonempty fixed boundary. The connecting homomorphism is consequently the fundamental-group extension of the bundle of tangent-circle framings at the marked points. Obstruction to a section of that circle bundle is its Euler class, proving the class identification with its sign convention. In complex notation this is the tangent-line first Chern class; the usual cotangent ψ_i has its opposite sign.

For the annulus the identity component of the free-boundary group is S^1. Its fundamental-group map to the two boundary-rotation groups is the diagonal when both circles use the same angular coordinate, or the antidiagonal when both use boundary orientations. The cokernel is $\mathbb{Z}$. For the disc this map is an isomorphism $\mathbb{Z} \to \mathbb{Z}$, so the cokernel is zero. These are the exact boundary models underlying the two abelian computations on scans 303–304. □

C.21. [Proved] The four small extension classes. With the positive boundary-twist convention of C.20, the special fixed-boundary extension classes for $(0,3)$ and $(0,4)$ are zero. For $(1,1)$ the class is a generator e of $H^2(\mathrm{SL}_2(\mathbb{Z});\mathbb{Z}) = \mathbb{Z}/12$. For $(1,2)$ the vector class is (q^*e, q^*e), where $q : \Gamma^{!+}_{1,2} \to \mathrm{SL}_2(\mathbb{Z})$ forgets the second marked point; its order is 12.

Manuscript: §25; folios 153–154; scans 304–306.

Proof. The pure oriented mapping class group of a three-punctured sphere is trivial. Filling one puncture of a four-punctured sphere gives the point-pushing group F_2, since the target pure group is trivial and the three-punctured surface is hyperbolic. A graph is a classifying space for F_2, so $H^2(F_2;\mathbb{Z}^4) = 0$. This proves the first two class computations, including the noncanonical splitting in type $(0,4)$.

The integral matrices

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

have orders 4 and 6 and satisfy $S^2 = R^3 = -I$. Their action on the Farey tessellation gives $\mathrm{PSL}_2(\mathbb{Z}) = C_2 * C_3$: the dual bipartite tree has one edge as fundamental domain, vertex stabilizers C_2, C_3, and trivial edge stabilizer. Connectivity and the absence of circuits follow from the Euclidean algorithm for adjacent reduced fractions. Lifting the central element $-I$ gives $\mathrm{SL}_2(\mathbb{Z}) = C_4 *_{C_2} C_6$. The cohomology sequence of this tree of groups identifies H^2 with

$$\{(r,s) \in \mathbb{Z}/4 \oplus \mathbb{Z}/6 : r \equiv s \pmod 2\} :$$

the preceding $H^1(C_2;\mathbb{Z})$ and $H^1(C_4;\mathbb{Z}) \oplus H^1(C_6;\mathbb{Z})$ vanish, while the restrictions in degree two reduce modulo 2. This group has twelve elements and is cyclic. The tangent-circle bundle restricts on the square torus fixed by S to the rotation representation by $\pi/2$, and on the hexagonal torus fixed by R to the representation by $\pi/3$. Lifting these circle rotations to $\mathbb{R}$ gives fourth and sixth powers equal to one positive full turn. By the convention of C.20 the Euler class therefore restricts to $(1,1)$, an element of order 12. This proves its generator assertion without an unproved presentation for the boundary mapping class group. The supplied code verifies the matrix orders and the finite abelian-group calculation.

In a complex torus translation canonically identifies the tangent spaces at any two points. These identifications vary with the marked complex torus and commute with all isomorphisms. Hence the two tangent circle bundles on the two-pointed moduli orbifold are isomorphic, and both are pulled back from the one-pointed tangent bundle. This proves the vector formula. It

has order dividing 12. A square torus with marked points 0 and $(1+i)/2$ admits the order-four rotation preserving both; a hexagonal torus admits an order-three rotation with at least two marked fixed points. The restrictions of the class to these cyclic subgroups are generators, as can be seen from their nontrivial local rotations, or from C.22 below. Its order is divisible by 4 and 3, and thus is exactly 12. The existence of the displayed fixed points is also verified in the integral torus matrices supplied with the code. □

C.22. [Proved] Restriction over a finite puncture stabilizer. For a hyperbolic type, a nontrivial finite subgroup $G \subset \Gamma_i^+$ is cyclic. The extension by the ith copy of $T = \mathbb{Z}$ obtained from the special group is nonsplit over G; its preimage is infinite cyclic and the kernel is the subgroup of index $|G|$.

Manuscript: §25; folios 153–154; scans 305–306.

Proof. Realize G by C.5. It fixes the marked point i and preserves orientation. The finite local action is a faithful rotation action: an element trivial on a small disc is the identity as a periodic surface homeomorphism, by the local classification and connectedness. Thus G embeds in the circle rotations and is cyclic. Delete that point and lift a small punctured quotient disc to its universal cover. Its group is $\mathbb{Z}$, while the upstairs punctured disc has subgroup $|G|\mathbb{Z}$. The exact normalizer interpretation proved in E.3 identifies this with the extension pushed out by the ith projection. An infinite cyclic group contains no nontrivial finite subgroup, so the extension cannot split. This proves the remark on scan 306 with its stated orientation and puncture-stabilizer hypotheses. □

C.23. [Proved] An infinite-group failure of existence. For every closed orientable surface X_g of genus $g \geq 2$, the identity homomorphism from the discrete group $\Gamma = \mathrm{Mod}^+(X_g)$ to its subgroup in $\pi_0 A_g$ has no homomorphic lift to A_g. Thus level 3 fails for an infinite operator group already on each such closed hyperbolic surface.

Manuscript: §23; folios 140–141; scans 279–280.

Proof. Marković–Šarić's Theorem 1.1 [42] states that the projection $\mathrm{Homeo}^+(X_g) \to \mathrm{Mod}^+(X_g)$ has no group section for $g \geq 2$. The theorem is about homeomorphisms, not merely diffeomorphisms; its first-page statement and definitions have been checked in the original paper. A lift of the displayed homomorphism would take every element to an orientation-preserving homeomorphism, since orientation is invariant under isotopy. Its composite with that projection would be the identity, giving the forbidden section. This is the exact reduction to the source theorem. The group is infinite: a twist about a nonseparating simple curve acts on $H_1(X_g;\mathbb{Z})$ by an integral transvection, whose distinct powers are distinct. □

3.6 The integral sphere comparison

C.24. [Refuted] The sphere comparison already fails for an order-three rotation. Let $A = \mathrm{Homeo}(S^2)$ with its compact-open topology, and let $\Gamma = C_3$ act on S^2 by rotation through $2\pi/3$ about a fixed axis. Give A the resulting conjugation action and base both fixed-point spaces at the identity. The actual comparison has

$$\pi_1(A^\Gamma, 1) = \mathbb{Z}, \qquad \pi_1(A^{h\Gamma}, 1) = \mathbb{Z}_3 \times \mathbb{Z}/2, \qquad n \longmapsto (n, n \bmod 2).$$

Consequently the requested integral homotopy equivalence fails in type $(0,0)$. This failure concerns the actual homeomorphism mapping space; it is independent of the earlier loss of the

sphere's π_2 in its fundamental groupoid. For every odd prime p, the same construction with C_p gives $\mathbb{Z}_p \times \mathbb{Z}/2$ and the same comparison formula with p in place of 3.

Manuscript: §23; folio 141; scans 280–281.

Proof. We give the comparison maps and compute their effect, including the integral rather than only completed homotopy group. Write $G = SO(3)$ and $\rho : C_p \hookrightarrow G$ for the rotation subgroup, with p odd. The inclusion $G \hookrightarrow A^+ = \mathrm{Homeo}^+(S^2)$ is C_p-equivariant and a weak equivalence. Indeed evaluation on three ordered distinct points is a locally trivial fibration onto $\mathrm{Conf}_3(S^2)$, by isotopy extension. Its fibre is the pure homeomorphism group of the three-punctured sphere. Its component group is trivial, and its identity component is contractible by the surface homeomorphism-group theorem used in C.7. The unique Möbius transformation carrying one ordered triple to another identifies this configuration space with $PSL_2(\mathbb{C})$; polar decomposition retracts this group onto $SO(3)$. The resulting evaluation diagram proves the stated weak equivalence for the actual inclusion. Homotopy fixed points preserve this objectwise weak equivalence, after applying singular complexes. A path through the identity in A^{hC_p} stays in A^+, because EC_p is connected and orientation takes values in the discrete group C_2. It therefore suffices to calculate $(G^{hC_p}, 1)$.

The identity component of the actual centralizer fixes the two poles individually. Quotient by $z \mapsto z^p$ identifies its descent map to $\mathrm{Homeo}^+(S^2; 0, \infty)^0$ as a covering of degree p. Every map of the quotient fixing both poles lifts: on the complementary annulus its action on the meridian is $+1$, so it preserves the subgroup $p\mathbb{Z}$ of the covering group. Each lift commutes with the deck group, since the induced automorphism of $\mathbb{Z}/p$ is $+1$; it extends uniquely over the two poles. The usual local lifting neighbourhoods establish that descent is a covering in the compact-open topology. Its total space is connected: the quotient group is connected, and all deck lifts lie on the connected circle of axis rotations. The annulus calculation of C.10 identifies the base with a space of homotopy type S^1. The rotation circle maps to its quotient rotation circle by $z \mapsto z^p$; its inclusion in the total centralizer therefore induces an isomorphism on every homotopy group. In particular the actual fundamental group is $\mathbb{Z}$, with generator one complete axis rotation.

Set $Y = BG = BSO(3)$. The adjoint bundle $EC_p \times_{C_p} G \to BC_p$, with action through ρ, has section space G^{hC_p}. It is also the bundle of gauge automorphisms of the principal G-bundle classified by $B\rho$. The universal bundle gives the natural weak equivalence

$$G^{hC_p} \simeq \Omega_{B\rho} \operatorname{Map}(BC_p, Y).$$

One way to verify the map is to send a gauge automorphism to the loop of its classifying map obtained by gluing the bundle over the two ends of $BC_p \times [0,1]$. Conversely a loop of classifying maps gives a bundle over this cylinder with its specified endpoint identifications; trivialization in the interval direction gives the gauge automorphism. The spaces of such trivializations are contractible. This construction is natural for the inclusion $C_G(\rho(C_p)) \hookrightarrow G^{hC_p}$.

We now compute π_2 of this mapping space using the arithmetic square of the simply connected finite-type space Y. Let

$$\widehat{Y} = \prod_q Y_q^\wedge, \qquad V = \Big(\prod_q \mathbb{Z}_q\Big) \otimes \mathbb{Q}.$$

Here q runs over primes and superscript $\wedge$ means homotopical q-completion, throughout this proof. We have $Y_{\mathbb{Q}} \simeq K(\mathbb{Q}, 4)$ and $(\widehat{Y})_{\mathbb{Q}} \simeq K(V, 4)$. This follows from $SO(3) \simeq \mathbb{RP}^3$: its rational homotopy is concentrated in degree three, and each of its higher homotopy groups is finite. The square

$$\begin{array}{ccc} Y & \longrightarrow & \widehat{Y} \\ \downarrow & & \downarrow \\ K(\mathbb{Q}, 4) & \longrightarrow & K(V, 4) \end{array}$$

is a homotopy pullback. For completeness its homotopy-group verification reduces on each free summand to

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \oplus \prod_q \mathbb{Z}_q \longrightarrow V \longrightarrow 0, \quad (a, (b_q)) \longmapsto (b_q - a)_q.$$

Surjectivity follows by clearing a common denominator and choosing its finitely many required residues by the Chinese remainder theorem; the kernel consists of rational numbers integral at every prime and is $\mathbb{Z}$. On finite summands completion gives the product of their primary parts and rationalization is zero. The long exact sequence of the homotopy pullback and Whitehead's theorem prove the square, also giving the usual arithmetic-square construction in the exact form needed here.

Apply $\mathrm{Map}(BC_p, -)$ to this homotopy pullback and take the component containing $B\rho$. For every rational vector space W, positive-degree cohomology $H^k(C_p; W)$ vanishes by averaging. The two rational mapping spaces consequently have vanishing π_2 and π_3. The homotopy-pullback exact sequence therefore gives

$$\pi_2 \,\mathrm{Map}(BC_p, Y)_{B\rho} \cong \prod_q \pi_2 \,\mathrm{Map}(BC_p, Y_q^\wedge)_{B\rho}.$$

For $q \neq p$, evaluation is a weak equivalence from this mapping-space component to $Y_q^\wedge$. To check it, use the Postnikov tower: positive-degree C_p-cohomology of every q-complete homotopy group vanishes since multiplication by p is invertible; at each tower stage only constant maps contribute, and homotopy inverse limits preserve the comparison. Thus only $q = 2$ contributes to π_2 among these primes, giving $\pi_2(Y_2^\wedge) = \mathbb{Z}/2$.

At $q = p$ we use the exact finite-p-group mapping-space theorem: for a finite p-group P, a compact Lie group K, and $\alpha : P \to K$, the map induced by the centralizer identifies

$$\pi_n \,\mathrm{Map}(BP, (BK)_p^\wedge)_{B\alpha} = \big(\pi_n BC_K(\alpha(P))\big)_p^\wedge \quad (n \geq 2).$$

This is the Dwyer–Zabrodsky–Notbohm formula, stated and derived in [32, Theorem 1.9 and its proof, using Theorem 3.2(iii) of their earlier paper]. Its hypotheses apply to the displayed finite cyclic group and $SO(3)$. Since a nontrivial odd-order rotation has centralizer exactly the axis circle $SO(2)$, the contribution is $\mathbb{Z}_p$. This proves $\pi_1(G^{hC_p}, 1) = \mathbb{Z}_p \times \mathbb{Z}/2$.

Naturality of the centralizer map makes the p-component of the map from the rotation circle the completion $\mathbb{Z} \to \mathbb{Z}_p$. At the prime 2 it is the map $\pi_1 SO(2) \to \pi_1 SO(3)$, namely reduction modulo 2: lifting a full rotation to unit quaternions gives the endpoint -1. This proves the stated formula for the exact comparison. For an explicit missing element at $p = 3$, put $\beta = \sum_{j \geq 1} 3^{j!} \in \mathbb{Z}_3$. Its base-three digits have infinitely many zeros and infinitely many ones, whereas an ordinary integer has eventually only zeros or eventually only twos. Hence $(\beta, 0)$ is outside the image. This also proves nonsurjectivity without a cardinality argument; for general odd p the same factorial series has digits zero and one, rather than eventually zero or $p - 1$. The explicit action, its finite order, the axis centralizer and the finite reductions of this comparison are checked in `code/verify_sphere_comparison.py`. The inverse-limit argument above, rather than a finite computation, proves the nonsurjectivity. □

C.25. [Refuted] Zero-level fidelity for a faithful infinite cyclic sphere action. Let $\Gamma = \mathbb{Z}$ act faithfully on S^2 by an irrational axis rotation, $u(1)(\theta, t) = (\theta + 2\pi\alpha, t)$ with $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and the usual collapse of the latitude circles at $t = \pm 1$. Let $M = A^\Gamma$ be its actual automorphism space, based at the identity, and let N be its weak automorphism space in the categorical group $\Pi_1 A$, where $A = \mathrm{Homeo}(S^2)$. The induced transporter map has

$$\pi_1(M, 1) = \mathbb{Z} \longrightarrow \pi_1(N, 1) = \mathbb{Z}/2, \qquad n \longmapsto n \bmod 2.$$

Its kernel is $2\mathbb{Z}$, so level 0 fails for this faithful action of an infinite group. This is the exact fundamental-groupoid comparison; C.24 separately retains the higher homotopy of A when computing the untruncated homotopy-fixed space. The hyperbolic zero-level question recorded in C.12 has its stated, stronger surface hypothesis.

Manuscript: §23; folios 140–141; scans 279–281.

Proof. Commutation with an irrational rotation implies commutation with every axis rotation: its powers are dense in S^1, and composition and inversion in the compact-open homeomorphism group are continuous. An equivariant homeomorphism preserves the two poles as a set and sends each other rotation orbit to another orbit. In the orientation- preserving identity component it has exactly the form

$$f(\theta, t) = (\theta + a(t), b(t)), \quad a \in C((-1,1), \mathbb{R}/2\pi\mathbb{Z}), \quad b \in \mathrm{Homeo}^+([-1,1]).$$

To prove that this parametrization is onto, equivariance forces the angular function on each orbit to be translation by a unique $a(t)$; the orbit quotient supplies b. Orientation forces b to be increasing. Conversely every such pair extends continuously at both poles, because those circle orbits collapse there, and the same argument applies to its inverse. Thus the parametrization is a homeomorphism of spaces with the compact-open topologies. Continuity at a pole imposes no limit on $a(t)$; on any compact subinterval of $(-1, 1)$ it is exactly ordinary compact-open convergence.

The homotopy $a_s(t) = a((1-s)t)$ contracts the angular-function factor to its evaluation circle at 0. The convex interpolation $b_s(t) = (1-s)b(t) + st$ contracts the increasing-homeomorphism factor to the identity: it remains strictly increasing and fixes both endpoints. These homotopies are continuous in the stated topologies and extend continuously to the sphere. The actual identity centralizer therefore has homotopy type S^1, with its axis-rotation circle giving the specified generator of $\pi_1 M$.

In the weak transporter groupoid, an automorphism of its identity object is a loop in A fixed by the conjugation action of Γ. This is exactly the groupoid calculation of C.9, now applied to $\Pi_1 A$ rather than asserting any higher truncation of A itself. By the evaluation and polar-decomposition comparison in C.24, $\pi_1(A, 1) = \pi_1 SO(3) = \mathbb{Z}/2$. Conjugation by an axis rotation acts trivially on this group: joining that rotation to the identity in $SO(3)$ gives a based homotopy of its conjugation map to the identity. Hence $\pi_1 N = \mathbb{Z}/2$.

The transporter comparison sends a loop of equivariant homeomorphisms to that same loop in A. Its effect is consequently the usual $\pi_1 SO(2) \to \pi_1 SO(3)$, reduction modulo two. More explicitly, a loop of two complete rotations has degree two under the evaluation $a \mapsto a(0)$ in the actual centralizer, so it is nonzero there. Its lift to the quaternion double cover $S^3 \to SO(3)$ is the closed loop $t \mapsto \cos(2\pi t) + \mathbf{i}\sin(2\pi t)$, which contracts in the simply connected S^3. It is therefore killed by the weak transporter map. This proves the exact nontrivial kernel and the zero-level failure, with no inference from disconnectedness. □

C.26. [Proved] Transitive centralizers on a closed hyperbolic surface. Let X be a closed connected orientable surface of genus $g \geq 2$, and give $A = \mathrm{Homeo}(X)$ the compact-open topology. For every subgroup $L \leq A$ acting transitively on X, the centralizer $C_A(L)$ is finite. Consequently, for an arbitrary strict action $u : \Gamma \to A$ and a based loop f_t in $C_{A^0}(u(\Gamma))$, the loop is equivariantly null-homotopic whenever the subgroup generated by its values acts transitively on X. There also exists a closed torsion-free Polish subgroup $G \leq \mathrm{Homeo}(D^2, \partial D^2)$ with $\pi_1 G = \mathbb{Z}$; extension by the identity embeds it closedly in every hyperbolic A^0, where its displayed generator becomes null-homotopic. This last subgroup is not asserted to be a centralizer.

Manuscript: 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

Proof. Put $K = \overline{L}$ and $N = C_A(K)$. Continuity of composition gives $C_A(L) = C_A(K)$, and K is a closed Polish subgroup of A that still acts transitively. Fix $x \in X$ and write K_x for its stabilizer. The open mapping principle for Polish actions gives a homeomorphism

$$K/K_x \longrightarrow X, \qquad kK_x \longmapsto kx. \tag{C.26.1}$$

Precisely, we use the ordinary-action case of [22, Theorem 3.5]: the acting group is Polish, X is nonmeagre Hausdorff, the action is continuous and transitive, and the domain K^x of the partial-action notation is all of K, hence open. The source proves this open mapping statement; no local section of (C.26.1) is being assumed.

Every $n \in N$ is determined by $n(x)$, since $n(kx) = kn(x)$. Its possible values form exactly the closed subset

$$N_K(K_x)/K_x \;\subseteq\; K/K_x = X. \tag{C.26.2}$$

Indeed, $n(x) = kx$ forces $K_x = K_{kx} = kK_xk^{-1}$. Conversely a normalizing k defines the homeomorphism $gK_x \mapsto gkK_x$, commuting with K. The normalizer of the closed subgroup K_x is closed: both inclusions in $kK_xk^{-1} = K_x$ are intersections of closed conditions. Its image under (C.26.1) is closed because its inverse image under the quotient map is that same normalizer. Thus (C.26.2) is compact.

Evaluation is a homeomorphism of N onto (C.26.2). Only inverse continuity needs proof. Fix $\varepsilon > 0$. For every $y = g_yx$ choose an identity neighborhood W_y in K such that each $g_ywg_y^{-1}$, $w \in W_y$, moves every point of X by less than $\varepsilon/3$. The open sets g_yW_yx cover X; take a finite subcover with elements $g_1, \dots, g_m$. When $n(x)$ is sufficiently close to $n_0(x)$, each $g_in(x)$ is within $\varepsilon/3$ of $g_in_0(x)$. At $z = g_iwx$ in the corresponding set, commutation gives

$$n(z) = g_iwn(x), \qquad n_0(z) = g_iwn_0(x).$$

The two uniformly small movements by $g_iwg_i^{-1}$ and the middle $\varepsilon/3$ bound imply $d(n(z), n_0(z)) < \varepsilon$. This is uniform in z and proves inverse continuity in the compact-open topology. In particular N is compact.

An element of N fixing one point fixes every point, by transitivity and commutation. Every element of A^0 has Lefschetz number $\chi(X) = 2 - 2g \neq 0$, hence has a fixed point. Therefore

$$N \cap A^0 = 1.$$

Projection to the discrete group π_0A is thus a continuous injection on N. Its image is compact and discrete, hence finite, as asserted.

For the stated loop, put $L = \langle f_t : 0 \le t \le 1 \rangle$. Every $u(\gamma)$ commutes with L, so the transitivity hypothesis and the first assertion place $u(\Gamma)$ inside a finite group. Write $H = u(\Gamma)$. The equality $C_{A^0}(u(\Gamma)) = C_{A^0}(H)$ is equality of subspaces, irrespective of the cardinality of Γ. C.7 applied to the finite group H contracts the loop in this exact centralizer. The assertion therefore does not replace the operator group by a different action without a comparison map. □

A proved obstruction to a closed-subgroup-only argument. There is a closed torsion-free Polish subgroup of $\mathrm{Homeo}(D^2, \partial D^2)$ with fundamental group $\mathbb{Z}$. We give the embeddings, because this example explains exactly why closedness and absence of compact circles do not by themselves settle C.12. It is not asserted to be a centralizer.

Let

$$E = \{f \in L^1([0,1], \mathbb{R}) : f \text{ is integer-valued almost everywhere}\}.$$

An L^1-convergent sequence has an almost-everywhere convergent subsequence; hence E is closed and Polish. For $0 < s \le 1$ define $D_sf(t) = f(t/s)$ on $[0, s]$ and zero on $(s, 1]$, and put $D_0f = 0$. These maps remain in E and satisfy $\|D_sf\|_1 = s\|f\|_1$. Strong continuity of dilation follows first

for step functions and then by L^1 approximation. Thus $(s, f) \mapsto D_s f$ contracts E and each of its norm balls at zero. In particular E is contractible and locally contractible.

The constant integer functions form a closed discrete subgroup $\mathbb{Z}$. The quotient $G = E/\mathbb{Z}$ is Polish, and $E \to G$ is a covering: translates of a norm ball of radius less than $1/3$ by distinct integers are disjoint. It is the universal cover, and

$$\pi_1(G) = \mathbb{Z}, \qquad \gamma(s) = [\mathbf{1}_{[0,s]}] \quad (0 \le s \le 1) \tag{C.26.3}$$

is a generator, because its lift joins zero to the constant function 1. Nevertheless G is torsion-free: $n[f] = 0$ says that nf is a constant integer, so the integer-valued f is constant and $[f] = 0$.

There is a closed isometric group embedding

$$G \longrightarrow V := L^1([0,1], \mathbb{R})/\mathbb{R}, \qquad [f]_{\mathbb{Z}} \longmapsto [f]_{\mathbb{R}}, \tag{C.26.4}$$

where the second denominator consists of all constant real functions. For integer-valued h, the convex function $c \mapsto \int |h - c|$ is linear between consecutive integers, tends to infinity at both ends, and attains a minimum at an integer. Hence

$$\inf_{c \in \mathbb{R}} \|h - c\|_1 = \inf_{n \in \mathbb{Z}} \|h - n\|_1,$$

proving the isometry. The quotient metric on G is complete: choose a Cauchy subsequence with successive quotient distances below 2^{-j} and adjust each representative by an integer so that the successive L^1 distances have that bound. The representatives converge in E, and the entire original Cauchy sequence then converges in G. This proves closedness in the Banach space V.

We spell out a closed linear isometric embedding $T : V \hookrightarrow C_0([0,1], \mathbb{R})$, where the subscript here means vanishing at both endpoints. The weak-star unit ball B of V^* is compact metrizable, and $v \mapsto (\ell \mapsto \ell(v))$ embeds V isometrically into $C(B)$. A continuous surjection from the Cantor set to B exists by nested finite covers of decreasing diameters. Pullback followed by linear interpolation across each complementary interval of the Cantor set embeds $C(B)$ linearly and isometrically in $C([0,1])$. Place this interval in the middle third and interpolate to zero at the two outer endpoints. The resulting map is the required T; its image is closed because V is complete.

For $F \in C_0([0,1])$, the formula

$$\Phi_F(x, y) = \big(\tanh(\operatorname{artanh}(x) + F(y)), y\big) \qquad (|x| < 1) \tag{C.26.5}$$

on $[-1, 1] \times [0, 1]$, extended by the identity on the vertical sides, fixes the entire boundary and satisfies $\Phi_F \Phi_Q = \Phi_{F+Q}$. It is a topological-group embedding: tanh is Lipschitz, giving continuity, and evaluation at $(0, y)$ recovers $\tanh(F(y))$, from which F is recovered uniformly near any given profile. Its image is closed. A homeomorphism limit fixes the boundary and preserves the horizontal leaves; the image of the compact segment $\{0\} \times [0, 1]$ stays a positive distance from the vertical sides. Applying artanh therefore recovers uniformly bounded, uniformly convergent profiles.

Composing (C.26.4), T, and (C.26.5) embeds G as a closed subgroup of $\mathrm{Homeo}(D^2, \partial D^2)$. Extension by the identity outside a coordinate disc is a closed topological-group embedding into A^0 for any hyperbolic surface: restriction recovers the maps, and being the identity off the disc is a closed condition. Its loop (C.26.3) is ambient-null but nontrivial in G. The precise enlargement killing it is already visible: if $F_s = T([\mathbf{1}_{[0,s]}]_{\mathbb{R}})$, then $F_0 = F_1 = 0$ and $(r, s) \mapsto \Phi_{rF_s}$ is a based contraction in the larger real shear group. No operator group cutting out a centralizer that excludes this contraction has been constructed here. Thus this example refutes the auxiliary assertion of fundamental-group injectivity for all closed subgroups, while C.12 retains its stated centralizer question.

C.27. [Proved] An order-natural contraction of line centralizer loops. Give $\mathrm{Homeo}(\mathbb{R})$ its compact-open topology, let $H \le \mathrm{Homeo}(\mathbb{R})$ be an arbitrary subgroup, and put

$$C = C_{\mathrm{Homeo}^+(\mathbb{R})}(H).$$

Every based loop in C has an explicit based null-homotopy in C. Orientation-reversing elements of H are allowed. The construction commutes with conjugation by every homeomorphism of the line and requires no finiteness, differentiability, or local compactness of H.

Manuscript: 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

Proof. Let $t \mapsto f_t$ be a compact-open continuous loop in C, with $f_0 = f_1 = \mathrm{id}$. For $0 \le r \le 1$ define

$$m_r(x) = \min_{0 \le u \le r} f_u(x), \qquad M_r(x) = \max_{0 \le u \le r} f_u(x). \tag{C.27.1}$$

Both are increasing homeomorphisms. Indeed, for $x < y$, compactness gives

$$\min_{0 \le u \le 1} \bigl(f_u(y) - f_u(x)\bigr) > 0,$$

which proves strict increase for each minimum and maximum. Their properness is uniform in r: compactness of the inverse family $\{f_u^{-1} : 0 \le u \le 1\}$ gives, for every bounded target interval, a common bound on all its inverse images. Continuity of (C.27.1), jointly in r and x, follows from uniform continuity on compact parameter rectangles. Proper continuous strictly increasing maps of $\mathbb{R}$ are homeomorphisms.

Set

$$F_{s,t}(x) = \mathrm{med}\bigl(m_{1-s}(x), f_t(x), M_{1-s}(x)\bigr) = \max\{m_{1-s}(x), \min\{f_t(x), M_{1-s}(x)\}\}. \tag{C.27.2}$$

Finite minima and maxima of increasing homeomorphisms are again increasing homeomorphisms, by the same strict-increase and properness arguments. Formula (C.27.2) is continuous in (s, t, x), uniformly on compact x-sets; it is therefore a continuous family in the compact-open homeomorphism group. At $s = 0$, the prefix interval is the entire loop, so $m_1 \le f_t \le M_1$ and $F_{0,t} = f_t$. At $s = 1$, $m_0 = M_0 = \mathrm{id}$, giving $F_{1,t} = \mathrm{id}$. Since $u = 0$ occurs in every prefix interval, $m_r \le \mathrm{id} \le M_r$, and hence $F_{s,0} = F_{s,1} = \mathrm{id}$ for every s.

It remains to check the exact equivariance. An increasing $h \in H$ fixes both envelopes under conjugation, because conjugation commutes with their pointwise minimum and maximum and fixes every f_u. A decreasing $h \in H$ interchanges m_r and M_r instead. Every order-preserving or order-reversing bijection preserves the median of three points. Thus both cases give

$$hF_{s,t}h^{-1} = F_{s,t} \qquad (h \in H, 0 \le s, t \le 1).$$

This proves that the entire based contraction lies in C. Replacing the loop by its conjugate by an arbitrary line homeomorphism gives the conjugate of the same formula, proving the asserted naturality. □

The surface problem in C.12 still requires a continuous construction across different orbit strata. Applying (C.27.2) separately after choosing coordinates on individual line orbits does not establish continuity on the surface. No such gluing statement follows from C.27, and none is used here.

C.28. [Proved] Orbit types of a loop-generated plane group. Let $t \mapsto f_t$ be a compact-open continuous based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, with $f_0 = f_1 = \mathrm{id}$, and put

$$L = \{f_t : 0 \le t \le 1\}, \qquad P = L \cup L^{-1}, \qquad G = \langle P \rangle.$$

Every G-orbit is one of the following: an open subset of the plane; a point; a Jordan curve; or the image of a continuous injective map $\eta : \mathbb{R} \to \mathbb{R}^2$ that restricts to a homeomorphism on every compact interval. In the last case its intrinsic arc order identifies it with the line, but no global homeomorphism from $\mathbb{R}$ to the orbit with its subspace topology is asserted. The loop restricts continuously in the compact-open topology to the intrinsically ordered line. Every homeomorphism commuting with G carries these orbit types to the same types and induces a homeomorphism between the corresponding intrinsic lines, allowing reversal of their orders.

Manuscript: 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

Proof. Fix $x \in \mathbb{R}^2$ and put

$$K_n = P^n x, \qquad O = Gx = \bigcup_{n \geq 1} K_n. \tag{C.28.1}$$

The sets K_n are nested compact Peano continua. Indeed, P is a continuous image of the wedge of two circles, using the given loop and its inverse; it is compact, connected and locally connected. A finite product of such spaces has the same properties, and the word-evaluation map has image K_n. To recall why local connectedness passes through this compact surjection, for an open neighborhood of a point in the image cover its compact inverse fibre by finitely many connected open subsets of its inverse image. Their images all meet that point and lie in its component. The complement of the image of the remaining compact set is a smaller open neighborhood in that component. Thus components of open sets in the image are open. In particular K_n is arcwise connected. The identity lies in P, which proves the nesting in (C.28.1). Every element of G can be joined to the identity by a path in G, by writing it as a finite word and using the specified paths to its factors.

Suppose first that O contains a Jordan curve J. For any $g \in G$ choose a path g_s from id to g, so that $g_s J \subseteq O$. Let D_s be the bounded Jordan domain of $g_s J$. If $gJ \neq J$, then $D_0 \triangle D_1$ contains a nonempty open set away from the two boundary curves. Every point q in this open set lies on some $g_s J$: otherwise the winding number of the continuously varying curve $g_s J$ around q would be constant, whereas its endpoint values are 0 and 1. Hence O contains a nonempty open set. Transitivity of G on O carries this open set to a neighborhood of each point of O, so O is open. If every g preserves J, then for a point $y \in J$ its orbit is contained in J; since $Gy = O$ and $J \subseteq O$, we obtain $O = J$. Thus a nonopen orbit containing a Jordan curve is itself that Jordan curve.

We may now consider a nonopen orbit containing no Jordan curve. Each K_n is a dendrite: a compact, connected, locally connected metric space containing no Jordan curve. Arcs between two points are unique, since two distinct arcs would contain a Jordan curve. This also proves uniqueness of arcs in all of O: two proposed arcs in O would give the same forbidden curve. Existence follows by putting their endpoints into one K_n.

Each K_n has countably many branch points. Choose a countable dense subset D_n. Three distinct components of the complement of a branch point are open there, so choose a point of D_n in each. The branch point is the unique common point of the three pairwise connecting arcs for this triple: two branches meet only at their point of separation. Thus each branch point is determined by one of the countably many triples from D_n.

A triod point of O is a branch point of some K_n. Put the three outer endpoints of its triod into one K_n; the unique connecting arcs in K_n must be the arcs of that triod. The triod points of O are consequently countable. Their set is invariant under every element of G. If it were nonempty, transitivity would make every point of O a triod point. A nontrivial arcwise connected orbit contains an arc and is uncountable, giving a contradiction. Thus a nontrivial O has no triod points, and each nondegenerate K_n is an arc.

The nested arcs give O a consistent linear arc order, unique up to reversal. To see the absence of additional branches directly, fix two distinct points a, b. Every other point is on $[a, b]$, beyond a, or beyond b, because any fourth alternative would give a triod. There are at most two endpoints of this ordered union. The endpoint set is invariant under G, so transitivity and uncountability exclude endpoints. We can therefore pass to a cofinal sequence of arcs A_j such that A_j lies in the relative interior of A_{j+1}: extend both endpoints in O and then put the two extensions and the next K_n into a later K_m.

Choose compatible homeomorphisms $[-j, j] \to A_j$, extending at each step along the two added arc intervals. Their union is a continuous injective map

$$\eta : \mathbb{R} \longrightarrow \mathbb{R}^2, \qquad \eta(\mathbb{R}) = O, \tag{C.28.2}$$

whose restriction to every compact interval is a homeomorphism onto its image. These assertions follow on each finite interval from one of the compact arc parametrizations. This supplies the stated intrinsic line structure. It does not prove that the inverse of (C.28.2) is continuous for the global subspace topology; accumulations from far along an orbit have not been excluded.

The restricted loop is continuous for that intrinsic line topology. A compact arc in O is contained in some K_m, and $f_t(K_m) \subseteq K_{m+1}$ for all t. The latter is a compact arc, on which the intrinsic and subspace topologies coincide. Joint continuity of $(t, z) \mapsto f_t(z)$ on each such compact arc therefore gives compact-open continuity of $\eta^{-1} f_t \eta \in \mathrm{Homeo}(\mathbb{R})$. Its order sign is positive throughout the loop: images of two distinct points vary continuously in the line and cannot cross. Finally a commuting homeomorphism h satisfies the exact equality

$$h(K_n(x)) = K_n(hx).$$

It therefore maps compact arcs homeomorphically to compact arcs and induces a homeomorphism of the intrinsic ordered lines; its inverse has the same property. It also preserves the point, Jordan-curve and open alternatives, completing the proof. □

Exact application and remaining step. The canonical lift in the paragraph after C.12 sends a hyperbolic centralizer loop to a closed based loop in $\mathrm{Homeo}^+(\widetilde{U})$, where $\widetilde{U} \cong \mathbb{R}^2$. The entire group $\widetilde{H}$ of lifts of the operator image commutes with it and hence with the G of C.28. Thus C.28 applies to this actual lift group, and C.27 provides an order-natural contraction on each intrinsic line orbit after restriction to its stabilizer in $\widetilde{H}$. These are typed restriction and lift maps, but they do not yet give a homotopy through homeomorphisms of the whole plane. Continuity across distinct orbit strata, compatibility with their frontiers, and a relative contraction on proper open orbits remain unproved. C.26 treats global transitivity on a closed hyperbolic surface; a proper open orbit in C.28 does not satisfy that hypothesis. The admissible infinite-group level-0 question therefore remains C.12.

C.29. [Proved] Circle centralizers and the winding of a plane loop. For every subgroup $H \leq \mathrm{Homeo}(S^1)$, put $C = C_{\mathrm{Homeo}^+(S^1)}(H)$, with the compact-open topology. The inclusion induces an injection

$$\pi_1(C, \mathrm{id}) \longrightarrow \pi_1(\mathrm{Homeo}^+(S^1), \mathrm{id}) = \mathbb{Z}, \tag{C.29.1}$$

where the integer is the winding of the image of one point. Its image lies in $\mathbb{Z}^H$, with H acting on $\mathbb{Z}$ by its orientation character. A based loop of winding zero has an explicit contraction in C that is natural under all homeomorphisms between circles. In particular, a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$ preserving a Jordan curve restricts to a winding-zero loop on that curve. Its restriction therefore has this natural contraction against every commuting circle homeomorphism.

Manuscript: 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

Proof. Write the circle as $\mathbb{R}/\mathbb{Z}$, and set

$$\widetilde{A} = \{F \in \mathrm{Homeo}^+(\mathbb{R}) : F(x+1) = F(x) + 1\}.$$

The quotient map $\widetilde{A} \to \mathrm{Homeo}^+(S^1)$ is a covering with deck transformations $F \mapsto F + n$, $n \in \mathbb{Z}$. Indeed, requiring $F(0)$ to lie in a fixed interval of length less than one gives continuous unique lifts over the corresponding evaluation neighborhood. The homotopy $(r, F) \mapsto (1-r)F + r\,\mathrm{id}$ contracts $\widetilde{A}$: strict increase, the translation equation, and properness hold at every parameter. Thus the target fundamental group in (C.29.1) is precisely $\mathbb{Z}$. For a based loop f_t, its unique lift F_t starting at id ends at $x \mapsto x + n$, where n is its point winding.

Suppose that $f_t \in C$ and $n = 0$. Every lift $\widetilde{h}$ of an element $h \in H$, including an orientation-reversing one, commutes with every F_t. The commutator projects to the identity on the circle, so it is an integer translation depending continuously on t; it is the identity at $t = 0$, and hence at all t. Consequently F_t is a based loop in the line centralizer of the entire lift group of H. This group also contains the integer translations. Apply C.27, explicitly

$$\begin{aligned} m_r(x) &= \min_{0 \le u \le r} F_u(x), \qquad M_r(x) = \max_{0 \le u \le r} F_u(x), \\ Q_{s,t}(x) &= \mathrm{med}\big(m_{1-s}(x), F_t(x), M_{1-s}(x)\big). \end{aligned} \tag{C.29.2}$$

Every $Q_{s,t}$ commutes with all lifts and with $x \mapsto x + 1$, so (C.29.2) descends to a contraction through elements of C. This proves injectivity in (C.29.1).

The winding of a conjugated loop hf_th^{-1} is the winding of f_t multiplied by the orientation sign of h. Since these loops agree for $h \in H$, the image is contained in $\mathbb{Z}^H$. For naturality, let q be a homeomorphism between two circles and choose any lift $\widetilde{q}$ to their universal lines. The based lift of the conjugated loop is $\widetilde{q}F_t\widetilde{q}^{-1}$. An increasing $\widetilde{q}$ carries the two extrema in (C.29.2) to the corresponding extrema; a decreasing one interchanges them. Every line homeomorphism preserves the median of three points. Thus (C.29.2) is carried to exactly the same formula for the conjugated loop. Changing the lift of q by an integer translation does not change the descended contraction.

It remains to prove the stated plane winding comparison. Conjugate by a Schoenflies homeomorphism so that the preserved Jordan curve is the unit circle. Choose two distinct points a, b on it. Evaluation and the difference direction give the continuous maps

$$\mathrm{Homeo}^+(\mathbb{R}^2) \longrightarrow \mathrm{Conf}_2(\mathbb{R}^2) \longrightarrow S^1, \qquad f \longmapsto (f(a), f(b)) \longmapsto \frac{f(b) - f(a)}{|f(b) - f(a)|}. \tag{C.29.3}$$

For the given circle-preserving loop choose continuous angular lifts $\alpha(t), \beta(t)$ with $\alpha(t) < \beta(t) < \alpha(t) + 2\pi$. They both acquire the increment $2\pi n$: their difference remains in $(0, 2\pi)$ and returns to its initial value modulo 2π. The identity

$$e^{i\beta(t)} - e^{i\alpha(t)} = 2i\, e^{i(\alpha(t)+\beta(t))/2} \sin \frac{\beta(t) - \alpha(t)}{2}$$

shows that the winding of (C.29.3) is exactly n. A null-homotopy of the ambient plane loop maps under (C.29.3) to a null-homotopy in S^1, forcing $n = 0$. This proves the exact comparison and the last assertion. □

Application to C.12. The canonical lift of a hyperbolic centralizer loop is null-homotopic as a loop of plane homeomorphisms: it is the image under the continuous canonical lift homomorphism of a loop in the contractible A^0. Every Jordan-curve orbit in C.28 is preserved throughout this loop. Its restriction therefore has winding zero by (C.29.3), and (C.29.2) contracts it naturally against its stabilizer in the full operator lift group. Commuting operators carrying one circle orbit to another carry these contractions to each other. This gives a circle-orbit contraction with

the exact required equivariance. C.34 subsequently proves joint continuity, including inverses, on the full nonopen locus. C.35 shows that this particular prescription need not admit an equivariant extension over an adjacent open orbit, and C.36 constructs compatible alternatives on finite unions of closures with two-dimensional proper operator image. The remaining global compatibility requirements are stated in C.12, which remains open.

C.30. [Proved] Properness of a transitive-action centralizer. Let V be a connected second-countable surface without boundary, and let K be a closed subgroup of $\mathrm{Homeo}(V)$ acting transitively on V, with the compact-open topology. Put $N = C_{\mathrm{Homeo}(V)}(K)$. Then N is locally compact, its action on V is free and proper, and for every $x \in V$ evaluation is a homeomorphism onto the closed subset

$$N \xrightarrow{\mathrm{ev}_x} N_K(K_x)/K_x \;\subseteq\; K/K_x \cong V. \tag{C.30.1}$$

Under the normalizer-quotient group structure, this is an anti-isomorphism of topological groups. For an open orbit O of the loop-generated group in C.28, the image in $\mathrm{Homeo}(O)$ of its operator stabilizer lies in such an N; the closure of that image acts freely and properly on O.

Manuscript: 23; folios 140–141; scans 279–281; Part C(b) infinite-operator request; exact partial reduction for C.12.

Proof. The group K is Polish. The open mapping theorem [22, Theorem 3.5], with the ordinary continuous transitive action on the nonmeagre Hausdorff surface V, identifies K/K_x homeomorphically with V. In particular the orbit map $k \mapsto kx$ is open. We will use the following exact consequence, requiring no local continuous section: for every sequence $x_j \to x$, there are $k_j \to 1$ in K with $k_j x = x_j$. To prove it, choose a decreasing identity-neighborhood basis W_m. The open sets $W_m x$ contain x, so successive tails of the sequence lie in them. Choose k_j from the corresponding $W_{m(j)}$, with $m(j) \to \infty$.

An element of N is determined by its value at x, since $n(kx) = kn(x)$. If $n(x) = ax$, equality of stabilizers gives $aK_xa^{-1} = K_x$. Conversely, for a in this normalizer the map $kK_x \mapsto kaK_x$ is a homeomorphism commuting with K. The normalizer is closed: both inclusions defining $aK_xa^{-1} = K_x$ are intersections of closed conditions. It is saturated for the quotient by K_x, so its image is a closed subset of $K/K_x = V$. This proves the claimed set-theoretic image and closedness in (C.30.1). If $n(x) = ax$ and $m(x) = bx$, then $(nm)(x) = bax$, proving the stated anti-isomorphism.

We prove inverse continuity of evaluation, including the issue of noncompact V. Suppose that $n_j(x) \to n(x)$, with $n_j, n \in N$. For any compact set $B \subset V$, a failure of uniform convergence on B would provide a subsequence and $y_j \in B$ with $y_j \to y$ witnessing that failure. Choose $a \in K$ with $ax = y$, and then $b_j \to 1$ with $b_j y = y_j$, using open evaluation at y. Commutation gives

$$n_j(y_j) = n_j(b_j a x) = b_j a\, n_j(x) \longrightarrow an(x) = n(y). \tag{C.30.2}$$

Also $n(y_j) \to n(y)$, a contradiction. Hence $n_j \to n$ uniformly on every compact set, which is compact-open convergence. Inversion can be checked directly as well. Put $c_j = n^{-1}n_j$, so $c_j x \to x$, and choose $a_j \to 1$ with $a_j x = c_j x$. Because c_j commutes with a_j, $c_j^{-1}x = a_j^{-1}x \to x$; applying (C.30.2) to c_j^{-1} gives convergence of the inverses. All the spaces are metrizable, so this sequential argument proves inverse continuity. Thus (C.30.1) is a homeomorphism, and N is locally compact as a closed subspace of the locally compact surface.

If $n \in N$ fixes one point, transitivity and commutation make it fix every point, proving freeness. To prove properness, consider sequences with

$$x_j \longrightarrow x, \qquad n_j x_j \longrightarrow y.$$

Choose $k_j \to 1$ with $k_j x = x_j$. Then

$$n_j x = k_j^{-1} n_j x_j \longrightarrow y. \tag{C.30.3}$$

The image of evaluation at x is closed, so $y = nx$ for a unique $n \in N$. Its inverse continuity gives $n_j \to n$. For any compact subset of $V \times V$, every sequence in its inverse image under $(n, z) \mapsto (z, nz)$ therefore has a convergent subsequence, with limit in the same inverse image. Metrizability turns this sequential compactness into compactness, proving that the action map is proper. Every closed subgroup of N inherits the free proper action.

Finally let G be the group in C.28 and let O be one of its open orbits. It is connected because every element of G is joined to the identity by a path in G. The closure

$$K = \overline{\{g|_O : g \in G\}}^{\mathrm{Homeo}(O)}$$

is a closed Polish group still transitive on O. Let $\widetilde{H}$ be any group commuting with G, and let $\widetilde{H}_O$ be its setwise stabilizer of O. The precise restriction homomorphism is

$$\widetilde{H}_O \longrightarrow \mathrm{Homeo}(O), \qquad h \longmapsto h|_O, \quad \ker = \{h \in \widetilde{H}_O : h|_O = \mathrm{id}\}. \tag{C.30.4}$$

Its image commutes with the restriction of G and hence with K. The closure of this image is a closed subgroup of N, and therefore acts freely and properly on O, as claimed. □

Precise remaining use. The closure in the last assertion has its subspace topology from Homeo(O). Formula (C.30.4) does not assert that the original discrete operator group acts freely, since its kernel may be nontrivial; nor does it equip a possibly dense abstract image with the discrete topology and claim properness for that topology. The result supplies the closed locally compact transformation group actually present on an open orbit. No Lie-group classification or quotient-bundle contraction is invoked here. A contraction relative to orbit frontiers, and its continuous assembly with C.27 and C.29 on the one-dimensional strata, remain unproved. Thus C.30 does not change the unresolved status of C.12.

C.31. [Proved] Track control and continuity at the fixed stratum. Let f_t be a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let G be its generated group as in C.28, and let Y be the union of the nonopen G-orbits. This is a closed subset of the plane. Define $Q_{s,t}$ on Y by C.27 on each intrinsic line orbit, by the descended formula of C.29 on each circle orbit, and by the identity on point orbits. Then every $Q_{s,t}$ is a bijection of Y preserving each orbit, and

$$Q_{s,t}^{-1} = Q_{s,t}[f^{-1}], \qquad Q_{s,t}(x) \in \{f_u(x) : 0 \le u \le 1\}, \qquad Q_{s,t}^{-1}(x) \in \{f_u^{-1}(x) : 0 \le u \le 1\}. \tag{C.31.1}$$

Here $Q[f^{-1}]$ means the same construction applied to the based inverse loop $u \mapsto f_u^{-1}$, which generates the same group. Both evaluation maps $(s, t, x) \mapsto Q_{s,t}^{\pm 1}(x)$ are continuous at every point of $[0, 1]^2 \times Z$, where $Z = \{x : f_u(x) = x \text{ for all } u\}$ is the point-orbit stratum. The construction commutes with every homeomorphism commuting with all f_u. No continuity at a nonfixed orbit is asserted here.

Manuscript: §23; folios140–141; scans279–281; Part C(b) infinite-operator request; fixed-stratum gluing reduction for C.12.

Proof. The union of open orbits is open, proving that Y is closed. C.28 supplies exactly the remaining point, circle and intrinsic-line types. The circle restrictions have zero winding by C.29, since the ambient loop is null-homotopic. Thus the formulas defining Q are available on every orbit in Y, and are independent of the chosen line or circle coordinates by their proved naturality. Each formula gives a homeomorphism of its single intrinsic line or circle, so together they give a bijection of Y.

We prove the inverse identity before making any continuity claim. For increasing homeomorphisms of a line, pointwise inversion reverses order and exchanges finite minimum and maximum:

$$(a \wedge b)^{-1} = a^{-1} \vee b^{-1}, \qquad (a \vee b)^{-1} = a^{-1} \wedge b^{-1}. \tag{C.31.2}$$

For example, at a point x the first inverse is the larger of $a^{-1}(x)$ and $b^{-1}(x)$: at this larger point one of a, b equals x, and the other is at least x. The same argument works for compact families, whose extrema and inverse extrema are attained. Therefore, with the notation of C.27,

$$m_r^{-1}(x) = \max_{0\le u\le r} f_u^{-1}(x), \qquad M_r^{-1}(x) = \min_{0\le u\le r} f_u^{-1}(x). \tag{C.31.3}$$

The median of three line homeomorphisms is the lattice expression $(a \wedge b) \vee (a \wedge c) \vee (b \wedge c)$; its dual expression $(a \vee b) \wedge (a \vee c) \wedge (b \vee c)$ has the same value. Applying (C.31.2) thus proves

$$\operatorname{med}(a, b, c)^{-1} = \operatorname{med}(a^{-1}, b^{-1}, c^{-1}). \tag{C.31.4}$$

Combining (C.31.3)–(C.31.4), and using symmetry of the median in its two envelope entries, gives the first identity in (C.31.1) on every line orbit. On a circle, the inverse of the unique based lift F_u is exactly the unique based lift of f_u^{-1}. The same calculation on the universal line descends to the circle. It also holds on point orbits.

At any point the median chooses one of its three entries. Each envelope entry is $f_u(x)$ for a parameter attaining its extremum, and the middle entry is $f_t(x)$ itself. This proves the second assertion of (C.31.1) on lines; projection of the same statement for the based lifts proves it on circles. The inverse identity then proves the third assertion.

Fix $p \in Z$ and an open neighborhood V of p in the plane. Joint continuity of the action, compactness of $[0, 1]$, and $f_u(p) = p$ give a neighborhood W of p such that $f_u(W) \subseteq V$ for every u. The compact inverse family gives the same assertion for all f_u^{-1} after shrinking W. By (C.31.1),

$$Q_{s,t}^{\pm 1}(W \cap Y) \subseteq V \cap Y \quad \text{for all } s, t \in [0, 1]. \tag{C.31.5}$$

Since $Q_{s,t}^{\pm 1}(p) = p$, this proves joint continuity at every (s, t, p), uniformly in the two homotopy parameters.

A homeomorphism commuting with all f_u permutes the G-orbits, preserves their four types, and maps intrinsic lines by line homeomorphisms by C.28. Naturality in C.27 and C.29 gives $hQ_{s,t}h^{-1} = Q_{s,t}$ on all of Y. This proves the asserted equivariance without any continuity assertion at the remaining points. □

Scope of the gluing step. Formula (C.31.5) resolves continuity at the fixed stratum for the orbitwise construction on Y. It does not yet make the bijections $Q_{s,t} : Y \to Y$ homeomorphisms, because continuity at the nonfixed line and circle strata remains unproved. Nor does it control a contraction on the omitted open orbits. These are the remaining global assembly problems for this route to C.12.

C.32. [Proved] Discrete proper operator groups on open orbits. Let O be a connected noncompact second-countable surface without boundary, and let a discrete group $Q \le \mathrm{Homeo}(O)$ act freely and properly. With compact-open topology on the homeomorphism groups, inclusion induces an injection

$$\pi_1\Big(C_{\mathrm{Homeo}(O)}(Q), \mathrm{id}\Big) \longrightarrow \pi_1(\mathrm{Homeo}(O), \mathrm{id}). \tag{C.32.1}$$

Moreover, a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$ preserving a connected open subset $O \subseteq \mathbb{R}^2$ restricts to a null-homotopic loop in $\mathrm{Homeo}(O)$. Consequently, for a hyperbolic centralizer loop and an open orbit in C.28 whose closed restricted operator image in C.30 is discrete, its restriction has a based equivariant null-homotopy on that orbit. The original restriction kernel is retained; no extension of this null-homotopy to the orbit frontier is asserted.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; discrete-closure open-orbit reduction for C.12.

Proof. First consider the free proper discrete action on O. It gives an actual covering

$$p : O \longrightarrow V = O/Q. \tag{C.32.2}$$

Here discreteness refers to the topology in the proper-action hypothesis. For completeness, a compact neighborhood of a point meets only finitely many of its translates, by properness. Shrink an open neighborhood inside it to avoid the nonidentity translates among this finite set, using freeness. Distinct translates of the result are disjoint, which gives the evenly covered neighborhoods for (C.32.2). The quotient V is Hausdorff by properness, is a surface without boundary by these local charts, and is second countable because images of a countable open basis of O form an open basis. The connected covering is regular, with deck group exactly Q.

We record the homeomorphism-group input precisely. Every second-countable surface admits a PL triangulation. Applied with the compact subpolyhedron consisting of one point v, the compact case of [60, Lemma 2.1] and the noncompact case of [60, Theorem 1.1] give

$$\mathrm{Homeo}(V, v)^0 \simeq * \quad \text{for } V \ncong S^2,\ \mathbb{R}P^2,\ \mathbb{R}^2. \tag{C.32.3}$$

These are all exceptions relevant to a surface without boundary with that one-point subpolyhedron. Evaluation $\mathrm{Homeo}(V)^0 \to V$ is a locally trivial bundle: in a coordinate disc a continuously chosen homeomorphism supported in that disc moves its center to a nearby point, giving a local section. Its identity fibre component is exactly the group in (C.32.3). The homotopy exact sequence therefore gives an injection

$$\pi_1(\mathrm{Homeo}(V), \mathrm{id}) \longrightarrow \pi_1(V, v) \quad (V \ncong S^2, \mathbb{R}P^2, \mathbb{R}^2). \tag{C.32.4}$$

The base V of (C.32.2) cannot be S^2 or $\mathbb{R}P^2$: their fundamental groups are finite, so every connected covering has finitely many sheets and compact total space, contrary to noncompactness of O. If $V = \mathbb{R}^2$, its connected covering is one-sheeted, so $Q = 1$ and (C.32.1) is the identity homomorphism.

Let f_t be a based loop in the centralizer that is null-homotopic in $\mathrm{Homeo}(O)$. It descends to a based loop b_t on V, defined by the exact equation

$$b_t p = p f_t. \tag{C.32.5}$$

The family is continuous: joint continuity descends through the open quotient map $[0,1] \times O \to [0,1] \times V$, and the compact-open exponential law applies since V is locally compact. For $x \in O$, the evaluation loop $f_t(x)$ is null-homotopic in O. Thus $b_t(p(x)) = p(f_t(x))$ is null-homotopic in V. By (C.32.4), b_t is a null-homotopic based loop of homeomorphisms of V. Choose a continuous filling $B : D^2 \to \mathrm{Homeo}(V)^0$ with the prescribed boundary b_t, and a point $z_* \in \partial D^2$ corresponding to the based endpoint, so that $B_{z_*} = \mathrm{id}$.

The lift is a filling through homeomorphisms and through the exact centralizer; we verify both assertions. Form

$$F : O \times D^2 \longrightarrow V, \qquad F(x, z) = B_z(p(x)). \tag{C.32.6}$$

At (x_*, z_*), the induced fundamental-group image is $p_*\pi_1(O, x_*)$, since D^2 is simply connected and $F(x, z_*) = p(x)$. The covering lifting criterion gives a unique continuous lift $\widetilde{B} : O \times D^2 \to O$ with $\widetilde{B}(x, z_*) = x$. Apply the identical construction to $F^-(x, z) = B_z^{-1}(p(x))$, giving a continuous lift $\widetilde{C}$ that is the identity at z_*. The maps $(x, z) \mapsto \widetilde{B}_z\widetilde{C}_z(x)$ and $(x, z) \mapsto x$ lift the same map $(x, z) \mapsto p(x)$ and agree at the basepoint. Uniqueness of lifts makes them equal; the reverse composition is equal for the same reason. Hence every $\widetilde{B}_z$ is a homeomorphism, with inverse $\widetilde{C}_z$, and the family is compact-open continuous.

For $q \in Q$, the two maps $\widetilde{B}_z q$ and $q\widetilde{B}_z$ lift the same map $B_z p$ and agree at z_*, so they are equal for every z. On $O \times \partial D^2$, both $\widetilde{B}_z$ and the specified f_t lift (C.32.5) and agree at the based

endpoint. They are therefore equal on that entire connected boundary product. Thus $\widetilde{B}$ fills the original loop in its centralizer. This proves (C.32.1).

Now let O be a connected open subset of the plane preserved by a based ambient-null loop. Its restriction is a continuous loop in $\mathrm{Homeo}(O)^0$. By [60, Theorem 1.1], with the empty subpolyhedron, this identity group is homotopically trivial unless O is the plane or the open annulus; the nonorientable and boundary exceptions in the source cannot occur for $O \subseteq \mathbb{R}^2$. In either remaining case its fundamental group is $\mathbb{Z}$. Choose model coordinates on O, a coordinate Jordan circle $J \subset O$, and two distinct points a, b of J. A Schoenflies homeomorphism h of the ambient plane sends J to the unit circle. The map

$$\Theta : \mathrm{Homeo}(O) \longrightarrow S^1, \qquad k \longmapsto \frac{h(k(b)) - h(k(a))}{|h(k(b)) - h(k(a))|} \tag{C.32.7}$$

has induced homomorphism $\pi_1 \mathrm{Homeo}(O) \to \mathbb{Z}$ that is an isomorphism. Indeed, the full rotation loop in the chosen plane or annulus model preserves J and has restriction winding one. The explicit computation in C.29 shows that its image under (C.32.7) also has winding one. A homomorphism $\mathbb{Z} \to \mathbb{Z}$ whose image contains 1 is an isomorphism. Formula (C.32.7) extends as a continuous map on all ambient plane homeomorphisms by the same two evaluations. The ambient-null loop therefore has zero image under it, so its restricted fundamental-group class is zero. This proves the second assertion in all cases.

Finally take an open orbit from C.28 and let Q be the closed restricted operator image supplied by C.30. When this group is discrete, C.30 gives a free proper action with precisely that discrete topology, so (C.32.2) applies. The restricted canonical plane loop commutes with Q and is null-homotopic in $\mathrm{Homeo}(O)$ by the second assertion. Its null-homotopy supplied by (C.32.1) commutes with Q, hence with the image of the operator stabilizer. The kernel in (C.30.4) acts identically on O throughout. This proves the stated equivariant conclusion with the actual restriction homomorphism. □

Transport and frontier control. One may choose the obtained contraction on one representative of an operator orbit of open G-orbits and transport it to the others by conjugation. Equivariance under the setwise stabilizer makes this independent of the transporting operator. This establishes well-defined orbitwise families. The covering lift above supplies no uniform estimate at orbit frontiers. C.33 subsequently treats all positive-dimensional proper closed operator images intrinsically, and C.34 proves joint continuity on the full nonopen locus. C.35 records the obstruction to extending its particular median prescription, while C.36 constructs compatible alternatives on finite unions of closures with two-dimensional proper operator image. The remaining frontier and accumulation requirements are stated in C.12, which remains open.

C.33. [Proved] All proper operator groups on an open orbit. Let O be a connected noncompact second-countable surface without boundary. Let Q be a closed locally compact subgroup of $\mathrm{Homeo}(O)$, with its compact-open topology, acting freely and properly. Then inclusion induces an injection

$$\pi_1\Big(C_{\mathrm{Homeo}(O)}(Q), \mathrm{id}\Big) \longrightarrow \pi_1(\mathrm{Homeo}(O), \mathrm{id}). \tag{C.33.1}$$

For every open orbit of the canonically lifted hyperbolic centralizer loop in C.28, its restriction therefore has a based null-homotopy commuting with the entire restricted operator stabilizer. This conclusion retains the restriction kernel in C.30 and imposes no finiteness or discreteness hypothesis on the original operator group. It supplies an intrinsic null-homotopy on each open orbit; continuity across orbit frontiers remains unproved.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; arbitrary proper-closure open-orbit reduction for C.12.

Proof. Write $N = C_{\mathrm{Homeo}(O)}(Q)$. The topological group Q acts faithfully on the connected three-manifold $O \times \mathbb{R}$ by

$$q(x, s) = (qx, s). \tag{C.33.2}$$

This action is continuous: on any compact subset of $O \times \mathbb{R}$ the first coordinate is controlled on its compact projection to O. Pardon proves that a locally compact group acting faithfully on a connected three-manifold is a Lie group [52, Theorem 1.5 and the reduction preceding it]. Thus Q is a Lie group. Since it is a subspace of the second-countable homeomorphism group, it is second countable. Set $A = Q^0$ and $d = \dim A$. Properness and freeness make $q \mapsto qx$ a proper embedding of A in O for each x. Invariance of dimension gives $d \leq 2$. The case $d = 0$ is precisely C.32.

If $d = 2$, invariance of domain makes every A-orbit open. Connectedness gives a single orbit, and evaluation $A \to O$ is a homeomorphism. For $q \in Q$, choose $a \in A$ with $qx = ax$. Then $a^{-1}q$ fixes x, so freeness gives $q = a$ and $Q = A$. Under $O \cong A$ with its left translation action, the elements of N are exactly the right translations. The correspondence is an anti-isomorphism of topological groups, and evaluation $N \to O$ is a homeomorphism. A loop in N null-homotopic in $\mathrm{Homeo}(O)$ has a null evaluation loop in O, and hence is null-homotopic in N. This proves (C.33.1) in dimension two.

It remains to treat $d = 1$. The connected one-dimensional Lie group A is $\mathbb{R}$ or $\mathbb{R}/\mathbb{Z}$, written additively. Its restricted action is still free and proper. Palais' slice theorem gives a locally trivial principal bundle

$$p_0 : O \longrightarrow B = O/A. \tag{C.33.3}$$

Here all hypotheses of the theorem hold: the surface is completely regular, A is Lie, and a relatively compact neighborhood has relatively compact transporter by properness, which is the Cartan condition. A slice at a point has trivial stabilizer. The resulting local trivialization is the map $(a, s) \mapsto as$ from $A \times S$ onto the open saturated neighborhood of that slice. This is the free-action specialization of [51, Theorem 2.3.3]; its maps and inverse are also given in [40, Theorem 3.7].

We verify the dimension and local topology of the base rather than assuming differentiability of this action. The elementary fact needed is that a metrizable space S for which $S \times (-1, 1)$ is a surface is a one-manifold without boundary. Indeed, projecting path-connected open neighborhoods in the product shows that S is locally path connected. It has no isolated point, since an isolated point would give an open interval in a surface. Every embedded open arc in S is open: its product with $(-1, 1)$ maps injectively into a surface, and invariance of domain applies.

Fix $p \in S$. Local path connectedness and the absence of isolated points give an embedded arc $\alpha : [0, 1] \to S$ with $\alpha(0) = p$. Choose a path-connected open neighborhood W of p avoiding $\alpha(1)$. If W contains a point outside $\alpha([0, 1])$, join p to that point by an embedded arc in W. This second arc cannot enter and subsequently leave $\alpha((0, 1))$: that open arc has boundary contained in $\{p, \alpha(1)\}$, and an embedded arc in W cannot revisit p or pass through $\alpha(1)$. Thus the two arcs meet only at p, and together contain an arc having p as an interior point. Such an open arc is an open neighborhood of p by the preceding paragraph. In the other case $W \subseteq \alpha([0, 1))$, and a smaller neighborhood of p is a half interval. Its product with $(-1, 1)$ has zero local second homology at $(p, 0)$, whereas a surface without boundary has local second homology $\mathbb{Z}$. This is impossible. The elementary fact follows. Applied to the slices in (C.33.3), it proves that B is a one-manifold without boundary. Properness gives a Hausdorff quotient, the open quotient map gives second countability, and connectedness is inherited from O. Consequently B is $\mathbb{R}$ or $\mathbb{R}/\mathbb{Z}$.

Every principal A-bundle over either of these one-manifolds is trivial. One can see the actual section as follows. Over an interval, local trivializations are joined successively across overlapping subintervals, adjusting one section by its continuous A-valued transition function. A locally finite interval exhaustion gives a section on $\mathbb{R}$. Over the circle, first cut at one point and choose a section on the resulting closed interval. Its endpoint discrepancy is an element of the connected

group A; multiplying the section by a path from 0 to the negative of that element makes the endpoints agree, so that it descends to the circle. The map $(a, b) \mapsto a\, s(b)$, with inverse given in each bundle chart, therefore identifies

$$O \cong A \times B, \qquad c(a, b) = (c + a, b) \quad (c \in A). \tag{C.33.4}$$

The pair $A = B = \mathbb{R}/\mathbb{Z}$ is excluded because O is noncompact.

Let $\Delta = Q/A$. This discrete group acts freely and properly on B. Freeness follows directly: an element of Q fixing a point of B sends any point of its fibre to an A-translate; after that translate is removed, it fixes a point of O and is the identity. For properness, suppose $b_j \to b$ and $\delta_j b_j \to b'$ in B. Local sections give $x_j \to x$ above b_j and $y_j \to y$ above $\delta_j b_j$. Choose a representative q_j of δ_j and $a_j \in A$ such that $a_j q_j x_j = y_j$. Properness of the Q-action gives a convergent subsequence of $a_j q_j$, so their components δ_j are eventually constant on that subsequence. This proves properness of the discrete action on the locally compact metrizable space B.

We spell out this discrete action. On $\mathbb{R}$, an orientation reverser has a fixed point and is excluded by freeness. If Δ is nontrivial, its orbit of a point is closed and discrete, with a nearest point to the right. The element taking the chosen point to that next point generates Δ: its iterates are unbounded in both directions, and any intervening orbit point would contradict the choice of the nearest one. Choosing coordinates on one fundamental interval and extending by its iterates identifies the action with

$$\Delta = \mathbb{Z}, \qquad \beta(b) = b + 1. \tag{C.33.5}$$

On $\mathbb{R}/\mathbb{Z}$, properness makes Δ finite and freeness again excludes orientation reversers. The action on one finite orbit preserves its cyclic order, so it identifies Δ with a cyclic group C_n. Coordinates on the successive complementary intervals identify a generator with

$$\beta(b) = b + 1/n \quad \text{in } \mathbb{R}/\mathbb{Z}. \tag{C.33.6}$$

The trivial group is also allowed on either base.

Suppose Δ is nontrivial and choose a representative $q \in Q$ of the indicated generator. Conjugation by q on A is multiplication by λ, where $\lambda \in \mathbb{R}^\times$ for $A = \mathbb{R}$, and $\lambda \in \{1, -1\}$ for $A = \mathbb{R}/\mathbb{Z}$. Equation (C.33.4) therefore expresses its action exactly as

$$q(a, b) = (\lambda a + \tau(b), \beta(b)) \tag{C.33.7}$$

for a continuous function $\tau : B \to A$. We next give a coordinate change removing this translation term, with all component relations retained.

For $B = \mathbb{R}$ and $\Delta = \mathbb{Z}$, choose a continuous function $c : [0, 1] \to A$ whose endpoints satisfy $c(1) = \lambda c(0) + \tau(0)$, and extend it in both directions by

$$c(b + 1) = \lambda c(b) + \tau(b). \tag{C.33.8}$$

The endpoint equation makes the extension continuous at every integer; invertibility of multiplication by λ gives the extension towards negative integers. In the new fibre coordinate $a' = a - c(b)$, (C.33.7) becomes $q(a', b) = (\lambda a', b + 1)$.

For $B = \mathbb{R}/\mathbb{Z}$ and $\Delta = C_n$, necessarily $A = \mathbb{R}$. Write $q^n = T_{a_0}$, where T_{a_0} is translation by a_0 on the fibre. Conjugation gives $\lambda^n = 1$ and $\lambda a_0 = a_0$. If $\lambda = 1$, replacing q by $T_{-a_0/n} q$ makes $q^n = 1$ and does not change the group generated with A. If $\lambda = -1$, then n is even and $a_0 = 0$, so $q^n = 1$ already. Thus the negative value of λ is retained. For this finite-order representative define the real-valued function

$$c(b) = \frac{1}{n} \sum_{k=0}^{n-1} \mathrm{pr}_A\Big(q^k (0, \beta^{-k} b)\Big)\,. \tag{C.33.9}$$

The affine nature of (C.33.7), followed by cyclic reindexing using $q^n = 1$, gives $c(\beta b) = \lambda c(b) + \tau(b)$. The same coordinate change $a' = a - c(b)$ now gives $q(a', b) = (\lambda a', \beta b)$. We have proved, by explicit changes of coordinates, that the nonidentity components can be represented as

$$q(a, b) = (\lambda a, \beta b). \tag{C.33.10}$$

When Δ is trivial no such additional equation is needed.

Commuting with all translations in (C.33.4) forces a homeomorphism of $A \times B$ to have the form

$$h(a, b) = (a + \phi(b), \psi(b)), \qquad \phi \in C(B, A), \quad \psi \in \mathrm{Homeo}(B). \tag{C.33.11}$$

The inverse is $(a, b) \mapsto (a - \phi(\psi^{-1}b), \psi^{-1}b)$. These formulas and evaluation at the zero section give mutually inverse continuous maps for the compact-open topologies. In the nontrivial-component case, commuting with (C.33.10) is exactly the pair of equations

$$\psi\beta = \beta\psi, \qquad \phi(\beta b) = \lambda\phi(b). \tag{C.33.12}$$

Thus the map (C.33.11) identifies N, as a space, with the product of the function space defined by the second equation and the homeomorphism centralizer defined by the first. In particular the two constraints can be contracted separately without changing the group Q being centralized.

Let h_t be a based loop in N null-homotopic in $\mathrm{Homeo}(O)$, and let (ϕ_t, ψ_t) be its coordinates. The path ψ_t preserves the orientation of the one-manifold B. When $A = \mathbb{R}$, multiplication of ϕ_t by $1 - s$ contracts the function-coordinate loop to zero and preserves its equation in (C.33.12), including $\lambda = -1$. For $B = \mathbb{R}$, C.27 contracts ψ_t in the required centralizer, with the trivial operator group when $\Delta = 1$. For $B = \mathbb{R}/\mathbb{Z}$, the evaluation loop $h_t(0, b) = (\phi_t(b), \psi_t(b))$ is null-homotopic in $\mathbb{R} \times (\mathbb{R}/\mathbb{Z})$; hence $\psi_t(b)$ has winding zero. C.29 contracts ψ_t in the centralizer of the finite rotation group in (C.33.6).

When $A = \mathbb{R}/\mathbb{Z}$, necessarily $B = \mathbb{R}$. The based family $\phi_t(b)$ has a unique continuous lift $\varphi : [0, 1] \times \mathbb{R} \to \mathbb{R}$ with $\varphi_0 = 0$. Evaluation of h_t in the annulus $A \times \mathbb{R}$ shows that $\varphi_1(b) = 0$: this endpoint is the winding number of the first coordinate of the null evaluation loop. If $\Delta = \mathbb{Z}$, equation (C.33.12) says that

$$\varphi_t(b + 1) - \lambda\varphi_t(b) \in \mathbb{Z}. \tag{C.33.13}$$

Its left side is a continuous function of (t, b), equal to zero at $t = 0$, and is therefore identically zero. Multiplying φ by $1 - s$ and descending to $\mathbb{R}/\mathbb{Z}$ contracts ϕ_t while preserving exact equivariance. The same scaling applies when $\Delta = 1$, without (C.33.13). C.27 contracts the remaining ψ_t on the line. All these contractions keep $t = 0, 1$ fixed. Equations (C.33.11)–(C.33.12), including the inverse formula, show that their product is a continuous based contraction in N. This completes the proof of (C.33.1).

Finally, for an open loop-group orbit from C.28, C.30 supplies the actual closed locally compact free proper operator image Q. The restricted canonical plane loop commutes with Q. It is null-homotopic in $\mathrm{Homeo}(O)$ by the second assertion of C.32, so (C.33.1) contracts it in the exact centralizer. Every original operator in the setwise stabilizer acts through its restriction to Q, and the restriction kernel acts identically on O. The stated conclusion follows with that kernel retained. □

What the open-orbit result provides. Equations (C.33.11)–(C.33.13) treat every component of the proper operator group, including a circle fibre with a nontrivial $\mathbb{Z}$-action on its line base. There is no replacement of the base-translation centralizer by the homeomorphism group of its quotient circle. As in C.32, the obtained contraction can be transported to other open orbits in the same operator orbit; exact stabilizer equivariance makes the transport independent of choices. This proof supplies no uniform control near the boundary of the open orbit. C.34 subsequently proves joint continuity, including inverses, on the entire closed union of nonopen orbits. C.35

exhibits an exact obstruction to extending that particular median prescription over an adjacent open orbit. C.36 constructs compatible contractions on closures with two-dimensional proper operator image and proves continuity on every finite union of those closures. C.37 supplies the uniform three-word estimate and gives a continuous contraction on the closed union X_2 of all those open orbits, including infinite accumulation. Extension across X_2 to its exterior and frontier-compatible choices for the dimension-zero and dimension-one operator images remain additional requirements. C.12 retains its unresolved status.

C.34. [Proved] The median contraction on all nonopen orbits. Let f_t be a based null-homotopic loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, and use P, G, Y from C.28 and C.31. The orbitwise bijections $Q_{s,t} : Y \to Y$ defined in C.31 are homeomorphisms. Both maps

$$[0,1]^2 \times Y \longrightarrow Y, \qquad (s,t,x) \longmapsto Q_{s,t}^{\pm 1}(x) \tag{C.34.1}$$

are continuous. Thus these formulas give a based compact-open null-homotopy on the entire closed nonopen locus, commuting with every homeomorphism commuting with all f_t. In particular, a hyperbolic centralizer loop whose canonically lifted generated group has no open orbit is null-homotopic in the original surface centralizer. No extension over the omitted open orbits is asserted.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; nonopen-stratum assembly in C.12.

Proof. We prove continuity at a nonfixed orbit. Continuity at the common fixed set, bijectivity on Y, equivariance, and the exact inverse identity have already been proved in C.31.

Short paths moving in either direction. Let q be a point of a nonopen nontrivial orbit. It lies on an intrinsic line or a circle by C.28. There are paths g_s from the identity, formed from two factors in P, which move q a nonzero amount in either chosen intrinsic direction while keeping its whole track inside any prescribed sufficiently small intrinsic arc about q. To see this, $f_t(q)$ is not identically q; otherwise every generator, and hence every element of G, would fix q. Choose a component (a,b) of $\{t : f_t(q) \neq q\}$ and a sufficiently short initial subinterval $[a, a+\eta]$. The path

$$h_s = f_{a+s\eta} f_a^{-1}, \qquad 0 \le s \le 1, \tag{C.34.2}$$

starts at the identity, and $h_s(q) = f_{a+s\eta}(q)$. C.28 gives compact-open continuity on an intrinsic line; the circle restriction is continuous as well. Consequently h_s and h_s^{-1} can be made uniformly close to the identity on an intrinsic neighborhood of q. Their endpoint displacements at q have opposite intrinsic signs. One of these two paths has the chosen sign, and its track stays in the prescribed arc. The paths also become uniformly small on each fixed ambient compact set as $\eta \to 0$. Every such g_s is a product of two elements of P. For any $z \in K_n(x) = P^n x$,

$$g_s(z) \in K_{n+2}(x). \tag{C.34.3}$$

In particular, the same compact-word argument as in C.28 proves intrinsic continuity of its track on every nonopen orbit, including when z varies along a compact intrinsic subarc.

A small cap cannot persist at a moving limit point. We give the local argument used below. Straighten a compact arc of a limit orbit, extending it a little at both ends, so that it is an interval of the horizontal axis in a plane chart. Suppose compact intrinsic arcs A_j in nearby nonopen orbits lie in strips of height tending to zero around this interval. Let $q_j \in A_j$ attain the minimum m_j of the horizontal coordinate on A_j, suppose $q_j \to q = (m, 0)$, and suppose both endpoints of A_j have horizontal coordinates at least $m + \delta$ for a fixed $\delta > 0$. Take a rectangle

$$U = (m-\rho, m+\rho) \times (-\varepsilon, \varepsilon), \qquad 0 < \rho < \delta/3, \tag{C.34.4}$$

whose closure stays in the chart. For large j, q_j lies in U, all of A_j has vertical coordinate of absolute value less than $\varepsilon/2$, and its horizontal minimum is greater than $m - \rho/2$. Starting at q_j in each intrinsic direction towards the two endpoints of A_j, take the first point outside U. Both first-exit points lie on its right edge. They are distinct points of the embedded arc. Let J_j be the closed subarc between these points. Its interior lies in U, and every point of J_j has horizontal coordinate at least m_j.

Use (C.34.2), taking the inverse when needed, to choose a path g_s whose q-track stays in U and whose endpoint is strictly to the left of q. This choice is possible because q is nonfixed. The track is a compact subset of U, so joint continuity gives $g_s(q_j) \in U$ for every s and every sufficiently large j. Its intrinsic continuity from (C.34.3) now matters: a continuous path in the intrinsic line or circle starting at an interior point of J_j cannot leave J_j without meeting one of its two endpoints. Both endpoints lie outside the open rectangle U. Thus the entire $g_s(q_j)$-track stays in J_j. At $s = 1$ its horizontal coordinate is at least m_j, whereas

$$g_1(q_j) \longrightarrow g_1(q), \qquad \mathrm{pr}_1(g_1(q)) < m = \lim_j m_j. \tag{C.34.5}$$

This contradiction excludes the cap. The argument with horizontal maxima and a path moving to the right is identical. The two first-exit points are the explicit obstruction to escaping the cap along another part of the same orbit.

The same argument excludes a whole Jordan-curve orbit contained in such a shrinking strip and having a nondegenerate interval as its limit: use a global minimum of the horizontal coordinate on the circle, and in its two directions take first exits towards a point whose coordinate is bounded above that minimum. The resulting subarc J_j again has two distinct endpoints outside U.

No excess in subarcs near an intrinsic line. Fix a point p on an intrinsic line orbit and take $x_j \to p$ in Y. For each fixed n, compactness of P^n and continuity of word evaluation give

$$K_n(x_j) \longrightarrow K_n(p) \quad \text{in Hausdorff distance.} \tag{C.34.6}$$

The limit is a nondegenerate compact arc. Straighten a slightly larger compact arc of the limit orbit. Then (C.34.6) places the nearby $K_n(x_j)$ in shrinking horizontal strips in that chart. Each $K_n(x_j)$ is an arc, a point, or a whole circle, by C.28 and connectedness. The whole-circle possibility is excluded for all sufficiently large j by the preceding cap argument.

Suppose $u_j, v_j \in K_n(x_j)$ tend to $u, v \in K_n(p)$. Write $[u_j, v_j]_j$ for the subarc contained in $K_n(x_j)$. Then

$$z_j \in [u_j, v_j]_j, \quad z_j \longrightarrow z \quad \Longrightarrow \quad z \in [u, v]. \tag{C.34.7}$$

Indeed, an alleged limit point strictly to the left of both u, v produces a horizontal minimum of $[u_j, v_j]_j$ whose two endpoints stay a fixed distance to its right. A subsequence of these minimum points converges, by (C.34.6), to a point of the nonfixed limit orbit. This is the forbidden cap. A point to the right is excluded by the maximum version. The assertion includes $u = v$: then all of the intervening subarc tends to u. This proves the needed continuity of subarcs using the path action; Hausdorff convergence alone has not been used to infer it.

It follows that medians of three points in $K_n(x_j)$ converge to the median of their limits. Each median lies in all three pairwise connecting subarcs. Every subsequential limit therefore lies, by (C.34.7), in all three corresponding limit intervals. Their intersection is the single limit median, including when two or three of the limiting points coincide.

A circle limit: the precise local covering. Let the orbit of p be a Jordan curve C. Choose a collar annulus W of C, with coordinates $(\mathbb{R}/\mathbb{Z}) \times (-1, 1)$ sending C to the middle circle, and write

$$\pi_W : \widetilde{W} = \mathbb{R} \times (-1, 1) \longrightarrow W \tag{C.34.8}$$

for its universal covering. The compact family f_t preserves C. Therefore a sufficiently small open disc V about p satisfies $f_t(V) \subset W$ for all t. The actual map

$$[0,1] \times V \longrightarrow W, \qquad (t,x) \longmapsto f_t(x) \tag{C.34.9}$$

has a unique continuous lift $\widehat{f}_t(x)$ after choosing a lift of the initial inclusion $V \to W$. Its endpoint differs from its starting point by an integer deck translation depending continuously on $x \in V$. It is zero at p, by C.29's exact plane-to-circle winding comparison, so it is zero on all of the connected disc V. Thus these are closed lifted point tracks. For $x_j \to p$ in $V \cap Y$ put

$$T_j(r) = \{\widehat{f}_u(x_j) : 0 \le u \le r\}, \qquad T(r) = \{\widehat{f}_u(p) : 0 \le u \le r\}. \tag{C.34.10}$$

Joint continuity on the compact parameter interval shows $T_j(r_j) \to T(r)$ in Hausdorff distance whenever $r_j \to r$. The set $T(1)$ is a nondegenerate compact interval on the line $\mathbb{R} \times \{0\}$.

We verify the intrinsic meaning of the lifted nearby tracks. An intrinsic line orbit, or a proper component of a circle orbit inside W, has an intrinsic interval as its component through x_j. Its continuous point track stays in that component and lifts to an injectively parametrized interval in $\widetilde{W}$, with embedded compact subintervals. If the entire circle orbit lies in W, it is a Jordan curve. Its winding in W is 0 or ± 1: the planar Jordan winding around the complementary inner disc has exactly these values. Its lifted components are therefore Jordan curves in the first case and intrinsic lines in the second. Consequently $T_j(1)$ is a compact arc, a point, or a whole lifted Jordan curve, with the corresponding intrinsic topology.

The cap argument applies in this covering chart with the original group paths. Here is the exact local map needed to justify that use. At any limiting extremum $q \in T(1)$ choose the small rectangle U of (C.34.4) with horizontal width less than one, so that $\pi_W|_U$ is injective. Choose a path (C.34.2) at $\pi_W(q)$ with the required intrinsic sign and sufficiently small that its images of a compact neighborhood of $\pi_W(q)$ stay in W. The map

$$(s,z) \longmapsto g_s(\pi_W(z)) \tag{C.34.11}$$

on that small lifted neighborhood has a unique continuous lift starting with $z \mapsto z$. Its q-track lies in U and has the required horizontal displacement. For $q_j \to q$ its whole q_j-track also stays in U, by compact parameter continuity. Downstairs it is an intrinsically continuous path in the original orbit, by (C.34.3). Its lift cannot leave the cap subarc without meeting one of its first-exit endpoints on ∂U. Thus (C.34.5) applies unchanged. No action of G on the entire annulus or its covering has been assumed.

This excludes a whole lifted Jordan curve for large j and proves (C.34.7), and hence median continuity, for compact subarcs of $T_j(1)$. These intrinsic interval operations are exactly the ones used by C.27 and C.29. For a line or a proper circle component the lift of its interval is order preserving or reversing. For an essential circle the annular covering is its universal line covering. For a contractible circle, the now proper arc $T_j(1)$ lies in a single interval of its universal line covering; a path leaving that interval would have to cover the whole circle. In all cases the based circle lift or intrinsic line formula, followed by projection through (C.34.8), gives the original $Q_{s,t}(x_j)$. Median invariance under reversal removes any choice of the interval orientation.

Continuity of the prefix-clamping formula. We can now handle a sequence $(s_j, t_j, x_j) \to (s,t,p)$ with p nonfixed. For a line limit use the actual point tracks $T_j(r) = \{f_u(x_j) : 0 \le u \le r\}$ inside $K_1(x_j)$. For a circle limit use (C.34.10). In either case $T_j(1)$ is eventually an intrinsic compact arc, its subarcs satisfy (C.34.7), and its prefix tracks have their stated Hausdorff limits. Set $r_j = 1 - s_j$ and $r = 1 - s$. Let a_j, b_j be the two unordered endpoints of the prefix arc $T_j(r_j)$, allowing equality. Every subsequence has a further subsequence on which both endpoints converge, say to $a, b \in T(r)$. By (C.34.7), the Hausdorff limit of their intervening arc is contained in $[a,b]$. It is also exactly $T(r)$ by the prefix-track continuity. Since a, b belong to $T(r)$, their interval is contained in $T(r)$. Therefore

$$T(r) = [a,b]. \tag{C.34.12}$$

In particular the limiting unordered pair is precisely the endpoint pair of the prefix interval, including a degenerate prefix.

The formula of C.27 or its C.29 lift is the median of a_j, b_j and the current track point at t_j. The median continuity just proved and (C.34.12) give its limit as the median of the limiting prefix endpoints and the current point at t. This is $Q_{s,t}(p)$, with projection through (C.34.8) in the circle case. Every subsequence has this same limit. The domains are metrizable, so this proves joint continuity on $[0,1]^2$ times the nonfixed part of Y. C.31 supplies the remaining fixed points.

Apply the same argument to the based inverse loop $u \mapsto f_u^{-1}$. It is ambient-null, generates the same group G, and has the same closed locus Y. The exact identity $Q[f]^{-1} = Q[f^{-1}]$ in C.31 proves joint continuity of the inverse family in (C.34.1). Thus all the orbitwise bijections are homeomorphisms of Y. The compact-open continuity follows from the evaluation exponential law, since Y is a closed, locally compact subspace of the plane. The based endpoints and equivariance are those already proved orbitwise in C.31.

Finally, when the canonical lifted group of a hyperbolic surface loop has no open orbit, $Y = \widetilde{U} \cong \mathbb{R}^2$. The obtained plane homeomorphisms commute with the full operator lift group, including all deck transformations. They descend by the exact equations

$$\overline{Q}_{s,t}\, p_U = p_U Q_{s,t} \quad (p_U : \widetilde{U} \to U) \tag{C.34.13}$$

to homeomorphisms of U commuting with every original operator. Their joint continuity and that of their inverses descend through the open covering quotient, giving compact-open continuity downstairs. The endpoints in (C.34.13) are the original loop and the identity, with both based edges fixed. Hence this is its null-homotopy inside the original centralizer. At every parameter it lies in the identity component by these same paths. Equivalently, its lift is the canonical one: two deck-commuting lifts of such a map differ by a central deck transformation, and the hyperbolic surface group has trivial center. This proves the stated corollary with the actual lift and descent maps. □

Remaining global extension. C.34 establishes the full nonopen-locus homeomorphism family and its inverse. C.33 provides intrinsic equivariant contractions on each open orbit. C.35 proves that this particular median family need not extend equivariantly over an adjacent open orbit. C.36 and C.37 construct compatible alternatives on the closed union X_2 of open orbits with two-dimensional proper operator image, including every infinite accumulation. Matching that boundary family to the exterior of X_2, and treating the other proper operator images at their frontiers, remain the assembly requirements in C.12, whose general status remains open.

C.35. [Proved] An obstruction to extending the fixed median prescription. There is a continuous $\mathbb{R}^2$-action on the plane and a based ambient-null loop in its centralizer such that the C.34 median homeomorphisms of the closed nonopen locus do not extend equivariantly over an adjacent open orbit at one parameter. The closed restricted operator image on that open orbit is the free proper translation group $\mathbb{R}^2$. The original loop nevertheless has an explicit equivariant null-homotopy on the entire plane. Thus this example refutes extension of the fixed C.34 median prescription; it is not a failure of the level-0 comparison.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; obstruction to extending a fixed contraction prescription in the C.12 proof route.

Proof. Use the closed simplex

$$\Delta = \{(x,y,z) \in \mathbb{R}^3_{\geq 0} : x + y + z = 1\}.$$

For $(a,b)\in\mathbb{R}^2$ define

$$T_{a,b}(x,y,z)=\frac{(e^a x,e^b y,z)}{e^a x+e^b y+z}. \tag{C.35.1}$$

The denominator is positive on Δ. Direct substitution gives $T_{a,b}T_{c,d}=T_{a+c,b+d}$ and $T_{a,b}^{-1}=T_{-a,-b}$, so (C.35.1) is a continuous action by homeomorphisms, fixing the three vertices and preserving each open edge. On the interior the coordinate homeomorphism

$$(x,y,z)\longmapsto(u,v)=\big(\log(x/z),\log(y/z)\big),\qquad (u,v)\longmapsto\frac{(e^u,e^v,1)}{1+e^u+e^v} \tag{C.35.2}$$

turns this action into $(u,v)\mapsto(u+a,v+b)$. Its restricted image is therefore the closed free proper translation group on $\mathbb{R}^2$.

Double Δ along its entire boundary. Since Δ is a closed disc, this double is a sphere. Apply (C.35.1) on both copies; the formulas agree on their identified boundary and hence give a continuous sphere action. Remove the fixed vertex $(0,0,1)$. The remaining surface M is a plane, and the action restricts to it. It is orientation preserving: each map preserves the orientation on either triangle interior, and preserves the two sheets of the oriented double.

Write $Q=\{T_{a,b}:(a,b)\in\mathbb{R}^2\}$ for this operator image. Its orbits in M are the two triangle interiors, the three open edges, and the remaining two vertices. Use the following coordinates on the edges:

edge	coordinate in $\mathbb{R}$	translation by $T_{a,b}$
$y=0$	$\log(x/z)$	a
$x=0$	$\log(y/z)$	b
$z=0$	$\log(x/y)$	$a-b$.

(C.35.3)

Thus the union Y of the three edges and the two vertices is exactly the closed nonopen orbit locus.

Let $w(t)=(a(t),b(t))$ be the piecewise linear based loop with successive values

$$(0,0),\quad (1,1),\quad (1,0),\quad (0,0)\quad \text{at } t=0,\tfrac{1}{3},\tfrac{2}{3},1, \tag{C.35.4}$$

and put $f_t=T_{a(t),b(t)}$. All its values commute with Q, and

$$F_{\sigma,t}=T_{(1-\sigma)a(t),(1-\sigma)b(t)} \tag{C.35.5}$$

is a based equivariant null-homotopy on the entire plane. Continuity, including that of the inverse family, follows from (C.35.1), which is already continuous on the compact double and preserves the removed point. The subgroup generated by the loop is exactly Q: its first segment contains all diagonal translations (r,r) for $0\le r\le 1$, and the differences of points of the second segment contain all $(0,r)$ for $-1\le r\le 1$. Integer sums and inverses give both full coordinate axes and hence $\mathbb{R}^2$. Consequently C.34 uses precisely the orbit locus Y just described.

At $s=2/3$ the prefix in the C.27 formula ends at $r=1-s=1/3$. By (C.35.3)–(C.35.4), the three prefix translation intervals are, in their displayed order,

$$[0,1],\quad [0,1],\quad \{0\}. \tag{C.35.6}$$

At $t=2/3$, the current edge translations are $(1,0,1)$. Taking the prescribed medians with the prefix endpoints gives

$$Q^{\mathrm{med}}_{2/3,\,2/3}\Big|_Y:\quad \text{edge translations } (1,0,0). \tag{C.35.7}$$

The vertices stay fixed. These edge maps fit continuously at the vertices, as also guaranteed by C.34.

Every Q-equivariant homeomorphism from one triangle interior to itself is a translation in (C.35.2). Indeed, commutation with all translations gives $h(u,v)=(u,v)+h(0,0)$. If it maps

one interior to the other, the same equation holds in the identical coordinates on the target sheet. The sheet exchange fixes the common boundary, so in either case any continuous boundary extension has edge translation parameters

$$R : \mathbb{R}^2 \longrightarrow \mathbb{R}^3, \qquad R(A, B) = (A, B, A - B). \tag{C.35.8}$$

This is the exact restriction morphism. Its image is the kernel of

$$\ell : \mathbb{R}^3 \longrightarrow \mathbb{R}, \qquad \ell(r_1, r_2, r_3) = r_1 - r_2 - r_3. \tag{C.35.9}$$

But $\ell(1, 0, 0) = 1$, so (C.35.7) does not lie in that image. It cannot extend equivariantly even over a single adjacent interior. This proves all assertions, while (C.35.5) gives the explicit global contraction of the original loop. □

Executed endpoint certificate and scope. The standard-library rational calculation `code/part_b/c35_median_extension_certificate.py` computes the prefix extrema, the three median values, the restriction matrix in (C.35.8), and the nonzero obstruction (C.35.9). Its JSON output records $(1, 0, 0)$ and residual 1. This certificate verifies the finite endpoint calculation; the preceding simplex, doubling, and restriction arguments prove the topological statement.

C.34 is a homeomorphism construction on the whole nonopen locus, and C.33 contracts the open-orbit restrictions. C.35 proves that a global argument must permit changing the nonopen-locus contraction to satisfy the actual boundary restriction morphisms of the open components. The remaining problem is therefore the existence of compatible equivariant contractions, not extension of the particular median family without modification. C.36 and C.37 meet the actual boundary equations on the closed union X_2 of open orbits with two-dimensional proper operator image, including infinite accumulation. Extending this family compatibly to the exterior of X_2 and the other proper operator cases remains required. The general C.12 comparison remains unresolved.

C.36. [Proved] Compatible contractions on two-dimensional proper closures. Let f_t be a based ambient-null plane loop, let G be its generated group, and let H be any group of plane homeomorphisms commuting with every f_t. For an open G-orbit O, let Q_O be the closure of the restricted setwise H-stabilizer, as in C.30. For every such O with $\dim Q_O = 2$, there is a based null-homotopy on the entire closure $\overline{O}$ through homeomorphisms of $\overline{O}$ that commute with its operator stabilizer. These closure contractions agree on every intersection $\overline{O}_1 \cap \overline{O}_2$ and are carried to one another by H. They therefore give a jointly continuous family, with jointly continuous inverse, on every finite union of these closures. No continuity across an infinite accumulating family of closures is asserted.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; frontier-compatible dimension-two operator closures in the C.12 proof route.

Proof. The actual quotient action on the closure. C.30 makes Q_O a closed locally compact free proper group of homeomorphisms of O, and C.33 makes it Lie. Its dimension-two argument shows that Q_O is connected and simply transitive on O, and that

$$N_O = C_{\mathrm{Homeo}(O)}(Q_O) \cong Q_O^{\mathrm{op}} \tag{C.36.1}$$

is also a connected, simply transitive two-dimensional Lie group. Evaluation $N_O \to O$ is a homeomorphism. The image of G in N_O is all of N_O: for any $n \in N_O$, transitivity of G supplies $g \in G$ with $gx = nx$ at one point $x \in O$, and the injectivity of evaluation on N_O then gives $g|_O = n$.

Put $L = \overline{G}$ in $\mathrm{Homeo}^+(\mathbb{R}^2)$. This is a Polish group. It preserves every open G-orbit O. Indeed, if $g_j \to l$ with $g_j \in G$, the identities $g_j(\partial O) = \partial O$ and the convergent inverse family

give $l(\partial O) = \partial O$. For $x \in O$, $lx \in \overline{O}$ by convergence, and $lx \notin \partial O$ because l preserves that boundary and is injective. Thus $lx \in O$; applying the same argument to l^{-1} proves $l(O) = O$. The elements of L still commute with H. Restriction consequently gives a continuous surjective homomorphism

$$r_O : L \longrightarrow N_O. \tag{C.36.2}$$

It is open by the open mapping theorem for Polish groups. Equivalently, apply the open orbit-map principle from [22, Theorem 3.5] to the transitive action of L on the Baire space N_O through (C.36.2).

The kernel of (C.36.2) fixes O pointwise, and hence fixes $\overline{O}$ pointwise. Therefore there is a well-defined action

$$N_O \times \overline{O} \longrightarrow \overline{O}, \qquad (n, z) \longmapsto lz \quad \text{for any } l \text{ with } r_O(l) = n. \tag{C.36.3}$$

This action is jointly continuous: $r_O \times \mathrm{id}$ is an open quotient map, and the continuous L-action factors through it. No continuous section of (C.36.2) is required. In fact each n has a preimage in G, as proved above. Hence (C.36.3) preserves every original G-orbit contained in $\overline{O}$, and its image on such an orbit is exactly the image of G. This conclusion does not require a claim that L preserves every other G-orbit in the plane.

The two-dimensional Lie contraction. We record the Lie groups and the maps being used. A two-dimensional real Lie algebra is either abelian or has a basis X, Y with $[X, Y] = Y$: in the nonabelian case its one-dimensional derived algebra is an ideal, and scaling a complementary basis vector gives that bracket. The simply connected groups are respectively $\mathbb{R}^2$ and the positive affine group of the line. In coordinates (a, b), the latter has product

$$(a, b)(a', b') = (a + a',\, b + e^a b'). \tag{C.36.4}$$

Its exponential map is

$$\exp(u, v) = \left(u, v\,\frac{e^u - 1}{u}\right), \quad \text{with } \frac{e^u - 1}{u} = 1 \text{ at } u = 0. \tag{C.36.5}$$

The scalar factor is positive for every real u, so this is a global diffeomorphism. For $\mathbb{R}^2$ the exponential is the identity. The affine group has trivial center, so its connected groups with this Lie algebra have no further covering quotients. In the abelian case the connected group is a quotient of $\mathbb{R}^2$ by a discrete central subgroup. The compact torus is excluded here because $N_O \cong O$ is noncompact. Thus in every case the universal covering group $\widetilde{N}_O$ has a globally defined inverse log to its exponential.

Let $n_O(t) = f_t|_O \in N_O$. By the restriction assertion of C.32 this loop is null-homotopic in $\mathrm{Homeo}(O)$. Evaluation on O and the homeomorphism $N_O \to O$ show that it is null-homotopic in N_O. Its unique based lift $\widetilde{n}_O(t)$ to $\widetilde{N}_O$ is therefore a closed loop. With $p_O : \widetilde{N}_O \to N_O$ the covering homomorphism, define

$$R^O_{s,t} = p_O\big(\exp((1 - s)\log \widetilde{n}_O(t))\big) \quad \text{acting on } \overline{O} \text{ by (C.36.3).} \tag{C.36.6}$$

This is a continuous family of homeomorphisms with continuous inverse, starts with f_t on $\overline{O}$, ends with the identity, and keeps both based edges fixed. On O it lies in the exact centralizer of Q_O. Its commutation with the original setwise operator stabilizer extends to $\overline{O}$ by continuity and density.

The common boundary image and its topology. Take two of these closures and a G-orbit B in their intersection. It is a point, an intrinsic line, or a circle: an open orbit cannot lie in the boundary of another open orbit. The point case is fixed throughout. For a nontrivial B, each $N_i = N_{O_i}$ acts transitively on B by (C.36.3), and both action images are exactly $G|_B$. We

verify that this common image has a single Lie topology, rather than identify abstract quotients without their topology.

For $p \in B$, its stabilizer in N_i is closed. The homogeneous space $N_i/(N_i)_p$ maps continuously and bijectively to B. Its dimension is one: dimension zero would make the connected orbit a point, and dimension two would make B open in the plane by invariance of domain. Its topology agrees with the intrinsic topology of B. Indeed, a compact interval in this homogeneous one-manifold maps to an embedded compact arc. In an intrinsic line orbit that arc is the unique connecting arc supplied by C.28; on a circle the same conclusion holds locally. The interiors of these arcs give intrinsic interval neighborhoods, so the bijective orbit map is a local, and therefore global, intrinsic homeomorphism. Thus the action is continuous also in the intrinsic topology.

Divide N_i by its closed kernel on B. The resulting connected Lie group M_i acts faithfully and transitively on that one-manifold, and has dimension at most two. In dimension one it is abelian, so a point stabilizer is normal and is therefore the kernel; faithfulness makes the stabilizer trivial. Evaluation then identifies its action with the translation group of $\mathbb{R}$ or the rotation group of $\mathbb{R}/\mathbb{Z}$. An abelian group of dimension two would give the same freeness conclusion and an orbit of dimension two, so cannot occur faithfully here. In the remaining case M_i is the positive affine group (C.36.4). The identity component of a point stabilizer is a one-dimensional subgroup. It cannot be the translation subgroup, which is normal and would lie in the action kernel. Every other such subgroup is conjugate to the dilation subgroup. Its normalizer is that dilation subgroup itself, as is seen by conjugating its unique fixed point on the line. The full point stabilizer is consequently the dilation subgroup, and the homogeneous action is the usual positive affine action on $\mathbb{R}$.

These descriptions prove topological embedding of each actual image in $\mathrm{Homeo}(B)$ with its intrinsic compact-open topology. For translations or rotations one evaluation recovers the parameter. For the affine group, the images of two distinct ordered points recover the positive slope and the intercept continuously. They also prove that the image groups are closed in that homeomorphism group. Since both images are the same set $G|_B$, we have obtained a common topological Lie group M_B, and continuous quotient homomorphisms

$$\rho_i : N_i \longrightarrow M_B. \tag{C.36.7}$$

The universal cover of M_B is $\mathbb{R}$ or the positive affine group, and its exponential is again a global diffeomorphism.

Equality on overlaps. The homomorphisms (C.36.7) lift uniquely to based homomorphisms

$$\widetilde{\rho}_i : \widetilde{N}_i \longrightarrow \widetilde{M}_B. \tag{C.36.8}$$

The two paths $\widetilde{\rho}_i(\widetilde{n}_i(t))$ are based lifts of the identical loop $f_t|_B$, so covering uniqueness makes them equal to one path $\widetilde{b}(t)$. Naturality of the Lie exponential and its global injectivity give

$$\begin{aligned} d\widetilde{\rho}_i(\log \widetilde{n}_i(t)) &= \log \widetilde{b}(t), \\ \widetilde{\rho}_i\big(\exp\big((1-s)\log \widetilde{n}_i(t)\big)\big) &= \exp\big((1-s)\log \widetilde{b}(t)\big). \end{aligned} \tag{C.36.9}$$

After projection to M_B, this says precisely that $R^{O_1}_{s,t}$ and $R^{O_2}_{s,t}$ agree on B. The intersection of the two closures is a union of such G-orbits, so they agree everywhere on the intersection. This includes circle orbits with their actual based universal-cover lifts; no choice of an unrecorded integer rotation is made.

For $h \in H$, conjugation carries Q_O onto Q_{hO} and N_O onto N_{hO}, as follows from the exact restriction of the setwise stabilizers. This is a topological Lie-group isomorphism. It carries $n_O(t)$ to $n_{hO}(t)$, since h commutes with f_t. Its based lift and exponential naturality show

$$R^{hO}_{s,t} = hR^{O}_{s,t}h^{-1}. \tag{C.36.10}$$

Thus the compatible families have the asserted operator equivariance.

Finally, on a finite union of these closures, their continuous maps agree on every overlap. The finite closed-set pasting lemma gives joint continuity. Each intersection is preserved, because its G-orbits are preserved; hence the inverse maps also agree there. Applying the same pasting argument to the inverse families gives the stated family of homeomorphisms on the finite union. This proves the claim. □

Exact scope of the closure construction. The quotients (C.36.2)–(C.36.3) provide continuity on each individual closure, and (C.36.7)–(C.36.9) enforce the actual common boundary restriction relations. In particular they respect the three dependent edge characters in C.35. The present proof uses finite closed-set pasting. C.37 subsequently proves a uniform three-word estimate and extends these contractions continuously, with their inverses, to the closed union X_2 of all such open orbits, including infinite accumulation. Extension across X_2 to its exterior, including frontier contractions for $\dim Q_O = 0, 1$ and compatible choices elsewhere on Y, remains required for a global conclusion in C.12.

C.37. [Proved] Uniform control on all two-dimensional operator closures. Let f_t, $0 \le t \le 1$, be an ambient-null based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let $G = \langle f_t : 0 \le t \le 1 \rangle$, and let H be any subgroup commuting with every f_t. For an open G-orbit O, let Q_O be the actual closed restricted operator image of C.30. Put

$$U_2 = \bigcup_{\dim Q_O = 2} O, \qquad X_2 = \overline{U_2}. \tag{C.37.1}$$

The contractions of C.36 extend to a jointly continuous based contraction through homeomorphisms of X_2, with jointly continuous inverses, commuting with H. This includes arbitrary accumulation of the open orbits in (C.37.1).

The uniform estimate used in the proof is the following exact word statement. In either simply connected two-dimensional Lie group $\mathbb{R}^2$ or $\mathrm{Aff}^+(\mathbb{R})$, every continuous path $k : [0, t] \to N$ from 1 to g satisfies

$$\exp(r \log g) = k(b)k(a)^{-1} \quad \text{or} \quad \exp(r \log g) = g\, k(b)k(a)^{-1} \quad (0 \le r \le 1) \tag{C.37.2}$$

for suitable $a, b \in [0, t]$. The same assertion is used in the universal covering group when the actual group is $\mathbb{R} \times S^1$. Consequently every contraction value on an open orbit in U_2 is represented by a word of length at most three in the original loop and its inverse, with equality of the specified based universal-cover lifts.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; uniform dimension-two open-orbit closure step for C.12.

Proof. The three-word estimate. For a continuous path $\gamma : [0, 1] \to \mathbb{R}^2$ with $\gamma(0) = 0$, $\gamma(1) = v$, one has

$$rv = \gamma(b) - \gamma(a) \quad \text{or} \quad rv = v + \gamma(b) - \gamma(a) \qquad (0 \le r \le 1). \tag{C.37.3}$$

Here is an exact measure-theoretic proof that applies to arbitrary paths. For a polygonal path apply the signed-measure splitting theorem of Grandoni–Ravi–Singh–Zenklusen [24, Theorem 4.7 and §4.3.2] to the two signed measures with densities γ_1' and γ_2'. A single circular interval has fraction r of both total measures. Its integral is either a path difference or the total vector plus a path difference, giving (C.37.3). Approximate an arbitrary continuous path uniformly by polygonal paths with the same endpoints. Compactness of the two endpoints and of the two alternatives passes the equation to the limit. The planar case of the cited splitting proof uses at most two cuts of an interval; the two equal-mass halves together have at most three pieces, so one is a single interval. This is the one-circular-interval count required here.

For $N = \mathbb{R}^2$, (C.37.2) is (C.37.3). Write the affine group in the exact coordinates of C.36:

$$(a, b)(a', b') = (a + a',\, b + e^a b'). \tag{C.37.4}$$

Every nonidentity $g = (a, b)$ lies on its unique exponential one-parameter subgroup. If $a \neq 0$, conjugation by $(0, b/(e^a - 1))$ sends g to $(a, 0)$. The global coordinate map

$$(a', b') \longmapsto (a'/a,\, e^{-a'} b') \tag{C.37.5}$$

then sends left multiplication by g^r to translation by $(r, 0)$, and sends $1, g$ to $(0, 0), (1, 0)$. If $g = (0, b)$ with $b \neq 0$, use $(a', b') \mapsto (b'/b, a')$ instead. Apply (C.37.3) in these coordinates. A difference $(r, 0)$ says $k(b) = g^r k(a)$; the other alternative says $k(a) = g^{1-r} k(b)$. These are exactly (C.37.2). The identity case uses the identity word. This proves the estimate also for the simply connected covering group of every group in C.36.

The compact family of actual word isotopies. Use $r = 1 - s$ for the contraction parameter, and set

$$R^O_{r,t} = p_O \exp(r \log \widetilde{n}_O(t)), \qquad n_O(t) = f_t|_O, \tag{C.37.6}$$

as in C.36. Consider the following compact family of actual plane isotopies, with the compact-open topology on maps and inverses:

$$W_{\epsilon,a,b,t}(u) = f^{\epsilon}_{ut} f_{ub} f^{-1}_{ua}, \quad 0 \le a, b \le t \le 1, \quad \epsilon \in \{0, 1\}, \quad 0 \le u \le 1. \tag{C.37.7}$$

Compactness follows from continuity of the original loop and of multiplication and inversion. Formula (C.37.2), applied to the based lift of its prefix $[0, t]$, represents every $R^O_{r,t}$ by the endpoint of one of (C.37.7). More precisely, the based lift of that word isotopy in N_O has endpoint $\exp(r \log \widetilde{n}_O(t))$. Thus the representation retains the lift, not only the projected homeomorphism.

The exact groups on nonopen limit orbits. Each N_O in (C.37.6) is metabelian. Therefore the identity

$$[[g_1, g_2], [g_3, g_4]] = 1 \tag{C.37.8}$$

holds on U_2 for all $g_i \in G$, and hence on X_2 by continuity. The set X_2 is G-invariant, so it is a union of whole G-orbits. Its open G-orbits are precisely those already in U_2: any other open orbit has an open neighborhood disjoint from U_2.

Let $B \subset X_2$ be a nontrivial nonopen orbit, endowed with the intrinsic line or circle topology of C.28. The image G_B in $\mathrm{Homeo}^+(B)$ is transitive, metabelian, and path connected. For the last assertion, every finite word has a path from the identity obtained from the original paths and their inverses; C.28 gives intrinsic compact-open continuity of all these finite-word paths. We now prove the exact classification needed at B:

$$G_B \cong \begin{cases} (\mathbb{R}, +) \text{ or } \mathrm{Aff}^+(\mathbb{R}), & B \text{ a line}, \\ S^1, & B \text{ a circle}, \end{cases} \tag{C.37.9}$$

with the usual transitive actions after conjugacy.

First, a transitive abelian subgroup A of $\mathrm{Homeo}^+(B)$ has abelian closure $\overline{A}$, and this closure acts freely: an element fixing one point commutes with A, so fixes its whole transitive orbit. The closure is Polish and transitive. The open orbit-map theorem used in C.26 identifies it homeomorphically with B by evaluation, since its point stabilizer is trivial. In particular it is locally compact. Its faithful product action on the three-manifold $B \times \mathbb{R}^2$ makes it a Lie group by the same Hilbert–Smith input as C.33. It is the one-dimensional connected group $\mathbb{R}$ or S^1, and evaluation conjugates its action to translation. Since A was already transitive, injectivity of evaluation also proves $A = \overline{A}$.

If G_B is nonabelian, its derived subgroup D is nontrivial, abelian, normal, and path connected. Path connectedness follows by taking commutator paths and finite products of them. The common

fixed set of D is G_B-invariant. Transitivity and nontriviality make it empty. Every D-orbit is therefore a connected nontrivial interval or circle. Such an orbit has no included endpoint, since that endpoint would be preserved by its transitive orientation-preserving action. All these orbits are open in B; connectedness of B forces D to be transitive. The preceding paragraph identifies D with the full translation group.

The orientation-preserving normalizer of translations on the line consists exactly of $x \mapsto cx+d$, $c > 0$. Indeed normalization says $h(x+a) = h(x)+\chi(a)$; continuity and additivity give $\chi(a) = ca$. The multiplier image of the path-connected group G_B is a connected subgroup of $\mathbb{R}_{>0}$. It is nontrivial in the present case, so it is all $\mathbb{R}_{>0}$; because G_B contains every translation, it is the full affine group. On the circle, the orientation-preserving normalizer of the full rotation group is just that rotation group: the induced continuous automorphism of S^1 has degree $+1$. This would contradict nonabelianness. These arguments prove (C.37.9), including the topology of its image in the intrinsic homeomorphism group.

For a line B, define $R^B_{r,t}$ by the logarithmic formula in this simply connected group. For a circle, use the unique based real lift of $f_t|_B$ to the cover of its rotation group and scale that lift by r. Its endpoint is zero by the exact winding comparison C.29. On a point orbit define $R^B_{r,t} = 1$. These definitions are independent of the conjugacies in (C.37.9), since Lie homomorphisms commute with exponential maps and with based universal-cover lifts. They agree with C.36 on every boundary orbit to which that construction already applies.

Limits from the dense open union. Take

$$x_i \in O_i \subset U_2, \quad x_i \longrightarrow x \in X_2, \qquad (r_i, t_i) \longrightarrow (r, t). \tag{C.37.10}$$

Choose word isotopies W_i from (C.37.7) representing $R^{O_i}_{r_i,t_i}$ with their specified based lifts. After a subsequence all word parameters converge. Then W_i and their inverses converge uniformly on compact sets, uniformly in u, to a word isotopy W from the same family. Write $w = W(1)$ and $g = f_t$. If x is in an open orbit of X_2, eventually x_i is in that same open orbit, where (C.37.6) is continuous. If x is a point orbit, $w(x) = x$ for every possible limiting word, which gives the required limit immediately. It remains to identify w on the intrinsic line or circle $B = Gx$.

We will repeatedly use the following exact passage of identities. For every fixed $h \in G$, $hx_i \in O_i$ and $hx_i \to hx$. An identity of word homeomorphisms on O_i, whose word isotopies converge, therefore passes to every point hx of B. In particular

$$[w|_B, g|_B] = 1, \tag{C.37.11}$$

because a fractional power commutes with its endpoint on O_i.

Fix a positive integer n and set $m_i = \lfloor nr_i \rfloor$. In the universal covering group of N_{O_i} the exact residual is

$$\widetilde{w}_i^{\,n}\, \widetilde{n}_{O_i}(t_i)^{-m_i} = \exp\big((nr_i - m_i) \log \widetilde{n}_{O_i}(t_i)\big). \tag{C.37.12}$$

Its exponent fraction lies in $[0, 1)$. By (C.37.2), the right side is again represented by one of the compact family (C.37.7), with equality of based lifts. Passing to a further subsequence for this fixed n, the integer m_i is constant, say m_n, and the residual word isotopies converge to Z_n. Hence

$$(w|_B)^n (g|_B)^{-m_n} = z_n, \qquad |m_n - nr| \le 1, \qquad z_n = Z_n(1)|_B. \tag{C.37.13}$$

All z_n belong to the same compact set of restricted three-word endpoints, independently of n. No simultaneous subsequence for all n is required.

For a line, (C.37.9) and (C.37.11) allow logarithms in $\mathbb{R}$ or $\mathrm{Aff}^+(\mathbb{R})$. In the latter group the centralizer of every nonidentity element is its exponential one-parameter subgroup, as is seen directly from (C.37.4). Thus logarithms add for commuting elements in both groups. The compact set in (C.37.13) has bounded logarithms, giving

$$n \log(w|_B) - m_n \log(g|_B) = \log z_n, \qquad \sup_n \| \log z_n \| < \infty. \tag{C.37.14}$$

Divide by n and let n tend to infinity. It follows that $\log(w|_B) = r\log(g|_B)$. This includes $g|_B = 1$, in which case boundedness forces $w|_B = 1$. The limiting word therefore has exactly the required fractional action on the line.

The lifted residual bound on a circle. Identify the action in (C.37.9) on a circle with rotations, and let $\theta(u)$ be the continuous real angle of $f_u|_B$, normalized by $\theta(0) = 0$. The normalization uses rotations $\mathbb{R}/\mathbb{Z}$. C.29 gives $\theta(1) = 0$. Every isotopy in (C.37.7), restricted to B, has a canonical real endpoint angle

$$\epsilon\theta(t) + \theta(b) - \theta(a). \tag{C.37.15}$$

Let A be this angle for W, let $T = \theta(t)$, and let C_n be the angle for Z_n. Writing $M = \max_{0\le u\le 1}|\theta(u)|$ gives $|C_n| \le 3M$. We must prove the real, rather than merely circle-valued, identity

$$nA - m_nT = C_n. \tag{C.37.16}$$

The word isotopy

$$\Lambda_i(u) = W_i(u)^n f_{ut_i}^{-m_i} Z_i(u)^{-1} \tag{C.37.17}$$

restricts to a based loop in N_{O_i}. Equation (C.37.12), with the specified lifts of W_i and Z_i, says that its universal-cover lift ends at the identity. Hence that loop is null-homotopic in N_{O_i}. Choose $h \in G$ with $hx \ne x$, and evaluate (C.37.17) at the two points x_i, hx_i of O_i. For large i they are distinct. The resulting loop in $\mathrm{Conf}_2(\mathbb{R}^2)$ has difference-direction winding zero. The word isotopies converge uniformly on compact sets to $\Lambda(u) = W(u)^n f_{ut}^{-m_n} Z_n(u)^{-1}$. The two-point loops therefore converge uniformly to its evaluation at x, hx. This limiting pair stays distinct, so winding is preserved. On the Jordan circle B its winding, by C.29, is exactly $nA - m_nT - C_n$ (up to a common orientation sign). It is zero, proving (C.37.16).

Using (C.37.13) and $|C_n| \le 3M$ now gives

$$|A - rT| \le \frac{3M + |T|}{n} \quad (n \ge 1). \tag{C.37.18}$$

Consequently $A = rT$. This determines the based lifted fractional action even when $f_t|_B = 1$ but $\theta(t)$ is a nonzero integer. Thus no hidden winding survives the limit from the open orbits.

Joint continuity, inverses, and equivariance. We have shown that every convergent subsequence of word parameters in (C.37.10) gives the same endpoint $R^B_{r,t}(x)$. Compactness of (C.37.7) proves

$$R^{O_i}_{r_i,t_i}(x_i) \longrightarrow R^B_{r,t}(x). \tag{C.37.19}$$

This holds for every sequence from the dense set U_2 approaching every point of X_2, with parameters also varying. It proves joint continuity on all of X_2. Explicitly, for a sequence of points of X_2, approximate each by a point of U_2 so closely that both its distance and its image distance for that fixed parameter are less than $1/i$. Such approximations exist by (C.37.19) with parameters fixed. Applying (C.37.19) to the chosen approximating sequence gives the full continuity assertion.

The inverse loop f_t^{-1} has the same group G, the same sets U_2, X_2, and the same actual operator images. Its construction gives precisely $(R^B_{r,t})^{-1}$, by inversion of the based lifts. Applying the preceding argument to it proves joint continuity of the inverse family as well. Each member is an orbitwise bijection, hence a homeomorphism of X_2. The identities $R_{0,t} = 1$, $R_{1,t} = f_t|_{X_2}$, and $R_{r,0} = R_{r,1} = 1$ give the based contraction.

Finally every $h \in H$ permutes the open G-orbits and conjugates their actual closed operator images. It therefore preserves U_2 and X_2. C.36 gives exact transported equivariance on U_2. The established joint continuity extends that equality to its closure X_2. This proves all assertions. □

The remaining global boundary requirement. C.37 treats the whole closed set X_2, including infinite accumulation of its open orbits. Its boundary contraction is the based Lie-group fractional action forced by (C.37.14) and (C.37.16). Extending this family to the complement of X_2, compatibly with the dimension-zero and dimension-one proper operator cases, remains a separate requirement. Even where that complement contains only nonopen orbits, its median contraction from C.34 has not been identified with this boundary family. C.35 proves that imposing the fixed median prescription can violate the actual restriction map. Thus the uniform estimate proved here closes the dimension-two accumulation step, while the global centralizer conclusion in C.12 remains open.

C.38. [Proved] Discrete cyclic operators on nonabelian line and circle orbits. Let B be a line or circle and let $T \le \mathrm{Homeo}^+(B)$ be transitive and nonabelian. Its full compact-open centralizer

$$N_B = C_{\mathrm{Homeo}(B)}(T) \tag{C.38.1}$$

is discrete and acts freely and properly. On a line it is trivial or infinite cyclic; on a circle it is finite cyclic.

In the plane-loop setting of C.28, let B be a nonopen orbit whose intrinsic image $G|_B$ is nonabelian, and let $H_B = \{h \in H : hB = B\}$ be the operator stabilizer. Its actual restriction has the exact sequence

$$1 \longrightarrow K_B \longrightarrow H_B \xrightarrow{\rho_B} P_B \longrightarrow 1, \quad K_B = \{h \in H_B : h|_B = 1\}, \quad P_B = \rho_B(H_B) \le N_B. \tag{C.38.2}$$

In the intrinsic compact-open topology, P_B is already closed and discrete, acts freely and properly, and

$$p_B : B \longrightarrow B/P_B \tag{C.38.3}$$

is an actual regular covering. Its deck transformations are exactly P_B. The quotient is a line for trivial P_B on a line, and otherwise a circle. On a circle P_B is finite and the covering degree is $|P_B|$. The kernel K_B acts identically on B throughout this comparison.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact one-dimensional operator-image reduction at the boundary remaining in C.12.

Proof. Let L be the closure of T in the compact-open homeomorphism group of B, choose $b \in B$, and put $J = L_b$. The group L is Polish and transitive, and $C(L) = C(T)$ by continuity of composition. The open orbit-map theorem used in C.26 gives $L/J \cong B$. The normalizer $N_L(J)$ is closed, and the same exact evaluation construction as in C.30 gives

$$\mathrm{ev}_b : N_B \xrightarrow{\cong} N_L(J)/J \ \subset B. \tag{C.38.4}$$

The right side is closed. The inverse sends the coset represented by k to the map $lb \mapsto lkb$. The normalizer equation makes this well defined. Its compact-open continuity follows by the same microtransitivity argument as C.30: convergent points of B lift locally through the open orbit map to elements of L tending to the identity. No surface-dimensional step occurs in that argument. The construction also proves freeness, local compactness, and properness. Explicitly, if $n_i b_i = c_i$, $b_i \to b$, and $c_i \to c$, choose $l_i \to 1$ with $l_i b = b_i$. Then $n_i b = l_i^{-1} c_i \to c$, and the closed image and homeomorphism (C.38.4) give a convergent subsequence in N_B. This is the required properness test.

Every orientation-reversing homeomorphism of a line or circle has a fixed point. Freeness therefore places N_B in $\mathrm{Homeo}^+(B)$. Its faithful product action on the three-manifold $B \times \mathbb{R}^2$ makes it a Lie group by the verified Hilbert–Smith theorem used in C.33. Equation (C.38.4) bounds its dimension by one. If that dimension were one, every orbit of N_B^0 would be open in B: its one-dimensional orbit map is injective and hence open by invariance of domain. Connectedness

of B would make N_B^0 transitive. Freeness would then give $N_B = N_B^0$, since an element of N_B^0 with the same evaluation as an element of N_B equals it. This is the translation group $\mathbb{R}$ or the rotation group S^1 in evaluation coordinates. Its centralizer is that same abelian group of right translations. But $T \le C(N_B)$, contradicting the assumed nonabelianness of T. Consequently N_B is a zero-dimensional Lie group, so is discrete.

A free proper discrete action supplies evenly covered neighborhoods, and its orbit space is a connected one-manifold without boundary. For a line this quotient is a line or a circle. Since the line is simply connected, the first case has trivial deck group and the second has deck group $\mathbb{Z}$. For a circle the quotient is a circle and the covering has finitely many sheets; its deck group is finite cyclic. This proves the first assertion.

In the stated plane-loop application, C.28 proves that each operator stabilizing B induces an actual homeomorphism for its intrinsic line or circle topology. Commutation with G puts its image in (C.38.1). Every subgroup of this discrete group is closed; restriction of a proper free action to a closed subgroup is proper and free. These facts give (C.38.3). Its fibers are precisely the P_B-orbits. A deck transformation is determined by one point in the connected covering, and each point of a fiber is obtained by exactly one element of P_B, so the whole deck group is exactly P_B. The same one-manifold covering classification gives the stated quotient and degree. Equation (C.38.2) retains the original stabilizer and its restriction kernel. □

Topology and the remaining relative extension. The topology asserted for P_B is its intrinsic compact-open image topology. The operator group may be regarded as discrete as in the original problem; no assertion about continuity of the restriction map from an inherited plane topology to a recurrent intrinsic line topology is needed. C.38 gives exact covering quotients on every nonabelian one-dimensional orbit. It supplies no transverse control between different orbits and no relative extension of the C.37 family from X_2 to its exterior. Those requirements remain in C.12.

C.39. [Proved] A global contraction for loops with commuting values. Let f_t, $0 \le t \le 1$, be a based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$ with $f_s f_t = f_t f_s$ for every s, t. Suppose the loop is null-homotopic in that ambient group. For every subgroup H commuting with all f_t, the loop has a based null-homotopy in $C_{\mathrm{Homeo}^+(\mathbb{R}^2)}(H)$. The homotopy and its inverse family are jointly continuous on the whole plane, including every change and accumulation of orbit types.

Consequently, on every connected admissible surface U of C.1, an element of the kernel of the exact inclusion map

$$\pi_1 C_{\mathrm{Homeo}(U)}(H) \longrightarrow \pi_1 \mathrm{Homeo}(U) \tag{C.39.1}$$

that is represented by a loop with pairwise commuting values is zero. On a hyperbolic surface every such based loop in $C_{\mathrm{Homeo}(U)^0}(H)$ is therefore null-homotopic in that centralizer.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; complete global commuting-value subcase of the exact C.12 level-zero comparison.

Proof. The actual image on every orbit. Set $G = \langle f_t \rangle$. This is an abelian group. The orbit classification C.28 applies. On an open orbit O, the closure K_O of $G|_O$ in $\mathrm{Homeo}(O)$ is Polish, abelian, connected, and transitive. It acts freely: an element fixing a point commutes with the transitive subgroup $G|_O$ and hence fixes every point. The Polish open orbit-map theorem identifies evaluation $K_O \to O$ homeomorphically, with trivial stabilizer. Thus K_O is locally compact. Its faithful action on $O \times \mathbb{R}$ and the Hilbert–Smith input of C.33 make it a two-dimensional abelian Lie group. Since O is a noncompact open subset of the plane, the possibilities are $\mathbb{R}^2$ and $\mathbb{R} \times S^1$. The original image is already the whole group: for each $k \in K_O$, transitivity of $G|_O$ gives an element with the same evaluation, and freeness makes the two elements equal.

On an intrinsic line or circle orbit B, C.28 gives compact-open continuity of the restricted original paths. The same abelian closure and free evaluation argument, now in the intrinsic topology, identifies $G|_B$ with the full translation group $\mathbb{R}$ or rotation group S^1. This is also the abelian case of (C.37.9). On a point orbit the image is trivial. Denote the resulting actual Lie image on any orbit B by K_B. Its simply connected covering group is a vector group of dimension at most two.

The original restricted loop has a closed based lift $\widetilde{n}_B(t)$ to this vector group. For an open orbit, C.32 gives its null-homotopy in Homeo(O); evaluation and the homeomorphism $K_O \cong O$ then make its class in $\pi_1 K_O$ zero. For a line there is no covering kernel. For a circle the based lift closes by the exact winding comparison C.29. Define on each orbit

$$R^B_{r,t} = p_B\big(r\,\widetilde{n}_B(t)\big), \qquad 0 \le r, t \le 1. \tag{C.39.2}$$

Here p_B is the Lie-group covering homomorphism, with the trivial interpretation for a point orbit. The action of the right side on B is its actual homeomorphism action. Its endpoints are the identity at $r = 0$ and the original loop at $r = 1$; both based edges $t = 0, 1$ are fixed.

One compact word family for all orbit types. In dimension two, C.37 represents every value of (C.39.2) by a word isotopy

$$W_{\epsilon,a,b,t}(u) = f^{\epsilon}_{ut} f_{ub} f^{-1}_{ua}, \quad \epsilon \in \{0, 1\}, \quad 0 \le a, b \le t \le 1, \tag{C.39.3}$$

with equality of its specified based universal-cover endpoint. In dimension one the intermediate value theorem already finds $b \le t$ with $\widetilde{n}_B(b) = r\widetilde{n}_B(t)$. This is a member of (C.39.3) with $\epsilon = 0, a = 0$. The identity word handles point orbits. Thus the same compact family of original plane isotopies works for every orbit, with the based lifts retained.

Take arbitrary points $x_i \to x$ in the plane and parameters $(r_i, t_i) \to (r, t)$, and write $B_i = Gx_i$, $B = Gx$. Choose representing word isotopies W_i for $R^{B_i}_{r_i,t_i}$. Compactness of the parameter set in (C.39.3) gives a subsequence converging, together with inverses and uniformly in u, to a word isotopy W. Put $w = W(1)$. If B is open, all sufficiently large x_i are in that same orbit and continuity follows from its Lie action. If B is a point, every possible word limit fixes x, giving the required convergence. It remains to identify the limiting word on a line or circle. This argument explicitly allows the B_i to have different types.

For each fixed positive integer n, set $m_i = \lfloor nr_i \rfloor$. In the vector covering group of the actual image K_{B_i}, one has

$$n\,\widetilde{w}_i - m_i\,\widetilde{n}_{B_i}(t_i) = (nr_i - m_i)\widetilde{n}_{B_i}(t_i). \tag{C.39.4}$$

The right side is represented by another word isotopy Z_i from (C.39.3), with equality of based lifted endpoints. For this fixed n, pass to a further subsequence so that $m_i = m_n$ is constant and $Z_i \to Z_n$. Then $|m_n - nr| \le 1$. Identities on B_i pass to the whole orbit B by evaluation at $hx_i \to hx$, for any fixed $h \in G$. In particular, with $g = f_t$ and $z_n = Z_n(1)$,

$$(w|_B)^n (g|_B)^{-m_n} = z_n|_B. \tag{C.39.5}$$

For a line, let $\theta(u)$ be the real translation parameter of $f_u|_B$, and put $M = \max_u |\theta(u)|$. The real parameters A of $w|_B$ and C_n of $z_n|_B$ come from their word paths: each has the form $\epsilon\theta(t) + \theta(b) - \theta(a)$. Consequently $|C_n| \le 3M$, and (C.39.5) is the real identity $nA - m_n\theta(t) = C_n$. Division by n gives $A = r\theta(t)$, as required.

For a circle, use the based real rotation lift θ of the original loop, so $\theta(0) = \theta(1) = 0$. The word paths again specify real endpoint angles A, C_n with $|C_n| \le 3M$, but (C.39.5) initially gives their relation only modulo integers. The exact lift in (C.39.4) removes this ambiguity. Indeed,

$$\Lambda_i(u) = W_i(u)^n f^{-m_i}_{ut_i} Z_i(u)^{-1} \tag{C.39.6}$$

is a based loop in the actual group K_{B_i} and has a closed based universal-cover lift. It is therefore null-homotopic in that group. Choose $h \in G$ with $hx \neq x$. For large i the two points x_i, hx_i of B_i are distinct. Evaluation of (C.39.6) at them has difference-direction winding zero in $\mathrm{Conf}_2(\mathbb{R}^2)$, regardless of whether B_i is an open, line, or circle orbit. This follows from the actual null-homotopy in K_{B_i} and its continuous evaluation into the plane; C.28 supplies that evaluation for intrinsic line orbits.

The word isotopies and the two evaluation points converge. Their limiting pair stays distinct. The winding therefore remains zero. On the limiting Jordan circle its value, by C.29, is $nA - m_n\theta(t) - C_n$, up to a common orientation sign. Thus the real identity

$$nA - m_n\theta(t) = C_n, \qquad |A - r\theta(t)| \leq \frac{3M + |\theta(t)|}{n} \tag{C.39.7}$$

holds for every n. It forces $A = r\theta(t)$. This also treats an identity endpoint rotation whose based angle $\theta(t)$ is a nonzero integer. All possible subsequential word limits therefore give the prescribed value (C.39.2).

Compactness of (C.39.3) now proves joint continuity of (C.39.2) on the entire plane. The inverse loop has the same orbits and image groups, and its construction gives precisely the inverse of (C.39.2); the same proof gives joint inverse continuity. Each map is an orbitwise bijection, so these are plane homeomorphisms. Every commuting operator carries the actual abelian Lie image on one orbit isomorphically to that on the transported orbit. These are continuous Lie-group isomorphisms in the specified intrinsic or open-orbit topology. Their based covering lifts are linear, and hence commute with multiplication by r. Formula (C.39.2) is therefore exactly H-equivariant. Taking $r = 1 - s$ proves the plane assertion.

The precise surface lift and descent. Every connected admissible surface in C.1 has universal covering $p : \widetilde{U} \to U$ with $\widetilde{U} \cong \mathbb{R}^2$. Let f_t represent an element of the kernel in (C.39.1), with its commuting-value property. Choose an ambient filling $A : D^2 \to \mathrm{Homeo}(U)$ whose boundary is this loop and whose specified based boundary point maps to the identity. The actual map

$$D^2 \times \widetilde{U} \longrightarrow U, \qquad (z, x) \longmapsto A(z)(p(x)) \tag{C.39.8}$$

has a unique lift equal to the identity on the specified based fiber. Its domain is simply connected. Lifting the inverse family as well, and using uniqueness, proves that its slices are homeomorphisms with jointly continuous inverses. It is an ambient filling of a closed based plane loop $\widetilde{f}_t$.

For every lift $\widetilde{h}$ of an operator, $[\widetilde{f}_t, \widetilde{h}]$ is a deck transformation. It depends continuously on t and starts at the identity, so it is identically the identity. Similarly $[\widetilde{f}_s, \widetilde{f}_t]$ is a deck transformation depending continuously on (s, t); it is the identity on the axes, and hence everywhere. Thus the lifted loop has commuting values and centralizes the full group of operator lifts, including the deck group. Apply the plane assertion. Its equivariant contraction descends through p, with both joint continuities, to the original centralizer on U. Its slices lie in the identity component by the displayed paths. This proves (C.39.1). For a hyperbolic U the ambient identity component is contractible, so the ambient kernel condition is automatic. □

Scope of the global subcase. This theorem treats loops with commuting values on every orbit at once, including open orbits whose operator image has dimension zero or one. It does not identify every fundamental-group class of an arbitrary centralizer with such a loop. No deformation of a general centralizer loop to this subcase has been constructed here. C.12 remains open for general loops.

The earlier abelian closed subgroup cannot be a centralizer. Fix the specific embedding $j : B \hookrightarrow A^0$ of C.26, where $B = L^1([0, 1], \mathbb{Z})/\mathbb{Z}$, $A = \mathrm{Homeo}(U)$ and U is hyperbolic. For every

$H \leq A$ commuting with $j(B)$, the induced map $\pi_1(B) \to \pi_1(C_A(H), 1)$ is zero. Indeed the displayed generator of $\pi_1(B) = \mathbb{Z}$ has pairwise commuting values and is ambient-null, so C.39 contracts its image; the induced homomorphism therefore vanishes on the whole group. The based contraction lies in A^0, so the same conclusion holds with target $\pi_1(C_{A^0}(H), 1)$. Consequently $j(B)$ cannot equal either of these centralizers, since its topological-group identification with B induces an isomorphism on the nonzero fundamental group. This rules out that exact realization as a zero-level counterexample.

C.40. [Proved] The locally compact loop-closure case. Let U be any connected admissible surface of C.1, let $A = \mathrm{Homeo}(U)$ with its compact-open group topology, and let $H \leq A$ be arbitrary. Put $C = C_A(H)$. For a based loop $f : [0, 1] \to C$, form the actual closed subgroup

$$L_f = \overline{\langle f_t : 0 \leq t \leq 1 \rangle}^{A}$$

with the topology induced from A. Every class in the kernel of

$$i_* : \pi_1(C, 1) \longrightarrow \pi_1(A, 1)^H \tag{C.40.1}$$

that has a representative with locally compact L_f is zero. Here $i : C \hookrightarrow A$ is the inclusion; its induced map lands in the displayed fixed subgroup because each $h \in H$ fixes the entire representative loop under conjugation.

In particular, the same conclusion holds for loops contained in any closed locally compact subgroup of C. If C itself is locally compact, (C.40.1) is injective. On a hyperbolic surface every based loop in $C_{\mathrm{Homeo}(U)^0}(H)$ with locally compact L_f is null-homotopic in that centralizer.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact inclusion map from the centralizer fundamental group to the H-invariant ambient fundamental group.

Proof. The exact subgroup and its Lie topology. Centralizers are closed in the Hausdorff topological group A. Consequently $L = L_f$ is a closed subgroup of C. The subgroup generated by the loop is path connected: each finite product of loop values and their inverses can be joined to the identity by the corresponding product of paths. Its closure L is connected. Thus no identity-component replacement or change of topology is being made. The topology on L is the subspace topology from A, and its action on U is continuous and faithful.

Suppose L is locally compact. The product action

$$L \times (U \times \mathbb{R}) \longrightarrow U \times \mathbb{R}, \qquad (l, (x, a)) \longmapsto (lx, a) \tag{C.40.2}$$

is a faithful continuous action on a connected three-manifold. The verified Hilbert–Smith theorem used in C.33, [52, Theorem 1.5 and the introductory reduction], therefore makes L a Lie group with precisely this topology. It is a connected Lie group. All homotopies constructed inside L below are consequently homotopies through actual elements of C, continuous in its given topology.

A based deformation into a compact subgroup. Borel's principal theorem gives a connected maximal compact subgroup K and closed one-parameter subgroups $H_1, \ldots, H_d$, each isomorphic to $\mathbb{R}$, such that multiplication is a homeomorphism

$$K \times \mathbb{R}^d \longrightarrow L, \qquad (k, a_1, \ldots, a_d) \longmapsto k\, h_1(a_1) \cdots h_d(a_d). \tag{C.40.3}$$

This is the actual product assertion in [10, §1, p. 271; §§3–4, pp. 277–278]: the product notation is explicitly defined there by uniqueness and continuous dependence of all factors. The maps $h_j : \mathbb{R} \to H_j$ are topological group isomorphisms with $h_j(0) = 1$. The identity has the unique coordinates $(1, 0, \ldots, 0)$. Thus

$$D_s(k h_1(a_1) \cdots h_d(a_d)) = k h_1((1 - s)a_1) \cdots h_d((1 - s)a_d) \tag{C.40.4}$$

is a continuous strong deformation retraction $D : L \times [0,1] \to L$ onto K, fixing the identity. In particular $D_s(f_t)$ is a based homotopy in C from f to a based loop β in K.

A commuting representative of the same based class. Average an inner product on the Lie algebra of K over its compact adjoint action. The resulting positive definite inner product is adjoint invariant, so left translation gives a bi-invariant Riemannian metric on K. Its pullback to the universal covering group $p : \widetilde{K} \to K$ is also bi-invariant. For left-invariant vector fields its Levi-Civita connection satisfies

$$\nabla_X Y = \tfrac{1}{2}[X, Y]. \tag{C.40.5}$$

Indeed the Koszul formula and adjoint invariance give this identity directly. Hence all geodesics through the identity are the one-parameter subgroups, and their left translates are defined for all real time. This proves geodesic completeness of the lifted metric. The Hopf–Rinow theorem therefore joins the identity to every point of $\widetilde{K}$ by a geodesic.

Lift β from the identity to a path $\widetilde{\beta}$ with endpoint $z \in \ker p$. Choose X in the Lie algebra of $\widetilde{K}$ with $\exp_{\widetilde{K}}(X) = z$, using the geodesic just obtained. The paths $\widetilde{\beta}(t)$ and $\exp_{\widetilde{K}}(tX)$ have the same endpoints in the simply connected space $\widetilde{K}$, so they are homotopic relative to both endpoints. Projection through p is a based homotopy in K from β to

$$\gamma(t) = p\big(\exp_{\widetilde{K}}(tX)\big). \tag{C.40.6}$$

This is a based loop, since $p(z) = 1$. It has pairwise commuting values because it is the restriction of a one-parameter subgroup. Every step from f to γ stays in C.

Now let $[f]$ be in the kernel of (C.40.1). The inclusion of the fixed subgroup $\pi_1(A,1)^H$ into $\pi_1(A,1)$ is injective, so f is ambient-null. The based centralizer homotopy just constructed makes γ ambient-null as well. C.39 applies on this same admissible surface to this actual commuting loop and the same operator subgroup H. It supplies a based contraction of γ inside C. Concatenating the two based homotopies contracts f inside C.

A loop contained in a closed locally compact subgroup $M \le C$ has $L_f \subseteq M$ closed; it is therefore locally compact. Taking $M = C$ proves the assertion about a locally compact centralizer. In the hyperbolic case $\mathrm{Homeo}(U)^0$ is contractible by the earlier surface theorem, so the ambient kernel condition holds for every based loop. The constructed contraction lies in the identity path component because it is a based homotopy; it consequently lies in the displayed centralizer inside $\mathrm{Homeo}(U)^0$. □

Exact remaining scope. C.40 proves a reduction to commuting values for the specified locally compact closed subgroup. No such based homotopy has been constructed for a general loop whose generated closure is not locally compact. The unrestricted comparison C.12 remains open. In particular, the proof does not replace the induced topology on that closure by a different locally compact topology.

C.41. [Proved] A dense operator orbit settles the zero-level kernel. Let U be a connected admissible surface of C.1, let $A = \mathrm{Homeo}(U)$ with its compact-open group topology, and let $H \le A$ have a dense orbit in U. Then the exact inclusion-induced comparison

$$\pi_1(C_A(H), 1) \longrightarrow \pi_1(A, 1)^H \tag{C.41.1}$$

is injective. In particular, on a hyperbolic surface $\pi_1(C_{A^0}(H), 1) = 0$. No local compactness assumption is imposed on H or on the subgroup generated by the values of the loop.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact centralizer inclusion on fundamental groups for a dense operator orbit.

Proof. The plane argument. Let f_t be a based ambient-null loop of plane homeomorphisms centralizing a subgroup H with a dense orbit D. Put $G = \langle f_t \rangle$. Since the loop is based, its maps preserve orientation, and the orbit results C.28–C.37 apply. Every $h \in H$ transports G-orbits according to the exact identity

$$h(Gx) = G(hx). \tag{C.41.2}$$

If there is no open G-orbit, C.34 already gives the required global equivariant contraction.

Otherwise let V be the nonempty union of the open G-orbits. It is open and H-invariant. Density makes D meet V, and invariance then gives $D \subseteq V$. Consequently V is dense in the plane.

Fix an open G-orbit O and choose $y \in D \cap O$. Let $H_O = \{h \in H : hO = O\}$ be its operator stabilizer. The actual orbit equality is

$$Hy \cap O = H_O y. \tag{C.41.3}$$

Indeed, $hy \in O$ implies that the two G-orbits $hO = G(hy)$ and O meet, hence are equal. The reverse inclusion is immediate. Since $Hy = D$ is dense, (C.41.3) is dense in O.

Retain the exact restriction homomorphism $H_O \to \mathrm{Homeo}(O)$, with its pointwise-fixing kernel. Let Q_O be the closure of its image in that homeomorphism group. C.30 proves that Q_O is locally compact and acts freely and properly on O; C.33 makes it a Lie group of dimension at most two. Properness makes every Q_O-orbit closed. In particular $Q_O y$ is a closed subset of O containing the dense set $H_O y$, so

$$Q_O y = O. \tag{C.41.4}$$

Thus this free proper action is transitive. Its orbit map $Q_O \to O$ is a homeomorphism, and $\dim Q_O = 2$.

This holds for every open G-orbit. In the notation of C.37, therefore,

$$U_2 = V, \qquad X_2 = \overline{U_2} = \mathbb{R}^2. \tag{C.41.5}$$

The contraction proved there is defined on the whole plane, with jointly continuous inverses, and centralizes every element of the original H. This settles both alternatives of the plane argument.

The actual surface comparison. Let a based loop f_t represent a kernel class in (C.41.1). The inclusion of the fixed subgroup in $\pi_1(A, 1)$ is injective, so choose an ambient filling $F : D^2 \to A$ of that loop. For the universal covering $p : \widetilde{U} \to U$, every admissible $\widetilde{U}$ is a plane. Lift the actual map

$$D^2 \times \widetilde{U} \longrightarrow U, \qquad (z, x) \longmapsto F(z)(p(x)) \tag{C.41.6}$$

with the identity prescribed on the based boundary fiber. Its domain is simply connected. As in C.39, lifting the inverse family and using uniqueness gives an ambient filling by plane homeomorphisms with jointly continuous inverses. Its boundary is a closed based plane loop $\widetilde{f}_t$.

Let $\widetilde{H}$ be the group of all lifts of elements of H, including the deck group. For every $\widetilde{h} \in \widetilde{H}$, the commutator $[\widetilde{f}_t, \widetilde{h}]$ is a deck transformation depending continuously on t and equal to the identity at $t = 0$. It is therefore always the identity. The lifted loop centralizes precisely the same operator lifts required for descent.

Choose $x \in U$ with dense Hx and $\widetilde{x} \in p^{-1}(x)$. One has the exact equality

$$\widetilde{H}\,\widetilde{x} = p^{-1}(Hx). \tag{C.41.7}$$

One inclusion follows by projection. For the other, choose a lift of the specified $h \in H$ and then a deck transformation taking its value at $\widetilde{x}$ to the specified point over hx. The right side is dense because p is open: every nonempty open set upstairs projects to a nonempty open set meeting Hx.

Apply the plane result to $\widetilde{H}$. Its contraction commutes with all deck transformations and all operator lifts. The equation $\overline{R}_{s,t}p = pR_{s,t}$ therefore defines a based contraction downstairs inside $C_A(H)$. The open covering quotient and its local sections give joint continuity of these maps and their inverses. It contracts the original class in the source of (C.41.1), proving injectivity. For a hyperbolic U, the ambient identity component is contractible, so its based centralizer loops all satisfy the kernel condition. The based contraction stays in A^0. □

Remaining scope. C.41 settles every operator action with a dense orbit. It does not assert that a general operator group has such an orbit or supply an equivariant decomposition into subsets on which this argument applies. The general operator problem without a dense orbit in C.12 remains open.

C.42. [Proved] The actual boundary quotient has proper line components upstairs. Let G be generated by a based plane loop as in C.28, let O be an open G-orbit, and put $L = \overline{G} \leq \mathrm{Homeo}(\mathbb{R}^2)$. Fix $x \in O$ and $J = L_x$. For $z \in \mathrm{Fix}(J)$ whose G-orbit $B = Gz$ is a nontrivial intrinsic line or a circle, the formula

$$\Phi_z(lx) = lz$$

defines the actual continuous open quotient $O = L/J \to B = L/L_z$, with the intrinsic topology on B. For the universal coverings $p : \widetilde{O} \to O$ and $q : \mathbb{R} \to B$, every lift $F : \widetilde{O} \to \mathbb{R}$ of $\Phi_z p$ is open and onto, and every connected component of every fiber of F is a properly embedded line in the covering plane $\widetilde{O}$. Every nondegenerate compact connected component of a fiber of Φ_z is a circle. Whole fibers of Φ_z can be disconnected, including for quotients obtained as limits of the original commuting operators.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact boundary quotients in the remaining centralizer extension problem.

Proof. The quotient and its topology. Every element of L preserves O: forward and inverse limits preserve its boundary, while a forward limit sends an interior point into $\overline{O}$ and cannot send it to that boundary. The Polish orbit-map theorem used in C.26 and C.30 therefore makes $l \mapsto lx$ an open quotient $L \to O$. If $l_1x = l_2x$, then $l_2^{-1}l_1 \in J$ fixes z, so Φ_z is well defined and continuous. More precisely, the continuous action map

$$\mathrm{Fix}(J) \times L \longrightarrow \mathbb{R}^2, \qquad (z, l) \longmapsto lz$$

descends through the open quotient $\mathrm{id} \times (L \to O)$. Consequently

$$\mathrm{Map}_L(O, \mathbb{R}^2) \xrightarrow[\cong]{\mathrm{ev}_x} \mathrm{Fix}(J) \tag{C.42.1}$$

is a homeomorphism in the compact-open topology: equivariance proves bijectivity, and the displayed joint action proves continuity of the inverse. For $l \in L$, choose $g \in G$ with $gx = lx$. Then $g^{-1}l \in J$, hence $lz = gz$. Thus the range is exactly $Lz = Gz = B$.

The map into an intrinsic line is continuous in its intrinsic topology. Indeed, the image of a compact coordinate disk in O is a compact Peano continuum in B. By C.28 it contains neither a Jordan curve nor a triod, and hence is a point or a compact arc; on such an arc the intrinsic and ambient topologies agree. These disks give neighborhoods at every point of O. For a circle the two topologies already agree. Each $l \in L$ restricts to an intrinsic homeomorphism of B, because compact arcs are carried to compact arcs. For $b = gz$ one has

$$lb = \Phi_z(lgx). \tag{C.42.2}$$

This makes every point evaluation of the restriction $L \to \mathrm{Homeo}(B)$ continuous. Finite ordered partitions of compact intervals show that pointwise convergence of monotone homeomorphisms to

a homeomorphism is uniform on compact intervals; apply the same argument to inverses. Finite cyclic partitions give the circle assertion. Thus restriction is continuous in the compact-open group topology. All these restrictions preserve orientation, since the original loop does and the restriction map is continuous.

Apply the Polish orbit-map theorem to this transitive action on the Polish one-manifold B. It identifies B with L/L_z in its stated topology. Since $J \subseteq L_z$, the map Φ_z is the open quotient $lJ \mapsto lL_z$. In particular, if operators h_n commute with G and $h_n x \to z$, then they commute with L and (C.42.1) gives $h_n|_O \to \Phi_z$ uniformly on compact subsets of O.

Both covering maps and the compatible group. The universal cover of the connected open planar surface O is a plane. Fix a lift

$$F : \widetilde{O} \longrightarrow \mathbb{R}, \qquad qF = \Phi_z p. \tag{C.42.3}$$

For a line B, take q to be an identification. Local covering charts and openness of $\Phi_z p$ show that F is open. Consider the following subgroup, with the topology induced from the indicated product:

$$\begin{aligned} \mathcal{L} = \{(l,a,b) \in L \times \mathrm{Homeo}(\widetilde{O}) \times \mathrm{Homeo}^+(\mathbb{R}) : \quad & pa = (l|_O)p, \\ & qb = (l|_B)q, \qquad Fa = bF\}. \end{aligned} \tag{C.42.4}$$

The three equations are closed conditions, as may be checked by evaluation on countable dense sets. Hence $\mathcal{L}$ is a Polish group, and its actions by a and by b are continuous. Every lift a of $l|_O$ has a unique compatible lift b of $l|_B$: the maps Fa and bF lift the same map to B, and the target deck translation is chosen to make them agree at one point. Uniqueness of lifts then gives equality everywhere. For a line target there is no target deck ambiguity.

The action of $\mathcal{L}$ on $\widetilde{O}$ is transitive. Choose l carrying the projections of two specified points to one another and adjust a source lift by a deck transformation to carry the specified points themselves. The image of F is a nonempty open interval I. It is invariant under every compatible b. If B is a circle and I has a finite endpoint, every increasing b fixes that endpoint, so its image under q would be fixed by every element of the transitive L-action on B. This is impossible. Therefore $I = \mathbb{R}$; for a line target surjectivity is already immediate from (C.42.3).

A component that separates the plane. Fix $t \in \mathbb{R}$ and write $E = F^{-1}(t)$. Choose u, v with $F(u) < t < F(v)$. The closed set E separates u from v. Frankel's planar separation lemma [21, Lemma 2.7, p. 1258] gives a connected component C of E separating them. The lemma applies to unbounded closed sets. For completeness, take the complementary domain U of E containing u, and then the component V of $\widetilde{O} \setminus \overline{U}$ containing v. The planar Brouwer property makes ∂V connected. It lies in E and separates u, v, so the component of E containing ∂V has the required property.

Let $\mathcal{K}$ be the closed subgroup of $\mathcal{L}$ whose b-coordinate fixes t. It acts transitively on E: transitivity on the covering plane finds an element carrying any two points of E to one another, and $Fa = bF$ forces $b(t) = t$. Thus all components of E are ambiently homeomorphic to C, and all separate the plane. The stabilizer $\mathcal{K}_C$ is a closed Polish subgroup and acts transitively on C. Closedness follows by applying forward and inverse limits to the closed set C.

Accessibility from each fixed complementary domain. Let D be any component of $\widetilde{O} \setminus C$. Its stabilizer in $\mathcal{K}_C$ is open: for a fixed $d \in D$, every sufficiently small group element sends d into D and therefore sends the whole complementary domain onto D. The open orbit-map theorem for $\mathcal{K}_C \curvearrowright C$ makes every orbit of this open stabilizer open in C. Since C is connected, the stabilizer acts transitively on C.

There is an accessible point of ∂D: an interior point and a closest boundary point give a straight end-cut in a coordinate copy of the plane. Its boundary endpoint belongs to C. Transporting this end-cut by the domain stabilizer proves that every point of C is accessible from this same D. This holds for every complementary domain.

There are exactly two such domains. There are at least two because C separates. For three domains, choose three distinct points $c_1, c_2, c_3 \in C$. Inside each domain, join end-cuts to these points to an embedded tree with these three boundary leaves and one interior branch point. Such a tree is obtained by joining two end-cuts to an arc and stopping a third connecting arc at its first intersection. The three trees have disjoint interiors and share just their specified boundary leaves. Their union embeds $K_{3,3}$ in the plane, contradicting the bipartite planar Euler inequality $9 \le 2 \cdot 6 - 4$.

The proper line conclusion. If C is unbounded, put $C^* = C \cup \{\infty\}$ in the sphere; if C is bounded, put $C^* = C$. This is a compact connected proper subset with exactly two complementary domains. Every point except possibly the added point ∞ is accessible from both. Choose a stereographic pole in one complementary domain. Deleting that pole leaves that domain connected, and accessibility at its boundary is unaffected, since end-cuts can be chosen to avoid the pole. In the resulting plane both inaccessible subsets of C^* have at most one point. Polulyakh's converse Jordan theorem [54, Theorem 1.1 and Corollary 1.1] therefore makes C^* a Jordan curve. The bounded alternative is impossible: the real-valued open map F is constant on that Jordan curve, and its restriction to the bounded disk either has an interior extremum or is constant on a nonempty open set. Both contradict openness. Thus C is a Jordan curve with the compactification point removed, a properly embedded line. Transitivity of $\mathcal{K}$ transfers this conclusion to every component of every fiber.

Downstairs, a fiber of Φ_z is homogeneous under L_z, and each of its component stabilizers is transitive by the same orbit argument. A nondegenerate compact component is therefore homogeneous by ambient neighborhood homeomorphisms. Brechner–Erkama [12, Theorem 2] makes it a Jordan curve.

An exact disconnected-fiber example. On $r > 1$ in the plane put $s = \log(r-1)$. Let β_b be the time-b flow of $\sin\delta\,\partial_\delta$ on $\mathbb{R}$. It fixes $\pi\mathbb{Z}$, is 2π-periodic as a lifted circle map, and $|\beta_b(\delta) - \delta| \le \pi$. Define

$$\begin{aligned} A_a(r,\theta) &= (r, \theta + a) && (r \le 1),\\ A_a(r,\theta) &= (1 + e^a(r-1), \theta + a) && (r > 1),\\ B_b|_{\{r\le 1\}} &= \mathrm{id},\\ B_b(s,\theta) &= (\theta + \beta_b(s-\theta), \theta) && (r > 1). \end{aligned} \tag{C.42.5}$$

The last first coordinate is s. Its expression is independent of the chosen representative of θ. The bound on $\beta_b - \mathrm{id}$ gives continuity at $r = 1$ and properness at infinity for B_b and B_{-b}. The A-formulas have these properties directly. They are continuous orientation-preserving one-parameter actions and commute, since A_a preserves $s - \theta$.

The open spiral strip

$$(t,\delta) \longmapsto (1 + e^{t+\delta},\ t \bmod 2\pi), \qquad t \in \mathbb{R}, \quad 0 < \delta < \pi, \tag{C.42.6}$$

is an open orbit O. This parametrization is injective because an angular ambiguity is $2\pi k$, while two δ-values differ by less than π. In these coordinates A_a translates t and B_b sends δ to $\beta_b(\delta)$. The resulting group is $\mathbb{R}^2$, simply transitive on O and closed in the plane homeomorphism group. Indeed, the open-orbit preservation argument at the start of the proof shows that any homeomorphic limit preserves O, using both the forward and inverse limits. Its evaluation at $(0, \pi/2)$ therefore remains inside O and recovers finite parameters $a = t$ and $b = \log\tan(\delta/2)$ continuously. The parameter limits recover the limiting plane homeomorphism, proving closedness. A triangular parameter loop $0 \to (1,0) \to (0,1) \to 0$ generates this group and contracts by scaling both parameters.

For the actual commuting operator subgroup $H = \langle A_{-2\pi}\rangle$, its powers $h_n = A_{-2\pi n}$ converge on compact subsets of O to

$$\Phi(t,\delta) = (1,\ t \bmod 2\pi). \tag{C.42.7}$$

This is exactly the quotient associated to the limit of $h_n(0, \pi/2)$. Its fiber over θ_0 is the disjoint union of the lines $t = \theta_0 + 2\pi k$, $k \in \mathbb{Z}$. Its real lift is t. All maps and the ambient contraction have been specified, so this example concerns disconnected whole fibers rather than a nonzero centralizer fundamental-group class. The executed symbolic certificate `code/part_b/c42_spiral_fibre_certificate.py` checks the flow law and differential equation in the exact half-angle coordinate, the A–B commutation identities, the integer-labelled whole fibers and their real lift, and the based triangular parameter loop. It uses no numerical approximations; the topological conclusions are proved above. □

Scope of the reduction. C.42 proves the component statement for the specified real-valued lift. It does not identify a lifted component with a whole downstairs connected component without checking the actual covering restriction. It does not yet give local product charts for the component decomposition, simultaneous charts for several boundary quotients, or the frontier-compatible contraction needed in C.12. Those global extension questions remain open.

C.43. [Proved] The lifted boundary quotient is a product projection. For the actual quotient and covering maps in C.42, every fiber of the real lift $F : \widetilde{O} \to \mathbb{R}$ is connected. There is a homeomorphism

$$T : \mathbb{R} \times \mathbb{R} \longrightarrow \widetilde{O}, \qquad F(T(u,t)) = t. \tag{C.43.1}$$

In particular the components in C.42 form a foliation with genuine local product charts. The original map $\Phi_z : O \to B$ also has local product charts with its specified target coordinate. The construction treats one quotient map at a time; its auxiliary flow and the homeomorphism T carry no asserted equivariance under the original operator family.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; actual boundary-quotient maps in the remaining relative centralizer extension.

Proof. Write $X = \widetilde{O}$ and let $\mathcal{L}$ be the closed Polish group (C.42.4), acting transitively on the covering plane X. For $v \in X$, let C_v be the connected component of $F^{-1}(F(v))$ containing v. By C.42 these sets partition X into properly embedded lines. The equation $Fa = bF$ for an element $(l, a, b) \in \mathcal{L}$ implies

$$a(C_v) = C_{a(v)}. \tag{C.43.2}$$

Thus this is the actual component partition preserved by the given compatible action.

Regularity and local products. Let $I \subset C_v$ be a compact arc containing v, and let $\epsilon > 0$. In a fixed metric on the covering plane, the condition

$$\sup_{w \in I} d(a(w), w) < \epsilon$$

defines an identity neighborhood in $\mathcal{L}$. The Polish open orbit-map theorem makes its image at v a neighborhood of v. Consequently, for every sufficiently near v', some such a sends v to v'. Restricting a to I gives an ϵ-close embedding of I into $C_{v'}$ carrying v to v'. This verifies precisely the regularity condition of Aarts–Oversteegen [1, Definition 5.3]. Their local-section and flowbox theorems [1, Theorems 5.6–5.7] therefore provide open product neighborhoods $S \times (-1, 1) \to X$, with each vertical plaque contained in one C_v.

The transverse factor is an interval locally. Indeed it is metrizable, and its product with an interval is a surface without boundary. The elementary product-cancellation argument proved in C.33 applies: local path connectedness descends to S, embedded open arcs in S are open by invariance of domain, and an endpoint would have zero local second homology in the product. Thus S is locally a one-manifold without boundary. Shrink to an interval and write one of these charts as

$$\chi : J \times (-1, 1) \longrightarrow V \subset X.$$

Because every plaque lies in a fiber, $F\chi(s,u) = f(s)$ for a continuous function f. For each open $W \subset J$, the set $\chi(W \times (-1,1))$ is open in X, so the openness of F proves that $f(W)$ is open. A continuous open map between intervals is strictly monotone: failure of strict monotonicity produces a local maximum, local minimum, or a constant interval, each contradicting openness. It follows that f is a homeomorphism onto an open interval. Replacing the transverse coordinate by $f(s)$ gives actual charts in which F is the projection.

The component quotient and its topology. Let Y be the set of components C_v, equipped with the quotient topology, and let $\pi : X \to Y$ be the quotient map. This map is open. To verify this directly, let $U \subset X$ be open and let v belong to the saturation of U. Choose $w \in C_v \cap U$ and the compact arc in C_v joining v to w. Apply the regularity argument above with an error small enough that the image of w remains in U. For every v' in a neighborhood of v the resulting image arc joins v' to a point of U in $C_{v'}$. The saturation of U is therefore open, which proves the assertion about π.

In a chart where F is projection, distinct plaques have distinct F-values. They therefore belong to distinct global components. The transversal maps homeomorphically onto $\pi(V)$: its inverse is the restriction of the continuous factor map $\overline{F}$ in

$$X \xrightarrow{\pi} Y \xrightarrow{\overline{F}} \mathbb{R}, \qquad F = \overline{F}\pi. \tag{C.43.3}$$

Here $\pi(V)$ is open by the preceding paragraph, and the forward inverse in the chosen target coordinate is the continuous transversal section. Thus Y is a one-manifold without boundary in the sense that it is locally an interval; no Hausdorff property has been assumed. The same charts show that $\overline{F}$ is a local homeomorphism.

The quotient Y is connected and second countable: an open quotient carries a countable base to a countable base. Every element of $\mathcal{L}$ induces a homeomorphism of Y by (C.43.2), and transitivity on X makes Y homogeneous. Baillif–Gabard's theorem [3, Theorem 1.1, proved in Section 4] now proves that Y is Hausdorff. The topological hypotheses here are exact: second countability gives Lindelöfness, and the actual quotient charts give the local Euclidean structure required in that theorem.

One can also see the separation mechanism directly from their proof. In a locally Hausdorff space, Zorn's lemma supplies a maximal Hausdorff open set through each point; maximality makes it dense. A countable subcover exists by Lindelöfness. A manifold is locally Baire and hence Baire, so those dense open sets have a common point. That point is separated from every other point inside one member of the subcover. Homogeneity transports this property to every point.

A connected second-countable Hausdorff one-manifold without boundary is a line or a circle. The circle alternative is impossible because $\overline{F}$ is open and has image $\mathbb{R}$ by C.42. Thus Y is a line. Its local homeomorphism $\overline{F} : Y \to \mathbb{R}$ is globally strictly monotone: the local direction cannot change on a connected line. It is onto, hence a homeomorphism. Each fiber of F is consequently one of the proper lines C_v.

An exact global product for this map. Orient each leaf using the orientation of the covering plane together with the increasing transverse coordinate F. On an overlap of two orientation-preserving charts (u, F), the leaf-coordinate transition is increasing. This supplies the coherent orientation required for the Aarts–Oversteegen global-flow theorem [1, Main Theorem 0.5, its proof in Section 4, and Remark 4.4]. It gives a continuous flow

$$R : \mathbb{R} \times X \longrightarrow X$$

whose orbits are exactly the specified leaves. The flow is free, since those leaves are lines rather than circles, and a nontrivial period would make the corresponding orbit a circle.

Take any small transversal section $\sigma : I \to X$ with $F\sigma(t) = t$. The map

$$\Theta_\sigma : \mathbb{R} \times I \longrightarrow F^{-1}(I), \qquad (u,t) \longmapsto R(u, \sigma(t)) \tag{C.43.4}$$

is continuous and bijective: every fiber is exactly one flow orbit, and freeness gives uniqueness of u. Both spaces are surfaces, so invariance of domain makes (C.43.4) a homeomorphism. These are actual local trivializations of a principal $\mathbb{R}$-bundle over $\mathbb{R}$.

For completeness, its global section can be constructed with the group coordinate retained. Choose a locally finite interval cover (I_i) with sections σ_i and a subordinate partition of unity (ρ_i) whose supports lie in I_i. On overlaps, let the continuous real transition functions be defined by

$$\sigma_j(t) = R(a_{ij}(t), \sigma_i(t)).$$

They obey $a_{ij} = a_{ik} + a_{kj}$. Set $b_i(t) = \sum_j \rho_j(t) a_{ij}(t)$ on I_i, extending the individual terms by zero outside their supports. The sum is locally finite and $b_i = a_{ik} + b_k$ on overlaps. Consequently $\sigma(t) = R(b_i(t), \sigma_i(t))$ is a well-defined continuous global section with $F\sigma(t) = t$. The same map as (C.43.4), now over all $\mathbb{R}$, gives (C.43.1).

Finally, the identities $qF = \Phi_z p$ from C.42, together with local charts for the two covering maps p and q, carry the projection charts to the original map Φ_z. This proves its stated local product property with the actual target coordinate. □

Scope of the reduction. The auxiliary flow in the proof is used to trivialize one specified quotient. It has not been shown to commute with the original operators, preserve another boundary quotient, or extend across the frontier of O. C.43 therefore strengthens the structure of the necessary boundary maps without completing the simultaneous relative extension in C.12.

C.44. [Proved] Exact downstairs forms of a nontrivial boundary quotient. For the actual quotient $\Phi_z : O \to B$ of C.42, the open planar orbit O is a plane or an annulus. Choose a coordinate $B = \mathbb{R}$ when B is a line, or a coordinate $B = \mathbb{R}/\mathbb{Z}$ when it is a circle, and use the corresponding target covering in the definition of F. There are source homeomorphisms giving one of the following normal forms for the actual map:

O	B	Φ_z	whole fiber
$\mathbb{R} \times \mathbb{R}$	$\mathbb{R}$	$(u,t) \mapsto t$	$\mathbb{R}$
$\mathbb{R} \times \mathbb{R}$	$\mathbb{R}/\mathbb{Z}$	$(u,t) \mapsto t + \mathbb{Z}$	$\coprod_{\mathbb{Z}} \mathbb{R}$
$(\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$	$\mathbb{R}$	$([u],t) \mapsto t$	$\mathbb{R}/\mathbb{Z}$
$(\mathbb{R}/\mathbb{Z}) \times \mathbb{R}$	$\mathbb{R}/\mathbb{Z}$	$([u],t) \mapsto t + \mathbb{Z}$	$\coprod_{\mathbb{Z}} (\mathbb{R}/\mathbb{Z})$
$\mathbb{R} \times (\mathbb{R}/m\mathbb{Z})$	$\mathbb{R}/\mathbb{Z}$	$(u,[t]_m) \mapsto [t]_1$	$\coprod_{j=0}^{m-1} \mathbb{R}, \quad m \geq 1.$

(C.44.1)

In the last row the actual induced image $\Phi_{z*}(\pi_1 O)$ is $m\mathbb{Z} \subset \pi_1 B = \mathbb{Z}$; in all other rows that image is zero. Every downstairs fiber component is a properly embedded line or a circle, and the source covering restricts onto each such component as the line covering specified by these formulas.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; exact quotient and covering constraints in the remaining frontier-extension problem.

Proof. Let Γ be the deck group of $p : \widetilde{O} \to O$. Use C.43 to identify the covering plane with $\mathbb{R}_u \times \mathbb{R}_t$ so that $F(u,t) = t$. The equation $qF = \Phi_z p$ implies, for each $\gamma \in \Gamma$,

$$F\gamma = F + \eta(\gamma), \tag{C.44.2}$$

where $\eta(\gamma) = 0$ for a line target and $\eta(\gamma) \in \mathbb{Z}$ for a circle target. For the circle, the difference is a continuous integer-valued function on the connected covering plane and is therefore constant. Composition shows that $\eta : \Gamma \to \mathbb{Z}$ is a homomorphism. Its value is exactly the winding number of the image under Φ_z of the source loop whose lift ends at γ times its initial point. Thus its image is the actual induced fundamental-group image.

Write $K = \ker \eta$. Every deck map preserves the orientation pulled back from the planar surface O. It has the form

$$\gamma(u,t) = (h_{\gamma,t}(u),\, t + \eta(\gamma)), \tag{C.44.3}$$

and each $h_{\gamma,t}$ is increasing: the base coordinate is increasing, and reversing the fiber direction would reverse the surface orientation. The functions in (C.44.3) and their fiberwise inverses are jointly continuous, by the deck map and its inverse.

The exact kernel on a real fiber. The action of K on any fiber $\mathbb{R} \times \{t\}$ is free and proper. Freeness follows from deck freeness. For properness, compact subsets of this fiber are compact in the covering plane, and the deck action has finite transporters of compact pairs. The restriction is faithful, since a deck transformation fixing one point is the identity. An increasing free proper group on a line is trivial or infinite cyclic. Here is the direct argument. Its orbit of a point is closed and discrete. In the nontrivial case choose the element taking that point to its nearest orbit point on the right. Its iterates tend to the two ends, because a finite limiting point would be fixed. Every other orbit point is one of these iterates: an intervening point could be translated back between the original point and its nearest right neighbor. Freeness then identifies every element with the corresponding power. Consequently

$$K = 1 \quad \text{or} \quad K \cong \mathbb{Z}. \tag{C.44.4}$$

Suppose $K = \langle k \rangle \cong \mathbb{Z}$, choosing the generator to move one fiber point to the right. Since the displacement $h_{k,t}(u) - u$ is continuous and nowhere zero on the connected plane, it is positive everywhere. Put $k_t = h_{k,t}$. For $v = n + r$, where $n \in \mathbb{Z}$ and $0 \le r \le 1$, define

$$H(v,t) = \big(k_t^n(r\, k_t(0)),\, t\big). \tag{C.44.5}$$

At integral v the two formulas agree. The intervals with endpoints $k_t^n(0)$ exhaust the line in both directions, so H is a continuous bijection of the plane. Invariance of domain makes it a homeomorphism. It preserves F and satisfies

$$H(v+1,t) = kH(v,t). \tag{C.44.6}$$

Thus the actual intermediate quotient is

$$\widetilde{O}/K \cong (\mathbb{R}/\mathbb{Z}) \times \mathbb{R}, \qquad F_K([v],t) = t. \tag{C.44.7}$$

The compactness obstruction retaining the quotient map. The case $K \cong \mathbb{Z}$ cannot coexist with a nonzero image of η. Indeed that image would be $m\mathbb{Z}$ for some $m \ge 1$. The quotient Γ/K is then infinite cyclic, and its chosen generator acts on (C.44.7) by a circle homeomorphism in the first coordinate and by $t \mapsto t + m$ in the second. The compact strip $(\mathbb{R}/\mathbb{Z}) \times [0,m]$ maps onto the full quotient by Γ/K: each point can be translated into that strip using its second coordinate. It follows that

$$O = \widetilde{O}/\Gamma = (\widetilde{O}/K)/(\Gamma/K)$$

is compact, contrary to its being a nonempty open subset of the plane. Therefore $K \cong \mathbb{Z}$ forces $\eta = 0$ and $\Gamma = K$. Equations (C.44.5)–(C.44.7) give the third and fourth rows of (C.44.1), according to the target.

If $K = 1$ and $\eta = 0$, then $\Gamma = 1$, so C.43 gives the first two rows.

The remaining nonzero covering degree. It remains that $K = 1$ and $\eta(\Gamma) = m\mathbb{Z}$, with $m \ge 1$. Then Γ itself is infinite cyclic. Choose its generator γ with $\eta(\gamma) = m$, and write

$$\gamma(u,t) = (h_t(u), t + m). \tag{C.44.8}$$

We give a coordinate change preserving the real target coordinate and removing h_t. For $0 \le t \le m$ set

$$\rho_t(u) = \left(1 - \frac{t}{m}\right) u + \frac{t}{m} h_0(u). \tag{C.44.9}$$

Each ρ_t is an increasing homeomorphism of the line, with $\rho_0 = \mathrm{id}$ and $\rho_m = h_0$. Extend to all real t in both directions by the exact recursion

$$\rho_{t+m} = h_t \circ \rho_t. \tag{C.44.10}$$

The endpoint identity makes the extension jointly continuous at every multiple of m. The map $H_1(u,t) = (\rho_t(u), t)$ is a continuous bijection of the plane and hence a homeomorphism. Equations (C.44.8)–(C.44.10) give

$$H_1^{-1} \gamma H_1(u,t) = (u, t+m). \tag{C.44.11}$$

The quotient is therefore $\mathbb{R} \times (\mathbb{R}/m\mathbb{Z})$, and the original map, still satisfying $qF = \Phi_z p$, is precisely $(u, [t]_m) \mapsto [t]_1$. This is the final row of (C.44.1).

All changes of source coordinates preserved F; the target was changed only by its initially specified coordinate. The whole-fiber descriptions now follow directly. For a circle target, levels $F = t_0 + j$ are indexed by $j \in \mathbb{Z}$. The deck group identifies their labels by $\eta(\Gamma)$ and acts within each level by K. Thus the component labels are $\mathbb{Z}/\eta(\Gamma)$ and each component is $\mathbb{R}/K$ in the explicit coordinates above. For a line target there is only one real level and the component is again $\mathbb{R}/K$. These formulas also prove surjectivity of the restricted source covering onto each downstairs component and give its deck group. In particular, they establish the component assertion without assuming component-surjectivity for an arbitrary restricted covering. □

Scope of the reduction. C.44 classifies each of the actual quotient maps separately, including its whole fibers and covering labels. Several boundary points can impose several such maps on the same open orbit. No common product coordinates or homotopy preserving all those maps have been constructed here. That simultaneous compatibility and frontier extension remain part of C.12.

C.45. [Proved] Two actual boundary quotients give explicit planar coordinates. Let $\Phi_1 : O \to B_1$ and $\Phi_2 : O \to B_2$ be two actual nontrivial boundary quotients of the same open orbit as in C.42. Use the same source universal covering $p : X \to O$ and the real target coverings $q_i : \mathbb{R} \to B_i$, and choose lifts $q_i F_i = \Phi_i p$. Exactly one of the following alternatives holds. Either there is a homeomorphism $f : \mathbb{R} \to \mathbb{R}$ with

$$F_2 = f F_1, \tag{C.45.1}$$

or the pair (F_1, F_2) is a homeomorphism onto an open domain $\Omega \subset \mathbb{R}^2$ whose horizontal and vertical sections are intervals and whose two projections are onto $\mathbb{R}$. In the second alternative, explicit independent changes of the two real coordinates, and a possible exchange of them, carry Ω onto one of

$$\mathbb{R}^2, \qquad \{(x,y) : x < y\}, \qquad \{(x,y) : x < y < x+1\}. \tag{C.45.2}$$

The full increasing coordinate-pair groups preserving these three domains are, respectively,

$$\mathrm{Homeo}^+(\mathbb{R})^2, \qquad \{(g,g) : g \in \mathrm{Homeo}^+(\mathbb{R})\}, \qquad \{(g,g) : g \in \mathrm{Homeo}^+(\mathbb{R}),\ g(x+1) = g(x)+1\}. \tag{C.45.3}$$

The actual compatible lifted action maps into the corresponding group. These statements retain both original covering equations and make no identification of their downstairs component labels.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; simultaneous actual boundary-quotient constraints in the remaining extension problem.

Proof. A common compatible group. Use the closed subgroup

$$\mathcal{P} = \{(l, a, b_1, b_2) \in L \times \mathrm{Homeo}(X) \times \mathrm{Homeo}^+(\mathbb{R})^2 :$$
$$pa = (l|_O)p, \tag{C.45.4}$$
$$q_i b_i = (l|_{B_i})q_i, \qquad F_i a = b_i F_i \quad (i = 1, 2)\}.$$

The equations are closed in the product of the indicated Polish groups. For every lift a of $l|_O$, each target lift b_i exists uniquely after its deck translation is chosen at one point, exactly as in C.42. Consequently $\mathcal{P}$ acts transitively on the source covering plane. Both F_i are open onto $\mathbb{R}$ with connected proper-line fibers, by C.43.

Fix $t \in \mathbb{R}$ and set $C = F_1^{-1}(t)$. The closed subgroup of $\mathcal{P}$ with $b_1(t) = t$ acts transitively on this whole line. Its action on the interval $I = F_2(C)$ is also transitive, by the equality $F_2 a = b_2 F_2$. All these target maps are increasing. A nondegenerate interval admitting such a transitive action has no included endpoint, so it is an open interval. The Polish open orbit-map theorem, applied to the actions on C and I, makes $F_2|_C : C \to I$ the open map between their homogeneous-space quotients: the point stabilizer in C is contained in the stabilizer of its image in I. It is therefore a strictly monotone homeomorphism.

If instead I is a point, transitivity on the source plane transports this constancy to every F_1-fiber. There is then a function f satisfying (C.45.1). Since F_1 is an open quotient, f is continuous. For every open $W \subset \mathbb{R}$,

$$f(W) = F_2(F_1^{-1}(W))$$

is open. Thus f is open and onto, hence a homeomorphism of the line. This is the first alternative.

In the nonconstant case, (F_1, F_2) is injective, since F_2 is injective on each F_1-fiber. Invariance of domain makes it a homeomorphism onto an open domain Ω. Each vertical section is the open interval I just proved. Interchanging the two indices gives the horizontal statement; the constant alternative in that direction would contradict the already proved two-dimensional open image. Surjectivity of the two F_i gives both full coordinate projections.

The endpoint maps are actual homogeneous quotients. The coordinate action (b_1, b_2) of $\mathcal{P}$ preserves Ω and is transitive on it. Write its vertical intervals as

$$\Omega_x = (\alpha(x), \beta(x)), \quad -\infty \le \alpha(x) < \beta(x) \le +\infty. \tag{C.45.5}$$

The two endpoints satisfy the exact equations

$$\alpha(b_1 x) = b_2(\alpha(x)), \qquad \beta(b_1 x) = b_2(\beta(x)), \tag{C.45.6}$$

where increasing homeomorphisms fix the appropriate infinite endpoints. Transitivity of the first coordinate action implies that α is either everywhere finite or identically $-\infty$, and that β is either everywhere finite or identically $+\infty$.

Each finite endpoint map is a homeomorphism of the real line. For example, choose x_0 with $\alpha(x_0) \in \mathbb{R}$. The formula

$$\alpha(b_1 x_0) = b_2(\alpha(x_0))$$

descends the continuous second evaluation through the open first orbit map $\mathcal{P} \to \mathbb{R}$. It proves continuity of α in its actual topology. Its image is nonempty and invariant under the transitive second coordinate action, hence is all $\mathbb{R}$. The stabilizer of x_0 in the first action fixes $\alpha(x_0)$ in the second by (C.45.6). The two open orbit maps identify α with the associated open coset-space map. It follows that α is continuous, open, and onto, and therefore strictly monotone and a homeomorphism. The same proof applies to finite β. This verifies the endpoint topology without assuming regularity of a boundary graph.

The three domains and the coordinate changes. If both endpoints are infinite, $\Omega = \mathbb{R}^2$. If precisely one endpoint is finite, use its inverse as the second coordinate. The defining inequality becomes $x < y$ or $y < x$, according to which endpoint and monotonicity occur. In the latter case exchange the two coordinates.

Suppose both endpoints are finite. They have the same monotonicity: an increasing homeomorphism and a decreasing homeomorphism cannot satisfy $\alpha(x) < \beta(x)$ for all real x, by their limits at the two ends. When both increase, replace y by $\alpha^{-1}(y)$ and put $\tau = \alpha^{-1}\beta$. When both decrease, replace y by $\beta^{-1}(y)$ and put $\tau = \beta^{-1}\alpha$. In either case the domain is now exactly

$$x < y < \tau(x), \qquad \tau \in \mathrm{Homeo}^{+}(\mathbb{R}),\ \tau(x) > x. \tag{C.45.7}$$

Choose x_0 and any increasing homeomorphism $h : [x_0, \tau(x_0)] \to [0, 1]$. Extend it by $h(\tau^n x) = h(x) + n$. The iterates of τ exhaust the two ends, since τ has no fixed point, and the endpoint formulas agree. Thus h is an increasing homeomorphism of $\mathbb{R}$ with $h\tau h^{-1}(x) = x+1$. Applying h to both coordinates in (C.45.7) gives the final domain in (C.45.2). These are the asserted explicit source coordinate maps; a reversal in one coordinate still conjugates increasing coordinate actions to increasing actions.

Finally, an increasing coordinate pair preserving $x < y$ must preserve its boundary graph $y = x$, which says $g = h$. For $x < y < x + 1$, its lower and upper endpoint graphs are separately preserved, which says $h(x) = g(x)$ and $h(x + 1) = g(x) + 1$. Conversely these identities preserve the indicated domains exactly. The unrestricted plane case has no additional equation. This proves (C.45.3).

All induced actions used above arise from (C.45.4). An element acting trivially on the covering plane may still act elsewhere under its l-coordinate; no faithfulness of that global coordinate has been imposed. The original equations $q_i F_i = \Phi_i p$ remain in force, so the separate deck shifts and component labels of C.44 are retained. □

Scope of the reduction. C.45 gives the relation between two specified actual boundary maps, including all coordinate changes. A third quotient can impose an additional equation on the group in (C.45.3). Preserving all such equations and obtaining a global frontier-compatible contraction are not consequences of this pairwise classification.

C.46. [Proved] Two boundary foliations admit a compatible intrinsic contraction. Use the actual based null-homotopic plane loop f_t, its generated group G, the open orbit O, and L, x, J of C.42. For every $z \in \mathrm{Fix}(J) \cap \partial O$ with nontrivial intrinsic line or circle orbit, lift its quotient Φ_z to the common source universal covering $p : X \to O$ as in C.42. Let $\mathscr{F}$ be the set of fiber partitions of these real lifts; changing their target coordinates or their lift constants does not change this set. For $|\mathscr{F}| = 2$, the loop $f_t|_O$ has a based compact-open null-homotopy

$$K_{s,t} \in \mathrm{Homeo}(O), \qquad K_{0,t} = f_t|_O, \quad K_{1,t} = K_{s,0} = K_{s,1} = \mathrm{id}\,. \tag{C.46.1}$$

It commutes with every plane homeomorphism commuting with all f_t and preserving O. For every indicated point z, it also satisfies the exact quotient relation

$$\Phi_z K_{s,t} = Q_{s,t}^{B_z} \Phi_z, \qquad B_z = Gz, \tag{C.46.2}$$

where Q^{B_z} is the natural median contraction of the original restricted loop on that intrinsic line or circle, as in C.27–C.29. Thus all the specified boundary quotient relations hold simultaneously. The asserted homotopy is intrinsic to O; continuity at its ambient frontier is a further assertion.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison; simultaneous boundary relations in C12.

Proof. Choose representatives $F_1, F_2 : X \to \mathbb{R}$ of the two partitions. C.45 supplies a homeomorphism from X onto one of its three domains D, with these two coordinates changed only by independent line homeomorphisms and a possible exchange. Lift the isotopy $f_t|_O$ from the identity to a_t on X. For each original boundary target, lift $f_t|_{B_i}$ to its based real path $b_{i,t}$. Covering uniqueness gives

$$F_i a_t = b_{i,t} F_i. \tag{C.46.3}$$

Each $b_{i,t}$ is increasing. For a line target its endpoint is the identity. For a circle target C.29 identifies its endpoint translation with the winding of the original ambient-null plane loop and proves that translation zero. Hence both $b_{i,1}$ are the identity. Since the pair (F_1, F_2) is injective, $a_1 = \mathrm{id}$. We have obtained an actual based loop in the corresponding coordinate-pair group (C.45.3), without discarding a source or target deck shift.

In these coordinates write $a_t = (g_{1,t}, g_{2,t})$. Apply C.27 to each coordinate loop and set

$$\begin{aligned} m_{i,r}(u) &= \min_{0 \le v \le r} g_{i,v}(u), \qquad M_{i,r}(u) = \max_{0 \le v \le r} g_{i,v}(u), \\ R_{i,s,t}(u) &= \mathrm{med}\big(m_{i,1-s}(u), g_{i,t}(u), M_{i,1-s}(u)\big), \\ \widetilde{K}_{s,t}(u,v) &= (R_{1,s,t}(u), R_{2,s,t}(v)). \end{aligned} \tag{C.46.4}$$

The coordinate maps and their inverses are jointly continuous by C.27 and C.31. They preserve the actual domain D. For $D = \mathbb{R}^2$ this follows directly from their being line homeomorphisms. For $D = \{u < v\}$, C.45 gives $g_{1,t} = g_{2,t}$, so the two new coordinate maps are identical and preserve the strict inequality. For $D = \{u < v < u + 1\}$ the same equality holds and each $g_{i,t}$ commutes with translation by one. Minima, maxima and the median preserve that equation, so the new equal coordinate maps preserve both inequalities. The inverse family has the same property. Formula (C.46.4) is therefore a based null-homotopy in $\mathrm{Homeo}(X)$ with the asserted endpoint identities.

We establish the operator comparison before descending. Let h commute with all f_t and preserve O. It commutes with L by continuity of composition and inversion. The stabilizers of x and hx in L are equal: an element fixes one precisely when it fixes the other, by commutation with h and h^{-1}. Choose $a_h \in L$ with $a_h x = hx$. The exact stabilizer identity gives $a_h J a_h^{-1} = J$. For every specified z and every $l \in L$, evaluation gives

$$\Phi_z h(lx) = \Phi_z(l a_h x) = l a_h z = \Phi_{a_h z}(lx). \tag{C.46.5}$$

The point $a_h z$ lies in $\mathrm{Fix}(J) \cap \partial O$: normalizing J proves the first property, and a_h preserves O and its boundary. Its orbit is the same $Lz = Gz$ as that of z, so it has the same intrinsic type. Thus pullback by h permutes the full set $\mathscr{F}$. The inverse h^{-1} gives the inverse permutation. After lifting to X, the two sides of (C.46.5) differ at most by the specified target deck translation, which leaves their fiber partitions identical.

Every lift $\widetilde{h}$ of $h|_O$ consequently permutes the two coordinate foliations of D. It is a coordinate pair of line homeomorphisms, possibly followed by exchange of the coordinates. For clarity, preservation of a fiber partition defines its map on the corresponding real quotient; the open quotient projection proves continuity of that map and of its inverse. Surjectivity of both coordinate projections then makes it a line homeomorphism. It may be increasing or decreasing. These quotient maps give exactly the stated coordinate form of $\widetilde{h}$.

Moreover $\widetilde{h}$ commutes with a_t. Their commutator projects to the identity on O and is a continuous deck-valued function of t, equal to the identity at $t = 0$. The discrete deck group makes it the identity for every t. C.27's naturality under every line homeomorphism, together with the identical construction in both coordinates, now gives

$$\widetilde{h} \widetilde{K}_{s,t} \widetilde{h}^{-1} = \widetilde{K}_{s,t}. \tag{C.46.6}$$

In particular this holds for every source deck transformation, which is a lift of $h = \mathrm{id}$. Consequently the homotopy descends to $K_{s,t}$ on O. The evaluation maps and their inverse maps descend continuously through the open covering quotient. This proves compact-open continuity, the endpoint identities, and the asserted commutation with the original operators.

Finally, take any other specified z. Its lifted fiber partition equals one of the two chosen ones. The first alternative of C.45 identifies its real map as a line homeomorphism composed with that coordinate. Naturality of the median formula under this exact reparametrization proves

$$F_z \widetilde{K}_{s,t} = \widetilde{Q}^{B_z^z}_{s,t} F_z.$$

For a circle target the formula commutes with its integer deck translations, so it descends; for a line target it already is the stated target map. Using $q_z F_z = \Phi_z p$ proves (C.46.2). No ambient limit of points approaching ∂O has been inferred from these quotient equations. □

C.47. [Proved] Three independent boundary quotients force a Lie action. Let $\Phi_i : O \to B_i$, $1 \le i \le 3$, be three actual boundary quotients of the same open orbit as in C.42. Use their common source universal covering and real target coverings, with

$$p : X \longrightarrow O, \qquad q_i F_i = \Phi_i p, \qquad F_i : X \longrightarrow \mathbb{R}. \tag{C.47.1}$$

Suppose the three lifted fiber partitions are pairwise distinct; equivalently, none of the pairs satisfies the first alternative of C.45. The full group

$$\mathcal{W} = \{a \in \mathrm{Homeo}(X) : F_i a = b_i F_i \text{ for some } b_i \in \mathrm{Homeo}^+(\mathbb{R}),\ 1 \le i \le 3\} \tag{C.47.2}$$

is a closed, locally compact subgroup in the actual compact-open topology. It is a Lie group of dimension two or three and acts transitively on X. If any pair (F_i, F_j) has image different from $\mathbb{R}^2$, this action is simply transitive and has dimension two. In particular, the closure in $\mathrm{Homeo}(X)$ of the compatible lifted loop group is locally compact. All three covering equations in (C.47.1) and the kernel of the global-to-restricted action are retained. The conclusion concerns the action on the source covering plane; it does not assert an extension of an intrinsic homotopy across the frontier of O.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; simultaneous actual boundary-quotient constraints in the remaining frontier extension problem.

Proof. The actual group topology. Include all three target lifts in the single closed Polish group of tuples (l, a, b_1, b_2, b_3) satisfying

$$pa = (l|_O)p, \qquad q_i b_i = (l|_{B_i}) q_i, \qquad F_i a = b_i F_i \quad (1 \le i \le 3). \tag{C.47.3}$$

For each source lift, the three compatible target lifts are obtained exactly as in C.42–C.45. This group acts transitively on X, so its image in $\mathcal{W}$ does also. Each b_i in (C.47.2) is unique because F_i is onto. C.43 supplies a continuous section $\sigma_i : \mathbb{R} \to X$ of each F_i. Consequently

$$b_i = F_i a \sigma_i, \qquad b_i^{-1} = F_i a^{-1} \sigma_i. \tag{C.47.4}$$

These formulas are continuous for uniform convergence on compact sets, for maps and inverses.

They also prove closedness. If $a_n \in \mathcal{W}$ converges to a homeomorphism a together with its inverse, (C.47.4) gives locally uniform limits b_i and d_i of the target maps and their inverses. The identities $b_{i,n} d_{i,n} = d_{i,n} b_{i,n} = \mathrm{id}$ pass to the limit, since the images of each compact set remain in a common compact set. Thus $d_i = b_i^{-1}$; each b_i is an increasing homeomorphism, and $F_i a = b_i F_i$. Hence $\mathcal{W}$ is a closed Polish subgroup of $\mathrm{Homeo}(X)$.

C.45 identifies X by any two of the F_i with an open domain having interval coordinate sections. In coordinates $(x, y) = (F_1, F_2)$, write $F_3 = w(x, y)$. Every restriction of w to a horizontal or vertical section is strictly monotone. The signs are constant over the connected domain: for two fixed nearby values in one coordinate, the continuous difference of their w-values cannot change sign. The coordinate representation of $\mathcal{W}$ is therefore exactly the triples (g_1, g_2, k) satisfying

$$(g_1, g_2)\Omega = \Omega, \qquad w(g_1 x, g_2 y) = k(w(x, y)). \tag{C.47.5}$$

The map from $\mathcal{W}$ to these triples and its inverse are continuous, by (C.47.4) and the pair-coordinate homeomorphism. All topological group assertions below use this topology.

A strictly interior mean makes an ordered-pair stabilizer trivial. We first prove an elementary fact used in the non-plane cases. Let

$$D = \{(a, b) : a < b\} \quad \text{or} \quad D = \{(a, b) : a < b < a + 1\}. \tag{C.47.6}$$

Suppose a group of increasing line homeomorphisms preserves D diagonally and preserves a continuous function $M : D \to \mathbb{R}$ with

$$a < M(a, b) < b, \qquad M \text{ injective in each argument separately.} \tag{C.47.7}$$

A group element fixing one pair $(a, b) \in D$ is the identity. Indeed, it fixes all the points obtained by repeatedly inserting M between consecutive fixed endpoints. These subdivisions have mesh tending to zero. To verify this, fix $\varepsilon > 0$. On the compact set of pairs

$$a \le u < v \le b, \qquad v - u \ge \varepsilon,$$

both $M(u, v) - u$ and $v - M(u, v)$ have a positive minimum. Every such pair is still in D, including in the band case because $v \le b < a + 1 \le u + 1$. Every subdivision of an interval of length at least ε therefore shortens each child by a fixed positive amount. No sufficiently deep subdivision can have that length. Continuity now makes the homeomorphism fix $[a, b]$ pointwise.

Let I be the maximal interval of fixed points containing $[a, b]$. A finite right endpoint c belongs to the fixed set. Choose $x < c$ in the interior of I with $(x, c) \in D$. For $y > c$ sufficiently close to c, the pair (x, y) remains in D and $M(x, y)$ remains in the interior of I. Equivariance gives

$$M(x, gy) = M(gx, gy) = gM(x, y) = M(x, y).$$

Injectivity in the second argument forces $gy = y$, extending I past c. The analogous argument, with a fixed $y > c$ and $x < c$ close to a finite left endpoint, uses injectivity in the first argument. Both finite endpoints are impossible. Thus $I = \mathbb{R}$ and $g = \mathrm{id}$.

A non-plane pair and two full remaining pairs. Suppose some pair image is not $\mathbb{R}^2$. Relabel and apply the explicit coordinate changes of C.45 so that $\Omega = D$ is one of (C.47.6). The first two target actions then coincide: $g_1 = g_2 = g$; in the band case $g(x + 1) = g(x) + 1$. First suppose the images of (F_1, F_3) and (F_2, F_3) are both the full plane. Each w-level is then the graph

$$y = r_z(x), \qquad z \in \mathbb{R}, \tag{C.47.8}$$

where $(x, z) \mapsto r_z(x)$ is continuous and every r_z is a homeomorphism of $\mathbb{R}$ onto $\mathbb{R}$. It is increasing, since $r_z(x) > x$ for all x, which a decreasing real-line homeomorphism cannot satisfy. Reverse the z-coordinate when necessary. Then $z \mapsto r_z(x)$ is strictly increasing, and its range is $(x, +\infty)$ in the half-plane case and $(x, x + 1)$ in the band case. The sign is the same for every x by connectedness. The action equation is exactly

$$g r_z g^{-1} = r_{kz}. \tag{C.47.9}$$

For $(a, b) \in D$, there is a unique $s = s(a, b)$ with

$$r_s^2(a) = b. \quad \text{Define} \quad M(a, b) = r_s(a). \tag{C.47.10}$$

Here are the endpoint and continuity checks. The left side is continuous and strictly increasing in s. As $s \to -\infty$, $r_s(x) \to x$, and hence $r_s^2(a) \to a$. This last limit follows by bounding $r_s(a)$ between a and $a + \delta$ and then using monotonicity of r_s at these two fixed points. As $s \to +\infty$ the square tends to $+\infty$ in the half-plane case. In the band case $r_s(x) \to x + 1$, and the same fixed-point bracketing gives $r_s^2(a) \to a+2$. The required value b lies strictly between the endpoint limits. Strict monotonicity gives uniqueness, and two nearby s-values bracketing a solution continue to bracket it when (a, b) varies slightly; this proves continuity.

The value in (C.47.10) lies strictly between a and b. It is strictly increasing in b for fixed a. For fixed b and $a_1 < a_2$, the corresponding parameters satisfy $s_1 > s_2$, because r_s^2 increases in its argument. If $r_{s_1}(a_1) \geq r_{s_2}(a_2)$, applying these increasing maps once more and using $s_1 > s_2$ gives

$$b = r_{s_1}(r_{s_1}(a_1)) > r_{s_2}(r_{s_2}(a_2)) = b,$$

a contradiction. Thus M is strictly increasing in its first argument as well. Equation (C.47.9) and uniqueness in (C.47.10) prove $M(ga, gb) = gM(a, b)$. The preceding mean argument therefore makes every source point stabilizer trivial: once $g = \mathrm{id}$, surjectivity of w in (C.47.5) also gives $k = \mathrm{id}$.

A further non-plane pair identifies all three line actions. It remains, with $\Omega = D$, to consider the case where at least one of the other pair images is not the full plane. One of its finite endpoint homeomorphisms from C.45 gives an exact equation

$$fg = kf. \tag{C.47.11}$$

Replace the third coordinate by $f^{-1}F_3$. Now all three target actions are the same g; this remains true when f is decreasing. Denote the new third coordinate again by $w(x, y)$. An equality $w = x$ or $w = y$ at one point would hold everywhere by transitivity and contradict independence. The relative order of the three values is therefore constant.

In the half-plane, the projected group of g's acts transitively on all ordered pairs. Its centralizer in $\mathrm{Homeo}(\mathbb{R})$ is trivial: the stabilizer of a point x acts transitively on each of $(-\infty, x)$ and $(x, +\infty)$, so its only common fixed point is x. A commuting map must send x to that same fixed point. It follows that every finite endpoint of either remaining pair domain, now an equivariant self-homeomorphism of the same line, is the identity.

When $x < w(x, y) < y$, take $M = w$. When $w(x, y) > y$, the (x, w) image is exactly $\{(x, z) : x < z\}$: its finite lower endpoint must be the identity, and a finite upper endpoint would be the same map and leave an empty interval. For each $a < b$, define $M(a, b) = c$ by

$$w(a, c) = b. \tag{C.47.12}$$

The restriction in c is increasing and maps $(a, +\infty)$ onto $(a, +\infty)$; its increase also follows from $w(a, c) > c$ and the limit as $c \to +\infty$. The unique solution satisfies $a < c < b$. When $w(x, y) < x$, the (y, w) image is exactly $\{(y, z) : z < y\}$; define instead

$$w(c, b) = a. \tag{C.47.13}$$

The restriction in c is increasing and maps $(-\infty, b)$ onto $(-\infty, b)$, so again $a < c < b$. Both inverse constructions are continuous by the pair-coordinate homeomorphisms. They are injective in each argument, by the strict injectivity of the corresponding restrictions of w. Equation (C.47.5), with all three actions equal, proves their equivariance.

In the band, g commutes with unit translation. The equalities $w = x + n$ and $w = y + n$, for any integer n, are excluded by the same transitivity and independence argument. There is consequently one integer n, constant on the connected domain, such that

$$x < w(x, y) - n < x + 1.$$

Either $w(x, y) - n < y$ everywhere, or the reverse inequality holds everywhere. In the first case take $M(x, y) = w(x, y) - n$. In the second case use the exact domain homeomorphism

$$(u, v) = (y, x + 1), \qquad M(u, v) = w(v - 1, u) - n. \tag{C.47.14}$$

It sends the band onto itself, and $u < M(u, v) < v$. Both constructions are continuous and injective in each argument. They are equivariant because every g commutes with the integer shifts. We have obtained a mean satisfying (C.47.7) in all cases.

The mean argument proves simple transitivity of $\mathcal{W}$ whenever a pair image is not the full plane. The Polish open orbit-map theorem makes evaluation $\mathcal{W} \to X$ a homeomorphism in this case. Thus $\mathcal{W}$ is locally compact and is a topological two-manifold. Its faithful product action on $X \times \mathbb{R}$ and the Hilbert–Smith theorem in dimension three [52, Theorem 1.5 and the introductory reduction] make it a two-dimensional Lie group.

Three full pair images give a global ordered loop. Suppose now that all three pair images are $\mathbb{R}^2$. The function $w : \mathbb{R}^2 \to \mathbb{R}$ is then continuous, and each partial map is a homeomorphism onto $\mathbb{R}$. Reverse input coordinates as necessary to make it increasing in both arguments. At a chosen source point (a, b) put $c = w(a, b)$ and use the explicit coordinates

$$u = w(x, b) - c, \qquad v = w(a, y) - c, \qquad z = w(x, y) - c. \tag{C.47.15}$$

They give a continuous multiplication $m : \mathbb{R}^2 \to \mathbb{R}$ with

$$m(u, 0) = u, \qquad m(0, v) = v, \tag{C.47.16}$$

strictly increasing in both variables, each partial map onto $\mathbb{R}$. Its two division maps are continuous: solving a strictly increasing continuous equation is continuous by fixed bracketing on either side of the solution.

Let S be the stabilizer of $(0, 0)$ in $\mathcal{W}$. Setting either argument equal to zero in its action equation shows that the three target maps coincide. Thus, in its actual subspace topology, S is exactly

$$\mathrm{Aut}^+(m) = \{f \in \mathrm{Homeo}^+(\mathbb{R}) : f(0) = 0, \quad f(m(x, y)) = m(fx, fy)\}. \tag{C.47.17}$$

This is a closed subgroup. We next prove its local compactness by uniform estimates for both maps and inverses.

Fix $a > 0$ and $0 < A < B$. Set

$$\mathcal{F}_{A,B} = \{f \in \mathrm{Aut}^+(m) : A \le f(a) \le B\}. \tag{C.47.18}$$

Define $D(x) = m(x, x)$ and $T_c(x) = m(c, x)$. The map D is an increasing homeomorphism, $D(x) > x$ for $x > 0$, and $D(x) < x$ for $x < 0$. The numbers

$$v_n = D^{-n}(a) > 0, \qquad u_n \text{ defined by } m(u_n, v_n) = 0 \tag{C.47.19}$$

therefore converge to zero, with $u_n < 0$. Every automorphism respects these formulas, and hence

$$D^{-n}(A) \le f(v_n) \le D^{-n}(B). \tag{C.47.20}$$

The two bounds tend to zero. If $\ell(t)$ is the unique solution of $m(\ell(t), t) = 0$, then $f(u_n) = \ell(f(v_n))$; continuity of ℓ makes this tend to zero uniformly over $\mathcal{F}_{A,B}$ as well.

For $c > 0$, T_c is increasing, onto, and satisfies $T_c(x) > x$. Its integer iterates of zero tend to both ends of $\mathbb{R}$. For every integer j,

$$f(T_a^j(0)) = T_{f(a)}^j(0). \tag{C.47.21}$$

To bound f on a given compact interval, enclose it between two such iterates with $j = -N, N$. The right sides of (C.47.21) range over a compact set as $f(a)$ ranges over $[A, B]$; the negative iterates depend continuously on c by the division-map continuity. Monotonicity then bounds all f uniformly on that interval.

For equicontinuity, let I be a compact interval. For fixed n, both gaps in

$$m(u_n, x) < x < m(v_n, x) \qquad (x \in I)$$

have positive minima on I. Thus a single $\delta_n > 0$ ensures

$$|y - x| < \delta_n,\ x \in I \quad \Longrightarrow \quad m(u_n, x) < y < m(v_n, x).$$

Applying f bounds $f(y)$ between $m(f(u_n), f(x))$ and $m(f(v_n), f(x))$. The already proved compact bound on $f(I)$, the uniform limits $f(u_n), f(v_n) \to 0$, and continuity of m show that both bounds approach $f(x)$ uniformly in x and f. This is equicontinuity on every compact interval.

The inverse family has the same estimates with new fixed positive bounds. Choose n so large that $D^{-n}(B) < a$. Then $f(v_n) < a$ and $f^{-1}(a) > v_n > 0$. Choose N with $T_A^N(0) > a$. Since $T_c^N(0)$ increases with $c > 0$, (C.47.21) gives $f(T_a^N(0)) \geq T_A^N(0) > a$, and hence

$$v_n < f^{-1}(a) < T_a^N(0). \tag{C.47.22}$$

Applying the same compact bounds and equicontinuity argument to these positive bounds controls all inverses uniformly.

Arzelà–Ascoli on a countable compact exhaustion now gives, from any sequence in $\mathcal{F}_{A,B}$, locally uniform limits f and h of a common subsequence of maps and inverses. Their compact image bounds permit passage to the composition identities, so $fh = hf = \mathrm{id}$. The multiplication equation and the bounds in (C.47.18) pass to the limit. Thus $\mathcal{F}_{A,B}$ is compact in the homeomorphism-group topology. Taking $A < a < B$ gives a compact neighborhood of the identity in S, and proves its local compactness.

Evaluation $f \mapsto f(a)$ is also injective. An automorphism fixing a fixes every v_n and therefore every $T_{v_n}^j(0)$, for $j \in \mathbb{Z}$. For fixed x, choose the first such iterate at or to the right of x. It lies in $[x, T_{v_n}(x)]$, whose length tends to zero since $v_n \to 0$. These fixed points tend to x, so the automorphism fixes every x. The compactness of (C.47.18) proves that this continuous evaluation is a proper embedding

$$S \hookrightarrow (0, \infty). \tag{C.47.23}$$

From the stabilizer to the full group, without changing topology. The locally compact group S acts faithfully and continuously on $\mathbb{R}^3$ by its action on the first coordinate. Pardon's theorem makes S a Lie group. Its embedding (C.47.23) gives $\dim S \leq 1$. The full group $\mathcal{W}$ is still the closed Polish group already defined; it has not yet been assumed locally compact.

Consider the free right action of S on $\mathcal{W}$. It is a Cartan action in its actual topology. Indeed, choose an identity neighborhood U in $\mathcal{W}$ such that $U \cap S$ has compact closure in S, using the proved local compactness and the subspace topology of S. Choose an identity neighborhood V with $V^{-1}V \subset U$. Then

$$\{s \in S : Vs \cap V \neq \varnothing\} \subseteq V^{-1}V \cap S$$

has compact closure in S, and translates provide the required neighborhoods at every point. The space $\mathcal{W}$ is metrizable and completely regular. The free-action form of Palais' slice theorem therefore gives local sections and local trivializations for

$$\mathcal{W} \longrightarrow \mathcal{W}/S \quad [51, \textit{Theorem } 2.3.3].$$

For clarity, the slice map has the form $\sigma(u)s \mapsto (u,s)$; its inverse is multiplication. If θ is the equivariant slice function to S, the inverse coordinates of w are $(w\theta(w)^{-1},\theta(w))$, as in the explicit free-action statement [40, Theorem 3.7]. The open orbit-map theorem identifies $\mathcal{W}/S$ homeomorphically with $X=\mathbb{R}^2$. These actual local products of the locally compact spaces S and an open subset of X prove that $\mathcal{W}$ is locally compact. Its faithful action on $X\times\mathbb{R}$ then makes it a Lie group by Pardon again. The same local products give $\dim \mathcal{W} = 2+\dim S$, which is two or three.

Finally, every compatible lifted loop value belongs to $\mathcal{W}$, and its closure in $\mathrm{Homeo}(X)$ is a closed subgroup of this locally compact group. It is therefore locally compact with precisely the induced homeomorphism-group topology. None of the arguments replaces the l-coordinate of (C.47.3) by its action on X: the kernel of that restriction can still act elsewhere. The coordinate changes act on the specified real target lifts and leave the original equations (C.47.1) available throughout. This proves the asserted intrinsic reduction and leaves the frontier extension step explicit. □

C.48. [Proved] One boundary foliation admits a compatible intrinsic contraction. Use the based ambient-null plane loop and the full family $\mathscr{F}$ of lifted boundary fiber partitions in C.46. For an open orbit O with $|\mathscr{F}|=1$, there is a based compact-open null-homotopy $K_{s,t}$ of $f_t|_O$ which commutes with its entire original operator stabilizer and satisfies, simultaneously for every specified boundary point,

$$\Phi_z K_{s,t} = Q^{B_z}_{s,t}\Phi_z. \tag{C.48.1}$$

Here Q^{B_z} is precisely the natural median homotopy of C.27–C.29. The original source and target covering maps are retained. This is an intrinsic assertion on O; ambient continuity at its frontier is an additional requirement.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and actual boundary quotient compatibility.

Proof. The actual lifted operator group. Let $p: X\to O$ be the universal covering and choose one real quotient $F: X\to\mathbb{R}$ representing the unique partition. By C.43, X is a plane, F has connected proper-line fibers, and a product chart identifies it with the second projection. Let Q be the closure, in $\mathrm{Homeo}(O)$, of the restrictions of the original O-stabilizing operators. C.30 proves that this exact closed group acts freely and properly. Define its group of all lifts by

$$\widetilde{Q} = \{a\in \mathrm{Homeo}(X) : pa = qp \text{ for some } q\in Q\}. \tag{C.48.2}$$

It includes the source deck group. Its topology is the subspace homeomorphism-group topology, with the maps and their inverses controlled on compact subsets.

We give the needed covering comparisons explicitly. A lift is determined by its value at one point once its downstairs map is specified. For a convergent sequence of downstairs maps and a convergent compatible choice of those values, the lifts converge on compact subsets: cover a compact path from that point by finitely many covering charts and continue the chosen sheet successively; the same argument on finitely many such neighborhoods controls a compact set. Apply it to the inverse maps as well. Conversely a convergent sequence of lifts gives convergence downstairs on each compact set by finitely many local sections of p. These observations prove that (C.48.2) is closed and that its projection to Q has the stated topology of lifts.

The action of $\widetilde{Q}$ on X is free. A lift fixing a point has downstairs map fixing a point, hence equal to the identity; the resulting deck map fixing a point is the identity too. It is proper in this topology. Indeed, from $x_n\to x$ and $a_nx_n\to y$, properness downstairs gives a subsequence $q_n\to q$. There is a unique lift a of q with $ax=y$, and the covering-chart argument gives $a_n\to a$. The same argument makes evaluation at a point a homeomorphism onto a closed subset of X:

convergence of its values supplies convergence of the corresponding downstairs maps and then of the selected lifts. Thus $\widetilde{Q}$ is locally compact. Its faithful product action on $X \times \mathbb{R}$ makes it a Lie group by the already checked Hilbert–Smith theorem of C.33. Its dimension is at most two, since each orbit is properly embedded in X.

The original operators preserve the unique partition by (C.46.5). This property passes to their restrictions' closure and to all lifts. More explicitly, choose lifts of an approximating sequence whose values at one point converge to the chosen lift. Equality of F-values on a fiber passes to the limit, and the inverse maps give equality of the resulting fiber partitions. Since F is an open quotient, every $a \in \widetilde{Q}$ therefore has a unique induced line homeomorphism $\bar{a}$ with

$$Fa = \bar{a}F. \tag{C.48.3}$$

A continuous section of F gives $\bar{a} = Fa\sigma$, and the inverse formula uses a^{-1}; hence this induced homomorphism is continuous in the actual topologies.

Lift the original restricted isotopy from the identity to a_t on X. C.32 proves its null evaluation loop in O, so $a_1 = \mathrm{id}$. Every element of (C.48.2) commutes with a_t: its commutator is a continuous deck-valued function of t, initially the identity. Write

$$Fa_t = b_t F. \tag{C.48.4}$$

These are the based target lifts, and $b_0 = b_1 = \mathrm{id}$ by C.29 for a circle target and directly for a line target. The group generated by the paths b_t acts transitively on $\mathbb{R}$. Its orbits are intervals, because every word is joined to the identity by a path of words. A finite endpoint of a proper orbit interval would be fixed by every word. Such a common fixed point would project to a fixed point of the nontrivial transitive boundary orbit, which is impossible. Thus the orbit interval is all of $\mathbb{R}$.

The base operator image and its kernel. Take the closed transitive group generated on this real line. The evaluation proof of C.30 applies to the line without change and shows that its full centralizer is locally compact, free and proper. Its product action on $\mathbb{R}^3$ makes it Lie. No decreasing element is possible: it would have a fixed point, contrary to freeness. In dimension one its identity component acts transitively on the line, and freeness excludes additional components; it is the real translation group. In dimension zero the nearest-orbit-point argument of C.44 gives either the identity group or a cyclic translation group. Choose the resulting line coordinate. We have thereby proved, with the comparison (C.48.3), that every base operator is a translation. Write

$$Fa = F + \alpha(a), \qquad \alpha : \widetilde{Q} \longrightarrow (\mathbb{R}, +), \qquad K = \ker \alpha, \tag{C.48.5}$$

after this specified change of target coordinate.

Restriction of K to any fiber is faithful, free and proper. Evaluation gives it the same topology as its closed orbit in that line. The preceding one-dimensional classification therefore gives exactly

$$K = 1, \quad K \cong \mathbb{Z}, \quad \text{or} \quad K \cong \mathbb{R}. \tag{C.48.6}$$

All coordinate changes below are homeomorphisms of the source and reparametrizations of this same target. We construct them and keep the action of every component.

The real kernel. For $K \cong \mathbb{R}$, its orbits are the entire fibers. A global section of its principal bundle over the line, constructed as in C.33, gives coordinates (u, r) with $F(u, r) = r$ and

$$k_c(u, r) = (u + c, r). \tag{C.48.7}$$

The quotient $\widetilde{Q}/K$ acts freely and properly on $X/K = \mathbb{R}$, by the exact local-section argument in C.33. Its image under α is consequently closed, and is 0, $d\mathbb{Z}$ with $d > 0$, or $\mathbb{R}$. For image zero (C.48.7) is the complete operator action. For image $d\mathbb{Z}$, choose a label-d element. Its action is

$$q(u, r) = (\lambda u + \tau(r), r + d), \lambda \in \mathbb{R}^{\times}. \tag{C.48.8}$$

The scalar is constant because conjugation on the actual group K is a single continuous group automorphism. Choose c on $[0,d]$ with $c(d)=\lambda c(0)+\tau(0)$ and extend it by $c(r+d)=\lambda c(r)+\tau(r)$. In the coordinates $u'=u-c(r)$ the complete additional operator is

$$q(u',r)=(\lambda u',r+d). \tag{C.48.9}$$

This retains negative λ as well.

For image $\mathbb{R}$, the group $\widetilde{Q}$ acts transitively and freely on X. Both its kernel and quotient are connected lines, so it is connected. A Lie-algebra element mapping to 1 integrates to a section s_r with $\alpha(s_r)=r$. Conjugation on K has the form $s_r k_u s_r^{-1}=k_{e^{cr}u}$ for a real constant c. Evaluation gives the exact group coordinates and product

$$X\cong\widetilde{Q}\cong\mathbb{R}^2,\qquad (u,r)(v,z)=(u+e^{cr}v,r+z),\qquad F(u,r)=r. \tag{C.48.10}$$

The formula proves directly that these coordinates are global; $k_u s_r$ is the unique element with the specified base and fiber values. The operator action is left multiplication, and its full centralizer consists of right multiplications.

The cyclic kernel. For $K\cong\mathbb{Z}$, the construction (C.44.5) supplies coordinates with $F(u,r)=r$ and generator $k(u,r)=(u+1,r)$. The quotient is the actual cylinder $X/K=(\mathbb{R}/\mathbb{Z})\times\mathbb{R}$. The action of $\widetilde{Q}/K$ on it is free and proper: lift convergent point sequences by local sections and correct by elements of K, then apply properness upstairs, exactly as in C.33. The compact circle fibers show that its base image is closed. Indeed, a sequence of base translations in a compact interval sends one fixed point into a compact cylinder strip; properness gives a convergent subsequence of the quotient elements. Thus again the image is 0, $d\mathbb{Z}$, or $\mathbb{R}$.

For image $d\mathbb{Z}$, the additional generator has the form

$$q(u,r)=(g_r(u),r+d),\qquad g_r(u+1)=g_r(u)+\epsilon,\quad \epsilon\in\{1,-1\}. \tag{C.48.11}$$

Here g_r is increasing for $\epsilon=1$ and decreasing for $\epsilon=-1$. Choose an increasing unit-translation equivariant path ρ_r on $0\le r\le d$ from $\rho_0=\mathrm{id}$ to $\rho_d=g_0\circ(\epsilon\,\mathrm{id})$. The convex interpolation of these two increasing maps is such a path. Extend it in both directions by

$$\rho_{r+d}=g_r\rho_r(\epsilon\,\mathrm{id}). \tag{C.48.12}$$

The endpoint equation makes this extension continuous. Every ρ_r remains increasing and commutes with unit translation. The homeomorphism $(u,r)\mapsto(\rho_r(u),r)$ therefore changes the two generators into exactly

$$k(u,r)=(u+1,r),\qquad q(u,r)=(\epsilon u,r+d). \tag{C.48.13}$$

For base image $\mathbb{R}$, the projection with discrete kernel is a covering homomorphism of Lie groups. Its identity component maps isomorphically onto the simply connected line. It centralizes K, since its conjugation map to $\mathrm{Aut}(\mathbb{Z})=\{1,-1\}$ is continuous and is initially the identity. Thus $\widetilde{Q}=\mathbb{Z}\times\mathbb{R}$. Use this real subgroup to transport the fiber $F^{-1}(0)$ to every other fiber, and use the nearest-orbit-point coordinate for K on that first fiber. These are global coordinates in which

$$k(u,r)=(u+1,r),\qquad s_z(u,r)=(u,r+z). \tag{C.48.14}$$

For base image zero only k remains.

Trivial kernel and the possible dense base image. For $K=1$, the group $\widetilde{Q}$ embeds algebraically in the translation group and is abelian. A positive dimensional identity component maps isomorphically onto $\mathbb{R}$; injectivity excludes other components. The same transport from $F^{-1}(0)$ gives just s_z in (C.48.14). In dimension zero, $X\to X/\widetilde{Q}$ is a regular surface covering with

simply connected source. Its deck group is the actual fundamental group of the quotient. A noncompact surface has free fundamental group, and an abelian free group is trivial or infinite cyclic. For a compact quotient the surface classification gives only the torus with abelian torsion-free fundamental group: the sphere and projective plane do not have plane universal cover, the Klein-bottle group is nonabelian, and every remaining closed surface group is nonabelian. Consequently the discrete possibilities are exactly 1, $\mathbb{Z}$, and $\mathbb{Z}^2$.

The cyclic case has a nonzero base translation; choose its positive generator. The construction (C.48.11)–(C.48.13), now without the unit-translation condition, changes it to $(u, r) \mapsto (\epsilon u, r + d)$. The trivial case has no equations. For $\widetilde{Q} \cong \mathbb{Z}^2$ the quotient is a torus, and its base image is dense: an injective homomorphism from $\mathbb{Z}^2$ into $\mathbb{R}$ has two generators with irrational ratio. The foliation descends to an oriented minimal topological foliation of that torus. To verify minimality, the saturation of any lifted fiber under the deck group is $F^{-1}(r + \alpha(\widetilde{Q}))$, which is dense because F is open and the displayed set of levels is dense. The quotient torus is orientable, and its deck maps preserve both the surface and the transverse orientation, so the leaf orientation descends as asserted.

We give the elementary foliation conjugacy needed here, including its maps. It is the minimal-torus conclusion recorded in [43, Proposition 2.5]; no bi-foliation classification or asymptotic-cycle statement is needed. The oriented local flowboxes give a nonsingular continuous flow with these leaves by the construction used in C.43. Choose a small closed transverse interval, with endpoints off the orbit of an interior point, and a first-return arc from that point to a different point of the interval. A flowbox excludes sufficiently short positive returns; minimality supplies a later return, and compactness gives a first one. The endpoints chosen off the orbit put that return in the interval's interior. The arc's interior avoids the interval, and its endpoint differs from its initial point because there is no periodic orbit. Its union with the intervening transverse subinterval is a simple closed curve. In a thin flow strip along the arc, tilt the arc strictly in the transverse direction pointing from the return point back to the initial point. The tilt may be chosen arbitrarily small, because the two endpoint transverse coordinates are distinct; choose it so that the new terminal point remains strictly between the old endpoints in the return interval. Join the new endpoints inside the transverse interval. Both pieces now have the same strict transverse direction; straightening their two corners in flowboxes gives an embedded closed transversal γ.

Every leaf meets γ, since it is dense and a flowbox around γ is open. Forward and backward returns exist: on a compact minimal flow each forward and backward limit set is a nonempty closed invariant set, hence the entire torus. Flowboxes and avoidance of γ on a compact interior return arc show that the first return time is continuous and positive. The return map $h : \gamma \to \gamma$ is a minimal orientation-preserving circle homeomorphism. The map which follows each point for a fraction of its return time gives the leaf-preserving homeomorphism from its mapping torus

$$[0, 1] \times \gamma/((1, z) \sim (0, hz)) \longrightarrow X/\widetilde{Q}. \tag{C.48.15}$$

Surjectivity uses the preceding returns; uniqueness of the last intersection gives injectivity, and the compact source and Hausdorff target give the inverse continuity.

For completeness, choose a probability measure invariant under h by taking a weak limit of finite orbit averages. Minimality gives full support, and the absence of periodic points gives no atoms. Its cumulative distribution is therefore a circle homeomorphism. It conjugates h to a Lebesgue-measure-preserving increasing homeomorphism, hence to a rotation: preservation of the measure of every interval makes the displacement constant. The angle θ is irrational by minimality. After this conjugacy, the explicit map

$$[s, z] \longmapsto [(s, z + s\theta)] \quad \text{in } \mathbb{R}^2/\mathbb{Z}^2 \tag{C.48.16}$$

is well defined at the identified endpoints and sends the suspended leaves to the lines of slope θ. This proves the required foliation conjugacy.

Lift (C.48.15)–(C.48.16) to the universal covers. It intertwines the original deck action with translations by $\mathbb{Z}^2$, through the induced isomorphism of their fundamental groups. Each original F-fiber becomes a whole affine line of slope θ. A linear transverse coordinate r and a complementary linear coordinate u thus give

$$q(u, r) = (u + \beta(q), r + \alpha'(q)) \quad (q \in \widetilde{Q}), \tag{C.48.17}$$

where these translation vectors form a lattice, and $\alpha'(\widetilde{Q})$ is dense. The original function F is a line homeomorphism composed with r: both are open quotients with exactly the same connected fibers. This gives the exact target reparametrization, rather than identifying the two quotient functions without a comparison map.

Contraction in the established normal forms. Except for the separately displayed transitive case (C.48.10), write the actual lifted loop as

$$a_t(u, r) = (h_t(u, r), b_t(r)). \tag{C.48.18}$$

Each $h_t(\,\cdot\,, r)$ is an increasing line homeomorphism. Its endpoints are the identity family. Let $B_{s,t}$ be the natural median contraction of b_t from C.27, and set

$$H_{s,t}(u, r) = \big((1 - s)h_t(u, r) + su,\ B_{s,t}(r)\big). \tag{C.48.19}$$

We check every operator equation rather than changing the operator group. For $k(u, r) = (u+1, r)$ the equation is $h_t(u + 1, r) = h_t(u, r) + 1$, which (C.48.19) preserves. For $q(u, r) = (\epsilon u, r + d)$ it is

$$h_t(\epsilon u, r + d) = \epsilon h_t(u, r), \qquad b_t(r + d) = b_t(r) + d. \tag{C.48.20}$$

Convex interpolation preserves the first identity and median naturality preserves the second. For the real kernel (C.48.7), commutation says $h_t(u, r) = u + \phi_t(r)$; with (C.48.9) the further equation is $\phi_t(r + d) = \lambda\phi_t(r)$. Scaling ϕ_t by $1 - s$ preserves it, including negative λ. For the real base translations in (C.48.14), the equations are that h_t is independent of r and $b_t(r) = r + c_t$. Both are preserved by (C.48.19).

For the lattice (C.48.17), the dense set of base translations and continuity force $b_t(r) = r + c_t$. The fiber equation is exactly

$$h_t(u + \beta(q), r + \alpha'(q)) = h_t(u, r) + \beta(q). \tag{C.48.21}$$

It is preserved by convex interpolation with u. The median of translations is again a translation, so the base equation is preserved as well. The trivial operator case has no extra equation. These cases exhaust (C.48.6).

In (C.48.10), write the loop as right multiplication by (v_t, c_t). Its action is $(u, r) \mapsto (u + e^{cr}v_t, r + c_t)$. The median base contraction is translation by some $C_{s,t}$. Right multiplication by

$$((1 - s)v_t, C_{s,t}) \tag{C.48.22}$$

is the required contraction and still commutes with every left multiplication.

Every map in (C.48.19) is a homeomorphism: its base map is a line homeomorphism, and its fiber map is an increasing convex combination of two increasing onto maps. The maps are jointly continuous. Their inverses are jointly continuous as well: on compact parameter sets, monotonicity and two fixed bracketing fiber values bound the inverse, and continuity and uniqueness identify every limit. This is the inverse continuity argument of C.31 applied to the displayed fiber family and the continuous inverse base family. Formula (C.48.22) has the same properties directly from the group law. All endpoint identities are preserved. We have therefore obtained a based null-homotopy commuting with all of $\widetilde{Q}$.

In particular it commutes with the original source deck group, so it descends to a compact-open null-homotopy $K_{s,t}$ on O, with continuous inverse family. Its commutation with every original operator follows from (C.48.2), retaining the restriction kernel, which acts identically on O. Finally each other boundary real quotient is an exact reparametrization of F by the first case of C.45. Median naturality under that line homeomorphism gives its required quotient identity. For a circle target it commutes with the target integer translations and descends. The equations $q_z F_z = \Phi_z p$ now give (C.48.1) for every z. No limit across the ambient frontier has been deduced from this intrinsic construction. □

C.49. [Proved] Lie contractions on closures with three boundary foliations. Let f_t be an ambient-null based loop in $\mathrm{Homeo}^+(\mathbb{R}^2)$, let G be its generated group, and let H commute with every f_t. For every open G-orbit O having at least three distinct actual lifted boundary-quotient fiber partitions of C.42, the group generated by the based lifts of $f_t|_O$ to the universal covering $X \to O$ is closed in $\mathrm{Homeo}(X)$. With this actual topology it is one of

$$\mathbb{R}^2, \qquad \mathrm{Aff}^+(\mathbb{R}), \qquad \mathcal{A}_2 = \mathbb{R}^2 \rtimes \mathbb{R}, \tag{C.49.1}$$

where the last factor acts on $\mathbb{R}^2$ by positive scalar dilations. Each group is simply connected and has a globally invertible exponential map.

There is a jointly continuous based contraction on the whole $\overline{O}$, with jointly continuous inverse, obtained from these actual lifted groups by fractional powers. These contractions preserve every actual boundary-quotient equation, agree on intersections of such orbit closures and with the contraction of C.37 on X_2, and are transported to one another by H. They consequently give a continuous homeomorphism family on every finite union of these closures and X_2. The conclusion does not assert continuity across an arbitrary infinite accumulating family of the new three-dimensional cases.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; actual simultaneous boundary-quotient and frontier-extension obligations.

Proof. The actual based source lifts are closed loops and act transitively. Choose three partitions and lifts $F_i : X \to \mathbb{R}$ as in C.47. Lift the isotopy $f_t|_O$ from the identity to $\widetilde{f}_t \in \mathrm{Homeo}(X)$. The covering homotopy property gives continuity for maps and inverses in their compact-open topology. Every $\widetilde{f}_t$ commutes with every deck transformation: the commutator is a continuous deck-valued function of t and is the identity at $t = 0$. Its target lifts satisfy

$$F_i \widetilde{f}_t = b_{i,t} F_i. \tag{C.49.2}$$

For a real target, $b_{i,1} = \mathrm{id}$ directly. For a circle target, the exact two-point winding comparison of C.29 makes the based real lift return to the identity, since the original plane loop is ambient-null. Thus $b_{i,1} = \mathrm{id}$ for all i. The injectivity of (F_1, F_2) from C.45 now gives $\widetilde{f}_1 = \mathrm{id}$.

Let $N_{\mathrm{alg}} = \langle \widetilde{f}_t : 0 \le t \le 1 \rangle$. This actual subgroup is path connected: each finite word is joined to the identity by simultaneously shortening its factor paths. Put

$$P = \{\widetilde{f}_t : 0 \le t \le 1\} \cup \{\widetilde{f}_t^{-1} : 0 \le t \le 1\}.$$

It is compact and contains the identity. The source deck group Δ is countable, because a fiber of a covering of a second-countable surface is a discrete countable subset of its second-countable covering plane. For $x \in X$, transitivity of G on O gives

$$X = \bigcup_{\delta \in \Delta} \bigcup_{n \ge 1} \delta P^n x. \tag{C.49.3}$$

All sets in this countable union are compact. Baire's theorem gives interior to one of them, hence to $N_{\mathrm{alg}}x$. Transitivity on that orbit then makes the entire orbit open. Every other orbit is a deck translate of it, since the generated lifted group commutes with Δ. These disjoint open orbits partition the connected plane; there is only one. Thus N_{alg} is transitive on X, not just on its projection O.

The connected three-web groups and their exact actions. Let $\mathcal{W}$ be the full group in C.47 and write $W = \mathcal{W}^0$. The group W is open in $\mathcal{W}$. The open orbit map makes every W-orbit open in X; normality of W and connectedness of X imply transitivity. For a point x, let $J = W_x$. The Lie principal bundle $W \to W/J = X = \mathbb{R}^2$ shows that J is connected: the exact sequence on components has $\pi_1(X) = 0$ and $\pi_0(W) = 0$.

If $\dim W = 2$, then J is discrete and connected, hence trivial. Evaluation identifies W homeomorphically with the simply connected plane. Its Lie algebra and group are the two explicit possibilities $\mathbb{R}^2$ and $\mathrm{Aff}^+(\mathbb{R})$ in C.36. Since $N_{\mathrm{alg}} \subset W$ is already transitive and this action of W is simply transitive, one has $N_{\mathrm{alg}} = W$.

Suppose $\dim W = 3$. Here C.47 puts us in the case where all three pair images are full planes. The connected one-dimensional isotropy J is a subgroup of the ordered-loop automorphism group in that proof. It cannot be a circle. Indeed, every compact group of increasing line homeomorphisms is trivial: the compact orbit of any point has a largest element, which all group elements must fix, so transitivity on that orbit makes it a singleton. Thus $J \cong \mathbb{R}$. The principal bundle over the contractible plane now gives $\pi_1(W) = 0$.

We identify its Lie algebra and homogeneous action explicitly. Let $\mathfrak{w} = \mathrm{Lie}(W)$ and $\mathfrak{j} = \mathrm{Lie}(J)$. For each i, the stabilizer K_i of $F_i(x)$ in the induced real target action is a closed subgroup containing J. The open orbit map identifies W/K_i with the actual real target. The same principal-bundle component argument makes K_i connected, and its Lie algebra $\mathfrak{k}_i$ has dimension two. The three $\mathfrak{k}_i$ are distinct: otherwise the corresponding connected subgroups, and hence the quotient fibers through x, would coincide, contradicting C.45.

For $0 \neq Z \in \mathfrak{j}$, the induced linear map $\operatorname{ad} Z$ on $\mathfrak{w}/\mathfrak{j}$ preserves the three distinct lines $\mathfrak{k}_i/\mathfrak{j}$. A real linear map on a two-dimensional space preserving three distinct lines is scalar; write that scalar λ. If $\lambda = 0$, then $[\mathfrak{j}, \mathfrak{w}] \subset \mathfrak{j}$, so $\mathfrak{j}$ is an ideal. Its connected subgroup J is then normal in W. As a normal point stabilizer in a transitive action it fixes all of X, contrary to the faithful inclusion $W \subset \mathrm{Homeo}(X)$. Therefore $\lambda \neq 0$.

The adjoint map on $\mathfrak{w}$ has the eigenspace decomposition

$$\mathfrak{w} = \mathfrak{j} \oplus V, \qquad \dim V = 2, \qquad [Z, v] = \lambda v \quad (v \in V).$$

For $v, w \in V$, the derivation identity gives $[Z, [v, w]] = 2\lambda[v, w]$. The value 2λ is absent from the spectrum $\{0, \lambda\}$, so $[V, V] = 0$. Replacing Z by Z/λ identifies the Lie algebra with scalar homotheties:

$$[Z, v] = v, \qquad [V, V] = 0. \tag{C.49.4}$$

Since W is simply connected, the corresponding group is exactly $\mathcal{A}_2$, with multiplication

$$(a, v)(b, w) = (a + b,\ v + e^a w), \qquad a, b \in \mathbb{R}, \quad v, w \in \mathbb{R}^2. \tag{C.49.5}$$

Under this isomorphism J is the dilation subgroup $\{(a, 0)\}$. The orbit identification $W/J \to X$ therefore gives an actual equivariant homeomorphism of the source plane with the standard action

$$z \longmapsto v + e^a z. \tag{C.49.6}$$

The generated subgroup is closed, rather than merely analytic. Yamabe's theorem makes the path-connected subgroup $N_{\mathrm{alg}} \subset W$ an analytic Lie subgroup [61, pp. 13–14]. This statement initially gives its analytic subgroup structure; it does not identify that structure with the induced topology. We now establish the latter point by classifying the transitive subgroups of (C.49.6).

A transitive analytic subgroup has dimension at least two. In dimension two its Lie algebra is either the translation plane V, or has a nonzero projection to the dilation coordinate. In the latter case it is

$$\mathbb{R}(Z + v_0) \oplus L, \tag{C.49.7}$$

where $L \subset V$ is a one-dimensional translation subspace. On the quotient line V/L, the generator $Z + v_0$ has flow

$$\bar{z} \longmapsto e^t \bar{z} + (e^t - 1)\bar{v}_0.$$

It fixes $-\bar{v}_0$, while L acts trivially there. Thus the analytic subgroup in (C.49.7) preserves the proper affine line $-v_0 + L$ in the source plane, contradicting transitivity. The remaining two-dimensional subgroup is exactly all translations V. A three-dimensional analytic subgroup is open and equals the connected group $\mathcal{A}_2$. Hence, in this case, $N_{\mathrm{alg}} = V$ or $N_{\mathrm{alg}} = \mathcal{A}_2$.

Both are closed with their actual homeomorphism-group topology. One evaluation recovers a translation vector. For a homothety, evaluation at two fixed distinct points recovers its positive scale and translation continuously; a limit that is still a homeomorphism has positive scale. Together with the dimension-two case already treated, this proves that $N = N_{\mathrm{alg}}$ is closed and is one of (C.49.1), with its induced topology. It also proves that choosing a different triple of partitions does not change N: it is the same subgroup generated by the same based lifted loop.

For the semidirect products in (C.49.1), with vector dimension one or two, the exponential and its inverse are given by

$$\exp(u, v) = \left(u, \frac{e^u - 1}{u}\, v\right), \qquad \frac{e^u - 1}{u} = 1 \quad (u = 0). \tag{C.49.8}$$

The displayed scalar factor is positive everywhere. For $\mathbb{R}^2$ the exponential is the identity. This proves the asserted global exponential coordinates without an unspecified choice of logarithm.

The exact Polish quotient extending the action to the closure. Put $L = \overline{G}$ in the plane homeomorphism group. It preserves O by the forward-and-inverse boundary argument of C.36. Define the closed Polish subgroup

$$E_O = \{(l, n) \in L \times N : pn = (l|_O)p\}. \tag{C.49.9}$$

Projection $E_O \to N$ is onto: each $n \in N$ is an actual finite word in the based lifted loop, so the identical word in the original plane loop supplies its $l \in G$. This continuous surjective homomorphism of Polish groups is open. Its kernel consists of pairs $(l, 1)$ with l fixing O pointwise, and therefore fixing $\overline{O}$ pointwise. Consequently

$$N \times \overline{O} \longrightarrow \overline{O}, \qquad (n, z) \longmapsto lz \quad \big((l, n) \in E_O\big) \tag{C.49.10}$$

is well defined and jointly continuous. Indeed, the product of the open projection with $\mathrm{id}_{\overline{O}}$ is an open quotient, and the continuous E_O-action factors through it. This argument keeps the exact source covering and the possibly nontrivial global restriction kernel.

Every n has its chosen preimage in G, so (C.49.10) preserves every original G-orbit in $\overline{O}$ and has on that orbit exactly the image of G. It commutes with the original setwise H-stabilizer of O, since those preimages commute with that stabilizer. All source deck transformations commute with N because they commute with its generating based lifts, so the restriction of (C.49.10) to O is also the exact descended source action.

The required closure homotopy is now

$$R^O_{s,t} = \exp\big((1 - s) \log \widetilde{f}_t\big) \quad \text{acting by (C.49.10).} \tag{C.49.11}$$

The source loop is closed and based, as proved above. Thus (C.49.11) starts with $f_t|_{\overline{O}}$, ends with the identity, and fixes both based edges. The action and group inversion give joint continuity of the family and its inverse.

All actual boundary quotients and common boundary images. Consider any nontrivial nonopen G-orbit $B \subset \overline{O}$, with its intrinsic line or circle topology from C.28. The action (C.49.10) is transitive on its underlying set. Its point stabilizer N_z is closed. The continuous bijective orbit map $N/N_z \to B$ has one-dimensional domain. Dimension zero is excluded by connectedness and nontriviality. If its dimension were at least two, restricting to a small two-dimensional coordinate disk and applying invariance of domain in the ambient plane would give interior to B, making this whole transitive orbit open.

The bijection agrees with the intrinsic topology. A compact interval in the homogeneous one-manifold maps to an embedded arc. On a non-circular orbit that arc is the unique connecting arc of C.28; on a circle the same statement holds in a sufficiently small interval. Its interior is therefore an intrinsic interval neighborhood. This proves that the orbit identification is a homeomorphism, and that the N-action is continuous in the intrinsic topology as well.

Every group in (C.49.1) is metabelian. Its image on B is path connected and equals the actual $G|_B$. The exact metabelian image argument of C.37 therefore identifies this common image, with its induced intrinsic compact-open topology, as

$$M_B = \begin{cases} \mathbb{R} \text{ acting by translations, or } \mathrm{Aff}^+(\mathbb{R}), & B \text{ a line}, \\ \mathbb{R}/\mathbb{Z} \text{ acting by rotations}, & B \text{ a circle}. \end{cases} \tag{C.49.12}$$

These groups are closed in the intrinsic homeomorphism group: one evaluation recovers the translation or rotation, and two recover the affine parameters. The action supplies a continuous surjective Lie homomorphism

$$\rho_B : N \to M_B. \tag{C.49.13}$$

In particular every actual boundary quotient retains its original equivariance equation under the closure action. To verify this directly for $\Phi_z(lx) = lz$, take the preimage $g \in G$ of $n \in N$. The equality $\Phi_z(gy) = g\Phi_z(y)$ follows from the defining quotient formula; replacing g by another preimage changes neither side because its restriction kernel fixes $\overline{O}$. Thus

$$\Phi_z(R^O_{s,t} y) = R^B_{s,t}(\Phi_z(y)) \tag{C.49.14}$$

with the same original line or circle target and its covering equation.

Since N is simply connected, (C.49.13) has a unique based lift $\widetilde{\rho}_B : N \to \widetilde{M}_B$. The latter group is $\mathbb{R}$ or $\mathrm{Aff}^+(\mathbb{R})$, whose exponential is globally invertible. The path $\widetilde{\rho}_B(\widetilde{f}_t)$ is precisely the based lift $\widetilde{b}_t$ of the original $f_t|_B$. Naturality and injectivity of the exponential give the exact identity

$$\widetilde{\rho}_B\left(\exp\big((1-s)\log \widetilde{f}_t\big)\right) = \exp\big((1-s)\log \widetilde{b}_t\big). \tag{C.49.15}$$

This defines $R^B_{s,t}$ in (C.49.14) after projection. On a point orbit the action is the identity.

Intersections, operators, and the finite-union conclusion. For two orbit closures, their intersection consists of nonopen G-orbits. On each nontrivial one, both images (C.49.12) are the identical group $G|_B$ with the identical intrinsic topology. The two based target-cover paths are the same lift of $f_t|_B$. Formula (C.49.15) therefore makes the closure contractions equal there. It is also exactly the boundary formula in C.36–C.37, so it agrees on every nonopen orbit in the intersection with X_2. If O itself belongs to U_2, evaluation identifies the simply transitive group of C.36 with O and its universal cover with X. The lifted translations are the unique based source lifts used here; their transitive generated group is exactly N. The two fractional-power formulas are consequently identical on O and its closure. An open orbit not in U_2 is disjoint from X_2, because it is an open neighborhood disjoint from U_2. These observations prove agreement on the entire intersection with X_2.

For $h \in H$ with $hO = O'$, lift $h|_O$ to a source-cover homeomorphism $\widetilde{h} : X \to X'$. Covering uniqueness for the two based isotopies gives

$$\widetilde{h}\,\widetilde{f}_t\,\widetilde{h}^{-1} = \widetilde{f}'_t. \tag{C.49.16}$$

Conjugation is consequently a continuous group isomorphism $N \to N'$. It intertwines their exponential maps and the unique logarithms in (C.49.8). It therefore carries (C.49.11) to the corresponding homotopy on O'. Equality extends to the closures. Changing the lift of h by a deck map does not change this identity, since deck maps commute with the generated lifted groups. The actual boundary-quotient families are transported at the same time: with basepoint hx, their exact formula is $\Phi^{O'}_{hz}(lhx) = h\Phi^{O}_{z}(lx)$. Thus the class of open orbits under consideration is H-invariant; no arbitrary selected triple is assumed to be preserved by the operators.

Every closure and X_2 is closed in the plane. The matching families, and separately their matching inverse families, paste continuously over a finite union. Each preserves its member closures and is inverted by the pasted inverse, so it is a homeomorphism of that union. These assertions include all original covering and operator labels. Uniform control for infinitely many new homothety orbits, and extension to the remaining one- and two-foliation regions, are not supplied by this finite-pasting argument. □

C.50. [Proved] Uniform contraction through all the new Lie orbit closures. In the setting of C.49, let U_3 be the union of the open G-orbits having at least three distinct actual lifted boundary-quotient fiber partitions, and retain U_2 from C.37. Set

$$U_* = U_2 \cup U_3, \qquad X_* = \overline{U_*}. \tag{C.50.1}$$

The contractions of C.36 and C.49 extend to a jointly continuous based contraction through homeomorphisms of X_*, with jointly continuous inverse, commuting with H. This includes arbitrary accumulation of the three-dimensional homothety orbits.

The uniform algebraic bound is exact. For every continuous path $k : [0, t] \to \mathcal{A}_2$ with $k(0) = 1$, put $P = k([0, t]) \cup k([0, t])^{-1}$ and $g = k(t)$. Then

$$\{\exp(r \log g) : 0 \le r \le 1\} \subseteq P^{1913}. \tag{C.50.2}$$

The identity belongs to P, so shorter words can be padded. For $\mathbb{R}^2$ and $\mathrm{Aff}^+(\mathbb{R})$ the three-letter bound of C.37 applies. All word equalities used on an orbit hold in its specified simply connected source group. The bound does not identify an additive vector sum with a noncommutative product: the latter's complete error is corrected explicitly in the proof.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; uniform frontier control for the actual three-quotient Lie reductions.

Proof. Two additive path identities. The planar identity of C.37 represents any fraction of the endpoint of a based path in $\mathbb{R}^2$ by at most three signed path values. We also require its three-dimensional version: for a continuous $\gamma : [0, t] \to \mathbb{R}^3$ from 0 to v,

$$rv = \sum_{j=1}^{m} \varepsilon_j \gamma(s_j), \qquad m \le 5, \quad \varepsilon_j \in \{1, -1\}, \quad 0 \le s_j \le t. \tag{C.50.3}$$

For a polygonal path, apply the signed-measure theorem [24, Theorem 4.7 and §4.3.2] to its three coordinate derivatives. The chosen set is a union of at most two circular intervals. Its integral is a sum of two path differences, with at most one additional total vector from an interval crossing the identified endpoint. This gives five signed path values. Zero terms $\gamma(0)$ permit padding to a nonempty word when necessary. The interval count in the halving proof is the same corrected count used in C.37: two intervals cut at at most three points give at most five pieces, so one half uses at most two. For the whole interval, three cuts give at most four pieces, so one of the two halves uses at most two intervals; these are also circular intervals. Complementation, halving, and compactness give every fraction. Uniform polygonal approximation with the same

endpoints, followed by compactness of the five parameters and the finite sign set, proves (C.50.3) for arbitrary continuous paths.

A maximizing anchor and its exact translation commutators. Use the coordinates

$$(a, v)(b, w) = (a + b, v + e^a w), \qquad v, w \in \mathbb{R}^2,$$

and write $k(s) = (a_s, v_s)$. Let $A = \max_{0\le s\le t} |a_s|$. When $A = 0$, the path lies in the translation plane and the three-letter bound already proves (C.50.2). For $A > 0$, choose u attaining the maximum and let $h = k(u)$ or $k(u)^{-1}$, choosing its sign so that its logarithmic scale is A. Thus h is one letter from P, not a power of unbounded length. Conjugate all coordinates by the translation which sends its unique fixed point to zero. This is one group automorphism applied to both sides of every forthcoming identity; it adds no letters to any word. In the new coordinates $h = (A, 0)$ and still $|a_s| \le A$. Put

$$D = e^A - 1, \qquad C_s = [h, k(s)] = hk(s)h^{-1}k(s)^{-1} = (0, Dv_s). \tag{C.50.4}$$

The path C_s is continuous in the translation plane and begins at the identity. Every one of its values and inverses uses at most four letters of P.

For any fixed s and $|\beta| \le B$, where B is a positive integer, the translation C_s^β uses at most $12B$ letters of P. Indeed, write it as B copies of $C_s^{\beta/B}$. For nonnegative $\beta/B \le 1$, the planar identity applied to the prefix $C|_{[0,s]}$ gives at most three signed commutator values. For a negative fraction invert the resulting word. Substituting (C.50.4) costs at most four letters per commutator. No continuous choice of the parameters in these words is required.

The large-scale case. Suppose $A \ge \log 2$, so $D \ge 1$. For $0 \le q \le 1$, the translation T_{qv_s} equals $C_s^{q/D}$ and therefore uses at most twelve letters. Writing $D_a = (a, 0)$ gives

$$D_{a_s} = T_{v_s}^{-1} k(s), \tag{C.50.5}$$

which uses at most thirteen. Let $a = a_t$ and $v = v_t$. Define

$$\theta(a, r) = \begin{cases} (e^{ra} - 1)/(e^a - 1), & a \ne 0, \\ r, & a = 0. \end{cases} \qquad g^r = (ra, \theta(a, r)v). \tag{C.50.6}$$

The function θ is continuous and lies in $[0, 1]$. The intermediate value theorem supplies s with $a_s = ra$, since the path starts at logarithmic scale zero and ends at a. Then $g^r = T_{\theta(a,r)v} D_{a_s}$ uses at most $12 + 13 = 25$ letters.

The small-scale case and its full noncommutative correction. Now suppose $0 < A \le \log 2$. Apply (C.50.3) to $\gamma(s) = (a_s, v_s)$, obtaining

$$ra = \sum_{j=1}^{m} \varepsilon_j a_{s_j}, \qquad rv = \sum_{j=1}^{m} \varepsilon_j v_{s_j}, \qquad m \le 5. \tag{C.50.7}$$

Form the actual product $w = k(s_1)^{\varepsilon_1} \cdots k(s_m)^{\varepsilon_m}$. For an inverse letter, its translation is $-e^{-a_{s_j}} v_{s_j}$. Consequently, with

$$p_j = \begin{cases} \sum_{\ell<j} \varepsilon_\ell a_{s_\ell}, & \varepsilon_j = 1, \\ \sum_{\ell\le j} \varepsilon_\ell a_{s_\ell}, & \varepsilon_j = -1, \end{cases} \tag{C.50.8}$$

the product is exactly

$$\begin{aligned} w &= \left(ra, \sum_{j=1}^{m} \varepsilon_j e^{p_j} v_{s_j} \right), \\ \mathrm{pr}_v(w) &= rv + \sum_{j=1}^{m} \varepsilon_j (e^{p_j} - 1) v_{s_j}. \end{aligned} \tag{C.50.9}$$

Every $|p_j|$ is at most mA. The pure translation which changes (C.50.9) to (C.50.6) by left multiplication is

$$c = (\theta(a,r) - r)v - \sum_{j=1}^{m} \varepsilon_j (e^{p_j} - 1) v_{s_j}. \tag{C.50.10}$$

Using the translation commutators (C.50.4), this is the exact group equation

$$g^r = C_t^{\beta_0} \left(\prod_{j=1}^{m} C_{s_j}^{\beta_j} \right) w, \quad \beta_0 = \frac{\theta(a,r) - r}{D}, \qquad \beta_j = -\varepsilon_j \frac{e^{p_j} - 1}{D}. \tag{C.50.11}$$

The powers before w commute because they are translations.

Here are uniform bounds, including $a = 0$. The integral formula

$$\theta(a,r) = \frac{\int_0^r e^{aq}\, dq}{\int_0^1 e^{aq}\, dq} \tag{C.50.12}$$

holds at zero as well as away from it. For $|a| \le A$, the inequality $|e^{aq} - 1| \le D$ holds on $0 \le q \le 1$, and the denominator is at least $e^{-A} \ge 1/2$. Subtracting r in (C.50.12), the numerator is the difference of an integral over $[0,r]$ and r times an integral over $[0,1]$, each with integrand $e^{aq} - 1$. Its absolute value is at most $2D$. Therefore $|\beta_0| \le 4$. For every other coefficient,

$$\frac{|e^{p_j} - 1|}{D} \le \frac{e^{mA} - 1}{e^A - 1} = \sum_{\ell=0}^{m-1} e^{\ell A} \le 2^m - 1 \le 31. \tag{C.50.13}$$

The same bound covers negative p_j because $1 - e^{-mA} \le e^{mA} - 1$. The word construction following (C.50.4) now applies with $B = 4$ and $B = 31$. Equation (C.50.11) uses at most

$$5 + 12(4 + 5 \cdot 31) = 1913 \tag{C.50.14}$$

original letters. This proves (C.50.2) in all regimes. The executable certificate accompanying this claim checks the inverse-letter exponents in (C.50.8), the complete product correction, the commutator and conjugacy formulas, and the integer count by exact symbolic arithmetic. The measure selection and the analytic coefficient bounds are proved above, independently of that certificate.

One compact family of actual word isotopies. Set $L_0 = 1913$. For the original plane loop, all words just proved belong to the compact parameter family

$$W_{\boldsymbol{\varepsilon},\boldsymbol{s}}(u) = \prod_{j=1}^{L_0} f_{us_j}^{\varepsilon_j}, \qquad \varepsilon_j \in \{1,-1\}, \quad 0 \le s_j \le t \le 1, \quad 0 \le u \le 1. \tag{C.50.15}$$

Identity letters are obtained by $s_j = 0$. Continuity of finite multiplication and inversion in the plane homeomorphism group gives compactness for maps and inverses, uniformly in u. On a C.49 orbit, the word equality is in its simply connected actual group N. On a C.36 orbit, use the based path in its specified universal covering group and the three-letter C.37 bound, then pad it. Thus the endpoint of the based lift of (C.50.15) is exactly the required fractional power, not just a homeomorphism with an unspecified deck shift.

Use $r = 1 - s$ for the contraction parameter and write $R_{r,t}^{O}$ for these fractional actions on an open orbit in U_*. C.36 and C.49 already prove their individual closure continuity and finite compatibility. Their groups are metabelian, so $[[g_1,g_2],[g_3,g_4]] = 1$ on U_* for every $g_i \in G$, and hence on X_*. Every nontrivial nonopen orbit $B \subset X_*$ therefore has the exact intrinsic image $M_B = \mathbb{R}$, $\mathrm{Aff}^+(\mathbb{R})$, or the circle rotation group, by the path-connected metabelian argument of C.37. Define $R_{r,t}^{B}$ by its unique line-group logarithm or its based real rotation lift, and use the

identity on point orbits. The based circle lift ends at zero by C.29. These are the same boundary prescriptions as in C.36 and C.49.

Limits, with the bounded residual in the specified cover. Take

$$x_i \in O_i \subset U_*, \quad x_i \to x \in X_*, \qquad (r_i, t_i) \to (r, t). \tag{C.50.16}$$

Choose representations (C.50.15) for $R^{O_i}_{r_i,t_i}$ and pass to convergent word parameters. Let W be the limiting plane isotopy, $w = W(1)$, and $g = f_t$. If x belongs to an open orbit of X_*, then that orbit already lies in U_* and eventually all x_i lie in it. Its intrinsic continuity applies. If x is fixed by G, every limiting word fixes it. For a remaining intrinsic line or circle orbit $B = Gx$, identities on O_i pass to every point $hx \in B$ by evaluating them at hx_i for fixed $h \in G$. In particular $w|_B$ commutes with $g|_B$.

For a fixed positive integer n, set $m_i = \lfloor nr_i \rfloor$. In each specified simply connected group, the residual is exactly

$$\widetilde{w}_i^{\,n} \widetilde{g}_i^{\,-m_i} = \exp\big((nr_i - m_i) \log \widetilde{g}_i\big). \tag{C.50.17}$$

Its exponent fraction is in $[0, 1)$, so it has another representation from the same compact word family. For this fixed n, pass to a subsequence on which $m_i = m_n$ and the residual word isotopies converge to Z_n. Then

$$(w|_B)^n (g|_B)^{-m_n} = z_n, \qquad |m_n - nr| \le 1, \qquad z_n = Z_n(1)|_B, \tag{C.50.18}$$

and all z_n belong to the same compact family of intrinsic restricted word endpoints. For a line, logarithms of this compact family are bounded. In the affine group commuting nonidentity elements belong to the same exponential one-parameter subgroup; the identity case gives the same additive logarithm equation. Thus

$$n \log(w|_B) - m_n \log(g|_B) = \log z_n.$$

Division by n proves $\log(w|_B) = r \log(g|_B)$.

For a circle, let $\theta(u)$ be the based real angle of $f_u|_B$, and put $M = \max_u |\theta(u)|$. Let A_W be the real endpoint angle of $W|_B$, $T = \theta(t)$, and C_n that of $Z_n|_B$. The fixed word length gives $|C_n| \le L_0 M$. The real, rather than merely circle-valued, equality is

$$nA_W - m_n T = C_n. \tag{C.50.19}$$

To prove it, use the plane word isotopies $W_i(u)^n f_{ut_i}^{-m_i} Z_i(u)^{-1}$. Their restrictions lift to loops ending at the identity by (C.50.17). The specified source groups are simply connected, so these loops are null-homotopic there. Evaluate at the two points x_i, hx_i, where $hx \ne x$. Their difference-direction winding is zero. The resulting loops converge uniformly to the analogous two-point loop at x, hx, which stays off the diagonal; its winding is also zero. The exact circle comparison of C.29 identifies that winding, up to the common orientation sign, with $nA_W - m_n T - C_n$. This proves (C.50.19). Consequently

$$|A_W - rT| \le \frac{L_0 M + |T|}{n} \quad (n \ge 1),$$

so $A_W = rT$. This also covers an identity endpoint on the circle whose specified real angle T is a nonzero integer.

Joint continuity and the remaining scope. Every convergent subsequence of the compact word parameters in (C.50.16) has therefore the prescribed endpoint action at x. This proves convergence to $R^B_{r,t}(x)$ for the entire sequence. It gives joint continuity from the dense set U_*. For a general sequence in X_*, approximate its ith point by a point of U_* with both point distance and image distance at that fixed parameter less than $1/i$. The just-proved dense-set limit supplies those approximations and proves full joint continuity.

Apply the same proof to f_t^{-1}. It has the same generated group, the same open-orbit families, and the same boundary partitions. Its fractional actions are the orbitwise inverses, so the inverse family is jointly continuous as well. Every member is an orbitwise bijection inverted by this family and is therefore a homeomorphism of X_*. The equations $R_{0,t} = 1$, $R_{1,t} = f_t|_{X_*}$, and $R_{r,0} = R_{r,1} = 1$ give the based contraction. The operator transport statements of C.36 and C.49 make U_* and X_* invariant under H and give commutation on U_*. Continuity extends that equality to X_*.

This settles infinite accumulation for the orbit closures treated by C.49. The remaining global step is a compatible extension from X_* through the regions with fewer actual boundary-quotient foliations and their frontiers. The particular median choice of C.34 cannot be imposed there without proving compatibility, as C.35 demonstrates. The general conclusion in C.12 consequently remains open. □

C.51. [Proved] Cocompact proper operators control the entire frontier. Let f_t be the based ambient-null plane loop of C.49, let $G = \langle f_t \rangle$, and let H commute with all its values. For an open G-orbit O, put

$$H_O = \{h \in H : hO = O\}, \qquad Q_O = \overline{\{h|_O : h \in H_O\}}^{\mathrm{Homeo}(O)}. \tag{C.51.1}$$

Consider all such O for which O/Q_O is compact and the full family $\mathscr{F}_O$ of actual lifted boundary quotient fiber partitions in C.46 has at most two members. For every one of these orbits the restricted loop has a based contraction on its entire closure,

$$\overline{K}^O_{s,t} \in \mathrm{Homeo}(\overline{O}), \qquad \overline{K}^O_{0,t} = f_t|_{\overline{O}}, \qquad \overline{K}^O_{1,t} = \overline{K}^O_{s,0} = \overline{K}^O_{s,1} = \mathrm{id}, \tag{C.51.2}$$

jointly continuous with its inverse. It commutes with H_O and equals exactly the C.34 median contraction on every nonopen orbit in ∂O. When $\mathscr{F}_O$ is empty, every point of ∂O is fixed by G.

The contractions can be chosen with exact transport under H. They agree on all common frontiers and with C.34, and hence paste over the closed nonopen locus together with any finite collection of these closures. This assertion does not prescribe their values on the closed set X_* of C.50: the comparison there with its exponential prescription remains to be proved.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b), infinite-operator zero-level comparison and ambient frontier extension.

Proof. Write $Q = Q_O$ and $L = \overline{G}$ in the plane homeomorphism group. C.30 proves that the actual group Q in (C.51.1) is closed, locally compact, free and proper on O, with the specified compact-open topology. The restriction kernel of $H_O \to Q$ acts trivially on O and therefore on its closure. We retain that map; we do not replace an element of Q by an unspecified global extension.

A compact set meeting every operator orbit. The quotient map $O \to O/Q$ is open. For each point of O, choose a relatively compact open neighborhood inside O. The images of these neighborhoods cover the compact quotient. Choose finitely many whose images still cover it, and let $C \subset O$ be the union of their closures. Then C is compact and

$$QC = O. \tag{C.51.3}$$

This argument supplies the exact compact fundamental set required below; it uses no fundamental-domain boundary or continuous choice of orbit representatives.

Every frontier sequence produces an actual quotient. Take a sequence $x_n \in O$ converging in the plane to $z \in \partial O$. By (C.51.3), write $x_n = q_n y_n$ with $q_n \in Q$ and $y_n \in C$. Pass to a subsequence with $y_n \to y \in C$. For the moment choose $h_n \in H_O$ with

$$|h_n y_n - q_n y_n| < 1/n. \tag{C.51.4}$$

Such a choice follows from the definition of Q as the closure of the actual restrictions in (C.51.1). Here and below $|\cdot|$ is the Euclidean distance in the original ambient plane.

Every h_n commutes with L: it commutes with all generators of G, and commutation passes to its closure in the plane homeomorphism group. The orbit map

$$L \longrightarrow O, \quad l \mapsto ly \tag{C.51.5}$$

is open by the same Polish orbit-map theorem used in C.30 and C.42. Thus there are $l_n \to 1$ in L with $l_n y = y_n$. For clarity, take a decreasing countable identity neighborhood basis V_k in L. Each $V_k y$ is a neighborhood of y, so $y_n \in V_k y$ eventually. Choose an increasing sequence of such eventual indices and then $l_n \in V_{k(n)}$ with $k(n) \to \infty$. This proves the stated selection. Commutation now gives the exact equality

$$h_n y = l_n^{-1} h_n y_n \longrightarrow z. \tag{C.51.6}$$

Convergence follows from (C.51.4), $x_n \to z$, and compact-open convergence of l_n^{-1} to the identity on a compact neighborhood of z.

Let $J_y = L_y$. For $j \in J_y$, $j h_n y = h_n j y = h_n y$, so (C.51.6) gives $jz = z$. Consequently $z \in \mathrm{Fix}(J_y)$, and C.42's actual evaluation comparison gives

$$h_n|_O \longrightarrow \Phi_z^y \quad \text{uniformly on compact subsets of } O, \qquad \Phi_z^y : O \longrightarrow \mathbb{R}^2, \quad \Phi_z^y(ly) = lz. \tag{C.51.7}$$

This uses the homeomorphism $\mathrm{Map}_L(O, \mathbb{R}^2) \to \mathrm{Fix}(J_y)$ of (C.42.1), including its compact-open topology. In particular its assertion applies when Gz is a single point, in which case Φ_z^y is constant. There is no convergence claim for arbitrary proper operator maps outside their actual global approximants.

We compare (C.51.7) with the quotient family defined at the fixed basepoint x in C.46. Choose $a \in L$ with $ax = y$, and put $z_0 = a^{-1} z$. Then $z_0 \in \mathrm{Fix}(L_x) \cap \partial O$. For every $l \in L$ the equality

$$\Phi_{z_0}^x(ly) = \Phi_{z_0}^x(lax) = laz_0 = lz = \Phi_z^y(ly) \tag{C.51.8}$$

identifies the actual maps, with the same target orbit. The orbit is contained in ∂O, since L preserves O and its boundary. It is nonopen; an open orbit meeting ∂O would be an open neighborhood disjoint from O, contradicting the definition of its closure. Thus C.28 makes a nontrivial such orbit an intrinsic line or circle. For that case (C.51.8) is exactly one of the original boundary quotients whose lifts define $\mathscr{F}_O$. It follows in particular that an empty $\mathscr{F}_O$ forces every frontier orbit to be a point: apply the construction to a sequence approaching any prescribed frontier point.

The intrinsic family and every boundary equation. For $|\mathscr{F}_O| = 1$ use the contraction $K_{s,t}^O$ of C.48, and for $|\mathscr{F}_O| = 2$ use C.46. Both commute with H_O, hence with its restricted closure Q, and both have jointly continuous inverses. They retain every quotient equation

$$\Phi K_{s,t}^O = Q_{s,t}^B \Phi, \tag{C.51.9}$$

where Q^B is the C.34 median family on the actual boundary orbit B, with its based real covering lift when that orbit is a circle. For an empty family, C.32 makes $f_t|_O$ null in $\mathrm{Homeo}(O)$ and C.33 supplies a based null-homotopy K^O in its exact Q-centralizer. All frontier orbits are then points, as just proved, so (C.51.9) for each constant Φ is automatic.

Define $\overline{K}^O$ to be K^O on O and the C.34 family on ∂O. We prove its full joint continuity, rather than continuity only for fixed parameters. Let

$$x_n \in O, \quad x_n \to z \in \partial O, \qquad (s_n, t_n) \to (s, t). \tag{C.51.10}$$

Use $x_n = q_n y_n$ and a subsequence $y_n \to y$ as above. The set

$$S = C \ \cup \bigcup_{(u,v)\in[0,1]^2} (K^O_{u,v}(C) \cup (K^O_{u,v})^{-1}(C)) \tag{C.51.11}$$

is compact in O, by joint continuity on a compact parameter space. Refine the choice of actual operators in (C.51.4) to satisfy

$$\sup_{w\in S} |h_n w - q_n w| < 1/n. \tag{C.51.12}$$

For each n this is a compact-open neighborhood condition on $q_n \in Q$, so density of the specified restrictions supplies $h_n \in H_O$. The previous argument still proves (C.51.6)–(C.51.8).

Commutation with Q and (C.51.12) give

$$\begin{aligned} K^O_{s_n,t_n}(x_n) &= q_n K^O_{s_n,t_n}(y_n), \\ |K^O_{s_n,t_n}(x_n) - h_n K^O_{s_n,t_n}(y_n)| &< 1/n. \end{aligned} \tag{C.51.13}$$

The points $K^O_{s_n,t_n}(y_n)$ converge in O to $K^O_{s,t}(y)$ and lie in the compact set S. Uniform convergence (C.51.7), followed by the exact quotient identity (C.51.9), therefore yields

$$\begin{aligned} h_n K^O_{s_n,t_n}(y_n) &\longrightarrow \Phi^y_z K^O_{s,t}(y) \\ &= Q^{Gz}_{s,t}(\Phi^y_z(y)) = Q^{Gz}_{s,t}(z). \end{aligned} \tag{C.51.14}$$

For a point orbit the same computation uses the constant map and gives z. Every subsequence in (C.51.10) has a further subsequence of this form because C is compact. The value on the right is independent of all choices. Thus (C.51.14) holds for the entire sequence and proves continuity from O to the prescribed frontier value. Continuity for sequences contained in ∂O is precisely C.34. Splitting a sequence according to its two pieces proves joint continuity everywhere on $[0,1]^2 \times \overline{O}$.

The inverse maps have the same argument, using the inverse images already included in S. Equation (C.51.9) implies $\Phi(K^O_{s,t})^{-1} = (Q^B_{s,t})^{-1}\Phi$ by composing with these actual inverse maps. The inverse boundary family is jointly continuous by C.34. This proves the inverse version of (C.51.14), and hence joint continuity of the piecewise inverse on the entire closure. The maps preserve O and each frontier orbit and are bijections there; the piecewise inverses are their actual inverses. They are homeomorphisms. The endpoint and based-edge identities and commutation with H_O hold on both pieces.

Transport and finite pasting. The class of orbits in the claim is H-invariant: conjugation by $h \in H$ identifies the two exact groups Q_O, Q_{hO}, their quotient spaces, and their full boundary-quotient families by C.46. Choose one orbit from each H-orbit of these open components, make the construction there, and set

$$\overline{K}^{hO}_{s,t} = h\overline{K}^O_{s,t}h^{-1}. \tag{C.51.15}$$

The equality with every stabilizing operator makes this independent of the choice of h carrying the selected representative to a given component. Median naturality in C.27–C.34 makes its frontier restriction the identical prescribed C.34 family.

Distinct open-orbit closures meet only in the closed nonopen locus Y. Every one of the new families and its inverse equals the corresponding C.34 map there. The finite closed-set pasting lemma thus gives the asserted maps and inverse maps on Y together with finitely many closures. For every finite collection $\mathcal{I}$, the homeomorphism h carries this union to the union indexed by $h\mathcal{I}$ and intertwines their homotopies by (C.51.15). This is the exact operator-transport comparison on those finite unions.

The proof uses one fixed compact fundamental set inside each treated orbit. It supplies no uniform control over infinitely many different such sets, and it does not identify the median boundary value with the exponential value forced on X_* by C.50. These are explicit remaining comparisons in C.12. □

C.52. [Proved] Cocompact frontier extensions relative to the exponential region. Use the actual based ambient-null plane loop f_t, the groups $G = \langle f_t \rangle$ and H, and the closed set X_* with its contraction $E_{s,t}$ from C.50. For every nonopen G-orbit B, define a based family $R^B_{s,t}$ as follows:

$$R^B_{s,t} = \begin{cases} \text{id}, & B \text{ a point}, \\ \text{the exponential fraction with parameter } 1-s, & G|_B \text{ metabelian}, \\ \text{the C.27–C.29 median family}, & G|_B \text{ nonmetabelian}. \end{cases} \tag{C.52.1}$$

The exponential fraction uses the actual intrinsic Lie image classified in C.37. On a circle it uses the entire based lift of the original loop to the covering group, including its real angle at each intermediate time.

For every open orbit O disjoint from X_* whose full family $\mathscr{F}_O$ of actual lifted boundary partitions has cardinality at most two, there is an intrinsic based null-homotopy $K^O_{s,t}$ commuting with its original operator stabilizer and satisfying every actual boundary quotient equation

$$\Phi_z K^O_{s,t} = R^{Gz}_{s,t} \Phi_z. \tag{C.52.2}$$

For each such O with compact proper-operator quotient O/Q_O, this family extends to a based contraction through homeomorphisms of $\overline{O}$, jointly continuous with its inverse, and equal to (C.52.1) on its entire frontier. It agrees with $E_{s,t}$ on $\overline{O} \cap X_*$.

These extensions can be chosen with exact transport under H. They agree on common frontiers. Consequently they paste with the prescribed C.50 contraction on X_* over every finite union

$$X_* \cup \bigcup_{O \in \mathcal{I}} \overline{O}, \tag{C.52.3}$$

where $\mathcal{I}$ is a finite collection of the cocompact orbits just specified. No continuity of (C.52.1) on the entire nonopen locus, and no extension over an infinite collection or over the other noncocompact open orbits, is asserted.

Manuscript: Section 23, folios 140-141, scans 279-281; Part C(b) infinite-operator request; relative frontier extension preserving the already prescribed exponential region.

Proof. The precise intrinsic target family. For a nontrivial intrinsic line or circle orbit, $G|_B$ is path connected and transitive. The proof of (C.37.9) classifies every such metabelian image, with its actual intrinsic homeomorphism-group topology: on a line it is conjugate to all translations or to the full positive affine group, and on a circle to all rotations. That proof uses only these three stated properties of the image, so it applies to every metabelian orbit in (C.52.1), without requiring that orbit to lie in X_*.

For a real lift $F : X \to \mathbb{R}$ of an actual boundary quotient, let b_t be its based target path, characterized by $Fa_t = b_t F$ for the based source lift a_t. The exact source and target coverings are those of C.42–C.44. The path b_t is a based loop: for a circle target this is the zero-winding conclusion of C.29, and for a line target it follows from the identity endpoint of f_t. Its generated group

$$T = \langle b_t : 0 \le t \le 1 \rangle \tag{C.52.4}$$

is transitive on $\mathbb{R}$. Indeed its orbits are intervals, as each word is joined to the identity by a path of words. A finite endpoint of such an orbit interval is a common fixed point, which would project to a common fixed point of the nontrivial transitive target action. Thus no proper orbit interval is possible.

For a line target, (C.52.4) is the conjugate of the actual image $G|_B$. For a circle, projection gives a surjection $T \to G|_B$. When the latter image is metabelian, its rotation conjugacy lifts

to a real coordinate in which every b_t is a translation. Transitivity then makes T the full translation group. Conversely a metabelian T has metabelian quotient $G|_B$. This proves the exact equivalence used in the case selection below.

Define, in the specified real lift,

$$B_{s,t} = \begin{cases} \exp_T((1-s)\log_T b_t), & T \text{ metabelian}, \\ \text{the C.27 median of the path } b_t, & T \text{ nonmetabelian}. \end{cases} \tag{C.52.5}$$

In the first case T is a simply connected translation or positive affine Lie group with a global exponential and logarithm. Formula (C.52.5) and its inverse are jointly continuous, with all four required based-edge identities. They commute with every target homeomorphism commuting with the original path: in the exponential case each new value lies in T, and in the median case this is C.27.

This rule is natural under any line reparametrization, including a decreasing one. Such conjugation identifies the actual groups (C.52.4) topologically, preserves metabelianness, and, in the Lie cases, intertwines exponential maps and their unique logarithms. The median case has the same naturality by C.27. The source of a real lift of a circle may also be changed by a covering conjugacy; the same argument retains the specified based target path. In particular the resulting map on B is exactly (C.52.1), independent of all these coordinates.

For clarity, at a time when $f_t|_B = \mathrm{id}$ on a circle, its based real lift can still be a nonzero integer translation. Formula (C.52.5) scales that real translation. It does not replace it by zero. The endpoint $t = 1$ is zero by C.29. This is the same covering convention as C.50.

Replacement in the two-foliation construction. For $|\mathscr{F}_O| = 2$, retain exactly the source cover, the two real quotient coordinates, and the domain D of C.45–C.46. Replace each of the two coordinate median families in (C.46.4) by (C.52.5). On $D = \mathbb{R}^2$ this is a product of increasing line homeomorphisms, with a jointly continuous inverse. On $D = \{u < v\}$ the original two coordinate loops are identical. Their generated groups and their selected families are therefore identical, so the strict inequality is preserved. On $D = \{u < v < u + 1\}$ the same argument applies, and the selected maps commute with unit translation. In the exponential case this holds because that translation centralizes T; in the median case it is the original median naturality. Thus both inequalities are preserved, as are they by the inverse family.

Every original stabilizing operator, and every one of its source lifts, permutes the full two-element family of partitions by the exact comparison (C.46.5). In these coordinates it is a pair of line homeomorphisms, possibly followed by exchange. Commutation with the based source loop identifies the corresponding target paths by these same conjugacies. Rule (C.52.5) is natural under them and makes the same choice on conjugate paths. Consequently the modified source family commutes with every such lift, including all source deck maps. It descends to O with continuous inverse.

Every other actual real quotient is a line homeomorphism composed with one of the two selected coordinates, by C.45. The preceding naturality therefore proves its exact lifted equation too. Descending through $q_z F_z = \Phi_z p$ proves (C.52.2), retaining the original source and target covers.

Replacement in the one-foliation construction. For $|\mathscr{F}_O| = 1$, keep the actual closed proper operator group, its group of all source lifts, and every coordinate change of C.48. In (C.48.19) replace only the median base map by (C.52.5), leaving the fiber interpolation as displayed:

$$(u, r) \longmapsto ((1-s)h_t(u, r) + su,\ B_{s,t}(r)). \tag{C.52.6}$$

Here the choice of target coordinate is the exact reparametrization already established in C.48. We check why all of its operator equations survive this replacement.

For the unit fiber translation, the fiber equation $h_t(u + 1, r) = h_t(u, r) + 1$ is unchanged by the base replacement. For an operator $(\epsilon u, r + d)$, the equations are $h_t(\epsilon u, r + d) = \epsilon h_t(u, r)$

and $b_t(r+d) = b_t(r) + d$. The first is preserved by the same convex interpolation; the second says that translation by d centralizes T, so it holds for (C.52.5). For the real fiber kernel and a further $(\lambda u, r+d)$, the fiber function is $h_t(u,r) = u + \phi_t(r)$ with $\phi_t(r+d) = \lambda\phi_t(r)$. Scaling ϕ_t preserves this equality for either sign of λ, and the base comparison is again the same one.

In the normal forms with all real base translations or a dense lattice base image, the original base loop is $b_t(r) = r + c_t$. Its selected family is $B_{s,t}(r) = r + (1-s)c_t$ and commutes with every base operator. The fiber equations, respectively independence of r or the exact affine lattice relation (C.48.21), are preserved by (C.52.6). The trivial operator case imposes no extra equation. These are the C.48 normal forms outside its transitive two-dimensional case. That latter case already lies in $U_2 \subset X_*$ and is absent from the open orbits of the present claim.

Thus the modified family commutes with the exact group of all operator lifts, with its original restriction kernel retained downstairs. The fiber and base inverse-continuity argument following (C.48.22) applies unchanged to (C.52.6): the base family has continuous inverses by (C.52.5), and fixed bracketing fiber values bound the increasing convex-interpolation inverses. The homotopy descends to O, with the asserted endpoint identities. All other actual quotients are reparametrizations of the one selected partition; naturality proves (C.52.2) for them.

For an empty family, C.32–C.33 supply a based null-homotopy in the exact proper-operator centralizer. There is then no nontrivial actual quotient equation to check. This finishes the intrinsic assertion for all three cardinalities.

Cocompactness and the entire frontier. Now assume O/Q_O compact and put $Q = Q_O$. The first part of C.51 supplies a compact $C \subset O$ with $QC = O$. It also proves that every nontrivial frontier orbit gives an actual quotient in the specified family, after the explicit basepoint comparison (C.51.8). In particular an empty family makes all frontier points fixed.

Define $\overline{K}^O$ by the just-constructed K^O on O and by (C.52.1) on ∂O. We first prove its limit along points of O; we do not assume transverse continuity of (C.52.1). Let

$$x_n \in O, \quad x_n \to z \in \partial O, \qquad (s_n, t_n) \to (s,t). \tag{C.52.7}$$

Write $x_n = q_n y_n$ with $y_n \in C$ and pass to a subsequence $y_n \to y \in C$. The compact set consisting of C and all its images and inverse images under K^O is exactly the type of set used in (C.51.11). Approximate q_n uniformly on this set to accuracy $1/n$ by restrictions of actual $h_n \in H_O$, as in (C.51.12). The open evaluation map of $L = \overline{G}$ gives $l_n \to 1$ with $l_n y = y_n$. Exact commutation gives $h_n y = l_n^{-1} h_n y_n \to z$.

The C.42 evaluation homeomorphism, with the basepoint comparison (C.51.8), now gives $h_n|_O \to \Phi_z^y$ uniformly on compact subsets. For a nontrivial target this is an actual quotient covered by (C.52.2); for a fixed target it is constant. Commutation with Q therefore gives precisely

$$\begin{aligned} K^O_{s_n,t_n}(x_n) &= q_n K^O_{s_n,t_n}(y_n) \\ &\longrightarrow \Phi_z^y K^O_{s,t}(y) = R^{Gz}_{s,t}(z). \end{aligned} \tag{C.52.8}$$

The approximation error in the second line is less than $1/n$ before passage to the limit. Compactness of C and uniqueness of the value give (C.52.8) for the entire sequence. This repeats C.51's exact operator comparison with the new quotient equation; no element of the restricted closure is assigned an unproved global extension.

The diagonal argument at frontier points. Take now $z_n \in \partial O$ with $z_n \to z$ and $(s_n, t_n) \to (s,t)$. For each fixed triple (s_n, t_n, z_n), (C.52.8) lets us choose $x_n \in O$ such that

$$|x_n - z_n| < 1/n, \qquad \left|K^O_{s_n,t_n}(x_n) - R^{Gz_n}_{s_n,t_n}(z_n)\right| < 1/n. \tag{C.52.9}$$

Then $x_n \to z$, so applying (C.52.8) to this new sequence proves $R^{Gz_n}_{s_n,t_n}(z_n) \to R^{Gz}_{s,t}(z)$. This establishes the needed joint continuity on this entire frontier, even across different target types

in (C.52.1). Splitting a sequence into its interior and frontier terms proves joint continuity on $[0,1]^2 \times \overline{O}$.

The same proof applies to the inverse family. The compact approximation set already contains all inverse images, and (C.52.2) implies the exact inverse equation $\Phi_z(K^O_{s,t})^{-1} = (R^{Gz}_{s,t})^{-1}\Phi_z$. Use this equation in (C.52.8), then repeat the diagonal choice (C.52.9) for inverses. Thus the piecewise inverse is jointly continuous. All maps are bijections on O and on each frontier orbit, with these actual inverses; they are homeomorphisms of the closure.

Agreement with the fixed exponential region. Every group image on an open orbit of $U_2 \cup U_3$ is metabelian, by C.36 and C.49. Hence, for all $g_1, g_2, g_3, g_4 \in G$, the plane homeomorphism $[[g_1, g_2], [g_3, g_4]]$ fixes $U_2 \cup U_3$ pointwise. Continuity makes it fix X_* pointwise. The set X_* is G-invariant. Therefore every nonopen orbit $B \subset X_*$ has metabelian actual image $G|_B$.

An open orbit outside $U_2 \cup U_3$ is disjoint from X_*, because it is an open neighborhood disjoint from that union. Thus the intersections $\overline{O} \cap X_*$ occurring here contain only nonopen orbits. On a point orbit both contractions are the identity. On every other such orbit, (C.52.1) and C.50 use the same actual intrinsic Lie image and the same based target-cover path. Exponential naturality therefore gives equality, including the intermediate real circle angles described above.

Distinct open-orbit closures meet only in the nonopen locus. The present construction assigns there the same rule (C.52.1), so their contractions and inverses agree. Conjugation by $h \in H$ identifies the actual target images and their based paths, preserves metabelianness, and intertwines both branches of (C.52.1). Choose the intrinsic construction on one representative of each H-orbit of components and transport it by h. Commutation with the full stabilizer makes this choice independent of the transporting element, exactly as in (C.51.15).

Finite closed-set pasting now gives the homeomorphism families and inverse families on (C.52.3). A general h intertwines this union with the union indexed by $h\mathcal{I}$; when it preserves the union, it commutes with the displayed contraction. The family on X_* is exactly the prescribed one throughout. The diagonal proof concerns a single fixed closure. It supplies no uniform control over infinitely many different compact fundamental sets, and no compact fundamental set for the remaining noncocompact quotients. These are still retained parts of C.12. □

C.53. [Proved] Global quotients and finitely many operator representatives. Use the ambient-null based plane loop f_t, its generated group G, the commuting operator group H, and $L = \overline{G}$ of C.42. For every open G-orbit O, choose $x \in O$ and put $J = L_x$. Define $\mathscr{F}^{\mathrm{all}}_O$ to be the family of fiber partitions on the common universal covering of O of all real lifts of

$$\Phi^x_z : O \longrightarrow Gz, \qquad \Phi^x_z(lx) = lz, \quad z \in \mathrm{Fix}(J), \quad Gz \text{ nonopen and nontrivial.} \tag{C.53.1}$$

The target points need not belong to ∂O. Let $U^\dagger$ be the union of all open orbits with either $\dim Q_O = 2$ or $|\mathscr{F}^{\mathrm{all}}_O| \geq 3$, and set $X^\dagger = \overline{U^\dagger}$. There is a jointly continuous based contraction on $X^\dagger$, with jointly continuous inverse and exact H-commutation, extending the prescribed contraction on X_* in C.50. On every nonopen orbit it has the corresponding exponential value of C.52.1.

Now choose finitely many open orbits $O_1, \ldots, O_m$ disjoint from $X^\dagger$, each with compact O_j/Q_{O_j}, and put

$$U = \bigcup_{j=1}^{m} \bigcup_{h \in H} hO_j, \qquad X = \overline{U}. \tag{C.53.2}$$

The original loop has a based contraction through homeomorphisms of $X^\dagger \cup X$, jointly continuous with its inverse, commuting with H and equal to the specified exponential contraction on $X^\dagger$. On every nonopen orbit in X its value is exactly the hybrid prescription $R^B_{s,t}$ of C.52.1. This assertion allows infinitely many different open components in (C.53.2), with arbitrary accumulation. It

retains the stated finite number of H-representatives and compactness of their actual operator quotients.

Manuscript: Section23, folios140-141, scans279-281; Part C(b), infinite-operator zero-level comparison, global quotient maps and arbitrary accumulation under finitely many operator representatives.

Proof. The global quotient family and its exact comparisons. C.42 is already stated for every $z \in \mathrm{Fix}(J)$ with nontrivial line or circle orbit; it imposes no frontier condition on z. It gives the actual continuous open map (C.53.1), its intrinsic target topology, the common covering construction, and its proper fiber lines. C.43–C.45 use these maps and their transitive compatible action to give the connected real fibers, product coordinates and the exact three pairwise coordinate domains. Thus each of those conclusions applies to (C.53.1) with the same maps.

Changing the source basepoint also preserves the complete family. If $y = ax$ for $a \in L$, then $L_y = aJa^{-1}$ and, for $z \in \mathrm{Fix}(L_y)$,

$$\Phi^y_z(ly) = lz = \Phi^x_{a^{-1}z}(lax). \tag{C.53.3}$$

The targets are the same actual G-orbit, because $Lz = Gz$ for each such quotient, as proved in C.42. This proves the asserted identification of partitions, including the original target coverings.

For $h \in H$ stabilizing O, choose $a_h \in L$ with $a_h x = hx$. Exact commutation gives $L_{hx} = L_x$, hence $a_h J a_h^{-1} = J$, and

$$\Phi^x_z h = \Phi^x_{a_h z}. \tag{C.53.4}$$

The point $a_h z$ is fixed by J and lies on the same nonopen orbit as z. Therefore (C.53.4) permutes the entire family $\mathscr{F}^{\mathrm{all}}_O$. For $hO = O'$, use basepoint hx in O'; then

$$\Phi^{hx}_{hz} h = h\Phi^x_z. \tag{C.53.5}$$

These equalities identify the families under operator transport and prove that their cardinalities are well defined and H-invariant.

Three global quotients give the same actual Lie group. Choose three distinct partitions in $\mathscr{F}^{\mathrm{all}}_O$. The full three-foliation group of C.47 is constructed exclusively from the real maps in C.43–C.45 and their common transitive action. Its proper-pair or ordered-loop argument therefore applies to these exact maps and gives the same locally compact Lie group of dimension two or three. The based target lifts in C.49.2 are loops: for circle targets C.29 uses the ambient-null plane loop and two points of that circle, independently of the position of O; real target endpoints are the identity. The Baire transitivity proof C.49.3, its isotropy calculation C.49.4–C.49.7 and its closedness argument now apply to the same based source lifts. Their actual generated group N is consequently one of $\mathbb{R}^2$, $\mathrm{Aff}^+(\mathbb{R})$, or $\mathcal{A}_2$, with exactly the topology, global exponential and unique logarithm proved in C.49.

We verify the action comparison for every global quotient, rather than only for frontier targets. Use the closed Polish group

$$E_O = \{(l, n) \in L \times N : pn = (l|_O)p\}. \tag{C.53.6}$$

Its projection to N is onto, since every n is an actual word in the lifted loop; the identical global word supplies $l \in G$. It is an open homomorphism of Polish groups. If $(l, 1)$ is in its kernel, l fixes O pointwise. Equivariance of (C.53.1) then makes l fix its entire image $B = Gz$ pointwise:

$$l\Phi^x_z(u) = \Phi^x_z(lu) = \Phi^x_z(u). \tag{C.53.7}$$

C.42 proves continuity of $L \to \mathrm{Homeo}(B)$ in the intrinsic compact-open topology. Its restriction to E_O therefore factors continuously through the open quotient to an actual action of N on B. That image is exactly $G|_B$: actual loop words prove both inclusions. It is metabelian and hence

is the translation, affine, or rotation Lie image of C.37. The resulting continuous homomorphism and its unique based covering lift give exactly

$$\Phi_z^x K_{s,t}^O = R_{s,t}^B \Phi_z^x, \qquad K_{s,t}^O = \exp_N((1-s)\log_N \widetilde{f}_t) \text{ descended to } O. \tag{C.53.8}$$

Here R^B is the exponential branch of C.52.1, with the full based real angle for a circle. The original source covering, every original quotient, and every restriction kernel are retained.

The enlarged closed exponential region. For a dimension-two proper-operator orbit use the based lifted contraction of C.36–C.37. For an orbit with three global partitions use (C.53.8). Where both constructions apply, their source groups are the same actual group generated by the same based lifted loop, as in C.49, so the maps agree. Operator conjugation transports these groups and intertwines their exponential maps, by (C.53.5) and the based covering identity C.49.16.

Every source group just specified has the uniform fractional-word bound 1913: C.37 gives the stronger bound three for the first two groups, and C.50.1–C.50.14 give the stated bound for $\mathcal{A}_2$. Each word is a word in the original based path and its inverse. In particular the full argument C.50.15–C.50.19 applies to this enlarged union with precisely the same compact set of actual global word isotopies. We spell out the transfer of its remaining inputs. All the source images are metabelian, so each global commutator-of-commutators fixes $U^\dagger$, hence its closure. At a nonopen limit orbit the actual transitive image is therefore the same translation, affine or rotation group classified in C.37. For every integer n, the residual after removing $\lfloor n(1-s)\rfloor$ endpoint powers is a fractional power in the specified simply connected source group, and thus has the same uniform word bound. On a line the bounded residual logarithms force the exponential limit. On a circle the source residual word loop is null in that same simply connected group; its exact two-point winding passes to the limiting orbit, giving the real equality C.50.19, including its integer part. These are all inputs of C.50's limit identification and joint continuity proof. Applying it also to the inverse loop gives a jointly continuous inverse on $X^\dagger$.

The original $U_2 \cup U_3$ lies in $U^\dagger$. On that original union the maps are the same source fractional powers. Continuity extends this equality to X_*. The operator identities also extend from the dense union. This proves the first assertion and fixes its prescription for the rest of the proof.

Compatible intrinsic maps on the remaining representatives. Each O_j in (C.53.2) has $|\mathscr{F}_{O_j}^{\mathrm{all}}| \le 2$ and $\dim Q_{O_j} \le 1$, by its disjointness from $X^\dagger$. For a family of size one or two, apply the construction of C.52 to this full global family. Here is the exact transfer. Its source coordinates and operator normal forms use C.42–C.45, already verified above for (C.53.1). Its full-partition invariance is (C.53.4). The base-family replacement C.52.5 is natural under every target reparametrization and exchange, and retains every affine operator equation C.52.6. No step of those intrinsic constructions uses the position of the target point on a frontier. They therefore give a based intrinsic contraction $K_{s,t}^j$, jointly continuous with its inverse, commuting with H_{O_j} and with its actual restricted closure Q_{O_j}, and satisfying

$$\Phi_z K_{s,t}^j = R_{s,t}^{Gz} \Phi_z \quad \text{for every map in (C.53.1).} \tag{C.53.9}$$

For an empty family, C.32–C.33 give the based intrinsic contraction in that exact proper-operator centralizer. Every constant equivariant quotient also satisfies (C.53.9), since its target is fixed.

Discard repetitions among the finitely many H-orbits represented by O_j, and transport K^j to each hO_j by hK^jh^{-1}. Commutation with the full stabilizer makes this well defined. Its union gives a jointly continuous homeomorphism family K on the open set U; components are open, and each carries its intrinsic family and inverse. It commutes with H exactly.

One compact comparison set for every accumulating sequence. For each representative, compactness of O_j/Q_{O_j} supplies a compact $C_j \subset O_j$ with $Q_{O_j}C_j = O_j$, by C.51's finite-cover argument. Suppose

$$x_n \in U, \quad x_n \longrightarrow z \in X \setminus U, \qquad (s_n, t_n) \longrightarrow (s,t). \tag{C.53.10}$$

The orbit Gz is nonopen. Indeed an open orbit containing z would contain every sufficiently late x_n, and so would itself be one of the components in U, contradicting $z \notin U$. There are only finitely many representatives. On a subsequence, write

$$x_n = h_n q_n y_n, \quad h_n \in H, \quad q_n \in Q_{O_j}, \quad y_n \in C_j, \quad y_n \longrightarrow y \in C_j \tag{C.53.11}$$

for one fixed j. Fix the compact subset of O_j

$$S_j = C_j \ \cup \bigcup_{(u,v)\in[0,1]^2} (K^j_{u,v}(C_j) \cup (K^j_{u,v})^{-1}(C_j)). \tag{C.53.12}$$

Both joint continuity assertions prove compactness.

Choose actual $a_n \in H_{O_j}$ so that

$$\sup_{w\in S_j} |h_n a_n w - h_n q_n w| < 1/n, \qquad g_n = h_n a_n \in H. \tag{C.53.13}$$

For each fixed n, the first condition is an open neighborhood of q_n in $\mathrm{Homeo}(O_j)$: composition with the fixed ambient homeomorphism h_n and restriction to the compact set S_j are continuous. The defining density of the actual restrictions $H_{O_j} \to Q_{O_j}$ supplies a_n. Thus the estimate is made *after* the possibly distorting operator h_n; no uniform modulus of continuity for the sequence h_n is assumed.

The open L-orbit map supplies $l_n \to 1$ with $l_n y = y_n$, exactly as in C.51. Since g_n commutes with L, (C.53.10)–(C.53.13) give

$$g_n y = l_n^{-1} g_n y_n \longrightarrow z. \tag{C.53.14}$$

Every element of L_y fixes this limit. The evaluation homeomorphism C.42.1 therefore gives

$$g_n|_{O_j} \longrightarrow \Phi^y_z \quad \text{uniformly on compact sets.} \tag{C.53.15}$$

The exact basepoint change (C.53.3) identifies this map with one of the global quotients in (C.53.1) when its target is nontrivial; otherwise it is constant. This is the reason the enlarged family is required in this argument.

Transport and commutation with Q_{O_j} now yield

$$\begin{aligned} K_{s_n,t_n}(x_n) &= h_n q_n K^j_{s_n,t_n}(y_n), \\ |K_{s_n,t_n}(x_n) - g_n K^j_{s_n,t_n}(y_n)| &< 1/n. \end{aligned} \tag{C.53.16}$$

The inner points converge in S_j to $K^j_{s,t}(y)$. Equations (C.53.15) and (C.53.9) consequently give

$$K_{s_n,t_n}(x_n) \longrightarrow \Phi^y_z K^j_{s,t}(y) = R^{Gz}_{s,t}(z). \tag{C.53.17}$$

Every subsequence admits a further subsequence of this form, by the finite number of j and the compactness of their C_j. The value in (C.53.17) is independent of every such choice. Hence the entire sequence has that limit. If a representative has an empty global family, (C.53.15) also proves that every limit reached through its translates is a fixed point.

Define the extension on $X \setminus U$ to be $R^{Gz}_{s,t}(z)$. The diagonal approximation C.52.9 now applies with the dense set U in X: for each varying boundary point and fixed parameter pair choose a point of U close to it and with close image, using (C.53.17). Applying (C.53.17) to the resulting sequence proves joint continuity on all of X. This argument does not require prior transverse continuity of the hybrid rule.

Use the inverse images already present in S_j and the exact inverse version of (C.53.9) to repeat (C.53.13)–(C.53.17) for inverse maps. The same diagonal approximation then proves joint inverse continuity on X. The maps preserve U and each nonopen limit orbit, with the displayed

maps as mutual inverses. Thus they are homeomorphisms of X. Their based-edge identities and H-commutation extend from U by continuity; the restriction to every nonopen orbit is the prescribed R^B.

Finally, U is disjoint from $X^\dagger$. Every point of $X \cap X^\dagger$ lies on a nonopen orbit; otherwise its open orbit would meet U and be a component of U. The metabelian identity on $X^\dagger$ and the first part of the proof show that both contractions there use precisely the same exponential branch of C.52.1, with the same based covering path. They and their inverses agree on the full intersection. Pasting the two closed sets X and $X^\dagger$ gives the asserted contraction on their union, preserving the prescribed family on $X^\dagger$.

The finite-representative argument does not supply compact comparison sets for arbitrary noncocompact quotients or infinitely many H-inequivalent representatives. Nor does it extend the family through an exterior region of nonopen orbits. These remain exact parts of the general C.12 obligation. □

4 D. Profinite extensions and the limits of the realization problem

We write $P = \widehat{\pi}_0$, where $\pi_0 = \pi_1(U, p)$ for a connected orientable surface of finite type with $\chi(U) < 0$. All maps between profinite groups in this section are continuous. A lift of an outer action means a homomorphism inducing the prescribed outer homomorphism, with its labels in the operator group retained. For a finite group G acting faithfully on U, E denotes the group of lifts to the universal cover, not a mapping class group. The completion map $E \to \widehat{E}$ is injective: E is a finitely generated linear hyperbolic orbifold group and is residually finite. We use E^+ for its orientation-preserving subgroup. The passages answered here are §31, folios 198–199, scans 391–394; the inserted $G = G^+$ on scan 391 is material.

D.1. [Proved] Exact completion of the lift extension. The completion of $1 \to \pi_0 \to E \xrightarrow{q} G \to 1$ is the exact sequence

$$1 \longrightarrow P \longrightarrow \widehat{E} \xrightarrow{\widehat{q}} G \longrightarrow 1.$$

It is the extension attached to the completed outer action. An invariant discretization therefore supplies a canonical comparison of extensions, with identity maps on P and G.

Manuscript: 31; 198–199 verso; 391–394.

Proof. Every subgroup of finite index in π_0 contains a subgroup normal and of finite index in E: take its core in E, using $[E : \pi_0] < \infty$. Thus π_0 has its full profinite topology induced from E, and its closure is P. The kernel of $\widehat{q}$ is that closure, since $\pi_0 = q^{-1}(1)$ is open. Conjugation in the completed extension is the continuous extension of conjugation in the dense subgroup E. The uniqueness of a center-free kernel extension, proved in Part A, identifies this extension with the pullback of $\mathrm{Aut}(P) \to \mathrm{Out}(P)$. The identification fixes the named kernel and quotient, so the assertion of canonicity has no unmentioned conjugator. □

D.2. [Proved] Finite subgroups of completed surface orbifold groups, including reflections. Every finite subgroup of $\widehat{E}$ is conjugate to a finite subgroup of E. Two finite subgroups of E conjugate in $\widehat{E}$ are conjugate in E. These finite subgroups are cyclic rotation groups or dihedral groups, with a single reflection included as the dihedral case of order two.

Manuscript: 31; 198–199 verso; 391–394.

Proof. We specify the two inputs and the reduction. Bridson–Conder–Reid, Theorem 5.1 [13], applies to the finitely generated Fuchsian group E^+: every finite subgroup of $\widehat{E^+}$ is conjugate

into E^+. Boggi–Zalesskii, Theorem 6.3 [9], gives conjugacy and normalizer comparison for finite p-subgroups of E:

$$N_{\widehat{E}}(A) = \overline{N_E(A)}.$$

Here its hypotheses hold because E has a finite-index surface group (a free group in the noncompact case), hence is virtually compact special; its good-group and finiteness hypotheses are precisely those in Theorem 6.3. We use only this prime-power theorem, not an assertion obtained by commuting arbitrary intersections with closure.

Put $K^+ = K \cap \widehat{E^+}$ for a finite $K \le \widehat{E}$. The first input makes K^+ cyclic. When $K^+ = 1$, the orientation character embeds K in C_2, and the prime-power input handles K. Otherwise choose the unique subgroup A of order p in K^+, for one prime $p \mid |K^+|$. It is characteristic in K^+ and normal in K. Conjugate A into E^+. A nontrivial elliptic rotation of the hyperbolic plane has one fixed point x. Its normalizer in the discrete group E stabilizes x, and conversely every element of E_x normalizes the unique subgroup of order p in its cyclic rotation subgroup. Thus $N_E(A) = E_x$ is finite cyclic or dihedral. The displayed comparison gives $N_{\widehat{E}}(A) = E_x$, so $K \le E_x \le E$.

For injectivity, suppose $K, L \le E$ and $tKt^{-1} = L$, with $t \in \widehat{E}$. The orientation character gives $tK^+t^{-1} = L^+$. In the case $K^+ = 1$ apply prime-power conjugacy separation. Otherwise choose $A \le K^+$ as above and $B = tAt^{-1} \le L^+$. Prime-power conjugacy separation supplies $e \in E$ with $eAe^{-1} = B$. Now $e^{-1}t$ normalizes A, hence belongs to $N_E(A) \subset E$. Therefore $t \in E$, after this adjustment. This proves the required conjugacy assertion. Finally finite discrete hyperbolic-plane isometry groups stabilize a point and embed in $O(2)$; the listed cyclic and dihedral possibilities follow. □

D.3. [Proved] The comparison of full and partial section classes. For every subgroup $H \le G$, completion induces a bijection

$$\mathrm{Sec}(E_H \to H)/\pi_0 \;\longrightarrow\; \mathrm{Sec}(\widehat{E}_H \to H)/P, \qquad E_H = q^{-1}(H).$$

Both sets are finite. In particular the first conjecture on scan 393 holds for geometrically realized actions, with orientation-reversing operators allowed.

Manuscript: 31; 198–199 verso; 391–394; 18; 99–101; 197–203.

Proof. We prove the change from total-group conjugacy to kernel conjugacy, since omission of this step changes the question. A profinite section has a finite image K mapping isomorphically to H. Apply D.2 to E_H, and choose $t \in \widehat{E}_H$ with $tKt^{-1} \le E_H$. Choose $e \in E_H$ with $q(e) = \widehat{q}(t)$. Then $e^{-1}t \in P$, and conjugation by this element puts K in E_H. The restriction of q identifies the resulting subgroup with H, so it defines a discrete section with the correct labels.

Conversely, let discrete section images K, L be conjugate by an element of P. D.2, applied in E_H, gives an $e \in E_H$ with $eKe^{-1} = L$. Let $l \in L$ be the unique element satisfying $q(l) = q(e)$. Then $l^{-1}e \in \pi_0$ still conjugates K to L. Because the projection is injective on each section image, this equality of subgroups is equality of the labelled section homomorphisms after conjugation. This proves injectivity.

There are finitely many E_H-conjugacy classes of finite subgroups: they occur in the finite list of cone stabilizers, corner stabilizers, and mirror components of a finite-type quotient orbifold, and each finite stabilizer has finitely many subgroups. Each such class splits into at most $[E_H : \pi_0] = |H|$ kernel-conjugacy classes. The asserted finiteness follows. □

D.4. [Proved] The exact profinite fixed subgroup of a section. Let $s : H \to E_H$ be a nontrivial discrete section. Its fixed subgroup $P^{s(H)} = C_P(s(H))$ is trivial when $s(H)$ contains a nontrivial rotation. For a reflection $s(H) = \langle t \rangle$, it is 1 for a noncompact real component and is

the closure of the primitive oval subgroup, isomorphic to $\widehat{\mathbb{Z}}$, for a compact real component. The same assertions hold for every profinite section by D.3.

Manuscript: 31; 199; 393; 33; 222 verso–225; 440–445.

Proof. For a rotation choose A of prime order inside $s(H)^+$. D.2's proof gives $N_{\widehat{E}}(A) = E_x$, a finite group. Thus $C_P(s(H)) \leq E_x \cap P = 1$, because P is torsion-free. For a reflection the prime-power centralizer comparison gives

$$C_{\widehat{E}}(t) = \overline{C_E(t)}, \qquad C_P(t) = \overline{C_{\pi_0}(t)}.$$

The second equality uses that P is open and $P \cap E = \pi_0$: the intersection of the closure of a subgroup with an open subgroup is the closure of their intersection. On the universal cover, $C_{\pi_0}(t)$ is the group of deck transformations preserving and commuting with reflection in its fixed line. It acts freely on that line. Its quotient is the corresponding real component of U. The deck group is trivial for a line and infinite cyclic for a circle; the circle is an essential primitive simple curve. Its induced profinite topology is full, so its closure is $\widehat{\mathbb{Z}}$.

This proves, rather than assumes, the correction to the expected answer in Part D(a): a dihedral section with a nontrivial rotation has fixed subgroup 1. The value $\widehat{\mathbb{Z}}$ belongs to a reflection in an oval. It is not the fixed subgroup of a larger dihedral stabilizer containing that reflection. □

D.5. [Refuted] Unique maximal sections require a rotation. A finite subgroup of $\widehat{E}$ containing a nontrivial rotation is contained in a unique maximal finite subgroup. The assertion for every nontrivial finite subgroup is false, already for type $(0, 3)$ with reflections.

Manuscript: 18; 99–101; 197–203; 31; 198–199; 391–393.

Proof. For the positive assertion conjugate the subgroup into E using D.2. Its rotation part has a subgroup A of prime order. Every finite overgroup has cyclic orientation-preserving part, so normalizes this same A. All finite overgroups therefore lie in the finite group $N_{\widehat{E}}(A) = E_x$, which is itself the unique maximal one.

For the counterexample use the $(2, 3, \infty)$ reflection group

$$W = \langle s_0, s_1, s_2 \mid s_i^2 = 1, (s_0 s_1)^2 = (s_1 s_2)^3 = 1 \rangle = V_4 *_{C_2} S_3.$$

Map it onto $S_3 \times C_2$, with $r^3 = a^2 = \tau^2 = 1$, $ara = r^{-1}$ and τ central, by $s_0 \mapsto \tau$, $s_1 \mapsto a\tau$, $s_2 \mapsto ra\tau$. The vertex groups inject. The kernel is an orientable torsion-free group of index 12, Euler characteristic -1, with three cusps; hence it is $\pi_{0,3}$. The two maximal finite subgroups $\langle s_0, s_1 \rangle$ and $\langle s_1, s_2 \rangle$ contain the same reflection s_1 and have distinct images in the quotient. They remain distinct and are not kernel-conjugate after completion. Their generated subgroup is W, so no finite overgroup contains both. Part B supplies the finite certificate and the geometric kernel computation. □

D.6. [Proved] A well-defined transfer of the geometric rotation invariant. For a realized E, its rotation invariant on finite-order elements transfers uniquely to the torsion of $\widehat{E}$ by conjugating each torsion element into E. This assertion uses the orientation module, so that conjugation by an orientation-reversing element acts by -1 on its values. Its values embed canonically in $\widehat{T} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$, where $\widehat{T} = T \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}}$.

Manuscript: 17; 77–79 verso; 155–160; 31; 198–199; 391–393.

Proof. Use D.2 for existence. Explicitly, for torsion $h \in \widehat{E}$ choose $t \in \widehat{E}$ with $u = tht^{-1} \in E$ and set

$$\mu_{\widehat{E}}(h) = \varepsilon(t)\mu_E(u),$$

where $\varepsilon : \widehat{E} \to \{\pm 1\}$ is the orientation character, acting on $\widehat{T}$. Suppose $u, v \in E$ and $v = tut^{-1}$ in $\widehat{E}$. For a rotation apply the proof of D.2 to an order-p subgroup of $\langle u \rangle$: first align it by discrete conjugation, then put the remaining conjugator in the finite discrete stabilizer. Thus the elements themselves, and not just their cyclic subgroups, are discretely conjugate, with the same orientation character of the conjugator. For reflections apply prime-power element conjugacy separation, or D.2 because an order-two subgroup has a unique nonidentity element. This establishes independence with the indicated action on T. A rotation of order n has a value in $T/nT \subset T \otimes \mathbb{Q}/\mathbb{Z}$; the identification $T/nT = \widehat{T}/n\widehat{T}$ defines the claimed inclusion. □

D.7. [Proved] Equivariance of the profinite rotation invariant. Let E be the lift group of a finite action on an orientable hyperbolic surface of finite type, and put $P = \widehat{\pi} \triangleleft \widehat{E}$. Every automorphism α of $\widehat{E}$ preserving P and its indexed peripheral structure satisfies

$$\mu_{\widehat{E}}(\alpha(u)) = \alpha_T(\mu_{\widehat{E}}(u)) \qquad (u \in \widehat{E} \text{ of finite order}).$$

Here α_T is the induced automorphism of the orientation line $\widehat{T}$ defined in A.2, and $\mu_{\widehat{E}}$ is the invariant of D.6. The assertion holds for compact and noncompact surfaces, and for actions containing orientation-reversing elements.

Manuscript: 17; 77–79 verso; 155–160; 31; 198–199; 391–393.

Proof. Choose an orientation, so that α_T is multiplication by some $\lambda \in \widehat{\mathbb{Z}}^\times$. Conjugation by an element of $\widehat{E}$ acts on $\widehat{T}$ by its geometric orientation sign. Naturality of this action and commutativity of the automorphism group of a rank-one module show that α preserves that sign. Thus α restricts to an automorphism of $\widehat{B}$, where $B = E^+$ is the orientable Fuchsian subgroup. A nonidentity negative torsion element is a reflection and has invariant $1/2$; multiplication by a profinite unit fixes this value. It remains to treat torsion in $\widehat{B}$.

We first make explicit the behavior of orientation under finite covers. For a degree-d connected finite cover of oriented surfaces, the map on orientation homology takes the fundamental class upstairs to d times the fundamental class downstairs. For punctured surfaces this follows directly in A.2's kernel model: the peripheral factors above any one puncture map by their positive local degrees, whose sum is d. For closed surfaces it is the degree formula for fundamental classes. The same formulas hold over $\widehat{\mathbb{Z}}$, either by this peripheral matrix or by the natural finite-coefficient cohomology comparison. Consequently, if α carries an open surface subgroup V to V', the multipliers on their orientation lines and on P satisfy $d\lambda_V = d\lambda$. Multiplication by a positive integer is injective on $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$, so $\lambda_V = \lambda$. This proves the required compatibility with further finite covers; it does not divide by d in a finite coefficient ring.

The noncompact case. Let L be a peripheral subgroup of π, and let K be the primitive cusp subgroup of B containing it. Write $K = \langle w \rangle$ and $L = \langle w^n \rangle$, with compatible positive generators. We have

$$N_{\widehat{B}}(\overline{L}) = \overline{K} \simeq \widehat{\mathbb{Z}}.$$

Here is the reduction from Part A, rather than an assumption that arbitrary normalizers commute with completion. For an element of the normalizer, choose a representative in B of its coset modulo P. That coset stabilizes the profinite peripheral class. Distinct discrete peripheral classes remain distinct in P, so the same coset stabilizes the discrete peripheral class. Adjust its representative by an element of π to put it in $N_B(L) = K$. The remaining factor belongs to $N_P(\overline{L}) = \overline{L}$, as proved in Part A. This proves the displayed equality.

The loop condition on $\alpha|_P$ sends a positive generator of $\overline{L}$ to a P-conjugate of the λth power of the corresponding generator of another peripheral subgroup $\overline{L}'$. After making this conjugation, α sends $\overline{K}$ to $\overline{K}'$ and has the form $w \mapsto (w')^a$. The indices $[\overline{K} : \overline{L}] = [\overline{K}' : \overline{L}'] = n$ are equal because α preserves P. Thus $na = n\lambda$ in $\widehat{\mathbb{Z}}$, and hence $a = \lambda$. Every primitive orbifold cusp therefore has precisely the same multiplier λ.

Write the signature of B as $(h; m_1, \dots, m_r; s)$ with $s \geq 1$. Choose cone generators c_j, of orders m_j, and cusp generators w_i in a common orientation. The abelianized presentation is

$$\widehat{B}^{\mathrm{ab}} = \left(\widehat{\mathbb{Z}}^{2h} \oplus \bigoplus_{j=1}^{r} \mathbb{Z}/m_j \, c_j \oplus \bigoplus_{i=1}^{s} \widehat{\mathbb{Z}} \, w_i \right) / \left\langle \sum_j c_j + \sum_i w_i \right\rangle .$$

D.2 applied to B makes α permute the maximal cone subgroups. Hence, up to conjugacy in $\widehat{B}$, $\alpha(c_j) = c_{\tau(j)}^{k_j}$, where $m_{\tau(j)} = m_j$ and k_j is a unit modulo m_j. The preceding paragraph gives $\alpha(w_i) \sim w_{\sigma(i)}^{\lambda}$. Apply α to the displayed relation and subtract λ times that relation. This yields

$$\sum_j (k_j - \lambda) c_{\tau(j)} = 0.$$

Eliminating w_s exhibits the cone factors as an independent direct sum, so $k_j \equiv \lambda \pmod{m_j}$ for every j. The local formula of B.12 now proves the assertion on every cone subgroup and, by conjugation in $\widehat{B}$, on all its torsion.

The compact case. First suppose that there are at least two cone points. We use Wilkes's cohomological calculation with its exact scope: for a closed orientable hyperbolic orbifold group B with cone generators c_j of orders m_j, an automorphism of $\widehat{B}$ sending each c_j to a conjugate of $c_j^{k_j}$ acts on $H^2(\widehat{B}; \mathbb{Z}/n)$, for every n, by one scalar $\kappa \in \widehat{\mathbb{Z}}$ satisfying $\kappa \equiv k_j \pmod{m_j}$. This is [57, Proposition 5.7], numbered Proposition 6.6 in arXiv:1512.05587v1. Its proof concerns arbitrary automorphisms of the indicated form; no lifting to a Seifert group is used in that calculation. In particular the scalar is independent of n.

For transparency, the part of the calculation that supplies both facts is as follows. After taking coinvariants, the relation cells y_0, y_j have boundaries $\partial y_0 = \sum_j x_j$ and $\partial y_j = m_j x_j$. The automorphism sends x_j to $k_j x_j$, y_j to $k_j y_j$, and y_0 to $\kappa y_0 + \sum_j \rho_j y_j$. Commutation with the boundary gives $\kappa + m_j \rho_j = k_j$ in $\widehat{\mathbb{Z}}$. The dual coboundaries are $y_0^* + m_j y_j^*$, which shows that the induced action is multiplication by κ for every finite coefficient group. The completed resolution and the naturality of this calculation are established in the cited proposition and its use of Proposition 2.9 there.

The cone permutation of our α is realized by an orientation-preserving homeomorphism of the underlying surface carrying the marked cone points to the correspondingly ordered points. It induces an orbifold automorphism β with the same permutation and power 1 on each cone. Put $\psi = \beta^{-1}\alpha$ and apply the preceding result to ψ. Let

$$\eta_n \in H^2(\widehat{B}; \mathbb{Z}/n)$$

be the image of the oriented fundamental cohomology class of the underlying closed surface under the canonical map from the orbifold classifying space to its underlying space. Equivalently, in the displayed resolution it is $[y_0^*]$, up to the common orientation sign. The finite-coefficient class is transported to continuous cohomology by goodness of Fuchsian groups, as in Wilkes's calculation. Since β preserves orientation, $\beta^* \eta_n = \eta_n$. Thus $\alpha^* \eta_n = \psi^* \beta^* \eta_n = \kappa \eta_n$. This argument does not require β to preserve the chosen subgroup P.

Set $N = [B : \pi]$. Restriction to the surface subgroup gives

$$\mathrm{res}_P(\eta_n) = N \omega_n, \qquad \omega_n \in H^2(P; \mathbb{Z}/n)$$

with ω_n the positive surface class: the underlying map from the covering surface to the orbifold's underlying surface is an orientation-preserving branched map of degree N. Naturality of restriction and A.2's duality yield

$$N\kappa\,\omega_n = N\lambda\,\omega_n \quad \text{for every } n.$$

Indeed $\alpha_*[U] = \lambda[U]$ implies $\langle \alpha^*\omega_n, [U]\rangle = \langle \omega_n, \alpha_*[U]\rangle = \lambda$. Taking all finite coefficients gives $N(\kappa - \lambda) = 0$ in $\widehat{\mathbb{Z}}$, hence $\kappa = \lambda$. The local congruences are therefore $k_j \equiv \lambda \pmod{m_j}$, exactly as required.

We remove the single-cone case from the resolution input explicitly. When $r = 1$, the underlying genus h is positive and $B^{\mathrm{ab}} = \mathbb{Z}^{2h}$, because the one cone generator is zero in abelianization. The characteristic subgroup

$$B_1 = \ker(B \longrightarrow B^{\mathrm{ab}}/2B^{\mathrm{ab}})$$

has index 2^{2h} and contains every conjugate of the cone group. Its orbifold has 2^{2h} cone points of the same order. The subgroup $\widehat{B_1}$ and $P \cap \widehat{B_1}$ are α-invariant. The argument for at least two cones applies there, and the finite-cover calculation at the beginning identifies its surface multiplier with λ. For $r = 0$ there is no nonidentity torsion in $\widehat{B}$, by D.2. Thus no exceptional compact signature remains.

Finally, choose positive local cone generators, for which $\mu(c_j) = 1/m_j$ in the chosen orientation line. The proved congruence says $\mu(\alpha(c_j)) = \lambda/m_j$; its powers have the same property. Every positive torsion element is conjugate in $\widehat{B}$ to one of these powers by D.2. Together with the reflection case this proves the stated equation on all torsion of $\widehat{E}$. □

Why the earlier rank-one argument was insufficient. Integral relative orbifold cohomology can have a torsion summand: for $C_3 * C_3$ with cusp $\langle ab\rangle$ it has the presentation $\langle u, v, w \mid 3u = w = 3v\rangle \cong \mathbb{Z} \oplus \mathbb{Z}/3$. The proof above uses the cusp relation in the noncompact case and the action on the full finite-coefficient cohomology in the compact case. It therefore supplies the missing local-power congruences without choosing an unproved canonical generator through that torsion summand.

4.1 Completion of geometric mapping class groups and full loop automorphisms

D.8. [Proved] All finite subgroups in the geometric completion in the three requested types. Let $M = \Gamma_{g,\nu}$ include puncture permutations and orientation reversal. For $(g, \nu) = (0, 3), (0, 4), (1, 1)$, completion induces a bijection on conjugacy classes of finite subgroups of M and $\widehat{M}$. Every finite subgroup of $\widehat{M}$ is solvable. For every finite $K \leq M$,

$$N_{\widehat{M}}(K) = \overline{N_M(K)}, \qquad C_{\widehat{M}}(K) = \overline{C_M(K)}.$$

These assertions concern $\widehat{M}$ with its given embedding in $\mathrm{Out}_{\mathrm{lac}}(P)$.

Manuscript: 30; 194 verso; 384; 30; 196 verso; 388.

Proof. For the three-punctured sphere $M = S_3 \times C_2$ is finite. For the punctured torus $M = \mathrm{GL}_2(\mathbb{Z})$; its principal congruence subgroup of level 3 is a torsion-free free group of finite rank. For the four-punctured sphere the subgroup fixing each puncture and preserving orientation is a free group of rank two and has finite index. Thus all three M are finitely generated virtually free groups.

Bridson–Piwek, Proposition 5.10 [14], states for every finitely generated virtually free group that finite subgroup conjugacy classes are preserved bijectively by profinite completion. Its proof

uses the finite graph of finite groups for M: completion gives the profinite fundamental group of the same graph, a finite subgroup fixes a vertex of its profinite Bass–Serre tree, and subgroup conjugacy separability distinguishes the resulting discrete subgroups. This applies without an orientation restriction and proves the first assertion. The normalizer equality is Ribes–Zalesskii, Theorem 2.6 [55], for finitely generated subgroups of finitely generated virtually free groups. To deduce the centralizer equality, the homomorphism $N_{\widehat{M}}(K) \to \mathrm{Aut}(K)$ has finite discrete target. Its kernel is open; the intersection of a dense subgroup with this open kernel is dense in the kernel. This is exactly $C_M(K)$.

For solvability, a finite subgroup of $M_{0,4}$ embeds in $S_4 \times C_2$ by the torsion-free pure oriented kernel; one in $\mathrm{GL}_2(\mathbb{Z})$ embeds in $\mathrm{GL}_2(\mathbb{F}_3)$ by the level-3 kernel. The former group is solvable. The latter has normal series with factors inside C_2, A_4, and the central C_2, since $\mathrm{PGL}_2(\mathbb{F}_3) = S_4$; it too is solvable. The finite case $S_3 \times C_2$ is immediate. Transfer by the proved conjugacy bijection gives solvability in $\widehat{M}$. □

D.9. [Proved] The nonsolvable discrete possibilities in genus at most two. A nonsolvable finite subgroup of the extended mapping class group of a punctured orientable surface of genus at most two is realized on a sphere and has group A_5 or $A_5 \times C_2$. Its puncture set is invariant under the corresponding icosahedral action. In particular it has at least twelve punctures. This classification does not assert the absence of additional nonsolvable finite subgroups in completions for all larger puncture numbers.

Manuscript: 30; 194 verso; 384.

Proof. Realize the finite subgroup by Part C. On the sphere its orientation-preserving subgroup is cyclic, dihedral, A_4, S_4, or A_5, by the finite rotation-group classification. The first four are solvable and an extension of one of them by an orientation group of order at most two is solvable. The orthogonal normalizer of the icosahedral rotation group is $A_5 \times \{\pm I\}$: a normalizing orthogonal transformation preserves its rotation axes and its icosahedral cellulation, and composition with an icosahedral rotation leaves either I or $-I$. A nonsolvable full group is consequently A_5 or $A_5 \times C_2$. The smallest nontrivial orbit has twelve vertices; an invariant puncture set has at least twelve points.

On a torus the kernel of the action on H_1 consists of a finite translation group, which is abelian. The orientation-preserving linear quotient is cyclic of order $1, 2, 3, 4$, or 6, and the extended linear quotient is cyclic or dihedral. These groups and their extensions by the translation group are solvable.

In genus two the hyperelliptic involution is central in the full automorphism group of the invariant Riemann surface. Adjoin it when it is absent. The quotient is a sphere with six branch points. A nonsolvable orientation-preserving quotient would contain A_5, whose smallest orbit has twelve points, contradicting preservation of the six branch points. The quotient is therefore solvable; so is its central extension by C_2 and its extension by orientation reversal. Punctures can only reduce the finite group. This proves the asserted list and its restriction to the discrete setting. □

D.10. [Proved] Finite torsion in the odd-prime normalized pro-p centralizer. Let p be odd, let $F^{(p)}$ be the free pro-p group on x, y, and define $GT_{0,p}$ by the normalized automorphisms

$$F_{\lambda,f}(x) = x^{\lambda}, \qquad F_{\lambda,f}(y) = f^{-1}y^{\lambda}f, \quad \lambda \in \mathbb{Z}_p^{\times}, \quad f \in (F^{(p)})',$$

satisfying the two-cycle and three-cycle equations. Every finite subgroup of $GT_{0,p}$ is cyclic of order dividing $p-1$. There is a subgroup of order $p-1$, and every finite subgroup is conjugate in $GT_{0,p}$ to a subgroup of it. For $p \geq 5$ this supplies finite loop-preserving outer automorphism groups which are not conjugate to geometric groups of the three-punctured sphere.

Manuscript: 25 bis; 157 verso–160; 312–317; 30; 194 verso; 384.

Proof. The normalized description is faithful: a conjugator between two normalized automorphisms first belongs to $C(x) = \langle x \rangle$; the normalization $f \in F'$ and the y equation remove its exponent. The same observation identifies these automorphisms with the pure outer centralizer of the three puncture permutations. The equations are closed conditions, composition is the usual substitution law, and their inverses satisfy the same equations. Hence this is a profinite group. The Galois action on $\mathbf{P}^1_{\overline{\mathbb{Q}}} - \{0, 1, \infty\}$ gives elements of this group with multiplier the p-adic cyclotomic character. This character surjects onto $\mathbb{Z}_p^\times$: the finite cyclotomic extensions have Galois groups $(\mathbb{Z}/p^n)^\times$, and their inverse limit is $\mathbb{Z}_p^\times$.

The kernel of $\mathrm{Aut}(F^{(p)}) \to \mathrm{Aut}(F^{(p)}/\Phi(F^{(p)}))$ is pro-p; the exact statement and its application are also given in Herfort–Zalesskii, proof of Theorem 6.3 [30]. For the normalized group the action on this Frattini quotient is the scalar λ mod p. Thus the kernel of

$$GT_{0,p} \longrightarrow \mathbb{F}_p^\times$$

is pro-p, and the map is surjective. The profinite Schur–Zassenhaus theorem supplies a complement C_{p-1} and conjugacy of complements in the inverse image of any subgroup of $\mathbb{F}_p^\times$. One can see the passage from the finite theorem directly: apply the finite theorem to every quotient by an open normal subgroup of the pro-p kernel, then take the inverse limit of the nonempty finite sets of compatible complements and conjugators. The finite intersection property follows by passing to a common finite quotient.

We check the necessary absence of torsion in the kernel of the multiplier. A free pro-p group has its injective Magnus map into $\mathbb{Z}_p\langle\langle X, Y \rangle\rangle^\times$, with $x \mapsto 1 + X$, $y \mapsto 1 + Y$. The map is injective because reduction modulo p is the usual faithful free pro-p Magnus map; it extends continuously from the free group for the (p, X, Y)-adic topology on the power-series algebra. The ordinary homogeneous degree is used only in the following leading-term calculation, not as the topology for that extension. An automorphism acting identically on $F^{(p)}_{\mathrm{ab}} = \mathbb{Z}_p^2$ acts identically on the degree-one quotient of this power-series algebra. For a nonidentity such automorphism choose the least degree $d \geq 2$ of a nonzero correction to X or Y. The degree-d correction to its nth iterate is n times that correction: substitution of the higher terms can change only degrees above d. Each homogeneous coefficient module is a torsion-free $\mathbb{Z}_p$-module, so no positive iterate is the identity. Consequently the multiplier is injective on a finite subgroup.

The torsion subgroup of $\mathbb{Z}_p^\times$ is cyclic of order $p - 1$. Therefore every finite subgroup maps injectively to $\mathbb{F}_p^\times$. It is a complement in the inverse image of its image, so the preceding conjugacy theorem proves the full assertion. Finally the geometric pure group of type $(0, 3)$ consists of 1 and complex conjugation, with multipliers 1 and -1. A generator with multiplier a primitive $(p-1)$st root cannot be conjugated into it; the multiplier is unchanged by conjugation even in the full loop-preserving outer group. The first example has order four at $p = 5$. □

D.11. [Proved] Executed finite checks and their exact scope. The program `code/root/finite_checks.py` verifies the two-cycle and three-cycle equations in every Heisenberg group $H(\mathbb{F}_p)$ for $p = 5, 7, 11$ and every $\lambda \in \mathbb{F}_p^\times$, using

$$f(a, b) = [a, b]^{(\lambda^2-1)/24}, \qquad m = (\lambda - 1)/2.$$

In particular it verifies an order-four normalized automorphism at $(p, \lambda \bmod p) = (5, 2)$ and an order-three one at $(7, 2)$. The multipliers of the corresponding infinite finite-order subgroups are Teichmüller lifts of these residues; the integer $2 \in \mathbb{Z}_5^\times$ itself has infinite order. D.10, not the finite checks alone, proves the existence of compatible infinite pro-p lifts.

Manuscript: 25 bis; 159 verso–160; 316–317; 30; 194 verso; 384.

Proof. The program defines all p^3 elements as triples and uses $(a,b,c)(a',b',c') = (a+a', b+b', c+c'+ab')$. It enumerates inverses and powers and evaluates each side of both equations in that group. Its output records every multiplier and both truth values. Since all commutators are central, the associated automorphism is $(a,b,c) \mapsto (\lambda a, \lambda b, \lambda^2 c)$; the program checks multiplication preservation and computes its order. Division by 24 is only used at the three displayed primes. No assertion of this formula at 2 or 3 is made. At $p=3$, D.10 gives only order-two prime-to-p torsion. At $p=2$, the Magnus argument still proves that a finite subgroup injects into $\{\pm 1\}$, but the complement argument does not settle conjugacy of involutions in $GT_{0,2}$. □

D.12. [Open] Torsion in the full profinite GT_0 and the full low-type groups. D.29 refutes the finite geometric-recognition assertion for $(g,\nu) = (0,4)$ by an actual involution with a complete preserved peripheral system. We do not settle whether every nonidentity torsion element of the full profinite GT_0 is conjugate to $(-1,1)$, or the full finite-subgroup recognition questions for $(g,\nu) = (0,3), (1,1)$. The geometric-completion assertion in all three low types is proved in D.8, and the odd-prime analogue is decided in D.10. D.32 proves that every finite subgroup of the full pure three-cusp group, including GT_0, has order at most two; it gives the faithful map $G \to S_3 \times C_2$ for the full three-cusp group. D.34, D.36, D.41 and D.42 identify the global orientation character of every such finite group: every involution has its global 2-adic sign, and every order-three element has multiplier 1. D.35 identifies the exact remaining third-cusp conjugator equation, proves full generation from it, and identifies its conjugator sets in the pure group and in GT_0. D.38 gives a complete characteristic A_5^{19} quotient test, including proved exclusions of finite candidates from full lifts. D.44 solves the coboundary and the first symmetry equation simultaneously in the full group; the third-cusp equation and hence existence of the conjugator for each actual full action remain unresolved. D.33 proves that full homology detects every finite subgroup in type $(1,1)$ and that its order divides 24. D.43 proves that every involution in this full punctured-torus group has its global orientation sign. D.45 identifies the exact characteristic binary-icosahedral refinement and its full finite solution counts; the same cusp-triangle survivors pass. D.46 proves that every cyclic order-four outer action on the punctured-torus group has an actual order-four lift, using the three full section classes of its positive square. D.47 checks the exact conjugator coordinates in the binary refinement: precisely 10,125,000 of the 324,000,000 coordinate pairs pass, with the count unchanged by finite invertibility and both symmetry relations. D.48 proves that a full punctured-torus order-three action has exactly two section classes and a torsion-free cusp normalizer whose cube exponent is a 3-adic unit. D.49 computes the standard pure involution centralizer and the exact two-element fibres of its conjugacy orbit map. D.50 classifies the finite two-subgroups of the full punctured-torus group as exactly $1, C_2, C_4, C_2 \times C_2, D_8$, exhibits all five, and proves the global multiplier character on every four-group or dihedral group of order eight. D.51 identifies the second torsion-free normal index-three kernel of every order-three torus extension, with its exact type-$(1,3)$ intersection cover and three labelled peripheral normalizers. D.52 gives the twelve possible abstract finite-subgroup types, realizes nine of them, and proves the reflection multiplier for every subgroup of type S_3, D_{12}, D_{24}. D.53 proves the unique full labelled order-four section class and its exact normalizer. D.54 excludes every nonzero prime-to-three invariant character for a cubic torus action by an explicit peripheral norm comparison in a cyclic cover. D.55 supplies the corresponding square comparison, proves that every positive involution acts as $-I$ at every prime, and proves faithfulness of every primary homology map on every finite subgroup. The finite torus abstract types are consequently exactly the nine realized types $1, C_2, C_3, C_4, C_2 \times C_2, C_6, S_3, D_8, D_{12}$; the three extra types in the earlier allowed list are excluded. D.56 computes the prime-section permutations of every cyclic-six action and proves that its full extension has no order-six subgroup or finite-order representative of the outer generator. D.57 proves actual labelled lifts for every four-group and order-eight dihedral action, with the exact three-or-one and one full section counts. D.58 determines the exact negative-involution section counts and their full fixed-subgroup mod-two cohomology, and proves that the two internal

cusp-reflection classes fuse under full conjugacy. Pure-involution orbit exhaustion, the remaining scalar cubic and quartic torus multipliers, and geometric conjugacy remain unresolved.

Manuscript: 25 bis; 157 verso–160; 312–317; 30; 194 verso; 384.

Here is the exact obstruction reached, rather than an extrapolation from computation. The homomorphism from a free profinite group to the product of its maximal pro-p quotients does not detect all its finite quotients. Our program constructs A_5 from a 5-cycle and a 3-cycle, enumerates its sixty elements, and verifies that its commutator subgroup has sixty elements. Thus A_5 is a nontrivial two-generated finite quotient invisible to every nilpotent quotient. The pro-p complements constructed in D.10 do not give a compatible solution of the defining equations in such quotients.

We also checked the $f = 1$ attack: the three-cycle relation becomes the power-product relation of Lochak–Schneps, Lemma 9 [41]; their finite modular quotients force the global alternatives $\lambda = 1$ or -1. A choice of independent signs in the factors of $\widehat{\mathbb{Z}}$ is not a solution of this relation. The finite nilpotent search therefore supplies no full-profinite counterexample. D.32 excludes every element of odd prime order in the full pure three-cusp group and every element of finite order greater than two. The smallest unresolved full-GT_0 case is therefore an additional involution up to conjugacy; D.34 proves that its multiplier is globally -1, supplies its three section classes and a commutator coboundary, and leaves the conjugator inside the specified loop group to be determined. D.23 supplies the exact order-bounded compatibility criterion over all isolated finite shadows. D.24 executes nonnilpotent refinement tests and proves that specific order-three shadows cannot lift with order three; D.25 computes the entire dihedral inverse limit and the exact residual multiplier-one norm equation. D.28 proves the additional prime-order section and fixed-subgroup constraint, including its exact maximal pro-p comparison. The finite-cover argument of D.32 now excludes all odd-order full lifts, beyond those bounded computations; the involution conjugacy problem retains its exact isolated-refinement formulation. D.29 uses an arbitrary Galois pants isomorphism and its inverse on two different pieces; this produces a finite involution in type $(0, 4)$ without requiring that pants isomorphism itself have finite order. For genus at most two and arbitrary puncture number, the solvable comparison is supplied by the checked argument in F.7. D.31 now proves that every nonsolvable finite subgroup in these completions is abstractly A_5 or $A_5 \times C_2$ on a punctured sphere, and gives exactly the geometric positive puncture-orbit sizes. Its proof also excludes every nonsolvable finite subgroup in the genus-one and genus-two completions. What remains is geometric conjugacy of the actual A_5 embeddings and the corresponding centralizer comparison of F.7.

The later tests also close two specific attempted constructions of a lower-type counterexample. D.39 constructs an actual rank-two Haran involution but proves, using its unique section class and the trivial centralizer of that section, that it preserves neither hyperbolic loop marking under any full profinite change of basis. D.37 gives the exact four-cusp descent condition for a negative transposition. D.40 excludes the required additional deck symmetry for the actual D.29 example by the unequal homomorphism counts 10200 and 4920 in $\mathrm{PSL}_2(\mathbb{F}_{11})$; the argument also excludes deck maps moving the original separator. These are proved obstructions for the specified constructions. They do not provide the conjugator still required in D.35.

4.2 Descent along a finite covering

Put $A(P) = \mathrm{Aut}_{\mathrm{lac}}(P)$ and let a discretization mean the image of a loop-preserving isomorphism $\widehat{\pi_0} \to P$ restricted to π_0. Thus it is an actual dense subgroup with its peripheral classes. Define $\mathrm{Discr}(P)$ using actual subgroups, and define $\mathrm{Discr}^{\mathrm{out}}(P)$ by quotienting this set by conjugation by P. This convention is necessary for an outer action: an outer automorphism acts canonically on the second set. Write $J = \mathrm{Aut}_{\mathrm{lac}}(\pi_0)$, $B = \mathrm{Out}_{\mathrm{lac}}(P)$ and $M = \mathrm{Out}_{\mathrm{lac}}(\pi_0)$ for their embedded images.

D.13. [Proved] Restriction and normal descent with the peripheral step included. For an open subgroup $V \leq P$, restriction of the corresponding finite covering defines an injective map

$$\mathrm{Discr}(P) \longrightarrow \mathrm{Discr}(V), \qquad \Delta \longmapsto \Delta \cap V.$$

For normal V, its image consists exactly of the lattices $\Lambda \subset V$ whose outer deck action $P/V \to \mathrm{Out}_{\mathrm{lac}}(V)$ is induced by orientation-preserving loop automorphisms of Λ. The cyclic peripheral subgroups of the descended lattice have closures equal to the specified peripheral subgroups of P.

Manuscript: 28; 178–179; 352–354.

Proof. We prove the self-normalizer input, including the class-preserving step. Paris [53, Theorems 1.7 and 3.2] proves conjugacy p-separability for closed orientable surface groups and free groups. Thus two elements of Λ conjugate in $\widehat{\Lambda}$ are already conjugate in Λ.

For a free group $F_r = \langle x_1, \dots, x_r \rangle$, let φ preserve every conjugacy class. After an inner adjustment we have $\varphi(x_1) = x_1$. Write $\varphi(x_i) = w_i x_i w_i^{-1}$. Choose a reduced representative w for the double coset $\langle x_1 \rangle w_i \langle x_i \rangle$, so that $w_i = x_1^a w x_i^b$ and w begins with neither x_1 nor x_1^{-1} and ends with neither x_i nor x_i^{-1}. The word $x_1 w x_i w^{-1}$ is cyclically reduced with length $2 + 2|w|$. Since $\varphi(x_1 x_i)$ is conjugate to $x_1 x_i$, we obtain $w = 1$. Hence $\varphi(x_i) = x_1^{n_i} x_i x_1^{-n_i}$. For distinct $i, j \geq 2$, conjugating $\varphi(x_i x_j)$ by $x_1^{-n_i}$ gives $x_i x_1^{n_j - n_i} x_j x_1^{n_i - n_j}$; cyclic length is two exactly when $n_i = n_j$. Thus all n_i agree, and φ is inner. The rank-two case has only one exponent and gives the same conclusion.

For a closed orientable surface group of genus $g \geq 2$, Dehn–Nielsen–Baer realizes the outer class of a class-preserving automorphism by a mapping class f. It fixes every oriented closed-curve class and acts identically on H_1, so preserves orientation. Choose a finite filling Alexander system of simple curves. The Alexander method ([20, §2.3]) makes the subgroup fixing all those curve classes finite: its possible permutations of intersection points and edges form a finite group; the kernel fixes the resulting graph and is isotopic to the identity on its complementary discs. Therefore f has finite order. Realize the cyclic group generated by f using Part C. For an action of order d, rational cochains of the quotient are the invariant cochains upstairs, by an equivariant triangulation and averaging over the finite stabilizers. The quotient's first Betti number is consequently $2g$, because the action on H_1 is trivial. Its underlying genus is g. Riemann–Hurwitz now gives

$$2 - 2g = d \left(2 - 2g - \sum_j (1 - 1/m_j) \right).$$

For $d > 1$ the right side is strictly smaller than $2 - 2g < 0$, a contradiction. Thus $f = 1$ and the automorphism is inner.

Now let $h \in N_{\widehat{\Lambda}}(\Lambda)$. Its conjugation automorphism is class-preserving by the first paragraph, hence is conjugation by an element of Λ by the two arguments just given. Dividing h by that element centralizes the dense lattice and therefore all of $\widehat{\Lambda}$. Part A gives trivial centre, so $h \in \Lambda$. This proves $N_{\widehat{\Lambda}}(\Lambda) = \Lambda$ with the exact comparisons needed in LM3 B.8.

For normal V, a descended lattice Δ normalizes $\Lambda = \Delta \cap V$. Conversely, an element of $N_P(\Lambda)$ has a coset in the finite group P/V. The intersection $N_P(\Lambda) \cap V$ is Λ by the just-proved input. Given that the deck action is induced on Λ, form the pullback discrete extension E_Λ of P/V by Λ inside $\mathrm{Aut}_{\mathrm{lac}}(\Lambda)$. We construct its map into P explicitly. An element of the pullback is a pair (g, φ), with $g \in P/V$, $\varphi \in \mathrm{Aut}_{\mathrm{lac}}(\Lambda)$, and prescribed outer class. Extend φ continuously to $\widehat{\varphi} \in \mathrm{Aut}(V)$ and choose a lift $p_g \in P$ of g. Equality of outer classes gives a $v \in V$ with $\widehat{\varphi} = \mathrm{Inn}(v p_g)|_V$. This v is unique, since $C_P(V) = 1$ by Part A. Send (g, φ) to $v p_g$. Changing p_g changes v inversely and leaves the product unchanged; multiplication is respected because both products induce the same conjugation on V and have the same quotient label. This defines an injective homomorphism: an element in its kernel has $g = 1$ and induces the identity on the

dense subgroup Λ. Its image is exactly $N_P(\Lambda)$, by restricting a normalizing conjugation to Λ and applying the same uniqueness. The discrete group is torsion-free, as a subgroup of P. The finite-extension realization theorem of Part C gives a surface lattice $\Delta = E_\Lambda$; its orientation is positive because the deck action on V has multiplier $+1$. Its completion is P by D.1.

We supply the peripheral argument left incomplete in LM3 B.10. Let $C \leq \Lambda$ be a primitive peripheral subgroup, and let $Q = C_P(\overline{C})$ be the maximal procyclic peripheral subgroup in P containing $\overline{C}$. Put $D = C_\Delta(C)$. The quotient surface supplied by Part C makes D its primitive cusp subgroup. We have $\overline{D} \leq Q$ and

$$\overline{D} \cap V = \overline{C} = Q \cap V.$$

The image of D in P/V is the stabilizer of the Λ-conjugacy class of C. To see this, adjust a representative of a stabilizing coset by an element of Λ to normalize C; orientation preservation changes normalization into centralization. Conversely a centralizing element fixes the class. The identical argument in the profinite group identifies the image of Q with the stabilizer of the V-conjugacy class of $\overline{C}$. The map from discrete cusp classes to their profinite classes is bijective and equivariant by peripheral noncommensurability and the finite covering construction of Part A. These two stabilizers are therefore equal. The two closed subgroups $\overline{D} \leq Q$ have the same intersection with the open subgroup V and the same image in P/V, so they are equal. This proves the promised loop identification.

In particular a descended lattice is forced to equal $N_P(\Lambda)$, proving injectivity for normal V. For arbitrary V, take $W = \bigcap_{p \in P} pVp^{-1}$, an open normal subgroup, and use the actual intersection map $\Lambda \mapsto \Lambda \cap W$. Equality after restriction to V implies equality after restriction to W, hence equality of the original lattices by the normal case. This proves injectivity without assuming V normal. □

D.14. [Proved] The image for a nonnormal finite covering. Let $V \leq P$ be open, $W = \mathrm{core}_P(V)$ and $\Lambda \in \mathrm{Discr}(V)$. Set $\Lambda_W = \Lambda \cap W$. The image condition for Λ is that the full deck action of P/W on W is induced by orientation-preserving loop automorphisms of Λ_W. This condition is independent of a choice of coset representatives. The unique extension is $N_P(\Lambda_W)$, and its intersection with V is Λ.

Manuscript: 28; 178–179; 352–354.

Proof. Necessity follows by restriction of an actual lattice in P. For sufficiency apply the normal descent statement D.13 to $W \triangleleft P$, obtaining $\Delta = N_P(\Lambda_W)$. Both $\Delta \cap V$ and Λ are discretizations of V with intersection Λ_W with the normal open subgroup W of V. The injectivity already proved inside V gives $\Delta \cap V = \Lambda$. The deck action is an outer action, so changing a representative multiplies its automorphism by an inner automorphism of W; its membership in the embedded geometric outer subgroup is unaffected. This proves the asserted independence and the exact nonnormal image description. □

D.15. [Proved] The explicit nonextending discretization in scan 354. Let $U = \mathbf{P}^1 - \{0, 1, \infty\}$ and $U' = \mathbf{P}^1 - \{0, \infty, 1, -1\}$, with the double cover $z \mapsto z^2$. Put $P = \widehat{\pi_1(U)}$, let $V \triangleleft P$ be its index-two subgroup, and let $\Lambda = \pi_1(U') \subset V$ be the geometric discretization. There is an explicit continuous loop automorphism γ of V for which $\gamma(\Lambda)$ is a discretization of V that does not extend to P, although it has the same prediscretization as Λ.

Manuscript: 28; 178–179; 352–354.

Proof. The deck involution $s : z \mapsto -z$ fixes $0, \infty$ and exchanges $1, -1$. Let a be a simple curve separating $\{0, 1\}$ from $\{\infty, -1\}$ and let $b = s(a)$; write T_a, T_b for positive Dehn twists. They

give a free basis of the pure oriented mapping class group $M_{0,4}^{!+} \simeq F_2$: in the ordered moduli space $\mathcal{M}_{0,4} \simeq \mathbf{P}^1 - \{0, 1, \infty\}$ they are monodromies around two distinct boundary points. One can verify the completed embedding directly from A.12: filling the fourth puncture identifies this pure group with the point-pushing group $\pi_{0,3}$, and the peripheral-normalization map sends its completion into $\mathrm{Aut}(\widehat{\pi}_{0,3})$ as $\mathrm{Inn}(\widehat{\pi}_{0,3})$. The latter is isomorphic to $\widehat{\pi}_{0,3}$ by triviality of its centre. The composite is injective, so the completion embeds already in $\mathrm{Out}_{\mathrm{lac}}(V)$. Thus the pure group's closure is its free profinite completion, with abelianization $\widehat{\mathbb{Z}}^2$ on the indicated twists.

Choose the fully specified profinite integer

$$\alpha_2 = 0 \in \mathbb{Z}_2, \qquad \alpha_\ell = 1 \in \mathbb{Z}_\ell \quad (\ell \text{ odd}), \qquad \alpha \in \widehat{\mathbb{Z}}.$$

It is not an integer: its 2-adic coordinate would force that integer to be zero, contrary to its 3-adic coordinate. Let $u = T_a^\alpha$ in the closure of $M_{0,4}^{!+}$. Since s preserves orientation, $sT_as^{-1} = T_b$, and the continuous power maps give

$$u^{-1}su = s\, T_b^{-\alpha} T_a^\alpha.$$

The pure factor on the right has abelianization $(\alpha, -\alpha)$. Every element of the discrete pure group has integral abelianization. Moreover $M_{0,4} \cap \overline{M_{0,4}^{!+}} = M_{0,4}^{!+}$: the puncture permutation and orientation sign are detected by finite quotients. Consequently $u^{-1}su$ does not belong to the discrete extended mapping class group.

Choose any continuous automorphism $\gamma \in \mathrm{Aut}_{\mathrm{lac}}(V)$ representing u. Its ambiguity is conjugation by an element of V, which does not change membership in the outer geometric subgroup. The geometric outer subgroup associated to $\gamma(\Lambda)$ is $uM_{0,4}u^{-1}$. The calculation says that the deck involution is absent from this subgroup. The necessary direction of D.13 therefore proves that $\gamma(\Lambda)$ does not extend to P. Since u lies in the closure of the pure geometric group, its coset modulo the closed geometric stabilizer is the identity, proving equality of the prediscretizations.

The code in `code/root/finite_checks.py` verifies compatible Chinese-remainder residues of α and the puncture partitions interchanged by s. Nonintegrality and the infinite completed-group conclusion follow from the coordinate argument above. This spells out and verifies the proposed element in LM3's review, §4.2. □

Manuscript: §28, folios 178–179, scans 352–354, especially *Un tel γ existe toujours.* We have proved the indicated index-two example; the word *toujours* is not used to assert the construction for every finite covering.

D.16. [Proved] A precise fixed-set formula for every actual action. Let A be a group, $J \leq A$, and $G \leq A$ a finite subgroup. Under the left action of G on A/J,

$$(A/J)^G = \{aJ : a^{-1}Ga \leq J\}.$$

It is the disjoint union of the $C_A(G)$-orbits

$$\coprod_{[a]} C_A(G)/(aC_J(a^{-1}Ga)a^{-1}),$$

where $[a]$ ranges over J-conjugacy classes of the labelled embeddings $g \mapsto a^{-1}ga$ occurring in J. This formula applies to actual discretizations with $(A, J) = (A(P), \mathrm{Aut}_{\mathrm{lac}}(\pi_0))$, to outer discretizations with $(A, J) = (B, M)$, and to prediscretizations with $(A, J) = (B, \overline{M})$.

Manuscript: 28; 178–181; 352–358.

Proof. The fixed-coset condition is $gaJ = aJ$ for every g, exactly the displayed inclusion. Two fixed cosets aJ, bJ are in the same centralizer orbit exactly when there are $c \in C_A(G)$ and $j \in J$ with $b = caj$. Substitution gives $b^{-1}gb = j^{-1}a^{-1}gaj$ for every labelled g. Conversely this equality makes $bj^{-1}a^{-1}$ centralize G. The stabilizer of aJ is $C_A(G) \cap aJa^{-1} = aC_J(a^{-1}Ga)a^{-1}$, proving the orbit formula.

The homogeneous-space identifications use the extension of each discrete loop automorphism to completion and its injectivity. The stabilizer of an actual lattice in $A(P)$ is exactly J. Its inner-conjugacy quotient is B/M. Since the closure of $\mathrm{Inn}(\pi_0)$ is $\mathrm{Inn}(P)$, the closed based stabilizer contains all inner automorphisms and its quotient is $\overline{M}$; hence the prediscretization set is $B/\overline{M}$. These identifications also show explicitly why a bare outer element is not assigned an action on the set of actual subgroups without choosing a lift. □

4.3 Torsion-free extensions and recognition

D.17. [Proved] Torsion-free extensions with an invariant discretization. Let $1 \to P \to \mathcal{E} \to G \to 1$ be a profinite extension with G finite, $P = \widehat{\pi}_0$ hyperbolic, and the outer action preserving the loop structure and a discretization. A torsion-free $\mathcal{E}$ is the profinite completion of a torsion-free discrete surface group, with the descended peripheral system. The quotient surface may be nonorientable.

Manuscript: 28; 178 verso; 353; 24; 147 verso; 293.

Proof. First the outer action is faithful. Its kernel is the image of $C_{\mathcal{E}}(P)$: an element acting inner can be divided by its unique inner element of P. The centralizer meets P trivially, so embeds in the finite quotient G and is finite. Torsion-freeness makes it trivial. Construct the discrete extension E of the stabilized lattice with this faithful outer action, using centre-freeness. By D.1, its completion is the given $\mathcal{E}$ with the identity on kernel and quotient. The discrete extension embeds in its completion because it is the finite extension realized by Part C and hence residually finite. Torsion-freeness of $\mathcal{E}$ therefore implies torsion-freeness of E. In a realization, a nontrivial point stabilizer gives a nontrivial finite-order element of the lift group. Thus G acts freely on the surface associated to π_0, and its quotient is a surface. Its fundamental group is E by lifting paths. The primitive cusp group in E is the centralizer of the covering cusp group; D.13 proves equality of their closures with the specified procyclic peripherals. Orientation-reversing free deck transformations yield the nonorientable quotient and its orientation character. This is the exact scope of the descent argument. □

D.18. [Proved] The unmarked pro-p extension statement for punctured surfaces. A finitely generated torsion-free pro-p group $\mathcal{E}$ containing an open free pro-p group P of rank r is free pro-p. Its rank is

$$1 + \frac{r-1}{[\mathcal{E} : P]}.$$

In particular the displayed quantity is an integer. This proves the underlying-group assertion for punctured pro-p surfaces, without identifying the peripheral system. More generally, every torsion-free profinite group with an open normal pro-p subgroup is itself pro-p. Thus D.19–D.20 apply to arbitrary finite-quotient extensions of their pro-p kernels; the quotient is forced to be a p-group.

Manuscript: 28; 178 verso; 353; 24; 147 verso; 293.

Proof. Herfort–Zalesskii, Theorem 1.1 [30], writes such a finitely generated virtually free pro-p group as the fundamental pro-p group of a finite graph of finite p-groups, with vertex groups

embedded in $\mathcal{E}$. Torsion-freeness makes all vertex and edge groups trivial. The fundamental pro-p group is therefore the free pro-p group on the cycles of the graph. The pro-p Schreier rank formula gives $r - 1 = [\mathcal{E} : P](\mathrm{rank}(\mathcal{E}) - 1)$, proving the formula. The theorem retains no designated peripheral subgroups in its graph-of-groups conclusion; their identification is a further assertion, not supplied by this calculation.

We prove the assertion about a general finite quotient. Write $G = \mathcal{E}/P$ and fix a prime $q \neq p$. For every open normal $U \triangleleft \mathcal{E}$ contained in P, the kernel of $\mathcal{E}/U \to G$ is a finite p-group. A Sylow q-subgroup of $\mathcal{E}/U$ therefore has order $|G|_q$ and maps isomorphically to a Sylow q-subgroup of G. Fix one such subgroup $Q \leq G$. The finite nonempty sets of Sylow q-subgroups above Q admit compatible choices under quotient maps: an image is again a Sylow subgroup, and any specified one lifts by taking a Sylow subgroup of its inverse image. Compactness gives an inverse-limit subgroup of $\mathcal{E}$ isomorphic to Q, since all these transition maps on the chosen subgroups are isomorphisms. Torsion-freeness forces $Q = 1$. Hence every prime divisor of $|G|$ is p, and every finite quotient of $\mathcal{E}$ is a p-group. This is exactly the assertion that $\mathcal{E}$ is pro-p. □

D.19. [Proved] Recognition of the peripheral pro-p extension. Let p be any prime, let P be the pro-p completion of an orientable hyperbolic surface group of type (g, ν) with $\nu > 0$, and let

$$1 \longrightarrow P \longrightarrow \mathcal{E} \longrightarrow G \longrightarrow 1$$

be an exact sequence of pro-p groups with G finite. Assume that $\mathcal{E}$ is torsion-free and its outer action preserves the P-conjugacy classes of the peripheral subgroups. Choose one L_j from each $\mathcal{E}$-orbit of these classes. Then

$$\mathcal{N}_j = N_{\mathcal{E}}(L_j) = C_{\mathcal{E}}(L_j) \simeq \mathbb{Z}_p,$$

and the pair $(\mathcal{E}; \{\mathcal{N}_j\})$ is continuously isomorphic, preserving the indexed peripheral conjugacy classes, to the pro-p completion of a compact surface with its boundary circles. Its interior is orientable for odd p. At $p = 2$ the interior may be nonorientable; its orientation character is the induced action on the orientation line of P.

Manuscript: 28; 178 verso; 353; 24; 147 verso; 293.

Proof. Each $\mathcal{N}_j$ is closed, and $\mathcal{N}_j \cap P = N_P(L_j) = L_j$ by Part A. Thus $\mathcal{N}_j$ is a torsion-free pro-p group with an open subgroup $\mathbb{Z}_p$. It is finitely generated, by lifting generators of its finite quotient and adjoining a generator of L_j. Apply D.18 to $\mathcal{N}_j$: the free pro-p rank formula gives rank one. Consequently $\mathcal{N}_j \simeq \mathbb{Z}_p$, and it centralizes L_j.

We identify the restricted peripheral family, including its multiplicities. The restriction of $(\mathcal{E}; \{\mathcal{N}_j\})$ to P contains the subgroups

$$P \cap e\mathcal{N}_j e^{-1} = eL_j e^{-1}, \qquad e \in P\backslash\mathcal{E}/\mathcal{N}_j.$$

The map from this double-coset set to the orbit of the P-conjugacy class of L_j is surjective by definition. It is injective: an equality $e'L_j e'^{-1} = aeL_j e^{-1}a^{-1}$ with $a \in P$ is equivalent to $e'^{-1}ae \in N_{\mathcal{E}}(L_j) = \mathcal{N}_j$, exactly equality of the double cosets. Thus the restricted family consists of all original peripheral classes, each once.

Wilkes, [58, Proposition 5.11], proves finite-index ascent and descent of Poincaré duality pairs at p for a p-torsion-free profinite ambient group, with compatible dualizing modules. The pair $(P; \{L_i\})$ is orientable of dimension two: this follows from its standard free pro-p surface presentation and the converse direction of [59, Theorem 3.3]. We have checked the exact restricted family and torsion hypothesis. Hence $(\mathcal{E}; \{\mathcal{N}_j\})$ is a Poincaré duality pair of dimension two.

Its dualizing module has underlying group $\mathbb{Q}_p/\mathbb{Z}_p$, with action through a continuous character $\chi : \mathcal{E} \to \mathbb{Z}_p^\times$ whose restriction to P is trivial. Naturality of duality under conjugation identifies

its character with the action on the surface orientation line, using the inverse convention on the dual cohomology line. The image of χ is a finite p-subgroup because it factors through G: it is trivial at odd p and is trivial or $\{\pm 1\}$ at $p = 2$. Thus inversion of the character does not affect the answer. Its restriction to each $\mathcal{N}_j$ is trivial, by the boundary-duality restriction in [58, Proposition 5.9] and the trivial duality character of $\mathbb{Z}_p$.

D.18 makes $\mathcal{E}$ free pro-p. Write the peripheral count as $b + 1$ and the rank as $n + b$. The condition $n \geq 0$ in the classification follows from the relative cohomology sequence: the cokernel of

$$H^1(\mathcal{E}; \mathbb{F}_p) \longrightarrow \bigoplus_{j=0}^{b} H^1(\mathcal{N}_j; \mathbb{F}_p)$$

has dimension one, since $H^2(\mathcal{E}; \mathbb{F}_p) = 0$ and the relative top group has dimension one. Hence $b \leq \operatorname{rank}(\mathcal{E})$. The classification in [59, Theorem 3.3] gives a free basis $x_1, \dots, x_n, s_1, \dots, s_b$, with each s_i generating a conjugate of $\mathcal{N}_i$, and a generator s_0 of $\mathcal{N}_0$ in one of these forms:

$$\begin{aligned}
&\chi = 1: && s_0 = [x_1, x_2]_- \cdots [x_{n-1}, x_n]_- \, s_1 \cdots s_b, \\
&\chi \neq 1, \ n \text{ odd}: && s_0 = x_1^2 [x_2, x_3]_- \cdots [x_{n-1}, x_n]_- \, s_1 \cdots s_b, \\
&\chi \neq 1, \ n \text{ even}: && s_0 = x_1^2 [x_1, x_2]_- [x_3, x_4]_- \cdots [x_{n-1}, x_n]_- \, s_1 \cdots s_b.
\end{aligned}$$

Here $[x, y]_- = x^{-1} y^{-1} x y$ matches that source's convention. The first line has n even. The other two have $p = 2$; their parameter 2^f is zero, since χ has no infinite principal-unit part.

These are surface words, with explicit changes of basis. Inverting each handle generator changes the commutator convention in the first line. The second line gives one crosscap and orientable handles. For the third, put

$$a = x_1 x_2^{-1}, \qquad b' = x_2, \qquad x_1 = a b', \quad x_2 = b'.$$

Then $x_1^2 [x_1, x_2]_- = a^2 (b')^2$, which gives two crosscaps and orientable handles. Replacing s_0 by its inverse gives the usual boundary relation and preserves its subgroup. Sending the discrete surface generators to these free pro-p generators induces the asserted isomorphism of completed pairs, with exactly the indexed peripheral conjugacies already specified. The free rank formula gives $\chi(\mathcal{E}) = \chi(P)/[\mathcal{E} : P] < 0$, so its interior is hyperbolic. This proves the peripheral statement at every prime, including 2 and 3. □

D.20. [Proved] Recognition of the closed pro-p extension. Let P be the pro-p completion of a closed orientable surface group of genus $g \geq 2$. A torsion-free pro-p group $\mathcal{E}$ containing P as an open normal subgroup is the pro-p completion of a closed surface group. The surface is orientable at odd p. At $p = 2$ a nonorientable surface is allowed, and its orientation character is the action on the orientation line of P.

Manuscript: 28; 178 verso; 353; 24; 147 verso; 293.

Proof. We give the absolute finite-index duality calculation. The orientable surface presentation makes P a Poincaré duality pro-p group of dimension two, with

$$H^i(P, \mathbb{Z}_p[[P]]) = \begin{cases} \mathbb{Z}_p, & i = 2, \\ 0, & i \neq 2, \end{cases}$$

and trivial right P-action on the top group. This is the trivial-character case of the Demushkin classification in [59, Theorem 1.9].

Serre's theorem [56, §1, Theorem] equates the p-cohomological dimensions of an open subgroup and a profinite group without elements of order p. Therefore $\operatorname{cd}_p(\mathcal{E}) = 2$. Wilkes [58, Proposition

1.26] gives finite-index ascent and descent of type p-FP_∞ for profinite groups. The surface resolution gives this property for P, and hence for $\mathcal{E}$.

At finite index, induction and coinduction of compact modules agree, and continuous Shapiro gives

$$\begin{aligned} H^i(\mathcal{E}, \mathbb{Z}_p[[\mathcal{E}]]) &= H^i\Big(\mathcal{E}, \operatorname{coind}_P^{\mathcal{E}} \mathbb{Z}_p[[P]]\Big) \\ &\simeq H^i(P, \mathbb{Z}_p[[P]]). \end{aligned}$$

The first coefficient isomorphism and compact Shapiro are [58, Propositions 1.16 and 1.21, and the remark after 1.21]. We make their module compatibility explicit. Put $\Lambda_H = \mathbb{Z}_p[[H]]$ and let $\mathrm{pr}_P : \Lambda_{\mathcal{E}} \to \Lambda_P$ retain the P-coset component. The coefficient isomorphism is

$$\Lambda_{\mathcal{E}} \longrightarrow \operatorname{Coind}_P^{\mathcal{E}} \Lambda_P, \qquad \lambda \longmapsto (x \mapsto \mathrm{pr}_P(x\lambda)).$$

Here coinduction consists of left-P-linear functions and its left $\mathcal{E}$-action is $(ef)(x) = f(xe)$. For representatives t of the finite set $P\backslash\mathcal{E}$, the inverse is $f \mapsto \sum_t t^{-1} f(t)$. These continuous maps are left-$\mathcal{E}$-linear and right-P-linear. Evaluation at 1 in Shapiro is natural for the right-P coefficient endomorphisms. Consequently the top right duality character restricts to that of P. The cohomology criterion for Poincaré duality now makes $\mathcal{E}$ a duality pro-p group of dimension two with dualizing module $\mathbb{Q}_p/\mathbb{Z}_p$. Its orientation character $\chi : \mathcal{E} \to \mathbb{Z}_p^\times$ has finite p-group image, trivial at odd p and contained in $\{\pm 1\}$ at $p = 2$.

Its one-dimensional degree-two cohomology and nondegenerate degree-one cup product make it an infinite Demushkin group. The classification [59, Theorem 1.9] gives, for these characters, exactly the three words of D.19 with the boundary letters omitted. The same invertible free-basis change converts the even-rank nonorientable word to the surface form. Hence an actual closed surface presentation induces a continuous isomorphism of its pro-p completion with $\mathcal{E}$. Finite-index multiplicativity for the Euler characteristic of the finite two-dimensional resolution gives

$$\chi(\mathcal{E}) = \frac{2 - 2g}{[\mathcal{E} : P]} < 0.$$

The resulting surface is hyperbolic, and conjugation-naturality of the dualizing module gives the stated orientation. □

D.21. [Refuted] A full profinite normalizer shortcut. The assertion that every torsion-free profinite group with an open subgroup $\widehat{\mathbb{Z}}$ is itself $\widehat{\mathbb{Z}}$ is false. The pro-p version used in D.19 is proved in D.18.

Manuscript: 28; 178 verso; 353.

Proof. Define

$$N = (\mathbb{Z}_3 \rtimes \mathbb{Z}_2) \times \prod_{\ell \neq 2,3} \mathbb{Z}_\ell, \qquad (a, b)(a', b') = (a + (-1)^{b \bmod 2} a',\ b + b')$$

on the first two factors, and use addition on the remaining factors. This is a profinite group with the open normal subgroup

$$L = (\mathbb{Z}_3 \times 2\mathbb{Z}_2) \times \prod_{\ell \neq 2,3} \mathbb{Z}_\ell \simeq \widehat{\mathbb{Z}}$$

of index two. A finite-order element projects trivially to the torsion-free quotient $\mathbb{Z}_2 \times \prod_{\ell \neq 2,3} \mathbb{Z}_\ell$, and then lies in the torsion-free kernel $\mathbb{Z}_3$; it is therefore the identity. Yet conjugation by $(0, 1)$ sends $(1, 0)$ to $(-1, 0)$, so N is nonabelian. The code in `code/root/finite_checks.py` checks its quotient $C_3 \rtimes C_4$, the index-two cyclic subgroup of order six, and this noncommutation.

The proof in the inverse limit establishes torsion-freeness; no finite quotient is claimed to be torsion-free.

The multiplier on L is -1 in its 3-adic coordinate and $+1$ in every other coordinate. Thus passing separately to pro-p groups loses precisely this interaction between two different primes. D.29 uses this N as an explicit retract in its full peripheral counterexample. Its cusp normalizers are all procyclic, so that example does not depend on making N itself a cusp normalizer. □

D.22. [Refuted] Full profinite recognition without an invariant discretization. The assertion that every torsion-free finite extension of a full profinite hyperbolic surface group with preserved loop structure is a completed surface pair is false. D.29 constructs a counterexample with kernel of type $(0, 4)$ and quotient C_2, including the exact peripheral comparisons and a finite quotient excluding every free profinite presentation. D.30 constructs a counterexample with closed genus-two kernel and quotient C_2; its degree-two cohomology at two odd primes excludes every surface completion. Recognition with an invariant discretization holds by D.17. The separate pro-p recognition theorems, with their precise group and peripheral hypotheses, are proved in D.18–D.20.

Manuscript: 28; 178 verso; 353; 24; 147 verso; 293.

Proof. The basis-derived four-cusp presentation, faithful outer involution, continuous twisted extension and torsion-free retraction are given in D.29. Its order-eighteen quotient needs three generators, whereas an index-two extension of a free rank-three group would have rank two if it were free. The exact finite-cover comparison excludes closed surface completions. Thus the counterexample satisfies the full peripheral hypothesis and fails the claimed conclusion.

For the closed case, D.30 gives the extension by its continuous twisted multiplication and by an equivalent profinite presentation. Its torsion-free maximal pro-2 quotient detects the parity of every element, proving that every torsion element lies in its torsion-free surface kernel. The two restriction isomorphisms give $H^2(\mathcal{E}; \mathbf{F}_3) = \mathbf{F}_3$ and $H^2(\mathcal{E}; \mathbf{F}_5) = 0$. The cohomology of every orientable, nonorientable, or punctured surface completion is computed in D.30 and excludes this pair. This settles the closed hyperbolic assertion in its smallest genus as well.

This also identifies the failed step in the earlier recognition attempt. Separate pro-p comparisons do not produce a single full profinite surface presentation. The mixed-prime multiplier in D.29 is an actual finite-order loop automorphism, and all its cusp normalizers are procyclic. The preserved global surface relation therefore does not restore the proposed conclusion. The first possible punctured nonabelian rank is three: D.27 excludes proper torsion-free finite extensions of $\widehat{F}_2$. The explicit free-basis comparison in D.29 settles the formerly undecided peripheral marking on D.26's rank-three extension.

We checked Collas–Philip–Yamaguchi [15], Definition 2.1, Theorems 1.3 and 2.4, against the supplied source. Their profinite F-groups already have the stated orbifold presentation. Theorem 2.4 constructs a torsion-free open normal subgroup, and Theorem 1.3 bounds the derived length of a finite quotient in its nonperfect case. The maps go from a prescribed presentation to an open subgroup; they do not assert the reverse reconstruction from an arbitrary finite extension. D.29 refutes that reverse assertion under the present full profinite hypotheses. The positive conclusions of D.17–D.20 retain their proved maps and hypotheses throughout. □

4.4 A full profinite lifting criterion and nonnilpotent finite tests

D.23. [Proved] Order-preserving lifting and conjugacy in the full shadow system. Let $\mathcal{I}$ be the cofiltered poset of isolated finite-index normal subgroups $N \trianglelefteq B_3$ contained in the pure braid group PB_3, and put $Q_N = GT(N)$. The finite groups and their reduction homomorphisms

$R_{N,H} : Q_N \to Q_H$ for $N \subseteq H$ are those of [17, Proposition 3.14, Theorem 5.2 and Proposition 5.5]. We use the continuous isomorphism

$$\Psi : GT_0 \xrightarrow{\sim} \varprojlim_{N \in \mathcal{I}} Q_N, \qquad (\lambda, f) \longmapsto \big([(\lambda-1)/2, f]_N\big)_N.$$

For a positive integer d, write $T_d(Q) = \{q \in Q : q^d = 1\}$. For each $H \in \mathcal{I}$, the image of $T_d(GT_0)$ in Q_H is exactly

$$\bigcap_{\substack{N \in \mathcal{I} \\ N \subseteq H}} R_{N,H}(T_d(Q_N)). \tag{D.23a}$$

An element of order exactly d in this intersection has a lift of order exactly d. Every element of order exactly d in GT_0 has an image of order exactly d at some isolated level. For $a, b \in GT_0$, conjugacy in GT_0 is equivalent to conjugacy of $\Psi(a)_N$ and $\Psi(b)_N$ in every Q_N.

Manuscript: §31; folios 198–199 verso; scans 391–394; §30; folio 194 verso; scan 384; Part D(b) of the task: full GT_0 torsion.

Proof. The cited theorem identifies the two profinite groups, including their topologies. Its Proposition 5.1 makes the coordinates cofinal in the finite quotients of $\widehat{\mathbb{Z}}$ and $\widehat{F}_2$. The map between the normalization in that paper and ours is exactly $(\widehat{m}, \widehat{f}) \mapsto (2\widehat{m}+1, \widehat{f})$; Proposition 5.5 checks the equations and the multiplication under this map. Thus its inverse limit concerns the full GT_0 of Part E.

Fix $q \in Q_H$ in the right side of (D.23a). For every isolated $N \subseteq H$ define the finite nonempty set

$$S_N = \{v \in Q_N : v^d = 1,\ R_{N,H}(v) = q\}.$$

Every reduction $R_{N,M}$ restricts to a function $S_N \to S_M$. The inverse limit of these sets is nonempty. To verify the exact compactness step, any finite list of coordinates and compatibility conditions is dominated by one isolated subgroup $L \subseteq H \cap N_1 \cap \cdots \cap N_t$; Proposition 3.14 supplies an isolated refinement of the intersection. Choose an element of S_L and project it to the specified coordinates. This satisfies all the finitely many conditions. The closed conditions in the compact product $\prod_N S_N$ consequently have the finite intersection property. A compatible element maps under Ψ^{-1} to $g \in GT_0$ with $g^d = 1$ and $g_H = q$. The reverse inclusion follows by taking the coordinates of such a g. The levels below H are cofinal, so the constructed family is a full inverse-limit point.

When q has order d, the order of g both divides d and is a multiple of the order of its image q, hence equals d. Conversely, for g of order d, choose, for each prime $p \mid d$, a finite coordinate detecting $g^{d/p} \neq 1$. A common isolated refinement detects all these powers. The order of its image divides d and divides none of the proper divisors d/p, so it is d.

For the conjugacy assertion, the sets

$$C_N(a, b) = \{v \in Q_N : v\Psi(a)_N v^{-1} = \Psi(b)_N\}$$

are finite nonempty sets with restriction maps. The same finite intersection argument gives a compatible conjugator, which Ψ^{-1} sends to a conjugator in GT_0. Necessity follows by projection. The construction does not assume surjectivity of a reduction map or that every element of Q_N comes from GT_0. □

D.24. [Proved] Dihedral tests with an order bound and the full dyadic quotient. The program `code/part_a/gt_shadow_refinement.py` executes the finite order-two and order-three tests below. Each entry counts nonidentity elements with order dividing d at the lower level, and then those admitting a lift with order dividing d at the upper level.

n	q	d	Lower elements	Elements admitting such a lift
6	18	2	7	7
6	18	3	2	0
18	54	3	8	0
54	162	3	8	0
14	98	3	14	14
4	8	2	3	1
8	16	2	7	2
16	32	2	11	4
32	64	2	19	8

The levels are isolated dihedral shadows $Q_{K^{(n)}}$. Their subsystem with $n = 2^\alpha$, $\alpha \geq 2$, has inverse limit

$$H_2 = \{(k, (-1)^a 5^b) \in \mathbb{Z}_2 \rtimes \mathbb{Z}_2^\times : a \in \{0,1\},\ b \in \mathbb{Z}_2,\ k \equiv b \pmod 2\}. \tag{D.24a}$$

Every finite subgroup of H_2 is trivial or cyclic of order two, and all its order-two subgroups are conjugate to $\langle(0,-1)\rangle$. There is a continuous surjection $GT_0 \to H_2$. The family of all dihedral levels is not cofinal in $\mathcal{I}$; an explicit A_5 quotient below proves this.

Manuscript: §31; folios 198–199 verso; scans 391–394; §30; folio 194 verso; scan 384; Part D(b) of the task: finite-quotient torsion program.

Proof. Write $D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1}\rangle$. The homomorphism

$$\psi_n : PB_3 = \langle x, y\rangle \times \langle c\rangle \longrightarrow D_n^3, \quad x \longmapsto (r,s,s), \quad y \longmapsto (rs, r, rs), \quad c \longmapsto 1$$

defines $K^{(n)} = \ker \psi_n$. For even $n \geq 4$, the explicit finite model of [11, Theorem 4.3 and Proposition 4.5] is

$$Q_{K^{(n)}} \simeq \left\{(k,u) \in (\mathbb{Z}/(n/2)) \rtimes (\mathbb{Z}/(2n))^\times :\ k \equiv \nu(u) \pmod 2 \text{ when } 4 \mid n \right\}, \tag{D.24b}$$

where $\nu(u) = (u-1)/4$ for $u \equiv 1 \pmod 4$ and $\nu(u) = (u+1)/4$ for $u \equiv 3 \pmod 4$. Multiplication is $(k,u)(l,v) = (k+ul, uv)$, with the indicated moduli. For $n \mid q$, reduction is $(k \bmod (q/2), u \bmod 2q) \mapsto (k \bmod (n/2), u \bmod 2n)$.

The code constructs the actual subgroup $G_n = \langle (r,s,s), (rs,r,rs)\rangle \leq D_n^3$ for $n = 4, 6, 8, 12, 14$. It computes its commutator subgroup and exhausts all pairs (m, f) with $0 \leq m < n$, $\gcd(2m+1, n) = 1$ and $f \in G_n'$. It tests the two cycle equations

$$\theta(f) f = 1, \qquad \tau^2(y^m f)\tau(y^m f) y^m f = 1,$$

where $\theta(x) = y$, $\theta(y) = x$, $\tau(x) = y$ and $\tau(y) = (xy)^{-1}$. It checks descent of these maps, and of $x \mapsto x^{2m+1}$, $y \mapsto f^{-1} y^{2m+1} f$, on every labelled edge of the finite Cayley graph, and checks that the resulting maps are bijective. The accepted pairs are exactly those in (D.24b), of respective orders $4, 12, 16, 24, 84$. This checks the reduction to the quoted finite model on the tested groups, with no reliance on floating point arithmetic. Projection of G_6, G_{12}, G_{14} onto D_3, D_3, D_7 exhibits finite quotients in which the odd-order rotation and order-two reflection do not commute; their nilpotence would force these coprime Sylow subgroups to commute. These tests therefore include nonnilpotent finite quotients of $\widehat{F}_2$.

For the refinement counts, the code enumerates every element of (D.24b), computes its dth power using the affine multiplication, reduces every upper root, and counts the exact fibres. For example, at $n = 6$ the group is $\mathrm{Aff}(\mathbb{F}_3) \times C_2$, and its two order-three elements are the nonzero translations. At $q = 18$, an element of order dividing three has multiplier $u \equiv 1, 4, 7 \pmod 9$ and translation divisible by 3. Both its affine coordinates reduce to the identity modulo 3. Thus

neither order-three element at $n = 6$ has an order-dividing-three lift, although the unbounded reduction map is surjective. At $n = 14$ the multipliers $2, 4$ modulo 7 have order three. Each lifts to its order-three root modulo 49, and each of the seven lifts of a translation coordinate is allowed because $1 + u + u^2 = 0$ there. This gives seven order-three lifts of each of the fourteen lower elements, as the exhaustive output records. The same program gives the remaining rows and the full table in its accompanying CSV.

The compatible affine coordinate reductions identify the dyadic inverse limit with (D.24a): the residue of b modulo 2 is the condition in (D.24b), and the inverse limits of the two coordinate rings are $\mathbb{Z}_2$ and $\mathbb{Z}_2^\times$. The surjection from GT_0 is [11, Theorem 5.7 and Proposition 5.8]; more precisely, that theorem proves surjectivity already after restriction along $G_{\mathbb{Q}} \to GT_0$.

The translation kernel $\mathbb{Z}_2$ is torsion-free, so a finite subgroup of H_2 injects into the finite-order subgroup of $\mathbb{Z}_2^\times$. This subgroup is $\{1, -1\}$: the parametrization $\mathbb{Z}_2^\times = \{\pm 1\} \times 5^{\mathbb{Z}_2}$ has torsion-free second factor, as follows from $v_2(5^{2^j} - 1) = j + 2$ and passage to the inverse limit. Every nonidentity finite-order element of H_2 is consequently $(k, -1)$ with $k \in 2\mathbb{Z}_2$. Put $t = k/2$ and choose $v = 5^{t \bmod 2}$. Then $(t, v) \in H_2$, and direct multiplication gives

$$(t, v)(0, -1)(t, v)^{-1} = (2t, -1) = (k, -1).$$

This proves the full finite-subgroup and conjugacy assertions for this infinite quotient, rather than extrapolating the finite dyadic counts.

Finally every G_n is metabelian: the commutator subgroup of D_n^3 lies in its abelian rotation subgroup, and G_n' is a subgroup of that rotation subgroup. Define the epimorphism $\eta : F_2 \to A_5$ by $x \mapsto (1\ 2\ 3\ 4\ 5)$ and $y \mapsto (1\ 2\ 3)$. The code enumerates its image and derived subgroup; both have order 60, proving directly that this image is a nontrivial perfect group. Proposition 5.1 of [17] gives an isolated N with $N \cap F_2 \subseteq \ker \eta$. An inclusion $K^{(n)} \subseteq N$ would give the surjection

$$G_n = F_2/(K^{(n)} \cap F_2) \longrightarrow F_2/\ker \eta = A_5.$$

Its image would be metabelian, contradicting the computed nontrivial perfect derived subgroup. Hence no such inclusion exists. This proves the asserted failure of cofinality by an explicit quotient and the exact factoring map. □

These tests give explicit finite obstructions for D.12. D.32 proves, by its full fixed-subgroup and finite-cover argument, that GT_0 has no nonidentity element of odd order and that every finite subgroup has order at most two, detected by the dyadic multiplier. D.34 further proves that a nonidentity involution has the global multiplier -1. The fourteen order-three shadows at $K^{(14)}$ survive the executed test through $K^{(98)}$. Combining D.32 with D.23 shows that each fails the order-three lifting condition at some isolated refinement; a common refinement excludes all fourteen. This existence conclusion does not identify that refinement or alter the recorded finite counts. D.25 explains why the entire dihedral subsystem alone cannot supply the exclusion. The remaining full GT_0 question is conjugacy of its involutions. D.35 gives the exact third-cusp equation and the finite conjugator criterion, and D.44 supplies a coboundary satisfying its first symmetry. D.42 also proves the global multiplier restriction for every finite subgroup of the full three-cusp loop-outer group.

D.25. [Proved] All-prime dihedral compatibility and the multiplier-one obstruction. The inverse limit D over all the dihedral levels of D.24 has the following explicit isomorphism of profinite groups:

$$D \xrightarrow{\sim} \left(\prod_{p \text{ odd}} \mathrm{Aff}(\mathbb{Z}_p) \right) \times H_2. \tag{D.25a}$$

There is a canonical continuous homomorphism $GT_0 \to D$. The group D contains an element of every prime order. In particular, imposing order three throughout this entire dihedral system

still permits a multiplier whose 7-adic coordinate is a primitive cube root and whose other prime coordinates are 1.

Put $F = \widehat{F}_2 = \langle x, y \rangle$ and define the continuous epimorphism and its closed kernel by

$$\kappa : F \longrightarrow \prod_p F_2^{(p)}, \qquad R = \ker \kappa.$$

Write $\lambda : GT_0 \to \widehat{\mathbb{Z}}^\times$ for the multiplier. For every integer $d \geq 2$, the elements $a \in \ker \lambda$ with $a^d = 1$ correspond exactly to the following elements $f \in R$. The continuous endomorphism

$$\alpha_f : F \to F, \qquad x \mapsto x, \quad y \mapsto f^{-1} y f$$

must be an automorphism and satisfy

$$\theta(f) f = 1, \qquad \tau^2(f)\tau(f) f = 1, \qquad \prod_{j=0}^{d-1} \alpha_f^j(f) = 1, \tag{D.25b}$$

where the product is in increasing order of j, $\theta(x) = y$, $\theta(y) = x$, $\tau(x) = y$, and $\tau(y) = (xy)^{-1}$. Every such finite-order element has identity image in D. The kernel R itself maps onto the explicit perfect group A_5 used in D.24. The finite-order solutions of (D.25b) in the stated automorphism domain are exactly $f = 1$, by D.32's subsequent full fixed-subgroup argument. Thus the nontrivial prime-order points of the dihedral inverse limit exhibited here do not lift with that order to GT_0 when the prime is odd.

Manuscript: §25 bis; folios 157 verso–160; scans 312–317; §30; folio 194 verso; scan 384; §31; folios 198–199 verso; scans 391–394.

Proof. It suffices to use the even levels $n = 2^\alpha n_0$ with $\alpha \geq 1$ and n_0 odd; every odd level agrees with its double. Apply the Chinese remainder isomorphism separately to the translation coordinate modulo $n/2$ and to the multiplier coordinate modulo $2n$ in (D.24b). The odd factors give $\mathrm{Aff}(\mathbb{Z}/n_0)$, and the powers of two give the finite groups defining H_2. These coordinate maps commute with every reduction $n \mid q$. Conversely, specified odd coordinates and a specified point of H_2 reconstruct every finite pair (k, u) by the Chinese remainder theorem. They satisfy the parity condition because that condition involves only the power-of-two coordinates. The resulting inverse maps are continuous, being continuous at every finite coordinate. This proves (D.25a), including its topology. Restriction of the full inverse-limit point in D.23 defines the indicated map from GT_0; surjectivity onto D is not used or asserted.

Let q be a prime, and choose a prime divisor p of $2^q - 1$. The residue of 2 in $\mathbb{F}_p^\times$ has order q: its order divides q, and order one would give $p \mid 2 - 1$. Hence $q \mid p - 1$, in particular p is odd and $p \neq q$. The element 2 mod p lifts to a unique qth root $\zeta \in \mathbb{Z}_p^\times$. Explicitly, after a root a modulo p^j is chosen, substitute $a + p^j t$, $0 \leq t < p$, in $X^q - 1$ modulo p^{j+1}. Its coefficient of t is $p^j q a^{q-1}$, and $q a^{q-1}$ is a unit modulo p; there is exactly one permitted value of t. The compatible limit has order exactly q because its residue already does. Put $(0, \zeta)$ in the p-factor of (D.25a), and put the identity in all the other factors. This gives the asserted element of order q in D.

For $q = 3$, take $p = 7$. The executed program `code/part_a/gt_shadow_all_prime.py` constructs ten successive roots of $X^2 + X + 1$ starting with 2 mod 7, and uses the exact Chinese remainder formulas to construct compatible shadow coordinates at sixteen levels through $n = 2744$. At $n = 14$ its affine pair is $(0, 9 \bmod 28)$; at $n = 98$ it is $(0, 177 \bmod 196)$. Their orders are three and the latter reduces to the former. The preceding root-lifting proof constructs the infinite point, while the program verifies the displayed finite coordinates. Thus the equations tested over all dihedral levels admit prime-dependent torsion multipliers. They do not force the global alternatives 1 and -1.

We next prove the statement about R. Each projection $F \to F_2^{(p)}$ is the canonical maximal pro-p quotient map. Their product is onto: in any finite collection of finite p-group quotients at distinct primes, its image surjects to each factor. Its order is consequently divisible by every factor order, hence by their product, since these orders are pairwise coprime. The image is the full product. These finite products form the finite quotients of $\prod_p F_2^{(p)}$, so the compact image of F is the whole product. Also $R \subset F'$, since the abelianization $F^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ is detected by the abelianizations of all these pro-p quotients.

Let $a = (1, f) \in GT_0$ have finite order dividing d. It induces a finite-order automorphism of each free pro-p quotient with trivial action on its abelianization. The Magnus leading-term argument in D.10, which also applies at $p = 2$, makes this automorphism the identity. Its normalized formula then gives $f_p \in C_{F_2^{(p)}}(y) \cap (F_2^{(p)})' = 1$ by A.7 and the primitive abelianization coordinate of y. Thus $f \in R$.

The two first equations of (D.25b) are precisely the defining cycle equations at multiplier one; the required invertibility is stated explicitly. To obtain the last equation, put $w_d = f\alpha_f(f) \cdots \alpha_f^{d-1}(f)$. Induction on d gives

$$\alpha_f^d(x) = x, \qquad \alpha_f^d(y) = w_d^{-1} y w_d.$$

Therefore $\alpha_f^d = 1$ is equivalent to $w_d \in C_F(y)$. But $w_d \in F'$ and $C_F(y) \cap F' = \overline{\langle y \rangle} \cap F' = 1$, again by A.7 and abelianization. This proves the necessity of the last equation and its sufficiency. The normalized identification of GT_0 with these automorphisms gives the asserted bijection. The kernel of the multiplier on D is a subgroup of the additive translation group $\prod_p \mathbb{Z}_p$, which is torsion-free. Consequently every finite-order multiplier-one element maps to the identity of D.

Finally let $\eta : F \to A_5$ be the epimorphism from D.24. The quotient $A_5/\eta(R)$ is a finite quotient of $F/R = \prod_p F_2^{(p)}$ and is therefore nilpotent. It is also perfect, since the executed permutation calculation in D.24 proves $A_5' = A_5$. A finite nilpotent perfect group is trivial: its lower central series would otherwise be constantly equal to the nontrivial group and could not terminate. Hence $\eta(R) = A_5$. This identifies a nontrivial part of the residual group in the exact equation (D.25b). The full three-cusp argument in D.32 proves that this equation, together with its stated cycle and invertibility requirements, has only the identity solution. The surjection $R \to A_5$ and the compatible points of D remain the exact quotient calculations proved above. □

D.26. [Refuted] Unmarked recognition at index two and the peripheral obstruction. The assertion that a torsion-free profinite group with an open free profinite subgroup of rank three is free is false, already at index two. We give the rank-three instance of [27, Proposition, pp. 350–352] as an explicit extension

$$1 \longrightarrow H = \widehat{F}(l, a, b) \longrightarrow E \longrightarrow C_2 \longrightarrow 1. \tag{D.26a}$$

It has no section. Its outer involution preserves no loop structure of type $(1, 2)$ on H. Every preserved type-$(0, 4)$ structure would have a double-transposition cusp permutation, multiplier $-\epsilon$, and canonical character $\chi(l) = 1$, $\chi(a) = \chi(b) = -1$ sending all four cusp generators to -1. Two explicit type-$(0, 4)$ candidates fail, with finite S_3 and S_4 certificates. D.29 supplies a different complete type-$(0, 4)$ marking, satisfying exactly these necessary conditions, and thereby turns this extension into a counterexample to the loop-preserving assertion D(d).

Manuscript: §28; folios 178–179; scans 352–354; Part D(d) of the task: full profinite recognition.

Proof. Choose $\epsilon, \kappa \in \widehat{\mathbb{Z}}$ by their prime coordinates:

$$\epsilon_3 = -1, \quad \epsilon_p = 1 \ (p \neq 3), \qquad \kappa_2 = 1, \quad \kappa_p = 0 \ (p \neq 2).$$

Thus ϵ is a unit, $\epsilon^2 = 1$ and $\epsilon\kappa = \kappa$. Define the continuous endomorphism of the free profinite group H by

$$\alpha(l) = l^{\epsilon}, \qquad \alpha(a) = b, \qquad \alpha(b) = l^{\kappa} a l^{-\kappa}, \qquad c = l^{\kappa}. \tag{D.26b}$$

Substitution on the three free generators gives $\alpha^2 = \mathrm{Inn}(c)$ and $\alpha(c) = c$. In particular $\mathrm{Inn}(c^{-1})\alpha$ is an inverse, so α is an automorphism.

Define E on the compact space $H \times \{0, 1\}$ by

$$(h, i)(h', j) = \big(h\alpha^i(h')c^{\lfloor (i+j)/2 \rfloor}, i + j \bmod 2\big). \tag{D.26c}$$

The two identities just checked prove associativity by the eight triples of quotient labels. When the first two labels are 1, the required interchange is $\alpha^2(h'')c = ch''$; the triple $(1, 1, 1)$ additionally uses $\alpha(c) = c$. The other triples follow directly. The identity and inverses follow by the same formulas. All operations are continuous, so this is a profinite group. Writing $t = (1, 1)$ gives $tht^{-1} = \alpha(h)$ and $t^2 = c$. The projection to the second coordinate gives (D.26a).

The closed subgroup $N = \overline{\langle l, t \rangle}$ has the explicit isomorphism

$$N \simeq (\mathbb{Z}_3 \rtimes \mathbb{Z}_2) \times \prod_{p \neq 2,3} \mathbb{Z}_p,$$

where the parity of the $\mathbb{Z}_2$ coordinate acts by inversion on $\mathbb{Z}_3$. Namely, the normal form $l^u t^i$ maps to $(u_3, 2u_2 + i, (u_p)_{p \neq 2,3})$. This is a continuous bijection preserving the displayed relations and multiplication; its inverse reads the parity i and then all coordinates of u. The projection of N to $\mathbb{Z}_2 \times \prod_{p \neq 2,3} \mathbb{Z}_p$ has torsion-free kernel $\mathbb{Z}_3$ and torsion-free target, hence N is torsion-free.

There is a continuous retraction $E \to N$ given on the generators by $l \mapsto l$, $t \mapsto t$, $a, b \mapsto 1$; (D.26b) verifies its compatibility with the group law. An element of finite order maps trivially under this retraction and therefore has trivial image in C_2. It lies in H, which is torsion-free by A.7, so it is the identity. Thus E is torsion-free and the extension has no section. The induced outer action is faithful: its nonidentity element acts nontrivially on H^{ab}.

For comparison with Haran's construction, the homomorphism $N \amalg \widehat{\mathbb{Z}} \to E$ from the free profinite product, identity on N and sending the other generator to a, is onto since $b = tat^{-1}$. In the opposite direction send l, a, b, t to l, a, tat^{-1}, t in that free product. Equations (D.26b) hold there, using the two relations in N, so (D.26c) gives a continuous homomorphism in the opposite direction. The composites fix every generator. These are mutually inverse isomorphisms.

To prove that E is not free, form the finite group

$$Q = (\mathbb{F}_3^2) \rtimes C_2, \qquad (v, i)(w, j) = (v + (-1)^i w, i + j).$$

The formulas $l \mapsto (e_1, 0)$, $a \mapsto (e_2, 0)$, $t \mapsto (0, 1)$, and $b \mapsto (-e_2, 0)$ give an epimorphism $E \to Q$: $\epsilon \equiv -1 \pmod 3$ and $\kappa \equiv 0 \pmod 3$ verify all relations. This group requires three generators. Two elements of $\mathbb{F}_3^2$ do not generate its C_2 quotient. When one of two proposed generators lies outside $\mathbb{F}_3^2$, it is an involution. Multiplying the other by it when necessary puts that other generator in $\mathbb{F}_3^2$. Conjugation then only inverts this one vector, so the generated subgroup has order at most six. Three generators, as displayed, do generate Q. Were E a free profinite group of rank r, its open index-two subgroup would have rank $2r - 1$ by the finite coset Schreier rewriting. Since H has rank three, this would give $r = 2$, contradicting its quotient Q. The program `code/part_a/haran_rank3.py` checks all 18^3 associativity instances and all 18^2 ordered pairs in Q. Their generated subgroup orders are $1, 2, 3, 6, 9$, with multiplicities $1, 27, 32, 216, 48$; none is 18.

We now check the peripheral condition rather than discarding it. On the basis $[l], [a], [b]$ of $H^{\mathrm{ab}} = \widehat{\mathbb{Z}}^3$, the outer involution has matrix and determinant

$$M = \begin{pmatrix} \epsilon & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \det M = -\epsilon. \tag{D.26d}$$

This determinant has value $+1$ at the prime 3 and -1 at every other prime, and is invariant under a change of basis. For a preserved type-$(1,2)$ peripheral structure, let W be the primitive rank-one cusp submodule in H^{ab}. Write λ for the action on the orientation line and $\delta \in \{1,-1\}$ for the sign of the permutation of the two punctures. The action on W is multiplication by $\lambda\delta$. The quotient H^{ab}/W is the first homology of the filled torus. The orientation comparison of A.2 identifies its determinant with λ. The square of the outer action is the identity, so $\lambda^2 = 1$. Multiplicativity of determinant in the exact sequence of these free modules gives

$$\det M = (\lambda\delta)\lambda = \delta.$$

This contradicts (D.26d). Thus no such type-$(1,2)$ peripheral structure is preserved.

For type $(0,4)$ this determinant argument permits a prime-dependent orientation multiplier. We test the specific free basis l, a, b^ϵ, which is a basis because the power substitution on b has inverse $b \mapsto b^\epsilon$. Set

$$P = lab^\epsilon, \quad L_1 = \overline{\langle l \rangle}, \quad L_2 = \overline{\langle a \rangle}, \quad L_3 = \overline{\langle b^\epsilon \rangle}, \quad L_4 = \overline{\langle P^{-1} \rangle}.$$

This is a full type-$(0,4)$ peripheral system. The first three classes transform as required with multiplier ϵ and permutation (2 3). The fourth would require $\alpha(P)$ to be conjugate to P^ϵ. The actual words are

$$\alpha(P) = l^\epsilon b l^\kappa a^\epsilon l^{-\kappa}, \qquad P^\epsilon = (lab^\epsilon)^\epsilon.$$

Map the free group H continuously onto S_3 by

$$l \longmapsto (1\ 2), \qquad a \longmapsto (2\ 3), \qquad b \longmapsto (1\ 2\ 3).$$

At its exponent six, $\epsilon \equiv 5$ and $\kappa \equiv 3$. Direct multiplication gives $P \mapsto 1$, $\alpha(P) \mapsto (1\ 2\ 3)$, and $P^\epsilon \mapsto 1$. These images are not conjugate, so the fourth loop class fails. The program checks these values and all 6^3 possible assignments, finding twenty-four nonconjugacy certificates.

There is a basis-independent restriction on any remaining type-$(0,4)$ structure. Write $\sigma \in S_4$ for its cusp permutation and $\lambda \in \widehat{\mathbb{Z}}^\times$ for its multiplier. The square of the outer action is the identity, so $\sigma^2 = 1$ and $\lambda^2 = 1$. Its action on H^{ab} is λ times the action of σ on $\widehat{\mathbb{Z}}^4/\widehat{\mathbb{Z}}(1,1,1,1)$. Comparing traces with (D.26d) gives

$$\epsilon = \lambda(|\operatorname{Fix}(\sigma)| - 1). \tag{D.26e}$$

The identity permutation would give $3\lambda = \epsilon$, which is impossible modulo 3. A single transposition forces $\lambda = \epsilon$; a double transposition forces $\lambda = -\epsilon$.

For each such peripheral structure there is a canonical character $\chi : H \to \{1,-1\}$ which is -1 on all four cusp generators. The product relation makes it well-defined, and a free peripheral basis makes it nonzero. It is preserved by α, since λ is odd and the cusp classes are permuted. In the basis l, a, b the three nonzero invariant characters are exactly

$$\chi_1 = (-1,1,1), \qquad \chi_2 = (1,-1,-1), \qquad \chi_3 = (-1,-1,-1). \tag{D.26f}$$

We compute the action on continuous cohomology with coefficients $\mathbb{Q}_p(\chi)$ for an odd prime p; equivalently one may use the inverse limit of the finite coefficient modules and then tensor with $\mathbb{Q}_p$. A crossed homomorphism u is determined freely by $(L, A, B) = (u(l), u(a), u(b))$. Put $\eta = \chi(l)$ and $\zeta = \chi(a) = \chi(b)$, and define

$$e = \begin{cases} \epsilon_p & \eta = 1, \\ 1 & \eta = -1, \end{cases} \qquad k = \begin{cases} 0 & \eta = 1, \\ 1 & \eta = -1. \end{cases}$$

Here $u(l^\epsilon) = eL$ and $u(l^\kappa) = kL$. For $\eta = 1$ this is multiplication by the p-coordinate of the exponent; for $\eta = -1$ the geometric sum is $(1 - (-1)^v)/2 = 1$ for either odd exponent $v = \epsilon, \kappa$. Substitution in (D.26b) gives

$$\alpha^* u \quad \longleftrightarrow \quad A_{\chi,p} \begin{pmatrix} L \\ A \\ B \end{pmatrix}, \qquad A_{\chi,p} = \begin{pmatrix} e & 0 & 0 \\ 0 & 0 & 1 \\ (1-\zeta)k & \eta & 0 \end{pmatrix}. \tag{D.26g}$$

The coboundary line is spanned by $(\eta - 1, \zeta - 1, \zeta - 1)$, and this nonzero vector is fixed by the displayed matrix. Thus

$$\det(\alpha^* | H^1(H, \mathbb{Q}_p(\chi_i))) = \begin{cases} 1 & i = 1, 3, \\ -\epsilon_p & i = 2. \end{cases} \tag{D.26h}$$

We identify the determinant required by the peripheral structure through actual restriction and filling maps. Let $K = \ker \chi$. Under the specified marking it is the completion of the double cover of the four-punctured sphere in which every cusp has nontrivial monodromy. The cover has four punctures and genus one, since its Euler characteristic is -4. The deck involution fixes each of the four peripheral homology classes. Their span is the three-dimensional invariant subspace of $H_1(K, \mathbb{Q}_p)$: restriction and averaging identify the invariant cohomology with $H^1(H, \mathbb{Q}_p)$, of dimension three. Filling the four punctures therefore identifies the anti-invariant part with the first homology of the filled torus. The restriction of crossed homomorphisms induces the precise comparison

$$H^1(H, \mathbb{Q}_p(\chi)) \xrightarrow{\sim} H^1(K, \mathbb{Q}_p)^- \xleftarrow{\sim} H^1(\widehat{\pi_1(T^2)}, \mathbb{Q}_p). \tag{D.26i}$$

The first isomorphism follows by averaging over the order-two deck group, which is invertible in $\mathbb{Q}_p$; the second is the dual of the filling quotient on the anti-invariant part. Both commute with the induced action of α. Upstairs cusp generators are squares of the downstairs generators. A conjugator in H can be moved into K by multiplying it by the corresponding downstairs cusp generator when necessary; that generator centralizes its own square. Thus α has the same multiplier λ on this covering pair. The transfer of the relative orientation class, followed by filling, is the comparison of A.2; it identifies the determinant on the filled torus with λ_p. Consequently the determinant in (D.26h) must be λ_p.

At $p = 3$ all three determinants in (D.26h) are 1. This excludes the single-transposition case, which by (D.26e) would have $\lambda_3 = \epsilon_3 = -1$. For the remaining double transposition, $\lambda = -\epsilon$; at $p = 5$ its value is -1. Formula (D.26h) then excludes χ_1 and χ_3, leaving exactly χ_2. This proves the necessary conditions stated at the beginning for every marking.

One can satisfy the first two cusp conditions in this remaining case by the explicit free basis

$$x = a, \qquad y = b^{-\epsilon}, \qquad z = b^\epsilon l, \qquad w = (xyz)^{-1} = (al)^{-1}. \tag{D.26j}$$

The inverse substitution is $a = x$, $b = y^{-\epsilon}$, $l = yz$, proving that this is a free basis. It has canonical character χ_2, and $\alpha(x)$ is $y^{-\epsilon}$ while $\alpha(y)$ is conjugate to $x^{-\epsilon}$. For permutation $(x\ y)(z\ w)$ and multiplier $-\epsilon$, the next condition would require

$$\alpha(z) \sim w^{-\epsilon}, \quad \text{equivalently} \quad a^\epsilon l^\epsilon \sim (al)^\epsilon. \tag{D.26k}$$

Indeed $\alpha(z) = ca^\epsilon c^{-1} l^\epsilon$ is conjugate to $a^\epsilon l^\epsilon$ because $c = l^\kappa$ commutes with l. In S_4 take $a = (2\ 3\ 4)$, $l = (1\ 2\ 3\ 4)$ and $b = 1$. Its exponent is twelve, with $\epsilon \equiv 5$. The left member of (D.26k) is the transposition $(1\ 4)$, whereas the right member is the four-cycle $(1\ 3\ 2\ 4)$. They cannot be conjugate. The supplied program checks all 24^2 pairs in S_4, finding 96 such nonconjugacy witnesses. It also checks all six matrices (D.26g) over $\mathbb{F}_3$ and $\mathbb{F}_5$, their fixed coboundary lines and the determinants on their two-dimensional cohomology quotients.

These calculations reduce the permissible type-$(0,4)$ markings to double transpositions with the specified multiplier and character. The second candidate (D.26j) is excluded, but D.29 constructs another marking by gluing a genuine three-cusp automorphism to its inverse. The exact basis change $b' = \tau(a)$ in that construction identifies its extension with (D.26a). Thus an invariant marking does exist; the finite failures above concern the two displayed candidates and the proved exclusions of the other permutation and character types. □

D.27. [Proved] The full-profinite rank divisibility and prime-order lifting test. For a torsion-free profinite extension

$$1 \longrightarrow \widehat{F}_r \longrightarrow E \longrightarrow G \longrightarrow 1, \qquad 2 \le r < \infty, \quad |G| < \infty,$$

the integer $|G|$ divides $r-1$. In particular a torsion-free finite extension of $\widehat{F}_2$ equals its prescribed kernel. Every homomorphism $C_p \to \mathrm{Out}(\widehat{F}_2)$, for any prime p, has a homomorphic lift to $\mathrm{Aut}(\widehat{F}_2)$; a faithful one has a lift of order exactly p. This last assertion holds in the full profinite category, without an invariant discretization and without first passing to a pro-p quotient.

Manuscript: §28; folios 178–179; scans 352–354; §31; folios 198–199 verso; scans 391–394.

Proof. We give the normal-subgroup version of Jarden's index argument [33, Theorems 1.2 and 4.1, pp. 496–498], with the quotient comparisons written out. Put $F = \widehat{F}_r$. For every prime q, $\mathrm{cd}_q(F) = 1$. The torsion-free finite-index dimension theorem of Serre [56, §1], whose torsion hypothesis applies to E itself, gives $\mathrm{cd}_q(E) = 1$. The same is true of every open subgroup of E. Thus these groups are projective profinite groups. Here we use the precise projectivity criterion recalled in [33, §2]: a profinite group is projective exactly when its cohomological dimension at every prime is at most one; equivalently, all its finite embedding problems have weak solutions. This is the criterion for the full profinite category, including finite embedding problems whose kernels are nonabelian.

Fix a prime divisor p of $|G|$, and let p^a be its exact p-part. Choose a Sylow p-subgroup $G_p \le G$ and write E_p for its inverse image in E. Define $O^p(J)$, for a profinite group J, as the kernel of its maximal pro-p quotient. The characteristic subgroup $O^p(F)$ of F is normal in E_p. There are exact continuous quotient maps

$$1 \longrightarrow F/O^p(F) \longrightarrow E_p/O^p(F) \longrightarrow G_p \longrightarrow 1.$$

The middle group is pro-p, since both its kernel and quotient are pro-p. Therefore $O^p(E_p) \le O^p(F)$. Conversely the restriction to F of the maximal pro-p quotient of E_p is a pro-p quotient of F, so $O^p(F) \le O^p(E_p)$. This proves

$$O^p(E_p) = O^p(F), \qquad [E_p^{(p)} : F^{(p)}] = p^a. \tag{D.27a}$$

The quotient $E_p^{(p)}$ is projective as a pro-p group. Indeed a finite embedding problem with p-group target and covering group pulls back to one for E_p; its projective solution factors through $E_p^{(p)}$ because its target is a p-group. A projective pro-p group is free pro-p. It has finite rank d_p because E_p is finitely generated. The subgroup $F^{(p)}$ in (D.27a) is free pro-p of rank r, with its full induced topology, as the exact quotient construction proves. Finite-coset Schreier rewriting therefore gives

$$r - 1 = p^a(d_p - 1). \tag{D.27b}$$

Consequently each exact prime-power divisor p^a of $|G|$ divides $r-1$. These divisors are pairwise coprime, proving the first assertion. At $r = 2$, this gives $|G| = 1$.

Now let $\rho : C_p \to \mathrm{Out}(\widehat{F}_2)$ be a prescribed continuous outer action, with the finite group given its discrete topology. Form the pullback

$$E_\rho = \{(a, g) \in \mathrm{Aut}(\widehat{F}_2) \times C_p : [a] = \rho(g)\}.$$

This is a profinite group. Centre-freeness identifies its kernel over C_p with $\mathrm{Inn}(\widehat{F}_2) = \widehat{F}_2$ by the injective map $x \mapsto (\mathrm{Inn}(x), 1)$. If E_ρ were torsion-free, the already proved rank-two assertion would force its quotient to be trivial, contradicting $|C_p| = p$. Hence E_ρ contains a nonidentity torsion element v. Its finite cyclic subgroup intersects the torsion-free kernel trivially, so projection embeds it in C_p. Its order is therefore exactly p, and it projects isomorphically onto C_p. The inverse of that projection is a section with the prescribed labels. Composing with the first projection $E_\rho \to \mathrm{Aut}(\widehat{F}_2)$ gives the asserted lift. For faithful ρ, the lift is faithful because its outer projection is. No choice of conjugacy class of this section is claimed; the argument proves existence.

Applied to a prime-order element of GT_0, this gives a genuine finite-order loop-preserving automorphism representing its outer class. It does not impose an unproved restriction on its multiplier, or identify its fixed subgroup. In particular it neither solves the residual norm equation D.25 nor proves conjugacy to the geometric involution. It does prove that the nontrivial torsion-free recognition test of D.22 first reaches free rank three, where (D.27b) permits index two and D.26 supplies an explicit unmarked example. □

D.28. [Proved] Prime-order sections and the full fixed subgroup. Let p be a prime and let $\rho : C_p \to GT_0$ be a continuous homomorphism whose multiplier has p-adic component 1. This condition is automatic for odd p. Let α be the normalized automorphism of $F = \widehat{F}(x, y)$ associated to the chosen generator of C_p, and put

$$E = F \rtimes_\alpha C_p, \qquad S = \{1\} \times C_p, \qquad C = C_F(S).$$

There is exactly one F-conjugacy class of sections of $E \to C_p$. Each of the three peripheral normalizers contains a section. The inclusion $C \hookrightarrow F$ induces an isomorphism of maximal pro-p quotients

$$C^{(p)} \xrightarrow{\sim} F^{(p)}. \tag{D.28a}$$

This assertion concerns the maximal pro-p quotient of the actual closed fixed subgroup. It does not identify that subgroup with F. An explicit order-three automorphism below has a proper fixed subgroup and exactly the isomorphism (D.28a), while its third peripheral class fails in a finite dihedral quotient.

Manuscript: §31; folios 198–199verso; scans 391–394; §30; folio 194verso; scan 384; Part D(a) of the task: actual full GT_0 torsion.

Proof. The normalized map $GT_0 \to \mathrm{Aut}(F)$ is a homomorphism: for the pair (λ, f) it sends $x \mapsto x^\lambda$, $y \mapsto f^{-1} y^\lambda f$, and composition gives precisely the product $f\alpha(g)$ in the second coordinate. Consequently $\alpha^p = 1$. On the maximal pro-p quotient its multiplier is 1; the Magnus argument of D.10 and D.25 makes this finite-order normalized automorphism the identity. Thus the quotient of the semidirect product is the explicit map

$$\pi : E \longrightarrow F^{(p)} \times C_p, \qquad (h, s^j) \longmapsto (\bar{h}, s^j). \tag{D.28b}$$

For odd p, a finite-order unit in $\mathbb{Z}_p^\times$ of order dividing p is 1: reduction modulo p eliminates the prime-to-p part, and the logarithm on $1 + p\mathbb{Z}_p$ eliminates the remaining torsion. This explains the automatic condition.

Write $c_1 = x$, $c_2 = y$, $c_3 = (xy)^{-1}$, and $L_i = \overline{\langle c_i \rangle}$. Preservation of the three peripheral classes gives an exact sequence

$$1 \longrightarrow L_i \longrightarrow N_E(L_i) \longrightarrow C_p \longrightarrow 1; \tag{D.28c}$$

its kernel is L_i by A.7. Choose a_i over the specified generator of C_p. Then

$$a_i c_i a_i^{-1} = c_i^{\lambda}, \qquad a_i^p = c_i^{m_i}, \qquad (\lambda - 1)m_i = 0. \tag{D.28d}$$

The first component of $\pi(a_i)$ centralizes $\bar{c}_i$, since $\lambda_p = 1$. The centralizer of this primitive element in the free pro-p group is $\overline{\langle \bar{c}_i \rangle}$. Hence $\pi(a_i) = (\bar{c}_i^{v_i}, s)$ for $v_i \in \mathbb{Z}_p$, and comparison of pth powers yields

$$(m_i)_p = p v_i. \tag{D.28e}$$

Let $N_\lambda = 1 + \lambda + \cdots + \lambda^{p-1}$. In the q-adic coordinate where $\lambda_q \neq 1$, the equation $\lambda_q^p = 1$ and the integral-domain property give $(N_\lambda)_q = 0$ and $(m_i)_q = 0$. Where $\lambda_q = 1$, the norm is p; division by p is possible for $q \neq p$, and (D.28e) supplies it at $q = p$. Define $u_i \in \widehat{\mathbb{Z}}$ coordinatewise by $(u_i)_q = -(m_i)_q/p$ in the latter coordinates and zero in the former. This gives $N_\lambda u_i = -m_i$. The element $b_i = a_i c_i^{u_i}$ has order p, by (D.28d), and projects to the prescribed generator. Thus every normalizer in (D.28c) contains a section S_i.

Let $e_p \in \widehat{\mathbb{Z}}$ be the idempotent with p-coordinate 1 and all other coordinates 0. Each S_i centralizes $c_i^{e_p}$. Its image in $F^{\mathrm{ab}} \otimes \mathbb{F}_p$ is the nonzero primitive vector $[c_i]$. It follows that

$$\dim_{\mathbb{F}_p} H^1(C_F(S_i), \mathbb{F}_p) \geq 1. \tag{D.28f}$$

This inference uses restriction of a linear functional on F^{ab}/p that is nonzero on the displayed element; it does not infer first homology from the mere presence of a pro-p subgroup.

We now use the finite-cohomology form of the profinite Brown theorem, quoted as Theorem 2 and applied in [37, §1.1, equations (1)–(6)]. There is no subgroup $C_p \times C_p$ in E, because every finite subgroup injects into $E/F = C_p$. The Lyndon–Hochschild–Serre spectral sequence and $\mathrm{cd}_p(F) = 1$ give, for every $n > 1$,

$$\dim_{\mathbb{F}_p} H^n(E, \mathbb{F}_p) \leq 1 + \dim_{\mathbb{F}_p} H^1(F, \mathbb{F}_p) = 3.$$

For each subgroup T of order p in E, projection to C_p shows that its normalizer equals its centralizer: a normalizing element must induce the identity on the generator of the quotient. Dividing by the unique element of T with the same quotient gives

$$N_E(T) = C_F(T) \times T.$$

The Brown restriction map is a dense embedding into the product of the cohomology of these normalizers. Its finite-dimensional source makes it an isomorphism and makes the number of factors finite. Each closed subgroup $C_F(T)$ has p-cohomological dimension at most one. The direct-product cohomology calculation therefore gives the exact finite sum and bound

$$\sum_{[T]} \Big(1 + \dim_{\mathbb{F}_p} H^1(C_F(T), \mathbb{F}_p)\Big) = \dim_{\mathbb{F}_p} H^n(E, \mathbb{F}_p) \leq 3, \tag{D.28g}$$

where $[T]$ runs through E-conjugacy classes of order-p subgroups. These are also their F-conjugacy classes: for any such T, the decomposition $E = FT$ gives the same conjugacy orbit under E and F.

By (D.28f), two distinct cusp-section classes would contribute at least four to (D.28g). Thus all three S_i are conjugate. Conjugating S_1 and S_2 to a single representative places conjugates of x^{e_p} and y^{e_p} in its fixed subgroup. Their images in F^{ab}/p are still the independent vectors $[x], [y]$. This gives two independent restricted linear functionals, so that common factor contributes at

least three. Bound (D.28g) is exhausted. There are no other classes, and its fixed subgroup has $\dim H^1(-, \mathbb{F}_p) = 2$. Every order-$p$ subgroup projects isomorphically onto C_p and determines one labelled section; hence there is exactly one conjugacy class of sections. In particular these conclusions apply to S.

The image of C in $F^{(p)}$ is closed and surjects on its Frattini quotient by the two independent cusp vectors, so it is all of $F^{(p)}$. Since C is a closed subgroup of a free profinite group, it is projective. Pulling a finite pro-p embedding problem back to C shows that $C^{(p)}$ is projective pro-p, and hence free pro-p. Its rank equals $\dim H^1(C, \mathbb{F}_p) = 2$. The resulting surjection from a rank-two free pro-p group onto $F^{(p)}$ is an isomorphism: after identifying free bases it is a surjective endomorphism of a finitely generated profinite group, which is injective. This proves (D.28a) in the claimed direction.

Here is an explicit limitation of that last comparison. Choose $u \in \widehat{\mathbb{Z}}^\times$ with $u_q = 1$ for $q \neq 7$, and u_7 the order-three root with $u_7 \equiv 2 \pmod 7$. The Hensel construction is given in D.25. The substitutions

$$\beta(x) = x^u, \qquad \beta(y) = y^u \tag{D.28h}$$

have inverse given by exponent u^{-1} and have order exactly three. The fixed subgroup C_β is proper, since $\beta(x) \neq x$ in the cyclic quotient of order seven. But it contains x^{e_3} and y^{e_3}, so its map to $F^{(3)}$ is onto. The same Brown bound, now using this one section and these two independent vectors, gives $\dim H^1(C_\beta, \mathbb{F}_3) = 2$. The preceding projectivity argument proves $C_\beta^{(3)} \cong F^{(3)}$, despite $C_\beta \neq F$.

The precise missing cusp condition is detected by

$$D_7 = \langle r, s \mid r^7 = s^2 = 1, srs = r^{-1} \rangle, \qquad x \mapsto s, \quad y \mapsto sr.$$

Here $xy \mapsto r$, $u \equiv 9 \pmod{14}$, and the two involutions are unchanged under exponent u. Thus $x^u y^u \mapsto r$ whereas $(xy)^u \mapsto r^2$. The conjugacy class of r consists of r, r^{-1}, so these are not conjugate. Consequently (D.28h) is not an element of the three-cusp loop-outer group, and in particular not a GT_0 torsion construction. The program `code/part_a/gt_torsion_sections.py` executes this finite certificate and the coordinate norm-solvability tests used above. D.32 uses the third cusp in a finite-cover rank argument to strengthen (D.28a) to $C = F$ under the three-cusp hypotheses, and thereby excludes the nontrivial prime-order actions considered here. □

D.29. [Refuted] A preserved four-puncture structure on the torsion-free Haran extension. The full profinite recognition assertion in D(d) is false, including its peripheral hypothesis. There is an exact sequence of profinite groups

$$1 \longrightarrow H = \widehat{\pi}_{0,4} \longrightarrow E \longrightarrow C_2 \longrightarrow 1 \tag{D.29.1}$$

in which E is torsion-free and its faithful outer action preserves a complete type-$(0, 4)$ loop structure, while E is the completion of no finite-type surface group, orientable or nonorientable. The extension is explicitly isomorphic to D.26's group. Its outer involution has a mixed-prime orientation multiplier and is conjugate to no subgroup of the geometric completion inside $\mathrm{Out}_{\mathrm{lac}}(\widehat{\pi}_{0,4})$. For this involution both the discretization and prediscretization fixed sets of F.1 are empty.

Manuscript: 28; 178–179; 352–354; 30; 194–196; 383–387; 31; 198–199 verso; 391–394.

Proof. Define units and an exponent by their prime coordinates:

$$\epsilon_3 = -1, \qquad \epsilon_q = 1 \ (q \neq 3), \qquad \kappa_2 = 1, \quad \kappa_q = 0 \ (q \neq 2), \qquad \lambda = -\epsilon. \tag{D.29.2}$$

Thus $\epsilon^2 = \lambda^2 = 1$ and $\epsilon\kappa = \kappa$. In particular $\lambda_3 = 1$ and $\lambda_q = -1$ for every other prime.

We first obtain the required pants isomorphism, without any finite-order requirement on it. The compatible assignments $\zeta_n \mapsto \zeta_n^{\lambda \bmod n}$ define an automorphism of the cyclotomic extension $\mathbb{Q}^{\mathrm{cyc}}/\mathbb{Q}$. An automorphism of an algebraic extension extends to its algebraic closure: extend its embedding along algebraic generators, or take a maximal extension of the embedding and adjoin a root of a missing minimal polynomial to prove maximality gives all of $\overline{\mathbb{Q}}$. We therefore choose an element $\sigma \in G_{\mathbb{Q}}$ with cyclotomic character λ. The Galois action on the three-punctured projective line commutes outerly with its rational S_3 action and preserves each cusp; on its local inertia it has exponent λ. Harbater–Schneps's normalized description [28, §0.2, Main Theorem (a), Theorem A] therefore supplies an element $(\lambda, f) \in GT_0$, with $f \in \widehat{F}(x,y)'$, whose automorphism is

$$x \longmapsto x^{\lambda}, \qquad y \longmapsto f(x,y)^{-1} y^{\lambda} f(x,y). \tag{D.29.3}$$

Its third cusp $(xy)^{-1}$ goes to a conjugate of its λth power. This is the only property of f needed below, besides invertibility of the displayed automorphism. Every choice of the extension σ has these properties, and every one gives the construction and conclusions that follow.

Let $H = \widehat{F}(l,a,b)$ and define its closed free factors

$$P_1 = \widehat{F}(l,a), \qquad P_2 = \widehat{F}(l,b), \qquad L = \overline{\langle l \rangle}.$$

Their shared-factor universal property is $H = P_1 *_L P_2$: homomorphisms out of the two free factors agreeing on l are exactly arbitrary continuous assignments of l, a, b. Give P_1 the ordered pants basis (l^{-1}, a) and P_2 the ordered pants basis (l, b). Formula (D.29.3), between these bases, defines a continuous isomorphism $\tau : P_1 \to P_2$ with

$$\begin{aligned} \tau(l) &= l^{\epsilon}, \\ \tau(a) &= f(l,b)^{-1} b^{\lambda} f(l,b), \\ \tau(a^{-1}l) &\sim_{P_2} (b^{-1}l^{-1})^{\lambda}. \end{aligned} \tag{D.29.4}$$

The symbol $\sim_{P_2}$ means conjugacy by an element of P_2; in particular it specifies a conjugacy in H. Because $\lambda^{-1} = \lambda$, the inverse isomorphism has the same peripheral multiplier.

Put $c = l^{\kappa}$ and prescribe

$$\alpha|_{P_1} = \tau, \qquad \alpha|_{P_2} = \mathrm{Inn}(c)\tau^{-1}. \tag{D.29.5}$$

Both maps send the shared generator l to l^{ϵ}: $\tau^{-1}(l) = l^{\epsilon}$ and c commutes with l. They consequently extend to a continuous endomorphism of H. Direct substitution on the two free factors gives

$$\alpha(c) = c, \qquad \alpha^2 = \mathrm{Inn}(c). \tag{D.29.6}$$

For example, on P_1 the second iterate is $\mathrm{Inn}(c)\tau^{-1}\tau$; on P_2 it is $\mathrm{Inn}(\alpha(c))\tau\tau^{-1}$. These identities also prove invertibility, with inverse $\mathrm{Inn}(c^{-1})\alpha$.

Here is the full peripheral structure, with its relation:

$$\begin{aligned} (u_1,u_2,u_3,u_4) &= (a,\ a^{-1}l,\ b,\ b^{-1}l^{-1}), \\ u_1u_2u_3u_4 &= 1. \end{aligned} \tag{D.29.7}$$

The first three entries form a free profinite basis, since $a = u_1$, $l = u_1u_2$, and $b = u_3$ are its continuous inverse substitutions. Thus (D.29.7) is precisely a type-$(0,4)$ presentation, not just four procyclic subgroups whose generators happen to satisfy a relation. Equations (D.29.4)–(D.29.5) give

$$\alpha(u_i) \sim_H u_{s(i)}^{\lambda}, \qquad s = (1\ 3)(2\ 4). \tag{D.29.8}$$

For the first two entries this is (D.29.4), and for the last two it is the inverse comparison, followed by conjugation by c. Hence the action preserves every required loop source, its conjugacy

ambiguity, and the common orientation multiplier. It permutes distinct cusp classes nontrivially, so its outer class is nonidentity. Equation (D.29.6) gives that class exact order two.

On the profinite space $H \times \{0,1\}$ define

$$(h,i)(k,j) = \big(h\alpha^i(k)c^{ij},\ i+j \bmod 2\big). \tag{D.29.9}$$

Associativity follows by the two carry identities in (D.29.6); checking the binary indices gives precisely $\alpha^2(k)c = ck$ and $\alpha(c) = c$. The element $t = (1,1)$ satisfies $t^2 = c$ and $tht^{-1} = \alpha(h)$. The inclusion $h \mapsto (h,0)$ and parity projection give (D.29.1), with the profinite product topology and continuous operations.

We make the comparison with D.26 explicit. Set $b' = \tau(a)$. The pair (l^ϵ, b') is a free basis of P_2, and replacing its first member by its ϵth power gives the free basis (l, b'). Thus (l, a, b') is a free basis of H, and

$$\alpha(l) = l^\epsilon, \qquad \alpha(a) = b', \qquad \alpha(b') = cac^{-1}. \tag{D.29.10}$$

The map from D.26's named free basis to (l, a, b'), with $t \mapsto t$, is a continuous isomorphism of its entire twisted extension onto E. It is an isomorphism of extensions with identity map on C_2. Formula (D.29.7) therefore supplies the previously missing preserved peripheral system on that exact group. In particular D.26's necessary double-transposition and double-cover character conditions are attained.

For clarity, we give the torsion and finite-quotient arguments also in the present basis. The retraction $r : H \to \widehat{\mathbb{Z}}$ sends l to 1 and a, b to 0. Since both $\tau(a)$ and $\tau^{-1}(b)$ lie in the normal closures of the respective nonshared generators, $r\alpha = \epsilon r$. It extends to a retraction of E onto the group $N = \widehat{\mathbb{Z}} \times \{0,1\}$ with law (D.29.9), action ϵ, and square parameter κ. An element $(z,1)$ in N has square $((1+\epsilon)z + \kappa, 0)$, whose 2-coordinate is $2z_2 + 1$ and is never zero. Since its open subgroup $\widehat{\mathbb{Z}}$ is torsion-free, every torsion element outside it would have order two; hence N is torsion-free. A torsion element of E has trivial image in N, so has even parity and lies in the torsion-free H. It is therefore the identity.

Let $V = \mathbb{F}_3^2$, with basis v_1, v_2, and send

$$l \longmapsto v_1, \qquad a \longmapsto v_2, \qquad b \longmapsto -v_2. \tag{D.29.11}$$

Here $\lambda_3 = 1$ and $\epsilon_3 = -1$; conjugators vanish in the abelian target. The induced action of α is -1 on V, and c has image zero because $\kappa_3 = 0$. Sending t to the involution consequently extends (D.29.11) to a continuous epimorphism

$$E \longrightarrow V \rtimes_{-1} C_2. \tag{D.29.12}$$

The target has order eighteen and requires three generators. Two positive elements do not generate the quotient C_2; one positive and one negative element generate at most one line of V; two negative elements reduce to the preceding case by taking their product. The three displayed images do generate it. The independent program `haran_quotient_independent_audit.py` checks all 18^3 associativity cases, all 18^2 ordered pairs, all ordered triples, and the actual matrix identity in (D.29.11). The ordered-pair subgroup-size histogram is

generated subgroup order	1	2	3	6	9
ordered pairs	1	27	32	216	48.

(D.29.13)

These calculations agree with the independently executed D.26 certificate under (D.29.10).

A free profinite E of finite rank d would have an index-two free subgroup of rank $1 + 2(d-1)$. The actual rank of H is three, forcing $d = 2$, contrary to (D.29.12). Thus E is not free. This excludes all punctured finite-type surface completions. For a putative isomorphism from a closed surface completion to E, the inverse image of H is an open index-two subgroup. Its

intersection with the dense surface group has index two, and completion of this intersection is exactly that open subgroup by the finite-index topology argument of D.1. It is the completion of the fundamental group of the corresponding closed double cover. Every aspherical closed surface completion has cohomological p-dimension two, whereas $\widehat{F}_3$ has dimension one. The spherical and projective-plane groups and their completions are finite and cannot give H. This excludes every closed case as well.

Finally A.2 supplies the continuous orientation homomorphism

$$\mathrm{or} : \mathrm{Out}_{\mathrm{lac}}(H) \longrightarrow \widehat{\mathbb{Z}}^{\times}.$$

It is unchanged by conjugation. Every discrete geometric class has value 1 or the global unit -1, and the closure of the geometric group still has values in that closed two-element subgroup. The value of $[\alpha]$ is the mixed unit λ in (D.29.2), which belongs to neither value. No conjugate of $\langle [\alpha] \rangle$ is therefore contained in the geometric completion. The transporter formulas of D.16 and F.1 now give both empty fixed sets claimed above.

The cusp normalizers themselves cause no exception to this argument. Since s fixes none of the four cusp classes, their normalizers in E map trivially to C_2; each equals its normalizer in H, namely its own procyclic loop group by A.7. The failure is present even with these procyclic normalizers retained. □

D.30. [Refuted] A closed genus-two counterexample to full profinite recognition. The assertion that every torsion-free finite extension of a full profinite closed hyperbolic surface group is a completed surface group is false. There is an exact extension

$$1 \longrightarrow H = \widehat{\pi}_1(\Sigma_2) \longrightarrow E \longrightarrow C_2 \longrightarrow 1 \tag{D.30.1}$$

with E torsion-free and with a faithful outer action, such that

$$H^2(E; \mathbb{F}_3) = \mathbb{F}_3, \qquad H^2(E; \mathbb{F}_5) = 0. \tag{D.30.2}$$

No profinite completion of a discrete surface group has these two cohomology groups. The exact maximal pro-p quotients are

$$E^{(2)} \cong \widehat{\pi_1(N_3)}^{(2)}, \qquad E^{(3)} \cong \mathbb{Z}_3^2, \qquad E^{(p)} \cong \widehat{F}_2^{(p)} \quad (p \neq 2, 3), \tag{D.30.3}$$

where N_3 is the closed nonorientable surface of genus three. Thus the counterexample also gives the precise maps through which its separate pro-p surface presentations arise. Its outer involution is conjugate to no element of the extended geometric completion. Its fixed discretization and prediscretization sets are both empty.

Manuscript: 28; 178 verso; 353; 24; 147 verso; 293; Part D(d) of the task.

Proof. We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$. Set

$$\epsilon_3 = -1, \quad \epsilon_p = 1 \ (p \neq 3), \qquad \kappa_2 = 1, \quad \kappa_p = 0 \ (p \neq 2), \qquad \lambda = -\epsilon \tag{D.30.4}$$

in $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. These elements satisfy $\epsilon^2 = \lambda^2 = 1$ and $\epsilon\kappa = \kappa$.

An actual punctured-torus isomorphism. Take the elliptic curve $Y^2 = 4X^3 - 4X$ over $\mathbb{Q}$, with its origin removed and its rational Weierstrass tangential base point. Nakamura's full profinite construction [50, §1, equation (1.2), and §2.5, p. 424] gives an actual homomorphism

$$G_{\mathbb{Q}} \longrightarrow \mathrm{Aut}\widehat{\langle x, y, z \mid [x, y]z = 1 \rangle}, \qquad \sigma(z) = z^{\chi(\sigma)}. \tag{D.30.5}$$

Here χ is the full cyclotomic character. The local coordinate at the rational puncture and its Kummer covers identify its inertia with $\widehat{\mathbb{Z}}(1)$, which explains the exact power equation. Every finite cyclotomic extension has Galois group $(\mathbb{Z}/n)^\times$; the resulting surjections and compactness show that $\chi : G_{\mathbb{Q}} \to \widehat{\mathbb{Z}}^\times$ is onto. Choose σ with $\chi(\sigma) = \lambda$. Renaming the target generators gives a continuous isomorphism

$$\tau : F_1 = \widehat{F}(x, y) \xrightarrow{\sim} F_2 = \widehat{F}(u, v), \qquad \tau([x, y]) = [u, v]^\lambda. \tag{D.30.6}$$

This uses a single Galois element of prescribed multiplier. The construction below proves its outer involution directly from τ and τ^{-1}.

The automorphism of the closed surface group. Use its defining profinite presentation

$$H = \langle x, y, u, v \mid [x, y][u, v] = 1 \rangle\widehat{\ }, \qquad l = [x, y]^{-1} = [u, v], \qquad c = l^\kappa. \tag{D.30.7}$$

Define α on the first pair by the two words of τ, and on the second pair by the two words of $\mathrm{Inn}(c)\tau^{-1}$. Equation (D.30.6) and $\lambda^{-1} = \lambda$ give

$$\alpha([x, y]) = l^\lambda, \qquad \alpha([u, v]) = c\,[x, y]^\lambda c^{-1} = l^{-\lambda}.$$

Their product is one, so this substitution defines a continuous endomorphism of the presented group H. It gives $\alpha(l) = l^\epsilon$ and $\alpha(c) = c$. Substitution on each of the four generators, using the actual inverse τ^{-1}, then gives

$$\alpha^2 = \mathrm{Inn}(c). \tag{D.30.8}$$

Consequently $\mathrm{Inn}(c^{-1})\alpha$ is its inverse. These identities use the universal property of the displayed presentation, so no assertion about efficiency of a profinite amalgam is required.

Define E on the compact space $H \times \{0, 1\}$ by

$$(h, i)(k, j) = \big(h\alpha^i(k)c^{\lfloor (i+j)/2 \rfloor}, i + j \bmod 2\big). \tag{D.30.9}$$

For quotient labels $(1, 1, 0)$ associativity is precisely $\alpha^2(k)c = ck$; for $(1, 1, 1)$ it also uses $\alpha(c) = c$. The other six label triples follow by direct substitution. The identity is $(1, 0)$; the inverse of $(h, 0)$ is $(h^{-1}, 0)$, and that of $(h, 1)$ is $(\alpha^{-1}(h^{-1}c^{-1}), 1)$. All operations are continuous. Thus E is a profinite group and its projection gives (D.30.1). Write $t = (1, 1)$, so that $tht^{-1} = \alpha(h)$ and $t^2 = c$.

An exact presentation and torsion-freeness. Set $u' = \tau(x)$ and $v' = \tau(y)$. This is a free basis of the formal second punctured-torus group, with

$$[u', v'] = [u, v]^\lambda = l^\lambda, \qquad u' = txt^{-1}, \quad v' = tyt^{-1}.$$

The relations for t on u', v' follow from $t^2 = c$. Eliminating these two generators therefore gives the exact profinite presentation

$$E \cong \langle x, y, t \mid tlt^{-1} = l^\epsilon, \quad t^2 = l^\kappa \rangle\widehat{\ }, \qquad l = [x, y]^{-1}. \tag{D.30.10}$$

For completeness, the homomorphism to (D.30.9) sends x, y, t to the named elements and is onto because their conjugates u', v' generate the second factor. In the reverse direction, recover u, v by applying the words of τ^{-1} to txt^{-1}, tyt^{-1}. Their commutator is $[txt^{-1}, tyt^{-1}]^\lambda = l$, by (D.30.10). Thus the surface relator holds, and the conjugation relations on both pairs follow from $t^2 = c$ and the defining words of α. The two maps fix all named generators and are inverse.

The universal property of the maximal pro-2 quotient substitutes $\epsilon_2 = \kappa_2 = 1$ into (D.30.10). Its first relation follows from the second, giving

$$E^{(2)} \cong \langle x, y, t \mid [x, y]t^2 = 1 \rangle_{\text{pro-2}}. \tag{D.30.11}$$

This is the rank-three Demushkin presentation with $q = 2$ and $f = \infty$: cyclically move t^2 to the beginning and replace x, y by their inverses to match the commutator convention of [59, §1.4 and Theorem 1.9]. The converse in that theorem proves the Demushkin assertion. Its abelianization is $\mathbb{Z}_2^2 \times C_2$, so it is infinite. The same source's preceding cohomological-dimension statement gives $\mathrm{cd}_2(E^{(2)}) = 2$. A subgroup of order two has infinite 2-cohomological dimension, so $E^{(2)}$ is torsion-free.

There is also an explicit free-basis identification with the three-crosscap word. The substitution

$$x \mapsto ad^{-1}b^{-1}a^{-1}, \qquad y \mapsto abdaba^{-1}, \qquad t \mapsto abd \tag{D.30.12}$$

sends $[x, y]t^2$ to $a^2b^2d^2$. Its inverse is

$$a \mapsto xt, \qquad b \mapsto t^{-1}x^{-1}t^{-1}yxt, \qquad d \mapsto t^{-1}x^{-1}y^{-1}t^2.$$

Free reduction checks the relator and both composites. The maps extend to mutually inverse pro-2 isomorphisms, proving the first comparison in (D.30.3).

The parity quotient $E \to C_2$ factors through $E^{(2)}$ and sends t to its nonidentity element and x, y to the identity. Every finite-order element of E maps trivially to the torsion-free $E^{(2)}$, hence has trivial parity and lies in H. The latter is torsion-free by A.7. This proves torsion-freeness of E itself and, in particular, the absence of a section of (D.30.1).

For an odd prime p, (D.30.10) gives $t^2 = 1$ in $E^{(p)}$, so $t = 1$. At $p = 3$ the remaining relation is $l = l^{-1}$, which forces $l = 1$, and the quotient is $\langle x, y \mid [x, y] = 1\rangle_{\text{pro-3}} = \mathbb{Z}_3^2$. At every other odd prime the remaining relation is the identity $l = l$. The universal properties in both directions give the free pro-p group on x, y. This proves all of (D.30.3).

The orientation line and the obstruction to every surface. Let $A \in \mathrm{GL}_2(\widehat{\mathbb{Z}})$ be the abelianization matrix of τ. In the finite Heisenberg group modulo n, the commutator of two elements is the central generator raised to the determinant of their two abelianization columns. Applying (D.30.6) there gives $\det A = \lambda \pmod n$ for every n, hence $\det A = \lambda$ in $\widehat{\mathbb{Z}}$. On the basis $([x], [y], [u], [v])$ of H^{ab}, inner conjugation disappears and the matrix of α is

$$M = \begin{pmatrix} 0 & A^{-1} \\ A & 0 \end{pmatrix}.$$

For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, the identity $A^t J A = (\det A)J$ gives

$$M^t \,\mathrm{diag}(J, J) M = \lambda \,\mathrm{diag}(J, J), \tag{D.30.13}$$

because $\lambda^{-1} = \lambda$. The companion identity $M \,\mathrm{diag}(J, J) M^t = \lambda \,\mathrm{diag}(J, J)$ also holds by $AJA^t = (\det A)J$. Since the matrix of α^* on degree-one cohomology is M^t, this last identity gives the multiplier of its cup pairing with all dual maps specified. The completed surface cup-product comparison of A.2 is natural for automorphisms. Its nondegenerate degree-one pairing therefore shows that α^* acts by λ_p on the one-dimensional $H^2(H; \mathbb{F}_p)$. The matrix M exchanges two nonzero direct summands, so α is not inner. Equation (D.30.8) thus gives a faithful outer C_2 action.

The orientation-line character is a continuous homomorphism

$$\mathrm{Out}(H) \longrightarrow \widehat{\mathbb{Z}}^{\times}.$$

Its restriction to the extended geometric mapping class group has image the two diagonal elements $+1, -1$, and continuity gives the same bound for its closure. Conjugation in $\mathrm{Out}(H)$ preserves this character because its target is abelian. Our element has value λ, with $\lambda_3 = 1$ and $\lambda_5 = -1$, so no conjugate lies in that closure. D.16's exact transporter formulas therefore give empty fixed discretization and prediscretization sets for this actual involution.

For odd p, averaging over C_2 makes all its higher cohomology with $\mathbb{F}_p$-coefficients vanish. The continuous Hochschild–Serre spectral sequence of (D.30.1) consequently gives

$$H^2(E;\mathbb{F}_p) \xrightarrow[\text{restriction}]{\sim} H^2(H;\mathbb{F}_p)^{C_2}. \tag{D.30.14}$$

At $p=3$ the scalar is $\lambda_3=1$, while at $p=5$ it is $\lambda_5=-1$. This proves (D.30.2).

Finally, the completion of a closed orientable surface group of positive genus has one-dimensional $H^2(-;\mathbb{F}_p)$ at every prime, by the surface comparison of A.2 (including its torus calculation). A closed nonorientable surface of genus at least two has an orientable double cover. Exact completion at index two and the same restriction map (D.30.14) compute its degree-two cohomology at odd primes as the invariants of the orientation reversal, hence zero. The sphere group is trivial and the projective-plane group is C_2, so both have zero degree-two cohomology at odd primes. A surface with boundary or a noncompact surface has a free fundamental group, whose profinite completion has cohomological dimension at most one. These cases exhaust the surface groups and exclude the pair of values in (D.30.2).

The independently executed program `code/part_b/closed_extension_d30_audit.py` checks the free-word isomorphism in (D.30.12), the intersection matrices and the finite cochain computation of the two invariant lines. The profinite construction, presentation comparisons and torsion-freeness are proved above; the finite computations verify the stated identities. □

D.31. [Proved] Abstract finite groups in the geometric completions of genus at most two. Let $M^+_{g,n}=\Gamma^+_{g,n}$ allow permutations of the n punctures, and let $M^\pm_{g,n}$ also allow orientation reversal. Every finite nonsolvable subgroup of $\widehat{M^+_{0,n}}$ is isomorphic to A_5. Its permutation action on the punctures has

$$n=12a+20b+30c+60m, \qquad a,b,c\in\{0,1\}, \quad m\geq 0, \tag{D.31.1}$$

where the summands are the actual orbit sizes. In particular, it has at least twelve punctures. Every finite nonsolvable subgroup of $\widehat{M^\pm_{0,n}}$ is isomorphic to A_5 or $A_5\times C_2$. Every finite subgroup of $\widehat{M^\pm_{g,n}}$ for $g=1,2$ is solvable.

These statements determine the abstract groups and, in the positive sphere case, their puncture-permutation representations. The surjectivity of the map from geometric A_5 conjugacy classes to completed A_5 conjugacy classes, and the completed A_5 centralizer comparison of F.7, remain to be proved.

Manuscript: §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D(b): finite subgroups of the geometric completion and full completion comparison.

Proof. For $n=0,1,2$ the positive sphere mapping class groups are respectively $1,1,C_2$; including orientation reversal gives groups of order at most four. Their completions are the same finite solvable groups. The sphere arguments below therefore take $n\geq 3$.

We give the completed exact sequences first, so that every use of a subgroup or quotient has its specified completion map.

A completion and configuration-group preparation. For the forgetting sequences used below, the kernel N is finitely generated and the quotient Q is good. These properties imply that $\widehat{N}\to\widehat{E}$ is injective. Here is the required argument. Take a characteristic finite-index subgroup N_0 of N; these are cofinal because N is finitely generated. In E/N_0 the kernel $B=N/N_0$ is finite. Its centralizer has finite index and fits into the central extension

$$1\longrightarrow Z(B)\longrightarrow C_{E/N_0}(B)\longrightarrow Q_0\longrightarrow 1,$$

where Q_0 has finite index in Q. Goodness passes to Q_0. The class in $H^2(Q_0, Z(B))$ is therefore represented by a continuous cohomology class of $\widehat{Q}_0$, hence is inflated from a finite quotient. Restriction to the kernel of that finite quotient kills it. The central extension splits over this finite-index subgroup; a chosen complement meets B trivially and has finite index in E/N_0. Its core is a finite-index normal subgroup still meeting B trivially. Pulling this core back to E gives a finite quotient whose restriction to N has kernel exactly N_0. This proves the full induced topology and thus

$$1 \longrightarrow \widehat{N} \longrightarrow \widehat{E} \longrightarrow \widehat{Q} \longrightarrow 1. \tag{D.31.2}$$

The last equality follows directly from the finite quotients of E/N.

The braid groups on the plane, on finitely punctured surfaces, and on the closed surfaces of genus one or two used here are good and have finite cohomological dimension. We recall how the ingredients apply here. The ordered configuration-space fibrations of Fadell–Neuwirth [19, Theorems 1 and 3] forget one point and have fibre the surface with finitely many points removed. The base surface and the fibres are aspherical; hence their configuration spaces are aspherical by the homotopy exact sequence. The fibres' groups are finitely generated free groups once a point has been removed. Starting with the good surface group, the preceding finite-kernel argument and the Lyndon–Hochschild–Serre comparison spectral sequences prove goodness inductively. The required cohomology of each fibre with finite coefficients is finite. For the plane the initial group is trivial; for the closed surfaces its goodness is the usual surface-group goodness also used in [9]. The unordered group has the ordered one as a finite-index subgroup, so goodness follows by the finite-index comparison and Shapiro's lemma. The configuration spaces are finite-dimensional aspherical manifolds, which give finite cohomological dimension. Goodness therefore makes each completed braid group torsion-free: its cohomology with every finite p-primary coefficient module vanishes above that dimension, whereas a subgroup C_p has infinite p-cohomological dimension. This also follows directly from the prime-subgroup comparison in [9, Theorem A].

Finite subgroups of a completed puncture stabilizer are cyclic. For $n \geq 4$, capping the boundary of a disc by a once-punctured disc gives the exact sequence

$$1 \longrightarrow \langle \Delta^2 \rangle \cong \mathbb{Z} \longrightarrow B_{n-1} \longrightarrow (M^+_{0,n})_p \longrightarrow 1. \tag{D.31.3}$$

Here the distinguished puncture is p, and the kernel is the central full twist; this is [5, Lemma 2.2], whose proof specifies the capping map. Braid abelianization sends each Artin generator to 1 and Δ^2 to $(n-1)(n-2)$. Thus the composite

$$\widehat{\mathbb{Z}} \longrightarrow \widehat{B}_{n-1} \longrightarrow \widehat{\mathbb{Z}}$$

is multiplication by this nonzero integer and is injective. Consequently completion of (D.31.3) is the exact central extension

$$1 \longrightarrow \widehat{\mathbb{Z}} \longrightarrow \widehat{B}_{n-1} \longrightarrow \widehat{(M^+_{0,n})_p} \longrightarrow 1. \tag{D.31.4}$$

The last group identifies with the stabilizer of p in $\widehat{M^+_{0,n}}$: its discrete source has finite index, and stabilizing p is the inverse image of a subgroup in the finite puncture-permutation quotient.

Let K be a finite subgroup of that completed stabilizer and let E_K be its inverse image in $\widehat{B}_{n-1}$. The subgroup $A = \widehat{\mathbb{Z}}$ is central and open in E_K, of index $d = |K|$. Continuous transfer over its finitely many cosets gives

$$\mathrm{Ver} : E_K \longrightarrow A, \qquad \mathrm{Ver}(a) = da \quad (a \in A). \tag{D.31.5}$$

Its kernel intersects A trivially, so injects into the finite group E_K/A. It is finite, and is trivial because E_K is a subgroup of the torsion-free completed braid group. Transfer is therefore a continuous injection into $\widehat{\mathbb{Z}}$. Its compact image is a closed procyclic subgroup. Hence E_K, and

then its finite quotient K, are procyclic and cyclic respectively. For $n = 3$ the same capping map is $1 \to 2\mathbb{Z} \to B_2 = \mathbb{Z} \to C_2 \to 1$, and the conclusion is immediate. For $n = 1, 2$ the positive point stabilizer is trivial.

It follows in particular that a finite subgroup of a completed positive sphere mapping class group preserving a specified set of $1 \le b \le 3$ punctures is solvable. The kernel of its permutation map to S_b fixes a puncture and is cyclic, while S_b is solvable. This assertion also applies to every finite-index decorated subgroup through its actual inclusion into the completed sphere group.

The prime-power inputs and elimination of a solvable radical. The good-group prime-power theorem [9, Theorem A] conjugates each finite p-subgroup of $\widehat{M^+_{0,n}}$ to a geometric one and gives its completed normalizer. The checked prime-power topology argument in F.7 identifies that normalizer with the full completion of its discrete normalizer. Thus, for a nontrivial normal p-subgroup A of a finite $G \le \widehat{M^+_{0,n}}$, conjugate A into $M^+_{0,n}$ and use the exact covering-normalizer sequence

$$1 \longrightarrow A \longrightarrow N_{M^+_{0,n}}(A) \longrightarrow Q_A \longrightarrow 1. \tag{D.31.6}$$

The group Q_A is a finite-index decorated subgroup of the mapping class group of the quotient sphere. Its specified branch set has two points for a cyclic rotation group, and three for a noncyclic rotation group. This branch set includes the images of ramification points that were already punctures upstairs; the ramification indices are retained in the covering decoration. Completion of (D.31.6) and the normalizer comparison give an actual injection

$$G/A \hookrightarrow \widehat{Q}_A. \tag{D.31.7}$$

It preserves that set of two or three branch points. The puncture stabilizer argument makes G/A solvable, and then G is solvable.

For a nontrivial finite solvable normal subgroup R of G, the last nonzero derived term of R is abelian. Its nontrivial p-primary component A is characteristic in R, hence normal in G. The preceding argument again makes G solvable. Consequently a nonsolvable finite G has trivial solvable radical. Only the prime-power comparison was used in this step.

The same prime-power theorem and the finite rotation-group list give cyclic odd Sylow subgroups of G, and cyclic or dihedral Sylow 2-subgroups, including C_2^2 in the latter case. There is also the precise normalizer constraint

$$\operatorname{im}(N_G(C_p) \longrightarrow \operatorname{Aut}(C_p)) \subset \{1, -1\}. \tag{D.31.8}$$

Indeed normalize a geometric rotation. A representative of a discrete normalizer either fixes its two poles or exchanges them. The oriented local rotation numbers are opposite, so the induced automorphism is respectively 1 or -1. The normalizer comparison preserves this image because $\operatorname{Aut}(C_p)$ is finite. The same argument with the local orientation sign proves (D.31.8) in the extended geometric completion as well. Equivalently one can apply the prime-power normalizer theorem to the good index-two extended mapping class group; the discrete representatives still induce only these two exponents.

We record the elementary finite-group reduction needed below. A finite group all of whose Sylow subgroups are cyclic is solvable. For the least prime p dividing its order, its cyclic Sylow group P has $N(P)/C(P)$ of order both prime to p and dividing $p^{a-1}(p-1)$, where $|P| = p^a$. Minimality of p forces this quotient to be trivial. Transfer to P has restriction $x \mapsto x^{[G:P]}$ and is onto. To check this formula without a fusion assumption, any two elements of the abelian Sylow group P conjugate in G are conjugate by $N(P)$: the two Sylow subgroups containing the conjugate element are conjugate inside its centralizer. Thus they are equal when $P \le Z(N(P))$. The coset-cycle formula for transfer now gives the stated exponent. Its kernel is a normal p-complement. Induction on the group order proves solvability. It follows that every odd-order subgroup here is solvable, and that a nonsolvable G has a noncyclic dihedral Sylow 2-subgroup.

The verified Gorenstein–Walter application. Let S be such a Sylow 2-subgroup and $z \in Z(S)$ an involution. The quotient of $C_G(z)$ by $\langle z \rangle$ in (D.31.7) preserves the two branch points of the rotation quotient. Its pointwise branch stabilizer is cyclic. Its inverse image $C_0 \leq C_G(z)$ has index at most two and is a central extension of a finite cyclic group by $\langle z \rangle$, hence is abelian: a lift of a quotient generator and the central kernel generate it. The odd-order part of C_0 is characteristic in C_0, normal in $C_G(z)$, and cyclic because all odd Sylow subgroups are cyclic. It is a normal abelian 2-complement of $C_G(z)$.

We can therefore apply the original, precisely matching theorem [23, Theorem I, p. 553]. It states that G has a normal odd-order subgroup K such that G/K is one of

$$\mathrm{PSL}_2(q), \quad \mathrm{PGL}_2(q) \quad (q \text{ odd}), \qquad A_7, \qquad \text{a Sylow 2-subgroup.} \tag{D.31.9}$$

The source explicitly includes C_2^2 among dihedral groups. All its hypotheses have just been verified. The odd group K is solvable by the preceding transfer argument, so the triviality of the solvable radical gives $K = 1$.

The group A_7 contains the commuting independent cycles (1 2 3) and (4 5 6), contradicting cyclic odd Sylow subgroups. A 2-group is solvable. Write $q = p^f$ in either matrix case. The unipotent subgroup $u(t) = \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ for $t \in \mathbb{F}_q$ is elementary abelian of order p^f, so $f = 1$. In $\mathrm{PSL}_2(p)$ the diagonal matrices $\mathrm{diag}(a, a^{-1})$ act on this C_p by $u(t) \mapsto u(a^2 t)$, supplying $(p-1)/2$ distinct automorphisms. In $\mathrm{PGL}_2(p)$ the matrices $\mathrm{diag}(a, 1)$ supply all $p - 1$ automorphisms. Constraint (D.31.8) leaves $p \leq 5$ in the first case and $p \leq 3$ in the second. At $p = 3$, the faithful projective-line action embeds these groups into S_4, so they are solvable. The remaining group is $\mathrm{PSL}_2(5) \cong A_5$.

For this last isomorphism the exact map is conjugation on the five Sylow 2-subgroups. The supplied matrix enumeration finds fifteen involutions, partitions them into the five four-groups, and verifies that the conjugation image has order sixty and consists of even permutations. With the recorded ordering of the five subgroups, the matrices

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

map respectively to (1 2 4 3 5) and (2 4)(3 5). Thus the map is injective and its image is exactly A_5, with its sources and targets fixed.

The puncture representation. The pure positive sphere group is the ordered configuration group of $n - 3$ points on the three-punctured sphere: fix the first three marked points and use their contractible identity homeomorphism component in the evaluation fibration. It is good of finite cohomological dimension by the preceding configuration argument, so its completion is torsion-free. This is exactly the kernel of the completed map to S_n, by the finite-index completion sequence. Thus every finite subgroup acts faithfully on the puncture set.

For a finite A_5 subgroup, each puncture stabilizer is cyclic by (D.31.4)–(D.31.5). The cyclic subgroups of A_5 have orders $1, 2, 3, 5$, so its transitive puncture orbits are precisely $A_5/1$, A_5/C_2, A_5/C_3, and A_5/C_5. The last three have sizes $30, 20, 12$. For each $d = 2, 3, 5$, an element generating C_d fixes $|N_{A_5}(C_d)/C_d| = 2$ points in the corresponding orbit and none in the other three types. Prime-power conjugacy makes its puncture permutation that of a rotation of the sphere, which has at most two fixed punctures. Each exceptional orbit occurs at most once. This proves (D.31.1), including its actual permutation representation, and gives the lower bound twelve.

The extended sphere group. Let $G \leq \widehat{M^{\pm}_{0,n}}$ be finite and nonsolvable. Its positive subgroup G^+ has index at most two and is nonsolvable, so is A_5. The centralizer $C_G(A_5)$ intersects A_5 trivially and injects into G/G^+, so has order at most two. With a nontrivial centralizer, its generator is negative and $G = A_5 \times C_2$. With a trivial centralizer, conjugation embeds G into $\mathrm{Aut}(A_5) \cong S_5$ with image containing the inner A_5, leaving A_5 or S_5. The automorphism isomorphism is induced

by the action on the five subgroups $A_4 \leq A_5$. An automorphism fixing all five makes g and its image induce the same permutation of them for every $g \in A_5$; the faithful natural action of A_5 then makes that automorphism the identity. Conjugation by S_5 is faithful as well and supplies one hundred twenty automorphisms, proving the isomorphism. In S_5, the normalizer of a 5-cycle acts on it by all four nonzero exponents modulo five. This violates the extended version of (D.31.8). Therefore only A_5 and $A_5 \times C_2$ occur as abstract nonsolvable groups.

Forgetting punctures in genus two. Evaluation on the unordered puncture set, with isotopy extension, gives the discrete forgetting sequence

$$1 \longrightarrow B_n(S_2) \longrightarrow M_{2,n}^{+} \longrightarrow M_{2,0}^{+} \longrightarrow 1. \tag{D.31.10}$$

The identity homeomorphism component of S_2 is contractible, as used in Part C, so there is no preceding evaluation-loop kernel. The quotient is good, the braid kernel is finitely generated, and (D.31.2) completes this sequence exactly. The completed kernel is torsion-free by the configuration argument. Every finite subgroup of $\widehat{M_{2,n}^{+}}$ therefore injects under forgetting into $\widehat{M_{2,0}^{+}}$. The hyperelliptic quotient is the exact sequence

$$1 \longrightarrow C_2 \longrightarrow M_{2,0}^{+} \longrightarrow M_{0,6}^{+} \longrightarrow 1; \tag{D.31.11}$$

see [5, Propositions 3.2–3.3]. The central involution is detected on homology modulo three, so it injects into the completion. Finite-kernel completion then makes (D.31.11) exact with hats. Every finite subgroup has solvable image in $\widehat{M_{0,6}^{+}}$ by (D.31.1), and kernel contained in C_2. It is solvable. The same holds in the extended group because the positive subgroup has index at most two. The case $n = 0$ uses (D.31.11) directly.

Forgetting punctures in genus one. For $n \geq 1$, evaluation gives

$$1 \longrightarrow N_n \longrightarrow M_{1,n}^{+} \longrightarrow \mathrm{SL}_2(\mathbb{Z}) \longrightarrow 1, \qquad N_n = B_n(T^2)/\mathbb{Z}^2. \tag{D.31.12}$$

Here $\mathbb{Z}^2$ is the image of the translation loops in the identity homeomorphism component, identified by evaluation as in C.9. It is central in the braid group: applying a loop of translations to every configuration of a braid gives the square homotopy interchanging the two loop products. Its injection can be checked on the ordered configuration space by evaluating the first strand. To verify its full topology in the unordered braid group, sum the homology classes of all strand paths:

$$B_n(T^2) \longrightarrow H_1(T^2, \mathbb{Z}) = \mathbb{Z}^2.$$

Endpoints are permuted, so the sum is a cycle and gives a homomorphism. On diagonal translation loops it is multiplication by n. The resulting completed injection proves the exact central extension

$$1 \longrightarrow \widehat{\mathbb{Z}}^2 \longrightarrow \widehat{B_n(T^2)} \longrightarrow \widehat{N}_n \longrightarrow 1. \tag{D.31.13}$$

The completed braid group is torsion-free by the configuration argument. For completeness the ordered torus configuration has the explicit product coordinates

$$(x_1, \ldots, x_n) \longmapsto (x_1, x_2 - x_1, \ldots, x_n - x_1) \in T^2 \times \mathrm{Conf}_{n-1}(T^2 \setminus \{0\}),$$

so its goodness also follows directly from that of $\mathbb{Z}^2$ and the iterated punctured-surface configuration groups.

For a finite $K \leq \widehat{N}_n$, pull it back in (D.31.13). Transfer to the central open $\widehat{\mathbb{Z}}^2$ has restriction $a \mapsto |K|a$. Its kernel again injects into K and is finite, so torsion-freeness makes transfer injective. The pullback, and hence K, is abelian. Since $\mathrm{SL}_2(\mathbb{Z})$ is good and N_n is finitely generated, (D.31.2) also completes (D.31.12) exactly. The image of every finite subgroup lies in a finite subgroup of $\widehat{\mathrm{SL}_2(\mathbb{Z})}$, which is cyclic of order $1, 2, 3, 4$, or 6 by D.8's virtually free comparison. The finite

kernel is abelian by (D.31.13), proving solvability. For orientation reversal the positive subgroup has index at most two, which preserves this conclusion. For $n = 0$ use D.8 directly with $\mathrm{GL}_2(\mathbb{Z})$.

The executed program `code/part_a/sphere_completion_restriction.py` checks the projective-matrix automizer calculations at $p = 3, 5, 7, 11, 13$, constructs the displayed $\mathrm{PSL}_2(5) \to A_5$ isomorphism, compares the C_5 normalizer actions in A_5 and S_5, and checks the exceptional coset fixed-point counts and the C_3^2 subgroup of A_7. The general reductions are the completed exact sequences and transfer arguments above. □

D.32. [Proved] The three-cusp rank test and finite pure loop groups. Let $F = \widehat{F}(x, y)$, with the three named peripheral generators $c_1 = x$, $c_2 = y$, and $c_3 = (xy)^{-1}$. Let P be the subgroup of $\mathrm{Out}(F)$ preserving each of their three procyclic conjugacy classes. Its common peripheral multiplier is denoted by λ. Every finite subgroup of P maps injectively, by λ_2, to $\{1, -1\} \le \mathbb{Z}_2^\times$. In particular it has order at most two, and a nonidentity element of finite order has order two and $\lambda_2 = -1$. These statements hold for $GT_0 \le P$ as well.

For the full three-cusp loop-outer group, allowing puncture permutations, every finite subgroup G has the explicit faithful representation

$$G \longrightarrow S_3 \times C_2, \qquad g \longmapsto (\sigma_g, \lambda_2(g)). \tag{D.32a}$$

This is an assertion about the displayed homomorphism and abstract finite groups. Conjugacy of its original action to the geometric action with this image is not asserted here. D.34 determines the global multiplier of a surviving involution; its geometric conjugacy remains to be proved.

Manuscript: 30; folios 194–196; scans 383–387; 31; folios 198–199 verso; scans 391–394; Part D(b) of the task.

Proof. We first prove the closed-subgroup fact which uses all three cusps. For a prime p, let $e_p \in \widehat{\mathbb{Z}}$ have p-coordinate one and all other coordinates zero. Suppose a closed subgroup $C \le F$ has $\dim_{\mathbb{F}_p} H^1(C, \mathbb{F}_p) = 2$ and contains, for each $i = 1, 2, 3$, a conjugate

$$g_i c_i^{e_p \mathbb{Z}_p} g_i^{-1}. \tag{D.32b}$$

We prove $C = F$.

The comparison with open subgroups is the actual restriction map

$$\operatorname*{colim}_{U \text{ open},\ C \le U \le F} H^1(U, \mathbb{F}_p) \xrightarrow{\sim} H^1(C, \mathbb{F}_p), \tag{D.32c}$$

where the transition maps are restrictions as U decreases. Indeed a continuous character of C kills $C \cap V$ for some open normal $V \triangleleft F$. It extends to CV by the quotient $CV/V = C/(C \cap V)$. Conversely a character on U which vanishes on C vanishes after restriction to its open kernel, which contains C. This proves both surjectivity and injectivity of (D.32c).

If C is open, finite-coset Schreier rewriting identifies it with a free profinite group of rank $1 + [F : C]$. Its given first-cohomology dimension therefore forces $[F : C] = 1$. Suppose instead that C has infinite index. The p-valuations of the finite indices $[F : U]$, for open U containing C, must be unbounded. To prove this, assume they were bounded and choose U_0 attaining their maximum. For $C \le V \le U_0$, every further restriction between such open subgroups has prime-to-p index and is injective on $H^1(-, \mathbb{F}_p)$: a nonzero character has an index-p kernel and cannot vanish on a subgroup of prime-to-p index. Thus (D.32c) contains the first cohomology of every such V. The indices $[F : V]$ are unbounded, since their intersection is C and $[F : C] = \infty$. Their Schreier dimensions $1 + [F : V]$ are therefore unbounded, a contradiction.

Choose an open $U \ge C$ for which $p \mid [F : U]$. Put $\Delta = F(x, y)$, the named dense discrete subgroup of F, and $U_0 = U \cap \Delta$. The finite covering of the pair of pants defined by $U_0 \le \Delta$ has degree $n = [F : U]$, and its completed fundamental group is exactly U: finite-index subgroups of

a discrete group carry the full induced profinite topology. For each g_i choose $h_i \in \Delta$ in its left U-coset and write $g_i = u_i h_i$, with $u_i \in U$. The subgroup

$$U \cap g_i \overline{\langle c_i \rangle} g_i^{-1} = u_i \overline{\langle h_i c_i^{d_i} h_i^{-1} \rangle} u_i^{-1} \tag{D.32d}$$

is consequently the completion, up to the displayed inner conjugation, of a particular boundary subgroup in this finite cover. Here d_i is its boundary covering degree. Inclusion (D.32b) proves $p \nmid d_i$.

The three selected boundaries lie over the three different original boundaries, so they are distinct. There is at least one more boundary. Otherwise each original boundary would have just one lift, and its degree would be n, contrary to $p \nmid d_i$ and $p \mid n$. If the cover has genus h and $b \geq 4$ boundary components, its presentation is

$$\left\langle a_1, b_1, \dots, a_h, b_h, d'_1, \dots, d'_b \;\middle|\; \prod_{j=1}^{h} [a_j, b_j]\, d'_1 \cdots d'_b = 1 \right\rangle.$$

Thus the only relation between its boundary homology classes over $\mathbb{F}_p$ is their sum. The three selected classes are independent, and evaluation on them gives a surjection $H^1(U, \mathbb{F}_p) \to \mathbb{F}_p^3$.

This remains a surjection after restriction of these characters to C and evaluation on the three elements in (D.32b). To check the comparison explicitly, the element $g_i c_i^{e_p} g_i^{-1}$ is the e_p/d_i power of the boundary generator in (D.32d). Inner conjugation does not change a character, and its evaluation is therefore multiplied by the nonzero scalar $d_i^{-1} \in \mathbb{F}_p$. We obtain three independent characters of C, contradicting its dimension two. This proves the closed-subgroup assertion.

We now apply it to finite loop actions, retaining the precise normalization map. If an outer class fixes the three peripheral classes, its exponents on c_1, c_2, c_3 are equal: in $F^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$, the relation $[c_1] + [c_2] + [c_3] = 0$ gives equality of the two coordinates. Conjugate a representative β to have $\beta(x) = x^\lambda$, and write $\beta(y) = k^{-1} y^\lambda k$. Write $[k] = (a, b)$ in the named abelian basis. Conjugating this representative by x^a gives the normalized pair

$$\alpha(x) = x^\lambda, \qquad \alpha(y) = f^{-1} y^\lambda f, \qquad f = y^{-b} k x^{-a} \in F'. \tag{D.32e}$$

Multiplication on the left of the conjugator by y^{-b} changes neither image of y nor the automorphism. Two normalized representatives differ by conjugation by an element centralizing x^λ, hence by x^v by A.7. Their two conjugators then differ by left multiplication by a power of y and right multiplication by x^{-v}. Since both are in F', their two abelian coordinates force both powers to be zero. Normalization is therefore unique. A product of normalized pairs has second coordinate $f\alpha(g) \in F'$, so (D.32e) is a homomorphic section into $\mathrm{Aut}(F)$. Its image is precisely the Belyi group defined in [28, p. 3120, equation (2)]; the argument has not identified that entire group with its subgroup GT_0.

Let an element of P have prime order p with $\lambda_p = 1$. Its normalized automorphism has order p. The proof of D.28 uses exactly the properties just established: finite order of the normalized automorphism, the common multiplier, and preservation of the three cusps. Neither of the extra GT_0 symmetry equations is used in its norm computation or Brown restriction argument. In particular that proof gives a single conjugacy class of sections in $F \rtimes C_p$ and the actual fixed subgroup $C = C_F(C_p)$ with $\dim H^1(C, \mathbb{F}_p) = 2$. Each of the three peripheral sections centralizes the whole $c_i^{e_p \mathbb{Z}_p}$; conjugating those sections to the specified C_p places the three groups (D.32b) in this same C. The closed-subgroup assertion gives $C = F$, so the normalized automorphism is the identity. This contradicts its prime order.

For odd p, an element of order dividing p in $\mathbb{Z}_p^\times$ equals one, by reduction modulo p and the torsion-free group $1 + p\mathbb{Z}_p$. Thus no odd prime occurs in the order of a finite subgroup of P. The finite image of λ_2 lies in $\{1, -1\}$, since $1 + 4\mathbb{Z}_2$ is torsion-free. Its kernel in any finite subgroup

has no element of order two by the preceding argument, and no element of odd prime order either. Cauchy's theorem makes that kernel trivial. This proves the first assertion.

For a class permuting cusps, compose it with the inverse of the corresponding positively oriented geometric permutation. The preceding abelian calculation gives its common multiplier too, and composition on cusp generators proves that both σ and λ are homomorphisms. On a finite subgroup, the kernel of (D.32a) is a finite subgroup of P on which $\lambda_2 = 1$, hence is trivial. All subgroups of the target already have their geometric actions on the three-punctured sphere: use an equilateral triple on the equator, the positive rotational S_3, and reflection in that equator for the commuting C_2. This proves the claimed abstract group bound and its attainment, while keeping the action-conjugacy question separate through the actual map (D.32a).

The program `code/part_c/three_cusp_rank_d32.py` enumerates the finite covers used for a validation through degree six. It computes the Schreier graph, every boundary cycle, the unique boundary-homology relation, and explicit dual characters for the selected boundaries, including the d_i^{-1} restriction scalars. The proof above applies to arbitrary degrees and closed subgroups; no bounded enumeration is used in place of it. □

Source check for the fixed-second-coordinate assertion. Lochak–Schneps, Lemma 9 on p. 583 of [41], asserts that a fixed f occurs only with the two multipliers λ and $-\lambda$. The displayed class-two identity $\lambda^2 = 24r + 1$ determines the squares of the prime coordinates. The second proof in that source adds the required full-group equation, whose elementary verification is as follows.

Comparing (λ, f) and (μ, f) by the exact group law gives $(\lambda^{-1}\mu, 1)$. Set $n = (\lambda^{-1}\mu - 1)/2 \in \widehat{\mathbb{Z}}$. The three-cycle relation is

$$x^n (xy)^{-n} y^n = 1. \tag{LS.1}$$

In the class-two Heisenberg quotients its central coordinate gives $n(n+1) = 0$. Thus in each integral domain $\mathbb{Z}_p$ one has $n_p = 0$ or $n_p = -1$. These choices are linked by actual finite quotients: for every odd prime q, evaluate (LS.1) in

$$D_q = \langle r, s \mid r^q = s^2 = 1,\ srs = r^{-1} \rangle, \qquad x = s, \quad y = r.$$

Both x and xy have order two. If $n_2 = 0$, (LS.1) is $r^{n_q} = 1$, forcing $n_q = 0$. If $n_2 = -1$, it is $r^{1+n_q} = 1$, forcing $n_q = -1$. Hence n is globally zero or globally minus one, proving $\mu = \lambda$ or $\mu = -\lambda$. Both occur, because multiplication on the right by $(-1, 1)$ leaves the second coordinate unchanged.

This verifies the source's fixed-f conclusion by exact quotient morphisms. Powers of a general torsion element have second coordinates $f\alpha(f) \cdots \alpha^{j-1}(f)$; the argument does not identify those coordinates with each other. D.32 supplies the separate argument restricting finite subgroups, and leaves the stated involution questions unchanged.

D.33. [Proved] Finite punctured-torus actions are detected by their primary homology. Put $F = \widehat{F}(x, y)$ and $c = [x, y]$, and let $T = \mathrm{Out}_{\mathrm{lac}}(F; \overline{\langle c \rangle})$ be the full once-punctured-torus loop-outer group. For every prime p, the homomorphism induced by the named abelianization,

$$A_p : T \longrightarrow \mathrm{GL}_2(\mathbb{Z}_p), \tag{D.33a}$$

is injective on every finite p-subgroup of T. The full abelianization homomorphism $T \to \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is therefore injective on every finite subgroup. Every such subgroup has order dividing 24.

The fixed-subgroup calculation underlying these assertions is also explicit. For an automorphism α of $\widehat{F}_r$ with $\alpha^p = 1$ whose induced automorphism on the maximal pro-p quotient is the identity, put $E = \widehat{F}_r \rtimes_\alpha C_p$. There is one kernel-conjugacy class of sections of $E \to C_p$, and for its fixed subgroup C the actual restriction map is an isomorphism

$$H^1(\widehat{F}_r, \mathbb{F}_p) \xrightarrow{\sim} H^1(C, \mathbb{F}_p). \tag{D.33b}$$

For the punctured-torus loop structure and $r = 2$, its cusp forces $C = F$ and hence $\alpha = 1$.

Manuscript: 30; folios 194–196; scans 383–387; 31; folios 198–199 verso; scans 391–394; Part D(b) of the task.

Proof. We first prove (D.33b), keeping the multiplicative information in the cohomology comparison. Write $P = \widehat{F}_r^{(p)}$ and $S = \{1\} \times C_p \leq E$. The quotient homomorphism is

$$\pi : E \longrightarrow P \times C_p, \qquad (h, s^i) \longmapsto (\bar{h}, s^i). \tag{D.33c}$$

Restriction along $\widehat{F}_r \to P$ is an isomorphism on mod-p cohomology: in degree one both sides are the same r-dimensional character group, and higher cohomology vanishes because both groups have p-cohomological dimension one. The Hochschild–Serre comparison for (D.33c) is consequently an isomorphism on its second pages and gives a ring isomorphism

$$\pi^* : H^*(P \times C_p, \mathbb{F}_p) \xrightarrow{\sim} H^*(E, \mathbb{F}_p). \tag{D.33d}$$

All maps here are induced by the specified quotient, not by an identification of the two groups. In particular $\dim H^n(E, \mathbb{F}_p) = r + 1$ for every $n > 1$.

Let $u \in H^2(C_p, \mathbb{F}_p)$ be the nonzero periodic class, and inflate it to either side of (D.33d). The Künneth calculation for the direct product gives, for every $m \geq 1$, an injective map

$$H^1(P, \mathbb{F}_p) \longrightarrow H^{2m+1}(E, \mathbb{F}_p), \qquad \chi \longmapsto \pi^*(\chi u^m). \tag{D.33e}$$

For a subgroup $Q \leq E$ of order p, its projection to C_p is an isomorphism, and $N_E(Q) = C_{\widehat{F}_r}(Q) \times Q$, exactly as in D.28. The first coordinate of $\pi(Q)$ is trivial because P is torsion-free. Thus every character $\pi^*\chi$ in (D.33e) vanishes on Q. Its restriction to this normalizer lies entirely in $H^1(C_{\widehat{F}_r}(Q), \mathbb{F}_p)$, and multiplication by u^m places it in

$$H^1(C_{\widehat{F}_r}(Q), \mathbb{F}_p) \otimes H^{2m}(Q, \mathbb{F}_p). \tag{D.33f}$$

Apply the profinite Brown restriction theorem in the precise form verified in [37, Theorem 2 and §1.1]. There is no $C_p \times C_p$ in E, since a finite subgroup injects into $E/\widehat{F}_r = C_p$. Its open subgroup $\widehat{F}_r$ has p-cohomological dimension one. The finite-dimensional source makes the theorem's dense restriction embedding an isomorphism onto a finite product. If k is the number of conjugacy classes and $d_Q = \dim H^1(C_{\widehat{F}_r}(Q), \mathbb{F}_p)$, its dimension formula is

$$k + \sum_{[Q]} d_Q = r + 1. \tag{D.33g}$$

The injection (D.33e), followed by this same restriction map, lands in the summands (D.33f). Therefore $\sum d_Q \geq r$. Since $k \geq 1$, (D.33g) forces $k = 1$ and $d_Q = r$. For the specified S, (D.33e)–(D.33f) then say exactly that restriction $H^1(\widehat{F}_r, \mathbb{F}_p) \to H^1(C_{\widehat{F}_r}(S), \mathbb{F}_p)$ is injective; both spaces have dimension r, proving (D.33b). As in D.28, conjugacy under E equals conjugacy under the kernel, because $E = \widehat{F}_r Q$ for any section Q.

Now take $r = 2$ and assume that α preserves the conjugacy class of $L = \overline{\langle c \rangle}$, where $c = [x, y]$. Write its peripheral multiplier as λ. In a class-two Heisenberg quotient, the image of the commutator under an abelianization matrix A is the $\det(A)$ power of the commutator. The finite Heisenberg quotients for all moduli therefore give the exact identity

$$\lambda = \det(\alpha^{\mathrm{ab}}). \tag{D.33h}$$

In the present calculation α is the identity on P, so $\lambda_p = 1$.

The cusp normalizer has the exact sequence

$$1 \longrightarrow L \longrightarrow N_E(L) \longrightarrow C_p \longrightarrow 1. \tag{D.33i}$$

The kernel is L by A.7. For a lift a of the chosen generator, write $aca^{-1} = c^{\lambda}$ and $a^p = c^v$, so $(\lambda - 1)v = 0$. The first coordinate of $\pi(a)$ centralizes the image $\bar{c}$ in P and belongs to $\overline{\langle \bar{c} \rangle}$, by the pro-$p$ peripheral centralizer assertion of A.7. Thus it is $\bar{c}^w$ and $v_p = pw$. This uses the whole cusp subgroup. Its primitive pro-p parameter can also be checked directly: in the mod-p Magnus expansion, $[x, y] - 1$ starts with $XY - YX$. A pth power starts in degree divisible by p. At $p = 2$, the possible degree-two square is $(aX + bY)^2$; its X^2 and Y^2 coefficients prevent it from equaling $XY + YX$. Hence the commutator is not a pth power.

Since $\alpha^p = 1$, its multiplier satisfies $\lambda^p = 1$. Set $N_\lambda = 1 + \lambda + \cdots + \lambda^{p-1}$. At a prime q with $\lambda_q \neq 1$, both $(N_\lambda)_q$ and v_q are zero. Where $\lambda_q = 1$ the norm is p; v_q is divisible by p, using $v_p = pw$ at $q = p$ and invertibility of p at all other primes. Consequently there is $z \in \widehat{\mathbb{Z}}$ with $N_\lambda z = -v$. The element ac^z has order p and centralizes $c^{e_p \mathbb{Z}_p}$. The unique section-conjugacy class just proved places a conjugate $gc^{e_p \mathbb{Z}_p} g^{-1}$ inside $C = C_F(S)$.

Every character of C vanishes on $gc^{e_p} g^{-1}$: by (D.33b) it is the restriction of a character of F, and c is a commutator. We prove that this forces $C = F$. Its mod-p first-cohomology dimension is two. If C is open, Schreier gives $1 + [F : C] = 2$. If C is proper and not open, the restriction-colimit proof of D.32 shows that its index has infinite p-part. Choose an open $U \geq C$ with $p \mid n = [F : U]$, and take the degree-n covering of the punctured torus corresponding to $U \cap F(x, y)$. The chosen cusp has a boundary lift of degree d prime to p, since its subgroup contains the full chosen $c^{e_p \mathbb{Z}_p}$. The comparison with a discrete boundary generator is exactly (D.32d), with a dense discrete representative for the profinite conjugator g. There must be at least two boundary lifts: a unique lift would have degree n, divisible by p. The selected boundary has nonzero homology in this cover, since the sole relation among the at least two boundary classes is their sum. A character of U detecting this boundary restricts to a character of C detecting $gc^{e_p} g^{-1}$; the value is multiplied by $d^{-1} \bmod p$, as computed in D.32. This is the required contradiction. Hence $C = F$ and $\alpha = 1$.

To apply this to (D.33a), let an outer element have order p and $A_p = I$. D.27 supplies an actual order-p automorphism representing it. Its induced pro-p automorphism has identity abelianization. The Magnus argument in D.10 and D.25 makes this finite-order automorphism the identity: a first nonzero homogeneous deviation in the $\mathbb{Z}_p$ Magnus algebra would be multiplied by p under the pth iterate, and the homogeneous module has no p-torsion. Thus (D.33c) applies to this lift directly; there is no subsequent conjugation which could change its order. The cusp argument proves that the lift, and hence its outer class, is the identity. The kernel of (D.33a) on a finite p-subgroup cannot contain an order-p element, so it is trivial. Likewise, the kernel of full abelianization on any finite subgroup would contain an element of some prime order and is trivial.

We finish with the numerical bound on these finite groups. For $p > 3$, a matrix of order p over $\mathbb{Q}_p$ has a primitive pth root among its eigenvalues. The polynomial $\Phi_p(t)$ is irreducible of degree $p - 1$ over $\mathbb{Q}_p$, as seen by Eisenstein after $t \mapsto t + 1$. It cannot be the factor of a degree-two characteristic polynomial. Thus no such prime divides the order of a finite subgroup of T. A finite 3-subgroup injects by A_3 into $\mathrm{GL}_2(\mathbb{Z}_3)$ and then injects modulo 3: the kernel $1 + 3M_2(\mathbb{Z}_3)$ is torsion-free by the matrix logarithm. Since $|\mathrm{GL}_2(\mathbb{F}_3)| = 48$, its order divides 3.

For completeness, a finite 2-subgroup K of $\mathrm{GL}_2(\mathbb{Q}_2)$ has order at most eight. The same Eisenstein argument for Φ_{2^a} excludes order 2^a for $a \geq 3$. Every matrix of order four has minimal polynomial $t^2 + 1$, so its square is $-I$. Consequently the projective image of K has exponent two. Extend the matrices to the algebraic closure $\overline{\mathbb{Q}}_2$; the scalar kernel is unchanged. An exponent-two subgroup of $\mathrm{PGL}_2(\overline{\mathbb{Q}}_2)$ has order at most four. To prove this last bound, conjugate a nonidentity element to $z \mapsto -z$. Every commuting transformation preserves its pair of fixed points, so is $z \mapsto az$ or $z \mapsto a/z$. The scaling subgroup has at most two elements under the exponent-two condition and has index at most two. The scalar kernel of K is contained in $\{I, -I\}$, the finite 2-torsion of $\mathbb{Q}_2^{\times}$. Thus $|K| \leq 8$. Apply this to a Sylow 2-subgroup through A_2. The Sylow bounds

and the exclusion of other primes give $|G| \mid 8 \cdot 3 = 24$.

The program `code/part_c/torus_homology_d33.py` checks the Magnus degree-two coefficients, the periodic cohomology multiplication used in (D.33e), and the one-cusp restriction argument on every connected finite cover through degree six. The matrices and boundary characters are obtained from actual Schreier walks. The bound and comparison maps above are proved for the full profinite groups, independently of the bounded checks. □

D.34. [Proved] Global signs and the cusp cycle of a pure involution. Every nonidentity finite-order element of the full pure three-cusp loop-outer group P of D.32 is an involution whose common multiplier is the global element $-1 \in \widehat{\mathbb{Z}}^{\times}$. In particular this holds in GT_0, and the multiplier-one kernel in either group is torsion-free.

For its normalized lift

$$\alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1}y^{-1}f, \qquad f \in F', \tag{D.34a}$$

there are precisely three F-conjugacy classes of sections of $F \rtimes_{\alpha} C_2 \to C_2$. Each of their fixed subgroups has zero first cohomology with $\mathbb{F}_2$ coefficients. The two section classes in each cusp normalizer are the endpoints of one edge of a triangle. Moreover there is $h \in F'$ with

$$f = h^{-1}\alpha(h). \tag{D.34b}$$

Thus α inverts both x and hyh^{-1}. The endomorphism $x \mapsto x$, $y \mapsto hyh^{-1}$ induces an automorphism of every maximal pro-p quotient. Its surjectivity on the full F, and conjugacy of the original involution to complex conjugation in the pure loop group, are not conclusions of this statement.

Manuscript: 30; folios 194–196; scans 383–387; 31; folios 198–199 verso; scans 391–394; Part D(b) of the task.

Proof. We first prove a genus-zero incidence calculation for an arbitrary number $n \geq 3$ of cusps. Let

$$F_n = \langle c_1, \ldots, c_n \mid \widehat{c_1 \cdots c_n} = 1 \rangle$$

and let β be an actual involution preserving each peripheral conjugacy class, with common multiplier μ and $\mu_2 = -1$. Put $E_n = F_n \rtimes_{\beta} C_2$. The equality $\beta^2 = 1$ gives $\mu^2 = 1$; hence each μ_q is 1 or -1.

For each cusp $L_i = \overline{\langle c_i \rangle}$, A.7 gives an exact sequence

$$1 \longrightarrow L_i \longrightarrow N_{E_n}(L_i) \longrightarrow C_2 \longrightarrow 1. \tag{D.34c}$$

Choose a lift a_i of the nonidentity element and write $a_i c_i a_i^{-1} = c_i^{\mu}$ and $a_i^2 = c_i^{v_i}$. Then $(\mu - 1)v_i = 0$. In every coordinate with $\mu_q = -1$ this forces $(v_i)_q = 0$, including $q = 2$. In a coordinate with $\mu_q = 1$ the prime q is odd, so division by two is possible. Thus $(1 + \mu)u_i = -v_i$ has a solution in $\widehat{\mathbb{Z}}$, and $b_i = a_i c_i^{u_i}$ is an involution.

The set of its further order-two translates by L_i, modulo conjugation by L_i, is exactly

$$\frac{\ker(1 + \mu : \widehat{\mathbb{Z}} \to \widehat{\mathbb{Z}})}{(1 - \mu)\widehat{\mathbb{Z}}} \;\simeq\; \mathbb{F}_2. \tag{D.34d}$$

At an odd prime with $\mu_q = -1$, multiplication by two is onto, and a coordinate with $\mu_q = 1$ contributes zero to the kernel. The remaining coordinate is $\mathbb{Z}_2/2\mathbb{Z}_2$. Representatives for the two classes are b_i and $c_i^{e_2} b_i$.

Since β is pure and the cusp classes span homology, its action on $V = H_1(F_n, \mathbb{F}_2)$ is the identity. Therefore the actual mod-two abelianization of the semidirect product is

$$H_1(E_n, \mathbb{F}_2) = V \oplus \mathbb{F}_2. \tag{D.34e}$$

The two involutions just exhibited have images differing by the nonzero vector $[c_i]$ in V. They cannot be conjugate in E_n or in F_n. Thus the two local classes in (D.34d) give two distinct global section classes.

Build a finite graph with one edge for each of the n cusps and with these two global classes as its endpoints. This graph uses some of the global section classes; we now bound all of them. The Brown calculation of D.28, with free rank $n-1$ and prime two, gives

$$\sum_{[Q]}(1+\dim_{\mathbb{F}_2} H^1(C_{F_n}(Q),\mathbb{F}_2)) = \dim_{\mathbb{F}_2} H^k(E_n,\mathbb{F}_2) \le n \qquad (k>1). \tag{D.34f}$$

Every order-two subgroup is a section, and its total-group and kernel conjugacy classes agree, because $E_n = F_nQ$. Hence there are at most n vertices available for the graph. An edge has no loop by (D.34e).

Assign to every vertex its image $(v,1)$ in (D.34e). The sum of the edge vectors around a graph cycle is zero, since its successive vertex differences cancel. But the n vectors $[c_1],\dots,[c_n]$ have just one nonzero linear relation over $\mathbb{F}_2$, namely $[c_1]+\cdots+[c_n]=0$. Consequently any simple cycle in the graph uses all n edges. The graph cannot be a forest: a forest with n edges needs at least $n+1$ vertices, more than (D.34f) permits. Thus it is exactly a simple n-cycle. It has n vertices, so there are no additional section classes outside this graph, and (D.34f) gives

$$H^1(C_{F_n}(Q),\mathbb{F}_2) = 0 \quad \text{for every section } Q. \tag{D.34g}$$

This construction is invariant under any automorphism of the extension preserving its kernel and peripheral classes, with any permutation of those classes: such an automorphism sends the normalizer and its two classes in (D.34c)–(D.34d) to those of the image cusp.

Return to a nontrivial finite-order element of P. D.32 gives its order two, its normalized actual lift α, and $\lambda_2 = -1$. We show that $\lambda_q = -1$ for every odd prime q. The only other possibility is $\lambda_q = 1$. Suppose this occurs and use the exact cyclic quotient

$$\chi : F = \widehat{F}(x,y) \longrightarrow C_q, \qquad \chi(x)=1, \quad \chi(y)=0, \qquad N = \ker\chi. \tag{D.34h}$$

The common multiplier gives $\chi\alpha = \lambda\chi = \chi$, since its reduction modulo q is one. In particular N is invariant. The corresponding degree-q covering of the pair of pants has one boundary over x, one over $z=(xy)^{-1}$, and q over y, with representative generators

$$x^q, \qquad z^q, \qquad x^iyx^{-i} \quad (0\le i<q). \tag{D.34i}$$

The two first degrees are q and all others are one. Its Euler characteristic is $-q$ and it has $q+2$ boundaries, so its genus is zero. Finite-index completion identifies its full profinite group with the specified subgroup N.

The restriction of α is pure on this entire peripheral structure, with the same multiplier λ. The first two boundaries are the unique lifts of their original boundaries. For the other q boundaries use the actual normalized formula

$$\alpha(x^iyx^{-i}) = x^{\lambda i}f^{-1}y^{\lambda}fx^{-\lambda i}. \tag{D.34j}$$

Here $\chi(f)=0$, because $f\in F'$, and $\lambda i \equiv i \pmod q$. The conjugator in (D.34j) therefore belongs to the same N-coset as x^i: explicitly $k_i = x^{\lambda i}f^{-1}x^{-i}\in N$ satisfies $\alpha(x^iyx^{-i}) = k_i(x^iyx^{-i})^{\lambda}k_i^{-1}$. This is the required conjugacy inside N. The action on $H_1(N,\mathbb{Z}_2)$ is consequently the scalar -1, so the restriction remains a nontrivial involution.

Put $E = F\rtimes_\alpha C_2$ and $E_N = N\rtimes_\alpha C_2$. The quotient maps give

$$E/N = C_q\times C_2. \tag{D.34k}$$

In particular E_N is normal in E. Conjugation by x defines an automorphism of E_N preserving its map to C_2, and thus acts on its N-conjugacy classes of sections. Its qth power is conjugation by $x^q \in N$, so this action has order dividing q on those classes and on the cusp incidence graph. On the cusp edges (D.34i), it fixes the x^q and z^q classes and cycles the q classes over y. The second fixed class is the unique boundary over z, or follows directly from its cyclic coset decomposition. Thus its graph action has order exactly q.

By the preceding calculation the graph is a cycle with $q+2$ edges. Its automorphism group has order $2(q+2)$: an image of a vertex and a choice of which neighbor is its successor determine the automorphism. The odd prime q does not divide $2(q+2)$. This contradicts the displayed order-q graph action. Therefore $\lambda_q = -1$ for every odd q, proving the global multiplier assertion. This argument used the whole pure group P, without an additional GT_0 symmetry assumption.

At $n = 3$ the incidence graph is the stated triangle. Write s for the chosen section in $E = F \rtimes C_2$. The normalized group law gives $f\alpha(f) = 1$, so fs is also an involution and it normalizes y by inversion. The three involutions

$$s, \quad x^{e_2}s, \quad y^{e_2}fs \tag{D.34l}$$

have the distinct images $(0,0,1)$, $(1,0,1)$, and $(0,1,1)$ in (D.34e). They represent all three classes. Since $f \in F'$, the image of fs is $(0,0,1)$, so it is conjugate to s by an element of F. Choose h with $fs = h^{-1}sh$. This is exactly (D.34b). In abelianization it says $0 = (-2)[h]$ in the torsion-free group $\widehat{\mathbb{Z}}^2$, hence $h \in F'$. Substitution gives

$$\alpha(hyh^{-1}) = (hf)(f^{-1}y^{-1}f)(f^{-1}h^{-1}) = hy^{-1}h^{-1}.$$

The two displayed inverted generators have the same abelian images as x, y. On each maximal pro-p quotient their images therefore generate the Frattini quotient and hence the whole group. The induced endomorphism of that finitely generated free pro-p group is surjective and thus an automorphism. No full-profinite surjectivity is inferred from those separate comparisons.

The program `code/part_c/pure_involution_d34.py` checks the normalizer orbit calculation, the unique binary cusp relation, and the exact cyclic-cover boundary and deck permutations for several odd primes. The exclusion of all odd primes is the unrestricted divisibility argument above. □

D.35. [Proved] One exact conjugator equation and its finite obstructions. Let P be the full pure three-cusp group of D.32, let $F = \widehat{F}(x,y)$, and put $\delta(x) = x^{-1}$, $\delta(y) = y^{-1}$. For a nonidentity involution $\alpha \in P$, use its normalized lift from D.34,

$$\alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1}y^{-1}f, \qquad f \in F'.$$

Define the compact solution space

$$\begin{aligned} \mathcal{S}_\alpha = \{(\mu, h, u) \in \widehat{\mathbb{Z}}^\times \times F' \times F : \quad & h^{-1}\alpha(h) = f, \\ & x^\mu h y^\mu h^{-1} = u(xy)^\mu u^{-1}\}. \end{aligned} \tag{D.35a}$$

The assignment

$$(\mu, h, u) \longmapsto [\beta_{\mu,h}], \qquad \beta_{\mu,h}(x) = x^\mu, \quad \beta_{\mu,h}(y) = hy^\mu h^{-1}, \tag{D.35b}$$

has image exactly

$$\mathrm{Conj}_P(\delta, \alpha) = \{\beta \in P : \ \beta\delta\beta^{-1} = \alpha\}. \tag{D.35c}$$

Every map (D.35b) is an automorphism of the full profinite group. The fibers of the assignment are exactly $(\mu, h, u) \mapsto (\mu, h, u(xy)^a)$ for $a \in \widehat{\mathbb{Z}}$. It therefore induces a homeomorphism from the corresponding quotient of $\mathcal{S}_\alpha$ to (D.35c).

For $\alpha \in GT_0$, the inclusion $GT_0 \hookrightarrow P$ gives the equality of actual conjugator sets

$$\mathrm{Conj}_{GT_0}(\delta, \alpha) = \mathrm{Conj}_P(\delta, \alpha). \tag{D.35d}$$

The defining Harbater–Schneps relations for the conjugating pair follow from the proof below; they are not discarded. Finally, the existence of a solution of (D.35a) is equivalent to the nonemptiness of its explicit finite solution sets (D.35h) at every characteristic finite quotient of F.

Manuscript: §31; folios 198-199verso; scans 391-394; §30; folio 194verso; scan 384; Part D(b): actual full GT_0 torsion and conjugacy.

Proof. For $(\mu, h, u) \in \mathcal{S}_\alpha$, the free universal property first gives a continuous endomorphism $\beta_{\mu,h} : F \to F$ with the displayed generator images. Write $H = \beta_{\mu,h}(F)$, a closed subgroup because F is compact. It is generated by two elements, so $\dim H^1(H, \mathbb{F}_2) \leq 2$. The restrictions of the two standard characters of F to these generators are the two coordinate characters: $\mu \bmod 2 = 1$ and conjugation does not change abelianization. These restrictions are independent, proving

$$\dim H^1(H, \mathbb{F}_2) = 2. \tag{D.35e}$$

The unit exponent μ shows that H contains $\overline{\langle x \rangle}$ and $h\overline{\langle y \rangle}h^{-1}$. The second equation in (D.35a) puts $u\overline{\langle xy \rangle}u^{-1}$ in H as well. In particular H contains the three full cusp $\mathbb{Z}_2$ subgroups required by D.32. Its closed-subgroup rank test, applied to (D.35e), gives $H = F$. A surjective endomorphism of a finitely generated profinite group is injective, so $\beta_{\mu,h}$ is an automorphism. Its images of the three cusps, including the exact displayed equation for the third, place its outer class in P.

The first equation in (D.35a) is $\alpha(h) = hf$. Substitution on the two free generators gives

$$\alpha\beta_{\mu,h}(x) = x^{-\mu} = \beta_{\mu,h}\delta(x), \qquad \alpha\beta_{\mu,h}(y) = hy^{-\mu}h^{-1} = \beta_{\mu,h}\delta(y). \tag{D.35f}$$

These are equalities of actual automorphisms, proving that the image of (D.35b) lies in (D.35c).

Conversely, let $\beta \in P$ satisfy (D.35c), and take its unique normalized representative from D.32. Write it as $\beta_{\mu,h}$ with $h \in F'$. The normalization is a homomorphic section $P \to \mathrm{Aut}(F)$, so the outer conjugacy equality lifts to the actual equality $\alpha\beta_{\mu,h} = \beta_{\mu,h}\delta$. Its value on y implies that $h^{-1}\alpha(h)f^{-1}$ centralizes $y^{-\mu}$. By A.7 it is a power of y. It also lies in F', so its abelianization forces this power to be zero. Thus $h^{-1}\alpha(h) = f$. The third-cusp condition on β gives a conjugator u in the second equation of (D.35a). This proves that the image is exactly (D.35c).

The normalized pair (μ, h^{-1}) is unique. For this fixed pair the two third-cusp conjugators u, u' differ by right multiplication by the centralizer of $(xy)^\mu$, which is $\overline{\langle xy \rangle}$ by A.7. These are precisely the asserted fibers. The assignment is continuous in the profinite automorphism topology, and its domain is compact. Its quotient is therefore homeomorphic to its Hausdorff image.

For an alternative form using D.34, fix $h_0 \in F'$ with $h_0^{-1}\alpha(h_0) = f$, and put $C = C_F(\alpha)$. All solutions of the first equation of (D.35a) are exactly $h = ch_0$ with $c \in C$: $\alpha(h) = hf$ and $\alpha(h_0) = h_0 f$ give $\alpha(hh_0^{-1}) = hh_0^{-1}$, and the reverse substitution proves the converse. Since α acts by -1 on F^{ab}, every $c \in C$ has $2[c] = 0$, hence $[c] = 0$. Thus $C \subset F'$ and this substitution respects the stated domain of (D.35a). The remaining exact equation is

$$x^\mu ch_0 y^\mu h_0^{-1} c^{-1} = u(xy)^\mu u^{-1}. \tag{D.35g}$$

We prove (D.35d) by identifying the relevant centralizer, rather than assuming that a conjugator in P lies in GT_0. Let $P^1 = \ker(\lambda : P \to \widehat{\mathbb{Z}}^\times)$. First, $C_{P^1}(\delta) = 1$. Indeed a normalized element $(1, k) \in P^1$ commuting with δ has $k = \delta(k)$, by uniqueness of the normalized

representative and its group law. The following explicit mutually inverse homomorphisms identify the semidirect product:

$$C_2 \amalg C_2 \amalg C_2 \xrightarrow{\sim} F \rtimes_\delta \langle s \rangle, \qquad (s_0, s_1, s_2) \longmapsto (s, xs, ys),$$
$$(x, y, s) \longmapsto (s_1 s_0, s_2 s_0, s_0).$$

The orders and conjugation relations follow by direct substitution. A free profinite factor has trivial intersection with a conjugate by an element outside that factor; the original full-profinite statement is recalled in [31, Theorem 5 of the inspected author version]. Consequently the centralizer of s_0 in this free product is $\langle s_0 \rangle$. Its intersection with F is trivial. Hence $C_F(\delta) = 1$, so $k = 1$ and $C_{P^1}(\delta) = 1$ as claimed.

Let $\Sigma \cong S_3$ be the geometric cusp-permutation subgroup of the full loop-outer group, generated by the outer classes $\theta(x) = y,\ \theta(y) = x$ and $\omega(x) = y,\ \omega(y) = (xy)^{-1}$. The exact identification $GT_0 = C_P(\Sigma)$ follows from [28, Propositions 3–4 and Theorem 2]: commutation with θ is relation (I), and, after (I), commutation with ω is relation (II). Both α and δ centralize Σ. For any $\beta \in \mathrm{Conj}_P(\delta, \alpha)$ and $\sigma \in \Sigma$, form the actual element of the outer group $r_\sigma = \beta^{-1}\sigma\beta\sigma^{-1}$. The pure subgroup P is normal in the full loop-outer group, and the common multiplier is a homomorphism there. Therefore $r_\sigma \in P^1$. Moreover

$$\begin{aligned} r_\sigma \delta r_\sigma^{-1} &= \beta^{-1}\sigma\beta(\sigma^{-1}\delta\sigma)\beta^{-1}\sigma^{-1}\beta \\ &= \beta^{-1}\sigma\alpha\sigma^{-1}\beta = \beta^{-1}\alpha\beta = \delta. \end{aligned}$$

The proved equality $C_{P^1}(\delta) = 1$ gives $r_\sigma = 1$. Thus β centralizes Σ, so its normalized pair satisfies both (I) and (II) and lies in GT_0. This proves (D.35d).

Here is the exact finite form of the remaining existence problem. For each open characteristic subgroup $N \triangleleft F$, put $Q_N = F/N$ and $m_N = \exp(Q_N)$. The automorphism α induces α_N. Use bars for the images of x, y, f in Q_N and define

$$\begin{aligned} \mathcal{S}_{\alpha,N} = \{(a, b, v) \in (\mathbb{Z}/m_N)^\times \times Q_N' \times Q_N :\ & b^{-1}\alpha_N(b) = \bar{f}, \\ & \bar{x}^a b \bar{y}^a b^{-1} = v(\bar{x}\bar{y})^a v^{-1}\}. \end{aligned} \tag{D.35h}$$

These are finite sets, obtained entirely from the specified finite quotient and its induced automorphism. Reduction sends $\mathcal{S}_\alpha$ to every $\mathcal{S}_{\alpha,N}$. Conversely, a point of a nonempty $\mathcal{S}_{\alpha,N}$ lifts to $\widehat{\mathbb{Z}}^\times \times F' \times F$: units lift by the Chinese remainder theorem, and the image of F' in Q_N is exactly Q_N'. The lifts satisfying the two equations modulo N form a nonempty closed subset of this compact space. Every finite collection of such conditions is implied by the one at a common characteristic refinement. Their finite intersection property therefore gives a point satisfying both equations in F itself. Open characteristic subgroups are cofinal for a finitely generated profinite group, so equality at all these quotients is equality in F. This proves the asserted finite criterion.

The need for the full third-cusp equation can be checked by a short executed finite certificate. Put $d = [x, y] = xyx^{-1}y^{-1}$ and take the actual commutator word

$$w = yd^{-1}y^{-1}x^{-1}d^{-1}x \in F'.$$

Consider the endomorphism $\rho(x) = x,\ \rho(y) = wyw^{-1}$. Its abelianization is the identity, so the Frattini argument makes its induced endomorphism on every maximal pro-p quotient an automorphism. In the finite quotient

$$F \twoheadrightarrow A_5, \qquad x \longmapsto (1\,2\,3), \quad y \longmapsto (3\,4\,5),$$

the program `code/part_a/pure_involution_conjugator.py` enumerates the generated group of order 60 and computes

$$\bar{w} = (1\,4\,3\,2\,5), \qquad \overline{wyw^{-1}} = (1\,2\,3).$$

The image of $\rho(F)$ in this quotient therefore has order 3, proving $\rho(F) \neq F$. Moreover $\overline{xy}$ has order 5, whereas $\overline{xywy^{-1}}$ has order 3; the third-cusp equation fails in this same quotient. This word is a certificate concerning the generation test. It is not asserted to solve the coboundary equation for a nonstandard involution. No nonemptiness or emptiness assertion for the spaces (D.35a) of arbitrary involutions is inferred from this finite example. □

D.36. [Proved] The global multiplier of a positive transposition. Let B be the full three-cusp loop-outer group, and let $g \in B$ be an involution whose 2-adic multiplier is $\lambda_2(g) = 1$. Its puncture permutation is a transposition, and its common multiplier is the global element $1 \in \widehat{\mathbb{Z}}^\times$.

Consequently a finite subgroup of B whose map (D.32a) has image $S_3 \times C_2$ has the usual global orientation character. The same is true when that image is a V_4 containing the pure involution, or is the positively oriented $S_3 \times \{1\}$. These statements identify the actual multiplier homomorphism; they do not assert conjugacy of those finite actions.

Manuscript: §30; folios194–196; scans383–387; §31; folios198–199verso; scans391–394; Part D(b) of the task.

Proof. We first record the genus-zero extension of the actual automorphism calculation already proved in D.32–D.33. Let $F_n = \widehat{\pi}_{0,n}$ with $n \geq 3$, and let β be a pure automorphism with $\beta^p = 1$ and p-adic peripheral multiplier one. Then $\beta = 1$. Indeed F_n is free of rank $r = n - 1$, and its cusp classes span its abelianization. Its induced pro-p automorphism has identity abelianization and finite order, so the $\mathbb{Z}_p$ Magnus argument of D.33 makes it the identity. The ring calculation (D.33c)–(D.33g) gives one conjugacy class of sections of $F_n \rtimes_\beta C_p \to C_p$, and its actual fixed subgroup C has

$$\dim H^1(C, \mathbb{F}_p) = n - 1. \tag{D.36a}$$

The norm calculation of D.28 applies to every cusp: its image is a primitive element of the free pro-p group, the first coordinate of its normalizer lift centralizes that element, and the lift's pth power has p-coordinate divisible by p. Coordinatewise division of the norm therefore supplies a section in each peripheral normalizer. Uniqueness of the section class places conjugates of all n full peripheral $\mathbb{Z}_p$ subgroups inside the same C.

The finite-cover rank proof of D.32 now applies with all n cusps. If C is open, its first-cohomology dimension is $1 + [F_n : C](n - 2)$, which equals $n - 1$ only at index one. If it has infinite index and finite p-part of that index, the same exact restriction colimit and eventual prime-to-p injections give unbounded first-cohomology dimension. Otherwise choose an open $U \geq C$ of p-divisible index. The n selected boundary lifts all have degrees prime to p and lie over different original boundaries. They cannot be all the boundaries: a single lift over any original boundary would have degree $[F_n : U]$. With at least $n + 1$ boundaries, the n selected boundary homology classes are independent. Their dual characters restrict to C with the nonzero multipliers $d_i^{-1} \bmod p$ computed in (D.32d). Thus $\dim H^1(C, \mathbb{F}_p) \geq n$, contradicting (D.36a). Hence $C = F_n$ and $\beta = 1$. This proof concerns an actual finite-order automorphism; it does not presume a lift for arbitrary finite outer actions in larger rank.

Return to g. D.32 excludes a nonidentity pure involution with $\lambda_2 = 1$, so its permutation must be a transposition. Relabel the original cusp generators as $x, y, z = (xy)^{-1}$ so that x is fixed and y, z are exchanged. D.27 provides an actual order-two lift α of this outer action. Its peripheral multiplier satisfies $\lambda^2 = 1$, so all its prime coordinates are 1 or -1.

Suppose $\lambda_q = -1$ for an odd prime q, and use the exact cyclic quotient

$$\chi : F \longrightarrow C_q, \qquad \chi(x) = 0, \quad \chi(y) = 1, \quad \chi(z) = -1, \qquad N = \ker \chi. \tag{D.36b}$$

The puncture permutation and the multiplier give $\chi\alpha = \chi$: on y the exponent -1 at q is multiplied by $\chi(z) = -1$, and similarly for z. Thus N is invariant and the induced action on its

cyclic deck quotient is the identity. The degree-q cover is a sphere with $q+2$ cusps. There are q degree-one cusps over x, and one degree-q cusp over each of y, z.

Here are the exact covering and filling generators. Finite-coset Schreier rewriting with coset representatives $1, y, \dots, y^{q-1}$ gives the free basis of N

$$a_i = y^i x y^{-i} \quad (0 \le i < q), \qquad b = y^q.$$

The remaining boundary generator is

$$d = z^q = b^{-1} a_{q-1}^{-1} \cdots a_0^{-1}, \qquad a_0 \cdots a_{q-1} b d = 1. \tag{D.36c}$$

For example the latter relation is the actual free-word identity $(xy)^q = a_0 \cdots a_{q-1} y^q$. Completing the discrete Schreier comparison identifies this presented free profinite group with the specified open subgroup N.

The lift α fixes the conjugacy class of every a_i inside N. To see this without choosing an unproved normalization, write $\alpha(x) = h^{-1} x^\lambda h$ for some $h \in F$. Since $\chi\alpha(y) = 1$, its action on the indices of these cusp classes is the translation $i \mapsto i - \chi(h)$ in C_q. Its square is the identity, so $2\chi(h) = 0$ and $\chi(h) = 0$. Explicitly

$$k_i = \alpha(y)^i h^{-1} y^{-i} \in N, \qquad \alpha(a_i) = k_i a_i^\lambda k_i^{-1}. \tag{D.36d}$$

The two unique boundary classes over y, z are exchanged, with multiplier λ.

Fill exactly these two cusps. The map is the quotient by the closed normal subgroup of N generated by b, d:

$$N \longrightarrow K = N/\overline{\langle\langle b, d \rangle\rangle}_N = \langle a_0, \dots, a_{q-1} \mid \widehat{a_0 \cdots a_{q-1}} = 1 \rangle \cong \widehat{\pi}_{0,q}. \tag{D.36e}$$

Both presentation maps are explicit: kill b and send the remaining a_i to the named peripheral generators; the inverse identifies a_{q-1} with $(a_0 \cdots a_{q-2})^{-1}$. These substitutions verify both universal properties and compositions. Since α exchanges the normal generators' peripheral classes, the closed normal subgroup in (D.36e) is invariant. Consequently α induces an actual automorphism $\bar{\alpha}$ of K with $\bar{\alpha}^2 = 1$. By (D.36d) it is pure, with the same multiplier λ.

The proved genus-zero calculation at $p = 2$ applies: $q \ge 3$ and $\lambda_2 = 1$, so $\bar{\alpha} = 1$. But its q-adic multiplier on each surviving nontrivial cusp is -1, which is not the identity on the cusp's $\mathbb{Z}_q$ subgroup. Equivalently, it acts as $-I$ on the nonzero q-adic abelianization of the genus-zero group K. This contradiction excludes every such q. Together with $\lambda_2 = 1$ it proves $\lambda = 1$ globally.

For the finite-group consequences, use the faithful map (D.32a). In a V_4 with pure kernel, the pure involution has multiplier -1 by D.34, and the positive transposition has multiplier 1 by the present proof. These generate it. The positively oriented S_3 is generated by its positive transpositions. Finally $S_3 \times C_2$ is generated by those transpositions and its pure involution. Multiplicativity of the common multiplier proves each asserted character identity.

The program `code/part_c/positive_transposition_d36.py` checks (D.36c) as an exact free-word identity, the finite cyclic cusp permutations and their involutive translations, and the two presentation substitutions in (D.36e) for several odd primes. The proof above supplies the all-prime assertion. □

D.37. [Proved] Negative transpositions and the specified four-cusp deck action. Every involution of the full three-cusp loop-outer group whose puncture permutation is a transposition and whose multiplier has $\lambda_2 = -1$ gives a faithful V_4 action on a specified four-cusp double cover. One generator is the standard free positive deck involution; another has double-transposition permutation and multiplier λ, and its center-free extension is torsion-free. Conversely every such

V_4 action containing the specified standard deck action descends to a negative transposition on the three-cusp group. The constructions are inverse as marked extensions.

For the double-transposition permutation of D.29, neither of the commuting single transpositions preserves any separating two-pants partition whose pants the double transposition exchanges. Thus an additional deck action in this construction necessarily moves the separator. This is an obstruction to preserving that decomposition, and asserts no obstruction to an action moving it.

Manuscript: Part D(b) and D(d) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394..

Proof. We specify both the markings and all the extension maps. Put

$$F = \widehat{F}(x,y), \qquad z = (xy)^{-1}, \qquad \chi : F \longrightarrow C_2, \quad (\chi(x),\chi(y)) = (0,1), \qquad N = \ker \chi. \tag{D.37a}$$

Schreier rewriting identifies N with the full free profinite group on

$$A_0 = x, \qquad A_1 = yxy^{-1}, \qquad B = y^2, \qquad D = z^2 = B^{-1}A_1^{-1}A_0^{-1}. \tag{D.37b}$$

Its four peripheral generators satisfy $A_0A_1BD = 1$. These are the actual degree-one lifts of x and the degree-two lifts of y, z; completion of the discrete index-two Schreier isomorphism gives precisely the specified inclusion $N \subset F$. Let $\delta \in \mathrm{Out}(N)$ be conjugation by y. Its named representative is

$$A_0 \longmapsto A_1, \qquad A_1 \longmapsto BA_0B^{-1}, \qquad B \longmapsto B, \qquad \delta^2 = \mathrm{Inn}(B). \tag{D.37c}$$

Here the final equality concerns the named automorphism; its outer class has order two, multiplier 1, and permutation $(A_0\ A_1)$. Its center-free extension is the explicitly identified group F, hence is torsion-free. In the converse below, "the specified deck action" means this action transported by a given peripheral isomorphism. It does not mean an arbitrary positive outer involution with a torsion-free extension.

Let g be a negative transposition, labeled to fix x and exchange y, z. Its center-free extension E contains F as an open subgroup of index two. The peripheral normalizer surjects onto $\langle g \rangle$, and A.7 gives its kernel $\overline{\langle x \rangle} = \widehat{\mathbb{Z}}$. For a lift v in this normalizer write

$$vxv^{-1} = x^{\lambda}, \qquad v^2 = x^a, \qquad (\lambda - 1)a = 0, \qquad \lambda^2 = 1. \tag{D.37d}$$

At every prime where $\lambda_p = -1$ the last commutation identity gives $a_p = 0$. At every prime where $\lambda_p = 1$ the prime is odd, since $\lambda_2 = -1$, so 2 is invertible. Choose $b_p = -a_p/2$ in these latter coordinates and $b_p = 0$ in the former. Then $s = x^b v$ has order two. Thus

$$E = F \rtimes_{\alpha} \langle s \rangle, \qquad \alpha^2 = 1, \qquad \alpha(x) = x^{\lambda}. \tag{D.37e}$$

This constructs the required finite-order lift inside the peripheral normalizer, without a further choice of inner normalization. The multiplier and the cusp permutation give $\chi\alpha = \chi$. Therefore N is normal in E, and $E/N = C_2 \times C_2$, with basis the images of y, s.

There is exactly one F-conjugacy class of order-two subgroups of E. Indeed the action on $H^1(F, \mathbb{F}_2)$ is the two-dimensional nontrivial unipotent module J_2, isomorphic to the regular $\mathbb{F}_2[C_2]$ module. Consequently $H^i(C_2, H^1(F, \mathbb{F}_2)) = 0$ for $i > 0$. The Hochschild–Serre spectral sequence, with the split inflation on its bottom row, gives

$$\dim H^n(E, \mathbb{F}_2) = 1 \qquad (n \geq 3). \tag{D.37f}$$

The full profinite Brown restriction theorem used in D.28 identifies this high-degree cohomology with the product over the order-two conjugacy classes. Each class contributes $1 +$

$\dim H^1(C_F(T), \mathbb{F}_2)$, since its normalizer is $C_F(T) \times C_2$; see [37, Theorem 2 and §1.1]. The section in (D.37e) exists, so (D.37f) forces exactly one class and zero first mod-two cohomology for its centralizer. Conjugacy by E agrees here with conjugacy by F: for any section T, one has $E = FT$, and the T factor centralizes that subgroup itself. Every finite-order element of E injects into E/F, because F is torsion-free, and consequently has order two. All nonidentity torsion therefore maps to the one line $\langle \bar{s} \rangle$ in the abelian group E/N.

Conjugation gives a faithful action of E/N on N in $\mathrm{Out}(N)$. To check the kernel, first $C_F(N) = 1$: an element centralizing both x and y^2 lies in $\langle x \rangle \cap \langle y \rangle = 1$ by A.7. For $e \in C_E(N)$ and $f \in F$, the commutator $[e, f]$ lies in $F \cap C_E(N)$ and hence is trivial. Thus e centralizes F; faithfulness of the original nonidentity outer action and $Z(F) = 1$ give $e = 1$.

Write a for the outer restriction of α to N. It fixes the classes of A_0, A_1 and exchanges B, D. For the second fixed class use the explicit conjugator $\alpha(y)y^{-1} \in N$:

$$\alpha(A_1) = (\alpha(y)y^{-1})A_1^{\lambda}(\alpha(y)y^{-1})^{-1}. \tag{D.37g}$$

The unique lifts of the other two cusps are exchanged inside N. Thus δ and a commute in $\mathrm{Out}(N)$, and

$$\tau = \delta a, \qquad \sigma(\tau) = (A_0\ A_1)(B\ D), \qquad \mathrm{or}(\tau) = \lambda. \tag{D.37h}$$

The preimage E_τ of the diagonal line $\langle \bar{y} + \bar{s} \rangle$ in E/N is the center-free extension of this outer involution. It is torsion-free: all nonidentity torsion has image $\bar{s}$, and its intersection with N is already torsion-free. For an explicit representative set

$$\beta = \mathrm{Inn}(y)\alpha|_N, \qquad c = y\alpha(y) \in N; \qquad \beta^2 = \mathrm{Inn}(c), \quad \beta(c) = c. \tag{D.37i}$$

Then E_τ is exactly the twisted extension generated by N and ys, whose square is c; the multiplication rule is the rule of (D.29.9) with these named β, c.

Conversely let a marked four-cusp group carry the specified δ and an involution τ with the permutation and multiplier in (D.37h), satisfying $[\delta, \tau] = 1$. The distinct permutations make $\Gamma = \langle \delta, \tau \rangle \cong V_4$. Take its center-free extension

$$K = \{(\varphi, \gamma) \in \mathrm{Aut}(N) \times \Gamma : [\varphi] = \gamma\}. \tag{D.37j}$$

Here N embeds by its inner automorphisms. The preimage K_δ of $\langle \delta \rangle$ is identified with the exact group F in (D.37a) by

$$f \longmapsto (\mathrm{Inn}(f)|_N, \delta^{\chi(f)}). \tag{D.37k}$$

Injectivity follows from $C_F(N) = 1$; surjectivity follows because an automorphism in either of these two outer classes differs from 1 or $\mathrm{Inn}(y)|_N$ by an inner automorphism from N. The quotient $K/F = C_2$ therefore gives an outer action on this very F. It is faithful, since any element of K centralizing F centralizes its inner copy of N, and an automorphism of the center-free N with that property is the identity.

We check descent of the peripheral powers, not only of the four unordered procyclic subgroups. The two classes A_0, A_1 become the one F-class of x. The other classes are $B = y^2$ and $D = z^2$. For a representative of τ, the induced automorphism ψ of F satisfies $\psi(y)^2 = hz^{2\lambda}h^{-1}$ for some $h \in N$. A.7 implies $C_F(hz^{2\lambda}h^{-1}) = h\langle z \rangle h^{-1}$, so $\psi(y) = hz^{\mu}h^{-1}$ and $2\mu = 2\lambda$. Multiplication by two is injective on $\widehat{\mathbb{Z}}$; hence $\mu = \lambda$. The same proof applies to z, and the x exponent is already given by A_0. Thus the descended action fixes x, exchanges y, z, and has exactly the original λ, in particular $\lambda_2 = -1$.

For completeness its normalizer construction (D.37d)–(D.37e) and (D.37f) also recover the torsion-free assertion upstairs, without making that assertion an extra converse hypothesis. An order-two lift in the normalizer of $\langle x \rangle$ fixes A_0, so its image in Γ is $\delta\tau$, not τ. All torsion has this same image by (D.37f). The preimage of $\langle \tau \rangle$ is therefore torsion-free. The conjugation

maps (D.37j)–(D.37k) identify the recovered extension with the original one on every element; conversely restriction to N recovers the given Γ and both named generators. This proves the asserted inverse correspondence of marked extensions.

We finish by making the obstacle for the D.29 gluing explicit. Write its ordered cusps as $1,2,3,4$, so that $\sigma(\tau) = (1\ 3)(2\ 4)$. The unordered two-pants partitions and their images are

partition	τ on its two blocks	(1 3) or (2 4)
$12 \mid 34$	exchanges	$14 \mid 23$
$13 \mid 24$	preserves each	$13 \mid 24$
$14 \mid 23$	exchanges	$12 \mid 34$.

(D.37l)

The only single transpositions commuting with (1 3)(2 4) are (1 3) and (2 4): a commuting permutation must send its two two-cycles to two-cycles, and a transposition can do this only by exchanging the members of one of them. Hence a deck involution of the required permutation cannot act on the same two-vertex pants graph as the inverse-pair gluing, under any relabeling or choice of its exchanged partition.

One can nevertheless write the required geometric deck map in the D.29 free basis. For $(u_1,u_2,u_3,u_4) = (a, a^{-1}l, b, b^{-1}l^{-1})$, the change of four-cusp marking

$$A_0 = a, \quad A_1 = (a^{-1}l)b(a^{-1}l)^{-1}, \quad B = a^{-1}l, \quad D = b^{-1}l^{-1} \tag{D.37m}$$

satisfies $A_0A_1BD = 1$. Its inverse substitutions are $a = A_0$, $l = A_0B$, $b = B^{-1}A_1B$. Transporting (D.37c) through these maps gives

$$\Delta(l) = a^{-1}lb, \qquad \Delta(a) = a^{-1}lbl^{-1}a, \qquad \Delta(b) = a, \qquad \Delta^2 = \mathrm{Inn}(a^{-1}l). \tag{D.37n}$$

It has multiplier 1 and permutation (1 3). The existence question for D.29 is the existence of a deck action of this marked geometric kind commuting outerly with its actual α; (D.37l) alone does not decide it.

In particular failure of commutation for one selected deck representative cannot settle the question. In the geometric specialization $\lambda = -1$, $\epsilon = 1$, $\kappa = 1$, $f = 1$, D.29 gives

$$\alpha(l) = l, \quad \alpha(a) = b^{-1}, \quad \alpha(b) = la^{-1}l^{-1}.$$

For (D.37n), the words $\alpha\Delta(l) = bl^2a^{-1}l^{-1}$ and $\Delta\alpha(l) = a^{-1}lb$ are not conjugate even in the full profinite group: the finite quotient $l \mapsto (1\ 2\ 3)$, $a, b \mapsto (2\ 3)$ in S_3 sends the first to 1 and the second to a nonidentity three-cycle. These two automorphisms therefore do not commute outerly. Yet another geometric deck does commute:

$$\begin{aligned} \Delta_0(l) = bla^{-1}, \qquad \Delta_0(a) = b, \qquad \Delta_0(b) = blal^{-1}b^{-1}, \\ \Delta_0^2 = \mathrm{Inn}(bl), \qquad \alpha\Delta_0 = \mathrm{Inn}(b^{-1})\Delta_0\alpha. \end{aligned} \tag{D.37o}$$

Its origin is also explicit. In the geometric extension $E = \widehat{F}(a,t)$, put $l = t^2$ and $b = ta^{-1}t^{-1}$. The involution $\eta(a) = a^{-1}$, $\eta(t) = a^{-1}t$ preserves the index-two kernel. Restricting $\mathrm{Inn}(t)\eta$ to that kernel gives exactly (D.37o). The formulas prove its peripheral permutation and multiplier as well as the two displayed composition identities. Thus this geometric specialization supplies an explicit example of the separator being moved. To verify directly that Δ_0 is a deck of the specified geometric kind, adjoin its lift r with $r^2 = bl$. The extension presentation eliminates to $\widehat{F}(a,r)$ by $b = rar^{-1}$ and $l = b^{-1}r^2 = ra^{-1}r$. The peripheral generators downstairs are $a, a^{-1}r, r^{-1}$, whose product is one. Their lifts are the two classes of a, b, the square $(a^{-1}r)^2 = a^{-1}l$, and the square r^{-2}, conjugate to $b^{-1}l^{-1}$. These substitutions identify its four-cusp inclusion with (D.37a)–(D.37b).

The executable `code/part_c/negative_transposition_d37.py` verifies the free-word covering and inverse substitutions, the deck squares, all the equations in (D.37o), the two-dimensional

mod-two cohomology calculation, and the complete permutation calculation (D.37l) under all relabelings. The extension and torsion arguments above prove the statements for the full profinite groups. □

D.38. [Proved] A nonnilpotent finite obstruction on the universal A_5 quotient. Let R be the intersection of the kernels of all continuous epimorphisms $\widehat{F}(x,y) \twoheadrightarrow A_5$. This is an open characteristic subgroup, and the quotient specified by the nineteen generating-pair representatives in the accompanying program is

$$Q = F/R \xrightarrow{\sim} A_5^{19}. \tag{D.38a}$$

Write $X, Y, Z = (XY)^{-1}$ for the images of the three cusps, and θ, ω for the automorphisms of Q induced by $x \leftrightarrow y$ and $x \mapsto y,\ y \mapsto (xy)^{-1}$. The image of every full GT_0 involution belongs to the following explicitly computed finite set:

$$\begin{aligned} \mathcal{T} = \{t \in Q : \quad & t^2 = 1, \quad tXt^{-1} = X^{-1}, \\ & \omega(t) = Y^{-1}\theta(t)\}. \end{aligned} \tag{D.38b}$$

Its normalized coordinates are $\alpha_Q = \mathrm{Inn}(t)$ and $f_Q = \theta(t)t^{-1}$. Every tuple in $\mathcal{T}$ satisfies the finite normalized relations (I), (II), and $f_Q \alpha_Q(f_Q) = 1$, and defines an actual automorphism of Q of order two.

The executed exact enumeration gives

finite set or test	cardinality
$\mathcal{T}$	10 125 000
tuples surviving the full cusp-triangle obstruction	5 062 500
tuples conjugate to standard inversion in the finite pure model (D.38f)	2 531 250
triangle survivors outside that finite conjugacy set	2 531 250.

(D.38c)

Thus 5 062 500 explicit finite tuples satisfying the finite involution equations have no full pure-involution lift. The last row is disjoint from the image of the standard full conjugacy class. No tuple in that last row is asserted to have a full lift.

Manuscript: §31; folios 198-199verso; scans 391-394; Part D(b): actual full GT_0 torsion and conjugacy.

Proof. The program enumerates all 60^2 ordered pairs in A_5, tests generation by closure under multiplication, and obtains 2 280 generating pairs. Simultaneous conjugation by all 120 elements of S_5 has nineteen free orbits on this set. These are all $\mathrm{Aut}(A_5)$-orbits: an automorphism permutes the five Sylow-2 subgroups, and equivariance with the faithful inner A_5 action makes that permutation representation of $\mathrm{Aut}(A_5)$ injective; conjugation by S_5 already supplies its 120 elements, as checked in D.31. Two epimorphisms to A_5 have the same kernel exactly when they differ by an automorphism of A_5. The nineteen kernels are therefore distinct. The joint quotient map is a subdirect product of nineteen copies of the nonabelian simple group A_5. The elementary subdirect-product argument shows that a proper subdirect product of copies of one simple group identifies two coordinates by an automorphism: induct on the number of factors and apply the normal-subgroup description of a direct product of nonabelian simple groups to the last projection. Such an identification would equate two of the nineteen kernels. This proves surjectivity of the specified map to A_5^{19} and hence (D.38a). The kernel is characteristic because every automorphism of F permutes all epimorphisms to A_5.

For each representative pair (X_i, Y_i) the program also records the ordered triple of conjugacy classes of X_i, Y_i, Z_i in A_5, allowing the simultaneous exchange of the two classes of elements of order five by odd conjugation in S_5. All nineteen resulting triples are different. Every triple contains an element of order five, and has at least two elements of odd order. These assertions

are checked by complete finite enumeration, with representatives and component permutations retained in the output.

By D.34 a full pure involution has multiplier -1. Inversion preserves each conjugacy class of A_5: this is immediate for involutions, and reversal of a 3-cycle or 5-cycle is achieved by an even permutation. An automorphism of A_5^{19} permutes its minimal normal simple factors and acts on each factor by an element of $\mathrm{Aut}(A_5) = S_5$. Preservation of the three cusp classes therefore forces its factor permutation to preserve the recorded class triple. The nineteen distinct triples force the permutation to be the identity. Since each triple contains an order-five entry, the component automorphisms must have even parity; odd conjugation exchanges its two order-five classes. Thus the induced automorphism of Q is $\mathrm{Inn}(t)$ for a unique $t \in Q$. The group Q has trivial center, so its involutivity gives $t^2 = 1$.

The exact Harbater–Schneps automorphism identities from [28, Propositions 3–4], with $m = (-1-1)/2 = -1$, descend to Q:

$$\theta\alpha_Q\theta^{-1}\alpha_Q^{-1} = \mathrm{Inn}(f_Q), \qquad \omega\alpha_Q\omega^{-1}\alpha_Q^{-1} = \mathrm{Inn}(Y^{-1}f_Q).$$

Center-freeness yields $f_Q = \theta(t)t^{-1}$ and $\omega(t)t^{-1} = Y^{-1}f_Q$. Together with the normalized equation on X these give precisely (D.38b).

Conversely take t in (D.38b) and define $f = \theta(t)t^{-1}$. Then $\theta(t)$ inverts Y, so

$$tYt^{-1} = f^{-1}Y^{-1}f, \qquad \theta(f)f = 1, \qquad f\,\mathrm{Inn}(t)(f) = 1.$$

The second relation in (D.38b) involving ω can be rewritten $f = Y\omega(t)t^{-1}$. Apply ω and ω^2 and use $\omega^3 = 1$:

$$\omega(f) = Z\omega^2(t)\omega(t)^{-1}, \qquad \omega^2(f) = Xt\omega^2(t)^{-1}.$$

Substitution now proves the complete finite three-cycle relation by cancellation,

$$\omega^2(f)Z^{-1}\omega(f)Y^{-1}fX^{-1} = 1. \tag{D.38d}$$

The tuple X has components of odd order, so $tXt^{-1} = X^{-1}$ excludes $t = 1$ in Q. Thus $\mathrm{Inn}(t)$ has exact order two. This verifies the assertions for every member of the finite set, independently of any lifting.

We next impose an obstruction from the full group. For each component the extension quotient is explicitly

$$A_5 \rtimes_{\mathrm{Inn}(t_i)} C_2 \xrightarrow{\sim} A_5 \times C_2, \qquad (g, s^j) \longmapsto (gt_i^j, s^j). \tag{D.38e}$$

The three vertices of the full cusp triangle in D.34 are represented by s, xs, and yfs. Their first coordinates under (D.38e) are t_i, X_it_i, and $Y_if_it_i$. None can be the identity in A_5. Indeed that would make the image of the corresponding section central in $A_5 \times C_2$. This section class is incident to two different cusp edges of the full triangle. At either edge choose its representative in the cusp normalizer. Its image is still central, because centrality is unchanged by conjugation. It must therefore both act trivially on the cusp image and act by inversion there. Both incident cusp images would have order at most two. No cusp image is trivial for a generating A_5 pair. Two involutions generate a dihedral group, while any two entries of the generating triple generate A_5. This is a contradiction. Possible fusion of distinct section classes in the finite quotient does not change this argument.

For all components together, this obstruction is exactly $t_i \notin \{1, X_i\}$ for every i. The third vertex condition follows from these: $f_it_i = \theta(t)_i$ and $\theta(X) = Y$, so $Y_i\theta(t)_i = 1$ is another occurrence of $t_j = X_j$ under the component action of θ.

Here are the complete finite counts, with the factorization used to execute them. The relation $\omega(t) = Y^{-1}\theta(t)$ joins components in twelve independent sets, using indices $0, \ldots, 18$:

$$\{0,2\},\{1\},\{3,5\},\{4\},\{6,7\},\{8\},\{9,17\},\{10\},\{11,14\},\{12,18\},\{13,15\},\{16\}.$$

Exhaustive propagation of each initial component value through its specified S_5 conjugation maps gives respectively

$$3, 3, 3, 3, 4, 2, 5, 5, 5, 5, 5, 5$$

assignments. The equations on a component set involve no other set, so their Cartesian product has exactly 10 125 000 elements. The triangle obstruction changes only the count for $\{6, 7\}$, from four to two. Its surviving cardinality is therefore 5 062 500. All forty-eight component assignments are executed and checked, including the original equations (D.38d) and the finite norm; the Cartesian-product count is exact.

For precision, the finite conjugacy model in (D.38c) consists of the following actual automorphisms of Q with a common unit exponent:

$$\begin{aligned}\mathcal{P}_Q = \{(\mu, \gamma) \in (\mathbb{Z}/30)^\times \times \mathrm{Aut}(Q) : \quad & \gamma(X) = X^\mu, \\ & \gamma(Y) \sim_Q Y^\mu, \quad \gamma(Z) \sim_Q Z^\mu\}.\end{aligned} \tag{D.38f}$$

Every normalized full pure automorphism maps to this set, since R is characteristic. The recorded class triples again imply that an element of (D.38f) acts on the ith factor by conjugation by some $g_i \in S_5$, without permuting factors. Its parity is even for $\mu \bmod 5 \in \{1, 4\}$ and odd for $\mu \bmod 5 \in \{2, 3\}$. The remaining condition is exactly $g_i X_i g_i^{-1} = X_i^\mu$; with that parity, the Y_i and Z_i class conditions follow.

The standard inversion induces $\mathrm{Inn}(d)$ on Q, where the program supplies each simultaneous inverter d_i of (X_i, Y_i). The conjugacy condition is the finite equation $t_i = g_i d_i g_i^{-1}$ for every i. The program tests all 120 choices of g_i for each of the eight units modulo thirty. There are exactly two distinct compatible patterns, each with component counts

$$3, 3, 3, 3, 1, 1, 5, 5, 5, 5, 5, 5.$$

They are disjoint, so their total cardinality is $2 \cdot 1\,265\,625 = 2\,531\,250$. This proves the last two rows of (D.38c). It proves an upper constraint on the image of the known full conjugacy class, not surjectivity from that class onto the finite model.

The complete implementation is `code/part_a/a5_involution_quotient.py`; its output retains the nineteen pairs, the θ, ω component maps, the standard inversion tuple, every component count, and an explicit rejected tuple. No family of lifts through all further characteristic quotients has been supplied for a surviving tuple. □

D.39. [Proved] A rank-two Haran involution admitting neither hyperbolic loop type. Let $\epsilon, \kappa \in \widehat{\mathbb{Z}}$ have the coordinates

$$\epsilon_3 = -1, \quad \epsilon_q = 1 \ (q \neq 3), \qquad \kappa_2 = 1, \quad \kappa_q = 0 \ (q \neq 2).$$

There is an actual involution of the rank-two free profinite group $H = \widehat{F}(a, b)$ given by

$$\alpha(a) = b^{-1} a^\epsilon b, \qquad \alpha(b) = b^{-1} a^\kappa. \tag{D.39a}$$

Its split extension has the explicit identification

$$H \rtimes_\alpha C_2 \xrightarrow{\sim} N \amalg C_2, \qquad N = (\mathbb{Z}_3 \rtimes \mathbb{Z}_2) \times \prod_{q \neq 2,3} \mathbb{Z}_q, \tag{D.39b}$$

where parity in $\mathbb{Z}_2$ acts by inversion on $\mathbb{Z}_3$. Its actual fixed subgroup $C_H(\alpha)$ is trivial. The outer involution preserves no three-cusp marking and no once-punctured-torus marking of H, under any full profinite change of free basis.

For the natural once-punctured-torus marking $\overline{\langle [a, b] \rangle}$, the peripheral condition also fails: a quotient of order 21 separates $\alpha([a, b])$ from every conjugate of $[a, b]^{-\epsilon}$.

Manuscript: §31; folios 198-199verso; scans 391-394; Part D: full rank-two loop actions and precise peripheral conditions.

Proof. The relations $\epsilon^2 = 1$ and $\epsilon\kappa = \kappa$ give, by substitution on the specified free generators,

$$\alpha^2(a) = (a^{-\kappa}b)(b^{-1}ab)(b^{-1}a^{\kappa}) = a,$$
$$\alpha^2(b) = (a^{-\kappa}b)(b^{-1}a^{\epsilon\kappa}b) = b.$$

Thus the universal continuous endomorphism defined by (D.39a) is an automorphism with square one. On full abelianization its matrix is

$$A = \begin{pmatrix} \epsilon & \kappa \\ 0 & -1 \end{pmatrix}, \qquad \det A = -\epsilon, \qquad \operatorname{tr} A = \epsilon - 1. \tag{D.39c}$$

It is nonidentity on abelianization, so both the automorphism and its outer class have exact order two.

For the exact identification (D.39b), use the normal form for N verified in D.26:

$$N = \overline{\langle l, t\rangle}, \qquad tlt^{-1} = l^{\epsilon}, \qquad t^2 = l^{\kappa}.$$

The map $l^u t^i \mapsto (u_3, 2u_2 + i, (u_q)_{q\neq 2,3})$, $i \in \{0,1\}$, is its displayed isomorphism with the group in (D.39b). The parity character sends l to zero and t to one. In $N \amalg \langle r \mid r^2 = 1\rangle$ extend this character by $r \mapsto 1$. The substitutions

$$a \longmapsto l, \qquad b \longmapsto tr, \qquad s \longmapsto r$$

define a homomorphism from $H \rtimes_\alpha \langle s\rangle$ to that free profinite product. To check its relations, note that $r = (tr)^{-1}t$, so

$$rlr = (tr)^{-1}l^{\epsilon}(tr), \qquad r(tr)r = (tr)^{-1}l^{\kappa}.$$

In the reverse direction send $l \mapsto a$, $t \mapsto bs$, and $r \mapsto s$. Here $(bs)^2 = b\alpha(b) = a^{\kappa}$ and $(bs)a(bs)^{-1} = b\alpha(a)b^{-1} = a^{\epsilon}$. The normal-form multiplication of N therefore gives its continuous homomorphism to the semidirect product, and the free product property extends it over r. The two composites fix every displayed generator, proving the isomorphism. In particular the parity kernel is exactly the free profinite group on the stated generators a, b, and the extension has the actual section $\langle r\rangle$.

The free factor $\langle r\rangle$ is malnormal by the full-profinite free-factor intersection property [31, Theorem 5 of the inspected author version]. Its centralizer in $N \amalg C_2$ is therefore precisely $\langle r\rangle$. Intersecting with the parity kernel proves $C_H(\alpha) = 1$.

In fact every involution $u \in H \rtimes_\alpha C_2$ has $C_H(u) = 1$. To prove this with the specified index-two kernel, reduce (D.39c) modulo two. The resulting matrix is $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Thus its dual action on $H^1(H, \mathbb{F}_2)$ is the regular two-dimensional $\mathbb{F}_2[C_2]$ module J_2. Its positive-degree C_2 cohomology vanishes: both differentials of the periodic resolution are $1 + \alpha$, and their image equals their kernel, both of dimension one. Since H has cohomological dimension one and the extension is split, the Hochschild–Serre spectral sequence gives

$$\dim H^n(H \rtimes_\alpha C_2, \mathbb{F}_2) = 1 \qquad (n \geq 3). \tag{D.39g}$$

The full profinite Brown restriction theorem [37, Theorem 2 and §1.1], applied exactly as in (D.37f), expresses this cohomology as the product over involution classes. Every class contributes $1 + \dim H^1(C_H(u), \mathbb{F}_2)$, because its normalizer is $C_H(u) \times C_2$. The existing section $\langle s\rangle$ therefore accounts for the unique conjugacy class. Conjugacy by the extension agrees with conjugacy by H: for any involution u the decomposition $E = H\langle u\rangle$ absorbs its second factor, which centralizes u. Every involution is consequently H-conjugate to s, and the already proved equality $C_H(s) = 1$ gives

$$C_H(u) = 1 \quad \text{for every involution } u \in E. \tag{D.39h}$$

There is a basis-independent obstruction to a three-cusp structure. An involution on three cusps has either the identity permutation or a transposition. In the former case its homology is a scalar matrix λI and has trace 2λ, whose 2-coordinate is nonzero. This contradicts the zero 2-coordinate of the trace in (D.39c). For a transposition the reduced permutation module has trace zero, and multiplication by a common unit keeps that trace zero. This contradicts the 3-coordinate -2 of (D.39c). Trace is unchanged by conjugating the homology matrix under a change of free basis. Hence neither peripheral permutation can occur.

There is also a basis-independent obstruction to a once-punctured-torus marking. Take any full profinite free basis a', b' of H and put $c' = [a', b']$, $L = \overline{\langle c' \rangle}$. Preservation of the conjugacy class of L would give the multiplier $\nu = \det A = -\epsilon$. The exact comparison is the one in (D.33h): in every finite class-two Heisenberg quotient the commutator exponent is the determinant of the abelianization matrix, so equality for all moduli identifies these elements of $\widehat{\mathbb{Z}}^\times$. Changing the free basis conjugates the matrix and leaves its determinant unchanged.

The normalizer $N_E(L)$ surjects onto $E/H = C_2$, and its kernel is L by A.7 transported through this free-basis isomorphism. For a lift v of the nonidentity element, write

$$vc'v^{-1} = (c')^{\nu}, \qquad v^2 = (c')^d, \qquad (\nu - 1)d = 0. \tag{D.39i}$$

Here $\nu_3 = 1$ and $\nu_q = -1$ for every $q \neq 3$. Therefore $d_q = 0$ for $q \neq 3$, including $q = 2$, since multiplication by two is injective on $\mathbb{Z}_q$. Choose $j \in \widehat{\mathbb{Z}}$ with $j_3 = -d_3/2$ and $j_q = 0$ otherwise. The lift $u = (c')^j v$ then satisfies

$$u^2 = (c')^{(1+\nu)j+d} = 1, \qquad u(c')^{e_3 w}u^{-1} = (c')^{e_3 w} \quad (w \in \mathbb{Z}_3), \tag{D.39j}$$

where e_3 is the idempotent supported at the prime three. The lift u has exact order two because its image in E/H is nonidentity. The peripheral injection $\widehat{\mathbb{Z}} \xrightarrow{\sim} L$ makes $(c')^{e_3\mathbb{Z}_3}$ a nontrivial subgroup of $C_H(u)$, contradicting (D.39h). This excludes every once-punctured-torus marking, including those obtained from arbitrary full profinite free bases.

For an additional finite certificate in the original basis, put $c = [a, b] = aba^{-1}b^{-1}$.

Use the finite group

$$Q = \mathbb{F}_7 \rtimes C_3, \qquad (v, i)(w, j) = (v + 2^i w,\ i + j), \tag{D.39d}$$

where the second coordinate is modulo three and 2 has order three in $\mathbb{F}_7^\times$. The assignment

$$a \mapsto (0, 1), \qquad b \mapsto (1, 1) \tag{D.39e}$$

is an epimorphism: a, b have order three, their commutator is $(1, 0)$, and these elements generate the normal $\mathbb{F}_7$ and the C_3 quotient. Since $\epsilon \equiv -1 \pmod 3$ and $\kappa \equiv 0 \pmod 3$, (D.39a) has the following exact images:

word	c	$\alpha(a)$	$\alpha(b)$	$\alpha(c)$	$c^{-\epsilon}$
image	$(1, 0)$	$(5, 2)$	$(3, 2)$	$(1, 0)$	$(6, 0)$.

(D.39f)

The last entry uses $(-\epsilon)_7 = -1$. Conjugacy on the normal $\mathbb{F}_7$ in (D.39d) is multiplication by a power of two. The two relevant orbits are exactly

$$\{(1, 0), (2, 0), (4, 0)\}, \qquad \{(3, 0), (5, 0), (6, 0)\}.$$

They are disjoint. Thus the images of $\alpha(c)$ and $c^{-\epsilon}$ are not conjugate, proving failure of the natural torus cusp.

The program `code/part_a/haran_torus_test.py` executes this order-21 quotient and an independent matrix check inside $\mathrm{PSL}_2(\mathbb{F}_7)$. For the latter it enumerates all 168 elements and uses

$$a = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix} \mod \{\pm I\}.$$

Both have order three, $c = \left(\begin{smallmatrix} 1 & 1 \\ 3 & 4 \end{smallmatrix}\right)$ has order seven, and $\alpha(c) = \left(\begin{smallmatrix} 0 & 3 \\ 2 & 2 \end{smallmatrix}\right)$. There are seven conjugators from c to this image and none from c^{-1}. The matrix calculation agrees with (D.39f). The finite calculation detects the original cusp explicitly. The argument (D.39g)–(D.39j) supplies the exclusion for every alternative marking. The program also verifies the regular mod-two module used in (D.39g). □

D.40. [Proved] A finite homomorphism-count obstruction to the extra deck symmetry. The mixed-multiplier four-cusp involution constructed in D.29 commutes with no single-transposition involution of multiplier 1 in its full loop-outer group. In particular it admits none of the specified geometric deck actions required for the descent of D.37, including actions moving the two-pants separator.

More precisely, for the exact profinite group

$$E = \overline{\langle a, l, t \mid tlt^{-1} = l^{\epsilon},\ t^2 = l^{\kappa} \rangle}, \qquad p = a^{-1}l, \tag{D.40a}$$

with ϵ, κ as in D.29 and $\lambda = -\epsilon$, there is no continuous automorphism sending the conjugacy classes of a, p to those of a^{λ}, p, respectively. There is also none sending those classes to a, p^{λ}. The obstruction is the exact inequality

$$\begin{aligned} \#\{\phi : E \to Q : \phi(a), \phi(p) \in C\} &= 10200, \\ \#\{\phi : E \to Q : \phi(a) \in C^{-1}, \phi(p) \in C\} &= 4920, \end{aligned} \tag{D.40b}$$

where all homomorphisms are continuous, $Q = \mathrm{PSL}_2(\mathbb{F}_{11})$, and C is the conjugacy class of $\left(\begin{smallmatrix} 0 & 1 \\ 10 & 2 \end{smallmatrix}\right)$ modulo scalar matrices $\{\pm I\}$.

Manuscript: Section 24; folio 147 verso; scan 293. Section 30; folios 194-196; scans 383-387. Section 31; folios 198-199 verso; scans 391-394. Part D(b) and D(d)..

Proof. We first identify the group and its peripheral marking. In the basis (l, a, b') of (D.29.10), adjoining the outer lift t gives the relations

$$tlt^{-1} = l^{\epsilon}, \quad tat^{-1} = b', \quad tb't^{-1} = l^{\kappa}al^{-\kappa}, \quad t^2 = l^{\kappa}.$$

Eliminating b' gives (D.40a): the third relation follows from the second and fourth. Conversely $b' = tat^{-1}$ defines the inverse homomorphism from (D.40a) to this presentation. Both composites fix the named generators. Thus this is an isomorphism of profinite groups, and

$$E = N_{\mathrm{bad}} \amalg \widehat{\mathbb{Z}}(a) \tag{D.40c}$$

with the first factor generated by l, t. The index-two subgroup H is the kernel of the character sending t to 1 and a, l to 0 in C_2.

The four H-peripheral classes of D.29 are represented by

$$a, \quad p = a^{-1}l, \quad b, \quad b^{-1}l^{-1}.$$

Since α interchanges the first and third classes, and the second and fourth, with multiplier λ, and conjugation by t is α, these are equally the classes of

$$a, \quad p, \quad (tat^{-1})^{\lambda}, \quad (tpt^{-1})^{\lambda}. \tag{D.40d}$$

Here $\lambda^2 = 1$, and the original cusp conjugators belong to H; thus (D.40d) records actual conjugacy classes under the specified inclusion into E. This description is independent of the Galois conjugator f used to construct the two pants isomorphisms.

For a finite group Q define the central function

$$F_Q(L) = \#\{T \in Q : TLT^{-1} = L^{\epsilon},\ T^2 = L^{\kappa}\}. \tag{D.40e}$$

Every assignment $a \mapsto A$, $p \mapsto P$ forces $l \mapsto AP$. The presentation (D.40a) then gives exactly $F_Q(AP)$ continuous homomorphisms to Q. For two conjugacy classes C, D put

$$K_Q(C, D) = \sum_{A \in C,\ P \in D} F_Q(AP). \tag{D.40f}$$

Conjugating L conjugates all its possible T, so F_Q is central. Since $PA = A^{-1}(AP)A$, this also proves $K_Q(C, D) = K_Q(D, C)$.

Precomposition by a continuous automorphism of E is a bijection on its continuous homomorphisms to Q. An automorphism with the first pair of class images in the claim would therefore give $K_Q(C, D) = K_Q(C^{\lambda}, D)$ for every pair C, D. The inverse map has the same power effect because $\lambda^2 = 1$, so this is equality of the two full sets, without a choice of conjugators in the target. The second proposed class effect would similarly give $K_Q(C, D) = K_Q(C, D^{\lambda})$.

We compute these integers in one completely specified finite group. Enumerate the determinant-one matrices over $\mathbb{F}_{11}$, and identify each matrix with its negative, using their lexicographically smaller representative. Matrix multiplication followed by this identification constructs all 660 elements of Q. The class C has size 60 and order 11. Its inverse class is distinct and has the same size. Since $\lambda_{11} = -1$, we have $C^{\lambda} = C^{-1}$. The following table gives the full product counts needed in (D.40f). The representative of each nonidentity column is $\left(\begin{smallmatrix} 0 & 1 \\ 10 & r \end{smallmatrix}\right)$; the last column is the identity. We write $N_{C,D}(L) = \#\{(A, P) \in C \times D : AP \in [L]\}$.

r	0	1	2	3	4	5	9	I
$\lvert L\rvert$	2	3	11	5	5	6	11	1
class size	55	110	60	132	132	110	60	1
$F_Q(L)$	0	6	1	1	1	0	1	56
$N_{C,C}(L)$	0	1320	120	660	660	0	840	0
$N_{C^{-1},C}(L)$	660	0	120	660	660	1320	120	60

(D.40g)

Multiplying either of the last two rows by the F_Q row and summing gives respectively

$$7920 + 120 + 660 + 660 + 840 = 10200, \qquad 120 + 660 + 660 + 120 + 3360 = 4920.$$

These unequal integers prove the first nonexistence assertion. Symmetry of K_Q proves the second.

All entries of (D.40g) are verified by the finite enumeration in `code/root/haran_pair_counts.py`. It constructs the multiplication, inverse, power and conjugacy tables directly from the displayed matrices, computes the residues of ϵ, κ modulo each element order, and tests both equations in (D.40e) for every pair (L, T). It then counts each product in the two displayed class pairs. Two independent computations, `code/part_a/haran_pair_count_audit.py` and `code/part_c/deck_obstruction_psl11_independent.py`, instead generate the faithful twelve-point permutation action by $z \mapsto z + 1$ and $z \mapsto -1/z$ on $\mathbf{P}^1(\mathbb{F}_{11})$. The first enumerates allowed (L, T) before setting $P = A^{-1}L$; the second also enumerates the triples directly. Both reproduce (D.40b). The attached outputs include the full integer data.

Finally suppose a positive single-transposition involution δ commuted with the D.29 involution τ. Their distinct peripheral permutations make $\langle \delta, \tau \rangle \simeq V_4$. Its center-free pullback extension inside $\mathrm{Aut}(H)$ contains E as the normal preimage of $\langle \tau \rangle$, using the canonical conjugation identification of center-free extensions. Conjugation by any lift of δ therefore defines a continuous automorphism η of this exact E. By D.37's permutation computation, δ is either (1 3) or (2 4) in the labeling (D.40d). Its positive multiplier is 1. In the first case η sends a to an E-conjugate of a^{λ} and p to an E-conjugate of p; in the second case the roles are reversed. Both have just

been excluded. The argument imposes no preservation of the original pants separator and no order condition on the chosen lift of δ. It proves the stated obstruction for this actual four-cusp construction; the general negative three-cusp conjugacy question remains in D.12. □

D.41. [Proved] The global multiplier of a negative transposition. An involution of the full three-cusp loop-outer group whose puncture permutation is a transposition and whose 2-adic multiplier is -1 has the global multiplier $-1 \in \widehat{\mathbb{Z}}^\times$. Together with D.34 and D.36, this identifies the multiplier of every involution in that full group with its global 2-adic sign. Consequently every finite subgroup generated by involutions has the usual global orientation character.

Manuscript: Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394..

Proof. Label the original peripheral generators as $x, y, z = (xy)^{-1}$ so that the involution fixes x and exchanges y, z. Write its common multiplier as λ. It satisfies $\lambda^2 = 1$, so every coordinate λ_p is 1 or -1. The peripheral-normalizer construction (D.37d)–(D.37e) supplies an actual order-two automorphism α of $F = \widehat{F}(x, y)$ representing it and satisfying $\alpha(x) = x^\lambda$.

Suppose a positive odd-prime coordinate exists, and let q be the least odd prime with $\lambda_q = 1$. Use the exact cyclic quotient

$$\chi : F \longrightarrow C_q, \qquad \chi(x) = -2, \quad \chi(y) = \chi(z) = 1, \qquad N = \ker \chi. \tag{D.41a}$$

The relation $xyz = 1$ is respected. Since $\lambda = 1$ modulo q and α exchanges the last two peripheral classes, $\chi\alpha = \chi$. The cover corresponding to N is the completion of the specified connected cyclic cover of the pair of pants. Each of its three original cusps has one lift of degree q: all three displayed monodromies are nonzero in C_q. Its Euler characteristic is $-q$, so its genus is

$$g = (q - 1)/2. \tag{D.41b}$$

Fill the two cusp lifts represented by y^q, z^q. The quotient and its deck extension are explicit:

$$\begin{aligned} R &= F/\overline{\langle\langle y^q, z^q \rangle\rangle}_F = \langle y, z \mid y^q = z^q = 1 \rangle_{\text{prof}} = C_q \amalg C_q, \\ K &= \ker(R \longrightarrow C_q,\ y, z \longmapsto 1) \\ &= N/\overline{\langle\langle y^q, z^q \rangle\rangle}_N \cong \widehat{\pi}_{g,1}. \end{aligned} \tag{D.41c}$$

Here $x = z^{-1}y^{-1}$, and the one remaining cusp is $c = x^q$. To verify the equality of normal subgroups used in (D.41c), note $F = N\langle y \rangle = N\langle z \rangle$. Conjugating y^q or z^q by any element of F therefore gives an N-conjugate of the same element. Thus its closed F-normal closure is its closed N-normal closure. The completion comparison is also exact: start with the discrete cyclic surface cover, fill those two boundaries, and complete the displayed quotient presentation. Finite-index subgroup topology identifies its completed kernel with the specified kernel in $C_q \amalg C_q$. The inverse presentation substitutions are the original y, z and $x = z^{-1}y^{-1}$.

The automorphism α preserves this normal subgroup, because it exchanges the two filled peripheral classes. It therefore induces an actual automorphism of R, preserves K, and induces an involution β of K with

$$\beta(c) = c^\lambda. \tag{D.41d}$$

Let T be conjugation by y on K. In R, $y^q = 1$, so $T^q = 1$ as an actual automorphism. The equality $\chi\alpha = \chi$ gives $\alpha(y)y^{-1} \in K$; hence $\beta T \beta^{-1}$ and T differ by an inner automorphism of K. In particular their maps on abelianization commute.

We compute that abelianization as a deck module, without requiring a comparison of traces. Reidemeister–Schreier rewriting with representatives $1, y, \ldots, y^{q-1}$ gives

$$u_i = y^i z y^{-(i+1)} \quad (i \in \mathbb{Z}/q), \qquad u_0 u_1 \cdots u_{q-1} = 1, \qquad T(u_i) = u_{i+1}. \tag{D.41e}$$

Exponents of y here are interpreted in R, where $y^q = 1$. The first $q-1$ elements form a free profinite basis; eliminate the last by $u_{q-1} = (u_0 \cdots u_{q-2})^{-1}$. These substitutions give both maps in the Schreier presentation. The remaining boundary also has the exact word

$$c = x^q = \prod_{j=0}^{q-1} u_{q-1-2j}^{-1}, \qquad \text{indices modulo } q. \tag{D.41f}$$

Indeed $z = u_0 y$ gives $x = z^{-1}y^{-1} = y^{-1}u_0^{-1}y^{-1} = u_{q-1}^{-1}y^{-2}$. Multiplying its q successive translates by y^{-2} gives exactly (D.41f), because $y^{2q} = 1$.

For any prime p put $V = H_1(K, \mathbb{Q}_p)$ and let A be the map induced by β. From (D.41e) we obtain the exact module isomorphism

$$V \cong \mathbb{Q}_p[C_q]/\mathbb{Q}_p(1 + T + \cdots + T^{q-1}) \cong \mathbb{Q}_p[X]/\Phi_q(X), \qquad A^2 = 1, \quad AT = TA. \tag{D.41g}$$

Every irreducible cyclotomic factor occurs with multiplicity one.

The one-boundary surface structure in (D.41c) supplies a nondegenerate alternating pairing, and its relation to (D.41d) is algebraic. In a surface free basis the boundary word is the product of the g commutators, up to inversion and conjugation. Its degree-two image in the class-two free pro-p quotient is a nondegenerate bivector $\omega \in \bigwedge^2 V$. Equation (D.41d) gives $(\bigwedge^2 A)\omega = \lambda_p \omega$. The deck map preserves its peripheral exponent, so $(\bigwedge^2 T)\omega = \omega$. If Ω is the nonsingular skew matrix of ω, these are precisely

$$A\Omega A^{\mathsf{t}} = \lambda_p \Omega, \qquad T\Omega T^{\mathsf{t}} = \Omega.$$

Consequently the alternating form with matrix Ω^{-1} satisfies

$$\langle Av, Aw \rangle = \lambda_p \langle v, w \rangle, \qquad \langle Tv, Tw \rangle = \langle v, w \rangle. \tag{D.41h}$$

This proves the multiplier comparison also at $p = 2$: one may first take the integral commutator calculation in each finite class-two quotient and then tensor with $\mathbb{Q}_2$. It uses no geometric realization of β.

Now let $p \neq q$ be a prime with $d = \operatorname{ord}_q(p)$ even. The irreducible factors of Φ_q over $\mathbb{Q}_p$ have root sets the Frobenius orbits

$$\{\zeta^{jp^k} : 0 \le k < d\}, \qquad j \in (\mathbb{Z}/q)^{\times}. \tag{D.41i}$$

For completeness these roots lie in the unramified extension with residue field $\mathbb{F}_{p^d}$: the primitive qth roots in that field lift uniquely since $p \neq q$, and Frobenius lifts their map $z \mapsto z^p$. The least such degree is d, proving the factor description. Since d is even, $p^{d/2} = -1$ modulo q, and each of these root sets is closed under inversion.

Write $V = \bigoplus V_i$ for those irreducible summands. By (D.41g), the commutant of T is the product of their fields $\mathbb{Q}_p(\zeta^j)$. Therefore $A|_{V_i}$ is multiplication by an element of a field whose square is one, and so is the scalar 1 or -1 on that entire summand. Each V_i is nondegenerate for the form in (D.41h). To see this explicitly after a splitting-field extension, a T-eigenvector with eigenvalue ζ^j can pair only with the eigenvalue ζ^{-j}, since the pairing is T-invariant. Both belong to V_i by (D.41i); different summands are orthogonal. Global nondegeneracy thus restricts to nondegeneracy on V_i. Multiplication by either scalar 1 or -1 preserves that restricted form. Comparing with (D.41h) gives

$$\lambda_p = 1 \quad \text{whenever } p \neq q \text{ and } \operatorname{ord}_q(p) \text{ is even.} \tag{D.41j}$$

There is a prime divisor p of $q-1$ having this even order. Otherwise every prime factor of $q-1$ would lie in the odd-order subgroup of the cyclic group $(\mathbb{Z}/q)^{\times}$. Their product, with

multiplicities, would also have odd order. But that product is $q-1=-1$ modulo q, of order two. This contradiction proves the assertion. Such a prime satisfies $p<q$. For $p=2$ the hypothesis gives $\lambda_p=-1$; for an odd $p<q$ the minimal choice of q gives the same value. Both contradict (D.41j).

Thus no positive odd-prime coordinate exists, and $\lambda=-1$ globally. D.34 handles a pure involution, while D.36 handles an involution with positive 2-adic multiplier. All involutions are therefore accounted for. Multiplicativity proves the last assertion for every subgroup generated by them. This conclusion determines these actual multiplier homomorphisms; it does not assert geometric conjugacy of the finite actions or settle the multiplier of an isolated order-three subgroup.

The program `code/part_c/negative_multiplier_d41.py` checks the exact Schreier boundary word, the induced cyclotomic deck matrices and the nondegenerate degree-two boundary form in examples, and checks the prime-factor and Frobenius-orbit calculation. The displayed argument proves the assertion for every odd prime, rather than only for the executed range. □

D.42. [Proved] Order-three pants actions have global multiplier one. Every element of order three in the full three-cusp loop-outer group has orientation multiplier $1\in\widehat{\mathbb{Z}}^{\times}$. Consequently every finite subgroup of that full group has multiplier image contained in the two global units $\{1,-1\}$, and its multiplier is determined by its 2-adic coordinate.

The closed-surface calculation used in the proof is also unconditional. For $g\geq 2$ and every prime p, the homomorphism

$$\mathrm{Aut}(\widehat{\pi}_{g,0})\longrightarrow \mathrm{GL}(H_1(\widehat{\pi}_{g,0},\mathbb{Z}_p)) \tag{D.42a}$$

is injective on every finite p-subgroup. Here the source consists of actual continuous automorphisms. The order-three outer action in the first assertion is supplied with an actual lift by the exact rank-two lifting result D.27.

Manuscript: §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D: full loop-outer actions of type (0,3).

Proof. We first prove (D.42a), keeping the two completion maps and the fixed subgroup explicit.

The pro-p automorphism calculation. Let $P=\pi_{g,0}^{(p)}$ be the pro-p completion. The closed lower central terms γ_nP satisfy

$$\bigcap_n \gamma_nP=1,\qquad P/\gamma_nP \text{ is torsion-free for every } n. \tag{D.42b}$$

These are precisely [48, Propositions 2.14–2.15, pp. 1515–1516], applied to the orientable surface presentation. The second proposition checks the primitive degree-two initial form of its single defining relation; the first passes the torsion-free nilpotent quotients to the pro-p completion by the stated canonical quotient maps. In particular every $\gamma_nP/\gamma_{n+1}P$, as a subgroup of $P/\gamma_{n+1}P$, is torsion-free.

Put $K_n=\ker(\mathrm{Aut}(P)\to\mathrm{Aut}(P/\gamma_{n+1}P))$ for $n\geq 1$. For $b\in K_n$, the formula

$$d_n(b)(\bar{z})=b(z)z^{-1}\pmod{\gamma_{n+2}P} \tag{D.42c}$$

defines a continuous homomorphism from P^{ab} to $\gamma_{n+1}P/\gamma_{n+2}P$. Indeed this last quotient is central modulo $\gamma_{n+2}P$, which makes the formula additive in z and trivial on commutators. An automorphism acting trivially on P^{ab} acts trivially on every lower central graded quotient: these quotients are generated by the iterated commutators of its abelianization generators. Consequently d_n is a group homomorphism on K_n, and its kernel is exactly K_{n+1}. Its target is

torsion-free by (D.42b). A finite-order element of K_1 therefore lies successively in every K_n and is the identity. Thus the kernel of $\mathrm{Aut}(P) \to \mathrm{GL}(H_1(P, \mathbb{Z}_p))$ is torsion-free.

The full fixed subgroup. Write $\Pi = \widehat{\pi}_{g,0}$ and let $a \in \mathrm{Aut}(\Pi)$ satisfy $a^p = 1$ and act trivially on $H_1(\Pi, \mathbb{Z}_p)$. Its induced automorphism on $P = \Pi^{(p)}$ is the identity by the preceding calculation. There is therefore a specified quotient homomorphism

$$E = \Pi \rtimes_a C_p \longrightarrow P \times C_p, \qquad (h, s^i) \longmapsto (\bar{h}, s^i). \tag{D.42d}$$

The natural map $\Pi \to P$ induces an isomorphism on mod-p cohomology, including products. Both groups have cohomological p-dimension two, first cohomology $\mathbb{F}_p^{2g}$, and second cohomology $\mathbb{F}_p$ with the standard nondegenerate surface cup pairing. The map is the identity on the generator characters, so the pairing shows that it is also an isomorphism in degree two; higher degrees vanish. These are the surface duality comparisons of A.2 and A.7, with the full and pro-p formations respectively. Hochschild–Serre comparison for the named map (D.42d) now gives a ring isomorphism

$$H^*(P \times C_p, \mathbb{F}_p) \xrightarrow{\sim} H^*(E, \mathbb{F}_p). \tag{D.42e}$$

In particular $\dim H^n(E, \mathbb{F}_p) = 2g + 2$ for $n > 2$.

Choose a nonzero surface class $\eta \in H^2(P, \mathbb{F}_p)$ and a nonzero periodic class $u \in H^2(C_p, \mathbb{F}_p)$. For every $m \geq 1$, the class $\eta u^m \in H^{2m+2}(E, \mathbb{F}_p)$ is nonzero under (D.42e). Every subgroup $S \leq E$ of order p projects isomorphically to C_p, because Π is torsion-free. Its first coordinate in $P \times C_p$ is trivial because P is torsion-free. Its normalizer is exactly

$$N_E(S) = C_\Pi(S) \times S. \tag{D.42f}$$

An element normalizing S fixes each of its elements by projection to the abelian quotient C_p; splitting off the uniquely determined S component proves (D.42f).

Apply the full profinite Brown restriction theorem [37, Theorem 2 and §1.1] with virtual cohomological dimension two. Its hypotheses hold: Π is an open subgroup of cohomological p-dimension two, and there is no $C_p \times C_p$ in E, since every finite subgroup injects into $E/\Pi = C_p$. The finite-dimensional source makes its dense restriction embedding an isomorphism onto a finite product of the normalizer cohomologies. Restriction of the nonzero class ηu^m is nonzero in at least one factor. For that factor put $C = C_\Pi(S)$. Because the first coordinate of S in (D.42d) is trivial, its restriction is exactly

$$(\eta|_C) \otimes (u^m|_S) \in H^2(C, \mathbb{F}_p) \otimes H^{2m}(S, \mathbb{F}_p). \tag{D.42g}$$

Thus $\eta|_C \neq 0$. The restriction $H^1(\Pi, \mathbb{F}_p) \to H^1(C, \mathbb{F}_p)$ is injective: for any nonzero v, nondegeneracy supplies w with $v \smile w$ a nonzero multiple of η, and its restriction cannot vanish. Hence $\dim H^1(C, \mathbb{F}_p) \geq 2g$ and $\dim H^2(C, \mathbb{F}_p) \geq 1$. For every normalizer (D.42f), the Künneth formula in degree $n > 2$ contributes

$$1 + \dim H^1(C_\Pi(S), \mathbb{F}_p) + \dim H^2(C_\Pi(S), \mathbb{F}_p). \tag{D.42h}$$

This holds also for odd p: the exterior generator of $H^*(C_p, \mathbb{F}_p)$ and its degree-two polynomial generator give one dimension in each nonnegative degree. At $p = 2$ the degree-one polynomial generator has the same dimension consequence. The one factor in (D.42g) already accounts for all $2g + 2$ dimensions. There is exactly one conjugacy class of sections, and for its centralizer

$$\dim H^1(C, \mathbb{F}_p) = 2g, \qquad H^2(C, \mathbb{F}_p) = \mathbb{F}_p. \tag{D.42i}$$

Conjugacy under E equals conjugacy under Π, since $E = \Pi S$ for each section S and the S factor centralizes that section. Consequently these conclusions apply to $C = C_\Pi(a)$ for the specified section in (D.42d).

We now prove directly that this closed subgroup equals Π. Continuous cohomology has the restriction comparison

$$H^j(C,\mathbb{F}_p) = \operatorname*{colim}_{C\le U\le_o\Pi} H^j(U,\mathbb{F}_p). \tag{D.42j}$$

For an explicit cochain comparison, choose an open normal subgroup V of Π so small that the given continuous finite-valued cochain on C factors through $C/(C\cap V)$ in every variable. The isomorphism $CV/V = C/(C\cap V)$ inflates it to the open subgroup CV. Cocycle equalities descend to that finite quotient; for a coboundary choose a common V also for its primitive cochain. This proves (D.42j) with the displayed restriction maps. Each U is the completion of its finite covering surface, of genus $g_U = 1 + [\Pi : U](g-1)$. For $V \le U$ the restriction on H^2 multiplies the fundamental class by $[U : V]$. A nonzero class surviving in (D.42j) in degree two therefore has a representative at some U for which every smaller open $V \ge C$ has $p \nmid [U : V]$. For these subgroups the restriction on H^1 is injective: corestriction composed with restriction is multiplication by this prime-to-p index. If C had infinite index, the indices $[\Pi : V]$ would be unbounded, making

$$\dim H^1(V,\mathbb{F}_p) = 2 + 2[\Pi : V](g-1)$$

unbounded inside the fixed finite-dimensional $H^1(C,\mathbb{F}_p)$, contrary to (D.42i). Thus C is open. The same genus formula and (D.42i) give $[\Pi : C] = 1$. It follows that $a = 1$. The kernel of (D.42a) on a finite p-subgroup cannot contain an element of order p, proving its asserted injectivity.

The order-three cyclic cover. Let g have order three in the full pants group, with multiplier λ. By D.32 its cusp permutation is a three-cycle. Label $x, y, z = (xy)^{-1}$ so that it sends their classes successively to y, z, x, with the common power λ. D.27 gives an actual order-three representative $\alpha \in \operatorname{Aut}(\widehat{F}(x,y))$. The equality $\lambda^3 = 1$ implies $\lambda_3 = 1$: an order-three unit of $\mathbb{Z}_3$ reduces to one, and for $v \in \mathbb{Z}_3^\times$, $r \ge 1$, the valuation of $(1+3^r v)^3 - 1$ is $r+1$.

Suppose a coordinate λ_q were nonidentity. Then $q \ne 3$, and its reduction $\zeta \in \mathbb{F}_q^\times$ has exact order three; in particular $q \equiv 1 \pmod 3$ and $q \ge 7$. Define the surjective character

$$\chi : F \longrightarrow C_q = \mathbb{F}_q, \qquad \chi(x) = \zeta, \quad \chi(y) = 1, \quad \chi(z) = \zeta^2. \tag{D.42k}$$

The relation is respected because $1 + \zeta + \zeta^2 = 0$. The three scalar identities $\zeta \cdot 1 = \zeta$, $\zeta \cdot \zeta^2 = 1$, and $\zeta \cdot \zeta = \zeta^2$ show exactly that $\chi\alpha = \chi$; all inner conjugators disappear under χ. Let $N = \ker\chi$. The specified degree-q cover has one cusp above each of the original three, since all three values in (D.42k) are nonzero. Its type is $(h, 3)$ with $h = (q-1)/2 \ge 3$, by $2 - 2h - 3 = -q$. Fill these three cusps and put

$$\Pi_h = N/\overline{\langle\langle x^q, y^q, z^q\rangle\rangle}_N = \widehat{\pi}_{h,0}. \tag{D.42l}$$

Completion of the finite-cover presentation gives this exact quotient map. The peripheral powers under α preserve the indicated closed normal subgroup, so it induces an actual automorphism b of Π_h with $b^3 = 1$. Conjugation by y gives the deck automorphism T of this quotient; $T^q = 1$, because y^q has been killed. The relation $\chi\alpha = \chi$ says that bT and Tb differ by an inner automorphism. Their actions on homology therefore commute.

Here is an explicit calculation of that deck module. First fill only y^q, z^q upstairs. The group downstairs is $D = C_q(y) \amalg C_q(z)$, with its map to C_q given by $y \mapsto 1$, $z \mapsto r = \zeta^2$. Its kernel K has Schreier generators

$$v_i = y^i z y^{-i-r} \quad (i \in \mathbb{Z}/q), \qquad v_0 v_r v_{2r} \cdots v_{(q-1)r} = 1. \tag{D.42m}$$

The displayed relation contains every generator once and eliminates one; these are the completed discrete Schreier presentation and maps. Conjugation by y sends v_i to v_{i+1}. The remaining cusp $x^q = (z^{-1}y^{-1})^q$ rewrites as

$$\prod_{j=0}^{q-1} v_{-r-j(r+1)}^{-1}. \tag{D.42n}$$

Since $r+1 \neq 0$, it also uses every index once. Its abelianization is the negative of the already imposed sum relation. Filling it changes no first homology. It follows, for the actual map $K \to \Pi_h$, that

$$H_1(\Pi_h, \mathbb{Q}_3) = \mathbb{Q}_3[C_q]/\mathbb{Q}_3(1 + T + \cdots + T^{q-1}) \simeq \mathbb{Q}_3[X]/\Phi_q(X). \tag{D.42o}$$

This is the regular augmentation module, with each irreducible factor occurring once.

Because $q \neq 3$, reduction of Φ_q modulo three is square-free. Hensel lifting its irreducible factors expresses the last algebra in (D.42o) as a product of finite unramified extensions of $\mathbb{Q}_3$. None contains a nonidentity cube root of one: reduction to the residue field, whose multiplicative group has order $3^d - 1$, first gives residue one; the same valuation calculation used for $\mathbb{Z}_3$ then applies, since three remains a uniformizer. The centralizer of T on its cyclic module (D.42o) is exactly that product algebra. Thus the commuting operator b_*, with $b_*^3 = 1$, equals one in every factor. The torsion-free lattice $H_1(\Pi_h, \mathbb{Z}_3)$ embeds in (D.42o), so b_* is also the identity on that lattice. The first part of this proof, applied with $p = 3$, now gives $b = 1$ on the full Π_h.

Finally the quotient filling comparison in A.2 identifies the relative orientation line of $(N; \langle x^q \rangle, \langle y^q \rangle, \langle z^q \rangle)$ with the closed orientation line of Π_h. The three cusp maps have the common power λ, so b acts on the latter line by λ as well. Its being the identity forces $\lambda_q = 1$, contradicting the chosen coordinate. There is no such q, proving $\lambda = 1$.

For the finite-subgroup assertion, D.32 embeds every finite subgroup in $S_3 \times C_2$. Its elements therefore have orders $1, 2, 3$, or 6. D.34, D.36, and D.41 give the global sign for every involution; the preceding proof handles order three. An element v of order six is the product of its commuting powers v^3 and v^4, of orders two and three respectively. Multiplicativity gives the assertion for every element, with its prescribed 2-adic sign.

The program `code/part_a/pants_order_three.py` executes the character invariance, both Schreier index cycles, the cyclic deck matrix and its cyclotomic polynomial, and the characteristic-three factorization checks for both primitive cube roots at every prime $q \equiv 1 \pmod 3$ through 97. The arguments above prove the result for every prime and for the full profinite groups. □

D.43. [Proved] Global orientation of punctured-torus involutions. Every involution in the full once-punctured-torus loop-outer group has the global orientation multiplier 1 or -1, with the same sign as its 2-adic determinant. Consequently the orientation character of every finite subgroup generated by involutions has values in the global two-element subgroup $\{1, -1\} \subset \widehat{\mathbb{Z}}^\times$.

Manuscript: Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394..

Proof. Write $F = \widehat{F}(x, y)$ and $c = [x, y]$ for the torus cusp. D.27 supplies an actual order-two representative α of the given outer involution. Its peripheral multiplier is

$$\lambda = \det(\alpha|_{F^{\mathrm{ab}}}) \in \widehat{\mathbb{Z}}^\times, \qquad \lambda^2 = 1, \tag{D.43a}$$

by the exact Heisenberg comparison of D.33. Thus every λ_p is 1 or -1.

The characteristic Heisenberg cover. For an odd prime q, take the characteristic quotient

$$F \longrightarrow G_q, \qquad G_q = \langle X, Y, Z \mid X^q = Y^q = Z^q = 1,\ [X, Y] = Z,\ [X, Z] = [Y, Z] = 1 \rangle, \tag{D.43b}$$

with $x \mapsto X$, $y \mapsto Y$, $c \mapsto Z$. Collection of these words identifies G_q with the order-q^3 Heisenberg group. Explicitly it is $\mathbb{F}_q^2 \times \mathbb{F}_q$, with multiplication

$$(v, z)(w, t) = (v + w, z + t + \tfrac{1}{2}\omega(v, w)), \qquad \omega((a, b), (a', b')) = ab' - ba'. \tag{D.43c}$$

Take $X = ((1,0),0)$, $Y = ((0,1),0)$ and $Z = ((0,0),1)$. The kernel N_q is the closed subgroup generated by the laws $\gamma_3 F$ and all qth powers, so it is characteristic. More precisely it is their closed normal closure: the exponent-q, class-two quotient has the normal forms $X^a Y^b Z^d$ in (D.43b), and (D.43c) verifies that all q^3 forms are distinct. This proves the characteristic-quotient identification.

The corresponding discrete cover of a once-punctured torus has q^2 cusps, one for each coset of $\langle Z \rangle$ in G_q, each of degree q. Fill all these cusps. Completing the finite-cover presentation gives the exact quotient

$$N_q \longrightarrow \Pi_q = N_q / \overline{\langle\langle hc^q h^{-1} : h \in F \rangle\rangle}_{N_q} \cong \widehat{\pi}_{g_q,0}, \qquad 2g_q = 2 + q^3 - q^2. \tag{D.43d}$$

Representatives for the q^2 cosets suffice in the normal closure. The genus formula follows by multiplying the punctured torus's Euler characteristic by q^3 and then adding its q^2 filled discs. The original deck group induces its actual homology representation on this closed surface. Since α preserves N_q and permutes all the filled peripheral classes, it induces an actual involution β_q of Π_q.

For every prime p, put $V_p = H_1(\Pi_q, \mathbb{Q}_p)$, let ρ be its G_q representation, and let A be the map induced by β_q. If θ is the automorphism induced by α on G_q, the precise deck comparison is

$$A^2 = 1, \qquad A\rho(h)A^{-1} = \rho(\theta(h)), \qquad \theta(Z) = Z^{\lambda \bmod q}. \tag{D.43e}$$

Indeed representatives in F for the two sides differ only by an element of N_q, whose conjugation is inner on Π_q and hence trivial on its first homology.

The nondegenerate alternating surface pairing satisfies

$$\langle Av, Aw \rangle = \lambda_p \langle v, w \rangle, \qquad \langle \rho(h)v, \rho(h)w \rangle = \langle v, w \rangle. \tag{D.43f}$$

Here is an exact verification of the first multiplier. The involution permutes the odd number q^2 of lifted cusps, so one cusp class is fixed. Fill every other cusp first. This is an invariant closed normal quotient, and gives a surface group of genus g_q with one cusp, which α preserves with power λ. Filling its last cusp does not change first homology. The degree-two boundary bivector calculation in (D.41h) therefore proves the first identity in (D.43f) for precisely A on V_p. The second identity comes from the orientation-preserving geometric deck maps of the specified finite surface cover. Thus no realization of α as a homeomorphism has been used.

We compute the whole deck module by finite cover chains. The cover before filling retracts onto the regular cover of a two-loop rose, so its first homology has representation class $1 + \mathbb{Q}_p[G_q]$. The relative second homology of the attached discs is $\mathbb{Q}_p[G_q/\langle Z \rangle]$. The exact homology sequence of this filling, including the invariant orientation class in degree two, gives

$$[V_p] = 2[1] + [\mathbb{Q}_p[G_q]] - [\mathbb{Q}_p[G_q/\langle Z \rangle]]. \tag{D.43g}$$

All these maps are deck-equivariant. They compute the homology of the completed surface groups through the finite-index and surface-homology comparisons already used in A.2 and D.42.

Over a characteristic-zero field containing a primitive qth root ζ, let V_j be the subspace where Z acts as ζ^j, for $j \in \mathbb{F}_q^\times$. The last permutation module in (D.43g) is trivial on Z, and the regular representation has each of its q central eigenvalues with multiplicity q^2. Semisimplicity therefore gives

$$\dim V_j = q^2. \tag{D.43h}$$

The G_q-invariant alternating form pairs V_j only with V_{-j}, and gives a nondegenerate pairing between them.

One negative odd-prime sign determines every sign. Suppose $\lambda_q = -1$ at an odd prime q. For any prime p, extend scalars to $\mathbb{Q}_p(\zeta)$. By (D.43e), A exchanges V_j and V_{-j}. Choose bases of

these two spaces and write P for their nonsingular pairing matrix, B for $A : V_j \to V_{-j}$ and C for $A : V_{-j} \to V_j$. Then $CB = BC = 1$, and alternation in (D.43f) gives

$$-B^{\mathsf{t}} P^{\mathsf{t}} C = \lambda_p P. \tag{D.43i}$$

The dimension q^2 is odd. Taking determinants in (D.43i) yields $-1 = \lambda_p^{q^2}$, hence $\lambda_p = -1$ by (D.43a). This works also at $p = q$ and $p = 2$, since the calculation is over characteristic-zero fields. Thus

$$\lambda_q = -1 \text{ for an odd } q \quad \Longrightarrow \quad \lambda = -1. \tag{D.43j}$$

The positive 3*-adic case.* It remains to prove that $\lambda_3 = 1$ forces $\lambda_2 = 1$. The matrix of α on $H_1(F, \mathbb{Z}_3)$ has square one and determinant one. Its two eigenvalues over $\mathbb{Q}_3$ are consequently both 1 or both -1, and it is diagonalizable, so the matrix is I or $-I$.

In the group G_3 of (D.43c), the resulting θ has the form

$$\theta(v, z) = (Mv, z + \ell(v)), \qquad M = I \text{ or } -I, \tag{D.43k}$$

with ℓ a linear functional. For $M = I$, $\theta^2 = 1$ implies $2\ell = 0$ and $\ell = 0$. For $M = -I$, conjugation of θ by the inner automorphism of $(u, 0)$ replaces $\ell(v)$ by $\ell(v) - 2\omega(u, v)$. Choose u with $\omega(u, v) = \ell(v)/2$. Lifting $(u, 0)$ to F and conjugating the actual automorphism α by that inner automorphism preserves its order two and its peripheral multiplier. It gives exactly

$$\theta = 1 \quad \text{or} \quad \theta = \iota, \qquad \iota(v, z) = (-v, z). \tag{D.43l}$$

This is conjugation of an actual automorphism, not multiplication by an arbitrary inner lift.

Work now over the unramified quadratic field $L = \mathbb{Q}_2(\zeta_3)$, with $\sigma(\zeta_3) = \zeta_3^{-1}$. For $j = 1, 2$ the irreducible G_3 module W_j with central character ζ_3^j has basis e_r, $r \in \mathbb{F}_3$, and matrices

$$Xe_r = e_{r+1}, \qquad Ye_r = \zeta_3^{-jr} e_r, \qquad Ze_r = \zeta_3^j e_r. \tag{D.43m}$$

These matrices satisfy (D.43b). The distinct eigenvalues of Y and the cyclic action of X show irreducibility. The diagonal projectors for Y, followed by powers of X, span the full matrix algebra. Hence the central block of $L[G_3]$ is $M_3(L)$. Together with (D.43g)–(D.43h), this gives

$$V_j = W_j \otimes_L U_j, \qquad \dim U_j = 3. \tag{D.43n}$$

The matrix $J = I$ implements $\theta = 1$ on W_j. For $\theta = \iota$, take $Je_r = e_{-r}$. This rational permutation matrix implements $X \mapsto X^{-1}$, $Y \mapsto Y^{-1}$, fixes Z, and has $J^2 = I$ and $J^{\mathsf{t}} J = I$. Thus (D.43e) and the full matrix-algebra commutant give

$$A|_{V_j} = J \otimes B_j, \qquad B_j^2 = I. \tag{D.43o}$$

Choose a multiplicity basis for U_1 and its σ-image for U_2. Since the original representation and A are defined over $\mathbb{Q}_2$, we then have $B_2 = \sigma(B_1)$. The canonical invariant pairing between W_1 and W_2 is the ordinary dot product in (D.43m). Consequently the nondegenerate cross-pairing in (D.43f) has the form $I_3 \otimes S$ for an invertible 3-by-3 matrix S on the multiplicity spaces. Equation (D.43f) becomes

$$B_1^{\mathsf{t}} S\, \sigma(B_1) = \lambda_2 S. \tag{D.43p}$$

Because $B_1^2 = I$, its determinant is 1 or -1, both fixed by σ. Taking determinants in (D.43p) therefore gives $1 = \lambda_2^3$. Together with $\lambda_2^2 = 1$, this proves $\lambda_2 = 1$.

If $\lambda_2 = -1$, the last conclusion forces $\lambda_3 = -1$, so (D.43j) gives $\lambda = -1$. If $\lambda_2 = 1$, (D.43j) excludes every negative odd-prime coordinate, and $\lambda = 1$. This proves the claim. Multiplicativity proves the stated consequence for subgroups generated by involutions; no conclusion about their conjugacy is inferred.

The program `code/part_c/torus_involution_d43.py` checks the characteristic Heisenberg presentations, the finite-cover chain ranks and central-character dimensions, and every G_3 involution with determinant one in (D.43k), including the exact inner-conjugacy formula in (D.43l). It also verifies the Schrödinger matrices and their rational implementer. The field and determinant arguments above prove the assertion for the full profinite groups. □

D.44. [Proved] A first-symmetry choice for the involution coboundary. Let $\alpha = (-1, f) \in GT_0$ be an involution in its normalized actual action on $F = \widehat{F}(x, y)$:

$$\alpha(x) = x^{-1}, \qquad \alpha(y) = f^{-1}y^{-1}f, \qquad f \in F'.$$

Write $\theta(x) = y$, $\theta(y) = x$. There exists $h \in F'$ satisfying both exact equations

$$f = h^{-1}\alpha(h), \qquad h\theta(h) = 1. \tag{D.44a}$$

Thus the coboundary in D.35 can be chosen with the first Harbater–Schneps symmetry. For every unit $\mu \in \widehat{\mathbb{Z}}^\times$, the associated continuous endomorphism $\rho_{\mu,h}(x) = x^\mu$, $\rho_{\mu,h}(y) = hy^\mu h^{-1}$ satisfies $\alpha\rho_{\mu,h} = \rho_{\mu,h}\delta$, where δ inverts x, y. Its full peripheral-conjugator condition is precisely D.35's remaining equation $x^\mu h y^\mu h^{-1} \sim_F (xy)^\mu$.

At the characteristic finite quotient A_5^{19} of D.38, this first-symmetry condition is soluble for every one of the 5 062 500 tuples that pass the triangle test. This finite equation therefore gives no further exclusion among them.

Manuscript: §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394.

Proof. The normalized involution and the first Harbater–Schneps equation give

$$f\alpha(f) = 1, \qquad \theta(f)f = 1. \tag{D.44b}$$

Checking the two free generators gives the identity of actual automorphisms

$$\theta\alpha\theta = \mathrm{Inn}(f)\alpha. \tag{D.44c}$$

Indeed both sides send x to $fx^{-1}f^{-1}$ and y to y^{-1}.

Form $E = F \rtimes_\alpha \langle s \mid s^2 = 1\rangle$. Define an automorphism Θ of this exact extension by

$$\Theta|_F = \theta, \qquad \Theta(s) = fs. \tag{D.44d}$$

The first equation of (D.44b) preserves $s^2 = 1$; (D.44c) preserves the conjugation relation; and the second equation of (D.44b) gives $\Theta^2(s) = \theta(f)fs = s$. Thus $\Theta^2 = 1$ on all of E. Adjoin its named involution τ and put

$$G = E \rtimes_\Theta \langle\tau\rangle, \qquad G/F = C_2 \times C_2. \tag{D.44e}$$

The two quotient coordinates are represented by s, τ.

The preimage of the τ line is the normal subgroup $G_\theta = F \rtimes_\theta \langle\tau\rangle$. It has the explicit free-product identification

$$G_\theta \xrightarrow{\sim} C_2(\tau) \amalg \widehat{\mathbb{Z}}(v), \qquad x \mapsto v, \quad y \mapsto \tau v \tau. \tag{D.44f}$$

The inverse sends v to x and τ to the named involution; both composites fix the indicated generators. The free-factor intersection theorem [31, Theorem 5 of the inspected author version] therefore gives $C_F(\tau) = 1$.

There is exactly one F-conjugacy class of involutions in G_θ. For a direct verification, its mod-two $H^1(F, \mathbb{F}_2)$ module is the regular two-dimensional C_2 module, since θ exchanges its

two basis characters. Its positive-degree C_2 cohomology vanishes. The split Hochschild–Serre calculation therefore gives $\dim H^n(G_\theta, \mathbb{F}_2) = 1$ for $n \geq 3$. Brown restriction [37, Theorem 2 and §1.1] gives one involution class, by exactly the dimension argument of (D.37f). Conjugacy by G_θ equals conjugacy by F, since $G_\theta = F\langle u\rangle$ for every involution u and its second factor centralizes that element.

Consider the closed normalizer $V = N_G(\langle\tau\rangle)$. For every $g \in G$, the conjugate $g\tau g^{-1}$ belongs to G_θ, and its image in G/F is still the τ coordinate. The preceding uniqueness gives an F-conjugator taking it back to τ. It follows that $V \to G/F$ is onto. Its kernel is $C_F(\tau) = 1$, so

$$V \xrightarrow{\sim} C_2 \times C_2. \tag{D.44g}$$

In particular V is a finite subgroup, contains τ, and has a unique involution u mapping to the s coordinate. That involution belongs to E and commutes with τ.

The three F-conjugacy classes of involutions in E were determined in D.34. They are represented by

$$s, \quad xs, \quad yfs. \tag{D.44h}$$

Their distinct mod-two abelianization vectors also identify the classes. Formula (D.44d) sends s to fs, which belongs to the class of s, and gives the exact transposition

$$\Theta(xs) = yfs, \qquad \Theta(yfs) = xs. \tag{D.44i}$$

Thus the class of s is the unique fixed class under Θ. Since u commutes with τ, its class is fixed, so $u = ksk^{-1}$ for some $k \in F$. Conjugate the finite subgroup (D.44g) by k^{-1}. It now contains s and

$$k^{-1}\tau k = h\tau, \qquad h = k^{-1}\theta(k). \tag{D.44j}$$

The square of this second involution gives $h\theta(h) = 1$. Its commutation with s gives $hf = \alpha(h)$, which is precisely the first equation of (D.44a). On abelianization, this equation reads $-2h^{\mathrm{ab}} = f^{\mathrm{ab}} = 0$. Multiplication by two is injective on $\widehat{\mathbb{Z}}^2$, hence $h \in F'$. Finally substitution in $\alpha(hy^\mu h^{-1})$ using $\alpha(h) = hf$ proves $\alpha\rho_{\mu,h}(y) = \rho_{\mu,h}(y^{-1})$; the x identity is immediate.

We check the finite scope explicitly. Write $Q = A_5^{19}$, and use D.38's component description of θ and its involution tuple t. The triangle test makes every component t_i a nonidentity involution of A_5. The equations to solve in this quotient are

$$h^{-1}th = \theta(t), \qquad h\theta(h) = 1, \qquad f = \theta(t)t^{-1}. \tag{D.44k}$$

They separate over the one-element and two-element orbits of θ on the factors.

For a two-element orbit, write $\theta(v)_i = gv_jg^{-1}$ and $\theta(v)_j = g^{-1}v_ig$, with $g \in S_5$. All nonidentity involutions of A_5 are conjugate within A_5. Choose $h_i \in A_5$ satisfying $h_i^{-1}t_ih_i = gt_jg^{-1}$, and set $h_j = g^{-1}h_i^{-1}g$. These substitutions give both equations of (D.44k) on that orbit.

For a fixed factor, θ is conjugation by an involution $g \in S_5$, allowing $g = 1$. Indeed its square is the identity and $C_{S_5}(A_5) = 1$. Choose an element v with $v^2 = 1$, commuting with t_i and having the same parity as g. When g is even choose $v = 1$. When g is odd, take either transposition in the disjoint-transposition expression for t_i. Then $h_i = vg^{-1} \in A_5$ satisfies both equations in (D.44k). Thus every triangle survivor has a finite first-symmetry solution, including the 2 531 250 survivors failing D.38's finite peripheral-conjugacy test. No lifting of these finite solutions is asserted.

The program `code/part_a/pure_involution_first_symmetry.py` executes the local construction for all 390 fixed-factor cases and all 27 000 two-factor cases. It checks both equations in (D.44k), which cover every factor orbit in D.38. □

D.45. [Proved] A characteristic binary-icosahedral refinement. Let $F = \widehat{F}(x, y)$ and let R be D.38's characteristic kernel with $F/R = A_5^{19}$. Intersecting the kernels of all epimorphisms $F \twoheadrightarrow S = \mathrm{SL}_2(\mathbb{F}_5)$ gives an open characteristic subgroup $\widetilde{R} \subset R$ and the following exact quotient and reduction map:

$$\widetilde{Q} = F/\widetilde{R} \xrightarrow{\sim} S^{19} \times C_2^2, \qquad \pi : \widetilde{Q} \longrightarrow A_5^{19}, \quad ((g_i), a) \longmapsto (g_i \bmod \{\pm I\})_i. \tag{D.45a}$$

For the 10 125 000 finite tuples t of D.38, impose at this specified refinement the normalized multiplier -1 action, both Harbater–Schneps cycle equations, and the exact norm $f\alpha(f) = 1$. Exactly 5 062 500 base tuples admit such a refined f-coordinate. They are precisely D.38's cusp-triangle survivors, and each has 64 admissible f-coordinates. Thus the total number of refined coordinate pairs is 324 000 000. For a fixed t, these 64 coordinates define the same actual automorphism of $\widetilde{Q}$, of order two. In particular this named characteristic refinement detects exactly D.38's first excluded half. All 2 531 250 triangle survivors outside D.38's finite pure conjugacy set still pass this refinement.

Manuscript: §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D of the task: full GT_0 torsion and involution conjugacy.

Proof. All matrix calculations below are in $\mathbb{F}_5$. The program enumerates the 120 determinant-one matrices and verifies that their commutator subgroup has order 120. Their center is $\{I, -I\}$, and their projective quotient is identified explicitly with D.38's A_5 by a generating pair of orders two and three with product of order five. The resulting bijection is checked against all 60^2 products. Conjugation by the 120 projective invertible matrices gives all automorphisms of S. Indeed $\mathrm{Aut}(S) \to \mathrm{Aut}(A_5)$ is injective: an element of its kernel has the form $g \mapsto c(g)g$, where $c : S \to \{\pm I\}$ is a homomorphism, and perfection makes $c = 1$. D.38 gives $|\mathrm{Aut}(A_5)| = 120$, and the enumerated projective conjugations already give 120 distinct automorphisms of S.

Choose lifts $X_i, Y_i \in S$ of the nineteen generating pairs in D.38, and put $Z_i = (X_iY_i)^{-1}$. Every pair of lifts generates S: its image generates A_5, so a proper image would have index two in S, contrary to perfection. The program also checks all $4 \cdot 19 = 76$ choices directly. Two maps to A_5 with a given kernel differ by an automorphism, and every automorphism lifts uniquely to S. Over each of the nineteen A_5 kernels there are exactly four S kernels, from the independent sign choices for the lifts of x and y. If two such lifts had the same kernel, the induced automorphism of S would act trivially on A_5; the proved injectivity would identify the lifts. Consequently these are exactly the 76 epimorphism kernels defining $\widetilde{R}$.

The joint map

$$x \longmapsto ((X_i), e_x), \qquad y \longmapsto ((Y_i), e_y), \qquad e_x, e_y \text{ the basis of } C_2^2, \tag{D.45b}$$

is onto. To see this without a rank inference, its projection D to S^{19} maps onto A_5^{19} by D.38. Thus $DZ(S^{19}) = S^{19}$; centrality gives $[S^{19}, S^{19}] = [D, D] \leq D$, so perfection gives $D = S^{19}$. The commutator subgroup of the image of (D.45b) therefore contains $S^{19} \times \{0\}$, and its projection to C_2^2 is onto. This proves surjectivity of (D.45b). Every other sign lift of the ith pair is obtained by composing (D.45b) with

$$((g_j), a) \longmapsto g_i(-I)^{\ell(a)}, \qquad \ell \in \mathrm{Hom}(C_2^2, C_2).$$

The intersection of these four kernels, for all i, is trivial: the zero character first makes each $g_i = I$, and all characters then make $a = 0$. This proves (D.45a), including the specified kernel. It is characteristic because every automorphism of F permutes its epimorphisms to S.

Here are the complete finite equations used in the refinement. The named word substitutions θ, ω of D.38 descend to $\widetilde{Q}$. On its characteristic perfect subgroup S^{19} they permute the factors

by D.38's recorded maps and apply the unique lifted S automorphisms just described. Write $\widetilde{t}_i \in S$ for any lift of t_i and put

$$f_{0,i} = \theta(\widetilde{t})_i \widetilde{t}_i^{-1}, \qquad f_i = (-I)^{b_i} f_{0,i}, \qquad b_i \in \mathbb{F}_2. \tag{D.45c}$$

These are all the 2^{19} lifts of $f_Q = \theta(t)t^{-1}$ into $\widetilde{Q}' = S^{19}$. For every such lift, set

$$\begin{aligned} c_{x,i} &= X_i^{-1}(\widetilde{t}_i X_i \widetilde{t}_i^{-1})^{-1}, \\ c_{y,i} &= (f_i^{-1} Y_i^{-1} f_i)(\widetilde{t}_i Y_i \widetilde{t}_i^{-1})^{-1}. \end{aligned} \tag{D.45d}$$

D.38's normalization implies $c_{x,i}, c_{y,i} \in \{I, -I\}$. The actual automorphism induced by the normalized pair is exactly

$$\alpha((g_i), (a_x, a_y)) = ((\widetilde{t}_i g_i \widetilde{t}_i^{-1} c_{x,i}^{a_x} c_{y,i}^{a_y})_i, (a_x, a_y)). \tag{D.45e}$$

This formula is a continuous automorphism; its restriction to the first factor is an automorphism, and its two correction characters are central. It sends the two generators (D.45b) to the required X^{-1} and $f^{-1}Y^{-1}f$. Since $\widetilde{t}_i^2$ is central and both correction characters have exponent two, its square is the identity. Its reduction is the nonidentity order-two automorphism of D.38, so its order is exactly two. Multiplying f_i by a central sign changes neither (D.45d) nor (D.45e). Conversely these images of the two generators determine the automorphism uniquely.

The program tests the following three exact equations in S^{19} for all choices (D.45c):

$$\theta(f)f = 1, \qquad \omega^2(f)Z^{-1}\omega(f)Y^{-1}fX^{-1} = 1, \qquad f\alpha(f) = 1. \tag{D.45f}$$

On f, which has zero C_2^2 coordinate, formula (D.45e) is conjugation by $\widetilde{t}$. Thus every operation in (D.45f) is a specified matrix multiplication or a specified factor map; the norm is tested as an equation, not inferred from the order of the finite automorphism.

The factor maps for θ, ω have six joint orbits. Exhaustion within these orbits gives the following complete counts, with D.38's zero-based factor indices:

factor orbit	base tuples	numbers of admissible central lifts
$\{0, 2, 10\}$	15	2 for each of 15
$\{1, 9, 17\}$	15	2 for each of 15
$\{3, 5, 6, 7, 11, 14\}$	60	0 for 30; 4 for 30
$\{4, 12, 18\}$	15	2 for each of 15
$\{8, 13, 15\}$	10	2 for each of 10
$\{16\}$	5	1 for each of 5.

(D.45g)

The rejected thirty assignments in the six-factor orbit are exactly its triangle failures; the other five orbits have no triangle failure. The equations on one joint orbit involve no other orbit, so their solution sets have the displayed Cartesian-product decomposition. The surviving count is $15 \cdot 15 \cdot 30 \cdot 15 \cdot 10 \cdot 5 = 5\,062\,500$, and the number of central lifts per survivor is $2 \cdot 2 \cdot 4 \cdot 2 \cdot 2 \cdot 1 = 64$. This proves every asserted count. The executed implementation is `code/part_a/a5_schur_refinement.py`; its output records each orbit, the lift histogram, and sample coordinates. These computations impose a necessary finite refinement on full involutions. They assert no full lift of a surviving finite coordinate pair. □

D.46. [Proved] Three positive involution sections and order-four lifting. Let T be the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$. For every involution $b \in T$ whose 2-adic orientation is positive, its centre-free extension

$$1 \longrightarrow F \longrightarrow E_b \longrightarrow \langle b \rangle \longrightarrow 1$$

has exactly three F-conjugacy classes of order-two sections. For every such section S,

$$H^1(C_F(S), \mathbb{F}_2) = 0. \tag{D.46a}$$

Every cyclic order-four subgroup of T has a homomorphic lift to $\mathrm{Aut}(F)$, with its prescribed outer action. Thus every element of order four in T has an actual order-four representative.

Manuscript: Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394..

Proof. The homology of a positive involution. D.27 gives an actual order-two representative β of b, and identifies E_b with $F \rtimes_\beta \langle s \rangle$. The map of D.33 on the finite 2-group $\langle b \rangle$ is faithful. The matrix of β on $H_1(F, \mathbb{Z}_2)$ is therefore nonidentity. Its square is the identity and its determinant is 1. Over $\mathbb{Q}_2$ it is diagonalizable; the determinant condition makes its two eigenvalues equal. It follows that

$$\beta_2^{\mathrm{ab}} = -I. \tag{D.46b}$$

This uses the punctured-torus statement D.33 to exclude the identity matrix; no analogous exclusion for an arbitrary unmarked rank-two action is assumed.

The full cohomology algebra and its Bockstein. Let $u : E_b \to \mathbb{F}_2$ be the quotient character, with $u(s) = 1$. Choose a basis χ_1, χ_2 of $H^1(F, \mathbb{F}_2)$ and extend these characters to E_b by declaring $\chi_i(s) = 0$. These extensions exist because (D.46b) is the identity modulo 2. The Hochschild–Serre spectral sequence has precisely the two rows

$$E_2^{a,0} = \mathbb{F}_2 u^a, \qquad E_2^{a,1} = H^1(F, \mathbb{F}_2)\, u^a. \tag{D.46c}$$

The transgression on $E_2^{0,1}$ is zero since the two characters have just been extended to E_b. Its other possible differentials are the products of this transgression with powers of u, and hence are zero as well. There are no further nonzero rows. Consequently, for every $n \geq 1$,

$$H^n(E_b, \mathbb{F}_2) = \langle u^n, u^{n-1}\chi_1, u^{n-1}\chi_2 \rangle_{\mathbb{F}_2}, \qquad \dim H^n(E_b, \mathbb{F}_2) = 3. \tag{D.46d}$$

Independence follows from the two associated graded pieces in (D.46c). Multiplication by u is an isomorphism in every degree at least one: it is an isomorphism on those associated graded pieces.

For each i, lift $\chi_i|_F$ to a continuous homomorphism $\varphi_i : F \to \mathbb{Z}/4$. Equation (D.46b) gives $\varphi_i(\beta(h)) = -\varphi_i(h)$. The actual continuous one-cochain on E_b

$$\widetilde{\chi}_i(hs^\epsilon) = \varphi_i(h) \quad (\epsilon = 0, 1)$$

therefore satisfies

$$\begin{aligned} \delta\widetilde{\chi}_i(hs^\epsilon, ks^\eta) &= \varphi_i(h) + \varphi_i(k) - \varphi_i(h\beta^\epsilon(k)) \\ &= 2\epsilon\, \varphi_i(k) \quad \text{in } \mathbb{Z}/4. \end{aligned} \tag{D.46e}$$

For the coefficient sequence $0 \to \mathbb{F}_2 \xrightarrow{2} \mathbb{Z}/4 \to \mathbb{F}_2 \to 0$, this says that the Bockstein of χ_i is $u\chi_i$. It is also χ_i^2. For completeness, the lift of any $\mathbb{F}_2$ character which takes its values in $\{0, 1\} \subset \mathbb{Z}/4$ has coboundary, divided by 2, equal to the cup square of that character; comparison of two lifts changes this cocycle only by a coboundary. Thus

$$\chi_i^2 = u\chi_i. \tag{D.46f}$$

Invert the central class u and take degree zero:

$$R = \big(H^*(E_b, \mathbb{F}_2)[u^{-1}]\big)^0.$$

It has dimension three, with basis $1, z_1 = \chi_1/u, z_2 = \chi_2/u$ by (D.46d). Equation (D.46f) makes both z_i idempotent. In particular R is a quotient of the algebra

$$\mathbb{F}_2[Z_1, Z_2]/(Z_1^2 - Z_1, Z_2^2 - Z_2) \cong \prod_{(a_1,a_2)\in\mathbb{F}_2^2} \mathbb{F}_2, \quad P \longmapsto (P(a_1, a_2))_{(a_1,a_2)}. \tag{D.46g}$$

The displayed evaluation map is an isomorphism: the four products of Z_i and $1 - Z_i$ are its four coordinate idempotents. Every ideal of a finite product of fields is a product of ideals in those fields. Every quotient is therefore reduced, so R has no nonzero nilpotent element.

Brown restriction and the three full sections. Every finite subgroup of E_b injects into $E_b/F = C_2$, since F is torsion-free. Thus the nontrivial elementary abelian 2-subgroups are exactly its order-two sections, with no inclusions between distinct ones. For such a section S put $C_S = C_F(S)$. Its normalizer is the exact direct product

$$N_{E_b}(S) = C_S \times S. \tag{D.46h}$$

Indeed every normalizer element centralizes S, since $\mathrm{Aut}(C_2) = 1$; multiplying it by its unique image in S leaves an element of C_S.

The full-profinite Brown restriction theorem used in D.28 and D.33 applies because F is open in E_b and has 2-cohomological dimension one [37, Theorem 2 and §1.1]. In sufficiently high degrees it identifies cohomology with the product of the cohomologies of (D.46h); the identification is the actual restriction map and preserves cup products. Each factor contributes a copy of $\mathbb{F}_2$ in every positive degree, so (D.46d) in particular makes the number of factors finite. Since C_S is a closed subgroup of F, it has $\mathrm{cd}_2(C_S) \le 1$. After the same localization, restriction identifies R as a ring with

$$\prod_{[S]} (\mathbb{F}_2 \oplus H^1(C_S, \mathbb{F}_2)\, u^{-1}). \tag{D.46i}$$

The second summand in each factor is an ideal with square zero, since $H^2(C_S, \mathbb{F}_2) = 0$. Here $u|_{C_S} = 0$ and $u|_S$ is the nonzero character, so the localization in (D.46i) is exactly the localization already used in (D.46g). Reducedness of R forces every second summand to vanish. Its dimension three then proves that there are exactly three section classes, and proves (D.46a).

These are also precisely the F-conjugacy classes. To see this directly, let $e \in E_b$ and choose $s' \in S$ with the same image as e in E_b/F. Then $eSe^{-1} = (es'^{-1})S(es'^{-1})^{-1}$, and $es'^{-1} \in F$.

An invariant section in the order-four extension. Now let $a \in T$ have order four, and form its centre-free extension

$$E = \{(\phi, j) \in \mathrm{Aut}(F) \times C_4 : [\phi] = a^j\}. \tag{D.46j}$$

Its kernel is identified with F by $h \mapsto (\mathrm{Inn}(h), 0)$. Let E_2 be the inverse image of $\langle a^2 \rangle$. The orientation of a^2 at 2 is positive: the only roots of unity of order dividing four in $\mathbb{Z}_2^\times$ are $1, -1$, whose squares are 1. Equivalently, D.33 embeds $\langle a \rangle$ in $\mathrm{GL}_2(\mathbb{Z}_2)$, where its square is $-I$. The first part of the proof thus applies to E_2.

Conjugation by E permutes the three F-conjugacy classes of sections of E_2. Every element of E_2 acts trivially on this class set, by the preceding paragraph, so the permutation action factors through $E/E_2 = C_2$. An action of C_2 on a set of three elements has a fixed element. Choose a section S in that class. If $e \in E$ projects to the specified generator of C_4, there is $f \in F$ with $eSe^{-1} = fSf^{-1}$. Hence $f^{-1}e \in N_E(S)$ still projects to that generator. Put $N = N_E(S)$ and $C = C_F(S)$. We have proved the exact sequences

$$\begin{aligned} 1 \longrightarrow C \longrightarrow N \longrightarrow C_4 \longrightarrow 1, \qquad S \subseteq Z(N), \\ 1 \longrightarrow C \longrightarrow J := N/S \longrightarrow C_2 \longrightarrow 1. \end{aligned} \tag{D.46k}$$

The embedding of C in the second line is injective because $C \cap S = 1$.

Projectivity supplies the order-four section. Equation (D.46a) implies

$$C^{(2)} = 1 : \tag{D.46l}$$

every nontrivial pro-2 group has a nontrivial finite 2-group quotient, which in turn has a quotient C_2. In particular every homomorphism from J to a finite 2-group kills C and factors through the last C_2 in (D.46k). That quotient is nontrivial, so

$$J^{(2)} = C_2. \tag{D.46m}$$

We show that J has a nonidentity torsion element. If it were torsion-free, its open subgroup C has $\mathrm{cd}_p(C) \le 1$ at every prime p, as a closed subgroup of the free profinite group F. Serre's torsion-free finite-index dimension theorem, in exactly the form used in D.27, would give $\mathrm{cd}_p(J) \le 1$ at every prime. Thus J would be projective as a full profinite group. Its maximal pro-2 quotient would then be a projective, hence free, pro-2 group: pull a finite 2-group embedding problem back to J, solve it by projectivity, and factor the solution through $J^{(2)}$. This contradicts (D.46m), since C_2 is not a free pro-2 group.

Let $j \in J$ be a nonidentity torsion element. Its cyclic subgroup injects into $J/C = C_2$, because C is torsion-free. Consequently j has order two and lies outside C. Choose $n \in N$ over j. Then $n^2 \in S$, whereas its image in C_4 is an odd element and has order four. It follows that n^2 is the nonidentity element of S, and n has order exactly four. After replacing n by n^{-1} if necessary, its projection is the specified generator of C_4. Thus $C_4 \to E$, sending that generator to n, is a section of (D.46j). Projection to $\mathrm{Aut}(F)$ gives the claimed homomorphic lift with its original labels.

The program `code/part_c/torus_order4_lift_d46.py` checks the finite cochain formula (D.46e), its comparison with the cup square, and the explicit Boolean-algebra evaluation in (D.46g). The dimension and normalizer arguments above prove the statement for the full profinite extensions. No claim of conjugacy to a geometric order-four action is made. D.43 applied to a^2 does also give $\lambda(a)^2 = 1$ globally, so the remaining orientation issue for a concerns mixed signs, rather than fourth roots of unity. □

D.47. [Proved] Conjugator coordinates at the binary refinement. Use the specified quotient $\widetilde{Q} = S^{19} \times C_2^2$, $S = \mathrm{SL}_2(\mathbb{F}_5)$, of D.45, and its 324 000 000 refined pairs (t, f). Every one of these pairs satisfies D.44's finite first-symmetry coboundary condition

$$h^{-1}\alpha(h) = f, \qquad h\theta(h) = 1, \qquad h \in S^{19}. \tag{D.47a}$$

Exactly 10 125 000 of the pairs satisfy the two finite conjugator equations of D.35 at this quotient. The count stays the same after requiring both actual invertibility of the finite substitution and both Harbater–Schneps symmetries for the conjugating pair. There are 5 062 500 pairs with admissible unit parameters congruent to a square modulo five, and 5 062 500 with parameters congruent to a nonsquare. These two sets are disjoint.

Thus 313 875 000 refined coordinate pairs lie outside the image of the standard full GT_0 involution conjugacy class. For each of D.38's 2 531 250 conjugator-compatible base tuples, exactly four of its 64 refined f-coordinates pass this test; for each of the other 2 531 250 base tuples, none does. These are statements about the exact normalized coordinates: D.45's 64 coordinates for a fixed t define the same finite automorphism. No surviving pair is asserted to have a full involution or conjugator lift.

Manuscript: §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D of the task: full GT_0 involution conjugacy.

Proof. Write $X_i, Y_i, Z_i, \widetilde{t}_i$ and α for the matrices and actual automorphism fixed in D.45. Since S has exponent dividing 60, the power parameter in D.35 factors through $(\mathbb{Z}/60)^\times$. For the second Harbater–Schneps symmetry we retain the finer, explicit parameter map

$$a \in (\mathbb{Z}/120)^\times \longmapsto (a \bmod 60,\ m = (a-1)/2 \bmod 60)\,. \tag{D.47b}$$

It is well-defined, and every such a is the reduction of a unit in $\widehat{\mathbb{Z}}$. Changing the representative of a by 120 changes m by 60, so all subsequent equations are independent of the choice of integer representative.

The image of F' is exactly S^{19} by D.45. For fixed (t, f) the first conjugator equation therefore separates into the nineteen exact equations

$$h_i^{-1}\widetilde{t}_i h_i \widetilde{t}_i^{-1} = f_i, \qquad h_i \in S. \tag{D.47c}$$

Each has exactly four solutions. The program verifies this by enumeration of all 120 matrices. It also follows from the order-four conjugacy class in S: $\widetilde{t}_i$ has square $-I$, and the norm equation in D.45 gives $(f_i\widetilde{t}_i)^2 = \widetilde{t}_i^2 = -I$. The thirty matrices with square $-I$ form one conjugacy class with centralizer of order four, as the same matrix enumeration verifies.

For each parameter a mod 60 and each of these four matrices, the second equation of D.35 is tested by the exact matrix conjugacy condition

$$X_i^a h_i Y_i^a h_i^{-1} \sim_S (X_i Y_i)^a. \tag{D.47d}$$

Indeed the C_2^2 coordinates of both sides are $e_x + e_y$, since a is odd, and that direct factor is central. Conjugacy in $\widetilde{Q}$ is therefore exactly these nineteen independent matrix conjugacies. The program computes every conjugacy class in S by conjugation by all 120 matrices; (D.47d) is tested against those complete classes.

The finite invertibility test is explicit as well. Reduce the ordered pair $(X_i^a, h_i Y_i^a h_i^{-1})$ to A_5 and test whether it belongs to the ith recorded $\mathrm{Aut}(A_5)$ orbit of generating pairs in D.38. This is sufficient: its unique component automorphism lifts to an automorphism γ_i of S, and the two central differences from $\gamma_i(X_i), \gamma_i(Y_i)$ are corrected by characters of C_2^2, exactly as in (D.45d)–(D.45e). The resulting automorphism of $\widetilde{Q}$ has the prescribed two generator images. It is necessary for a finite pure automorphism with these images: its reduction preserves the three specified power-conjugacy classes, and D.38's distinct passports force it to preserve each simple factor. Its pair must therefore be in that recorded orbit. This tests the actual finite substitution, with its characteristic quotient map retained.

For completeness the program also imposes both symmetries on the conjugating pair, not just its three peripheral images. Put $k = h^{-1}$ and use m from (D.47b). The equations tested are

$$h\theta(h) = 1, \qquad \omega^2(k) Z^m \omega(k) Y^m k X^m = 1. \tag{D.47e}$$

All h, k, f coordinates lie in the perfect factor S^{19}; the actions of θ, ω there are the specified lifted factor maps of D.45. The C_2^2 coordinate of the second equation is $m(e_x + e_y + e_z) = 0$. Thus these are exact equations at the indicated finite quotient, including the half-exponent required by relation (II).

The first equation of (D.47e) is exhausted without assuming independent choices where factors are exchanged. If $\theta(v)_i = G_i v_j G_i^{-1}$, a choice of h_i forces

$$h_j = G_i^{-1} h_i^{-1} G_i. \tag{D.47f}$$

For a fixed factor it must satisfy the same equation with $j = i$. The program enumerates one of the four candidates from (D.47c) for each orbit of θ, uses (D.47f), and checks membership in the corresponding candidate sets. It verifies that (D.47a) is soluble for all the refined pairs. For the

full conjugator test it then checks (D.47d), finite invertibility, and the second equation of (D.47e) for every remaining choice and all 32 units modulo 120.

Here is the entire factorization of the result. Let U_+ and U_- denote the subsets of $(\mathbb{Z}/120)^\times$ whose residues modulo five are respectively squares and nonsquares; each has 16 elements. For each of D.45's six joint factor orbits, the table gives the number of refined pairs having the specified set of admissible parameters.

factor orbit	U_+	U_-	$U_+ \cup U_-$	$\varnothing$
$\{0, 2, 10\}$	15	15	0	0
$\{1, 9, 17\}$	15	15	0	0
$\{3, 5, 6, 7, 11, 14\}$	30	30	0	60
$\{4, 12, 18\}$	15	15	0	0
$\{8, 13, 15\}$	10	10	0	0
$\{16\}$	0	0	5	0.

(D.47g)

The admissible global parameter set is the intersection of the six local sets, because the parameter is common while the factor equations have the exact orbit decomposition of D.45. It is nonempty precisely when the first five nonempty choices all select the same one of U_+, U_-. Each choice gives $15 \cdot 15 \cdot 30 \cdot 15 \cdot 10 \cdot 5 = 5\,062\,500$ pairs. Their sum is 10 125 000, and subtraction from D.45's 324 000 000 gives the stated 313 875 000 failures. The program records the same table when only (D.47c)–(D.47d) are imposed, and when finite invertibility and the first symmetry are added. Thus the asserted equality of these counts is itself checked at every factor orbit. It also records the counts for each base tuple before multiplying over the six orbits. Their Cartesian product gives the exact histogram

admissible f-coordinates per base tuple	number of base tuples
0	2 531 250
4	2 531 250.

(D.47h)

Every full conjugator in D.35 reduces to solutions of (D.47c)–(D.47d), and D.35's equality of the P and GT_0 conjugator sets also supplies (D.47e). Consequently failure of the displayed finite test excludes that exact coordinate pair from the image of the standard full conjugacy class. The executed implementation is `code/part_a/pure_involution_binary_conjugacy.py`. Its output retains all unit sets, the six local tables, the per-base histogram, and the separate first-symmetry counts. The proof uses only this finite refinement and makes no assertion that one of its surviving pairs lifts through all characteristic refinements of F. □

D.48. [Proved] The two sections of a punctured-torus order-three action. Let a have order three in the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$, and let E_a be its centre-free extension. There are exactly two F-conjugacy classes of order-three sections in E_a. Each such section S satisfies

$$H^1(C_F(S), \mathbb{F}_3) = 0. \tag{D.48a}$$

The normalizer in E_a of the torus cusp $L = \overline{\langle [x, y] \rangle}$ is torsion-free. If v is any element of this normalizer projecting to the prescribed generator a, and

$$v[x, y]v^{-1} = [x, y]^\lambda, \qquad v^3 = [x, y]^\kappa,$$

then its 3-adic coordinate satisfies $\kappa_3 \in \mathbb{Z}_3^\times$.

Manuscript: Part D(b) of the task; §24, folio 147 verso, scan 293; §30, folios 194–196, scans 383–387; §31, folios 198–199 verso, scans 391–394..

Proof. The actual action and its abelian quotient. D.27 gives an actual order-three representative α of a, so $E_a = F \rtimes_\alpha \langle s \rangle$. Let A be its matrix on $H_1(F, \mathbb{Z}_3)$. It has order exactly three by D.33. Over $\mathbb{Q}_3$ the polynomial $X^2 + X + 1$ is irreducible: a primitive cube root would give a square root of -3, whose 3-adic valuation is odd. Since $A^3 = I$ and $A \neq I$, it follows that

$$A^2 + A + I = 0, \qquad \det(A - I) = 3, \qquad \operatorname{coker}(A - I) \cong C_3. \tag{D.48b}$$

The last equality follows by Smith reduction over $\mathbb{Z}_3$, since the determinant has valuation one. Abelianizing the actual semidirect product gives the exact maximal pro-3 abelian quotient

$$E_a^{\mathrm{ab},(3)} \cong C_3^2. \tag{D.48c}$$

One factor is (D.48b), the other is the specified section quotient.

The reduction of A modulo 3 is a nontrivial unipotent Jordan block. Indeed, if $A = I + 3B$, then $A^3 = I$ implies $B(I + 3B + 3B^2) = 0$ after division by 9. The second factor is invertible, giving $B = 0$, contrary to $A \neq I$. The induced action on $M = H^1(F, \mathbb{F}_3)$ is therefore also one nontrivial Jordan block of dimension two. Writing D for the action matrix on this dual space, the exact cyclic resolution has successive operators

$$D - I, \quad 1 + D + D^2 = 0, \quad D - I, \quad 0, \ldots, \qquad \operatorname{rank}(D - I) = 1. \tag{D.48d}$$

Consequently $H^j(C_3, M)$ has dimension one for every $j \geq 0$.

Bockstein injectivity and periodicity. Let $u \in H^1(E_a, \mathbb{F}_3)$ be the quotient character, and let χ be a nonzero invariant character of F, extended to E_a by $\chi(s) = 0$. They form a basis of $H^1(E_a, \mathbb{F}_3)$ by (D.48c). For the coefficient sequence

$$0 \longrightarrow \mathbb{F}_3 \xrightarrow{3} \mathbb{Z}/9 \longrightarrow \mathbb{F}_3 \longrightarrow 0$$

write $\mathfrak{b}$ for the Bockstein. Every homomorphism $E_a \to \mathbb{Z}/9$ factors through (D.48c), so its values are in $3(\mathbb{Z}/9)$ and its reduction modulo 3 is zero. The coefficient long exact sequence therefore makes

$$\mathfrak{b} : H^1(E_a, \mathbb{F}_3) \longrightarrow H^2(E_a, \mathbb{F}_3) \quad \text{injective.} \tag{D.48e}$$

Put $v = \mathfrak{b}(u)$ and $w = \mathfrak{b}(\chi)$. The notation v in this cohomology calculation denotes a class, not the normalizer element in the statement. The class v is the inflation of the usual degree-two period class of C_3; it is nonzero on the section $\langle s \rangle$. The classes v, w are linearly independent by (D.48e).

The Hochschild–Serre spectral sequence has only rows zero and one. By (D.48d), at each positive total degree its E_2 page has two one-dimensional terms. Thus its degree-two abutment has dimension at most two. The two independent Bockstein classes make its dimension exactly two. The only possible differential is $d_2 : E_2^{j,1} \to E_2^{j+2,0}$. At $j = 0$ it vanishes because χ extends to E_a, and at $j = 1$ it vanishes because the one-dimensional term $E_2^{1,1}$ must survive to obtain dimension two in degree two.

Multiplication by the period class v identifies each cyclic-resolution term with the term two places later. On the row with coefficient M this also holds starting at $j = 0$, since the norm in (D.48d) is zero, so $H^0(C_3, M) \to H^2(C_3, M)$ is the identity on $\ker(D - I)$. On the row with trivial coefficients it is the usual period isomorphism. The multiplicative spectral sequence therefore makes every remaining d_2 a multiple of one of the two already vanishing differentials. It collapses, and multiplication by v is an isomorphism on its two associated graded pieces at every positive total degree. Hence

$$\begin{gathered} \dim H^n(E_a, \mathbb{F}_3) = 2 \quad (n \geq 1), \\ v : H^n(E_a, \mathbb{F}_3) \xrightarrow{\sim} H^{n+2}(E_a, \mathbb{F}_3) \quad (n \geq 1), \\ wv^m \neq 0 \quad (m \geq 0). \end{gathered} \tag{D.48f}$$

Exactly two full section classes. Every finite subgroup of E_a injects into its quotient C_3, because F is torsion-free. For an order-three section S the normalizer is the exact direct product

$$N_{E_a}(S) = C_F(S) \times S. \tag{D.48g}$$

Conjugation by a normalizer element induces the identity on S: projection to the abelian quotient C_3 identifies every element of S with its label, so that label cannot change under conjugation. Multiplying by the section element with the same quotient label then leaves precisely $C_F(S)$.

The full-profinite Brown restriction theorem [37, Theorem 2 and §1.1], applied with the open subgroup F of 3-cohomological dimension one, identifies sufficiently high cohomology with the product over (D.48g). Since $C_F(S)$ is a closed subgroup of F, its 3-cohomological dimension is at most one. Writing k for the number of section classes, the dimensions in (D.48f) therefore give

$$k + \sum_{[S]} \dim H^1(C_F(S), \mathbb{F}_3) = 2. \tag{D.48h}$$

In particular the class set is finite.

It cannot consist of a single class. Otherwise take that class to be the chosen $S = \langle s \rangle$. On its normalizer (D.48g), the character χ is pulled back from $C_F(S)$, since $\chi(s) = 0$. Its Bockstein therefore restricts to zero:

$$w|_{N_{E_a}(S)} = \mathrm{pr}^*_{C_F(S)}\, \mathfrak{b}(\chi|_{C_F(S)}) = 0, \tag{D.48i}$$

because $H^2(C_F(S), \mathbb{F}_3) = 0$. Every nonzero class wv^m from (D.48f) would then have zero Brown restriction, a contradiction for large m. Thus $k \geq 2$; (D.48h) makes $k = 2$ and makes both summands zero, proving (D.48a). As in D.46, E_a-conjugacy equals F-conjugacy: for any section S choose its unique element with the same quotient label as a given conjugator and multiply the conjugator by its inverse.

The peripheral normalizer. We spell out the boundary consequence of the vanishing in (D.48a). A closed subgroup $C \leq F$ with $H^1(C, \mathbb{F}_3) = 0$ cannot contain a conjugate of the full 3-part $\langle [x, y]^{e_3} \rangle \cong \mathbb{Z}_3$ of the torus cusp. First its supernatural index in F must have infinite 3-part. Otherwise choose an open $U_0 \geq C$ whose index has maximal 3-valuation. For every open V with $C \leq V \leq U_0$, the index $[U_0 : V]$ is prime to 3. Transfer makes $H^1(U_0, \mathbb{F}_3) \to H^1(V, \mathbb{F}_3)$ injective. The restriction-colimit comparison of D.32 would then inject the nonzero group $H^1(U_0, \mathbb{F}_3)$ into $H^1(C, \mathbb{F}_3)$, a contradiction.

Choose an open $U \geq C$ with $3 \mid [F : U]$. Its corresponding finite cover of the punctured torus has a lift of the selected cusp of degree prime to 3, because its completed boundary subgroup contains the selected full $\mathbb{Z}_3$. The exact discrete boundary comparison uses the dense discrete representative for the profinite conjugator, as in D.32 and D.33. There are at least two boundary lifts: if there were only one, its degree would be $[F : U]$. The selected boundary class is nonzero in $H_1(U, \mathbb{F}_3)$, since the only relation among the boundary classes is their sum. A boundary-dual character restricts nontrivially to the selected $\mathbb{Z}_3$ in C; the scalar comparison is the inverse of its covering degree modulo 3. This again contradicts $H^1(C, \mathbb{F}_3) = 0$ and proves the assertion.

For $L = \overline{\langle [x, y] \rangle}$, the peripheral normalizer has the exact sequence

$$1 \longrightarrow L \longrightarrow N_{E_a}(L) \longrightarrow C_3 \longrightarrow 1. \tag{D.48j}$$

Surjectivity follows from preservation of the cusp class, and its kernel is L by A.7. The multiplier satisfies $\lambda^3 = 1$, and $\lambda_3 = 1$, since $\mathbb{Z}_3^\times$ has no nontrivial root of unity of order three. An order-three section in (D.48j) would therefore centralize the full cusp $\mathbb{Z}_3$. Its fixed subgroup in F would contradict the boundary consequence of (D.48a). Every torsion subgroup of (D.48j) injects into C_3, so this proves that the normalizer is torsion-free.

Finally take the element called v in the statement, with $v^3 = [x, y]^\kappa$. Commuting v with its own cube gives $(\lambda - 1)\kappa = 0$. Where $\lambda_p \neq 1$, the coordinate κ_p and the norm $1 + \lambda_p + \lambda_p^2$ are zero. Where $\lambda_p = 1$, that norm is 3. If κ_3 were divisible by 3, the exact norm equation

$$(1 + \lambda + \lambda^2)z = -\kappa \quad \text{in } \widehat{\mathbb{Z}} \tag{D.48k}$$

would consequently have a solution: divide by 3 at the positive coordinates, and use zero at the other coordinates. The element $v[x, y]^z$ would then have order three and project to a, contradicting the torsion-free conclusion. Thus κ_3 is a unit, as asserted.

The program `code/part_c/torus_order3_sections_d48.py` checks the integral and mod-3 matrix calculations, the abelian character and Bockstein input in its finite semidirect quotient, and the cyclic resolution operators. The full restriction and finite-cover arguments above establish the section and normalizer assertions. They do not assert global triviality of the order-three multiplier or geometric conjugacy. □

D.49. [Proved] The standard centralizer and the two-element conjugator fibres. Let P be the full pure three-cusp group of D.32 and $\delta(x) = x^{-1}$, $\delta(y) = y^{-1}$. Its full centralizer and normalizer are

$$C_P(\delta) = N_P(\langle\delta\rangle) = \langle\delta\rangle. \tag{D.49a}$$

Define the actual conjugacy orbits

$$\mathcal{O}_P = \{\beta\delta\beta^{-1} : \beta \in P\}, \qquad \mathcal{O}_{GT_0} = \{\beta\delta\beta^{-1} : \beta \in GT_0\}.$$

Every fibre of $P \to \mathcal{O}_P$ is exactly the right coset $\{\beta, \beta\delta\}$, and

$$\mathcal{O}_{GT_0} = \mathcal{O}_P \cap GT_0. \tag{D.49b}$$

For $\alpha \in \mathcal{O}_P$, its two conjugators have the exact normalized formulas

$$\begin{array}{ll} x \mapsto x^\mu, & y \mapsto hy^\mu h^{-1}, \\ x \mapsto x^{-\mu}, & y \mapsto hy^{-\mu}h^{-1}, \end{array} \qquad \mu \in \widehat{\mathbb{Z}}^\times, \quad h \in F'. \tag{D.49c}$$

Here h is unique, and the unordered pair $\{\mu, -\mu\}$ is unique. In D.35's solution space, a choice (μ, h, u) therefore gives all the solutions over α as

$$(\mu, h, u(xy)^a), \qquad (-\mu, h, x^{-\mu}u(xy)^a), \qquad a \in \widehat{\mathbb{Z}}. \tag{D.49d}$$

These assertions concern the specified orbit map; they do not assert that it contains every full pure or GT_0 involution.

Manuscript: §30; folios 194–196; scans 383–387; §31; folios 198–199 verso; scans 391–394; Part D of the task: full GT_0 involution conjugacy.

Proof. Lochak–Schneps prove that complex conjugation is self-centralizing in the usual GT [41, Proposition 4(ii), with Lemma 9]. Their argument uses the first two defining relations and applies to GT_0 as well. We give the passage from the full group P explicitly, together with the normalized fibre formulas.

Let $\beta \in C_P(\delta)$ have normalized pair (μ, k), so $\beta(x) = x^\mu$ and $\beta(y) = k^{-1}y^\mu k$, with $k \in F'$. D.32's normalization is a homomorphic section into $\mathrm{Aut}(F)$, so the commutation is an actual automorphism equality. Its value on y makes $k\delta(k)^{-1}$ centralize y^μ. The peripheral centralizer is $\langle y\rangle$ by A.7, while $k\delta(k)^{-1} \in F'$. Abelianization therefore gives $k = \delta(k)$. The exact free-product identification of D.35, $F \rtimes_\delta C_2 \cong C_2 \amalg C_2 \amalg C_2$, gives $C_F(\delta) = 1$. Hence $k = 1$.

We also need the precise inclusion in GT_0. For each geometric cusp permutation $\sigma \in \Sigma$, form $r_\sigma = \beta^{-1}\sigma\beta\sigma^{-1}$. It lies in $P^1 = \ker\lambda$, and commutes with δ, because both β and σ do. D.35

proved $C_{P^1}(\delta) = 1$, so $r_\sigma = 1$. The exact identity $GT_0 = C_P(\Sigma)$ of D.35 therefore puts this β in GT_0.

Its second Harbater–Schneps relation, with $k = 1$ and $m = (\mu - 1)/2 \in \widehat{\mathbb{Z}}$, is

$$(xy)^{-m} y^m x^m = 1, \qquad \text{equivalently} \quad (xy)^m = y^m x^m. \tag{D.49e}$$

For every positive integer n use the explicit finite Heisenberg quotient with coordinates $(a, b, c) \in (\mathbb{Z}/n)^3$ and multiplication

$$(a, b, c)(a', b', c') = (a + a', b + b', c + c' + ab'), \quad x = (1, 0, 0), \quad y = (0, 1, 0).$$

The two sides of (D.49e) have coordinates

$$(m, m, m(m+1)/2), \qquad (m, m, 0), \tag{D.49f}$$

where the binomial expression is the continuous integer-valued function on $\widehat{\mathbb{Z}}$. These identities first follow for nonnegative integers by multiplication, and then for all $m \in \widehat{\mathbb{Z}}$ by continuity; modulo n they depend only on m mod $2n$. Equality in every such quotient gives $m(m+1)/2 = 0$ in $\widehat{\mathbb{Z}}$, and hence

$$\mu^2 = 1. \tag{D.49g}$$

Since $k = 1$, the actual square of β fixes both x and y by (D.49g). Thus $\beta^2 = 1$. D.34 gives the global multiplier -1 for a nonidentity pure involution; the identity has multiplier 1. The normalized pair is therefore $(1, 1)$ or $(-1, 1)$, proving $C_P(\delta) = \langle \delta \rangle$. A normalizer of its order-two subgroup must centralize its sole nonidentity element, proving the normalizer equality in (D.49a).

Two elements β, β' have the same image under the orbit map exactly when $\beta^{-1}\beta' \in C_P(\delta)$. This proves its two-element fibres. For an orbit element lying in GT_0, D.35's proved equality of conjugator sets places both conjugators in GT_0; the reverse inclusion is immediate. This proves (D.49b).

Write one conjugator in D.35's convention as $\beta_{\mu,h}$. Right composition with δ gives exactly $\beta_{-\mu,h}$, so (D.49c) follows with the same h. They are distinct because $\mu \neq -\mu$ in $\widehat{\mathbb{Z}}^\times$. Moreover actual normalized conjugation identifies $C_F(\alpha)$ with $\beta(C_F(\delta)) = 1$. The description of all coboundaries in (D.35g) then makes h the unique solution of $h^{-1}\alpha(h) = f$ in F'. The two-element fibre proves uniqueness of the unordered multiplier pair.

Finally put $A = x^\mu$, $B = hy^\mu h^{-1}$ and suppose $AB = u(xy)^\mu u^{-1}$. The exact word identity

$$A^{-1}B^{-1} = A^{-1}(AB)^{-1}A$$

shows that $x^{-\mu}u$ is a third-cusp conjugator for the opposite multiplier. D.35's centralizer calculation gives all other third-cusp conjugators by right multiplication by $(xy)^a$. This proves (D.49d). The program `code/part_a/standard_involution_centralizer.py` checks (D.49f) over sixteen finite moduli and checks the displayed word identity. The full conclusion (D.49a) uses the proved global-sign theorem D.34, beyond those finite checks. □

D.50. [Proved] The finite two-subgroups of the full punctured-torus group. Let $T = \mathrm{Out}_{\mathrm{lac}}(\widehat{F}(x, y); \overline{\langle [x, y] \rangle})$ be the full once-punctured-torus loop-outer group. The possible abstract isomorphism types of its finite 2-subgroups are exactly

$$1, \quad C_2, \quad C_4, \quad C_2 \times C_2, \quad D_8, \tag{D.50a}$$

where D_8 has order eight. All five types occur. For a subgroup of type $C_2 \times C_2$ or D_8, the full multiplier takes its values in the globally constant signs $\{1, -1\} \subset \widehat{\mathbb{Z}}^\times$. In $C_2 \times C_2$ there is one positive involution and two negative involutions. In D_8 the rotation subgroup has multiplier 1 and the four reflections have multiplier -1.

For an arbitrary cyclic subgroup of order four, a generator a satisfies the already proved restrictions

$$\lambda(a)^2 = 1, \qquad \lambda(a)_2 = 1, \qquad \lambda(a^2) = 1. \tag{D.50b}$$

The abstract classification (D.50a) does not establish conjugacy of arbitrary such subgroups, or replace (D.50b) by a global multiplier determination.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. D.33 makes the actual homology homomorphism

$$A_2 : T \longrightarrow \mathrm{GL}_2(\mathbb{Z}_2) \tag{D.50c}$$

injective on every finite 2-subgroup K and proves $|K| \leq 8$. We classify its image using that same specified injection.

No matrix in $\mathrm{GL}_2(\mathbb{Q}_2)$ has order eight: an eigenvalue of order eight has minimal polynomial X^4+1 over $\mathbb{Q}_2$, which is irreducible by Eisenstein after $X \mapsto X+1$. The same argument excludes higher powers of two. Every matrix R of order four has minimal polynomial X^2+1, since this polynomial is irreducible over $\mathbb{Q}_2$ and $R^2 \neq I$. Consequently

$$R^2 = -I. \tag{D.50d}$$

An exponent-two subgroup is abelian, because $(ab)^2 = a^2 = b^2 = 1$ implies $ab = ba$. Its matrices are commuting diagonalizable involutions over $\mathbb{Q}_2$, and simultaneous diagonalization embeds it in $\{\mathrm{diag}(\epsilon_1, \epsilon_2) : \epsilon_i = \pm 1\}$. Its order is at most four. Thus an image of order eight contains an R of order four.

The matrix centralizer of R is the multiplicative group of the field $\mathbb{Q}_2[R] \cong \mathbb{Q}_2(i)$. For completeness, choose a nonzero vector v: the vectors v, Rv are independent, since an eigenvalue in $\mathbb{Q}_2$ would contradict the irreducibility of X^2+1. A commuting endomorphism is determined by its value on v and is therefore $aI + bR$. Every element of the finite 2-subgroup in this field has order at most four by the preceding matrix argument. It is consequently one of the at most four roots of $X^4 - 1$. Since $\langle R \rangle$ already has four elements,

$$C_K(R) = \langle R \rangle \tag{D.50e}$$

for a subgroup of order eight.

Choose $S \in K \setminus \langle R \rangle$. The index-two subgroup $\langle R \rangle$ is normal; by (D.50e) conjugation by S induces its nonidentity automorphism. Hence

$$SRS^{-1} = R^{-1}, \qquad S^2 \in \{I, R^2\}. \tag{D.50f}$$

The second inclusion follows because $S^2 \in \langle R \rangle$ and S fixes its own square, whereas it inverts R. The alternative $S^2 = I$ gives the dihedral presentation of order eight. We exclude the alternative $S^2 = R^2 = -I$ explicitly.

In the basis (v, Rv) used above one has

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The equation $SR = -RS$ forces

$$S = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \qquad S^2 = (a^2 + b^2)I. \tag{D.50g}$$

Thus the second alternative would require $a^2 + b^2 = -1$ in $\mathbb{Q}_2$. Here is the exact valuation obstruction. Put $m = \min(v_2(a), v_2(b))$ and write $a = 2^m u$, $b = 2^m v$, with $u, v \in \mathbb{Z}_2$ and at least one a unit. If both are units, $u^2 + v^2 \equiv 2 \pmod 8$, so this sum has valuation one. If exactly one is a unit, the sum has valuation zero. On the other hand the equation says

$$u^2 + v^2 = -2^{-2m}; \tag{D.50h}$$

the valuation of its right side is even. Both-unit values are therefore impossible. The other case forces $m = 0$; reduction modulo four then gives $1 = -1$, also impossible. Zero entries cause no exception to this argument. This excludes the quaternion alternative and leaves only D_8 at order eight. Orders four, two, and one give the remaining groups in (D.50a).

We specify actual full-group realizations. On $\widehat{F}(x, y)$ define

	x	y
r	y	x^{-1}
s	y	x.

(D.50i)

The free universal property gives continuous endomorphisms, and the displayed relations $r^4 = s^2 = 1$, $srs = r^{-1}$, verified on x, y, give their inverses and make them automorphisms. For $c = [x, y]$ direct multiplication gives

$$r(c) = x^{-1}cx, \qquad s(c) = c^{-1}. \tag{D.50j}$$

Thus their outer classes lie in T. Their homology matrices are

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{D.50k}$$

and the eight matrices $R^i S^j$ $(0 \le i < 4,\ 0 \le j < 2)$ are distinct. These maps therefore exhibit an embedded D_8 in T. The subgroups $\langle r \rangle$, $\langle r^2, s \rangle$, $\langle s \rangle$, and the identity realize C_4, $C_2 \times C_2$, C_2, and 1.

We finish with the asserted multiplier data. D.43 makes every involution's multiplier a global sign. A positive nonidentity involution has determinant 1 in (D.50c). Its order-two matrix is diagonalizable in characteristic zero, and faithfulness excludes I; its matrix is consequently $-I$. There is at most one such involution in any finite 2-subgroup. For $C_2 \times C_2$, simultaneous diagonalization identifies its full image with the four sign matrices above, giving exactly one positive and two negative involutions. For D_8, the square of a rotation is positive by (D.50d), so every other involution is negative. The reflections generate D_8, and a rotation is a product of two reflections. The homomorphic multiplier is therefore globally 1 on the rotations and globally -1 on the reflections.

Finally let a have order four in an arbitrary cyclic subgroup. Its matrix in (D.50c) has minimal polynomial $X^2 + 1$ and determinant one. Thus $\lambda(a)_2 = 1$ and $\lambda(a^2)_2 = 1$. D.43 applied to the involution a^2 gives $\lambda(a^2) = 1$ globally. The multiplier homomorphism then gives (D.50b). No assertion of conjugacy follows from this matrix classification.

The finite certificate `code/part_c/torus_two_subgroups_d50.py` verifies the full-word maps (D.50i)–(D.50j), their eight distinct matrices, and the unit square computations modulo eight used in (D.50h). The passage from that finite calculation to all of $\mathbb{Q}_2$ is the valuation argument just given. □

D.51. [Proved] The second index-three kernel and its three peripheral normalizers. Let a have order three in the full punctured-torus loop-outer group, put $F = \widehat{F}(x, y)$, $c = [x, y]$,

and let E be its centre-free extension. Besides F, there is exactly one other torsion-free normal subgroup U of index three in E. Write $V = F \cap U$. Then

$$\begin{gathered}[F:V] = [U:V] = 3, \qquad E/V \cong C_3^2, \\ V = \widehat{\pi_1(S_{1,3})}, \qquad U \text{ is projective}, \qquad U^{(3)} \cong \widehat{F}_2^{(3)}.\end{gathered} \tag{D.51a}$$

The cusp normalizer $N = N_E(\langle c \rangle)$ is contained in U. The three cusp subgroups L_0, L_1, L_2 of the displayed degree-three cover have normalizers $N_i = N_U(L_i)$ with the exact sequences

$$1 \longrightarrow L_i \longrightarrow N_i \longrightarrow U/V \longrightarrow 1. \tag{D.51b}$$

They are three distinct U-conjugacy classes, cyclically permuted by E/U. All have the given multiplier λ under the same generator label in U/V.

For a prime $q \neq 3$ at which λ_q is a nonidentity cube root of unity, the U/V-character multiplicities of $H_1(V, \mathbb{Q}_q)$, ordered as $(1, \lambda_q, \lambda_q^2)$, are exactly one of

$$(1,3,0), \qquad (0,2,2). \tag{D.51c}$$

Correspondingly $U^{(q)}$ is free pro-q of rank one or zero. These are constraints on the actual subgroup U; they do not identify it with a full free profinite group.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. The section and cusp lines in the actual abelian quotient. Use D.27 to choose an actual order-three representative α and write $E = F \rtimes_\alpha \langle s \rangle$. Let $u : E \to \mathbb{F}_3$ be its quotient character, with $u(s) = 1$. D.48 gives exactly two full section classes, $S_0 = \langle s \rangle$ and S_1, and gives $E^{\mathrm{ab},(3)} \cong C_3^2$.

Let M be the reduction of the homology matrix of α modulo three. By D.48 it is a nonidentity unipotent matrix with $(M-I)^2 = 0$. On $\mathbb{F}_3^2$ write $\omega((a,b),(a',b')) = ab' - ba'$. Choose w with $e = (M-I)w \neq 0$. Then (e,w) is a basis, $Me = e$, $Mw = w + e$, and $d = \omega(e,w) \neq 0$. The linear functional

$$\chi(r) = \omega(e,r)/d \tag{D.51d}$$

is M-invariant. It gives a character of F, which we extend to E by $\chi(s) = 0$. Thus $(u,\chi) : E \to \mathbb{F}_3^2$ is the actual abelian quotient above.

The generator of S_1 labelled by $u = 1$ has χ-value $b_1 \neq 0$. Here is the required separation, using the maps already proved in D.48 rather than an assumption about section labels. The Bockstein of χ has nonzero high periodic multiples. On a section normalizer its restriction is the Bockstein of χ restricted to the section: the fixed-group first cohomology vanishes by D.48. If χ vanished on both section classes, all those multiples would have zero full Brown restriction. This contradicts D.48f. It follows that $b_1 \neq 0$.

To locate the cusp line, use the characteristic class-two exponent-three quotient

$$Q = \mathbb{F}_3^2 \times \mathbb{F}_3, \qquad (r,z)(r',z') = (r + r', z + z' + \tfrac{1}{2}\omega(r,r')). \tag{D.51e}$$

The images of x, y are the two standard vectors with central coordinate zero, and c maps to $Z = (0,1)$. Since $\lambda_3 = 1$, the induced automorphism is

$$\theta(r,z) = (Mr, z + \ell(r)) \tag{D.51f}$$

for a linear functional ℓ. For an element $(r,z)s$ of $Q \rtimes_\theta C_3$ with quotient label one, direct multiplication gives

$$((r,z)s)^3 = Z^{d\chi(r)^2 + \ell(e)\chi(r)}. \tag{D.51g}$$

To check the formula, write $r = be+tw$, so $\chi(r) = t$. The successive vector parts are $r, r+te, r+2te$, whose sum is zero. The sum of the three central correction terms is $\ell((2I + M)r) = t\ell(e)$. The three alternating cross terms, multiplied by $1/2$, sum to dt^2 in $\mathbb{F}_3$. This proves (D.51g) for every element, including the arbitrary central coordinate z.

The image of the actual generator of S_1 has cube one. Equation (D.51g) and $b_1 \neq 0$ therefore give $\ell(e) = -db_1$. Thus its cube polynomial is

$$db(b - b_1). \tag{D.51h}$$

Choose $v \in N$ with $u(v) = 1$, as in D.48. It satisfies $v^3 = c^\kappa$ with κ_3 a unit. Its image in (D.51e) has nonzero cube exponent. The two roots of (D.51h) are 0 and b_1; the only other element of $\mathbb{F}_3$ is $-b_1$. Consequently

$$(u, \chi)(S_0) = \langle (1, 0) \rangle, \quad (u, \chi)(S_1) = \langle (1, b_1) \rangle, \quad (u, \chi)(N) = \langle (1, -b_1) \rangle. \tag{D.51i}$$

The last equality uses that N is generated topologically by v and $\langle c \rangle$, and both characters kill c.

The full kernels and peripheral labels. Define the continuous characters and subgroups

$$\psi = \chi + b_1 u, \qquad U = \ker \psi, \qquad V = \ker u \cap \ker \chi. \tag{D.51j}$$

These are actual subgroups of E. The character ψ is nonzero on both section classes in (D.51i), and it kills N. Every nonidentity finite subgroup of E is one of the order-three section groups, up to conjugacy. Hence U is torsion-free. The four lines in $E^{\mathrm{ab},(3)} = \mathbb{F}_3^2$ give all normal index-three subgroups of E. Two contain respectively S_0 and S_1. The other two are exactly $\ker u = F$ and $\ker \psi = U$, proving the uniqueness assertion. Also $E = U \rtimes S_0$, since $\psi|_{S_0}$ is an isomorphism onto $\mathbb{F}_3$.

The restriction of χ to F is nonzero and kills its commutator cusp. Choose a word f in the dense discrete group $F(x, y)$ with $\chi(f) = 1$. The degree-three covering of the punctured torus corresponding to $\ker(\chi|_{F(x,y)})$ has three boundary lifts of degree one. Its Euler characteristic is -3; thus its type is $(1, 3)$, and its fundamental group has rank four. The induced profinite subgroup is exactly V, with cusp representatives

$$L_i = f^i \langle c \rangle f^{-i}, \qquad N_i = f^i N f^{-i} \quad (0 \leq i < 3). \tag{D.51k}$$

The full peripheral normalizer assertion A.7 gives $N_E(L_i) = N_i$. Since $N \subset U$ and U is normal in E, all N_i lie in U. Their kernels under $U \to U/V$ are exactly L_i, because $N_i \cap F = L_i$; their images are all of U/V, because $u(v) = 1$. This proves (D.51b), with the common generator label supplied by u.

For the class assertions, the actual quotient (u, χ) identifies E/V with $\mathbb{F}_3^2$, and the image of every N_i is the same line U/V. The three V-conjugacy classes of the cusp lifts are therefore the three cosets of this line in E/V. The subgroup U/V fixes each class, so they remain distinct under U. The quotient E/U acts regularly on these three cosets, proving its asserted cyclic permutation. This also proves that conjugation by a generator of U/V fixes each cusp class with multiplier λ.

The open subgroup V of U is free profinite of rank four and U is torsion-free. Serre's finite-index cohomological-dimension theorem, used in D.27, makes U projective. Its maximal pro-three quotient is projective as a pro-three group, by the finite embedding problem characterization, and is therefore free pro-three. The pro-three topology on V is exactly the one induced from U: taking the intersection of the three conjugates of a normal three-power-index subgroup of V produces one normal in U, still of three-power index. Thus $V^{(3)}$ embeds as an open subgroup of index three in $U^{(3)}$. Schreier's formula now reads

$$4 = 1 + 3(\operatorname{rank} U^{(3)} - 1),$$

and gives rank two. This proves (D.51a).

The exact character restriction at another prime. The three oriented boundary classes of V have their sole relation given by their sum. They form a two-dimensional submodule of $H_1(V, \mathbb{Q}_q)$ on which U/V acts by λ_q. Filling the three boundaries gives the compact degree-three cover of the original torus. Its map on homology is the inclusion of a rank-two sublattice of index three, hence an isomorphism over $\mathbb{Q}_q$, and over $\mathbb{Z}_q$ when $q \neq 3$. Conjugation by v projects to the original outer action a on the base; its inner factor acts trivially on that base homology. Therefore, if A_q denotes the original matrix, the exact boundary sequence and averaging over C_3 give

$$H_1(V, \mathbb{Q}_q) \cong \lambda_q^{\oplus 2} \oplus (\mathbb{Q}_q^2, A_q) \quad \text{as } U/V\text{-modules.} \tag{D.51l}$$

This comparison is induced by the stated boundary inclusion and covering homology map.

At the specified prime, $A_q^3 = I$ and $\det A_q = \lambda_q$. The three cube roots $1, \lambda_q, \lambda_q^2$ are distinct in $\mathbb{Q}_q$. Diagonalization gives exactly the two eigenvalue multisets

$$\{1, \lambda_q\}, \qquad \{\lambda_q^2, \lambda_q^2\}. \tag{D.51m}$$

Adding the two boundary characters in (D.51l) proves (D.51c). For $q \neq 3$, the same decomposition holds over $\mathbb{Z}_q$: the averaging idempotents have unit denominator three, and the cube-root differences are units. Hochschild–Serre for $V \triangleleft U$ then identifies $H^1(U, \mathbb{F}_q)$ with the invariant subspace of $H^1(V, \mathbb{F}_q)$, of dimension one or zero respectively. The projectivity of U makes its maximal pro-q quotient free pro-q, with this number of generators. This proves the last assertion.

The program `code/part_c/torus_order3_sibling_d51.py` checks (D.51g) for every central-linear automorphism above every nonidentity unipotent matrix over $\mathbb{F}_3$, the resulting section and cusp lines, and the finite eigenvalue patterns in (D.51m). These finite calculations validate the displayed formulas. The full subgroup, topology, and projectivity conclusions are proved by the maps above. □

Exact remaining recognition problem. The construction supplies a specified torsion-free projective group U with an open full free subgroup of rank four and three specified normalizers. It supplies no identification of U with $\widehat{F}_2$. At the primes in (D.51c), its maximal pro-q quotient has rank zero or one, whereas the maximal pro-three quotient has rank two. The existence or exclusion of such a full U with these three normalizers remains the step needed for this approach to determine every torus order-three multiplier.

D.52. [Proved] An abstract allowed list for every finite full torus subgroup. Let T be the full once-punctured-torus loop-outer group of D.50. Every finite subgroup of T has one of the following twelve abstract types:

$$\begin{gathered} 1,\ C_2,\ C_3,\ C_4,\ C_2 \times C_2,\ C_6,\ S_3,\ D_8, \\ C_{12},\ D_{12},\ \mathrm{Dic}_{12},\ D_{24}. \end{gathered} \tag{D.52a}$$

Here D_{2n} has order $2n$, and $\mathrm{Dic}_{12} = C_3 \rtimes C_4$, with a generator of C_4 inverting C_3. The nine types obtained by omitting $C_{12}, \mathrm{Dic}_{12}, D_{24}$ all occur. Occurrence of those three remaining types is not determined by this allowed list. The subsequent full covering arguments in D.54–D.55 exclude all three.

Every subgroup of type S_3, D_{12} or D_{24} has the global usual reflection multiplier: rotations have multiplier 1 and reflections have multiplier -1. In a subgroup of type Dic_{12}, the subgroup of order three has multiplier 1, and every generator b of a cyclic subgroup of order four satisfies

$$\lambda(b)^2 = 1, \qquad \lambda(b)_2 = 1, \qquad \lambda(b)_3 = -1. \tag{D.52b}$$

For a subgroup of type C_{12}, its order-four generator instead has $\lambda(b)_2 = \lambda(b)_3 = 1$, and still has $\lambda(b)^2 = 1$. These restrictions retain the unresolved cyclic order-three and order-four multiplier questions.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. The two primary representations and their actual kernels. For a finite subgroup $G \leq T$, D.33 gives

$$|G| \mid 24, \qquad A_p : G \longrightarrow \mathrm{GL}_2(\mathbb{Z}_p), \qquad \det A_p(g) = \lambda(g)_p. \tag{D.52c}$$

It makes A_p injective on every finite p-subgroup. Put $K = \ker(A_2|_G)$. An element of order two in K would contradict that injection. Cauchy's theorem and (D.52c) therefore give

$$K = 1 \quad \text{or} \quad K \cong C_3. \tag{D.52d}$$

We classify the actual image $H = A_2(G)$ and then the extension by this specified kernel. We do not identify G with a representation at one prime when the kernel is nontrivial.

Finite images over $\mathbb{Q}_2$. The two-subgroups of H have the types in D.50. Indeed a Sylow two-subgroup of the preimage of any such subgroup maps isomorphically onto it, since the kernel has odd order. The matrix arguments in D.50 also explicitly exclude the quaternion group of order eight.

Suppose 3 divides $|H|$. Its number of Sylow three-subgroups is either one or four. We exclude four. Conjugation gives a transitive homomorphism $H \to S_4$. Its kernel has order at most six. If that kernel had order divisible by three, its unique Sylow three-subgroup would be characteristic in the kernel and normal in H, contrary to the assumed four Sylow subgroups. Thus the transitive image has order divisible by both four and three, and is A_4 or S_4. The kernel consequently has order one or two.

There is no faithful two-dimensional representation of A_4 over $\mathbb{Q}_2$. Its normal Klein four subgroup would be simultaneously diagonalizable and, by faithfulness, would be the full group of diagonal sign matrices. One of its three involutions would therefore be $-I$. Conjugation fixes this scalar, whereas the order-three elements of A_4 cyclically permute all three involutions. This is a contradiction. It excludes the cases with trivial kernel, whose images are A_4 or S_4. In the remaining case $|H| = 24$ and the image is A_4 with a kernel of order two. The preimage P of its Klein four subgroup is a normal subgroup of order eight, hence is D_8 by D.50. But $|\operatorname{Aut}(D_8)| = 8$: a generator of its rotation subgroup has two possible images, and a chosen reflection has four. Every element of order three therefore centralizes P. Multiplication by a Sylow three-subgroup gives $H = P \times C_3$, which has only one Sylow three-subgroup. This contradiction proves that H has a normal subgroup P_3 of order three.

Let R generate P_3. Its minimal polynomial is $X^2 + X + 1$, irreducible over $\mathbb{Q}_2$ by reduction modulo two. The basis (v, Rv) for any nonzero vector v identifies the matrix centralizer with

$$L_2^{\times}, \qquad L_2 = \mathbb{Q}_2(\zeta_3). \tag{D.52e}$$

Under this identification, a commuting endomorphism is multiplication by $a + b\zeta_3$. The full normalizer consists exactly of these multiplication maps and their compositions with $\sigma(\zeta_3) = \zeta_3^{-1}$: conjugation must send R to R or R^{-1}, and division by σ in the second case gives a commuting map.

The field L_2 contains no element of order four. To check this directly, the equation $(a+b\zeta_3)^2 = -1$ has coefficient equations

$$a^2 - b^2 = -1, \qquad b(2a - b) = 0. \tag{D.52f}$$

They require either $a^2 = -1$ or $a^2 = 1/3$ in $\mathbb{Q}_2$. Both are impossible: these are units congruent respectively to seven and three modulo eight, whereas an odd square is congruent to one. Since $|H| \mid 24$, it follows that $C = H \cap L_2^\times$ is C_3 or C_6. Here its two-part is contained in $\{1, -1\}$, and its three-part is P_3, so no general classification of local roots of unity is required.

If $H = C$, these are the two possible groups. Otherwise choose $B = a\sigma \in H \setminus C$. Then

$$B^2 = N_{L_2/\mathbb{Q}_2}(a)\, I \in C \cap \mathbb{Q}_2^\times = \{I, -I\} \cap C. \tag{D.52g}$$

The last scalar restriction follows because a finite-order scalar of order dividing twenty-four in $\mathbb{Q}_2$ is a sign: there are no primitive cube or fourth roots in $\mathbb{Q}_2$. For $C = C_3$, equation (D.52g) gives S_3. For $C = C_6$, its two values give respectively D_{12} and Dic_{12}. Thus the images with order divisible by three have the complete allowed list

$$C_3,\ C_6,\ S_3,\ D_{12}, \mathrm{Dic}_{12}\,. \tag{D.52h}$$

The three-adic normalizer and its determinant. We will need a statement about any normal subgroup $P_3 \cong C_3$ of the actual G. The map A_3 is injective on P_3. Its generator has minimal polynomial $X^2 + X + 1$ over $\mathbb{Q}_3$, irreducible since its discriminant -3 has odd valuation. The same centralizer and normalizer calculation identifies

$$C_{\mathrm{GL}_2(\mathbb{Q}_3)}(A_3(P_3)) = L_3^\times, \qquad N_{\mathrm{GL}_2(\mathbb{Q}_3)}(A_3(P_3)) = L_3^\times \rtimes \langle\sigma\rangle, \quad L_3 = \mathbb{Q}_3(\zeta_3). \tag{D.52i}$$

This field also has no element of order four: (D.52f) would require $a^2 = -1$, impossible modulo three, or $a^2 = 1/3$, impossible by valuation. Thus every finite-order element in its centralizer with order dividing twenty-four lies in μ_6 and has norm one.

There is also no element of norm -1 in L_3. Here is an explicit calculation, including possible nonintegral coordinates. The norm of $a + b\zeta_3$ is $a^2 - ab + b^2$. Put $m = \min(v_3(a), v_3(b))$ and write $a = 3^m u, b = 3^m v$, where $u, v \in \mathbb{Z}_3$ and at least one is a unit. When $u + v \not\equiv 0 \pmod 3$, the norm of $u + v\zeta_3$ has valuation zero and residue $(u+v)^2 = 1$ modulo three. When $u + v \equiv 0 \pmod 3$, both are units; writing $u = -v + 3k$ gives

$$u^2 - uv + v^2 = 3(v^2 - 3vk + 3k^2), \tag{D.52j}$$

of valuation one. The norm -1 would have valuation zero. The second case cannot give that valuation, since $2m + 1 \neq 0$. The first forces $m = 0$ and residue one, whereas -1 has residue two. This proves the asserted norm obstruction.

An element of the finite image $A_3(G)$ outside the centralizer has the form $B = a\sigma$, and $B^2 = N(a)I$ is a finite-order scalar, hence a sign. The norm obstruction forces $B^2 = I$. In the basis $(1, \zeta_3)$, the determinant of σ is -1, so $\det B = -N(a) = -1$. Together with the centralizer calculation, this proves the exact character identity

$$\lambda(g)_3 = \begin{cases} 1, & g \text{ centralizes } P_3, \\ -1, & g \text{ inverts } P_3. \end{cases} \tag{D.52k}$$

The argument uses faithfulness only on P_3, not on all of G.

Reconstructing the actual finite subgroup. If $K = 1$ in (D.52d), the injection into H gives precisely D.50 and (D.52h). Now let $K = P_3 \cong C_3$. A Sylow two-subgroup Q of G has order $|G|/3$ and maps isomorphically onto H. The actual multiplication map

$$P_3 \rtimes Q \longrightarrow G, \qquad (a, q) \longmapsto aq, \tag{D.52l}$$

is an isomorphism, with action given by conjugation. D.50 lists the five possibilities for Q. For $Q = 1, C_2, C_4$, their homomorphisms to $\mathrm{Aut}(C_3) = C_2$ give C_3; C_6 or S_3; and C_{12} or Dic_{12}, respectively.

For $Q = C_2 \times C_2$ or D_8, D.50 already determines its global multiplier. Equation (D.52k) therefore makes its action on P_3 exactly the reflection character of D.50. Write a for a generator of P_3. For $Q = C_2 \times C_2$, write z for its positive involution and s for a negative one. The element az has order six, is inverted by s, and generates a, z through $(az)^4 = a$ and $(az)^3 = z$. This gives an explicit D_{12} presentation. For $Q = D_8 = \langle r, s\rangle$, where r is its order-four rotation, the element ar has order twelve, is inverted by s, and satisfies

$$(ar)^4 = a, \qquad (ar)^9 = r. \tag{D.52m}$$

It gives the explicit D_{24} presentation. These actual maps finish the allowed list.

Multipliers and realized cases. In every group just obtained that inverts its normal C_3, the multiplier on that C_3 is one: a conjugation-invariant homomorphism into the abelian group $\widehat{\mathbb{Z}}^\times$ takes an inverted element to its own inverse, and a value of order dividing three with square one is one. In S_3, D_{12}, D_{24} each reflection inverts C_3 and hence has three-adic multiplier -1 by (D.52k). D.43 makes its full multiplier -1. Every rotation is a product of two reflections, proving the stated global reflection character. For Dic_{12}, an order-four generator inverts C_3, so (D.52k) gives its negative three-coordinate. D.50 gives its positive two-coordinate and global square one. For C_{12} the same generator centralizes C_3, giving instead the positive three-coordinate. This proves all the claimed multiplier restrictions.

D.50 realizes its five types by explicit full profinite free-group automorphisms. The other four realized types have the following simultaneous exact surface model. On $\mathbb{R}^2/\mathbb{Z}^2$ the integral matrices

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{D.52n}$$

fix the origin and satisfy $R^6 = S^2 = I$, $SRS = R^{-1}$. Their twelve matrices $R^i S^j$ are distinct. Removing the origin therefore gives an actual D_{12} action on a once-punctured torus. Passing to its fundamental group, then to its full profinite completion and outer automorphisms, gives a homomorphism to T. A small peripheral circle is carried to an ellipse of local degree $\det R = 1$ or $\det S = -1$, so this homomorphism preserves the stated full cusp with the indicated multiplier. The induced homology matrices are exactly R, S in $\mathrm{GL}_2(\widehat{\mathbb{Z}})$; their distinctness proves that the homomorphism is injective. The subgroups $\langle R^2\rangle, \langle R\rangle, \langle R^2, S\rangle$, and $\langle R, S\rangle$ realize C_3, C_6, S_3, D_{12}.

The program `code/part_c/torus_finite_types_d52.py` checks the matrix relations, the local norm and square residues, the eight automorphisms of D_8, and the explicit finite semidirect-product identifications in (D.52l)–(D.52m). The prime-wise kernel and valuation arguments above prove the full allowed list. No assertion of occurrence or nonoccurrence of its three remaining types is inferred from those finite checks. □

D.53. [Proved] The unique labelled order-four section class. Let a have order four in the full punctured-torus loop-outer group, and let

$$1 \longrightarrow F \longrightarrow E \xrightarrow{\pi} C_4 = \langle a\rangle \longrightarrow 1, \qquad F = \widehat{F}_2,$$

be its centre-free extension. Its homomorphic sections form exactly one F-conjugacy class, with the generator label a retained. For an actual lift $b \in E$ with $b^4 = 1$ and $\pi(b) = a$, put $B = \langle b\rangle$. Then

$$N_E(B) = C_F(b) \times B, \qquad H^1(C_F(b), \mathbb{F}_2) = 0. \tag{D.53a}$$

Conjugation by a lift of a fixes exactly one of the three full section classes for a^2 and interchanges the other two. Consequently E has two conjugacy classes of subgroups of order two; exactly one of them is contained in its class of cyclic subgroups of order four.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. D.46 supplies the actual lift b with its prescribed label. Put $T = \langle b^2 \rangle$ and

$$E_2 = \pi^{-1}(\langle a^2 \rangle) = F \rtimes T. \tag{D.53b}$$

The action of b^2 on $H_1(F, \mathbb{Z}_2)$ is $-I$ by D.46. Thus

$$E_2^{\mathrm{ab},(2)} \cong C_2^3. \tag{D.53c}$$

Use coordinates $(v, u) \in \mathbb{F}_2^2 \times \mathbb{F}_2$, where u is the quotient character and the first two coordinates are the extended characters χ_1, χ_2 of D.46. The three full section classes of E_2 give three distinct points of the affine plane $u = 1$. Indeed, the degree-zero localized algebra in D.46g–i has dimension three, is generated by $\chi_1/u, \chi_2/u$, and restriction to the three section normalizers identifies it with $\mathbb{F}_2^3$. Each restriction is evaluation at $(\chi_1(t), \chi_2(t))$ for the nonidentity element t of that section. If two such points agreed, the two restriction coordinates would agree on both generators and hence on the entire algebra, contradicting the stated product isomorphism.

Let $R = A_2(a) \in \mathrm{GL}_2(\mathbb{Z}_2)$. D.50 gives $R^2 = -I$. Moreover $R \not\equiv I \pmod 2$. For otherwise $R = I + 2D$ with an integral matrix D, so $R^2 = -I$ would give $I + 2D + 2D^2 = 0$, an impossibility modulo two. Conjugation by b on (D.53c) fixes u and induces the actual affine map

$$v \longmapsto \overline{R}v + d \quad \text{on } u = 1 \tag{D.53d}$$

for a vector $d \in \mathbb{F}_2^2$. The linear part follows from the restriction of conjugation to the specified subgroup F; the translation is the image of a chosen point of $u = 1$. Conjugation by an element of E_2 acts trivially on the three F-conjugacy classes, as shown in D.46. Thus (D.53d) permutes their three points with order dividing two. Three distinct points of $\mathbb{F}_2^2$ affinely span it. Fixing all three would therefore make this affine map the identity, contrary to $\overline{R} \neq I$. Its permutation is consequently a transposition, with exactly one fixed class. The class of $T = \langle b^2 \rangle$ is that fixed class.

Put $C = C_F(T)$ and $N = N_E(T)$. Since b normalizes T, the image of N under π is all of C_4. Its kernel is exactly C, because an element of F normalizing T centralizes its nonidentity element. The chosen section gives the actual semidirect-product identification

$$N = C \rtimes \langle b \rangle, \qquad T \subseteq Z(N), \qquad J := N/T = C \rtimes \langle \bar{b} \rangle, \quad |\bar{b}| = 2. \tag{D.53e}$$

In the last expression the embedding of C is injective since $C \cap T = 1$. Conjugation by b on C has square the identity because b^2 centralizes C.

As a closed subgroup of the full free profinite group F, the group C is projective and has $\mathrm{cd}_2(C) \leq 1$. D.46 gives $H^1(C, \mathbb{F}_2) = 0$. All its positive-degree mod-two cohomology therefore vanishes. Hochschild–Serre for the last extension in (D.53e) identifies inflation as

$$H^*(C_2, \mathbb{F}_2) \xrightarrow{\cong} H^*(J, \mathbb{F}_2). \tag{D.53f}$$

This uses the displayed full group C, not an identification of C with its maximal pro-two quotient.

We spell out the full Brown hypotheses for this new normalizer quotient. The subgroup C is open in J and has finite 2-cohomological dimension. It is torsion-free. Every finite subgroup of J consequently injects into $J/C = C_2$; in particular J has no subgroup C_2^2. Theorem 2 of [37, §1.1] therefore supplies, in degrees $n > 1$, the dense restriction embedding into the product over the conjugacy classes of order-two subgroups of J. By (D.53f) its source has dimension one. Each product factor is nonzero: for a section S of $J \to C_2$, its normalizer is

$$C_C(S) \times S, \tag{D.53g}$$

and the restriction from the S factor already supplies one dimension. Thus the product has finitely many factors and the finite dense image is the entire product. Since $\mathrm{cd}_2(C_C(S)) \le 1$, the dimensions give the exact equality

$$1 = \sum_{[S]} (1 + \dim H^1(C_C(S), \mathbb{F}_2)). \tag{D.53h}$$

There is a section $\langle \bar{b} \rangle$. It follows that it is the unique conjugacy class and that $H^1(C_C(\bar{b}), \mathbb{F}_2) = 0$. These are also C-conjugacy classes: an element of J can be multiplied by the appropriate element of a given section to leave a conjugator in C.

Now let B' be any subgroup of order four in E. Its intersection with F is trivial, so it is cyclic and $\pi|_{B'}$ is an isomorphism onto C_4. Choose its unique generator b' with $\pi(b') = a$. Its square generates an order-two section T' of E_2. Conjugation by b' has the same action on the section classes of E_2 as conjugation by b, because $b'b^{-1} \in F$. The class of T' is fixed, and is therefore the unique class of T. An F-conjugation puts T' equal to T, and then $B' \subseteq N_E(T) = N$. Its image in $J = N/T$ has order two. By (D.53h), a further conjugation by an element of C puts that image equal to $\langle \bar{b} \rangle$.

The exact inverse image of this subgroup under $N \to N/T$ has order four, because its kernel is the specified subgroup T of order two. It contains $\langle b \rangle$ and hence equals $\langle b \rangle$. Thus the conjugated B' equals B. All conjugations used lie in F and preserve π, so its labelled generator b' is sent to b, proving uniqueness of the homomorphic section with labels.

The fixed group $C_C(\bar{b})$ in (D.53h) is exactly $C_F(b)$. For an element of C fixed by $\bar{b}$, its commutator with b lies in T and also in C, and $T \cap C = 1$. The converse is immediate, and an element commuting with b already centralizes T. This proves the cohomology statement in (D.53a). Finally, an element normalizing B must fix its generator b: conjugation preserves its label in the abelian quotient C_4, and $\pi|_B$ is injective. Multiplication by the matching power of b then leaves an element of $C_F(b)$, proving the direct-product normalizer in (D.53a).

The two orbits of the transposition in (D.53d) are exactly the two E-conjugacy classes of order-two subgroups. Only its fixed class can be the square of a subgroup of order four, and B realizes that containment. This proves the final assertion.

The finite program `code/part_c/torus_order4_sections_d53.py` checks the mod-four matrix obstruction and every invariant three-point affine configuration above a nonidentity order-two matrix in $\mathrm{GL}_2(\mathbb{F}_2)$. The normalizer and restriction arguments prove the full section result. No conjugacy of the outer action a to a prescribed geometric action is asserted here. □

D.54. [Proved] No fixed character for a cubic torus action. Let a have order three in the full once-punctured-torus loop-outer group. For every prime $q \neq 3$, it fixes no nonzero continuous character

$$\chi : F \longrightarrow C_q, \qquad F = \widehat{F}(x, y). \tag{D.54a}$$

Consequently the reduction of $A_q(a)$ modulo q has no eigenvalue one. For every prime p, the endomorphism $A_p(a) - I$ of $\mathbb{Z}_p^2$ is injective. The full fixed group of every actual order-three representative is therefore contained in the full commutator subgroup F'.

In particular, A_2 is injective on every finite subgroup of the full torus group. The allowed list of D.52 loses C_{12} and D_{24}. At a nontrivial cubic multiplier coordinate $q \neq 3$, the remaining homology matrix is exactly

$$A_q(a) = \lambda(a)_q^2 I. \tag{D.54b}$$

This statement does not exclude the remaining scalar cubic multiplier case.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. Use D.27 to choose an actual representative α of order three and put $E = F \rtimes_\alpha \langle s \rangle$, where s has quotient label a. Write $c = [x, y]$ and $L = \langle c \rangle$. D.48 supplies a labelled cusp-normalizer element v with

$$v \in N_E(L), \qquad v^3 = c^\kappa, \qquad vcv^{-1} = c^\lambda, \qquad \kappa_3 \in \mathbb{Z}_3^\times. \tag{D.54c}$$

It also gives $\lambda_3 = 1$ and

$$I + A_3(a) + A_3(a)^2 = 0 \quad \text{on } \mathbb{Z}_3^2. \tag{D.54d}$$

Suppose χ in (D.54a) were nonzero and invariant. Its extension

$$\widetilde{\chi} : E \longrightarrow C_q, \qquad \widetilde{\chi}(hs^j) = \chi(h) \tag{D.54e}$$

is an actual continuous homomorphism. Put

$$N = \ker \chi, \qquad E_N = \ker \widetilde{\chi} = N \rtimes \langle s \rangle. \tag{D.54f}$$

Both have their indicated index q in F and E. Since $\chi(c) = 0$, (D.54c) gives $3\widetilde{\chi}(v) = 0$. The prime q differs from three, so $v \in E_N$. Thus $v = ns$ for a specified $n \in N$.

The restriction of χ to the dense discrete free group is onto C_q. Its covering surface is the degree-q cover of the punctured torus with boundary monodromy zero. It has q boundary lifts of degree one, Euler characteristic $-q$, and hence type $(1, q)$. Its full fundamental-group completion is exactly N, by the finite-index completion map. Choose a discrete word f with $\chi(f) = 1$. Its boundary representatives are

$$c_i = f^i c f^{-i} \quad (0 \le i < q). \tag{D.54g}$$

Their homology classes have exactly the single relation $\sum_i [c_i] = 0$. In particular each $[c_i]$ is nonzero in $H_1(N, \mathbb{F}_3)$, because $q \ge 2$. Equivalently, omitting any other boundary from the usual genus-one presentation puts c_i into a free basis.

The action on every boundary class has multiplier λ and fixes its label. This follows from the actual $v \in E_N$: the element $n_i = (vf^iv^{-1})f^{-i}$ belongs to N, and $vc_iv^{-1} = n_i c_i^\lambda n_i^{-1}$. Conjugation by $s = n^{-1}v$ differs from this by an inner automorphism of N. Thus α fixes all q boundary homology classes over $\mathbb{Z}_3$.

We give the required splitting of the entire homology module, including its comparison with the base torus. Let $D = F/N = C_q$ be the deck group and set

$$M = H_1(N, \mathbb{Z}_3), \qquad e = \frac{1}{q} \sum_{d \in D} d. \tag{D.54h}$$

The action of α on M commutes with the deck action: it induces the identity on D, and two lifts of a deck transformation differ by an inner automorphism of N on this homology. Consequently e commutes with α.

Filling the q boundaries gives the compact degree-q torus cover. Explicitly its lattice is the kernel $\Lambda \le \mathbb{Z}^2$ of the character $\mathbb{Z}^2 \to C_q$ induced by χ, and its map to the base torus is $\mathbb{R}^2/\Lambda \to \mathbb{R}^2/\mathbb{Z}^2$. The deck maps are translations and act trivially on the filled homology. The boundary submodule

$$B_\partial = \mathbb{Z}_3[D]/\mathbb{Z}_3\Big(\sum_{d \in D} d\Big) \tag{D.54i}$$

has no deck-invariant vector, because q is a unit in $\mathbb{Z}_3$. The exact boundary sequence and (D.54h) therefore give

$$(1 - e)M = B_\partial, \qquad eM \cong H_1(\mathbb{R}^2/\Lambda, \mathbb{Z}_3) \cong H_1(\mathbb{R}^2/\mathbb{Z}^2, \mathbb{Z}_3). \tag{D.54j}$$

The last isomorphism is induced by the displayed covering: its lattice inclusion has index q, a 3-adic unit. These maps intertwine the induced automorphisms, so $\alpha|_{eM}$ is exactly $A_3(a)$. On $(1-e)M$ it is the identity by (D.54g). Hence

$$I + \alpha + \alpha^2 = 0 \quad \text{on } H_1(N, \mathbb{F}_3) : \tag{D.54k}$$

on the first summand this is (D.54d), and on the boundary summand the norm is multiplication by three.

The cube of the actual element ns now gives the contradiction:

$$\begin{aligned} [v^3] &= [n\,\alpha(n)\,\alpha^2(n)] \\ &= (I + \alpha + \alpha^2)[n] = 0 \quad \text{in } H_1(N, \mathbb{F}_3), \\ [c^\kappa] &= (\kappa_3 \bmod 3)[c] \neq 0. \end{aligned} \tag{D.54l}$$

Equations (D.54c) and (D.54l) concern the same two elements of N. Thus no invariant character (D.54a) exists.

For $q \neq 3$, a matrix and its transpose have the same rank, so a one-eigenvector would supply a nonzero invariant character. It has just been excluded. In particular $A_q(a) - I$ is invertible over $\mathbb{Z}_q$. At $q = 3$ its determinant is three, by D.48, so it is still injective. If h is fixed by an actual representative, its image in every $\mathbb{Z}_p^2$ is therefore zero. The common kernel of the actual abelianization map $F \to \widehat{\mathbb{Z}}^2 = \prod_p \mathbb{Z}_p^2$ is F', proving the fixed-group assertion.

At a nontrivial cubic multiplier coordinate the two eigenvalue multisets were exactly $\{1, \lambda_q\}$ and $\{\lambda_q^2, \lambda_q^2\}$ in D.51. The first is now excluded. A finite-order matrix in characteristic zero is diagonalizable, so the second gives the actual scalar matrix (D.54b).

Finally let G be any finite torus subgroup. D.33 gives $|G| \mid 24$. A nontrivial kernel of $A_2|_G$ would contain a prime-order element. Its prime can only be two or three. D.33 excludes an element of order two, and (D.54a), with $q = 2$, excludes an element of order three acting trivially on two-adic homology. Thus $A_2|_G$ is injective. The faithful-image part (D.52h), together with D.50, now excludes C_{12} and D_{24} from that allowed list.

The program `code/part_c/torus_cover_norm_d54_d55.py` builds the cyclic covering homology, its individual boundary vectors and the deck-average splitting for several primes $q \neq 3$, and verifies the forced norm contradiction over $\mathbb{F}_3$. The displayed maps prove it for every such prime. □

D.55. [Proved] Full primary faithfulness and the nine finite torus types. Every positive involution b in the full once-punctured-torus loop-outer group has

$$A_p(b) = -I \quad \text{for every prime } p. \tag{D.55a}$$

Consequently, for each prime p, the actual homology map A_p is injective on every finite subgroup of that group. The possible finite abstract subgroup types are exactly the following nine, all realized:

$$1,\ C_2,\ C_3,\ C_4,\ C_2 \times C_2,\ C_6,\ S_3,\ D_8,\ D_{12}. \tag{D.55b}$$

The groups D_8, D_{12} have orders eight and twelve respectively.

For an arbitrary order-four element a,

$$A_p(a)^2 = -I \quad \text{for every } p. \tag{D.55c}$$

At a negative multiplier coordinate q it is necessarily a scalar matrix $i_q I$ or $-i_q I$, where $i_q^2 = -1$ in $\mathbb{Q}_q$; thus $q \equiv 1 \pmod 4$. Together with (D.54b), this retains the precise scalar cubic and

quartic multiplier cases. The abstract classification (D.55b) does not settle those scalar cases or the conjugacy of arbitrary finite subgroups.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. The positive involution's full cusp normalizer. Let b be any positive involution. D.43 gives its global multiplier 1. D.27 gives an actual representative β of order two, and put $E_b = F \rtimes_\beta \langle s \rangle$. The first half of D.46 applies to every such positive involution; it does not require b to have first been exhibited as an order-four square. It gives

$$A_2(b) = -I, \qquad H^1(C_F(S), \mathbb{F}_2) = 0 \quad \text{for every order-two section } S. \tag{D.55d}$$

We recall the prime-independent part of the closed-subgroup argument in D.48. If $C \le F$ is closed and $H^1(C, \mathbb{F}_p) = 0$, it cannot contain the full subgroup $c^{e_p \mathbb{Z}_p}$ of the torus cusp, where $c = [x, y]$. If the p-part of $[F : C]$ were finite, choose an open neighborhood $W \ge C$ with maximal such part. All subsequent indices are prime to p, so transfer makes the restriction maps on first cohomology injective. Their direct limit is $H^1(C, \mathbb{F}_p)$, contradicting the nonzero first cohomology of the open full free group W. If the p-part is infinite, choose an open $W \ge C$ of index divisible by p. Its selected cusp lift has degree prime to p, because it contains the full cusp $\mathbb{Z}_p$. There must be at least two boundary lifts: a unique lift would have degree $[F : W]$. A boundary character of this finite cover then restricts nontrivially to that cusp $\mathbb{Z}_p$ and hence to C, again a contradiction. These are the actual open-subgroup restriction and finite-cover maps used in D.48, with an arbitrary prime p in place of three.

Apply this with $p = 2$ to (D.55d). An involution in $N_{E_b}(\langle c \rangle)$ would be a section and would centralize the entire cusp, since the global multiplier is 1. Its full fixed group would therefore contain the cusp $\mathbb{Z}_2$, which has just been excluded. Thus this cusp normalizer is torsion-free. Choose its labelled element v:

$$vcv^{-1} = c, \qquad v^2 = c^\kappa. \tag{D.55e}$$

If κ_2 were even, the equation $2z = -\kappa$ would be soluble in $\widehat{\mathbb{Z}}$: at two by that divisibility, and at every odd prime because two is a unit. Then $(vc^z)^2 = 1$, and its nontrivial quotient label would make it an involution in the cusp normalizer. Consequently

$$\kappa_2 \in \mathbb{Z}_2^\times. \tag{D.55f}$$

The use of the global multiplier 1 in (D.55e) is essential to this exact norm equation.

The primitive boundary rules out every odd fixed character. Let q be odd and suppose a nonzero β-invariant character $\chi : F \to C_q$ existed. Extend it by $\widetilde{\chi}(s) = 0$ to E_b, and put $N = \ker \chi$ and $E_N = \ker \widetilde{\chi} = N \rtimes \langle s \rangle$. Equation (D.55e) gives $2\widetilde{\chi}(v) = 0$, so $v \in E_N$. Write $v = ns$ with $n \in N$.

The exact cover and its maps are those of (D.54g)–(D.54j), now over $\mathbb{Z}_2$. It has type $(1, q)$. The deck-average idempotent $e = q^{-1} \sum_{C_q} d$ commutes with β, identifies $eH_1(N, \mathbb{Z}_2)$ with the base torus homology by the index-q lattice inclusion, and has boundary augmentation as its complement. On the first summand $\beta = -I$ by (D.55d). On the second it is the identity: the actual $v \in E_N$ fixes each cusp label and centralizes its cusp generator up to N-conjugacy, exactly as in (D.54g). Hence

$$I + \beta = 0 \quad \text{on } H_1(N, \mathbb{F}_2). \tag{D.55g}$$

The same two elements in (D.55e) then have incompatible homology:

$$[v^2] = [n\,\beta(n)] = 0, \qquad [c^\kappa] = (\kappa_2 \bmod 2)[c] \neq 0. \tag{D.55h}$$

The second class is nonzero because c is a primitive boundary in the cover with $q \ge 3$ cusps. This excludes the invariant character.

At every odd prime q the matrix $A_q(b)$ squares to I and has determinant 1. Its characteristic-zero eigenvalues are both 1 or both -1. The first possibility makes it I and supplies a nonzero invariant character modulo q, contrary to (D.55h). The second makes it $-I$. Together with (D.55d), this proves (D.55a).

Faithfulness at every prime and the exact abstract list. Fix a prime p and a finite subgroup G of the full torus group. By D.33 its order divides twenty-four. A nontrivial kernel of $A_p|_G$ would contain an element of order two or three. A positive involution has image $-I$ by (D.55a). A negative involution has determinant -1 at every prime by D.43, so also cannot lie in the kernel. An order-three element has nonidentity image at $p = 3$ by D.33, and at every other prime by D.54's fixed-character obstruction. The kernel is therefore trivial, as claimed.

D.54 already removed C_{12} and D_{24} from D.52. The only additional type in that allowed list was Dic_{12}. Its normal subgroup of order three would be faithfully represented over $\mathbb{Q}_3$, and an order-four element inverting it would also have order four by the just-proved faithfulness. But the exact field-normalizer calculation (D.52i)–(D.52k) proves that every finite-order matrix inverting this C_3 has square I: the alternative would require norm -1 in $\mathbb{Q}_3(\zeta_3)$. This contradiction excludes Dic_{12}. The nine remaining types all have the explicit realizations of D.50 and D.52, proving (D.55b).

For an arbitrary order-four a, its square is a positive involution, so (D.55a) gives (D.55c). The full multiplier still has square one by D.50. At a negative coordinate the two eigenvalues of $A_q(a)$ are roots of $X^2 + 1$ whose product is -1, so they are equal. The matrix is diagonalizable in characteristic zero, hence scalar $i_q I$ or $-i_q I$. The two-coordinate is positive by D.50. For an odd q a square root of -1 in $\mathbb{Q}_q$ reduces to one in $\mathbb{F}_q$, which requires $q \equiv 1 \pmod 4$. This proves the stated quartic restriction without excluding its remaining scalar case.

The program `code/part_c/torus_cover_norm_d54_d55.py` also checks the square norm and primitive-boundary comparison over $\mathbb{F}_2$ in explicit cyclic odd-degree covering homology modules. The full subgroup, covering and normalizer arguments above give the all-prime conclusion. □

D.56. [Proved] The prime sections of a cyclic-six torus action. Let a have order six in the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$, and let

$$E = \{(\phi, j) \in \mathrm{Aut}(F) \times C_6 : [\phi] = a^j\} \tag{D.56a}$$

be its centre-free extension, with kernel identified with F by the inner-automorphism map. There are exactly three F-conjugacy classes of order-two sections over $\langle a^3 \rangle$ and exactly two F-conjugacy classes of order-three sections over $\langle a^2 \rangle$. Conjugation by an element with quotient label a^2 cycles the three former classes; conjugation by an element with quotient label a^3 exchanges the two latter classes.

Consequently E has exactly one conjugacy class of subgroups of order two and one of order three, and these are its only nontrivial finite subgroup types. In particular the full extension (D.56a) does not split, and a has no finite-order representative in $\mathrm{Aut}(F)$. For each prime section S of order $p = 2$ or 3,

$$N_E(S) = C_F(S) \times S, \qquad H^1(C_F(S), \mathbb{F}_p) = 0. \tag{D.56b}$$

The assertions concern these specified extension and section maps; they do not identify E with a free profinite product.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), finite actions and actual lifts in type (1,1).

Proof. Put

$$E_2 = \pi^{-1}(\langle a^3 \rangle), \qquad E_3 = \pi^{-1}(\langle a^2 \rangle), \quad \pi : E \longrightarrow \langle a \rangle. \tag{D.56c}$$

The involution a^3 is positive. Indeed D.55 embeds $\langle a\rangle$ faithfully in $\mathrm{GL}_2(\mathbb{Q}_2)$. An order-six matrix there has its order-three part in the irreducible quadratic field $\mathbb{Q}_2(\zeta_3)$; its commuting involution is the nonidentity scalar $-I$. For completeness, the centralizer is that field because the cubic part has irreducible polynomial X^2+X+1, and its only elements squaring to one are $1,-1$. Faithfulness excludes the former for a^3. Thus a^3 has determinant one at two, and D.43 gives its global positive sign. D.27 supplies the actual prime-order sections of both groups in (D.56c).

The three involution classes and their affine coordinates. Choose an order-two section generator t in E_2 and identify $E_2 = F \rtimes \langle t\rangle$. By D.46 and D.55, its action on two-adic homology is $-I$, and its maximal elementary abelian two-quotient has coordinates

$$(u,\chi_1,\chi_2) : E_2 \longrightarrow \mathbb{F}_2^3, \quad u(t)=1, \quad \chi_1(t)=\chi_2(t)=0, \tag{D.56d}$$

where $\chi_1|_F, \chi_2|_F$ form a basis of $H^1(F,\mathbb{F}_2)$. Every section generator has $u=1$, so its coordinate is a point of the affine plane $\mathbb{F}_2^2$. The three F-conjugacy classes of D.46 give three distinct such points. One can read the distinctness directly from that proof: Brown restriction identifies the localized degree-zero ring with $\mathbb{F}_2^3$, and this ring is generated by χ_1/u and χ_2/u. Equal values on two section classes would prevent these two elements from generating the corresponding two coordinate factors.

Conjugation by an element $e \in E$ with $\pi(e)=a^2$ preserves u and acts on the plane (D.56d) by

$$r \longmapsto Rr + d, \tag{D.56e}$$

where the linear part is the reduction of $A_2(a^2)$ in the chosen homology coordinates. D.54, at $q=2$, says that $R-I$ is invertible. Also $R^3=I$, so R is a nonidentity cubic matrix over $\mathbb{F}_2$ and $I+R+R^2=0$. The affine map (D.56e) has one fixed point, and its other three points form a single orbit of length three: translate its unique fixed point to zero and use the corresponding linear statement. Its invariant three-point set of section classes must be that orbit. This proves the asserted cyclic permutation. The action on the class set factors through $E/E_2 = C_3$, because conjugation by E_2 is already F-conjugation on its sections, as proved in D.46.

The two cubic classes and their affine coordinates. Choose an order-three section generator s in E_3 with label a^2. Use the actual abelian quotient of D.48 and the separation proved in D.51:

$$(u_3,\chi) : E_3 \longrightarrow \mathbb{F}_3^2, \qquad u_3(s)=1, \quad \chi(s)=0. \tag{D.56f}$$

The two F-conjugacy classes of section generators with prescribed label $u_3=1$ have χ-values 0 and b, with $b\neq 0$. For an element $e\in E$ with label a^3, conjugation preserves u_3 because E/F is abelian. On the restriction of χ to F it acts by -1, since D.55 gives $A_3(a^3)=-I$. The difference between the conjugated character and $-\chi$ vanishes on F, so is a scalar multiple of u_3. On the line $u_3=1$ the action consequently has the exact form

$$r \longmapsto -r + d'. \tag{D.56g}$$

Preserving the two-point set $\{0,b\}$ forces $d'=b$: the alternative of fixing both would give $2b=0$ in $\mathbb{F}_3$. Thus this action exchanges the two classes. Conjugation by E_3 is F-conjugation on its sections, by D.48, so the action factors through $E/E_3 = C_2$.

The full finite subgroups and the failure of lifting. Every finite subgroup of E injects into $E/F = C_6$, because F is torsion-free. It is therefore cyclic of order one, two, three or six. A subgroup of order six would normalize its order-two subgroup. Its element with quotient label a^2 would then fix that subgroup's F-conjugacy class, contradicting (D.56e). Thus no subgroup of order six exists. The two affine actions show respectively that all order-two subgroups are conjugate in E and all order-three subgroups are conjugate in E.

For a chosen section S of order two, the stabilizer of its F-conjugacy class in E/F is exactly $\langle a^3\rangle$; for order three it is exactly $\langle a^2\rangle$. The image of $N_E(S)$ equals this stabilizer: a conjugator

whose class is fixed can be multiplied by an element of F to normalize S itself. Thus the full normalizer is already its normalizer in E_p. Equations (D.46h) and (D.48g) identify it as $C_F(S)\times S$ and give the cohomology assertion in (D.56b).

The first projection of (D.56a) is injective and identifies E with the inverse image of $\langle a\rangle$ in $\mathrm{Aut}(F)$. Indeed the outer class of the first coordinate determines its label j. A homomorphic lift of $\langle a\rangle$ would give a subgroup of order six in this exact inverse image, which has been excluded. More generally, a finite-order representative ϕ of a would satisfy $\phi^6 \in \mathrm{Inn}(F)$. The latter group is isomorphic to the torsion-free group F, so $\phi^6 = 1$, making ϕ a forbidden element of order six. This proves the labelled nonlifting assertion while preserving the actual prime-order lifts in (D.56c).

The program `code/part_a/torus_order6_sections_d56.py` checks every affine cubic action on $\mathbb{F}_2^2$ with nonidentity cubic linear part, both nonzero cubic section separations in (D.56f), and the corresponding quotient stabilizers. The extension and cohomological arguments above establish that these are the actions on the full section class sets. □

D.57. [Proved] Full lifts of the noncyclic torus two-groups. Let Q be a subgroup of the full once-punctured-torus loop-outer group, isomorphic to $C_2 \times C_2$ or to D_8 of order eight. Put

$$E_Q = \{(\phi, q) \in \mathrm{Aut}(F) \times Q : [\phi] = q\}, \qquad F = \widehat{F}(x, y). \tag{D.57a}$$

The projection $E_Q \to Q$ has a homomorphic section with exactly its specified outer labels. For $Q = D_8$ there is exactly one F-conjugacy class of such sections.

For $Q = C_2 \times C_2$, let z be its positive involution and let r be either of its two negative involutions. There are exactly three section classes when

$$A_2(r) \bmod 2 = I, \tag{D.57b}$$

and exactly one otherwise. This choice does not depend on which negative involution is called r. For the image H of every full section,

$$N_{E_Q}(H) = C_F(H) \times H, \qquad H^1(C_F(H), \mathbb{F}_2) = 0. \tag{D.57c}$$

Together with D.27 and D.46, this supplies actual lifts for every finite two-subgroup in the list of D.50. It is a statement about labelled automorphism lifts, not a geometric conjugacy classification.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), finite two-groups and actual labelled lifts in type (1,1).

Proof. The invariant prime-section classes for a four-group. Suppose first that $Q = C_2 \times C_2$. By D.50 and D.55 its faithful dyadic representation has exactly one nonidentity positive element z, with $A_2(z) = -I$; the two other involutions are negative. In the inverse image E_z of $\langle z\rangle$, D.46 supplies exactly three F-conjugacy classes of sections T of order two, each with

$$H^1(C_F(T), \mathbb{F}_2) = 0. \tag{D.57d}$$

As in (D.56d), their elementary abelian coordinates are three distinct points of the affine plane $\mathbb{F}_2^2$.

Conjugation by an element of E_Q with label r induces an affine map

$$v \longmapsto Mv + d, \qquad M = A_2(r) \bmod 2. \tag{D.57e}$$

Its square is the identity on the plane: the square of the conjugating element lies in F, and inner automorphisms act trivially on the elementary abelian quotient of E_z. Its three-point set of section classes is invariant.

If $M = I$, a nonzero translation would have only two-element orbits, so could not preserve a three-point set. Thus $d = 0$, and all three classes are fixed. If $M \neq I$, it is the unique two-dimensional unipotent type over $\mathbb{F}_2$, with

$$\ker(I + M) = \mathrm{im}(I + M), \quad \dim \ker(I + M) = 1.$$

The square identity gives $(I + M)d = 0$, so (D.57e) has two fixed points and one orbit of size two. An invariant three-point subset must contain that orbit and one fixed point. Thus exactly one section class is fixed in this case. Since $A_2(zr) = -A_2(r)$, the reductions of the two negative involutions are equal, proving the independence in (D.57b).

Choose any fixed class and one of its section groups T. Its normalizer $N = N_{E_Q}(T)$ maps onto Q: adjust a conjugator with label r by an element of F to normalize T, and use T itself for the label z. Its kernel is $C = C_F(T)$, since an element of F normalizing the order-two group T centralizes it. Moreover T is central in N. The quotient $J = N/T$ therefore has the exact sequence

$$1 \longrightarrow C \longrightarrow J \longrightarrow Q/\langle z\rangle \longrightarrow 1, \qquad Q/\langle z\rangle = C_2. \tag{D.57f}$$

The inclusion of C in this quotient is injective, because $C \cap T = 1$.

The invariant order-four class for a dihedral group. Now suppose that $Q = D_8$ and let $R \cong C_4$ be its rotation subgroup. By D.53 the inverse image E_R has exactly one F-conjugacy class of labelled order-four sections B, and

$$H^1(C_F(B), \mathbb{F}_2) = 0. \tag{D.57g}$$

Every element of E_Q preserves this class, since R is normal in Q. The normalizer $N = N_{E_Q}(B)$ thus maps onto Q, by the same adjustment of a conjugator by an element of F. Its kernel is $C = C_F(B)$: an element of F normalizing B fixes the label of every element of B, and therefore centralizes B. The subgroup B is normal in N, and the quotient $J = N/B$ has

$$1 \longrightarrow C \longrightarrow J \longrightarrow Q/R \longrightarrow 1, \qquad Q/R = C_2. \tag{D.57h}$$

Again C embeds, because $C \cap B = 1$. Conjugation on B in this construction is the actual conjugation on its specified rotation labels in D_8.

Torsion and exact finite preimages in the two constructed quotients. In each of (D.57f) and (D.57h), the group C is a closed subgroup of F, and hence is torsion-free with $\mathrm{cd}_p(C) \leq 1$ at every prime. Equations (D.57d) and (D.57g) give $C^{(2)} = 1$. Every map from J to a finite two-group therefore kills C, and the displayed nontrivial quotient gives

$$J^{(2)} = C_2. \tag{D.57i}$$

The same full cohomological-dimension argument used in D.27 and D.46 proves that J has torsion. Indeed, a torsion-free J would be projective, since C is open and has the stated cohomological dimensions. Its maximal pro-two quotient would then be a free pro-two group, contrary to (D.57i). Every finite subgroup of J injects into $J/C = C_2$. Consequently J has an involution j outside C.

Take the full inverse image H of $\langle j\rangle$ in N. In the four-group construction its kernel is T, so $|H| = 4$; in the dihedral construction its kernel is B, so $|H| = 8$. In either case $H \cap F = 1$. To check this in the actual maps, an element of $H \cap F$ has image in $\langle j\rangle \cap C = 1$ in J, and then lies in $T \cap F = 1$ or in $B \cap F = 1$ respectively. Thus the projection $H \to Q$ is injective. Its order equals $|Q|$, so it is an isomorphism with the prescribed labels. Its inverse is the required section. This exact injection also fixes the isomorphism type of the finite lift; no alternative central extension is substituted for Q.

Uniqueness above each invariant class. The section just constructed identifies $N = C \rtimes H$ and

$$J = C \rtimes \langle j \rangle. \tag{D.57j}$$

The Hochschild–Serre sequence, with $H^1(C, \mathbb{F}_2) = 0$ and $\mathrm{cd}_2(C) \leq 1$, gives the actual inflation isomorphism

$$H^*(J, \mathbb{F}_2) \cong H^*(C_2, \mathbb{F}_2). \tag{D.57k}$$

The full Brown restriction theorem [37, Theorem 2 and §1.1] applies: C is open of finite two-cohomological dimension and no finite subgroup of J can be a four-group, since it injects into C_2. In high degree the source of its dense restriction map is the finite one-dimensional space in (D.57k). Restriction to the displayed section in (D.57j) is already nonzero in every degree, so this map is injective. Its image is therefore closed as well as dense, so the restriction is an isomorphism onto the product of the section-normalizer cohomologies. For a section $S \leq J$ its normalizer is $C_C(S) \times S$, by the quotient labels. Each such factor contributes

$$1 + \dim H^1(C_C(S), \mathbb{F}_2)$$

in every sufficiently high degree. There is consequently exactly one conjugacy class of sections in J, and its fixed group has zero first cohomology over $\mathbb{F}_2$. These are also exactly the C-conjugacy classes: multiply a given conjugator by the section element with its quotient label.

Full sections of E_Q containing the chosen T or B lie in N, and are exactly the full inverse images of the order-two sections of J. This follows from their orders and the same kernel calculation used above. They therefore form one C-conjugacy class. Every full four-group section has a positive subgroup T in one of the invariant classes of (D.57e); distinct such classes cannot be identified by F-conjugacy. The numbers of full section classes are thus exactly three and one as asserted. Every full dihedral section has a rotation subgroup in the unique class of D.53, so the dihedral count is exactly one.

The fixed group $C_C(j)$ in the Brown calculation is precisely $C_F(H)$ in the original extension. For example, commuting with j modulo T or B means that a commutator lies in that finite subgroup. The commutator also lies in F, so is trivial. The elements of C already commute with T or B. This proves the cohomology part of (D.57c). Finally an element normalizing H can be multiplied by the inverse of the unique element of H with its outer label. The result lies in F and normalizes H, and therefore centralizes each element of H by preservation of labels. This gives the exact direct product in (D.57c).

The program `code/part_a/torus_two_group_lifts_d57.py` checks all affine involutions of $\mathbb{F}_2^2$ together with their invariant three-point sets and verifies the three-versus-one fixed-class count. The full projectivity, Brown and finite-preimage arguments above provide the labelled lifts and their uniqueness statements. □

D.58. [Proved] Negative torus involution sections and cusp fusion. Let b be a negative involution of the full once-punctured-torus loop-outer group of $F = \widehat{F}(x, y)$, put $c = [x, y]$, and let E_b be its centre-free extension. Write A_2 for its action on $H_1(F, \mathbb{Z}_2)$. The F-conjugacy classes of order-two sections and their full fixed-subgroup cohomology are exactly

A_2 mod 2	number of section classes	$\{\dim_{\mathbb{F}_2} H^1(C_F(S), \mathbb{F}_2)\}_{[S]}$
I	2	$\{1, 0\}$
$\neq I$	1	$\{0\}$.

(D.58a)

For every such section, $N_{E_b}(S) = C_F(S) \times S$. The cusp normalizer is the full profinite dihedral group

$$N_{E_b}(\overline{\langle c \rangle}) = \overline{\langle c \rangle} \rtimes \langle v \rangle, \qquad v^2 = 1, \quad vcv^{-1} = c^{-1}. \tag{D.58b}$$

Its two internal conjugacy classes of order-two sections become the same F-conjugacy class in E_b. In particular, for every choice of the cusp reflection v, there exists $g \in F$ with

$$gvg^{-1} = cv, \qquad c = [g, v] = g\,(vgv^{-1})^{-1}. \tag{D.58c}$$

The dimensions in (D.58a) describe continuous cohomology of the full closed fixed subgroups.

Manuscript: Section 30, folios 194-196, scans 383-387; Section 31, folios 198-199 verso, scans 391-394; Part D(b), full loop-outer finite actions in type (1,1).

Proof. The actual extension and the two integral module cases. D.27 supplies an actual order-two representative α and the exact identification $E_b = F \rtimes_\alpha \langle s \rangle$. D.43 gives its global multiplier $\lambda = -1$, so $A_2^2 = I$ and $\det A_2 = -1$. Let $u : E_b \to \mathbb{F}_2$ be the quotient character, $u(s) = 1$.

First suppose $A_2 \equiv I \pmod 2$. On the free $\mathbb{Z}_2$-module $H^1(F, \mathbb{Z}_2)$, the induced involution D also satisfies $D \equiv I \pmod 2$. The endomorphism

$$P = (I - D)/2 \tag{D.58d}$$

is integral and satisfies $P^2 = P$. Its image and kernel are direct summands of rank one, because D has the two eigenvalues $1, -1$ over $\mathbb{Q}_2$. Thus choose continuous characters $\varphi_+, \varphi_- : F \to \mathbb{Z}_2$ which form a basis and satisfy

$$\varphi_+\alpha = \varphi_+, \qquad \varphi_-\alpha = -\varphi_-. \tag{D.58e}$$

This is a decomposition of the specified cohomology module. All subsequent maps use these actual characters; no free generating pair is changed.

Their mod-2 reductions extend to characters $a, b : E_b \to \mathbb{F}_2$ by $a(s) = b(s) = 0$. The Hochschild–Serre spectral sequence has the two rows $\mathbb{F}_2 u^i$ and $\mathbb{F}_2^2 u^i$. The degree-one characters extend, so their transgression vanishes. All other possible differentials are its products with powers of u and also vanish. The associated graded pieces give, for every $n \geq 1$,

$$H^n(E_b, \mathbb{F}_2) = \langle u^n, u^{n-1}a, u^{n-1}b \rangle_{\mathbb{F}_2}. \tag{D.58f}$$

Multiplication by u is an isomorphism in degrees at least one, since it is an isomorphism on both associated graded pieces.

The two Bocksteins and the actual localized ring. Reduce (D.58e) modulo four and use the continuous one-cochains

$$\widetilde{a}(hs^\epsilon) = \varphi_+(h) \bmod 4, \qquad \widetilde{b}(hs^\epsilon) = \varphi_-(h) \bmod 4.$$

The first is a homomorphism, while the second has coboundary

$$\delta\widetilde{b}(hs^\epsilon, ks^\eta) = 2\epsilon\varphi_-(k) \pmod 4. \tag{D.58g}$$

For the coefficient sequence $0 \to \mathbb{F}_2 \xrightarrow{2} \mathbb{Z}/4 \to \mathbb{F}_2 \to 0$, the Bockstein of a degree-one character is its cup square. This follows directly by taking its $\{0, 1\}$-valued lift; changing the lift changes the divided coboundary by a coboundary. Thus (D.58g) gives

$$a^2 = 0, \qquad b^2 = ub. \tag{D.58h}$$

Write $ab = \epsilon ua + \delta ub + \gamma u^2$ using (D.58f), with coefficients in $\mathbb{F}_2$. Restriction to the chosen section $\langle s \rangle$ gives $\gamma = 0$. Multiplication by a and (D.58h) then give

$$0 = a^2 b = \delta\epsilon\, u^2 a + \delta^2 u^2 b.$$

The two displayed degree-three classes are independent by (D.58f), so $\delta = 0$. Consequently the degree-zero localization $R = (H^*(E_b, \mathbb{F}_2)[u^{-1}])^0$ has the exact presentation

$$R = \mathbb{F}_2[A, B]/(A^2,\ B^2 - B,\ AB - \epsilon A), \qquad A = a/u, \quad B = b/u, \quad \dim_{\mathbb{F}_2} R = 3. \tag{D.58i}$$

For either value of ϵ the map

$$\begin{aligned} R &\longrightarrow \mathbb{F}_2[\nu]/(\nu^2) \times \mathbb{F}_2, \\ 1 &\longmapsto (1, 1), \\ A &\longmapsto (\nu, 0), \\ B &\longmapsto (\epsilon, 1 - \epsilon) \end{aligned} \tag{D.58j}$$

is a ring isomorphism. It preserves the stated relations and sends the three-dimensional basis $1, A, B$ to a basis. In particular R has exactly two primitive idempotents, and its nilpotent ideal has dimension one.

The nonidentity action modulo two. Now suppose $A_2 \bmod 2 \neq I$. The dual two-dimensional module $M = H^1(F, \mathbb{F}_2)$ is the regular $\mathbb{F}_2[C_2]$-module: $D - I = D + I$ has rank one, with image equal to kernel. The periodic cyclic resolution therefore gives

$$\dim M^{C_2} = 1, \qquad H^i(C_2, M) = 0 \quad (i > 0).$$

Choose the nonzero invariant character on F and extend it to $\chi : E_b \to \mathbb{F}_2$ with $\chi(s) = 0$. This again makes the only possible transgression zero. The two-row spectral sequence now gives

$$H^1(E_b, \mathbb{F}_2) = \langle u, \chi \rangle, \qquad H^n(E_b, \mathbb{F}_2) = \mathbb{F}_2 u^n \quad (n \geq 2). \tag{D.58k}$$

Both $u\chi$ and χ^2 restrict to zero on $\langle s \rangle$, whereas u^2 restricts nontrivially. Thus $u\chi = \chi^2 = 0$. In this case the degree-zero localization is

$$(H^*(E_b, \mathbb{F}_2)[u^{-1}])^0 = \mathbb{F}_2. \tag{D.58l}$$

Full Brown restriction and the section labels. Every finite subgroup of E_b injects into $E_b/F = C_2$, because F is torsion-free. In particular E_b has no subgroup $C_2 \times C_2$. For a section S, conjugation in its normalizer induces the identity on S. Multiplying a normalizer element by the section element with its quotient label leaves an element of $C_S = C_F(S)$. This proves the exact direct product

$$N_{E_b}(S) = C_S \times S. \tag{D.58m}$$

The closed subgroup $C_S \leq F$ has $\mathrm{cd}_2(C_S) \leq 1$.

The full-profinite Brown restriction theorem applies with the open subgroup F of cohomological 2-dimension one [37, Theorem 2 and §1.1]. For $n > 1$ its restriction map to the product of the cohomologies of (D.58m) is a dense embedding. The source is finite by (D.58f) or (D.58k); therefore that map is an isomorphism onto a finite product, as in the cited proof. These are the actual restriction maps and preserve cup products. Since u restricts trivially on C_S and nontrivially on S, localization gives the ring isomorphism

$$R \xrightarrow{\text{res}} \prod_{[S]} \left(\mathbb{F}_2 \oplus H^1(C_S, \mathbb{F}_2) u^{-1} \right). \tag{D.58n}$$

The second summand in each factor has square zero, because $H^2(C_S, \mathbb{F}_2) = 0$. Each factor is local, with that summand as its nilpotent ideal and with residue field $\mathbb{F}_2$.

If $A_2 \bmod 2 \neq I$, (D.58l) and (D.58n) give one factor and a zero second summand. If $A_2 \bmod 2 = I$, (D.58j) gives two factors, one with a one-dimensional second summand and one with a zero second summand. This proves all the numbers in (D.58a). The classes in this

comparison are also precisely the F-conjugacy classes: for a conjugator $e \in E_b$, multiply it by the inverse of the unique element of S having the same quotient label. This replaces e by an element of F without changing eSe^{-1}.

In the identity-modulo-two case the two classes are distinguished by the value of b on their nonidentity section element. Indeed the two residue-field evaluations of (D.58j) give $B = 0$ and $B = 1$. Restriction to a section is exactly that residue-field evaluation in (D.58n). Also a vanishes on every section element, since $a^2 = 0$ whereas the square of the nonzero character of C_2 is nonzero.

The exact cusp reflections and their fusion. Put $L = \overline{\langle c \rangle} \cong \widehat{\mathbb{Z}}$. Preservation of the cusp class and A.7 give

$$1 \longrightarrow L \longrightarrow N_{E_b}(L) \longrightarrow C_2 \longrightarrow 1. \tag{D.58o}$$

Choose v over its nonidentity label. The global multiplier -1 gives $vcv^{-1} = c^{-1}$. Write $v^2 = c^\kappa$. Commutation of v with its own square implies $c^\kappa = c^{-\kappa}$, hence $2\kappa = 0$ in $\widehat{\mathbb{Z}}$ and $\kappa = 0$. This proves (D.58b), with the multiplication and specified label of its finite factor retained.

Every section of this dihedral group is $\langle c^t v \rangle$, $t \in \widehat{\mathbb{Z}}$. Conjugation by c^z changes t to $t + 2z$, and conjugation by v changes it to $-t$. Its two internal classes are therefore represented by v and cv. These elements have the same image under every abelian character of E_b, since $c = [x, y] \in [F, F]$. When there is only one full section class they are already F-conjugate. When there are two, their equal b-values and the preceding residue-field distinction again put them in the same F-class. An F-conjugator sends the unique nonidentity element to the unique nonidentity element, giving the first equation of (D.58c). Multiplication by v^{-1} gives its other two exact equations.

The executable `code/part_c/torus_negative_involution_d58.py` checks the mod-four one-cochains, the cyclic resolution modules, the two explicit localized algebra isomorphisms, and geometric word examples of the cusp equation. The full-group conclusions above follow from restriction and the actual normalizer maps. They assert no full triviality or procyclicity of the fixed subgroups from their first cohomology dimensions. □

5 E. Special groups and two operator groups

5.1 E(a). Special groups from conjugators and from framed diagrams

We give the constructions of §25, folios 149–154, scans 296–306, with their precise hypotheses. Let π be a discrete hyperbolic surface group or its full profinite completion, and let I be its finite peripheral index set. Choose representatives L_i and coherent parametrizations $c_i : T \to L_i$, where $T = \mathbb{Z}$ or $\widehat{\mathbb{Z}}$. The self-centralizing and self-normalizing properties needed below are the results of Part A. Put $\mathcal{E} = \mathrm{Aut}_{\mathrm{lac}}(\pi)$ and $\Gamma = \mathrm{Out}_{\mathrm{lac}}(\pi)$. Every $\alpha \in \mathcal{E}$ has a permutation σ_α of I and an automorphism χ_α of T.

E.1. [Proved] The conjugator construction and the product kernel. Define $S\mathcal{E}$ to consist of pairs $(\alpha, (g_i)_{i \in I})$ with $\alpha \in \mathcal{E}$, $g_i \in \pi$, and

$$\alpha(c_i(t)) = g_i^{-1} c_{\sigma_\alpha(i)}(\chi_\alpha t) g_i \quad (t \in T).$$

The multiplication is

$$(\beta, (h_i))(\alpha, (g_i)) = \big(\beta\alpha, (h_{\sigma_\alpha(i)}\, \beta(g_i))_{i \in I}\big).$$

There are exact sequences and a canonically embedded normal subgroup

$$1 \to T^I \to S\mathcal{E} \to \mathcal{E} \to 1, \qquad \iota : \pi \hookrightarrow S\mathcal{E}, \quad a \mapsto (\mathrm{Int}(a), (a^{-1})_i).$$

The groups T^I and $\iota(\pi)$ commute and intersect trivially. Consequently

$$S\Gamma = S\mathcal{E}/\iota(\pi), \qquad 1 \to T^I \times \pi \to S\mathcal{E} \to \Gamma \to 1, \qquad 1 \to T^I \to S\Gamma \to \Gamma \to 1.$$

The superscript ! takes the inverse image of the kernel of the permutation action on I. All maps in the profinite case are continuous maps of profinite groups, and all kernels and quotients are closed.

Manuscript: §25; folios 149–151; scans 296–300.

Proof. Apply β to the defining equation for α. The two conjugating factors multiply as $\beta(g_i)^{-1}h^{-1}_{\sigma_\alpha(i)}$, which is the inverse of the factor displayed in the product. Associativity follows either by substitution in that formula or by associativity of composition of the specified conjugations. The identity is $(\mathrm{id}, (1)_i)$, and the inverse has automorphism α^{-1} and jth conjugator $\alpha^{-1}(g^{-1}_{\sigma_\alpha^{-1}(j)})$.

Surjectivity to $\mathcal{E}$ is the definition of preserving the loop classes. In its kernel every g_i centralizes L_i, hence belongs to L_i; this identifies the kernel with T^I. Substituting $(\mathrm{Int}(a), (a^{-1})_i)$ proves that ι is a homomorphism and that conjugation by $(\alpha, (g_i))$ takes $\iota(a)$ to $\iota(\alpha(a))$. Its injectivity follows from $Z(\pi) = 1$, including when I is empty. To check commutation with the kernel, the two products with $(\mathrm{id}, (\ell_i))$, $\ell_i \in L_i$, both have conjugators $\ell_i a^{-1}$. Their intersection is trivial because projection to $\mathcal{E}$ sends $\iota(a)$ to $\mathrm{Int}(a)$. Finally an element lying over the identity in Γ has an inner automorphism component and can be multiplied by a unique $\iota(a)^{-1}$ to put that component equal to the identity. This proves that the whole kernel is the displayed direct product, not merely an abstract extension of its two factors.

For a finitely generated profinite π, its automorphism group has the usual profinite topology obtained from characteristic open quotients. The defining equations of $S\mathcal{E}$ are closed equations in $\mathcal{E} \times \pi^I$. Thus it is profinite, the multiplication and all maps above are continuous, and the compact subgroups giving the quotients are closed. The same algebraic proof gives the exact continuous sequences. □

E.2. [Proved] Agreement with autoequivalences of the framed diagram. The group $S\Gamma$ of E.1 is the group of isomorphism classes of autoequivalences of

$$B_I \longrightarrow \coprod_{i \in I} BL_i \longrightarrow B\pi$$

with the common orientation action retained. Adding a fixed base object in $B\pi$ gives $S\mathcal{E}$. Thus the two constructions on scans 296–300 agree, with the comparison maps and inner ambiguities exactly as in E.1.

Manuscript: §25; folios 150–151; scans 298–300.

Proof. An autoequivalence gives a permutation of the selected objects, the specified automorphism χ on their cyclic automorphism groups, and an automorphism α at the generic base object. The natural isomorphism making the second square commute is, at its ith object, an element $g_i \in \pi$. Its naturality equation is exactly the defining equation of E.1. Pasting the squares gives the product there, including the index $\sigma_\alpha(i)$.

Initially the first square also contains an isomorphism at each selected peripheral object. A natural transformation at that object changes this isomorphism by the corresponding element of L_i. Use it to make the first-square isomorphism the identity. After this gauge choice there is no remaining peripheral-object natural-transformation ambiguity. A natural transformation at the generic object is an element $a \in \pi$ and changes the representative by $\iota(a)$, with the convention of E.1. These are precisely the equivalences being divided out in $S\Gamma$. Fixing the generic base object and its comparison isomorphism removes that last ambiguity and gives $S\mathcal{E}$. Conversely every pair in E.1 defines the three functors and their two commutativity squares. The two constructions undo each other on the chosen representatives and on the specified natural transformations, so

the group isomorphisms are canonical relative to the initial peripheral representatives. Changing those representatives transports the diagrams by their explicit conjugating isomorphisms. □

E.3. [Proved] The puncture-stabilizer pushout is the normalizer group. Let $\Gamma_i \subset \Gamma$ stabilize i. Restrict the extension $S\Gamma \to \Gamma$ to Γ_i and push its kernel out by $\mathrm{pr}_i : T^I \to T$. The resulting extension is canonically

$$1 \to L_i \xrightarrow{\mathrm{Int}} \{\alpha \in \mathrm{Aut}_{\mathrm{lac}}(\pi) : \alpha(L_i) = L_i\} \longrightarrow \Gamma_i \to 1.$$

Manuscript: §25; folios 153–154; scans 305–306.

Proof. For a representative $(\alpha, (g_j))$ with $\sigma_\alpha(i) = i$, put

$$F_i[(\alpha, (g_j))] = \mathrm{Int}(g_i)\alpha.$$

The defining equation shows that this automorphism normalizes the actual subgroup L_i. Replacing the pair by its generic-basepoint change replaces α by $\mathrm{Int}(a)\alpha$ and g_i by $g_i a^{-1}$; the displayed value is unchanged. Hence the map descends to $S\Gamma_i$. The multiplication formula gives

$$\mathrm{Int}(h_i\beta(g_i))\beta\alpha = \mathrm{Int}(h_i)\beta\,\mathrm{Int}(g_i)\alpha,$$

so it is a homomorphism. An automorphism normalizing L_i has a representative with $g_i = 1$, proving surjectivity. On the kernel T^I the map is $\ell_i \mapsto \mathrm{Int}(\ell_i)$; its kernel is exactly the other factors because $Z(\pi) = 1$. This is the universal pushout claimed. The normalizer equality in Part A gives the last kernel L_i in the displayed sequence. This is the compatibility of scans 305–306, with its actual group of automorphisms rather than an unspecified outer class. □

E.4. [Refuted] The product-kernel claim requires the rigidity hypotheses. The formula $\ker(S\mathcal{E} \to \Gamma) = T^I \times \pi$ does not hold unchanged for all abelian types. For the unpunctured torus, $I = \varnothing$, $S\mathcal{E} = \mathrm{GL}_2(\mathbb{Z}) = \Gamma$ and the kernel is trivial, whereas $\pi = \mathbb{Z}^2$. For the annulus the inner basepoint subgroup intersects the peripheral kernel, and the special outer kernel is a quotient of T^2 by one diagonal copy of T.

Manuscript: §25; folios 149–153; scans 296–304.

Proof. For the torus, all inner automorphisms of $\mathbb{Z}^2$ are the identity and there are no conjugators indexed by punctures. Thus the pair construction contains only the automorphism itself, giving the stated identity map and its trivial kernel.

For the annulus, both peripheral images are the whole group $\pi = \mathbb{Z}$. The conjugator kernel is π^2, but the generic-basepoint change a is $(\mathrm{id}, (a^{-1}, a^{-1}))$ and lies in that same kernel. Dividing by these changes gives $\mathbb{Z}^2/\Delta\mathbb{Z} \cong \mathbb{Z}$, with the antidiagonal convention after orienting both boundary loops. Keeping the two end labels and the independent orientation data supplies the appropriate extended quotient; their omission would create an additional error. For the plane the peripheral homomorphism $\mathbb{Z} \to 1$ is not injective, so the normalizer calculation used in E.1 is unavailable. Its correct fixed-boundary positive special group is trivial, as C.20 computes. These explicit maps identify the failed hypotheses in each case. □

5.2 E(b). Two commuting operator groups

We retain the names θ and ω for the standard order-two and order-three lifts on $\widehat{F}_2$; Grothendieck's σ_0 and ρ denote the corresponding choices on scans 316–317, §25 bis, folios 159–160. A change of base path conjugates a lift by $\widehat{F}_2$. We keep this ambiguity in the statements below.

E.20. [Proved] The two lifting conditions cut out exactly the multiplier-1 subgroup modulo 3. In $B = \mathrm{Out}_{\mathrm{lac}}(\widehat{\pi}_{0,3})$ let S_3 be the geometric positive subgroup. Its centralizer is GT_0, with the normalized description (λ, f) of Harbater–Schneps. Every element of GT_0 has an automorphism representative commuting with the chosen order-two lift θ. The elements with an automorphism representative commuting with the chosen order-three lift ω are exactly

$$GT_0^{(3)} = \{(\lambda, f) \in GT_0 : \lambda \equiv 1 \pmod 3\}.$$

This subgroup has index two. The two representatives need not be the same representative; this is the two-condition formulation on scans 316–317.

Manuscript: 25 bis; 157 verso–160; 312–317.

Proof. Harbater–Schneps, Main Theorem (a) and Theorem A(i) [28], identify the centralizer and state precisely

$$[\theta, F] = \mathrm{Inn}(f), \qquad [\omega, F] = \mathrm{Inn}(y^m f), \qquad m = (\lambda - 1)/2,$$

for $F(x) = x^\lambda$, $F(y) = f^{-1} y^\lambda f$. Here $[a, b] = aba^{-1}b^{-1}$. A representative $\mathrm{Inn}(h)F$ commutes with θ precisely when

$$\theta(h) f = h,$$

since F_2 has trivial centre. It commutes with ω precisely when $\omega(h) y^m f = h$. These are the two exact coboundary equations. We determine their solvability without imposing the pentagon equation.

The positive lift extension of the S_3 action on the three-punctured sphere is

$$E = C_2 * C_3 = \langle s, r \mid s^2 = r^3 = 1 \rangle, \qquad E \longrightarrow S_3,$$

whose kernel is the peripheral free group F_2. Its cusp generator can be chosen as $e = sr$; e^2 is a primitive cusp in that kernel. Complete this exact sequence by D.1. An outer centralizer element, represented by F, acts on $\widehat{E}$ by conjugating its image in $\mathrm{Aut}(\widehat{F}_2)$ and induces the identity on S_3. It sends e^2 to a $\widehat{F}_2$-conjugate of $e^{2\lambda}$. After that conjugation it sends e into $C_{\widehat{E}}(e^{2\lambda}) = \overline{\langle e \rangle}$. To prove this equality, put $L = \overline{\langle e^2 \rangle}$. A centralizing element preserves its $\widehat{F}_2$-conjugacy class, so its image in S_3 stabilizes the associated puncture. That stabilizer is $\langle q(e) \rangle$ of order two. Multiplication by e when necessary puts the element in $\widehat{F}_2$, where Part A gives $C_{\widehat{F}_2}(e^{2\lambda}) = L$. This proves the equality and identifies the whole centralizer with $\widehat{\mathbb{Z}}$. Doubling is injective in that group, so the image of e is e^λ.

Consequently this automorphism acts on $\widehat{E}_{\mathrm{ab}} = C_6$ by multiplication by λ. Every order-two element is conjugate to s, and every order-three subgroup is conjugate to $\langle r \rangle$, by D.2 or the free profinite product theorem. A lift of the given transposition is F_2-conjugate to s: a total-group conjugator projects to $C_{S_3}(s) = \langle s \rangle$, and multiplication by s adjusts it into the kernel. Thus there is one kernel-conjugacy class for the order-two lift. It is fixed by every centralizer element, proving solvability of the first equation.

There are exactly two kernel-conjugacy classes of lifts of the specified three-cycle. They are represented by r and $sr^{-1}s^{-1}$. Indeed a total-group conjugator has either an even or an odd image in S_3; adjusting by $\langle r \rangle$ or by $s\langle r \rangle$ gives these two representatives. They are distinct because their images in the order-three quotient of E_{ab} are inverse nonidentity elements. The action of (λ, f) preserves the class of r exactly when $\lambda \equiv 1 \pmod 3$, by the computed action on C_6. When it does, the same adjustment by r puts its conjugator in the kernel. This is precisely solvability of the second coboundary equation. The argument treats elements, so it retains the label of the three-cycle.

For comparison, Lochak–Schneps, Proposition 3 and Theorem 2 [41], compute the same two nonabelian cohomology sets: one class for θ, two for ω, with the second class selected by $\lambda \equiv -1 \pmod 3$. The first two parts of their proof use only the two-cycle and three-cycle equations, so their computation also applies to GT_0. Finally $(-1, 1) \in GT_0$ exchanges the two classes. The homomorphism to $(\mathbb{Z}/3)^\times$ is therefore surjective and the asserted index is exactly two. The residue calculation is independently recorded in the executed program. □

E.21. [Proved] The arithmetic subgroup and the two fixed points. The intersection of the usual Galois image $G_{\mathbb{Q}} \to GT_0$ with $GT_0^{(3)}$ is the image of $G_{\mathbb{Q}(\sqrt{-3})}$. The order-three fixed points in $U = \mathbf{P}^1 - \{0, 1, \infty\}$ are $e^{\pm i\pi/3}$, and the order-two lift has one fixed point in U. Thus the requirement to fix the chosen order-three section is a proper finite-index restriction, even on the Galois subgroup.

Manuscript: 25 bis; 157 verso–160; 312–317.

Proof. The multiplier of the Galois action is the cyclotomic character, by its action on the tame inertia at 0. Its reduction modulo 3 is the action on μ_3 and has fixed field $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$. E.20 identifies precisely its kernel. The three-cycle $z \mapsto 1/(1 - z)$ has fixed-point equation $z^2 - z + 1 = 0$, and the transposition $z \mapsto 1 - z$ has fixed-point equation $z = 1/2$. These points lie in U. Complex conjugation exchanges the former pair and fixes the latter point, which also proves properness directly.

On scan 312 the relevant subgroup is obtained by fixing a finite set of critical data; it is therefore of finite index. An assertion that every element of $G_{\mathbb{Q}}$ lifts commuting with a specified one of the order-three lifts would contradict the displayed computation. We record the scan's wording as editorial evidence, not as a settled reading. □

E.22. [Proved] Normalizer, stabilizer, and the forced lift. Let $1 \to \pi \to E \xrightarrow{q} \mathcal{G} \to 1$ have centre-free kernel, and let $K \leq E$ be a finite subgroup on which q is injective. Assume $C_\pi(K) = 1$, as holds for a nontrivial rotation stabilizer in the anabelian geometric situation and its completion. Put $G_Q = q(K)$ and let Q denote the π-conjugacy class of K. The restriction

$$q : N_E(K) \longrightarrow \mathcal{G}$$

is injective and has image the stabilizer $\mathcal{G}_Q$ of Q. Its inverse on that image is a canonical homomorphism. In a two-group situation $\mathcal{G} = \Gamma G$ with Γ centralizing G, its restriction gives the forced lift of $\Gamma_Q = \Gamma \cap \mathcal{G}_Q$, and this lift centralizes K.

Manuscript: 25 bis; 155–157 verso; 307–312.

Proof. An element of $N_E(K) \cap \pi$ acts trivially on $q(K)$, and the restriction of q to K is injective. It therefore centralizes K, so the kernel is $C_\pi(K) = 1$. A quotient element stabilizes Q precisely when one, and hence every, lift carries K to a π-conjugate. Multiplying that lift by the appropriate element of π produces an element of $N_E(K)$ above it. This proves the image statement and uniqueness of the lift. The inverse is a homomorphism because it is the inverse of an injective homomorphism onto its image. For $\gamma \in \Gamma_Q$, conjugation of its lift on K projects to conjugation by γ on G_Q, which is trivial. Injectivity on K then makes this conjugation trivial. In the profinite case all these maps are continuous, and the inverse is continuous by compactness of the closed normalizer and the Hausdorff property of the quotient.

For the centralizer, the precise image is $\mathcal{G}_Q \cap C_{\mathcal{G}}(G_Q)$, rather than all of $\mathcal{G}_Q$. When G_Q is nonabelian dihedral, its own elements already distinguish these images. For a reflection in an oval the kernel is $\mathbb{Z}$ or $\widehat{\mathbb{Z}}$ by Part B and D.4, so the stated injectivity is replaced by an exact sequence with that kernel. These computations identify exactly the rigidity used in the manuscript's argument. □

E.23. [Proved] Explicit discrete centralizers and pointwise kernels. For the three actions in the question, the full extended geometric groups, their centralizers, their normalizers, and their pointwise kernels on the exceptional points are as follows. Here D_{2n} has order $2n$.

$(g,\nu), G$	M	$C_M(G)$	$N_M(G)$	$C_M(G)^\natural$
$(0,3), S_3$	$S_3 \times C_2$	C_2	$S_3 \times C_2$	1
$(1,1), C_4$	$\mathrm{GL}_2(\mathbb{Z})$	C_4	D_8	C_2
$(1,1), C_6$	$\mathrm{GL}_2(\mathbb{Z})$	C_6	D_{12}	1

The last column fixes every exceptional point individually, not merely its orbit under G.

Manuscript: 25 bis; 155–160; 307–317.

Proof. For type $(0,3)$, the geometric group is S_3 times complex conjugation. The exceptional points are the three transposition fixed points $-1, 1/2, 2$ and the two order-three fixed points of E.21. Complex conjugation fixes the first three and interchanges the last two. The centralizer of S_3 is therefore C_2 and its pointwise kernel is trivial; its normalizer is the whole product.

Use the matrices

$$R_4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R_6 = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

on $\mathbb{R}^2/\mathbb{Z}^2$, punctured at 0. A rational matrix commuting with either one is $aI + bR_n$. For an integral commuting matrix $a, b \in \mathbb{Z}$. Its determinant is $a^2 + b^2$ for R_4 and $a^2 + ab + b^2$ for R_6. Equating this nonnegative norm to 1 gives four and six units respectively, proving the centralizer assertions. Conjugation in the normalizer embeds its quotient by the centralizer in $\mathrm{Aut}(C_n) = C_2$ for $n = 4, 6$. Reflection in the square or hexagonal lattice realizes inversion, proving the two normalizer assertions.

For C_4 the exceptional set outside the puncture is exactly the three nonzero two-torsion points. The matrix $-I$ fixes all three, while R_4 interchanges $(1/2, 0)$ and $(0, 1/2)$. Thus the kernel is $\langle -I \rangle$. For C_6 the exceptional set consists of the same three two-torsion points and $(1/3, 1/3), (2/3, 2/3)$. The three two-torsion points force a power of R_6 to be I or $-I$, and the latter exchanges the two order-three points. Hence the kernel is trivial. These lists are exhaustive: each nonidentity power has its fixed points given by $(R_n^j - I)v \in \mathbb{Z}^2$; their union is the displayed finite set. The program enumerates these solutions, the kernels, and the matrix centralizers and normalizers. The norm calculation proves exhaustiveness beyond its bounded integer search. □

E.24. [Refuted] Recovery of an arbitrary commuting group from its pointwise subgroup. For an arbitrary specified $\Gamma \leq C_{\Gamma'}(G)$, its pointwise subgroup is

$$\Gamma^\natural = \Gamma \cap \Gamma'^\natural,$$

and it need not equal $C_{\Gamma'^\natural}(G)$. Nor is Γ determined as a normalizer by $\Gamma^\natural$. For the maximal choice $\Gamma = C_{\Gamma'}(G)$ the proposed equality holds, by the displayed intersection identity.

Manuscript: 25 bis; 157 verso; 312.

Proof. Both identities follow by testing separately the two defining conditions: membership in Γ and pointwise stabilization of each critical datum. To refute recovery, take the first row of E.23. The choices $\Gamma = 1$ and $\Gamma = C_2$ have the same pointwise subgroup 1. Its normalizer inside the centralizer C_2 is C_2, so the procedure cannot recover the first choice. For the proposed equality without the maximal-choice hypothesis, use the second row and take $\Gamma = 1$: its pointwise subgroup is 1, whereas the centralizer pointwise kernel is C_2. All groups and their kernels are included in the finite check. Thus the failure persists even in the specifically requested discrete cases and has a complete intersection formula as its replacement. □

6 F. Statements usable in the later sections

We use $\Gamma_{g,\nu}$ for the extended geometric group permitting puncture permutations and $\Gamma^{!+}_{g,\nu}$ for the pure oriented group, following §19, folio 102, scans 204–205. In each formula below the chosen variant is stated explicitly. The full profinite loop-outer group is $B = \mathrm{Out}_{\mathrm{lac}}(P)$; the closure $H = \overline{M}$ of a chosen geometric subgroup M is a subgroup of B. We do not identify that closure with the abstract profinite completion without the congruence assertion required for that identification.

F.1. [Proved] LM4-F.a: fixed discretizations and prediscretizations. For $M = \Gamma_{g,\nu}$ and $H = \overline{M} \leq B$, the sets of outer discretizations and prediscretizations are B/M and B/H. For a given finite $G \leq B$ their fixed sets are exactly D.16's transporter sets and centralizer-orbit decompositions. In the three types of D.8, every fixed prediscretization has a fixed outer discretization above it. The same lifting conclusion holds for a finite solvable G and genus at most two for the orientation-preserving geometric group covered by Boggi's Theorem C. Over a fixed coset represented by b, the fibre is

$$(H/M)^{b^{-1}Gb},$$

with its full labelled-embedding decomposition as in D.16.

Manuscript: 28; 178–181; 352–358.

Proof. The identifications and the fixed-set computation are proved in D.16, including their inner-automorphism ambiguities. A fixed coset bH has $K = b^{-1}Gb \leq H$. Its fibre consists of bhM, $h \in H$; this coset is fixed precisely when $h^{-1}Kh \leq M$, proving the formula. In the three low types D.8 conjugates every finite $K \leq H$ into M, giving a point in that fibre. In genus at most two and for solvable K, Boggi's Theorem C [7] gives the same conjugation into the oriented geometric group. The congruence subgroup property in these genera identifies its completion with H, so the theorem applies to the displayed subgroup. We retain the labelled embeddings in the fibre decomposition because subgroup conjugacy alone does not fix an automorphism of K.

The actual based version has $A(P)/J$ in place of B/M, as D.16 proves. A chosen genuine lift of G acts on it; an outer subgroup alone acts on its quotient by P. These are the exact maps through which the fixed-set assertions of §28, folios 178–181, scans 352–358 are used. For a finite group in the full B not known to stabilize a prediscretization, the same formula still determines the fixed set as a transporter; its nonemptiness is exactly the unresolved recognition question D.12, not an additional conclusion of this lemma. □

F.2. [Refuted] LM4-F.b, finiteness and equality of the two centralizers. The centralizer of a finite subgroup $G \leq \Gamma^{+}_{g,\nu}$ need not be finite, either discretely or in the geometric completion. In genus at most two, for finite solvable G, the normalizer comparison and the open-kernel argument below give

$$C_{\widehat{\Gamma^{+}_{g,\nu}}}(G) = \overline{C_{\Gamma^{+}_{g,\nu}}(G)}.$$

The corresponding equality with the full loop-outer group in place of the geometric completion is false already for $(g, \nu) = (0, 3)$ and $G = S_3$.

Manuscript: 30; 194–196; 383–387.

Proof. On a closed genus-two surface let G be generated by the hyperelliptic involution. This involution is central in Γ^{+}_{2}, so its centralizer is the entire group. A Dehn twist about a nonseparating curve has infinite order: its action on integral homology sends a transverse vector b to $b + a$, and its nth power sends it to $b + na$. This also gives infinite order in the

congruence image, since some reduction modulo an integer detects every nonzero n. Thus the completed centralizer is infinite as well. For finite solvable G, the normalizer comparison, with the source induction and its topology input audited in F.7, gives $N_{\widehat{M}}(G) = \overline{N_M(G)}$ for $M = \Gamma^+_{g,\nu}$. The continuous conjugation homomorphism from this profinite normalizer to the finite group $\mathrm{Aut}(G)$ has open kernel $C_{\widehat{M}}(G)$. Intersecting the dense subgroup $N_M(G)$ with that open kernel gives a dense subgroup of the kernel: every open subset of the kernel is open in the normalizer and meets $N_M(G)$. The intersection is exactly $C_M(G)$, proving the displayed centralizer equality. The map of the completion to the loop-outer group is injective in these genera by the congruence subgroup property. The cited sources state the equality for every finite G; F.7 records the precise missing intersection step for their nonsolvable assertion. The passage from any such equality to finiteness would still require the discrete centralizer to be finite, contradicted by the hyperelliptic example.

For the smallest hyperbolic type, $\Gamma^+_{0,3} = S_3$, and $C_{S_3}(S_3) = 1$. Harbater–Schneps identify $C_B(S_3) = GT_0$. This group is infinite: its multiplier contains the cyclotomic image $\widehat{\mathbb{Z}}^\times$ of $G_{\mathbb{Q}}$. These are actual homomorphisms on the same group $\widehat{F}_2$, with the geometric S_3 supplied by rational Möbius transformations, so the commutation with the Galois image is the naturality of the fundamental-group action. The two centralizers are therefore 1 and an infinite group. With orientation reversal included the geometric centralizer becomes C_2, and the same failure remains.

The statement denoted (c) on scans 383–385, §30, folios 194–195, concerns the centralizer of an arithmetic germ $\Gamma^\natural$. It cannot be replaced merely by a finite G: scan 385 itself discusses infinite centralizers of finite groups. The counterexamples above compute the result of the requested replacement explicitly and explain which datum that replacement removes. □

F.3. [Proved] LM4-F.b, the intrinsic open-image statement. Let $M \le B$ be an oriented geometric subgroup, $H = \overline{M}$, $N = N_B(H)$, and let $i : G \hookrightarrow H$ be a fixed finite embedding. Conjugation defines an action of N/H on the set of H-conjugacy classes of embeddings $G \hookrightarrow H$. The image of the continuous homomorphism

$$C_N(i(G)) \longrightarrow N/H$$

is exactly the stabilizer of $[i]$, and its kernel is $C_H(i(G))$. When the orbit of $[i]$ is finite, this image is open. The orbit is finite for solvable finite G in genus at most two and for every finite G in the three low types of D.8.

Manuscript: 30; 195 verso–196; 386–387.

Proof. An element nH stabilizes $[i]$ exactly when there is $h \in H$ with

$$ni(g)n^{-1} = hi(g)h^{-1} \quad (g \in G).$$

This is equivalent to $h^{-1}n \in C_N(i(G))$. It proves the image assertion, and the kernel is obtained by intersecting the centralizer with H. Because H is compact, the stabilizer is the image of a compact closed transporter and is closed in N/H. A closed subgroup of finite index in a profinite group is open: its finitely many closed cosets have open complements, so each coset is open.

For the claimed cases, finite subgroup conjugacy classes in H come from the finite list of geometric classes, by D.8 or Boggi's solvable statement. There are finitely many geometric classes for each fixed finite group: a finite action has an orbifold quotient of bounded signature by the Euler characteristic formula; there are finitely many epimorphisms from each finitely generated orbifold group to the fixed finite group. Passing from subgroup classes to labelled-embedding classes adds at most $|\mathrm{Aut}(G)|$ choices per subgroup class. Thus the orbit is finite and the image is open. F.6 constructs the arithmetic quotient directly from universal monodromy and proves the open-image assertion in that quotient for every finite geometric subgroup and every hyperbolic type. The inclusion of that quotient into N/H is retained there as the exact comparison map. □

F.4. [Proved] LM4-F.c: the exact canonical lift of a section stabilizer. Let π be a centre-free discrete surface group or its full profinite completion, and put

$$A = \mathrm{Aut}_{\mathrm{lac}}(\pi), \qquad B = \mathrm{Out}_{\mathrm{lac}}(\pi).$$

Let $G \leq B$ be finite and let $s : G \to A$ be a specified lift with $C_\pi(s(G)) = 1$. The natural map

$$C_A(s(G)) \longrightarrow C_B(G)$$

is injective. Its image is the stabilizer of the π-conjugacy class $[s]$ in the set of lifts of the labelled inclusion $G \hookrightarrow B$. Its inverse is the canonical homomorphism lifting that stabilizer to A and centralizing $s(G)$. In the profinite case it is continuous. For geometrically realized actions with a nontrivial rotation the section-class set is finite, so this stabilizer has finite index and is open in the profinite case.

Manuscript: 31; 197 verso–199; 390–393.

Proof. The kernel is the inner automorphisms centralizing $s(G)$, identified, by centre-freeness, with $C_\pi(s(G))$. For $b \in C_B(G)$ choose an automorphism representative v. The conjugate homomorphism $g \mapsto vs(g)v^{-1}$ still lifts the labelled inclusion, because b centralizes each g. It represents $[s]$ exactly when some $p \in \pi$ makes $\mathrm{Inn}(p)^{-1}v$ commute with every $s(g)$. This proves that the image equals the claimed stabilizer. The kernel computation proves uniqueness. The inverse is a homomorphism by uniqueness, since products of commuting lifts still commute with $s(G)$.

For a finitely generated profinite π, A and B are profinite, and the centralizer is closed. The map is a continuous injection from a compact space into a Hausdorff space, so its inverse on its image is continuous. In the discrete geometric situation Part B proves finiteness of the lift classes; D.3 proves their equality with the profinite classes. The orbit of a class is therefore finite. Its stabilizer is closed, as the image of the compact centralizer, hence open. This proves all topology assertions as well as the group assertion.

For the notation of scans 390–393, §31, folios 197–199, the target $\mathfrak{S}(\pi_0)^G$ is the centralizer of the chosen genuine lift inside the loop automorphism group. The source is the stabilizer of the corresponding section class inside $\mathcal{T}(\pi_0)^G = C_{\mathrm{Out}_{\mathrm{lac}}(\pi_0)}(G)$. A point P determines that section class after a base path; changing the path conjugates the resulting lift by π_0, exactly as displayed. Thus the canonicity belongs to the chosen section, and the geometric point supplies its conjugacy class. □

F.5. [Refuted] The unrestricted uniqueness assertion for reflection ovals. For a reflection fixing an oval, the lifting map in F.4 has kernel $\mathbb{Z}$ in the discrete case and $\widehat{\mathbb{Z}}$ after completion. Consequently it is not injective and its lifts are not unique without further data.

Manuscript: 31; 197 verso–199; 390–393; 33; 222 verso–225; 440–445.

Proof. Part B and D.4 identify $C_\pi(s(G))$ with the primitive oval subgroup and its closure. Every element of that subgroup induces an inner automorphism centralizing the chosen reflection lift and maps to the identity outer class. The kernel computation in F.4 therefore gives the displayed groups. A nonidentity primitive oval element is an explicit second lift of the identity outer element. The corrected general statement is the exact sequence with this kernel and with image the same section stabilizer, proved by the same representative-and-conjugator calculation. No splitting of that sequence is supplied by the point class alone. □

F.6. [Proved] LM4-F.b: open arithmetic image for every finite geometric subgroup. Let $2g-2+\nu>0$, let $M=\Gamma^{+}_{g,\nu}$ permit permutations of the ν punctures, and retain

$$P=\widehat{\pi}_{g,\nu},\qquad B=\mathrm{Out}_{\mathrm{lac}}(P),\qquad H=\overline{M}\leq B.$$

Write $A_{\mathrm{ar}}\leq B$ for the image of universal *arithmetic* étale monodromy of the moduli stack of smooth genus-g curves with an unordered reduced divisor of degree ν, over $\mathbb{Q}$. Then $H\triangleleft A_{\mathrm{ar}}$, and put $\Gamma_{\mathrm{ar}}=A_{\mathrm{ar}}/H$. For every finite subgroup $G\leq M$, the continuous homomorphism

$$C_{A_{\mathrm{ar}}}(G)\longrightarrow\Gamma_{\mathrm{ar}} \tag{F.6.1}$$

has open image. Its kernel is $C_H(G)$. In the notation $N=N_B(H)$ and $q:N\to N/H$, this is exactly the assertion that

$$q(C_N(G))\cap\Gamma_{\mathrm{ar}}=q(C_{A_{\mathrm{ar}}}(G))\quad\text{is open in }\Gamma_{\mathrm{ar}}. \tag{F.6.2}$$

The same assertion holds with ordered punctures and the pure oriented group throughout. No identification of H with the abstract completion of M is used.

Manuscript: 30; 195–196; 385–387.

Proof. We first type the arithmetic group and its comparison maps. Denote the indicated moduli stack over $\mathbb{Q}$ by $\mathscr{M}$, and choose a geometric base point and an orientation-preserving topological marking of its complex fiber. Its universal punctured curve supplies the outer monodromy homomorphism

$$R:\Pi_{\mathrm{ar}}=\pi_1^{\mathrm{et}}(\mathscr{M})\longrightarrow B.$$

The geometric comparison and the arithmetic exact sequence are

$$1\longrightarrow\widehat{M}\longrightarrow\Pi_{\mathrm{ar}}\longrightarrow G_{\mathbb{Q}}\longrightarrow 1,\qquad R(\widehat{M})=H. \tag{F.6.3}$$

These are the constructions in [6, §§7.1–7.2, equation (29) and the universal étale monodromy representation]. For the geometric comparison, finite étale covers analytify to finite topological covers; the inverse comparison is compatible on an étale atlas and its double and triple overlaps, so descends to the stack. The marked complex moduli stack has contractible Teichmüller space as its covering, with group M, which identifies its topological fundamental group. The universal curve then identifies its monodromy with the completed geometric action on P, rather than introducing a separate representation. All these maps preserve peripheral conjugacy classes, since the removed divisor is carried to itself in the universal family. Their base-point ambiguities are inner automorphisms of P, already removed in B; a change of topological marking conjugates the displayed subgroups by the corresponding element of M.

Set $A_{\mathrm{ar}}=R(\Pi_{\mathrm{ar}})$. It is compact and hence closed in B. Normality in (F.6.3) gives $H\triangleleft A_{\mathrm{ar}}\leq N$. The quotient map induces a continuous surjection

$$\overline{R}:G_{\mathbb{Q}}\longrightarrow\Gamma_{\mathrm{ar}}. \tag{F.6.4}$$

The proof uses this quotient of $G_{\mathbb{Q}}$ as defined. Its kernel and the kernel of $\widehat{M}\to H$ need not be computed.

We next construct a number-field curve with the specified *labelled topological* G-action. Finite realization C.5 supplies a compact Riemann surface X, a reduced marked divisor D, and a faithful holomorphic action of G preserving D whose marked mapping classes are the given inclusion $G\hookrightarrow M$. Choose projective equations for X, equations for D, and formulas for the finitely many automorphisms and their inverses. Their finitely many coefficients lie in a finitely generated $\mathbb{Q}$-subalgebra $T\subset\mathbb{C}$. Adjoin the coefficients needed for their identities and invert

finitely many nonzero elements. This gives a proper smooth family of curves $\mathcal{X} \to V$, a finite étale reduced divisor $\mathcal{D} \to V$, and the same labelled G-action over a nonempty smooth open subscheme $V \subset \operatorname{Spec} T$. Smoothness and disjointness are open conditions. Faithfulness can be kept on this open set as well: for each nonidentity $h \in G$, the equalizer of h and the identity is a proper closed subset of the generic fiber, and this property persists after shrinking. The original inclusion $T \hookrightarrow \mathbb{C}$ defines a point of every such nonempty open, since it has generic underlying point.

Base change to $\overline{\mathbb{Q}}$ and take the connected component whose complex base change contains that point. After a finite extension this component and the family are defined over a number field. It is a smooth connected variety; its complex points form a connected, locally path-connected complex manifold and hence a path-connected space. It contains an $\overline{\mathbb{Q}}$-point, by taking a closed point in any nonempty affine open. Choose a piecewise smooth complex path from the original fiber to this fiber.

For precision, parallel transport along this path preserves the labelled action. A smooth horizontal splitting of the relative tangent sequence can be chosen to agree with the tangent splitting on $\mathcal{D}$: locally the divisor is a disjoint union of sections, and a partition of unity extends their prescribed splittings. Average this splitting over the finite group G. Averaging retains its right-inverse property and its prescribed values on the divisor, and makes it G-equivariant. The horizontal lift along the compact path has complete transport, because the family is proper and its fibers are compact. Its transport is an orientation-preserving diffeomorphism of pairs commuting with each labelled element of G. Thus the endpoint curve has exactly the required topological action. This constructs a deformation and its comparison diffeomorphism; it does not assert that the original complex curve is isomorphic to a curve over $\overline{\mathbb{Q}}$.

All coefficients of the endpoint curve, divisor, and the finitely many labelled automorphisms lie in one finite extension $K/\mathbb{Q}$. Let $x_K : \operatorname{Spec} K \to \mathscr{M}$ be the resulting moduli point. Choose the geometric path to x_K supplied by the transport just constructed and the finite-cover comparison. The morphism x_K then supplies a continuous homomorphism

$$s_K : G_K \longrightarrow \Pi_{\mathrm{ar}}$$

whose composite with $\Pi_{\mathrm{ar}} \to G_{\mathbb{Q}}$ is the inclusion $G_K \hookrightarrow G_{\mathbb{Q}}$. Under the transported marking, $R \circ s_K$ is the Galois outer action on the punctured endpoint curve.

Every $h \in G$ is an automorphism over K of that curve and its marked divisor. Naturality of the fiber functor on finite étale covers gives

$$(R \circ s_K)(\sigma)\, h = h\, (R \circ s_K)(\sigma) \quad \text{in } B \qquad (\sigma \in G_K). \tag{F.6.5}$$

Indeed, pullback by h commutes with scalar conjugation by σ; choosing paths for the individual base points can alter the induced automorphisms only by inner automorphisms of P. Passing to B gives precisely the equality written. The equivariant transport aligns these h with the original labelled inclusion, so (F.6.5) places $R(s_K(G_K))$ in $C_{A_{\mathrm{ar}}}(G)$ itself.

The image of (F.6.1) therefore contains $\overline{R}(G_K)$. The latter has index at most $[K : \mathbb{Q}]$ in Γ_{ar} by the surjectivity of (F.6.4). It is compact and hence closed; a closed finite-index subgroup of a profinite group is open, as proved in F.3. Every subgroup containing it is a union of its open cosets. This proves openness in (F.6.1), with kernel $C_H(G)$ by intersection with H.

Finally $q^{-1}(\Gamma_{\mathrm{ar}}) = A_{\mathrm{ar}}$, since $H \subset A_{\mathrm{ar}}$ and $\Gamma_{\mathrm{ar}} = q(A_{\mathrm{ar}})$. An element of $C_N(G)$ whose quotient lies in Γ_{ar} therefore already belongs to $C_{A_{\mathrm{ar}}}(G)$. This proves both inclusions in (F.6.2). Ordering the divisor and using the pure moduli stack changes none of the maps or arguments when G belongs to the pure oriented group.

For §30, folios 195–196, scans 385–387, formula (F.6.2) is the arithmetic open-image assertion with its exact quotient made explicit. F.3 supplies the centralizer image in the entire quotient

N/H; F.6 supplies its openness after intersection with the arithmetic image for every finite geometric G. The inclusion $\Gamma_{\mathrm{ar}} \hookrightarrow N/H$ is the comparison map between the two statements. Its image has not been identified with all of N/H here, so no larger openness assertion is obtained by suppressing that inclusion. □

F.7. [Open] The nonsolvable centralizer comparison and its checked finite image. For an oriented mapping class group M of genus at most two, the normalizer and centralizer comparisons for finite solvable subgroups have the complete proof below. For a geometric $A_5 \le M = \Gamma^+_{0,\nu}$, the equality

$$C_{\widehat{M}}(A_5) = \overline{C_M(A_5)} \tag{F.7.1}$$

is stated in [7, Theorem C(iii)] and [9, Theorem 5.1(iii)]. The displayed proofs contain an additional closure–intersection equality for which we have not verified a justification. We do not assert that (F.7.1) is false.

There is nevertheless equality of their images in the finite puncture-permutation quotient. Write the invariant puncture set as a union of some of the exceptional orbits of sizes $12, 20, 30$ and m free orbits of size 60. Both images fix the exceptional punctures individually and induce the full group $A_5^m \rtimes S_m$ on the free orbits. In particular, with only exceptional orbits the unresolved part of (F.7.1) lies in the pure profinite mapping class group; its discrete right side is trivial.

Manuscript: §30; folios 194–196; scans 383–387; Part D(b) and Part F(b): nonsolvable completion centralizers.

Source audit and proved comparisons. The supplied preprint snapshot of [7] has since appeared in *São Paulo Journal of Mathematical Sciences*, volume 20, article 35, published 23 September 2026, DOI `10.1007/s40863-026-00558-y`. The publisher's full text was checked as well: Proposition 2.4 retains the same displayed closure–intersection step, and Lemma 3.2 still invokes that proposition. The audit below applies to both inspected versions; it is a proof-step audit and makes no claim based on the paper's publication status.

For every finite subgroup $J \le M$, there are canonical continuous maps

$$\widehat{C_M(J)} \longrightarrow \overline{C_M(J)} \hookrightarrow C_{\widehat{M}}(J). \tag{F.7.2}$$

The first is onto by density and compactness. Its injectivity states that the topology induced from M on the discrete centralizer is its full profinite topology. Surjectivity of the second arrow is the ambient centralizer comparison. These are two specific properties of the displayed maps; injectivity of the first arrow does not supply surjectivity of the second.

In [7, Proposition 2.4], choose the displayed generators $x_1, \dots, x_k$ of the finite group, each of prime-power order. The prime-power centralizer theorem does prove

$$C_{\widehat{M}}(J) = \bigcap_i C_{\widehat{M}}(x_i) = \bigcap_i \overline{C_M(x_i)}. \tag{F.7.3}$$

The source then identifies this last intersection with $\overline{\bigcap_i C_M(x_i)}$. The same final equality occurs in the proof of [9, Theorem 5.1(iii)], there indexed by all $x \in J$. Neither displayed argument supplies a closure–intersection hypothesis. This exact step is what is left unverified here. The later virtually compact special application in [9, §6.2] has a separate relatively quasiconvex product-separability input; no such input has been proved here for the punctured-sphere mapping class group.

A noncircular solvable proof. We first record the prime-power inputs used in the induction. Goodness and finite virtual cohomological type of mapping class groups of genus at most two, and their finite-index subgroups, permit the direct application of [9, Theorem A]. Thus for a finite p-subgroup A we have

$$N_{\widehat{M}}(A) = \overline{N_M(A)}, \qquad C_{\widehat{M}}(A) = \overline{C_M(A)}. \tag{F.7.4}$$

The proof of [7, Theorem A and Corollary B] also supplies the full induced profinite topology on $C_M(A)$ and hence on $N_M(A)$ in this prime-power case. Here is why this use avoids the preceding gap. Its free-action argument is the restriction-injectivity argument on the finite covering surface group. Its ramification-removal argument uses Proposition 2.4 only for the specified group A, now a p-group, in the good configuration-group extension. For that application the second equality in (F.7.4), directly from [9, Theorem A], replaces Proposition 2.4's intersection argument. Thus the two required inputs are justified without using any nonsolvable assertion. The passage from centralizer to normalizer topology uses its finite-index kernel under conjugation into $\mathrm{Aut}(A)$; this kernel is open in the induced topology, as is detected by any finite quotient in which A injects.

We prove the solvable normalizer comparison, together with its full induced topology, by induction on the order of the finite solvable group, simultaneously for all finite-index subgroups of these mapping class groups. Finite-index passage preserves the comparison: intersect the completed normalizer with the corresponding open subgroup and use density in an open set.

For a nontrivial finite solvable J, take a nontrivial p-primary subgroup A of the last nonzero term of its derived series. That term is abelian, so its p-primary subgroup is characteristic in J. Consequently every normalizer of J normalizes A. The discrete covering normalizer sequence is

$$1 \longrightarrow A \longrightarrow N_M(A) \longrightarrow Q \longrightarrow 1, \tag{F.7.5}$$

where Q is a finite-index subgroup of the mapping class group of the quotient surface with its branch and puncture decorations. Its genus is again at most two. The exact quotient map is induced by descent of the commuting covering diagram; the decorations record the original punctures, the removed branch points and the covering subgroup. This is the discrete normalizer sequence in [7, §2.1].

By the prime-power topology input, the completed normalizer in (F.7.4) is $\widehat{N_M(A)}$. Completing (F.7.5) therefore gives its quotient by the finite embedded kernel A as $\widehat{Q}$. The subgroup $J/A \leq Q$ is smaller and solvable. The induction hypothesis applies to its normalizer in $\widehat{Q}$. The inverse image of that normalizer under the quotient in (F.7.5), on either side, is exactly the normalizer of J: J is the full inverse image of J/A because it contains A. Pulling back the dense discrete normalizer proves

$$N_{\widehat{M}}(J) = \overline{N_M(J)}. \tag{F.7.6}$$

It also proves the topology assertion. Indeed the completion of this inverse image maps to $\widehat{N_M(A)}$; on the quotient by A it is injective by induction, and on A it is the specified inclusion. Its kernel is therefore trivial. This completes the induction.

Conjugation gives a continuous homomorphism

$$N_{\widehat{M}}(J) \longrightarrow \mathrm{Aut}(J)$$

to a finite group. Its kernel is the ambient centralizer and is open. The dense subgroup $N_M(J)$ from (F.7.6) meets that open kernel densely, and its intersection there is exactly $C_M(J)$. This proves

$$C_{\widehat{M}}(J) = \overline{C_M(J)}$$

for every finite solvable J, without taking an unjustified intersection of closures.

The finite image for A_5. Realize the geometric A_5 as the icosahedral rotation group. Its exceptional point stabilizers have orders $2, 3, 5$, and the corresponding transitive orbits are A_5/C_2, A_5/C_3, A_5/C_5. Every permutation commuting with this transitive action is right multiplication by an element of $N_{A_5}(C_d)/C_d$. This quotient has order two in each of the three cases; its nonidentity permutation exchanges the two poles of each relevant rotation axis.

An orientation-preserving homeomorphism commuting with an odd-order nonidentity rotation fixes its two poles individually. The local oriented rotation numbers at the two poles are opposite,

and an orientation-preserving commuting homeomorphism preserves this local rotation number. This excludes the nonidentity permutation on the 12- and 20-point orbits for the discrete cyclic centralizers. Their completed centralizers have the same finite permutation images by (F.7.4). The full A_5 centralizer lies in these cyclic centralizers, so it fixes these exceptional punctures individually.

For an order-two stabilizer choose a tetrahedral subgroup $A_4 \leq A_5$ containing it. An orientation-preserving homeomorphism commuting with A_4 fixes every ramification point of the A_4 quotient. To see this, its quotient fixes the branch point of order two. It also fixes each of the two branch points of order three, rather than exchanging them, by the preceding local rotation argument. Isotope the quotient map to the identity relative to these three branch points and lift the isotopy. The lift stays in the centralizer; its endpoint is a deck transformation in $Z(A_4) = 1$. The ramification set is finite and invariant along the isotopy, so every one of its points is fixed. The usual discrete covering-centralizer comparison provides a commuting representative for a centralizing mapping class. Since A_4 is solvable, the proved centralizer comparison transfers this fixed-puncture property to $C_{\widehat{M}}(A_4)$. It therefore holds for $C_{\widehat{M}}(A_5)$, excluding the last possible nonidentity permutation on the 30-point orbit.

On m free orbits the finite commutant is $A_5^m \rtimes S_m$. Every element of this commutant is already realized by the discrete centralizer. For a direct construction, remove the three branch points from the quotient sphere. Its unramified A_5-cover has surjective monodromy onto A_5. Move one of the m quotient punctures around a loop, keeping all the others fixed, and lift the isotopy starting at the identity. The lift commutes with A_5 and acts on that one free orbit by its specified monodromy, while fixing the other orbits and all ramification points. These loops realize each factor A_5. Half-twists interchanging the quotient punctures realize S_m modulo these factors. Thus the displayed wreath product is the exact finite permutation image on both sides of (F.7.1).

The program `code/part_a/a5_centralizer_permutations.py` constructs A_5 as the sixty even permutations of five letters. It checks its order-two and order-three generating pair, the three coset actions, normalizer orders $4, 6, 10$, and commutants of order two. The fixed-pole constraints leave only the identity in each exceptional block; the order-two block uses the six ramification points for the specified A_4.

For completeness, the residual intersection question has an exact finite-quotient formulation. Take $u = (1\ 2)(3\ 4)$ and $v = (1\ 3\ 5)$ in A_5, which generate it, and put $D_u = C_M(u)$, $D_v = C_M(v)$. For open normal subgroups $V \subset U$ of M, let $D_u(V), D_v(V)$ denote their images in M/V. Compactness of coset fibres proves

$$\operatorname{im}\bigl(C_{\widehat{M}}(A_5) \to M/U\bigr) = \bigcap_{V \subset U} \operatorname{im}\bigl(D_u(V) \cap D_v(V) \to M/U\bigr). \tag{F.7.7}$$

Indeed choose a lift $t \in \widehat{M}$ of a point in the right side. For each $V \subset U$, the set

$$(\overline{D_u}\,\overline{V}) \cap (\overline{D_v}\,\overline{V}) \cap t\overline{U}$$

is nonempty and compact. These sets are directed under refinement, so their finite intersections are nonempty. Compactness produces a single element in their intersection, which is $\overline{D_u} \cap \overline{D_v} \cap t\overline{U}$. Conversely any such element supplies every finite coordinate. Equations (F.7.3) and (F.7.7) identify the remaining assertion precisely: its right side must equal the image of $C_M(A_5)$ for every U. The finite permutation computation proves that equality at the puncture-permutation level. It does not establish it at every refinement. With only exceptional orbits the discrete centralizer is trivial, since the quotient sphere has just the three differently ordered branch points and $Z(A_5) = 1$; the still unproved assertion is then that the pure completed centralizer is trivial.

The minimal twelve-puncture case. There is a further exact localization of this remaining group. Write $P = P\Gamma_{0,12}$ and choose $A = A_4 \leq A_5$. The action of A on the twelve vertices is

free and transitive. Its quotient has one free puncture and three branch points of orders $2, 3, 3$. The labelled commuting homeomorphism group on the filled sphere is identified, by descent and the unique commuting lift, with the identity homeomorphism component of the quotient fixing those three branch points. It is contractible. Evaluating the free quotient puncture consequently identifies the mapping-class centralizer with the free group $\pi_1(S^2 \setminus \{3 \text{ branch points}\}) = F_2$. Its permutation on the free fibre is the surjective monodromy onto A_4. The pure centralizer is therefore the exact kernel

$$1 \longrightarrow C_P(A) = F_{13} \longrightarrow F_2 \longrightarrow A_4 \longrightarrow 1. \tag{F.7.8}$$

Indeed it is the fundamental group of $X = S^2 \setminus R_A$, where the A_4 ramification set R_A has $6+4+4 = 14$ points; its rank is also $1+12(2-1) = 13$ by Schreier. Label the vertices by $g \in A_4$ using a chosen vertex z_0. The inclusion is the configuration map $z \mapsto (gz)_{g \in A_4}$, followed by the quotient by Möbius transformations. The proved solvable comparison and full topology identify its completion with $C_{\widehat{P}}(A)$. In particular the unknown group $C_{\widehat{M}}(A_5)$ lies in this specified embedded $\widehat{F}_{13}$.

Each of the six subgroups $C_5 \leq A_5$ has two fixed vertices and two free five-cycles. Retain either one fixed vertex and one five-cycle, giving twenty-four six-point sets, or retain both fixed vertices and one five-cycle, giving twelve seven-point sets. For each retained set I, the continuous forgetting map r_I is defined on $\widehat{P}$ and is equivariant for the specified C_5. In either target the quotient has exactly its two branch points and one free puncture. Its discrete C_5 centralizer is therefore exactly the deck group C_5: the quotient mapping class group fixing these three decorated points is trivial. Prime-power comparison gives the same completed centralizer. Its intersection with the pure group is trivial, since the deck group permutes the five-cycle. It follows that

$$C_{\widehat{M}}(A_5) \subset C_{\widehat{P}}(A) \cap \bigcap_{I \text{ among these 36 sets}} \ker(r_I). \tag{F.7.9}$$

The six-point maps in (F.7.9) have a proved common kernel even on the discrete F_{13}. The antipodal permutation on the twelve vertices commutes with A_5. In the regular A_4 labelling it is right multiplication by an involution $h \in A_4$. Every retained six-point set contains exactly one member of each antipodal pair. Collisions of two entries $g_i z, g_j z$ at a ramification point with stabilizer $H \leq A_4$ occur precisely when $g_j^{-1} g_i \in H$; these are the right cosets gH. Thus there is no collision among the retained entries at either fixed point of h. Each six-point configuration map extends across both of these deleted points of X. Their nontrivial peripheral meridians in F_{13} map to the identity under all twenty-four maps. Completion retains these meridians as nontrivial primitive procyclic subgroups. Consequently the joint six-point map on the embedded $\widehat{F}_{13}$ is noninjective by these explicit elements.

Every seven-point set retains a collision at each of the fourteen ramification points: seven choices from six pairs force a pair, and seven choices from four triples force a triple collision. The program verifies all twelve sets and the preceding common invisible axis for the six-point sets, using right cosets throughout. It also computes the induced maps on $H_1(F_{13}, \mathbb{F}_{101})$. A meridian at an A_4 ramification point maps to the sum of the pair generators within its collision clusters. In the target pure sphere group the pair generators satisfy the vertex-sum and total-sum relations. The two fixed points of each cyclic stabilizer give the same homology column. Each seven-point map and their joint map have rank six, with seven-dimensional kernel; the joint six-point map has rank five. These are exact calculations of the indicated abelian quotients. They neither produce a nontrivial A_5 centralizer element nor prove injectivity of the joint seven-point map on $\widehat{F}_{13}$.

F.8 proves that all twelve seven-point kernels in (F.7.9) are the same, both discretely and after completion. It further proves that the discrete kernel contains no nonidentity reducible mapping class and that the completed kernel meets every conjugate of every full source cusp subgroup

trivially. The exact four-point characters detecting those cusps and all eighty-four finite subset–inertia checks are supplied there. F.9 subsequently constructs an explicit nontrivial common kernel element with a full procyclic closure, and proves that all twenty-four six-point restrictions factor through these seven-point maps. The thirty-six-fold kernel is therefore nontrivial. The full centralizer comparison and geometric conjugacy remain unproved; no converse to the inclusion (F.7.9) is inferred.

This audit concerns the comparison requested for §30, folios 194–196, scans 383–387. F.6's arithmetic open-image proof applies to A_5 without using (F.7.1), so that proved arithmetic assertion does not depend on the unresolved intersection. □

F.8. [Proved] The seven-point kernel and its peripheral restrictions. In the regular $A = A_4$ labelling of (F.7.8), all twelve seven-point subsets in (F.7.9) form one free orbit for left multiplication by A. Consequently their kernels on the specified F_{13} are equal; their kernels on its profinite completion are equal as well. The completed kernel meets every conjugate of every full source cusp subgroup $\widehat{\mathbb{Z}}$ trivially. The discrete kernel contains no nonidentity reducible mapping class under the specified index-twelve inclusion $F_{13} \leq P\Gamma_{0,4}$.

Manuscript: §30; folios 194–196; scans 383–387; Part F: nonsolvable finite centralizer comparison.

Proof. Here is the actual comparison of maps. Regard the target as $M_{0,I}$ with its seven labels $I \subset A$, and write

$$f_I : X \longrightarrow M_{0,I}, \qquad z \longmapsto [(gz)_{g\in I}].$$

For $h \in A$, let $r_h : M_{0,I} \xrightarrow{\sim} M_{0,hI}$ be the relabelling which assigns to the label hg the former coordinate with label g. The configuration defining $f_{hI}(z)$ has coordinate $h(gz)$ at that label. Applying the same Möbius transformation h^{-1} to every coordinate therefore gives the exact identity of maps

$$f_{hI} = r_h \circ f_I.$$

This identity also identifies the base configurations; it induces an isomorphism of the target fundamental groups intertwining the two homomorphisms from $\pi_1(X)$. Functoriality of profinite completion gives the same commuting diagram for their completions. Since r_h and its completion are isomorphisms, the two kernel equalities follow.

The exact label certificate is `code/part_a/seven_point_same_kernel.py`. It constructs the twelve subsets in the action A_5/C_5 and gives all twelve left multipliers and the corresponding seven-label permutations. With the certificate's zero-based vertex numbering, one subset is $I = \{0, 1, 2, 3, 4, 7, 10\}$. Its intersections with the three cosets of the normal subgroup $V_4 \leq A_4$ are respectively

coset	I retained in that coset
$\{0, 2, 7, 11\}$	$\{0, 2, 7\}$
$\{1, 4, 5, 9\}$	$\{1, 4\}$
$\{3, 6, 8, 10\}$	$\{3, 10\}$.

The program checks that none of the twelve subsets contains a full V_4 coset. Thus the four-label V_4-orbit cross-ratio is not one of the forgetful projections of any of these seven-point maps.

The intersection of the twelve kernels in (F.7.9) is therefore the kernel of any one of these named maps. F.9 below proves it nontrivial by an exact braid certificate. This discrete kernel contains no nonidentity reducible mapping class in the source subgroup of $P\Gamma_{0,4}$. To prove this, use the exact A_4 covering

$$X = S^2 \setminus R_A \longrightarrow S^2 \setminus \{b_2, b_3, b_3'\}$$

and its index-twelve injection on fundamental groups in (F.7.8). On the four-punctured quotient sphere, every essential curve separates the moving puncture q and one branch point b from the

other two branch points. Let $e = 2$ or 3 be the order at b. The preimage of the disk containing q, b consists of $12/e$ disks, each containing e of the twelve points above q. These assertions hold for every such curve: its monodromy subgroup is the conjugate of the inertia subgroup at b, so its connected components have degree e, and each is a disk branched only at the one point above b.

The lift of a nonzero twist power belonging to the index-twelve subgroup is a product of twists about these boundary curves. More precisely, its exponent is divisible by e, and each lifted twist has exponent equal to the original exponent divided by e. After forgetting five points, at least one of these boundaries remains essential: seven retained points distributed among six pairs or four triples force a disk containing at least two retained points, and its complement contains at least four. The surviving twists all have the same nonzero sign; parallel essential components add their exponents. Independence of twists about pairwise disjoint nonisotopic essential curves makes the resulting mapping class nonidentity. Finally, each nonidentity reducible pure mapping class of a four-punctured sphere is a nonzero power of one such twist: its invariant multicurve has just one component, and cutting along it gives two three-punctured spheres. This proves the asserted exclusion of reducible elements from the discrete kernel.

There is also an injective detector for each full profinite cusp. Let c be a primitive peripheral generator at any one of the fourteen punctures of X. Its image in $P\Gamma_{0,4}$ is a conjugate of T^e, where T is a twist as above. Thus its lift is the indicated product of twists of exponent one. Choose two retained labels in an essential lifted disk and two outside that disk. Forgetting all other labels gives a specified map to $P\Gamma_{0,4}$. The inside disk contributes one twist U. Among the other disks, another twist survives exactly when both outside labels lie in one disk; it then gives the same isotopy class and the same sign. Hence the image of c is U^d with $d \in \{1, 2\}$, up to conjugation.

There is a character $\ell : P\Gamma_{0,4} \to \mathbb{Z}$ with $\ell(U) = 1$. For example the three peripheral twists in the free group $P\Gamma_{0,4} = F_2$ have abelianization vectors $(1, 0), (0, 1), (-1, -1)$, and an essential twist is conjugate to one of these under the pure group. Choose the corresponding integral linear functional. Completing this character and the two forgetting maps gives a continuous homomorphism

$$\widehat{F}_{13} \longrightarrow \widehat{P\Gamma}_{0,7} \longrightarrow \widehat{P\Gamma}_{0,4} \longrightarrow \widehat{\mathbb{Z}}$$

whose restriction to $\overline{\langle c \rangle}$ is $a \mapsto da$. Multiplication by 1 or 2 is injective on $\widehat{\mathbb{Z}}$. The cusp is a full embedded $\widehat{\mathbb{Z}}$ in $\widehat{F}_{13}$, since its primitive generator is a member of a free basis after choosing the omitted sphere peripheral. The same final character detects every conjugate $gc^a g^{-1}$ with $g \in \widehat{F}_{13}$, because its target is abelian. This proves the asserted trivial intersection of the completed kernel with all conjugate cusp groups. The program gives the four retained labels and d in each of the $12 \cdot 7 = 84$ subset–inertia-partition checks; the two fixed points of each inertia subgroup have the same partition.

The exclusion just proved confines nonidentity discrete kernel elements to pseudo-Anosov mapping classes. F.9 supplies an explicit such element by an exact braid certificate. No assertion about a Nielsen–Thurston type for profinite elements is used.

The preceding rank-six calculation is a calculation of its induced abelian map, and does not identify the kernel of the fundamental-group homomorphism. The separate file `code/part_a/seven_point_braid_explore.py` contains a numerical crossing exploration; it is explicitly exploratory and is not used in this map comparison. F.9 replaces the discovery computation by a separate exact rational certificate of a kernel relation. □

F.9. [Proved] An explicit common kernel element for the seven-point maps. Every one of the twelve seven-point maps in F.7–F.8 has a nontrivial kernel, both on the specified discrete F_{13} and on its full profinite completion. With the source coordinate, basepoint, and

meridians specified below, the same element

$$w = m_{-1} m_i m_1^{-1} m_i^{-1} \tag{F.9.1}$$

lies in all twelve kernels. A source winding character sends w to 1, so its procyclic closure is a full $\widehat{\mathbb{Z}}$ contained in their common completed kernel. The joint kernel of all thirty-six six- and seven-point maps in (F.7.9) is exactly this same kernel. This refutes injectivity of these particular restriction maps; the residual A_5 centralizer comparison in F.7 remains unresolved.

Manuscript: Section 30, folios 194-196, scans 383-387; Part F, nonsolvable A5 centralizer comparison on the twelve-punctured sphere.

Proof. **The specified source and target.** Use the faithful tetrahedral action generated by

$$z \longmapsto -z, \qquad z \longmapsto \frac{z+i}{z-i}.$$

Its fourteen ramification points are infinity and the zeros of

$$D(z) = z(z^4 - 1)(z^8 + 14z^4 + 1).$$

Thus $X = \mathbf{P}^1 \setminus R_{A_4}$ has $\pi_1(X, b) = F_{13}$ for $b = 5 + 4i$. The seven labels $I = \{0, 1, 2, 3, 4, 7, 10\}$ of F.8, in that order, have the following coordinate functions:

$$(m_0(z), \dots, m_6(z)) = \left(z, \frac{i(z+1)}{z-1}, -\frac{1}{z}, -\frac{z-i}{z+i}, \frac{i(z-1)}{z+1}, -z, \frac{z-i}{z+i} \right). \tag{F.9.2}$$

Here the subscripts $0, \dots, 6$ denote positions in the displayed seven-set, not its original labels. The exact finite part of the certificate below verifies all 144 matrix products in this A_4, its correspondence with the labels in F.8, and the ramification polynomial D by multiplying the fixed-point polynomials of its seven nontrivial cyclic subgroups.

For $z \in X$, let

$$L_z(t) = \frac{(t - m_0(z))(m_1(z) - m_6(z))}{(t - m_6(z))(m_1(z) - m_0(z))}.$$

The seven orbit points are distinct, so this is the unique Möbius transformation carrying the first, second, and seventh points to $0, 1, \infty$. Define the ordered configuration map

$$\Phi : X \longrightarrow \mathrm{Conf}_6(\mathbf{C}), \qquad z \longmapsto (R_0(z), \dots, R_5(z)), \quad R_j(z) = L_z(m_j(z)). \tag{F.9.3}$$

The rational functions extend across any removable singularities in their displayed expressions; their only poles are among R_{A_4}. In particular $R_0 = 0$ and $R_1 = 1$. Appending the seventh point ∞ and taking the labelled Möbius quotient gives a map

$$q : \mathrm{Conf}_6(\mathbf{C}) \longrightarrow M_{0,I} \quad \text{with} \quad q \circ \Phi = f_I. \tag{F.9.4}$$

These are actual continuous maps, with the displayed base configurations. The induced map of fundamental groups therefore factors through the pure braid group $P_6 \leq B_6$. The inclusion here is induced by the cover from ordered to unordered configurations and is injective by path lifting. We shall prove $\Phi_*(w) = 1$ in B_6, which proves $f_{I,*}(w) = 1$ by this factorization.

The three based source meridians. Put $r = 27/1000$. For $c = -1, i, 1$, set

$$t_c = (\mathrm{Re}\, c - r) + 4i, \quad b_c = c - r - ri, \quad d_c = c + r - ri, \quad e_c = c + r + ri, \quad a_c = c - r + ri.$$

Let m_c be the loop obtained by traversing the straight polygonal segments with successive vertices

$$b,\ t_c,\ b_c,\ d_c,\ e_c,\ a_c,\ b_c,\ t_c,\ b. \tag{F.9.5}$$

The square is counterclockwise; its stem is followed by its reverse. Each square contains the single ramification point c. This also follows directly from the above polynomial: the other eight finite points have real and imaginary coordinates with absolute values $(\sqrt{3}-1)/2$ or $(\sqrt{3}+1)/2$, whereas the remaining finite points are $0, \pm 1, \pm i$. The exact segment check below also verifies that every stem and every square edge avoids all zeros of D. Source loop products in (F.9.1) are chronological concatenation.

Exact extraction of the three braids. Multiply every coordinate in (F.9.3) by $\eta = 1 + (173/1000)i$ and project to its real part. This multiplication is an orientation-preserving homeomorphism of the plane. At the basepoint the six coordinate positions, in increasing projected order, are

$$(0, 5, 3, 1, 4, 2).$$

Identify this chamber with the standard real ordered base configuration by a path within the chamber. The chamber of any fixed strict real ordering is convex, so the identification is unique up to homotopy. Read crossings chronologically. Write j for σ_j, and $-j$ for σ_j^{-1}; the positive sign means that the strand on the left immediately before the crossing has larger imaginary coordinate. This fixes both the numbering and the crossing convention.

The executed certificate `code/part_a/seven_point_kernel_f9.py` uses only rational and Gaussian-rational arithmetic. For a segment $z = p + (q-p)t$, $0 \le t \le 1$, and each pair of normalized coordinates, write their projected complex difference before taking real parts as $A(t)/B(t)$ with $A, B \in \mathbf{Q}(i)[t]$. It verifies that A, B, every coordinate denominator, and $D(p + (q-p)t)$ have no complex zero on the real segment: the real and imaginary parts have a greatest common divisor with no real root there. The possible crossings are precisely the roots in $(0, 1)$ of

$$P(t) = \mathrm{Re}(A(t)\overline{B(t)}). \tag{F.9.6}$$

Clearing rational denominators preserves its roots. Sturm isolation over $\mathbf{Q}$ enumerates all of them. Their rational isolating intervals are pairwise disjoint, their multiplicities are one, and $\mathrm{Im}(A\overline{B})$ has a certified constant nonzero sign on each interval. No endpoint is a crossing. The two strands are checked to be adjacent in the current order at every crossing, and the final order is checked against direct exact evaluation at the endpoint.

There are 18 different segments up to reversal, 270 pair checks, and 64 isolated crossings in these segments; the largest crossing-polynomial degree is 12. Reversed segments give the inverse words. Between successive crossings the path lies in a convex ordering chamber. At a crossing only the named adjacent pair changes order, with its certified nonzero imaginary separation. Straightening these intervals and crossing neighborhoods gives the usual elementary braid pieces. Thus the recorded words represent the actual monodromies, not just a linear representation of them. This is also the projection-isotopy argument of Artin's Theorem 3 [2].

The freely reduced words for m_{-1}, m_i, m_1 are, respectively, A, B, C below; a line break only separates consecutive parts of the same word:

$$\begin{aligned}
A = (&1, 4, 5, 2, 1, 3, 2, 3, 2, 1, 2, -5, -4, -5,\\
&-5, 4, 5, -2, -1, -2, -3, -2, -3, -1, -2, -5, -4, -1),\\
B = (&1, 4, 5, 2, 3, 1, 4, 2, 3, 4, -3, 1, 2, 3, 4, 5,\\
&-2, 5, 4, 3, 2, 1, -4, -3, -2, -4, -1, -3, -2, -5, -4, -1),\\
C = (&1, 4, 5, 2, 3, 4, 1, 2, 3, 4, 5, -1, 3, 2, -5, -4,\\
&-4, 5, -2, -3, 1, -5, -4, -3, -2, -1, -4, -3, -2, -5, -4, -1).
\end{aligned} \tag{F.9.7}$$

The braid identity and the source detector. Use the faithful Artin substitution on $F_6 = \langle X_1, \ldots, X_6 \rangle$:

$$\sigma_j(X_j) = X_j X_{j+1} X_j^{-1}, \qquad \sigma_j(X_{j+1}) = X_j, \qquad \sigma_j(X_k) = X_k \ (k \neq j, j+1). \tag{F.9.8}$$

Its inverse is used for a negative letter. Products of substitutions are composed from the displayed word order; the certificate implements composition by substitution and cancellation of adjacent inverse letters. Artin's Theorem 14, with the inverse generator formula on p. 113, proves faithfulness [2]. Simultaneously reversing all generator signs is an automorphism of the braid group, and reversing the stacking convention is an anti-automorphism; neither changes the identity test.

For the exact words (F.9.7), substitution gives

$$\mathrm{Artin}(AB) = \mathrm{Artin}(BC), \qquad \mathrm{Artin}(ABC^{-1}B^{-1})(X_j) = X_j \quad (1 \le j \le 6). \tag{F.9.9}$$

This is an integer free-word calculation. The output records all six freely reduced common images for the first equality and all six single-letter images for the second. Faithfulness gives $AB = BC$ in B_6. By the certified identification of the three words with the three source monodromies, (F.9.1) lies in $\ker \Phi_*$, and hence in $\ker f_{I,*}$ by (F.9.4).

The map $X \to \mathbf{C}^*$, $z \mapsto z + 1$, gives the integral winding character

$$\ell : \pi_1(X, b) \longrightarrow \mathbb{Z}, \qquad \ell(m_{-1}) = 1, \quad \ell(m_i) = \ell(m_1) = 0.$$

Therefore $\ell(w) = 1$ and $w \neq 1$. Completing this exact character gives $\widehat{\ell} : \widehat{F}_{13} \to \widehat{\mathbb{Z}}$ with $\widehat{\ell}(w) = 1$; this proves nontriviality in the full completion without a detection assumption. The continuous power map $\widehat{\mathbb{Z}} \to \widehat{F}_{13}$, $a \mapsto w^a$, has composite $a \mapsto a$ with $\widehat{\ell}$, so it is injective. Its image lies in the closed completed kernel.

Finally F.8 gives the actual map identities $f_{hI} = r_h \circ f_I$ for the twelve left translates hI, with r_h a labelled target isomorphism. They carry this same source word, and all its profinite powers, into the identity in every target.

The twenty-four six-point conditions in (F.7.9) follow from these twelve conditions by specified forgetting maps. For one of the six subgroups $C_5 \le A_5$, let p_+, p_- be its fixed poles and let O be either of its two five-cycles. Put

$$K = O \cup \{p_+, p_-\}, \qquad J_\pm = O \cup \{p_\pm\}.$$

Forgetting the other pole gives the labelled morphism $d_\pm : M_{0,K} \to M_{0,J_\pm}$ and the exact identity $f_{J_\pm} = d_\pm \circ f_K$ on the same based source. The twelve choices of (C_5, O) give all twelve seven-sets, and their two pole choices give all twenty-four six-sets. The separate executed certificate `code/part_a/seven_point_six_factorization_f9.py` lists all twenty-four containments, the omitted pole, and the six positions retained in each seven-tuple. Functoriality of fundamental groups and of full profinite completion proves on both sources that

$$\bigcap_{I \text{ among all } 36} \ker f_{I,*} = \bigcap_{K \text{ among the } 12 \text{ seven-sets}} \ker f_{K,*}. \tag{F.9.10}$$

For the completed equality the starred maps in (F.9.10) mean their completed homomorphisms. Consequently (F.9.1) and its full procyclic closure also satisfy every six-point condition.

The common-kernel injectivity route in F.7 therefore fails. The necessary seven-point restrictions of an element of the residual A_5 centralizer still vanish, but the converse implication has not been proved. No centralizer element is constructed here.

The discovery computation used numerical crossings and a finite Burau image to find the short candidate. Neither computation is an input to this proof. The file `code/part_a/seven_point_kernel_f9_output.json` contains the deterministic exact certificate: the coordinate functions, Gaussian-rational path vertices, every crossing polynomial, every rational root interval and sign, the three words, and the Artin identities. The companion stdout file records the successful execution. □

References

[1] J. M. Aarts and L. G. Oversteegen. Flowbox manifolds. *Transactions of the American Mathematical Society*, 327(1):449–463, 1991. Verified: the author-uploaded original published text was read, including Definition 1.1, Main Theorem 0.5, the construction in Section 4, Remark 4.4, and Definition 5.3 with Theorems 5.6–5.7 on page 460. The regularity condition is the existence of arbitrarily close arc embeddings through all nearby points. The cited Whitney local-section construction was also read in Section 3 of Aarts–Hagopian–Oversteegen, Pacific Journal of Mathematics 150 (1991), 1–12, in the publisher's primary PDF. The auxiliary global flow is not claimed to preserve the original operator action.

[2] Emil Artin. Theory of braids. *Annals of Mathematics, Second Series*, 48(1):101–126, 1947. Verified: the original article scan was opened. The projection-pattern isotopy lemma and Theorem 3 on p.103, Theorem 14 on p.112 (faithful braid substitutions), the generator and inverse formulas (14)–(15) on p.113, and the projection interpretation on pp.125–126 were read. F.9 uses exact rational crossing isolation and the faithful free-group substitution, not faithfulness of a linear braid representation.

[3] Mathieu Baillif and Alexandre Gabard. Manifolds: Hausdorffness versus homogeneity. *Proceedings of the American Mathematical Society*, 136(3):1105–1111, 2008. Verified: Theorem 1.1 on page 1106 and its full proof in Section 4, pages 1109–1110, were read in the published version deposited in the University of Geneva archive. The theorem allows non-Hausdorff manifolds initially and proves that a homogeneous Lindelof manifold is Hausdorff. The proof uses dense Hausdorff open sets, a countable subcover, the Baire property, and homogeneity.

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[5] Stephen J. Bigelow and Ryan D. Budney. The mapping class group of a genus two surface is linear. *Algebraic and Geometric Topology*, 1:699–708, 2001. Verified: original complete ten-page article opened. Lemma 2.2, pages 701–702, gives the capped-disc braid central extension for a sphere puncture stabilizer; Propositions 3.2–3.3, pages 703–704, give the genus-two hyperelliptic central extension. The full-twist abelianization is displayed on page 702.

[6] Marco Boggi. Automorphisms of profinite mapping class groups, 2026. verified: version6 of22April2026, the original53-page PDF and supplied TeX opened; Sections7.1–7.2 and equation(29) checked. F.6 uses the foundational arithmetic exact sequence, geometric comparison and universal etale monodromy construction, with its own spreading-out and equivariant-transport proof. No assertion from the main automorphism theorems is used.

[7] Marco Boggi. A congruence subgroup property for symmetric mapping class groups. *São Paulo Journal of Mathematical Sciences*, 20:Article 35, 2026. verified: supplied v2 source, original arXiv page, and publisher's full text opened; Theorem C and its induction inspected. F.7 records the unverified closure-intersection step in Proposition 2.4 and supplies the non-circular solvable normalizer/centralizer proof using the direct prime-power input. Published 23 September 2026. Publication status does not supply the missing nonsolvable intersection argument. The reflection argument in D.2 is proved separately.

[8] Marco Boggi and Pavel Zalesskii. A restricted magnus property for profinite surface groups. *Transactions of the American Mathematical Society*, 371:729–753, 2019. Verified: primary author preprint opened; Definition 1.3 and Proposition 3.6 read. Only positive integer exponents are used here, within the stated exponent range. Journal publication data cross-checked against a later primary author bibliography.

[9] Marco Boggi and Pavel Zalesskii. Finite subgroups of the profinite completion of good groups. *Bulletin of the London Mathematical Society*, 57:236–255, 2025. verified: complete supplied source and arXiv HTML opened; Theorems A and 6.3, including conjugacy and normalizer comparison for finite p-subgroups, checked. D.2 uses the geometric specialization for a characteristic prime-order rotation subgroup. F.7 identifies the unverified closure-intersection step in Theorem 5.1(iii); the prime-power comparison and the independent solvable proof do not use that step.

[10] Armand Borel. Sous-groupes compacts maximaux des groupes de Lie. In *Séminaire Bourbaki: années 1948/49–1949/50–1950/51, exposés 1–49*, number 1 in Séminaire Bourbaki, pages 271–279. Société mathématique de France, 1952. Verified: exposé 33, delivered December 1950. The principal theorem and definition of product notation in section 1, page 271, explicitly give a connected maximal compact subgroup K and a multiplication homeomorphism K times H1 times ... times Hs onto the connected Lie group, each Hj a closed one-parameter subgroup isomorphic to R. The displayed statement was visually checked; the reductions for semisimple covers and the general case in sections 3–4, pages 277–278, were read in the original scan.

[11] Ivan Bortnovskyi, Vasily A. Dolgushev, Borys Holikov, and Vadym Pashkovskyi. Accessing non-abelian quotients of the Grothendieck–Teichmueller group via elementary tools, 2026. verified: original complete arXiv v3 HTML opened 13 November 2026; version dated 6 August 2026. Theorem 4.3, Proposition 4.5, Theorem 4.6, Theorem 5.7 and Proposition 5.8 checked. Finite formulas independently verified on the quotient Cayley graphs in the attached code. Cited as a preprint.

[12] Beverly Brechner and Timo Erkama. On topologically and quasiconformally homogeneous continua. *Annales Academiae Scientiarum Fennicae, Series A I Mathematica*, 4(2):207–208, 1979. Verified: both pages of the original 1978/1979 volume were read. Theorem 2 classifies nondegenerate proper compact subcontinua homogeneous with respect to neighborhood extensions as simple closed curves. Only compact components are passed directly to this theorem.

[13] Martin R. Bridson, Marston D. E. Conder, and Alan W. Reid. Determining Fuchsian groups by their finite quotients. *Israel Journal of Mathematics*, 214:1–41, 2016. verified: supplied primary source and original arXiv version opened; Theorem 5.1 and Corollaries 5.2–5.3 checked. The proof below makes the passage from total-group conjugacy to kernel conjugacy explicitly.

[14] Martin R. Bridson and Paweł Piwek. Profinite rigidity for free-by-cyclic groups with centre. *Journal of the London Mathematical Society*, 111(6):e70181, 2025. verified: original full paper opened; Proposition 5.10 on finite subgroups in virtually free groups and its proof checked.

[15] Benjamin Collas, Séverin Philip, and Naganori Yamaguchi. Anabelian geometry for Deligne–Mumford curves, 2026. verified: supplied complete TeX source and primary arXiv record opened; Definition 2.1 and Theorems 1.3 and 2.4 checked. The direction of these assertions, including the prescribed orbifold presentation, is retained in D.22.

[16] Adrian Constantin and Boris Kolev. The theorem of Kerékjártó on periodic homeomorphisms of the disc and the sphere. *L'Enseignement Mathématique*, 40:193–204, 1994. verified: supplied complete TeX source opened; Lemma 2.5, Theorems 3.1 and 4.1, Corollary 4.2 and their proofs checked.

[17] Vasily A. Dolgushev and Jacob J. Guynee. GT-shadows for the gentle version of the Grothendieck–Teichmueller group, 2024. verified: original complete arXiv v2 PDF opened

13 November 2026. Proposition 3.14, Proposition 5.1, Theorem 5.2, Corollary 5.4 and Proposition 5.5 checked, including the topology and the change of coordinates lambda=2m+1. Cited as a preprint.

[18] David B. A. Epstein. Curves on 2-manifolds and isotopies. *Acta Mathematica*, 115:83–107, 1966. verified: original full paper opened; Theorem 5.8 for open and closed Mobius bands, Theorem 6.3 with its proper boundary-respecting homotopy hypothesis, and the topological Appendix checked. The annulus boundary interchange is explicitly excluded in C.10.

[19] Edward Fadell and Lee Neuwirth. Configuration spaces. *Mathematica Scandinavica*, 10:111–118, 1962. Verified: publisher's original eight-page scan opened; Theorems 1 and 3 on pages 111–113, the surface homotopy calculations in Section IV, and the aspherical configuration examples on pages 115–116 visually inspected. Publisher metadata confirms 1962, correcting OCR of the scanned year in one search extract.

[20] Benson Farb and Dan Margalit. *A Primer on Mapping Class Groups*, volume 49 of *Princeton Mathematical Series*. Princeton University Press, 2012. Verified: actual 489-page primary book copy opened. Theorems 1.2, 2.5, 4.6 (pp.97–98), the following extended-group and point-push convention remarks, Theorem 8.1 (p.221), Theorem 8.8 (p.234), and the comparison diagram on p.235 read. The lowercase URL path is required. A.12 retains the residual inner ambiguity correctly instead of using the informal uniqueness wording in the last paragraph of p.235.

[21] Steven Frankel. Quasigeodesic flows and sphere-filling curves. *Geometry & Topology*, 19(3):1249–1262, 2015. Verified: Lemma 2.7 and its proof on printed page 1258 apply to arbitrary closed planar separating sets, including unbounded sets. The connected frontier of the complementary domain V is contained in A and separates the specified points. The final printed phrase should select the component containing that frontier; the present proof writes this explicitly.

[22] J. Gómez, H. Pinedo, and C. Uzcátegui. The open mapping principle for partial actions of Polish groups, 2017. Verified: full primary text and Theorem 3.5 with its proof opened. For C.26 the action is ordinary, so the partial-action domains are the whole Polish group; the compact surface is nonmeagre Hausdorff. Only openness of the orbit evaluation map is used, with no local-section claim.

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[24] Fabrizio Grandoni, R. Ravi, Mohit Singh, and Rico Zenklusen. New approaches to multi-objective optimization. *Mathematical Programming*, 146(1–2):525–554, 2014. Verified: author version dated 17 July 2013, Theorem 4.7 and Section 4.3.2, including the signed-measure statement and proof. In the planar case used in C.37, one circular interval is obtained because at most two cuts leave at most three pieces in the two equal-mass halves. Journal metadata and pagination verified against the authors' institutional publication records. Application: manuscript Section 23, folios 140–141, scans 279–281.

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[26] Benedict H. Gross and Joe Harris. Real algebraic curves. *Annales scientifiques de l'École Normale Supérieure*, 14(2):157–182, 1981. verified: supplied PDF opened; Proposition 3.1 and its quotient-surface argument checked.

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[29] W. J. Harvey. On branch loci in Teichmüller space. *Transactions of the American Mathematical Society*, 153:387–399, 1971. verified: supplied PDF opened; Theorem 7 and the inverse-deck sign convention on pages 393–394 checked.

[30] Wolfgang Herfort and Pavel Zalesskii. Virtually free pro-p groups. *Publications Mathématiques de l'IHÉS*, 118:193–211, 2013. verified: original full article opened; Theorem 1.1 and the conclusions of Section 6 checked. D.18 supplies the torsion-free reduction and rank calculation.

[31] Wolfgang Herfort and Pavel A. Zalesskii. Virtually free pro-p groups whose torsion elements have finite centralizer. *Bulletin of the London Mathematical Society*, 40(6):929–936, 2008. verified: original complete nine-page author manuscript opened. Theorem 5 on manuscript page 4 explicitly states the malnormal free-factor intersection property for full free profinite products as well as pro-p products. D.35 uses only this full-profinite statement, not the paper's pro-p decomposition theorem. Section 5, Lemma 10, was also read: the full-profinite example S3 amalgamated with S3 over C2 explains why the paper's free-product conclusion cannot be imported wholesale. Publisher record independently opened to verify the 2008 publication, volume 40, pages 929–936, and DOI.

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[42] Vladimir Marković and Dragomir ˇ Sarić. The mapping class group cannot be realized by homeomorphisms, 2008. verified: original paper opened; definitions and Theorem 1.1, asserting no homeomorphism-group section for closed orientable genus at least two, checked. Only this theorem and its exact reduction are used.

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[52] John Pardon. The Hilbert–Smith conjecture for three-manifolds. *Journal of the American Mathematical Society*, 26(3):879–899, 2013. Verified: Theorem 1.5 together with the reduction in the introduction proves the Lie-group conclusion for faithful continuous locally compact group actions on connected three-manifolds. The present application is the explicit product action on a surface times the real line.

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[58] Gareth Wilkes. Relative cohomology theory for profinite groups, 2017. verified: original complete preprint opened; Propositions 1.16, 1.21, 1.26, 5.9 and 5.11 checked. The citation uses this exact preprint numbering.

[59] Gareth Wilkes. Classification of pro-p PD2 pairs and the pro-p curve complex. *Groups, Geometry, and Dynamics*, 14:917–948, 2020. verified: original arXiv v2 PDF opened; Theorem 1.9, Definition 3.2 and Theorem 3.3 read with every presentation parameter and the commutator convention. The finite-character specializations and basis changes are written out in D.19–D.20.

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[61] Hidehiko Yamabe. On an arcwise connected subgroup of a Lie group. *Osaka Mathematical Journal*, 2(1):13–14, 1950. Verified: the original two-page published proof and the University of Osaka version-of-record metadata were read. The theorem supplies an analytic Lie subgroup for an arcwise connected subgroup of a Lie group. The present application separately proves closedness and equality with the induced homeomorphism-group topology by an explicit classification of the transitive subgroups.

[62] Naganori Yamaguchi. Center-freeness of finite-step solvable groups arising from anabelian geometry. arXiv:2601.07112v4, 2026. Verified: supplied primary TeX and primary arXiv HTML opened. Proposition 1.3, Theorem 1.5 in finite rank, Corollary 1.6, Proposition 2.6, Theorem 2.9 and Corollary 2.10 checked. A.17 independently proves the precise finite-cover and transfer steps used. A.19 independently proves the completed Magnus embedding it needs and strengthens the closed characteristic-zero case to slimness. No infinite-rank argument from this preprint is used.